VISAYA

FoamDDI™

Foam Digital Detection Imaging Device

User Manual

August 2025

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Lubricant 1.17.8+
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Document Revisions

Date	Version Number	Document Changes
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1/28/2021	0.2	Gen 2 updates
7/23/2021	0.3	Added Standard Package Contents
2/8/2024	1.6	Added spares and consumables to end of manual
5/31/2024	1.7	Added Seq IV and revised Unpacking
6/3/2024	1.8	Updated software sections
8/15/2025	2.0	Full rewrite, including coolant and amines protocols

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1. General Information

The Foam Digital Detection Imaging Device (FoamDDI) is one of many high-tech optical-powered devices in Ayalytical's suite of visual testing instruments. The FoamDDI is powered by sophisticated optics and improves the visual rating of manual foam analysis methods like ASTM D892, ASTM D6082, and ASTM D1881.

Foam testing is a critical indicator of the system's operability for lubricants and coolants. The FoamDDI is a new standard of foam analysis for the lubricant, coolant, and aminic systems using a high-resolution camera with optical intelligence. The FoamDDI tracks static and dynamic foam throughout the test and uses a high contrast vision system to measure the foam height and collapse. Results are then digitally recorded and seamlessly integrated with LIMS software.

The FoamDDI's higher precision, patented design eliminates guesswork and operator bias from foam analyses. The digital detection imaging is achieved through a unique vision algorithm and dual camera system that records, calculates, and displays accurate foam levels and calculations. This standardized measurement of foam qualities improves on the current method and sample handling procedures performed in the current lab tests.

This user manual contains all of the essential information required for standard installation, operation, and maintenance. Please contact your local supplier or VISAYA for information not included in this manual. For the latest versions of documents and firmware, navigate to section 9 for instructions on retrieving updates, software, and documentation.

While every precaution has been taken to ensure adequate and sufficient preparation of this document, VISAYA does not assume any responsibility for technical or printing errors or omissions. Nor is any liability assumed for damages resulting from the use of the information contained within this manual.

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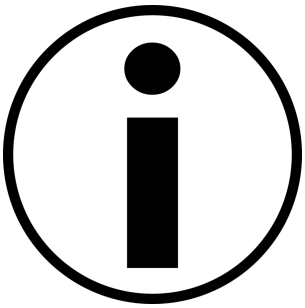
1.1. Safety

This document does not claim to address all of the safety concerns associated with the use of the FoamDDI and its samples. It is the responsibility of the operators to establish health and safety practices and determine the applicability of regulatory limitations prior to use.



Make sure you read and understand all instructions, guidelines, and safety precautions listed in this manual before installing or operating the equipment to ensure proper operation and avoid damage to the equipment or its accessories. If a malfunction occurs, consult the *Errors and Warnings* section. If the problem persists or if there are any questions regarding the operation of the equipment or information in this manual, please contact your local distributor or VISAYA.

Follow the precautions below for safe operation:

- The instrument described in this document should be used for foam analysis operations only. The device should only be operated in a laboratory or similar indoor environment for analytical purposes by qualified personnel. If an instrument is used in a manner not specified by VISAYA and this operation manual, the warranty provided by the manufacturer may be voided.
- Never operate equipment that is not correctly installed or fully operational. Ensure all operators are fully trained to use the analyzer correctly and safely. Unqualified personnel should not operate the equipment.
- It is not advised to remove the chassis to access internal components of the analyzer unless you are qualified to provide service to the equipment. Do not operate the analyzer without the chassis properly installed. Always disconnect the power cord of the equipment before handling internal components to ensure safe handling. Do not use any accessories or wearing parts other than those supplied or approved by the manufacturer. If a problem persists, please call the local distributor or VISAYA. Repair and service should only be carried out by authorized personnel or by the manufacturer.
- When the FoamDDI is functioning, it is necessary to check that nothing hinders the instrument parts in any way. Nothing should be placed in or around the IPC Logic Box Controller and the Testing Modules.
- Voltages present inside the instrument are potentially dangerous. If there is a problem with the instrument, the power cable should be removed until qualified service personnel have repaired it. This is to prevent anyone from inadvertently using the instrument, thus causing possible harm to themselves or damage to the

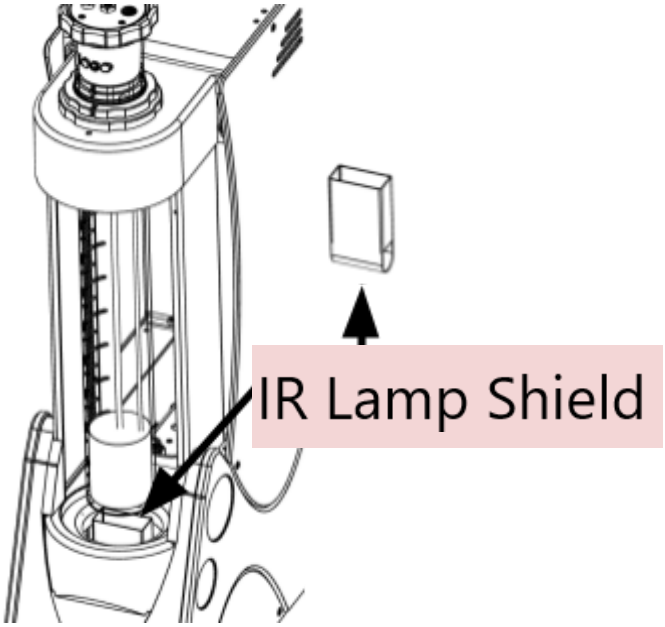
instrument itself.

- The leakage current of this instrument is within the limits allowed by international safety standards for laboratory equipment. An efficient ground connection is imperative for the physical protection of the user.
- Power supply cord reference 7080316106 is for use in France and Germany. Power supply cord reference 7080316105 is for use in the USA and Canada. For other countries, contact your local VISAYA distributor. You must only use the type of fuse described in this document: 4.0 Amp type “T” slow blow.
- Adequate protection, including clothing and ventilation, must be provided if dangerous liquids are used in the analytical work. In case of incidental spillage, carefully wipe with a dry cloth, considering the nature of the spilled liquid and the necessary safety precautions.
- Cleaning, dismantling, maintenance, adjustment, and repair should only be performed by personnel trained in such work and who are aware of the possible dangers involved. These instruments must not be sterilized using an autoclave or any other mechanical device. When you need to clean an instrument, use one of the three following methods:
    - a clean dry cloth,
    - a cloth dampened with water, or
    - a cloth dampened with soapy water.

  *If a cloth dampened with soapy water is used to clean the instrument, only domestic soap may be used. No other form of detergent or chemical may be used.

- The FoamDDI is supplied with a USB cable, which is specific to FoamDDI. The length is specified at 3’, and this cable does not have a +5V line. This cable should be used for proper operation and to conform to EMC and FCC directives and standards.
- The FoamDDI utilizes an IR Lamp to heat and maintain sample temperature. This IR source is located inside the base of the instrument, and shielded from direct line of sight - but some of this light is still visible while the instrument is actively heating. Exposure to this light is perfectly safe, even during long durations, while the instrument is assembled and operated correctly.

> ***WARNING***: Running the instrument without the IR Lamp Shield in place could allow the IR Lamp to become visible at some angles, in which direct eye contact could be harmful. Be sure to only run the instrument with the IR Lamp Shield in place.



- In addition to the IR Lamp, there are red-colored LEDs located at the base and the top of the instrument. This light source is completely safe, even for long durations, and even during direct eye contact.

1.1.1. Lighting Information

- IR Lamp (Heat Source)
  - 110VAC Wavelength 620nm to 3800nm (1120nm peak)
  - 220VAC Wavelength 620nm to 3800nm (1160nm peak)
- Red LED (Illumination)
  - Wavelength 630nm

1.2. Warranty

Any product manufactured by VISAYA is sold to the purchaser on the following basis and none other. All implied warranties of merchantability and fitness for a particular purpose are hereby expressly excluded.

(a) Products are warranted against defects in material and workmanship for one year (Warranty Period) from the date of purchase.

(b) VISAYA shall either replace or repair (at its discretion) free of charge any VISAYA-manufactured product (or component or part thereof) which shall be returned to VISAYA within one year from the date of delivery.

(c) VISAYA shall either replace or repair (at its discretion) free of charge any VISAYA-manufactured product (or component or part thereof) previously repaired or replaced under VISAYA warranty, which shall be returned to VISAYA before the later of the balance of the original one-year warranty period or three (3) months after the date of repair or replacement

(d) VISAYA shall either replace or repair (at its discretion) free of charge any part (sold as such) manufactured by VISAYA which shall be returned to VISAYA within three (3) months from the date of delivery, but only if such product, component, or part does not comply with one of the expressed warranties stated above

(e) None of the obligations assigned to VISAYA by this paragraph shall apply to any products which have been repaired, altered, or modified at the hand of the customer, unless the customer has disclosed the nature and full extent to any proposed repair, alteration, or modification to VISAYA in writing before it has been effected, and obtained an undertaking from VISAYA that any outstanding warranty obligations in respect of such products will continue to apply to such products despite such repair or alteration. VISAYA retains an unfettered discretion to grant or refuse any such undertaking.

(f) None of the obligations of VISAYA in this warranty shall apply to any products which have been subjected to misuse, neglect, accident, or (without restricting the generality of the foregoing) any extreme environmental condition or improper handling.

(g) If defective products are returned to VISAYA, the costs of such return will be borne by the customer, and those of delivering the repaired or replacement products to the customer will be borne by VISAYA. VISAYA‘s sole liability hereunder shall be to repair or replace any product that proves to be defective. In no event shall VISAYA be liable for any

special, incidental, consequential, indirect, or other similar damages arising from the failure of its products.

(h) Resale Products: Shall carry the warranty offered by VISAYA to the original purchaser, unless specified otherwise in the offering or separately and specifically agreed upon in writing by VISAYA.

1.2.1. Limitation of Remedies

Neither VISAYA nor the customer will be liable to the other for any special, incidental, consequential, indirect, or other similar damages arising from breach of warranty, indemnifications, breach of contract, negligence, strict liability, or any other kind of civil liability. VISAYA‘s total liability in connection with this agreement shall in no event exceed the amount paid or payable by the Customer in respect of the product that is the cause of such liability on the part of the company.

1.2.2. Obligation of the Customer

The customer agrees, as a condition of purchase of a product from VISAYA, not to reproduce any part of the product. The customer also agrees not to make unauthorized use of the proprietary information that belongs to VISAYA.

1.3. Copyright

The FoamDDI and this manual are protected by copyright ©. Reproduction of the unit will result in prosecution. All rights to the manual are reserved. Reproduction of this manual in any form, including in the form of excerpts, shall require express written permission from the copyright owner.

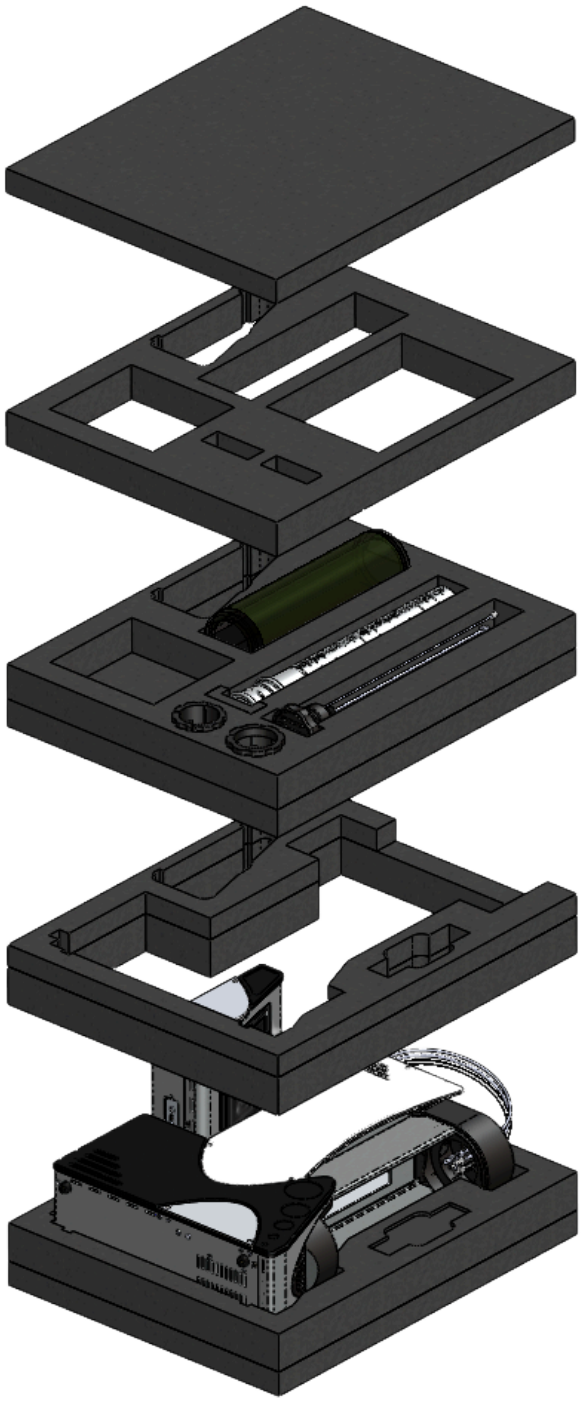
VISAYA reserves the right to make technical and content changes, including without prior notice.

1.4. Disposal

This product shall not be disposed of as municipal waste. Please return this product to VISAYA for disposal. For detailed information, please contact VISAYA before returning this instrument.

2. Unboxing

The standard FoamDDI package comes complete with all necessary accessories for easy installation and quick operation. During the unpacking procedure, ensure safe handling to not damage the instrument or injure the handler.



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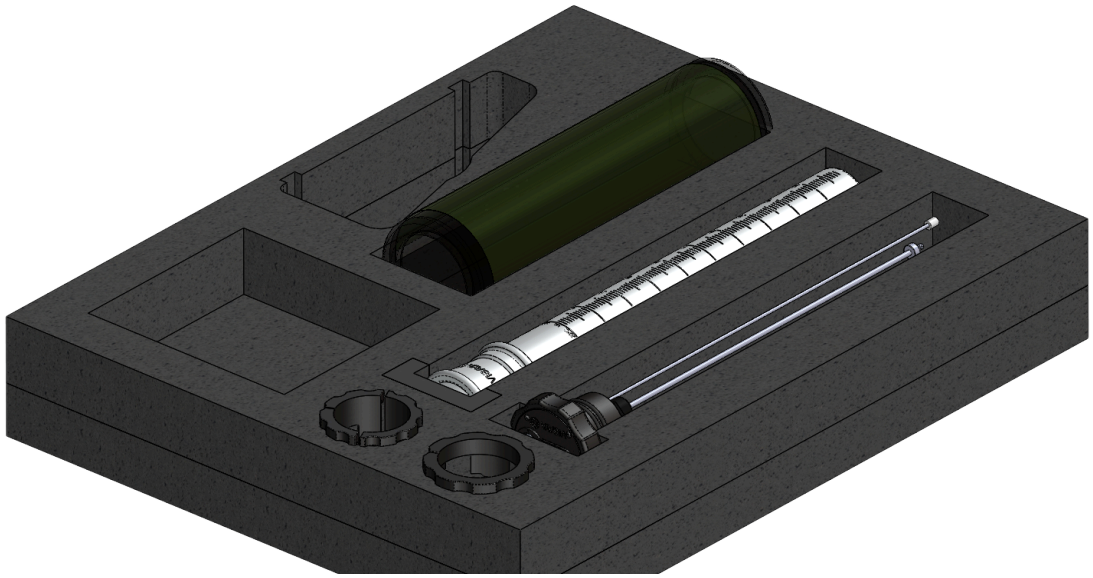
Inspect the shipping container for damage before opening.

- Place the box facing up
- Open the box and remove the top protective layer
- Remove the second protective layer and take out the lightest side (accessories) of the package.
- Remove the middle layer of foam, exposing the main unit and IPC Logic Box (if purchased).
- Take out main unit from the foam inserts and place it standing up.
- You will find the IPC logic box beside the FoamDDI unit. Remove it as you did the FoamDDI in the previous steps. Place it on a table.



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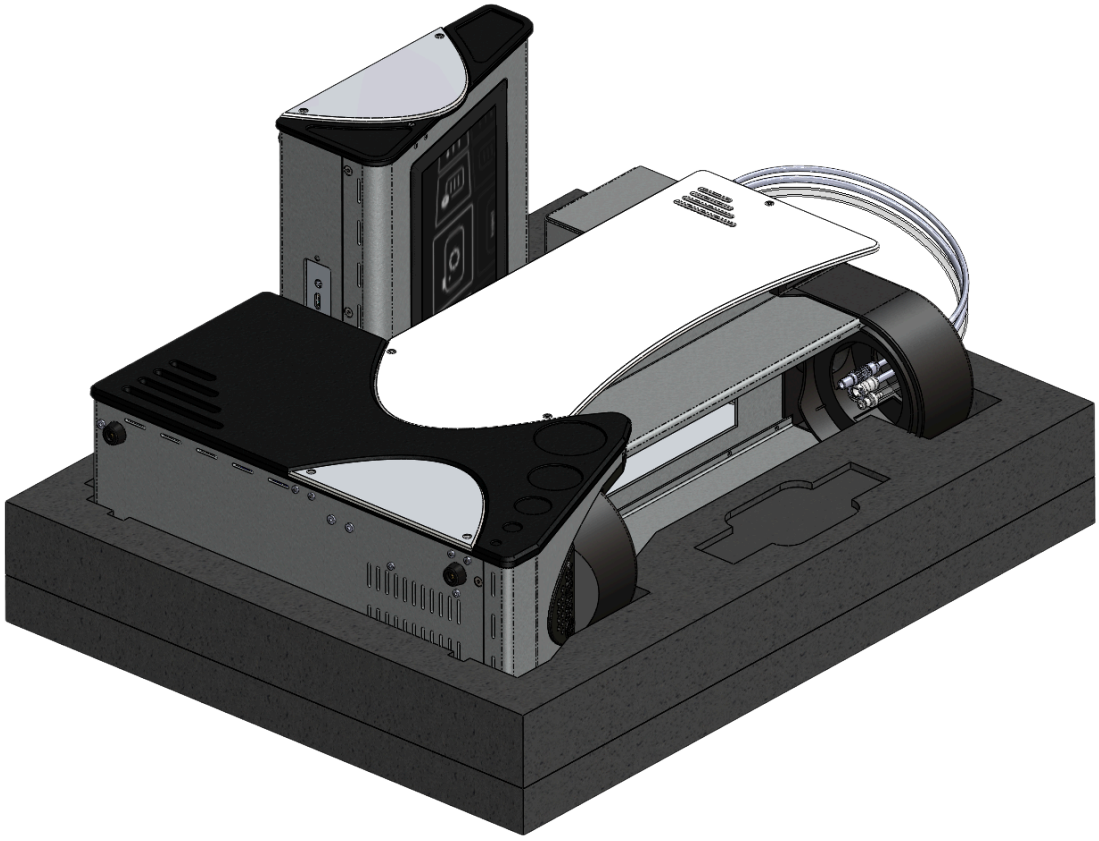

Note: Please save shipping boxes and custom foam inserts, as they may serve for future transportation and securing of contents.

2.1. Parts on Arrival

The FoamDDI should have arrived with the parts listed below:

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Part Number	Photo	Description
AYA-13-88696		Graduated Sample Cylinder (1000 mL) - D892, D6082, General
AYA-13-88683		Graduated Sample Cylinder (500 mL) - D1881, Amine

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AYA-13-88942		Stopper & Temperature Sensor Assembly (Lubricant, General, 1000 mL)
AYA-13-88944		Stopper & Temperature Sensor Assembly (Amine, Coolant, 500 mL)
AYA-13-88745		Outer Glass V2
AYA-13-88647		Adjustable Sample Tube Ring Part A
AYA-13-88844		Adjustable Sample Tube Ring Part B
AYA-13-88650		USB Cable (custom, 3', no +5V line)

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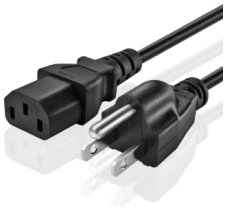
N/A		Power Cord
AYA-13-89200		Allen Wrench (3×25×127 mm)
AYA-10-88605		IPC Logic Box
AYA-13-89027		IPC Logic Box Power Supply
N/A		FoamDDI Manual
N/A		USB Memory Stick

If any of these items are missing or damaged, please contact your distributor right away.

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2.1.1. Optional Accessories and Consumables

Optional accessories and consumables are shown below:

Part Number	Photo	Description
AYA-52-50231		Spherical Diffusers
AYA-52-50230		Cylindrical Diffusers
AYA-10-88606		StoneDDI Automated Diffuser Stone Calibration Device
AYA-52-50233		Four Position Cylinder Rack
AYA-15-00050		Air Supply Pump for FoamDDI, 110VAC, 60Hz
AYA-15-00052		Air Supply Pump for FoamDDI, 220VAC, 50Hz

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AYA-15-00055		Filter, Regulator and Dryer System for FoamDDI
AYA-13-89000		Air Line Connection Kit

2.2. Handling Accessories

Store accessories in a clean, dry environment. Glassware should be wrapped or boxed to prevent scratches.

Diffusers are known to break down over time. Ensure the diffusers being used are within specification for analyses.

For more information on diffuser specifications see below or check out the associated method.

2.2.1. Diffuser Specifications

2.2.1.1. ASTM D892 & ASTM D6082

ASTM D892 and D6082 largely share the same specifications for diffusers, though D6082 is more restrictive on shape.

Property	***Specification***
Material	Sintered stainless steel (required for D6082, optional for D892) or fused crystalline alumina grain (optional for D892).
Shape	Cylindrical (required for D6082) or spherical (permitted for D892 only).
Diameter	25.4 mm (1 in.).

Maximum Pore Diameter	Not greater than 80 µm for the alumina stone; 15 to 60 µm for the stainless steel diffuser (as tested per the standard's annex method).
Permeability	3000 to 6000 mL of air/min at a pressure of 2.45 kPa (250 mm of water).

2.2.1.2. ASTM D1881

ASTM D1881 uses a different material and shape requirement from D6082 and specifies a slightly different permeability range than D892.

Property	***Specification***
Material	Fused crystalline alumina grain.
Shape	Spherical.
Diameter	25.4 mm (1 in.).
Maximum Pore Diameter	Not greater than 80 µm (as tested per the standard's annex method).
Permeability	3000 to 6400 mL of air/min at a pressure of 2.45 kPa (250 mm of water).

Note: Diffuser performance (porosity and permeability) can change with use, so the test methods require periodic verification of the specifications using a Gas Diffuser Verification Device.

2.3. Transportation

Although FoamDDI is designed to withstand general vibration, it is a sensitive optical instrument. Always reuse the original packaging materials when transporting to avoid misalignment of the camera optics and internal assemblies.

To ensure that the instrument is not damaged, it is recommended to retain and reuse the shipping boxes and foam packing materials, as they are designed to provide optimal protection for the equipment during transport in cases such as service.

3. Introduction

Foam, in this context, refers to a gas-liquid colloidal system containing bubbles. Despite its thermodynamically unstable nature, foam plays a crucial role in several manufacturing processes. However, its presence can be detrimental, particularly in industrial procedures where lubrication is essential for reducing metal-on-metal friction, where foam inhibits sweetening processes using amines, and where foam decreases the efficacy in systems where glycols operate as coolants. For instance, excessive foaming can cause operational inefficiencies and severe mechanical damage.

3.1. Basis for Development

Foam formation in liquids can cause operational issues, including overflow, reduced effectiveness, cavitation, and poor lubrication efficiency. Traditional ASTM methods rely on visual inspection, introducing subjectivity and operator bias.

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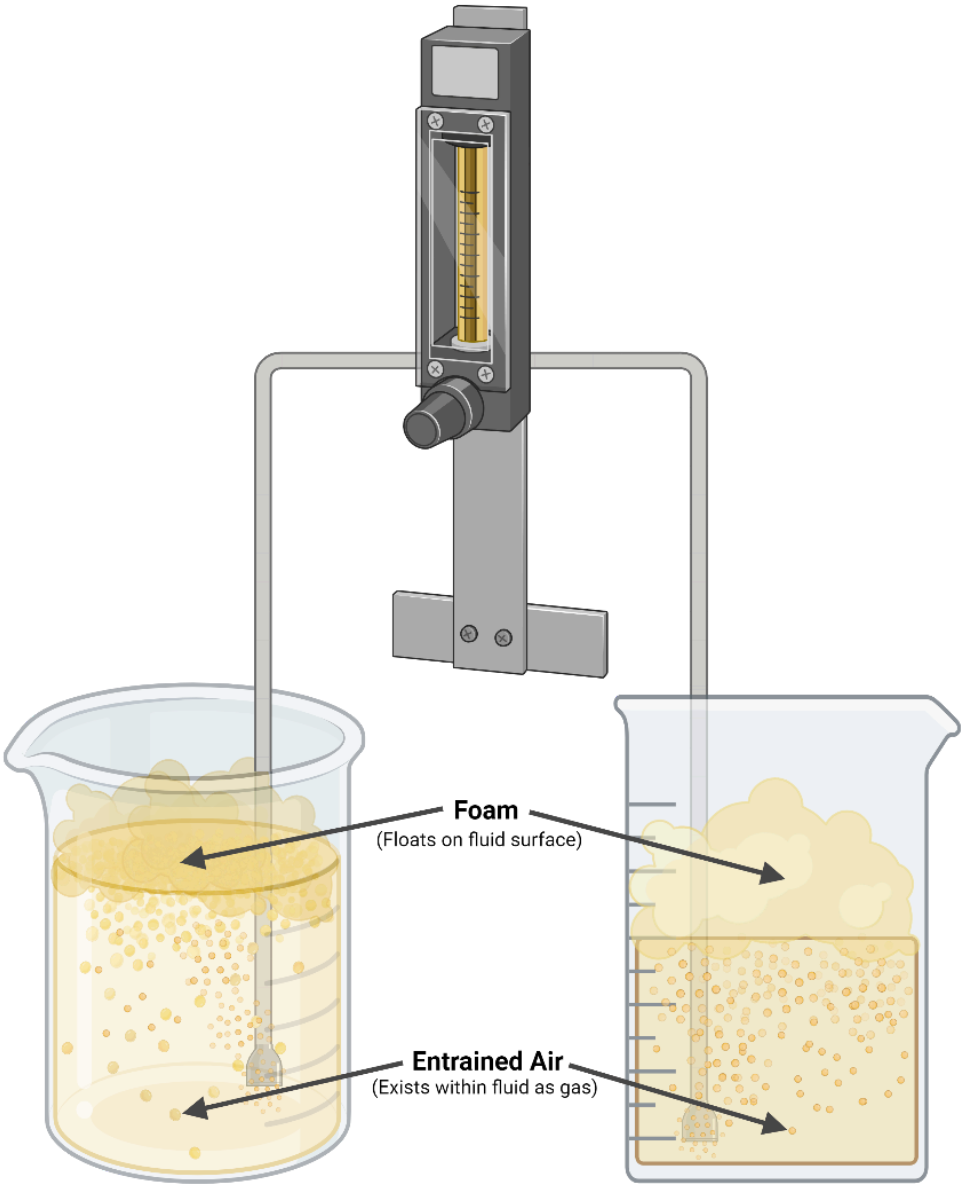

The FoamDDI works to eliminate this variability and subjectivity by using an automated imaging and algorithmic analysis system. VISAYA's advanced optics and proprietary recognition software provide precise measurement of foam tendency, stability, and collapse.

The system complies with ASTM D892 (Foaming Characteristics of Lubricating Oils) and ASTM D6082 (Foaming Characteristics at High Temperature), and ASTM D1881 (Foaming Tendencies of Engine Coolants), and is applicable for other liquids like amines in sweetening systems while extending capability with real-time video, playback, and LIMS-compatible reporting.

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Testing the foam in lubricants and transformer oils is crucial to ensure equipment longevity and safety by identifying problems that foam can cause, such as reduced lubrication, increased oxidation, and poor heat dissipation. Foaming can indicate issues like contamination, mechanical problems, or depleted additives, leading to system failures like cavitation, overheating, and component damage.

Testing coolant for foam using ASTM D1881 is crucial because excess foam can cause a loss of cooling efficiency and poor heat transfer, leading to engine overheating. Excessive foaming indicates the coolant has a higher tendency to foam, which can impair its ability to properly circulate and protect the engine. The test distinguishes between coolants that are stable and those with a tendency to foam excessively.

Testing amine foam is crucial to identify contamination and control operational problems in gas sweetening systems like reduced efficiency, excessive amine loss, and off-spec treated gas. By understanding the foaming tendency of the solution, operators can prevent and manage foam formation, which can be caused by hydrocarbons, suspended solids, and amine degradation, ensuring the system operates efficiently and reliably

Excess foam in any of these scenarios can lead to equipment failure or unoptimized processes. The FoamDDI was designed to provide comprehensive data to evaluate the foaming tendency of various solutions.

3.2. Measurement Principle

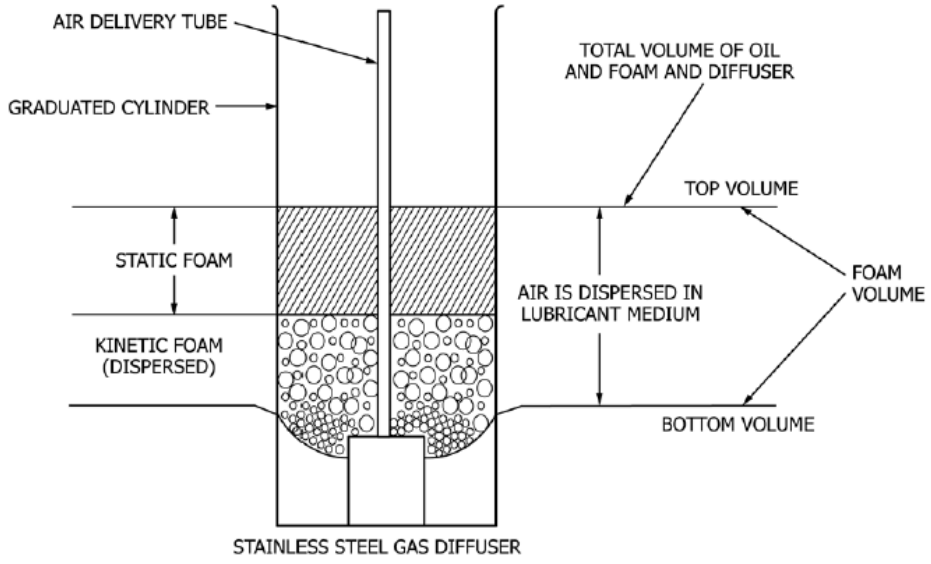
The FoamDDI operates by combining controlled air injection, precise temperature regulation, and automated optical analysis to measure foam generation and dissipation. There are two key variables to focus on in these tests.

Firstly, the foaming tendency or foamability, which assesses the quantity of foam produced and what remains immediately after air introduction.

Secondly, foam stability, which gauges the interfacial surface tension strength through the bubbles' resistance to combining and breaking.

Conforming strictly to ASTM D892, D6082, and D1881 standards, as well as incorporating other methods based on standards of operations for non-standardized methods in the industry, the FoamDDI accurately measures the foaming tendency and stability of liquids at varying temperatures. Its high-resolution dual camera system and state-of-the-art optics digitally measure static and dynamic foam heights with precision, thus minimizing biases and errors associated with human eye measurements. Moreover, the FoamDDI

offers optimized control of temperature, air flow, and sequences with minimal user intervention.



3.2.1. Analysis Process

The sample is introduced into a graduated cylinder and sealed with a specialized stopper.

The test is initiated via touchscreen interface.

LED backlight activates for consistent illumination.

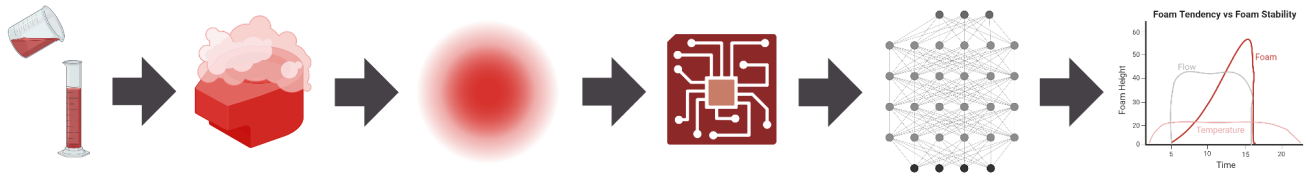
Controlled air is diffused through a submerged porous diffuser to generate foam.

High-resolution cameras and LED illumination capture foam height and stability in real-time.

VISAYA's vision algorithm identifies the foam/solution boundary using edge-contrast detection and predictive analytics.

Data tabulated in real-time. Foam tendency and stability are calculated automatically and tabulated in compliance with ASTM methods.

Results are displayed, stored, and optionally exported to LIMS.



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3.3. Specifications

Purpose	Foam Properties of a Fluid
Technique	Automated vision (edge contrast algorithm)
Test Methods	D892, D6082, D1881
Applications	Lubricants, Amines, Coolants
Temperature Control	up to 150 °C
Optical System	High-quality dual CCD camera system
Lighting Source	IR Lamp and Red LEDs
Heat Source	IR Lamp
Measurement Time	Sequence I about 40 minutes; II about 30 minutes; III about 60 minutes; II + III about 90 minutes; IV with cooling about 90 minutes at 10 min heating. Under Amine and Coolant settings, the test timelines is usually between 30 to 60 minutes. Other sequences are dependent on parameters set forth.
Air Supply	25-30 psi, clean, dry air
Interfaces	USB, HDMI, Ethernet
Software	Version 17.5
Storage Database	64 GB SSD Storage
Data Formats	PDF, CSV, TXT, XML, JPG, MP4
Output	Digital images and video, graphs, tabulated foam results
Export	PDF Summary, XML, images, videos, and graphs
Ambient Conditions	15–25 °C
Relative Humidity	30–80% (non-condensing)
Power Requirements	FoamDDI: 110/120VAC, 50/60Hz 3.2A or 220/230 VAC 50/60 Hz 1.6A, IPC Logic Control Box: Auto-switching 90 ~ 264VAC, 47 ~ 63Hz, 280 Watt Power Supply
Dimensions	Module: 20 x 9 x 25 IPC: 12 x 10 x 6 Box: 30 x 24 x 19
Weight	Gross: 53 lbs Net: 48 lbs

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4. Installation

The FoamDDI is designed for intuitive setup. Installation should be carried out by trained personnel to ensure safety and instrument performance.

4.1. Operating Environment

Operate the system in a stable, non-condensing environment with 30–80% relative humidity, at 15–25 °C, ensuring bench stability and protection against electrostatic discharge.

The FoamDDI system requires a power supply of either 110/120 VAC at 50/60 Hz, drawing 3.2 A, or 220/230 VAC at 50/60 Hz, drawing 1.6 A.

The IPC Logic Control Box is equipped with an auto-switching power supply that operates within the range of 90 to 264 VAC, 47 to 63 Hz, and provides 280 watts of power.

The system must be connected to a clean, dry, instrument-grade air supply with a pressure of 25 to 30 psi.

For safe operation, the FoamDDI should be placed in a fume hood or in a well-ventilated laboratory area.

4.2. Dimensions and Space Requirements

The FoamDDI system requires adequate bench space to accommodate both the FoamDDI unit and the Logic Box. Sufficient rear clearance must be allowed for tubing connections and power supply access to the FoamDDI unit, as well as power access for the Logic Box.

The FoamDDI module measures 20 × 9 × 25 inches. The IPC measures 12 × 10 × 6 inches. The shipping box measures 30 × 24 × 19 inches. The FoamDDI module and the Logic Box require a minimum space of 30 x 30 x 60 together.

4.3. Identifying Parts and Controls

FoamDDI consists of two main components, an IPC Logic Box Controller and FoamDDI modules. Each IPC box can control up to two FoamDDI modules and an additional

optional calibration module. Place FoamDDI and Logic Box devices on a stable bench in the designated laboratory space.

The IPC Logic Box Controller features an embedded grade computer with easy-to-use software. Interface to the software is achieved using an industrial-grade, large 10.5-inch LCD with an integrated touch controller. The computer, which runs an embedded version of Windows 10 Professional, allows for easy integration with network and modern LIMS systems. On the front view of the FoamDDI's IPC LogicBox is an auxiliary USB port. The USB port can be used to download data, software updates, or external peripheral devices. This port is USB 2.0 compatible only.

Each FoamDDI module is a self-contained single-position testing apparatus for Foam Testing. Each module is capable of heating and cooling within the same module. Internally, there is automated temperature control, air flow control, and sequencing as per ASTM methods.

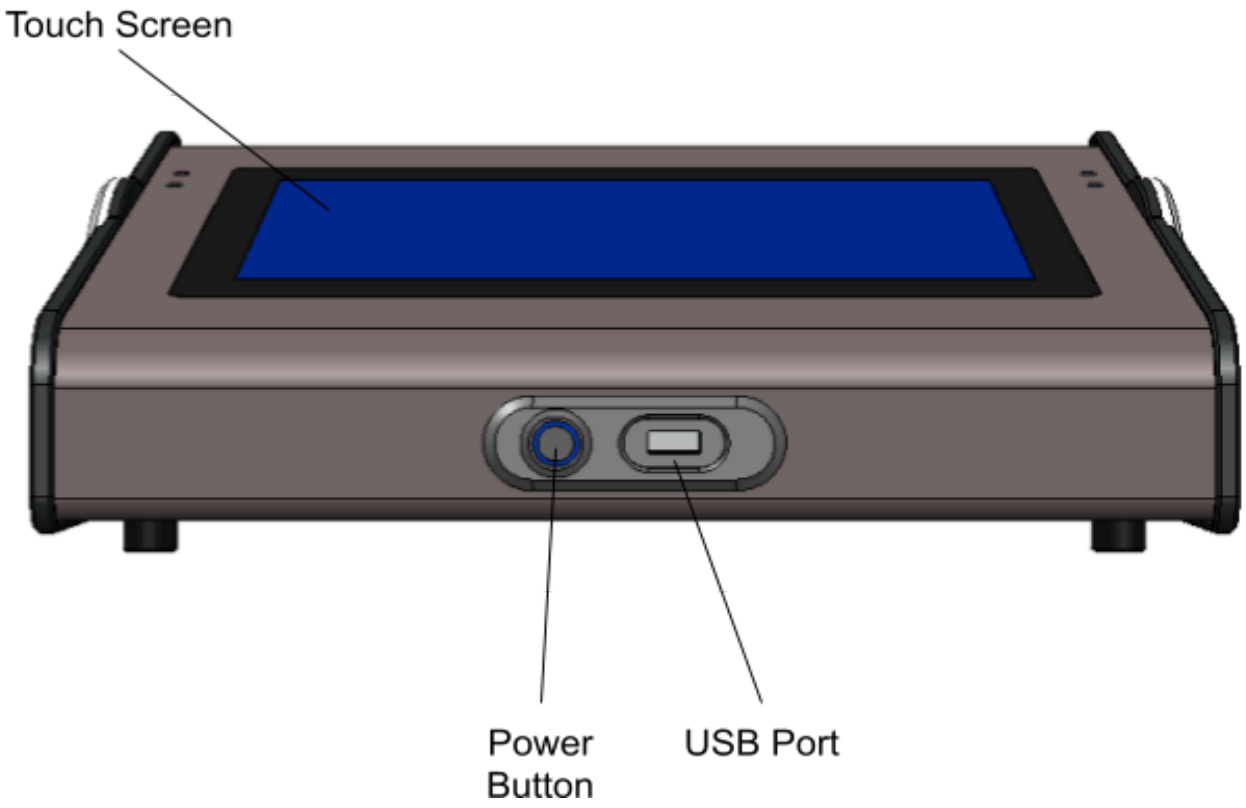
4.3.1. Logic Box

IPC LogicBox features two Ethernet (RJ-45) ports, HDMI display ports, an On/Off Button, a Front USB Port, and four Rear USB 3.0 ports. Dual Ethernet ports allow simultaneous dedicated LIMS network connectivity, while allowing the second port to be used for traditional network or internet access. Display ports can be used to output displays for external presentations or for dual monitor support. A FrontUSB Port can be used to connect external keyboards, printers, Wi-Fi adapters, memory sticks, and other USB

devices supported by Windows 10 Pro. The Rear USB 3.0 ports are reserved for connecting Visaya Modules (e.g., FoamDD).

Power supply to your IPC Box is achieved using an external voltage converter. This converter features a universal switching power supply, which can accept power from 90 - 264 VAC and frequencies ranging from 47 - 63 Hertz.

Ventilation for the IPC Box is achieved using multiple vents placed on the top rear panel, as pictured below, and on the bottom (not shown). Please allow a minimum clearance of 6” (15cm) for proper ventilation. The IPC Box uses this passive cooling system to maintain 24/7 operation.



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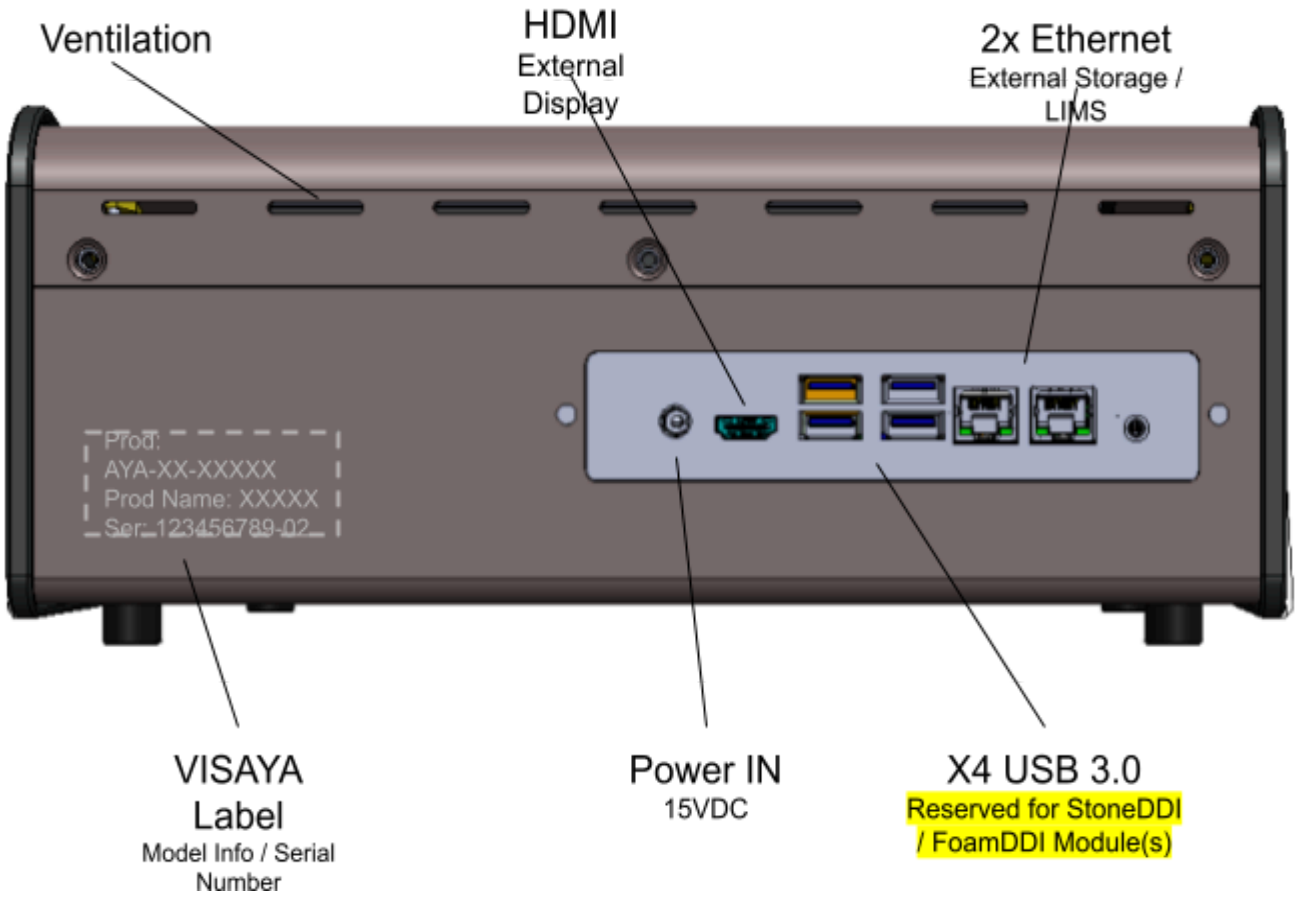
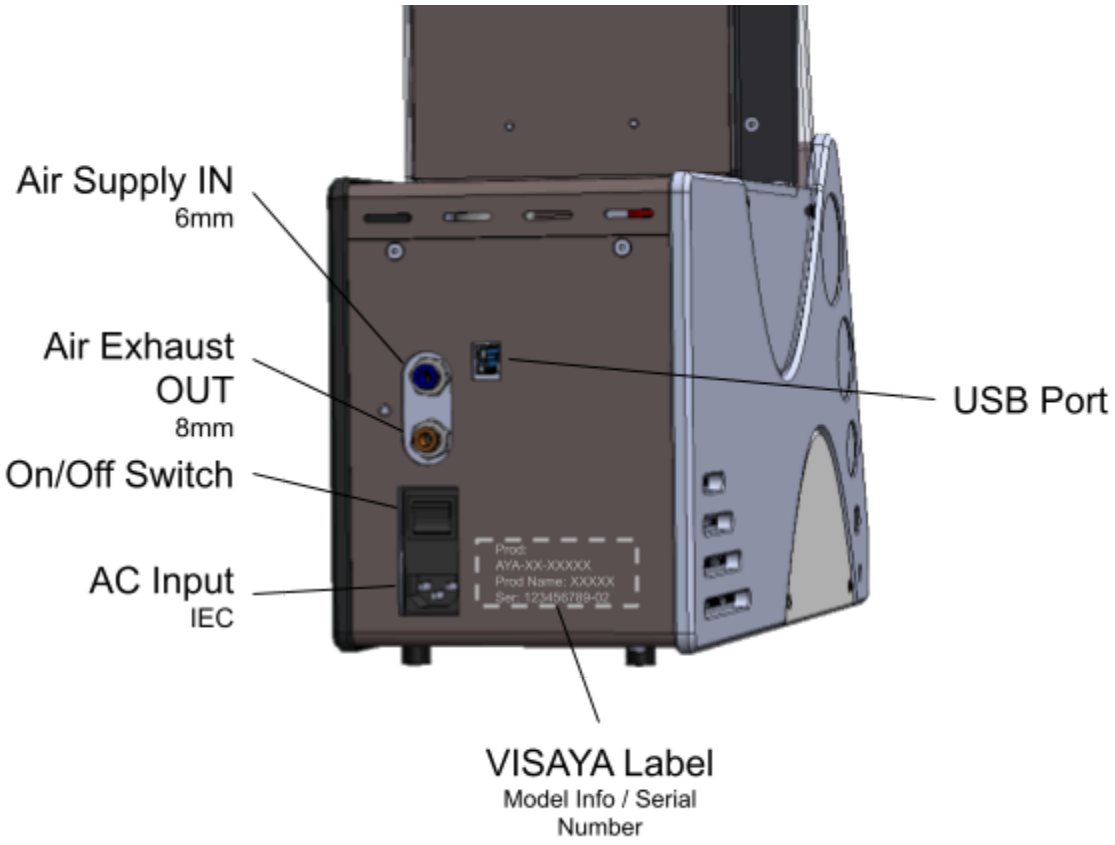


4.3.2. Module

FoamDDI Modules are powered using 110/120VAC, or 220/230VAC (50/60 Hz) main power supply. Proper fusing must be selected depending on the voltage. Please consult the image below for fuse selection based on voltage input. Air supply must be connected to the rear of the FoamDDI. Clean, dry instrument-grade air in the range of 20-25 psi pressure must be connected to the rear using 6mm OD Tubing. Optionally exhaust can be connected to the rear using 8mm tubing. Communication with the IPC Logic Box is achieved using the supplied USB cable.



Note: Changing voltage must be done by a trained VISAYA personnel. Damage may be caused by connecting the wrong voltage.

4.3.2.1. Power and Air Set Up

When setting up your new FoamDDI, select a location away from heat, smoke, electrical interference, and avoid any electrostatic discharges (ESD) to the analyzer unit. The bench top should be a level surface and free of vibration.

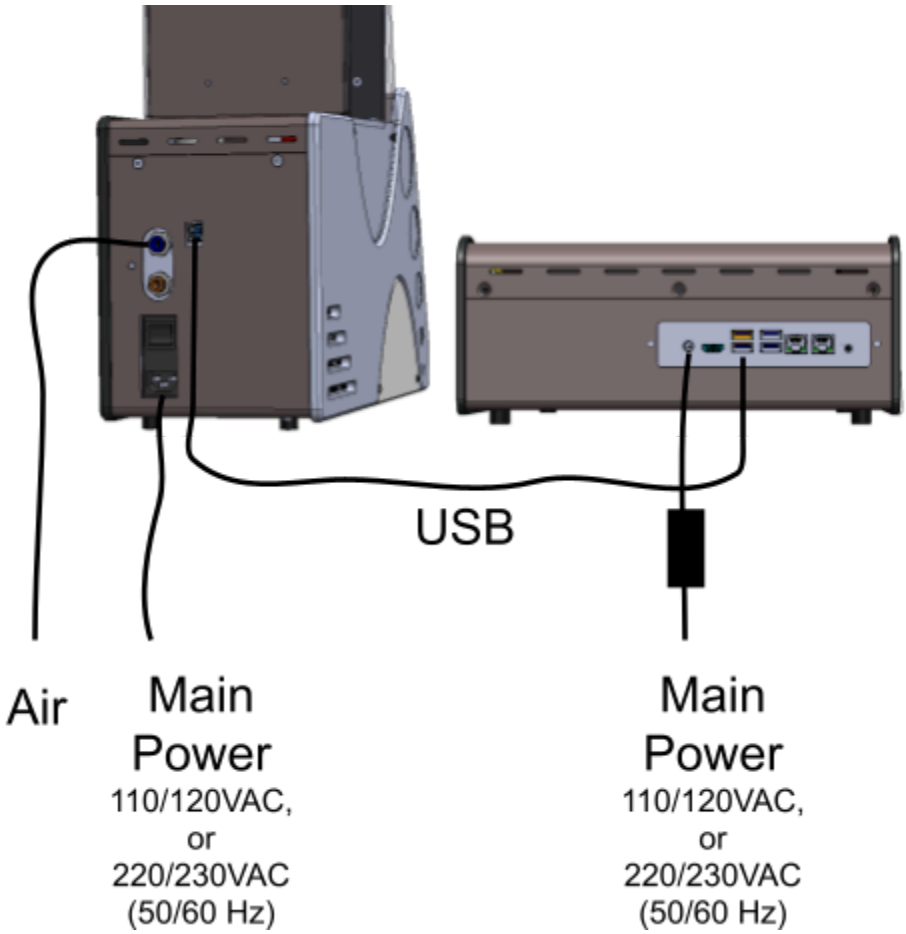
Ensure that the vent holes of the analyzer are not obstructed. The analyzer requires an environment with ambient room temperature in the range of 15°C to 25°C with 30-80% RH non-condensing.

CAUTION! Before making any connections, make sure that the power cable at the back of the analyzer is not plugged into an electrical outlet. Only one set-up utility is required: an A.C. electrical outlet. It is recommended that the analyzer be plugged into its own circuit to avoid power line interference and voltage surges or drops, which may occur if the circuit is shared with devices with high electrical draw.

1. Connect the FoamDDI to AC Power using the IEC Cordset provided.
2. Connect the IPC LogicBox to AC Power using the IEC/DC Power Supply provided.
3. Connect the FoamDDI to the IPC LogicBox using the provided USB Cable.
4. Connect a 25-30 psi pressure tube of dry, clean, instrument grade air to the rear of the FoamDDI module using 6mm OD rigid tubing. Optionally, exit air can be routed to fume hood or extraction using 8mm OD connection.
5. On the Rear of the FoamDDI, move the ON/OFF toggle switch to the ON Position. Power to the IPC LogicBox can be confirmed via the front Blue Pushbutton.

Additional connections such as Ethernet or printers can be done at this point.

Note: certain printers may require drivers, which can be loaded by an administrator. Please contact your distributor or VISAYA for more instructions.



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4.4. Getting Started

4.4.1. Installing Outer Vacuum Cylinder

Unwrap outer cylinder and stand upright.

Note: This part is shipped with a protective film over the polarizing film. Only remove this protective film prior to installation.

Locate top housing and wrench included in FoamDDI box.

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Loosen bolts halfway until the top housing can move up and down.

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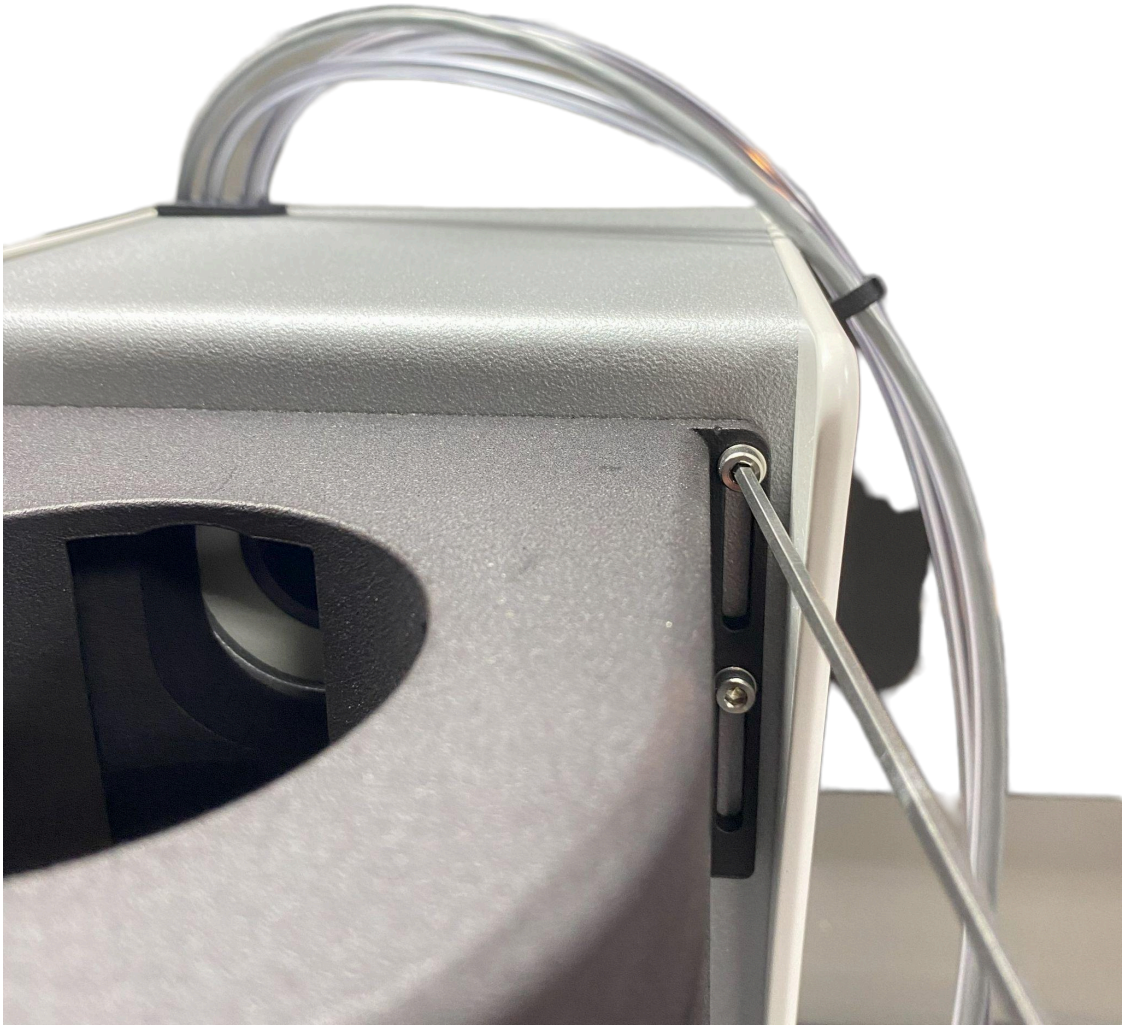

Note: DO NOT remove fully.

Place vacuum cylinder with rubber nipple up and facing inward.

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Note: Installing cylinder upside down will cause an obstruction of the camera.

4.4.2. Foam Tube Alignment

Locate the Foam Tube Alignment parts A and B.

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Insert and align Part A into Part B (Large or Small, depending on Grad Sample Cylinder size).

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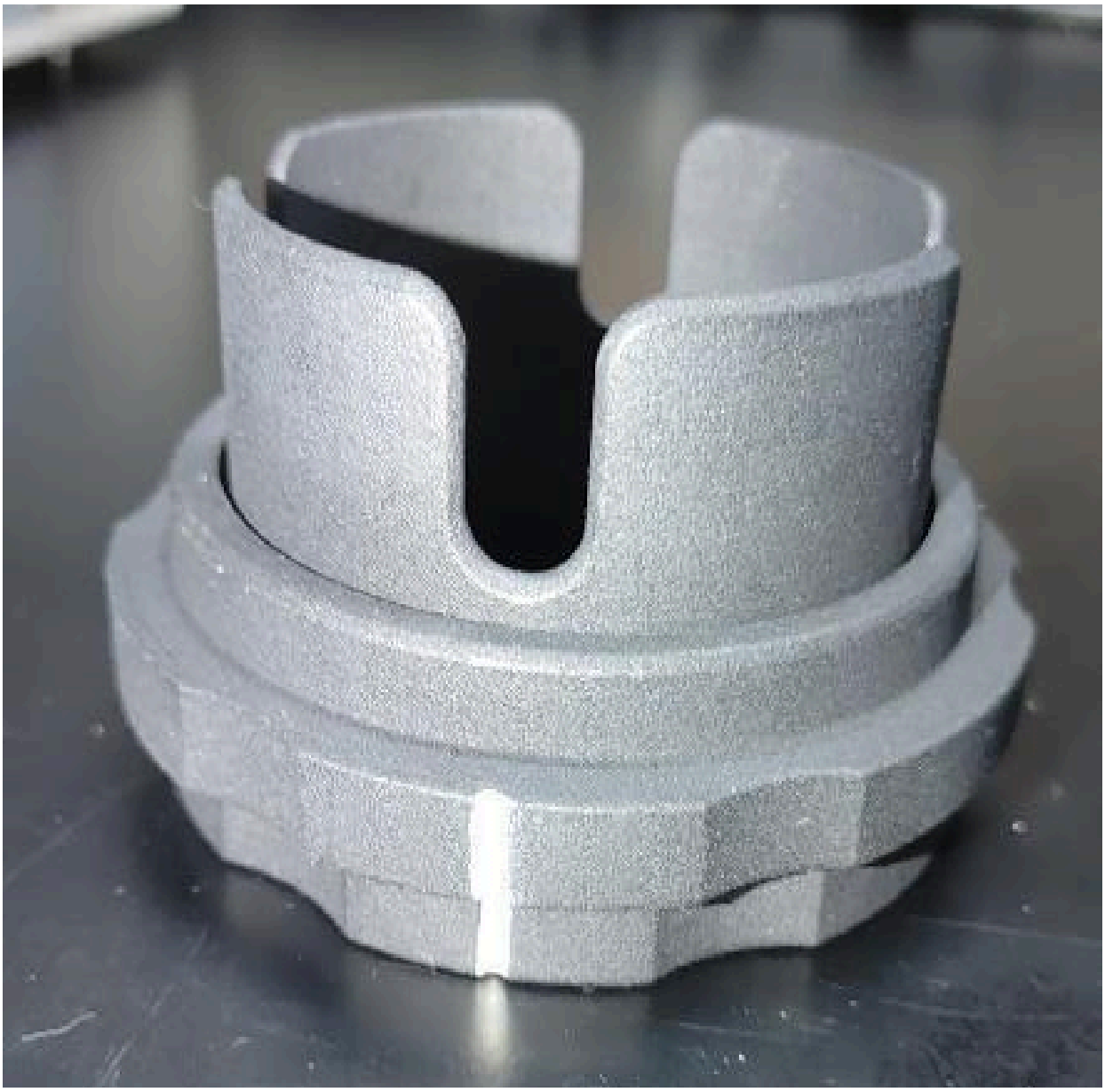

Place the Foam Tube Alignment with the longer side facing down on top of the cap aligned with the white mark.

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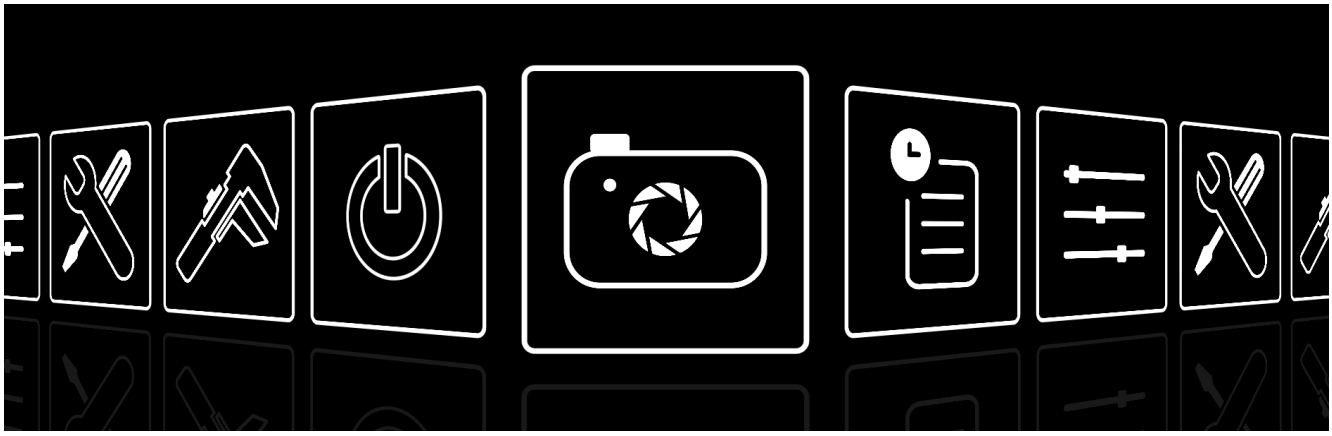
5. Software and Operation

The FoamDDI was designed with an internal high-end industrial computer, enabling advanced graphical interfaces and a smooth operator workflow. Using a touchscreen (P-Cap interface), the operator can navigate the menus with simple swipes or taps. External peripherals such as a mouse or keyboard are also supported.

After installation and power-up, the FoamDDI software automatically boots to the Main Screen. The interface is designed for ease of use and minimal training, featuring large, icon-based navigation similar to other VISAYA instruments.

Coolant, lubricant, and amine software all take the same format and navigation. Unless otherwise explicitly noted, all parts of this manual are the same across all versions.

5.1. Main Screen



The Main Screen is divided into six sections, each represented by an icon:

- Inspect: Begin inspection and run foam analysis.
- History: Archive and database of previous results.
- Settings: Adjust parameters such as date, time, language, LIMS, and printing.
- Diagnostics: Access mechanical, electrical, and I/O system checks.
- Calibration: Perform recalibration of flow, temperature, and camera systems.
- Power Options: Shut down or restart the FoamDDI IPC.

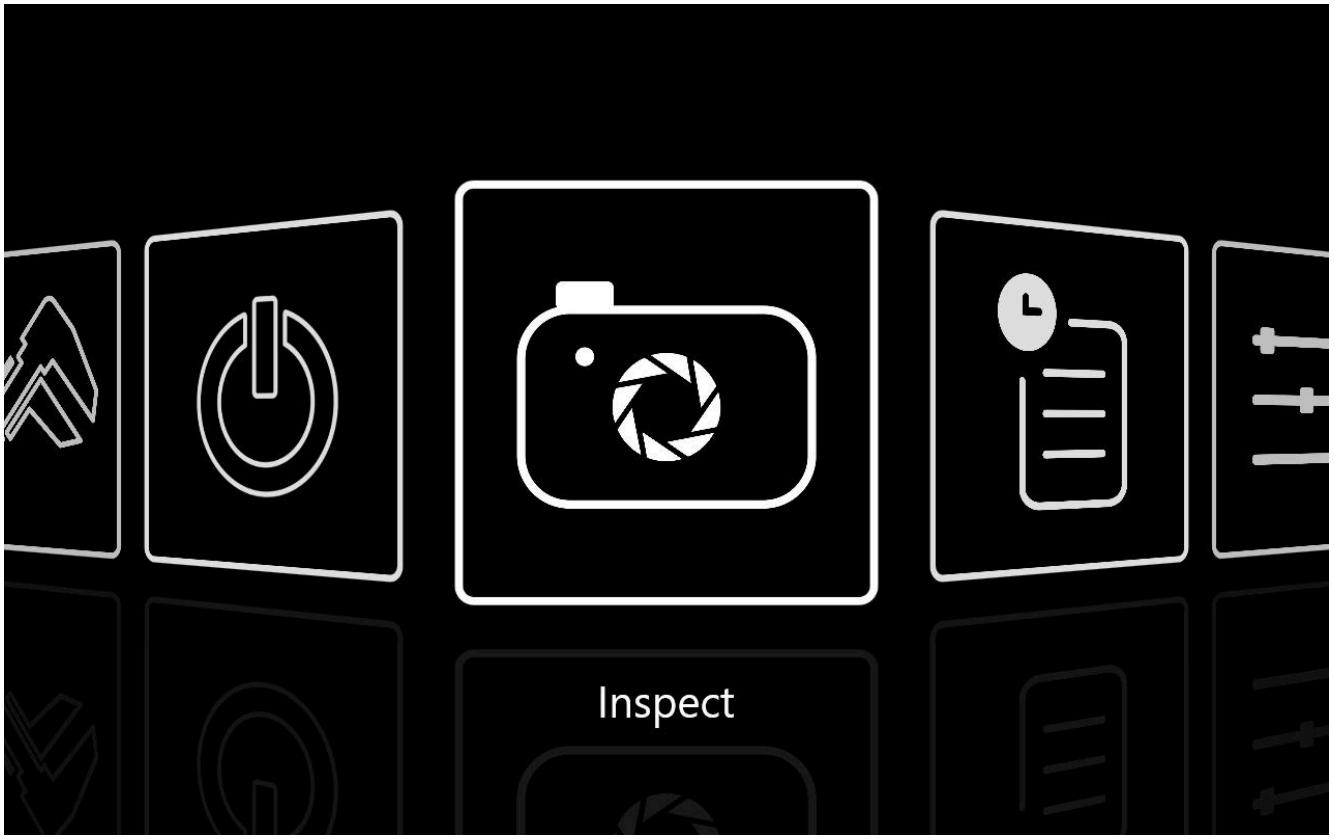
Navigation:

- Swipe left or right on the touchscreen to move between icons.
- Tap once to select a menu.
- Use the back arrow in the upper left corner to return to the Main Screen.
- For external peripherals (mouse, keyboard), the drag-and-drop and single-click functions are also supported.

This intuitive interface typically requires only minimal operator training, with most tasks guided by a wizard-driven setup that simplifies routine sample analysis.

5.2. Inspect

The Inspect screen is where an operator would perform the automated inspection of different samples. To start an inspection, simply tap on the Inspect icon from the main menu. All variations of foam analyses—lubricant, coolant, and aminic—can be started under this screen, depending on the analyzer configuration.



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5.2.1. Lubricants – ASTM D892 and D6082

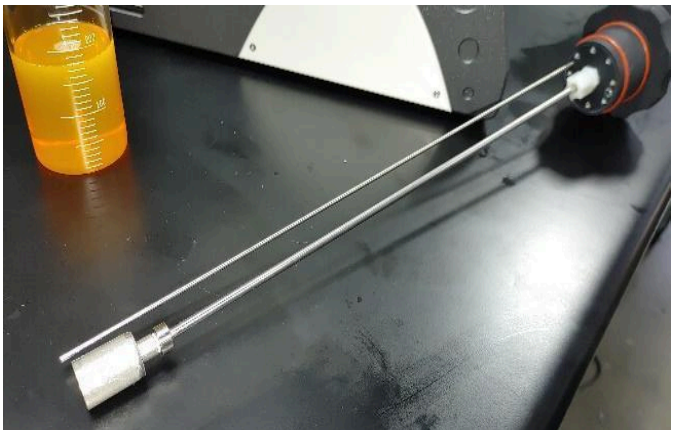
This section details the protocols and procedures to test to foamability of lubricants in accordance with ASTM D892 and D6082.

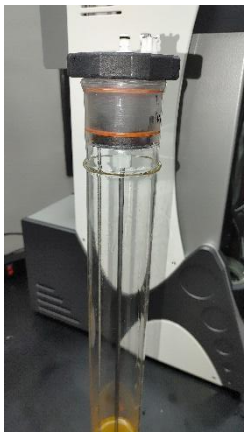
5.2.1.1. Sample and Cylinder Preparation - Oils

Ensure clean and dry glassware and wetted components (outer cylinder, graduated test cylinder, stopper). Add your lubricant sample to the 1000ml sample cylinder according to the associated test. Sequence-dependent volumes defined by ASTM D892 and D6082 are noted in the table below. Do not exceed ASTM-defined sample volumes.

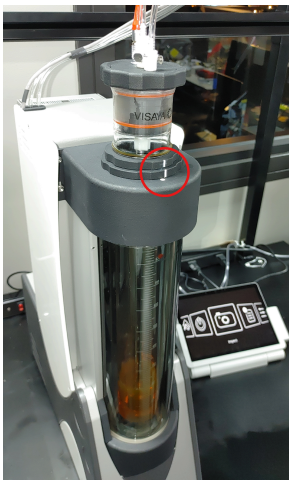
Seq I	Seq II	Seq III	Seq IV
190mL (+/- 5mL)	180mL (+/- 5mL)		200mL (+/- 5mL)

Screw the diffuser to the end of the stopper and insert the stopper into the sample cylinder. Ensure the stopper is well pressed (airtight) to the sample cylinder.

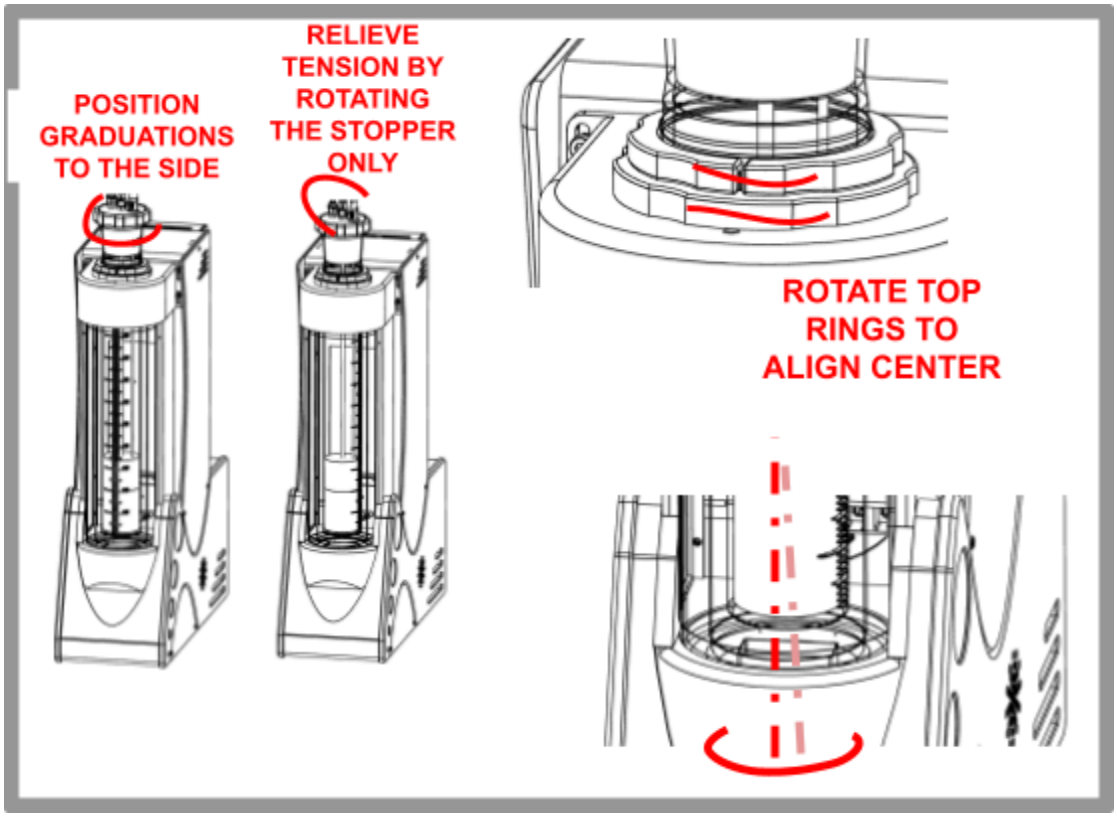




Insert the Sample Cylinder into the unit with white markings aligned. Make sure to connect the Probe and Air Tubes.



Adjust the cylinder such that the graduations are to the side. This is critical for the automatic foam analysis (vision system) to work properly. While doing so, you may have issues with the weight of the air and electrical, as they will tend to drop down and maybe rotate the cylinder back. To work around this, correct the glass rotation first, then rotate only the stopper until the weight of the connection is no longer working against you. Review the bottom of the cylinder inside the FoamDDI. The cylinder should be as close to the center as possible, and you can use the two stopper rings to correct it.



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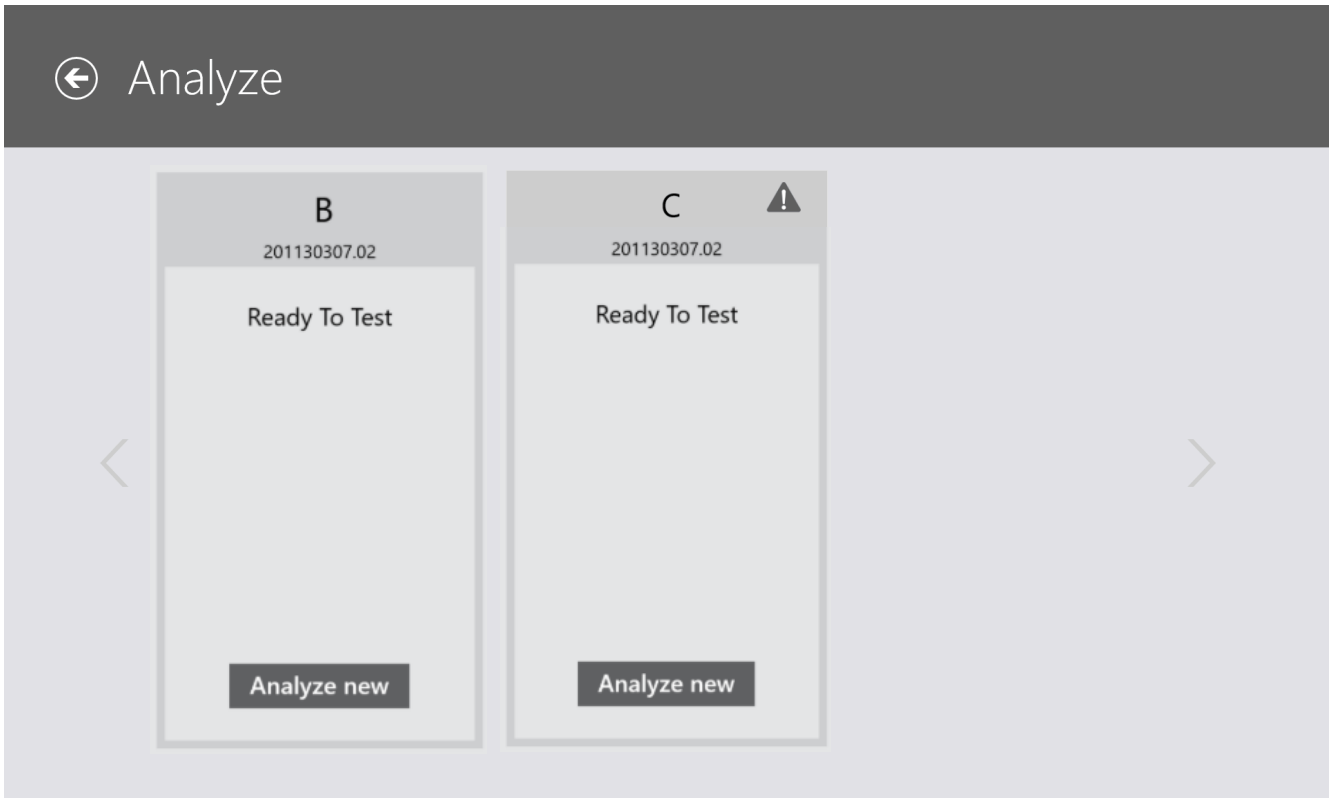
5.2.1.2. Running a Lubricant Analysis

Running an inspection is achieved using a simple Wizard-driven menu. Upon entering the Inspect menu, the user is presented with a box showing connected FoamDDI module(s).

Choose whichever module is to be used (if multiple are active) and select Analyze New. Checkout **Section 5.4.8** to connect multiple modules.

If two modules are connected, they will both appear side by side. Any warnings or errors will be indicated by the triangle containing an exclamation mark. Click on the warning icon or the Detail icon to find out more information. **Section 6** has more information on warnings and errors.



Once selecting Analyze New, two options will propagate: New or Continue. Choose New to run a completely new sample. The Continue option allows you to append to an existing sample ID, if one has started within the past 3 hours.

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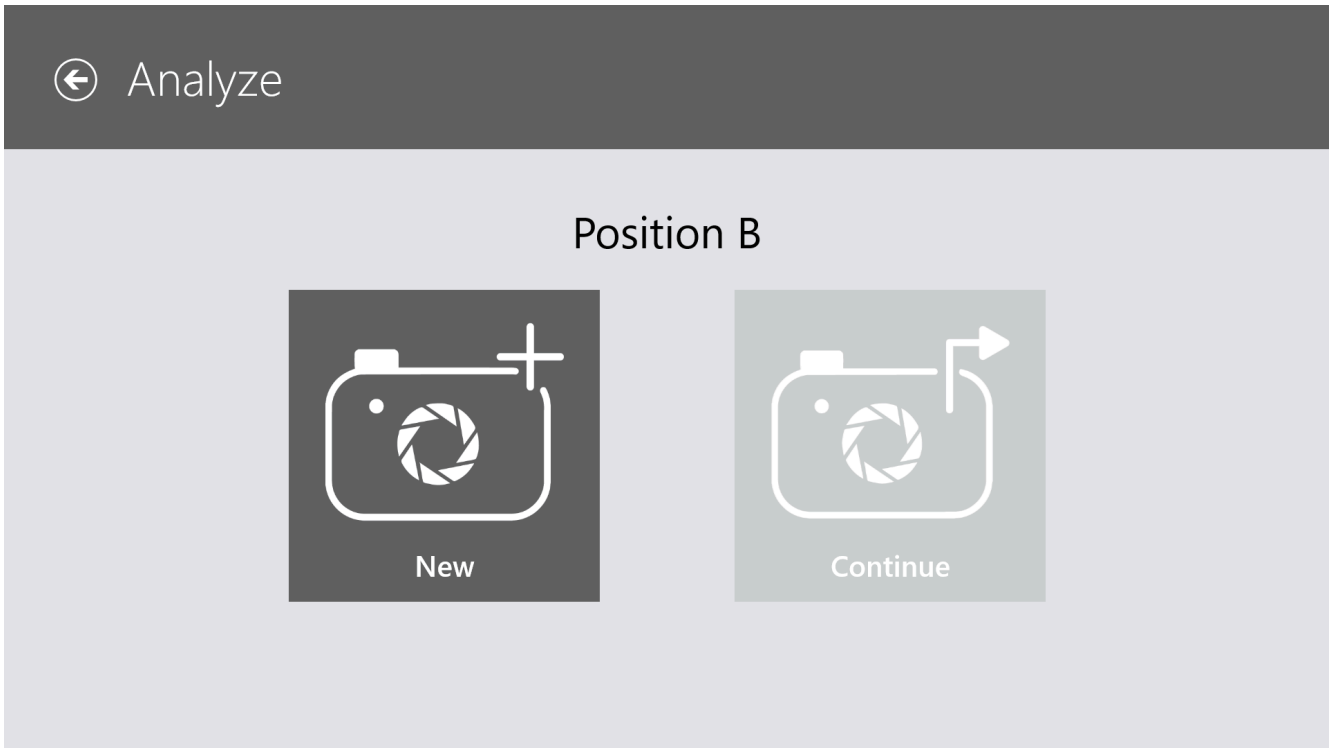

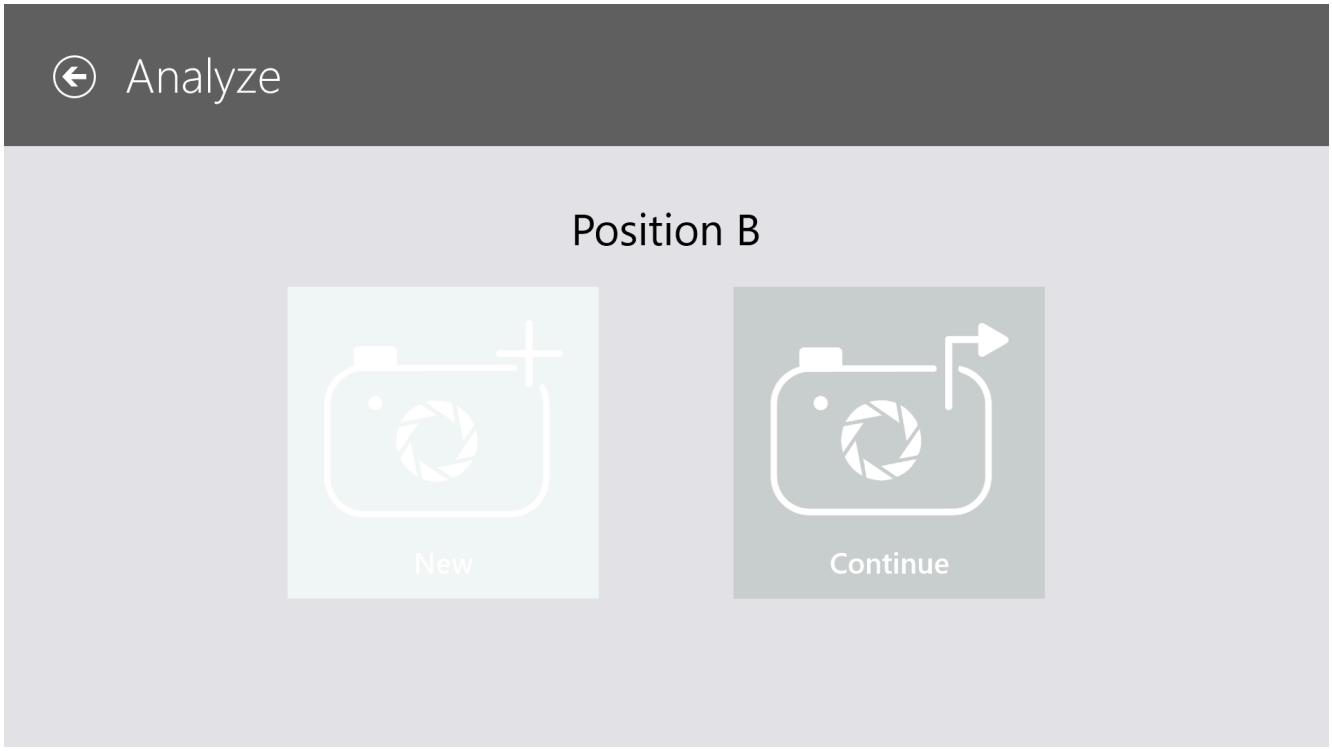

When starting a New sample, you will be asked to enter a sample name. Using the Continue option will propagate this information for the operator automatically. The user

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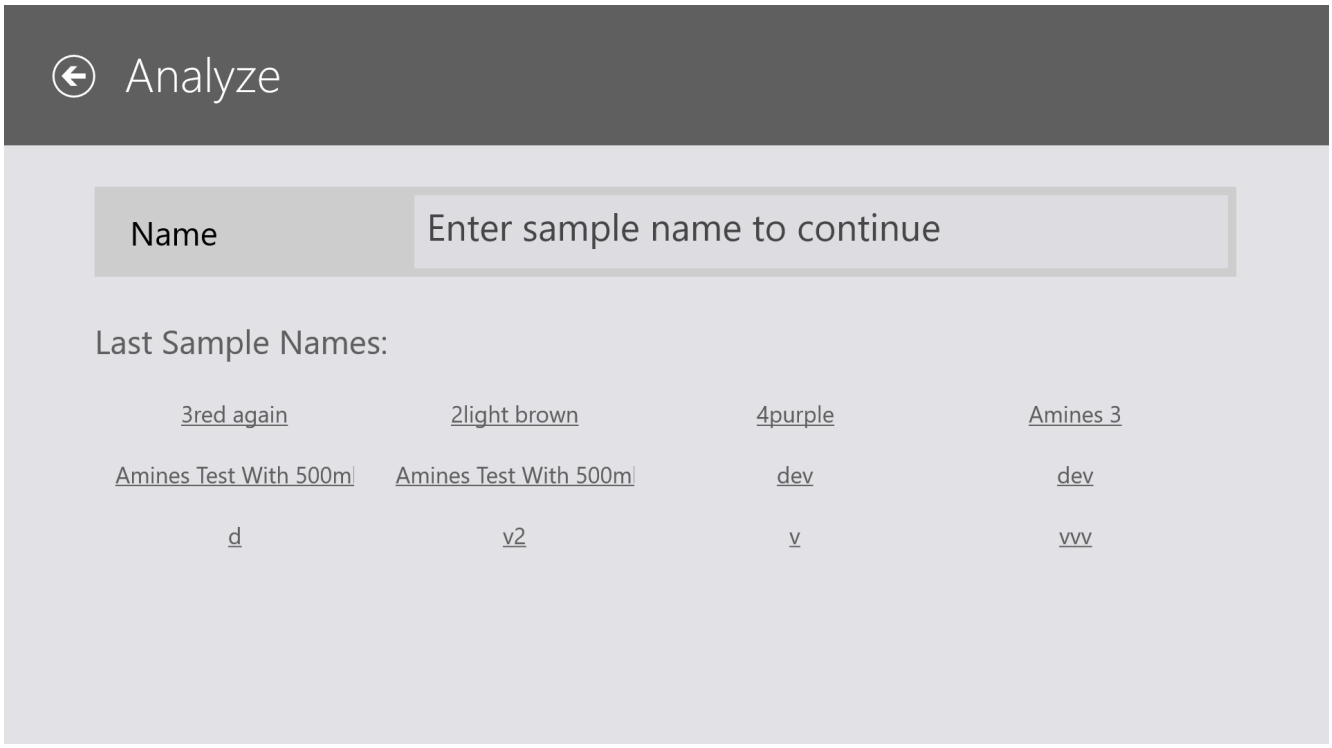
can click or tap in the sample name section to display an onscreen keyboard for entry of a new sample ID or select a sample name from the last twelve samples tested.

The keyboard allows the entry of up to 255 characters and accepts standard ASCII characters. Changes and deletions of text can also be made using the keyboard.



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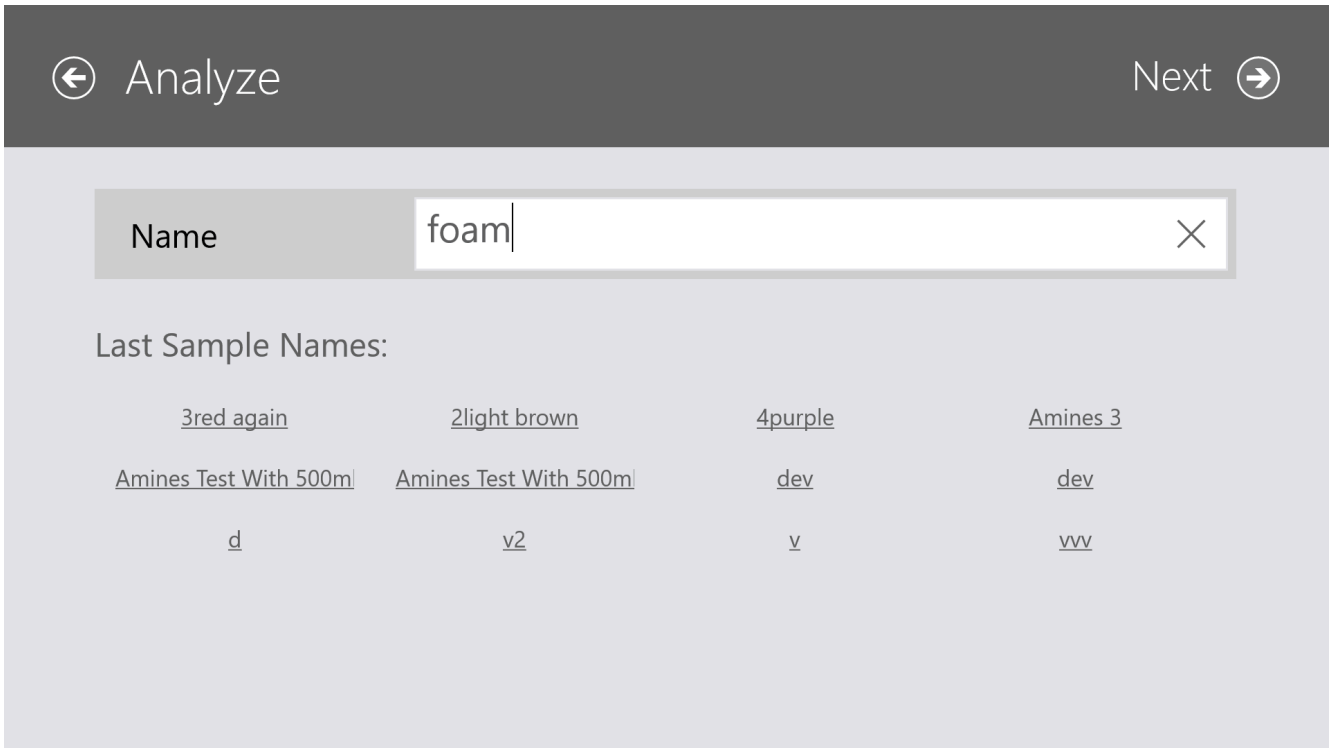
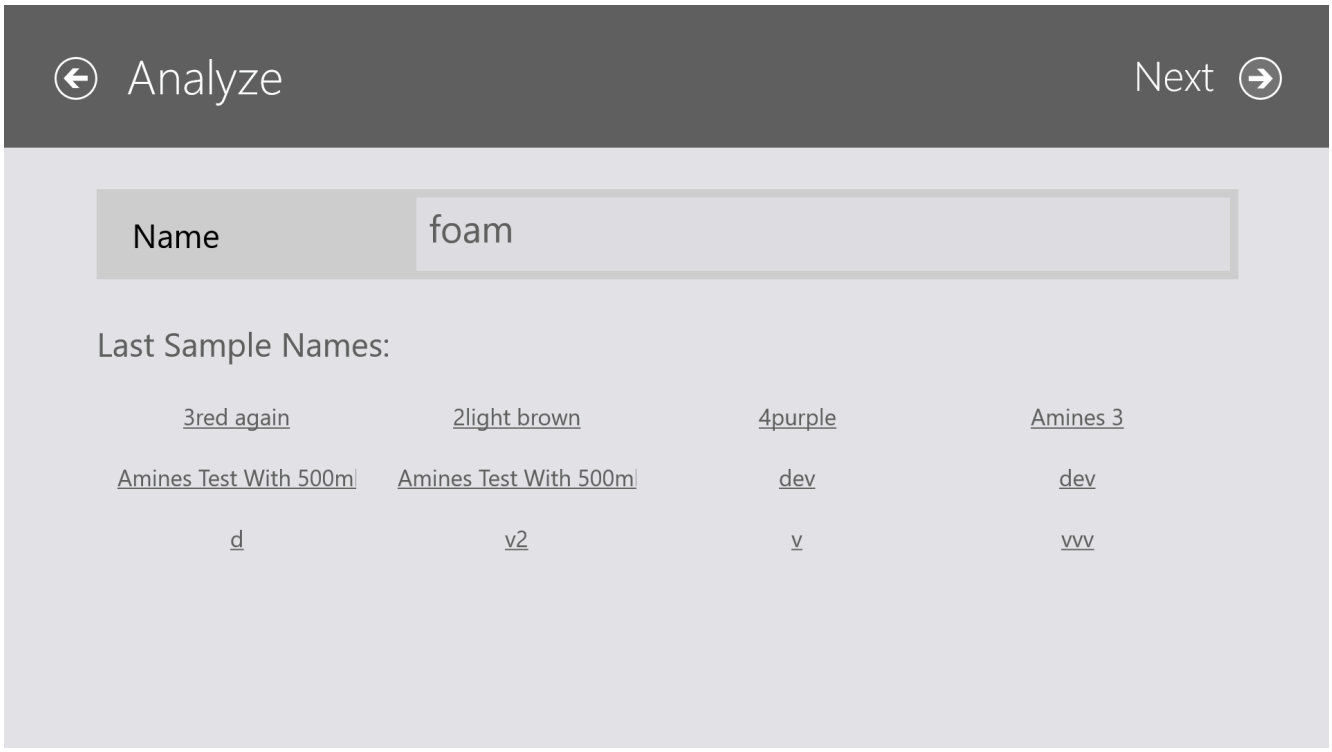


After successful entry of a valid sample ID, the Next button will become activated. The sample ID is confirmed using the Next button located at the top-right hand corner. Select Next.



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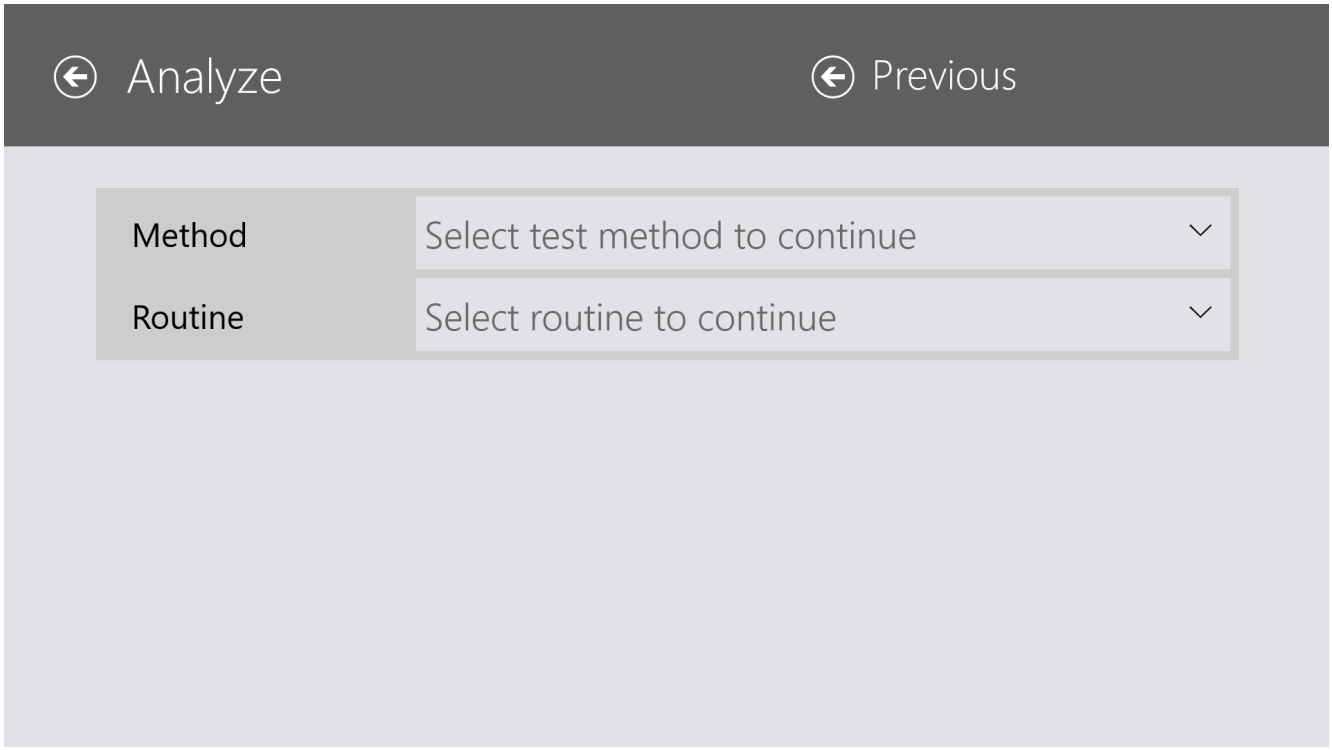
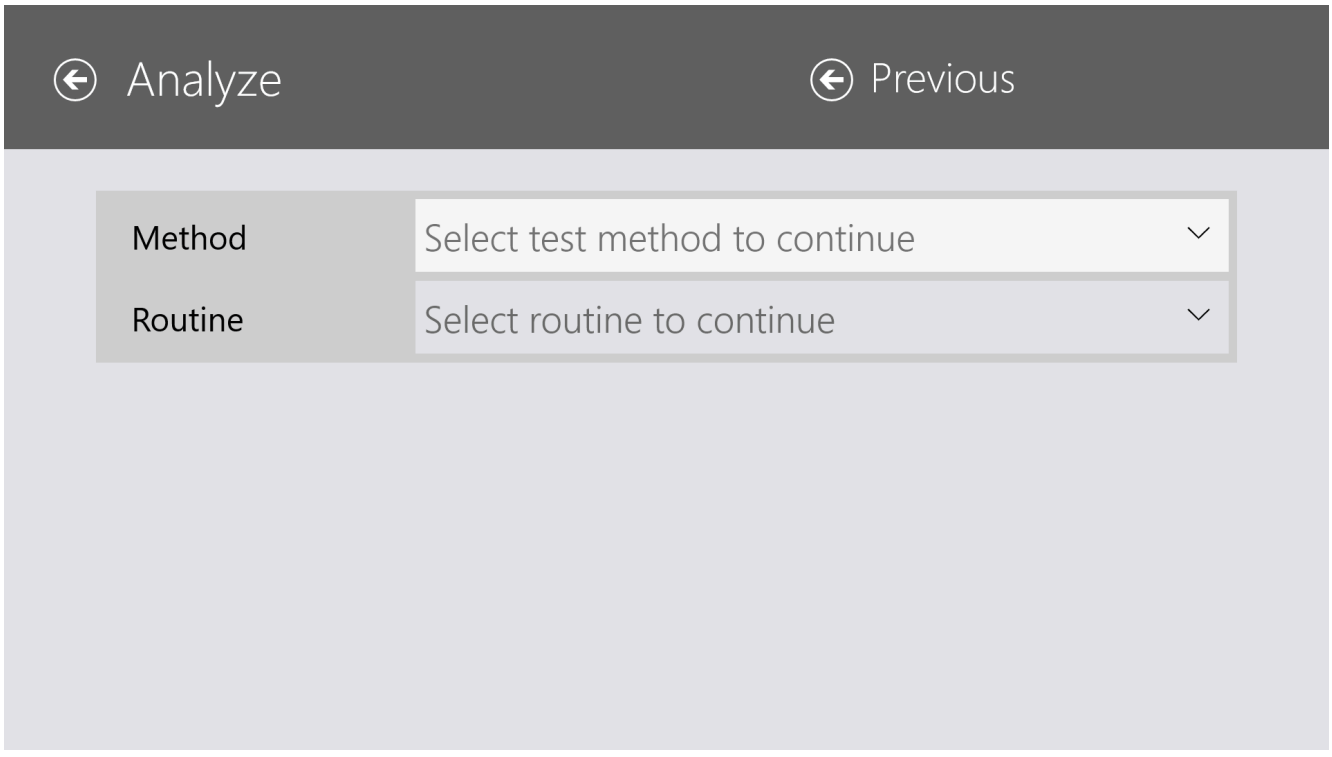
After pressing the Next button, the operator is presented with options to enter a Test Method and Routine.

Methods and Routines are fixed and selected using a dropdown box.





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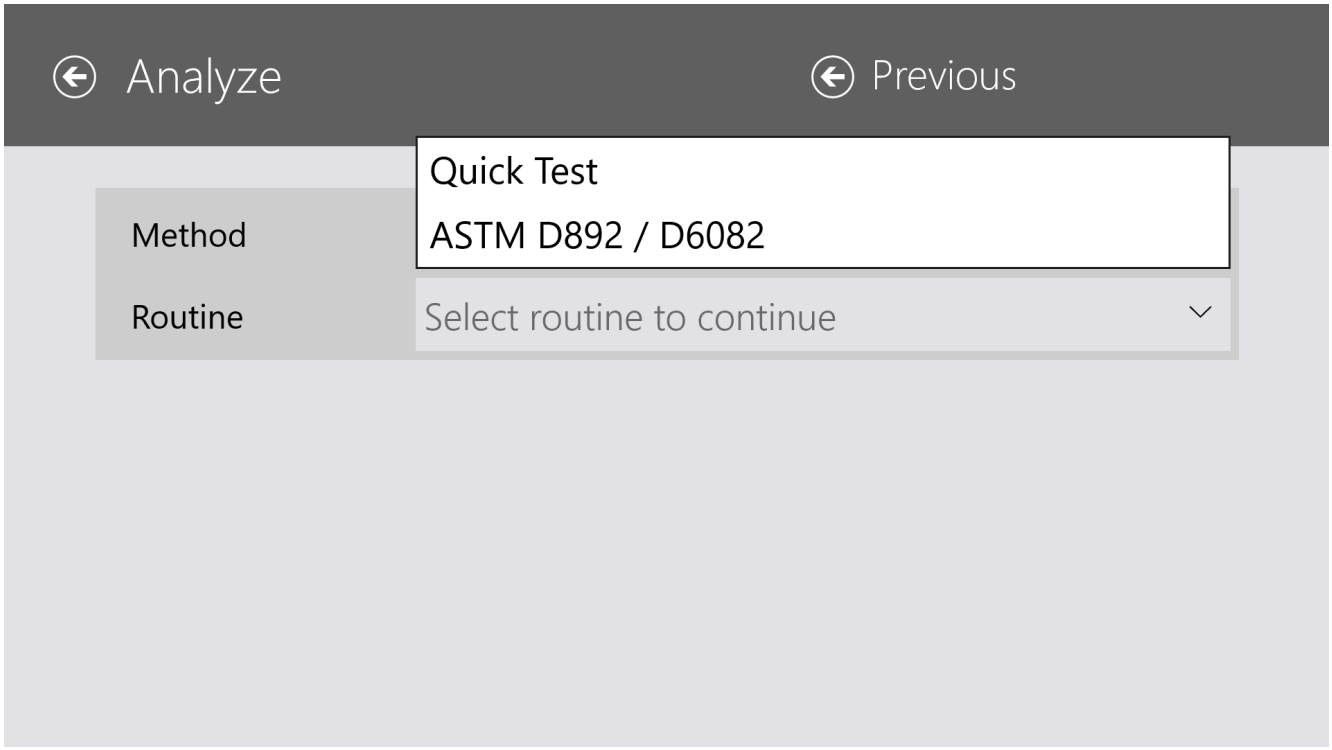
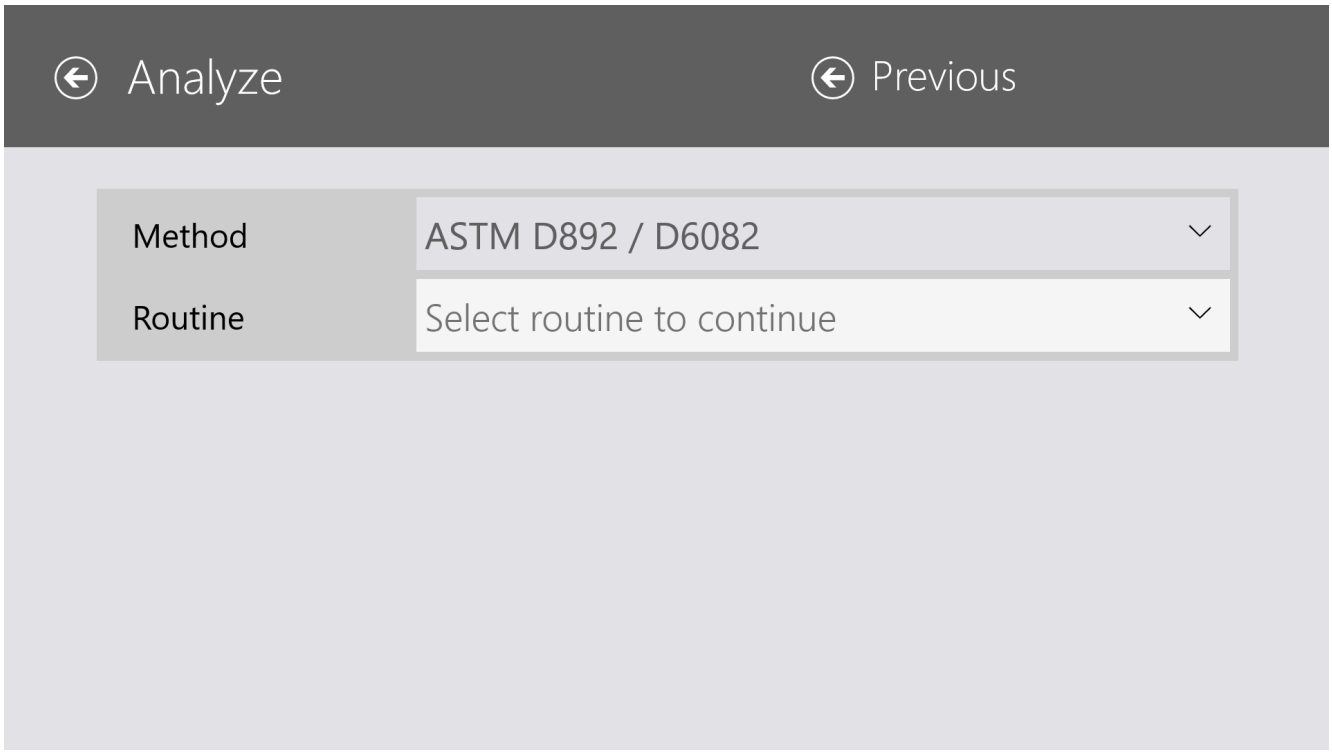
The operator will choose whether they want to run an ASTM D892/D6082 analysis or a Quick Test.

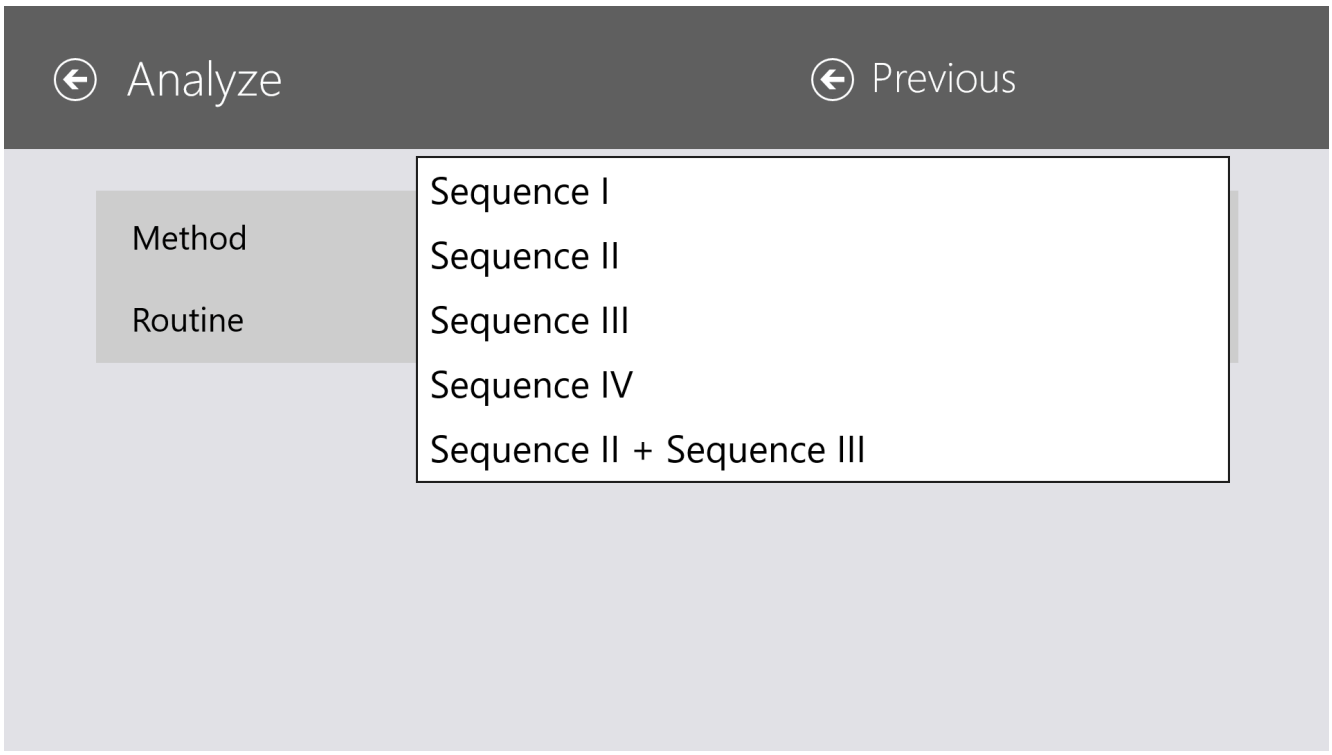


Then they will select the type of sequence to run from the ASTM D892 or D6082 test methods options or run a Quick Test:

Test	Duration	Description
Sequence I	40 minutes	Room temperature test (24℃)
Sequence II	30 minutes	Hot temperature test (93.5℃)
Sequence III	60 minutes	Room temperature stress test (24℃)
Sequence II + Sequence III	90 minutes	Room and hot temperature test (24℃ and 93.5℃)
Sequence IV	90 minutes at 10 min heating	Highest temperature maintained for specific intervals (5 sec, 15 sec, 1 min, 5 min, 10 min); (150℃)
Quick Test	5 minutes	Air flow test (24℃ at 94 mL / min)

The operator will have the option to run compliant sequence analyses with the ASTM methods for foam analysis. For more information, check out the published methods with the ASTM.





The operator will also have the choice to select a Quick Test. This option is a feature that conducts a room temperature demo analysis.

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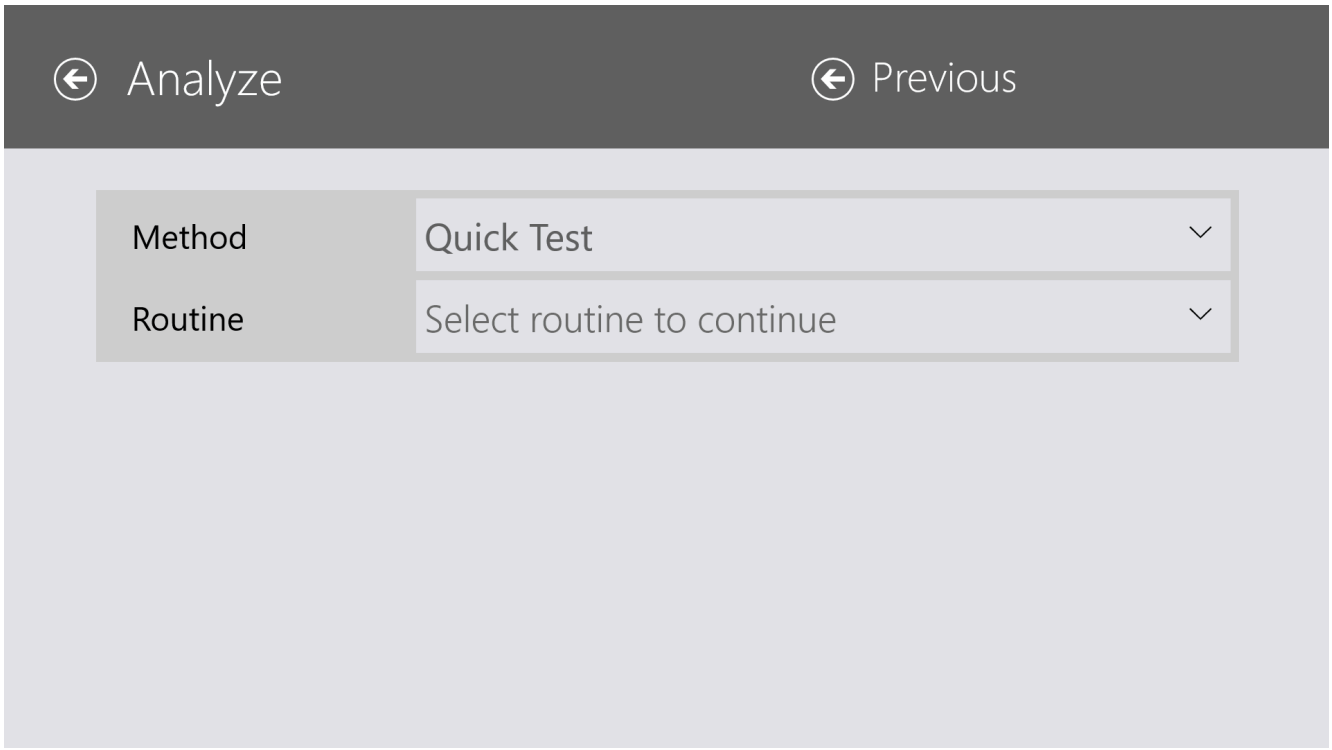
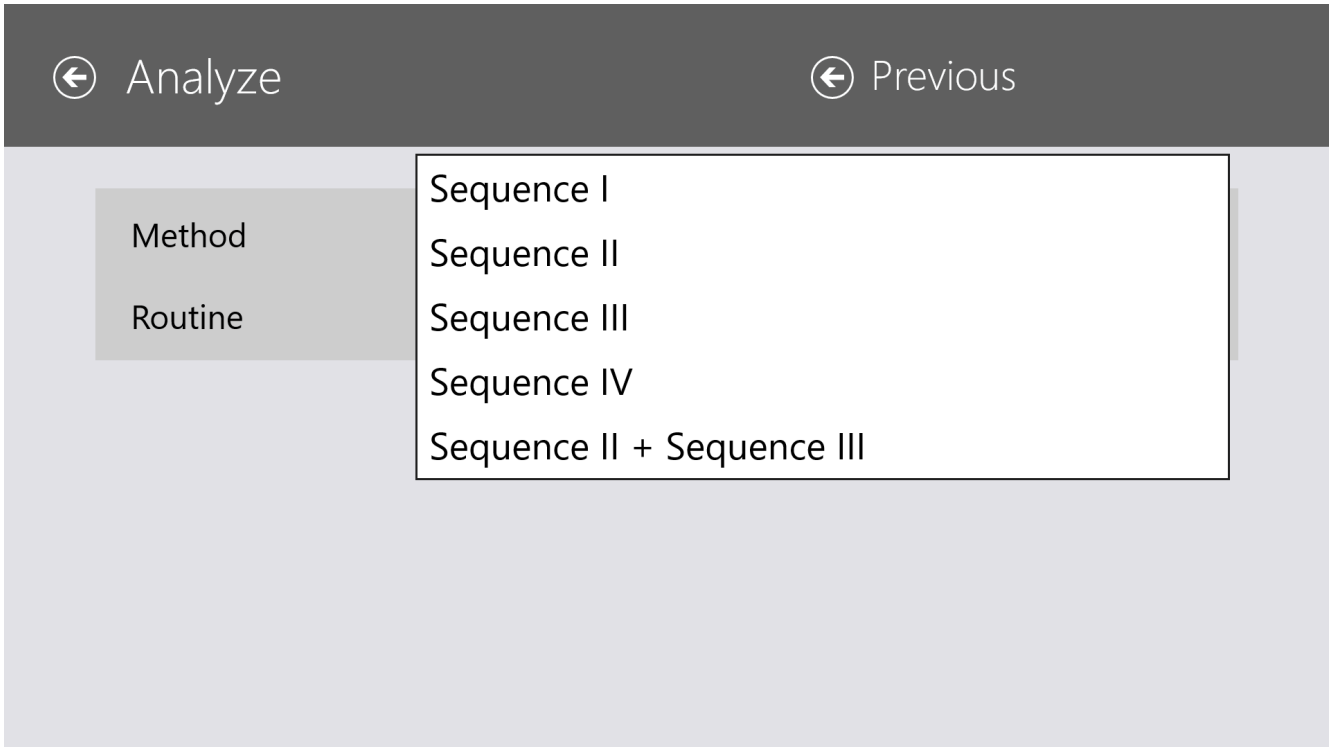


If a Quick Test is selected, the operator will need to select Sequence I as the routine.



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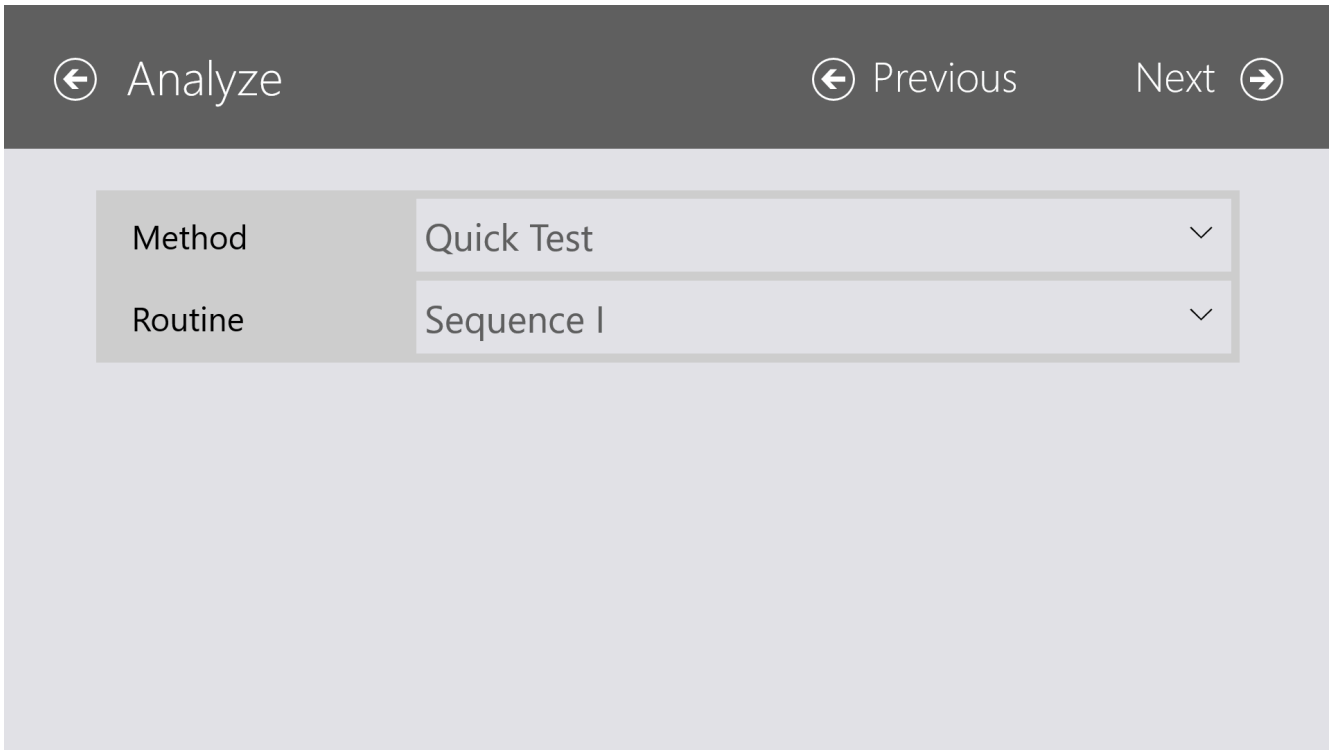

Once the method and routine are selected, Next will appear at the top-right hand corner. Select Next.

The next selection is the Operator. Operator names are fixed and selected using a dropdown box. To add new operator names, please consult the Settings section of this manual (**Section 5.4.2**).

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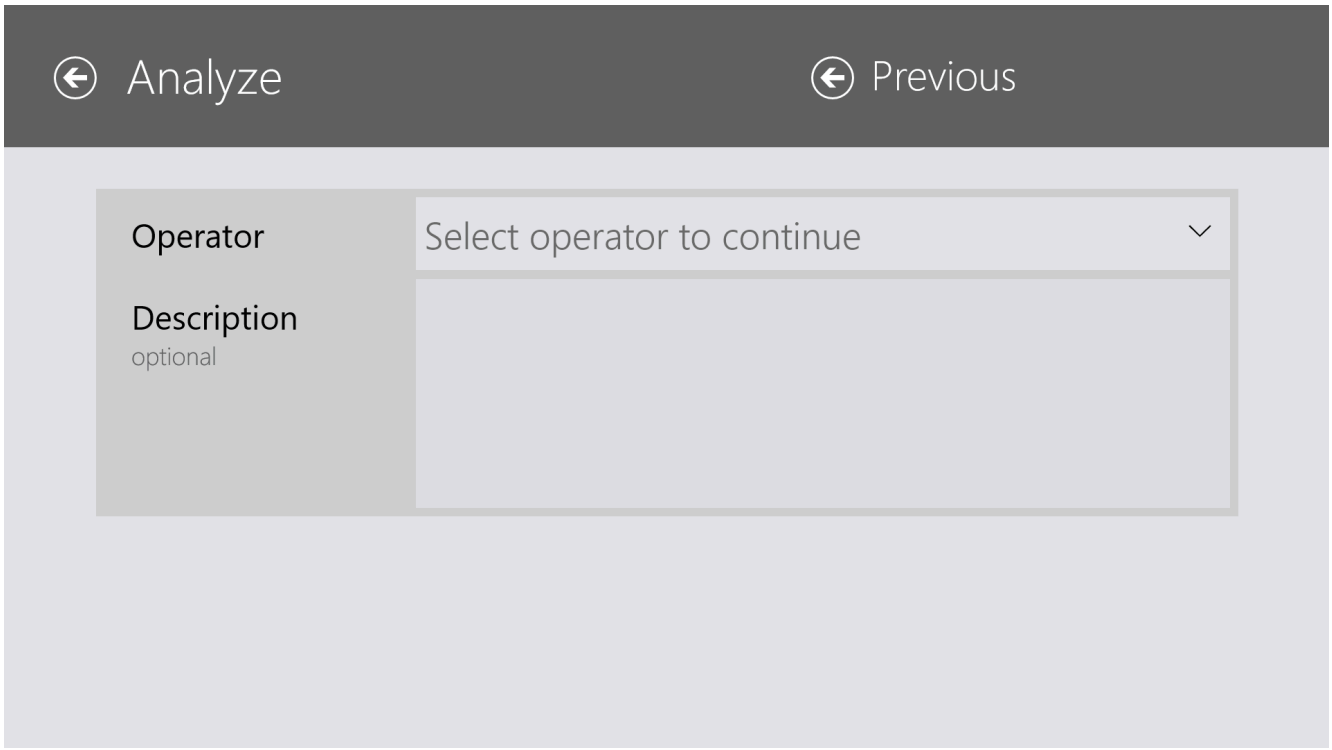

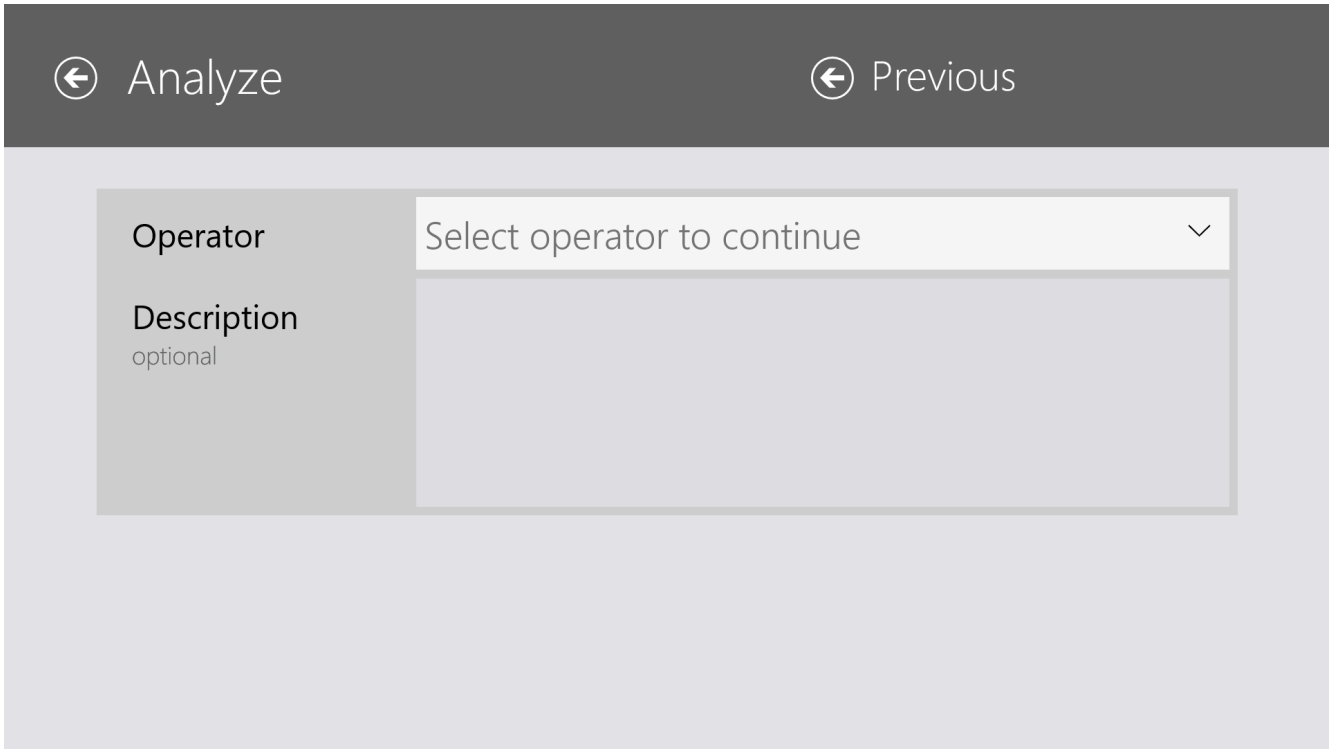

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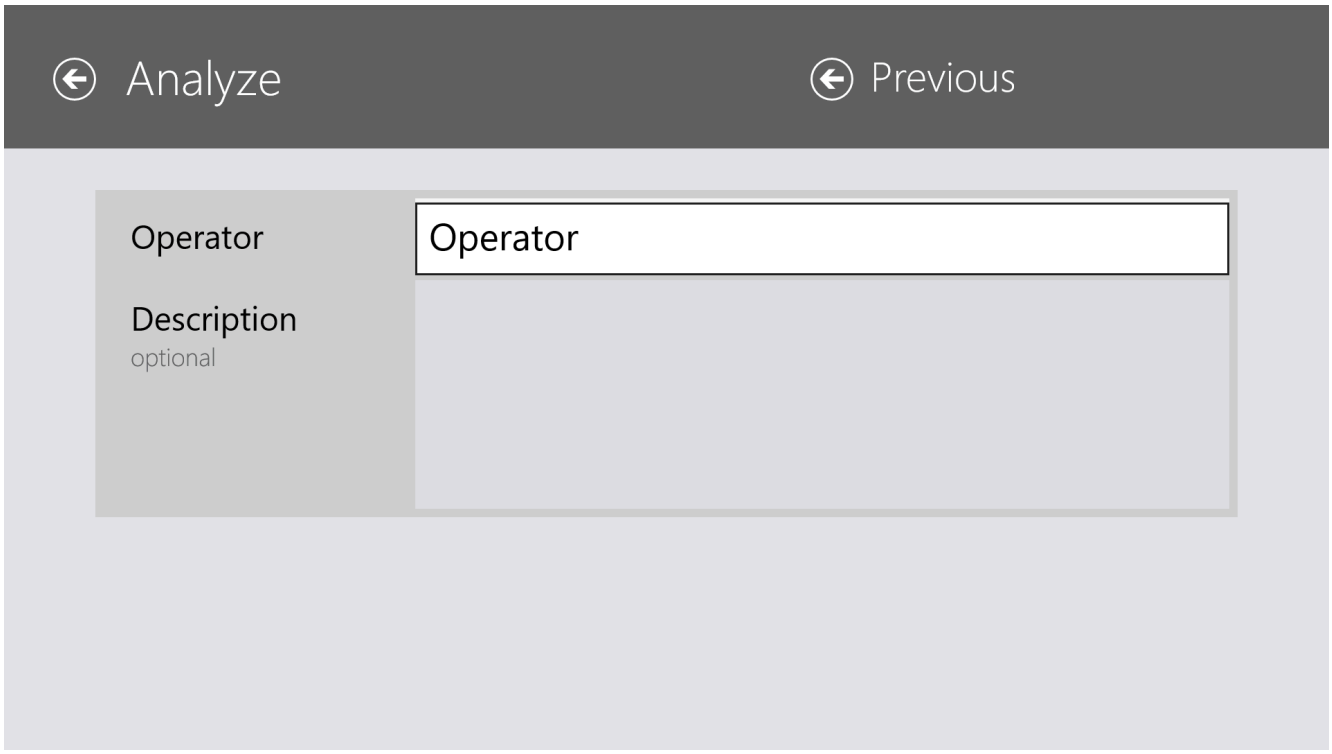
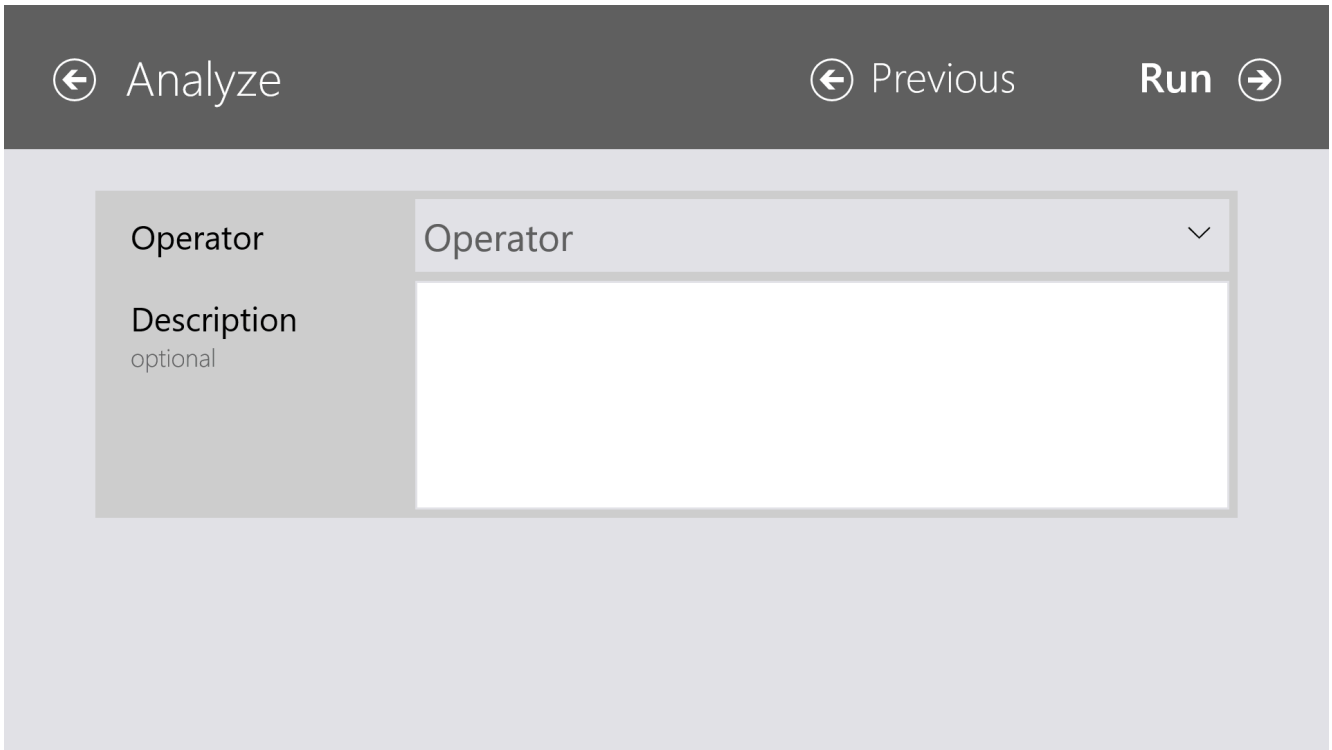


Below the Operator selection there is an optional Description or comments of a sample inspection. This field is used to enter other information, such as notes, batch ID, etc. To enter notes, simply tap or click on the white entry box to display an onscreen keyboard.



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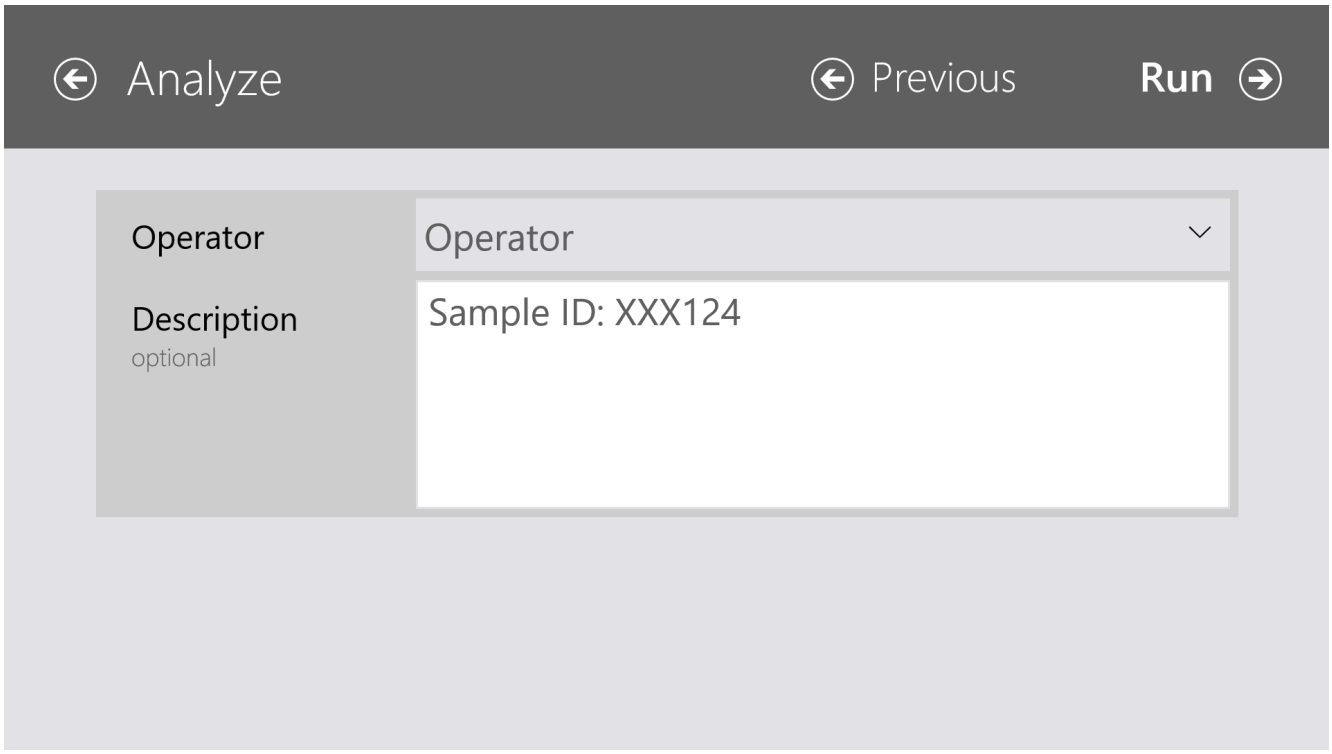
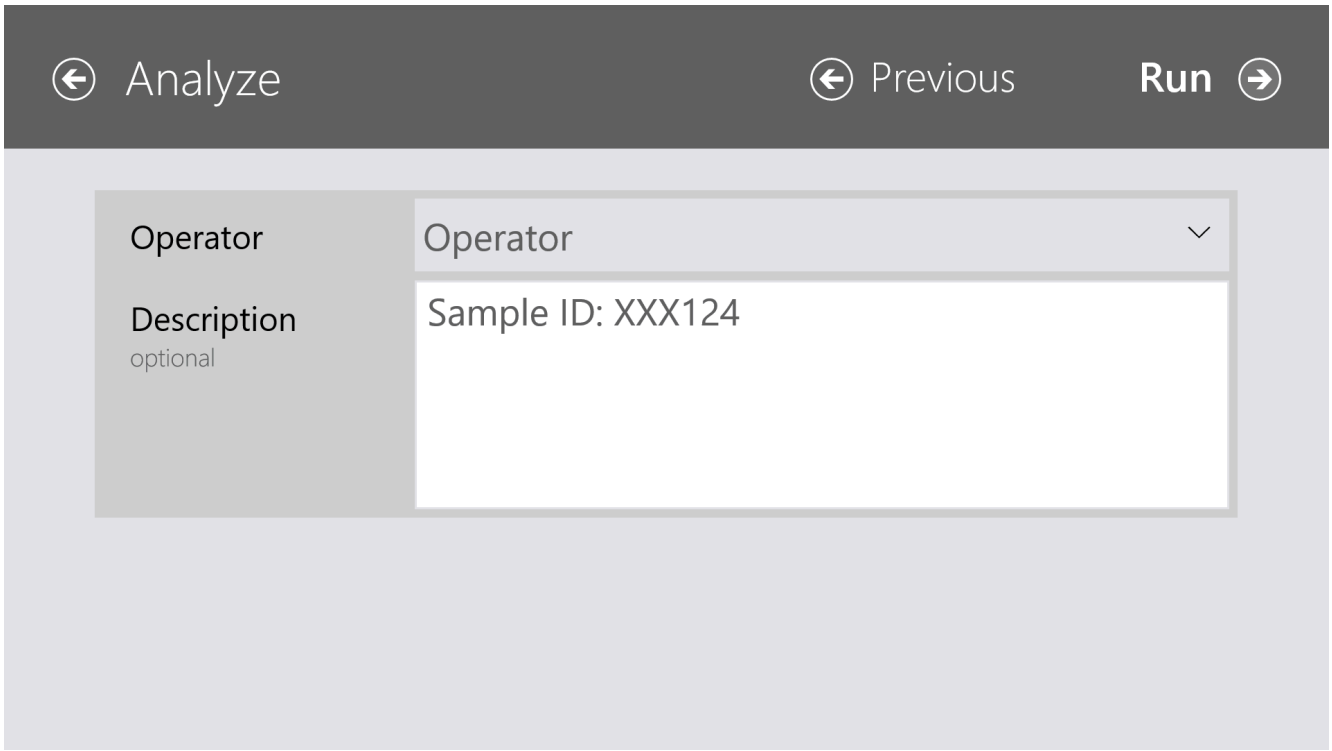


Once complete, press the Run button to proceed with your inspection.



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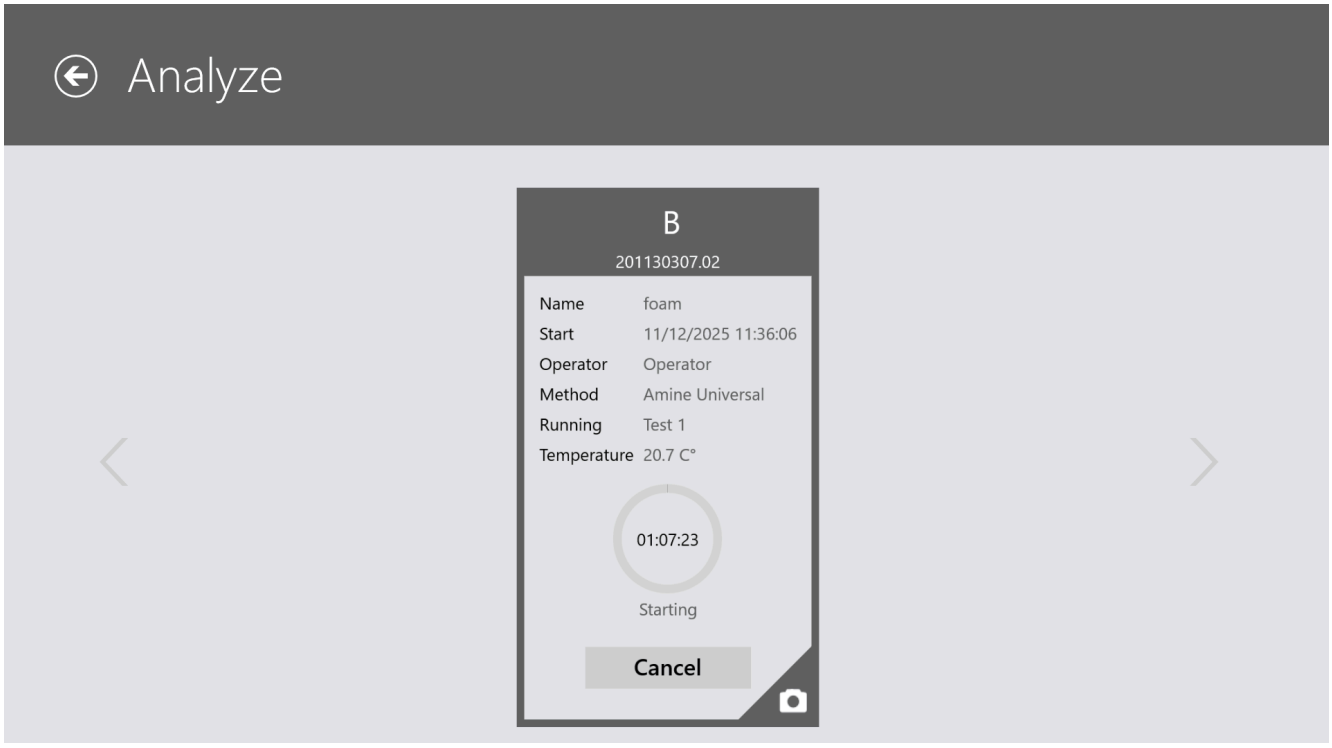
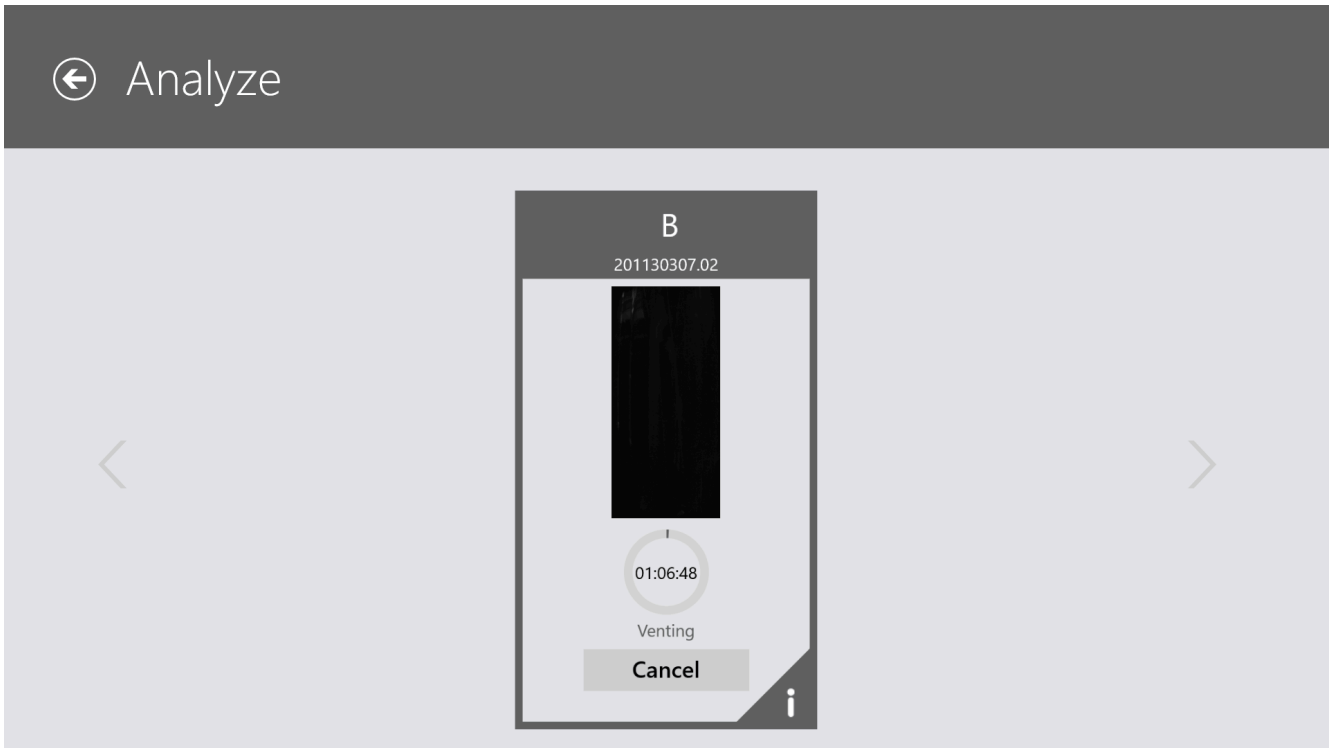
After completing all the necessary sample identifiers (sample ID, operator name, test method, and optional description) and clicking Run, the automated visual analysis starts.

A progress and information bar will appear and display the status of the analysis.

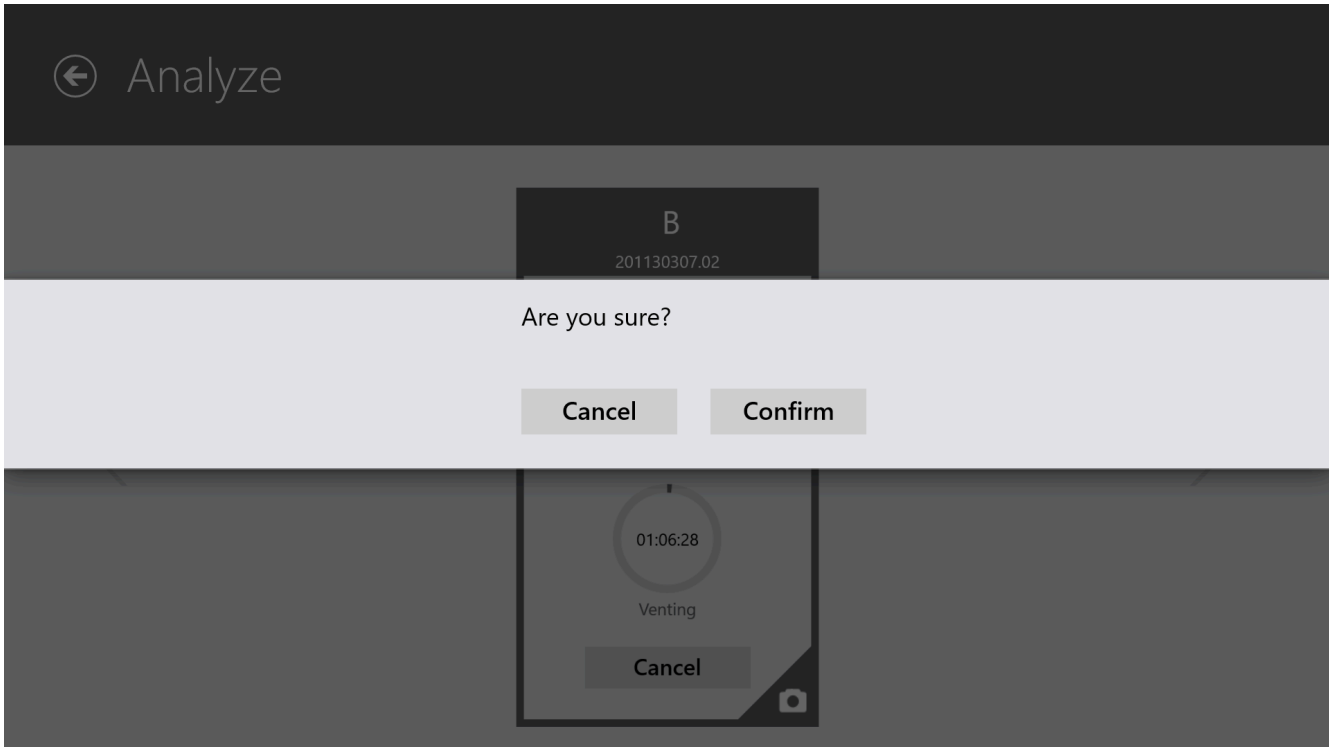


A live feed showing the sample can be accessed by clicking the camera icon on the bottom right. To return to the identification screen, simply click the information icon.



At this point, the operator may select a second module to start a second asynchronous analysis following the same procedures as described above.

An analysis can be cancelled at any time by Cancel and selecting confirm.



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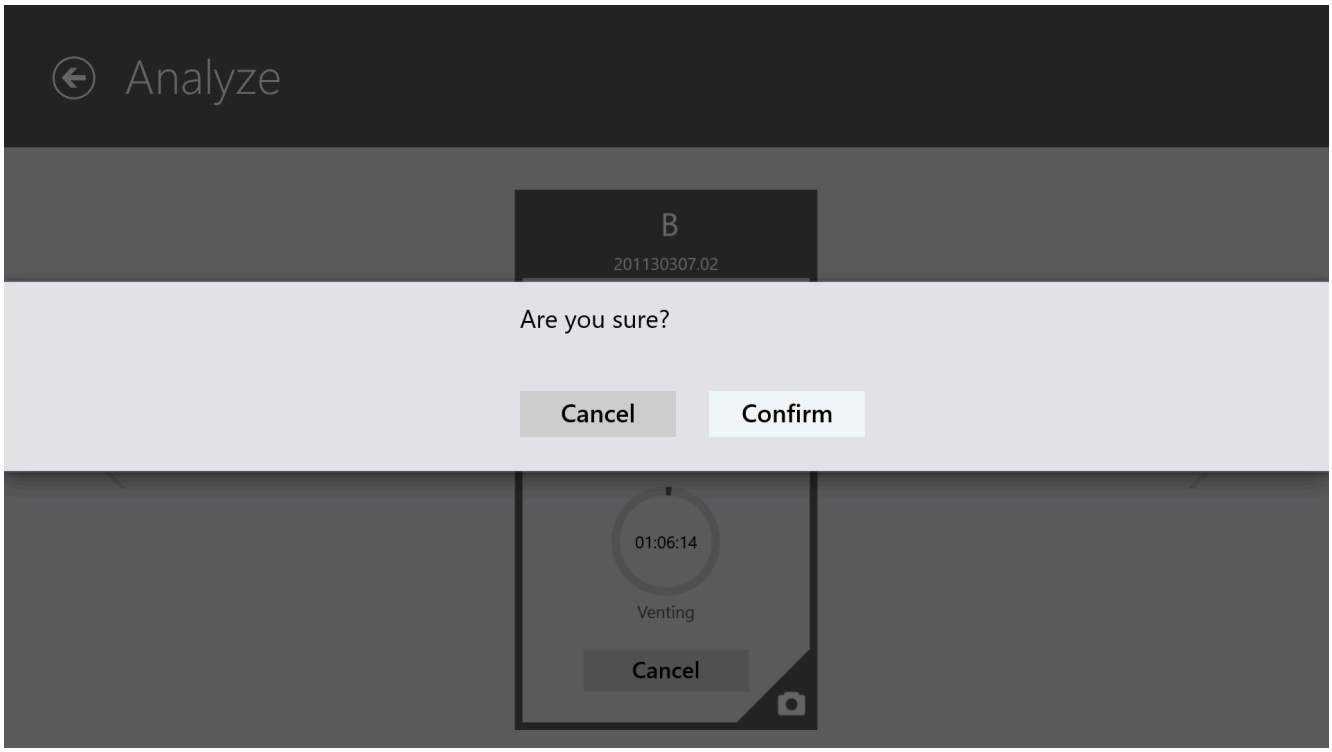


5.2.2. Coolants – D1881

This section details the protocols and procedures to test to foamability of coolants and glycol formulations in accordance with ASTM D1881.

5.2.2.1. Sample and Cylinder Preparation - Glycols

To prepare an engine coolant sample for ASTM D1881, prepare a 33% by volume solution using the coolant to be tested and Type II water, following the mixing procedure in Test Method D 1176 to ensure any insoluble materials are included. If testing an additive concentrate, dilute it to its recommended use concentration with Type II water first.

Prepare the solution: Create a 33% by volume solution of the coolant you are testing using Type II water.

Use correct glassware: Use the same glassware for preparing the test solution as you would for a reference coolant solution.

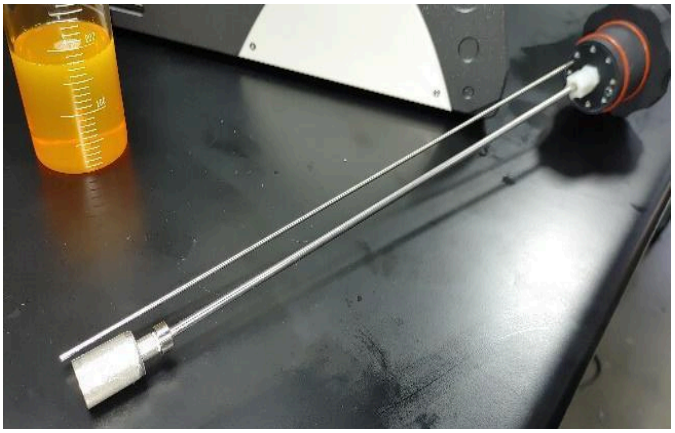
Rinse glassware: Rinse the glassware with Type II water between each sample preparation to avoid contamination.

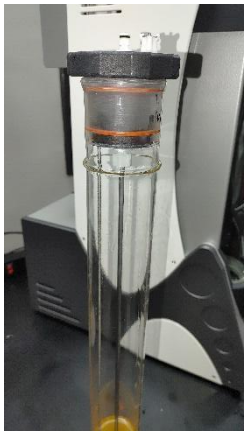
Dilute concentrates: If your sample is an additive concentrate, dilute it with Type II water to its recommended use concentration before preparing the 33% test solution.

Follow D 1176 procedures: The sample preparation must be done in accordance with the treatment of mixtures described in Practice D1176. This ensures that all insoluble materials are part of your representative sample.

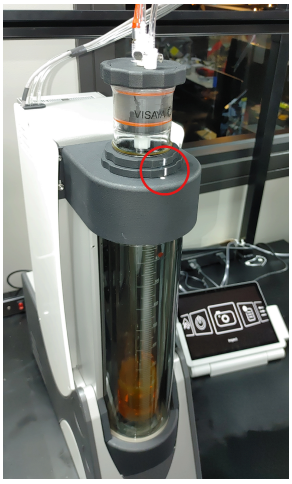
Ensure clean and dry glassware and wetted components (outer cylinder, graduated test cylinder, stopper). Add your coolant mixture to the 500ml sample cylinder.

Screw the diffuser to the end of the stopper and insert the stopper into the sample cylinder. Ensure the stopper is well pressed (airtight) to the sample cylinder.



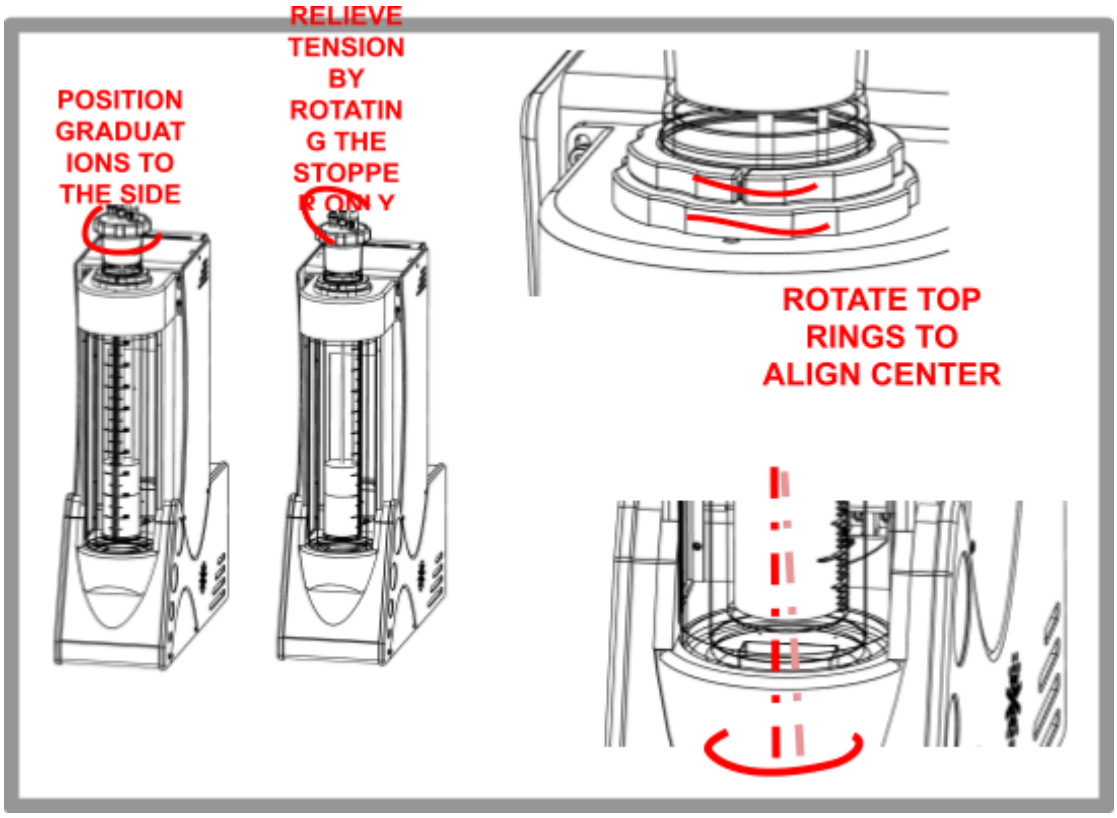


Insert the Sample Cylinder into the unit with white markings aligned. Make sure to connect the Probe and Air Tubes.



Adjust the cylinder such that the graduations are to the side. This is critical for the automatic foam analysis (vision system) to work properly. While doing so, you may have issues with the weight of the air and electrical, as they will tend to drop down and maybe rotate the cylinder back. To work around this, correct the glass rotation first, then rotate only the stopper until the weight of the connection is no longer working against you.

Review the bottom of the cylinder inside the FoamDDI. The cylinder should be as close to the center as possible, and you can use the two stopper rings to correct it.

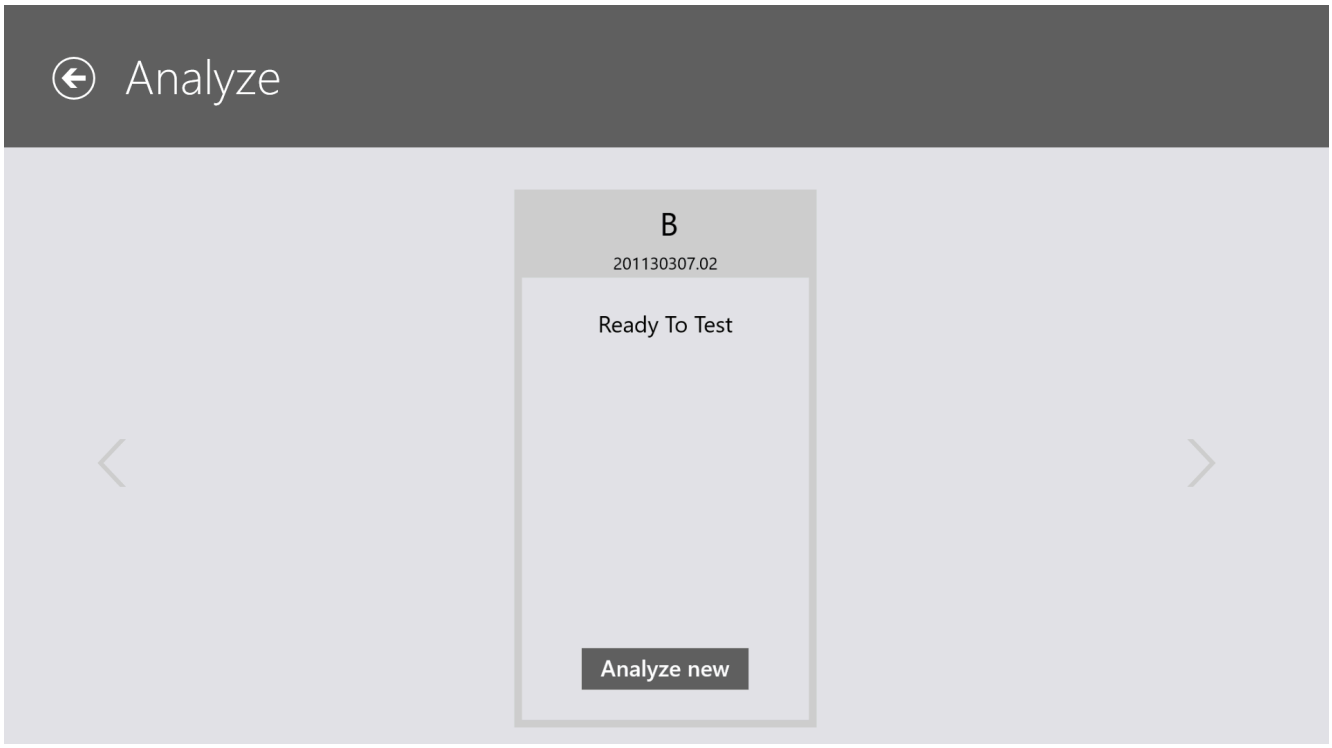


From the Main Screen, tap Inspect.

5.2.2.2. Running a Coolant Analysis

Running an inspection is achieved using a simple Wizard-driven menu. Upon entering the Inspect menu, the user is presented with a box showing connected FoamDDI module(s).

Choose whichever module is to be used (if multiple are active) and select Analyze New. Checkout **Section 5.4.8** to connect multiple modules.



If two modules are connected, they will both appear side by side. Any warnings or errors will be indicated by the triangle containing an exclamation mark. Click on the warning icon or the Detail icon to find out more information. **Section 6** has more information on warnings and errors.

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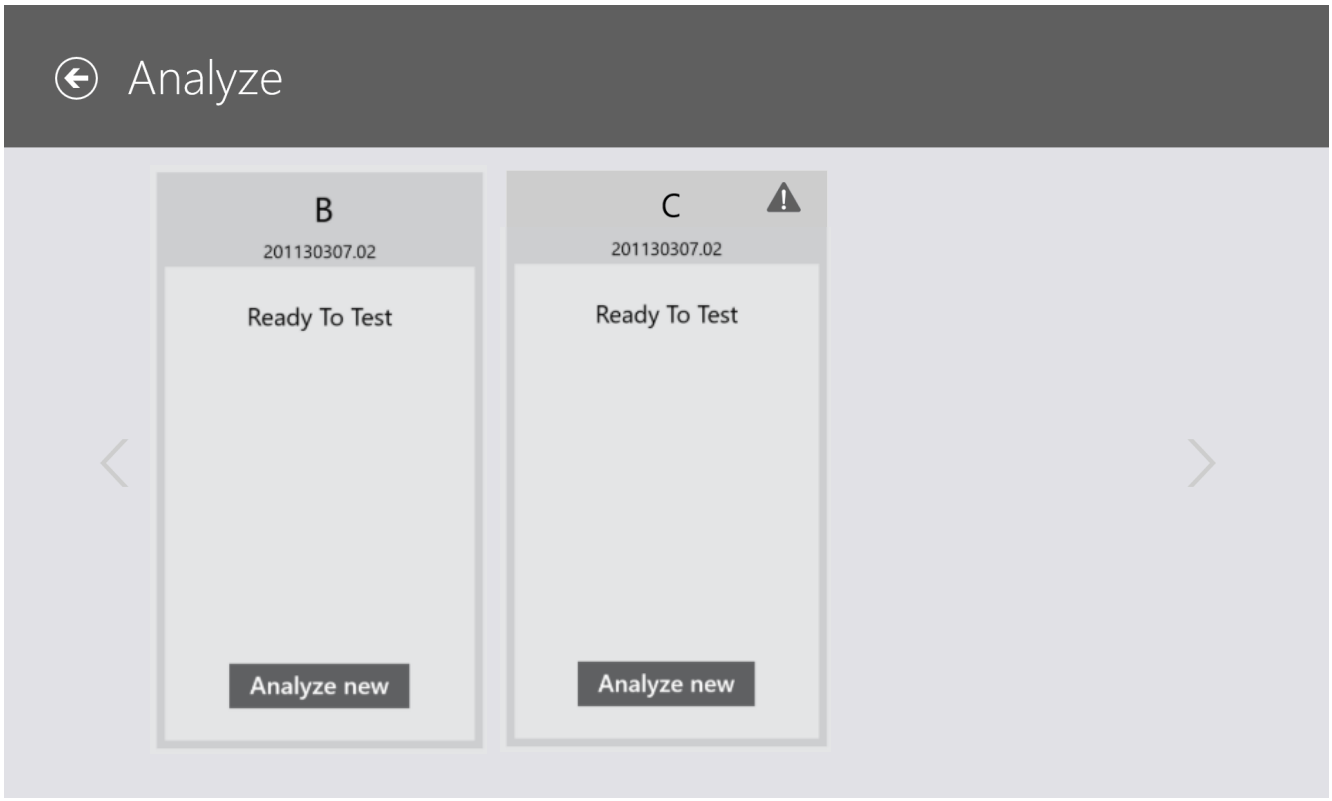

Once selecting Analyze New, two options will propagate: New or Continue. Choose New to run a completely new sample. The Continue option allows you to append to an existing sample ID, if one has started within the past 3 hours.

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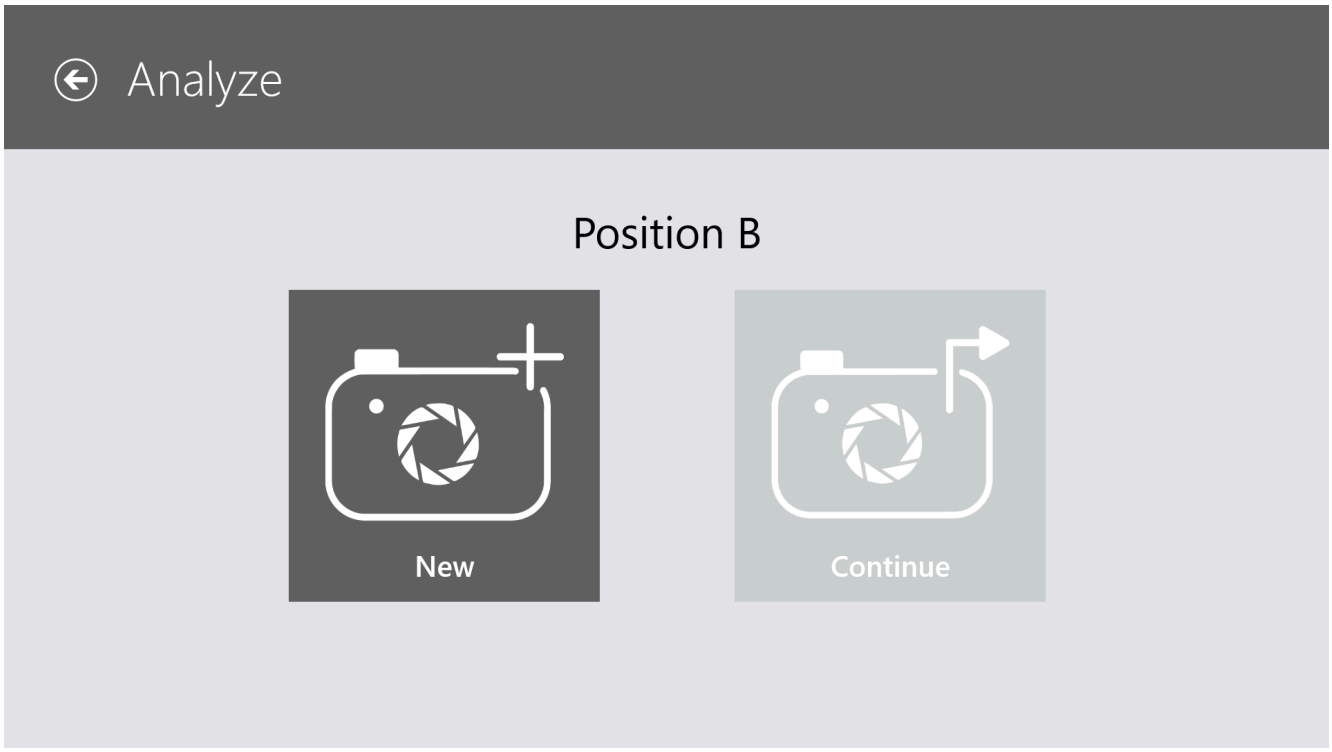

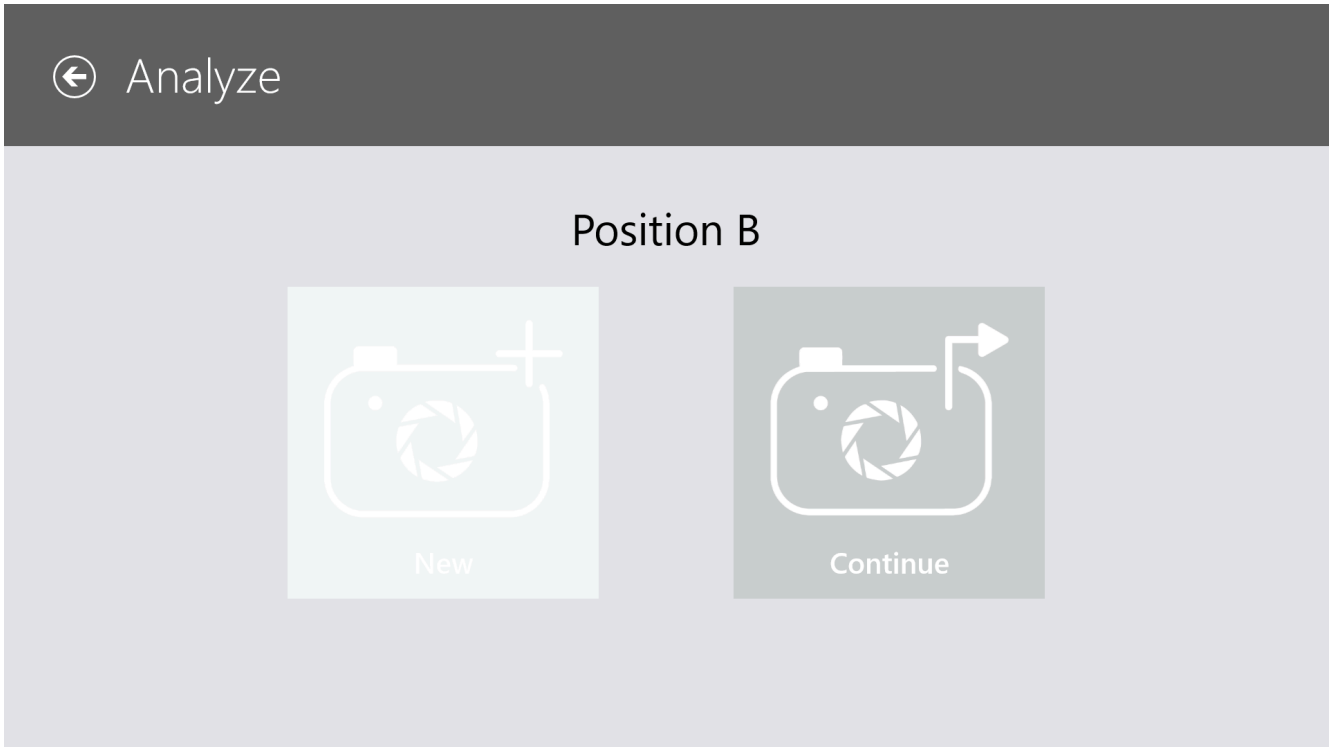

When starting a New sample, you will be asked to enter a sample name. Using the Continue option will propagate this information for the operator automatically. The user

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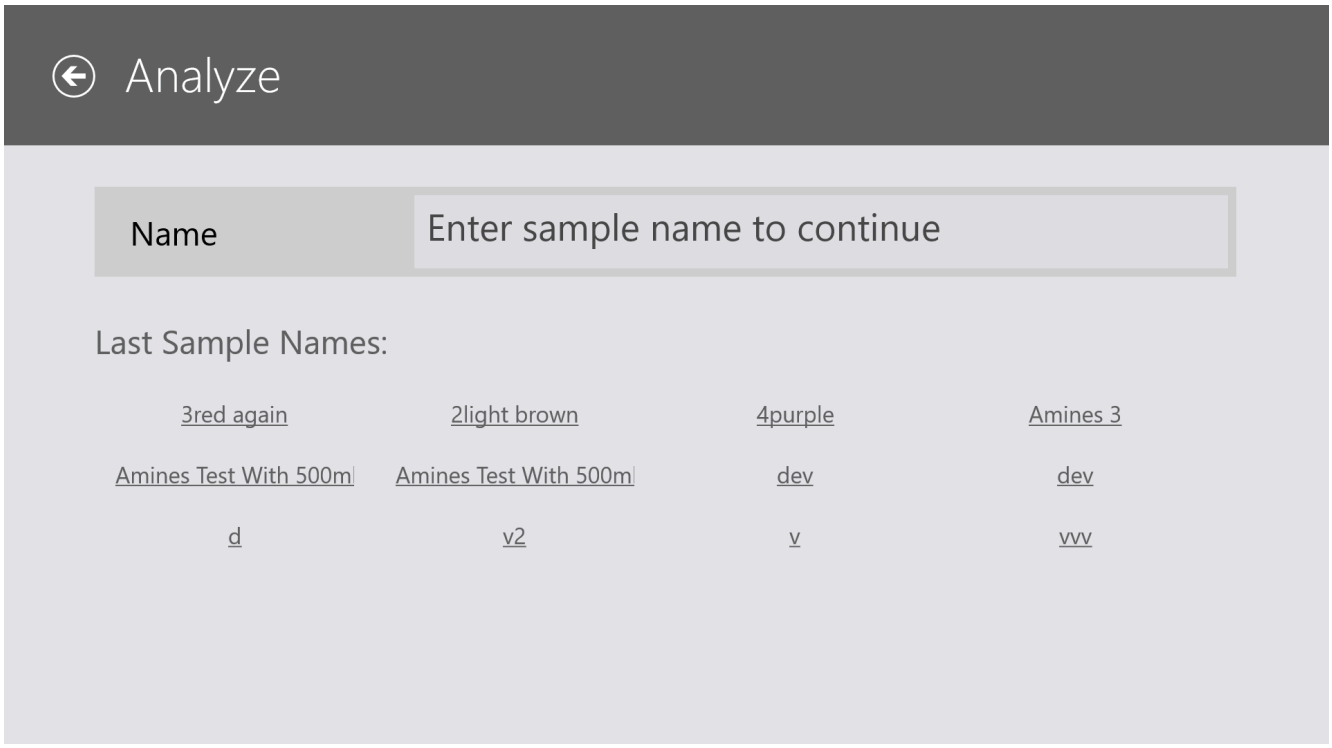
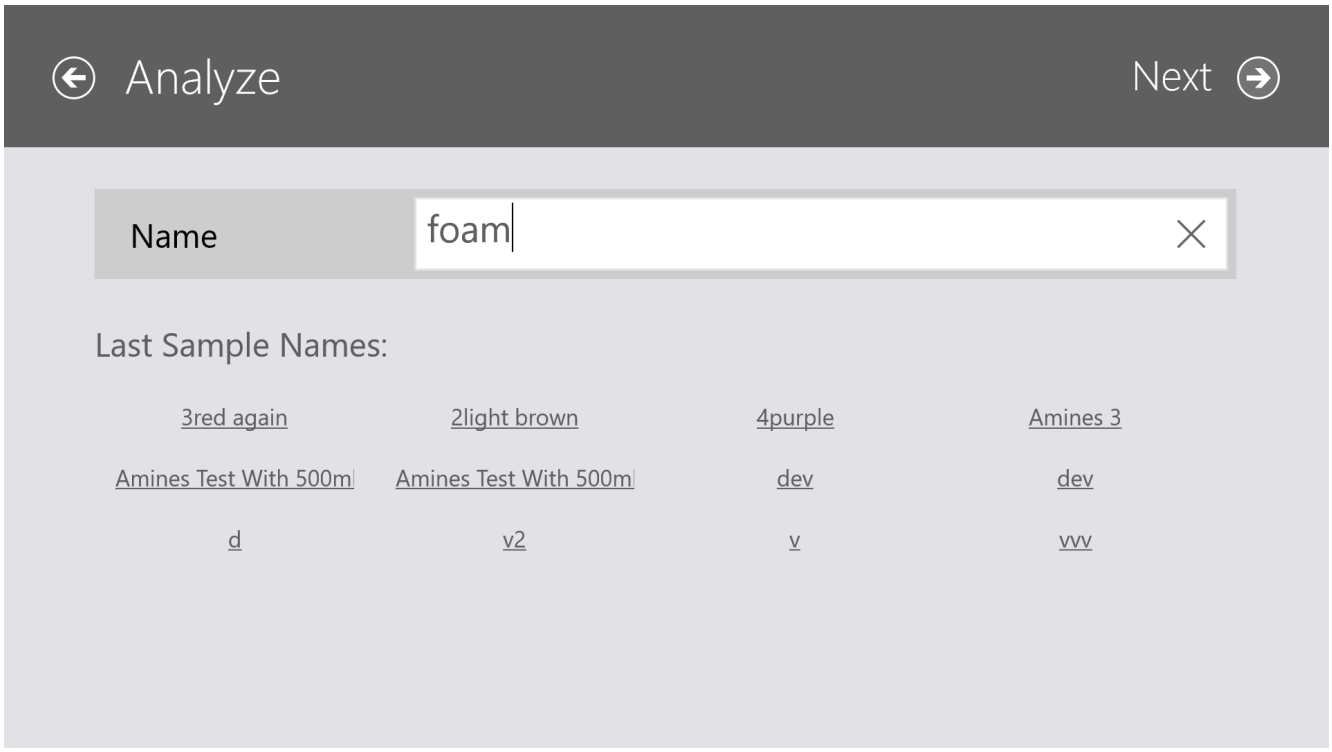
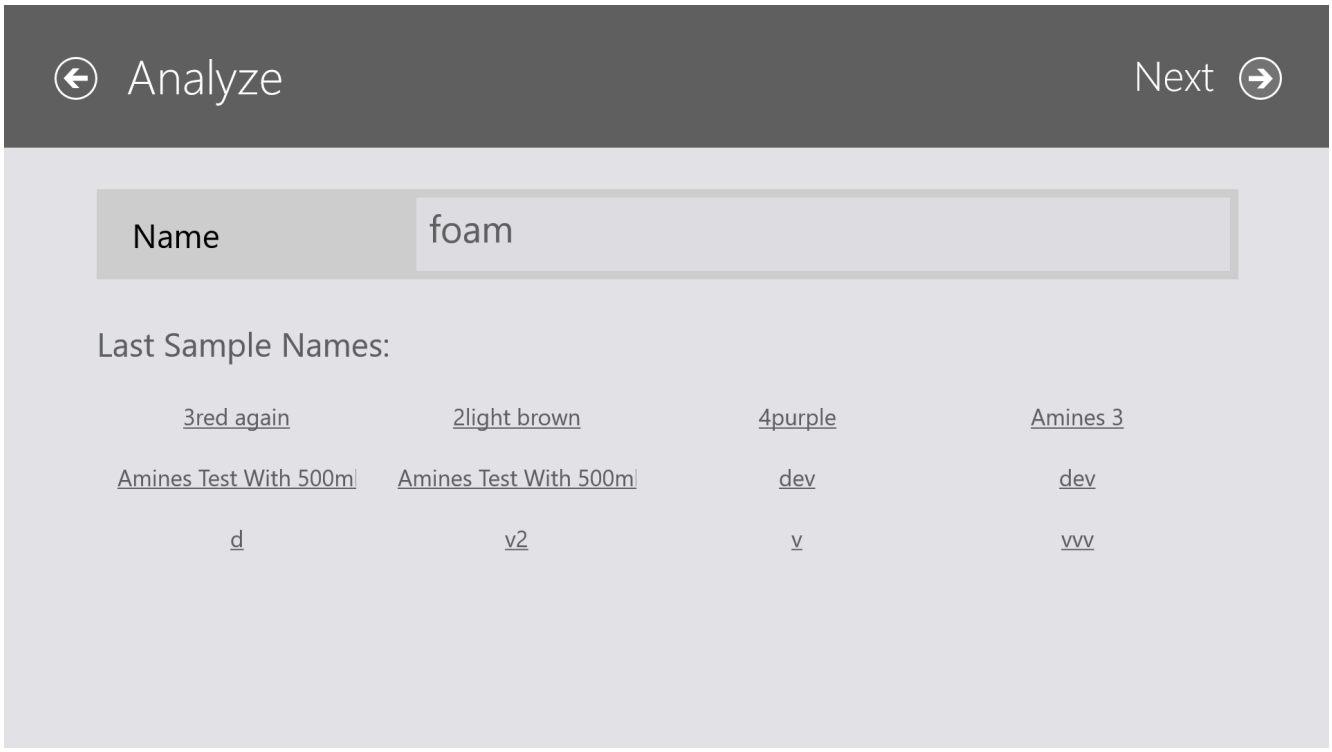
can click or tap in the sample name section to display an onscreen keyboard for entry of a new sample ID or select a sample name from the last twelve samples tested.

The keyboard allows the entry of up to 255 characters and accepts standard ASCII characters. Changes and deletions of text can also be made using the keyboard.





After successful entry of a valid sample ID, the Next button will become activated. The sample ID is confirmed using the Next button located at the top-right hand corner. Select Next.



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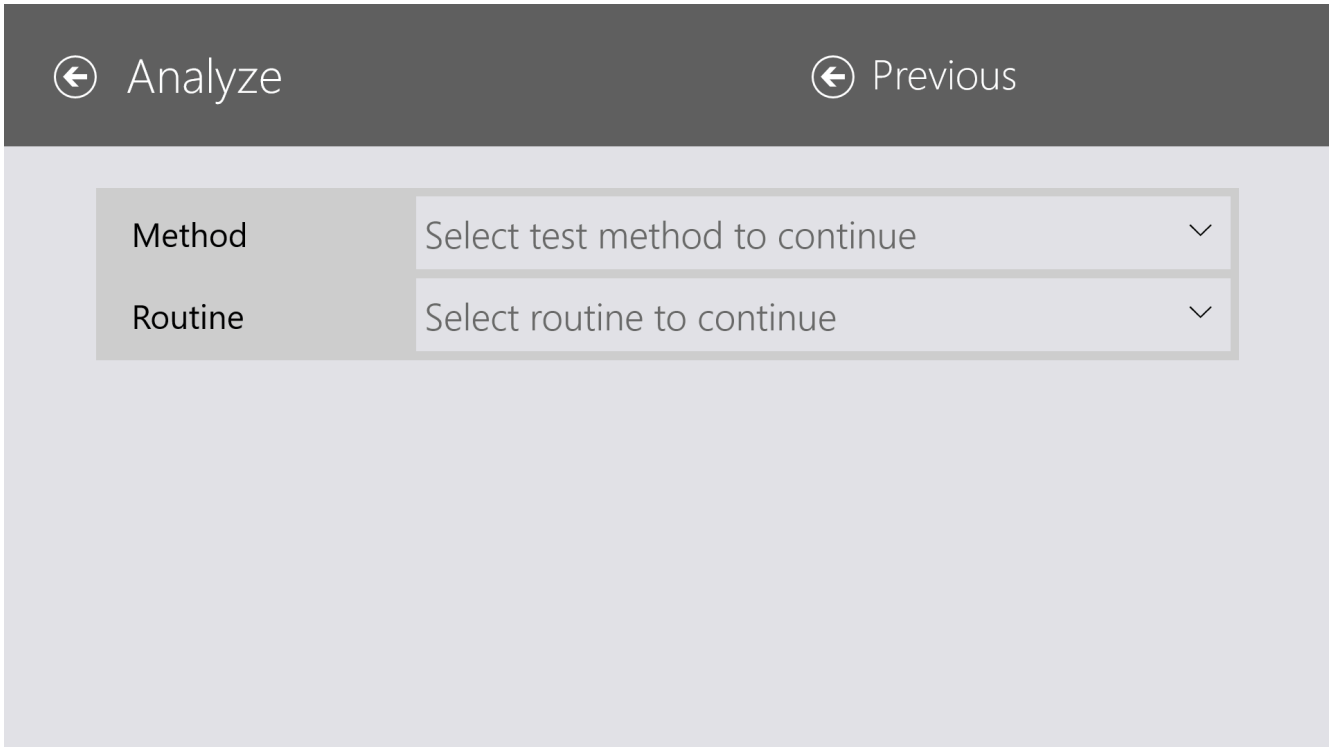
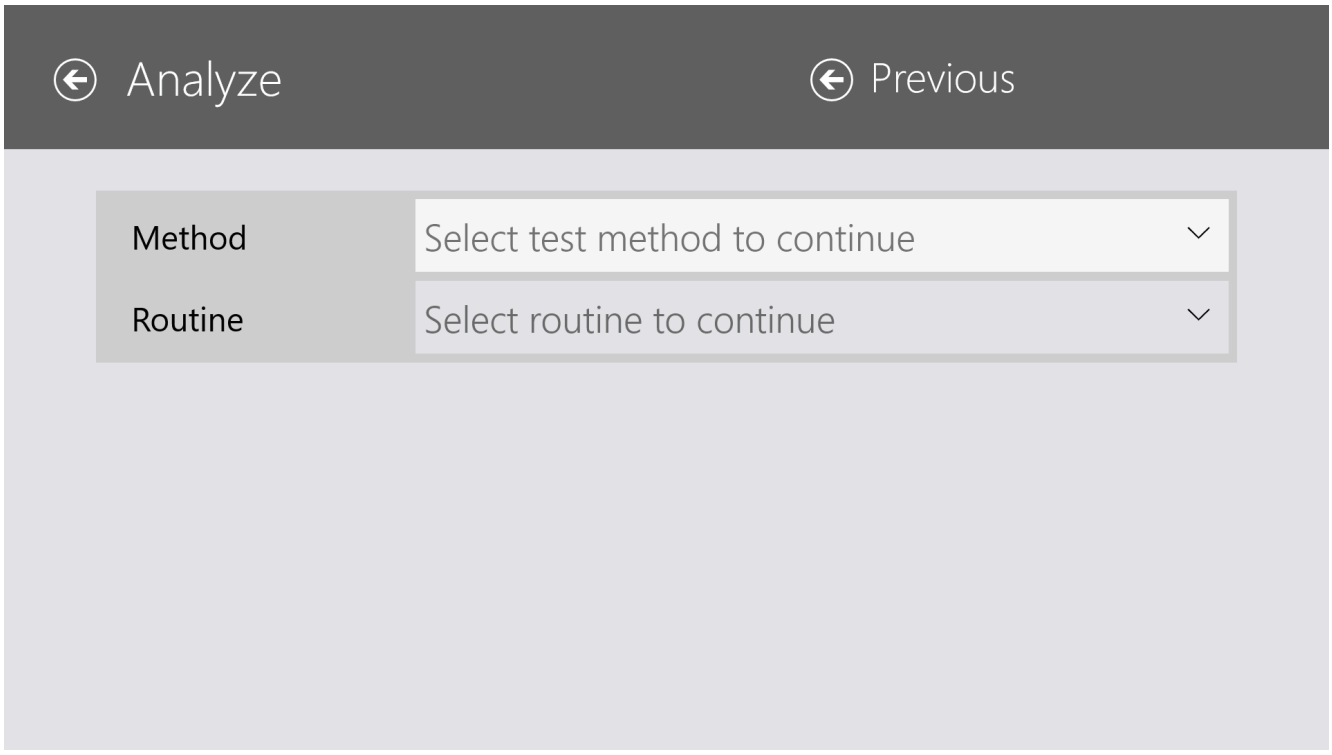
After pressing the Next button, the operator is presented with options to enter a Test Method and Routine.

Methods and Routines are fixed and selected using a dropdown box.





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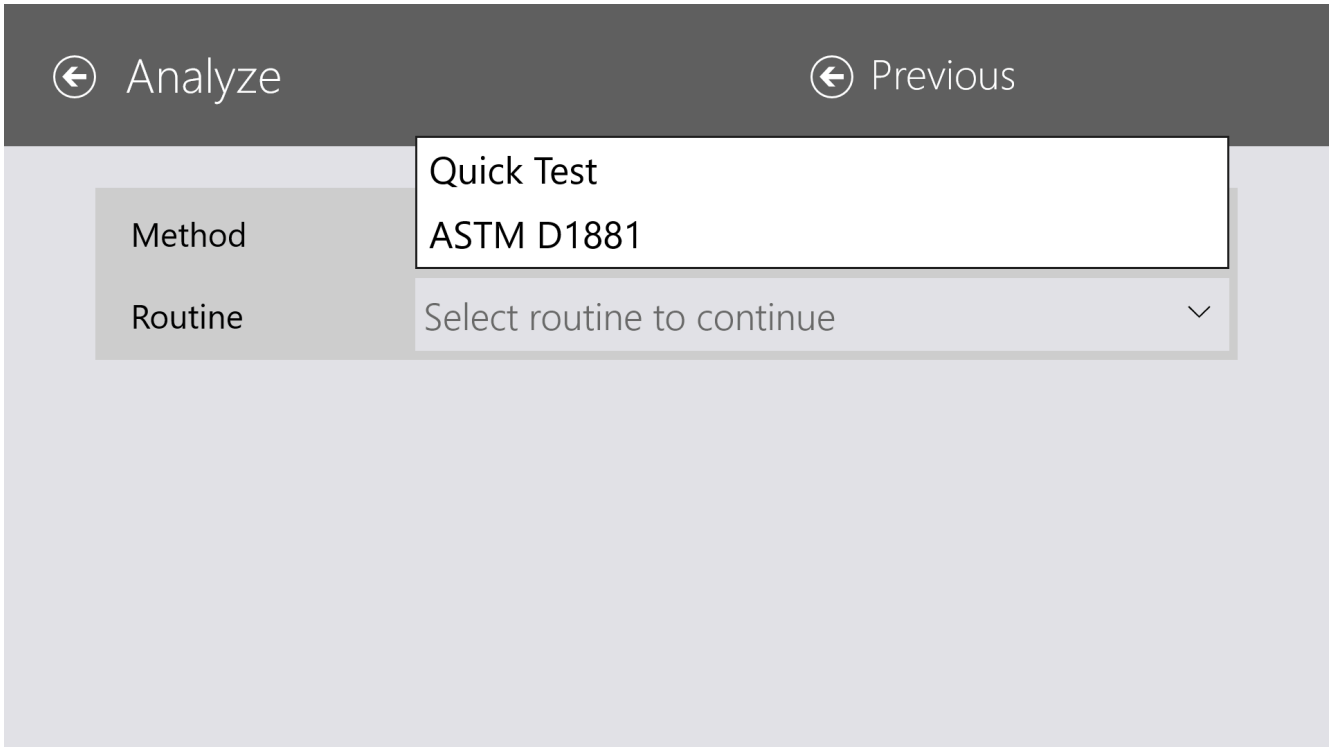
The operator will choose whether they want to run an ASTM D1881 analysis or a Quick Test.



Test	Parameters	Description
Test 1	88°C at 1000 mL / min	Initial Analysis
Test 2		Useful when using the 'Continue' option — Averages all three tests
Test 3		
Quick Test	24°C for 5 minutes	Air flow test at 1000 mL / min

The operator will have the option to run compliant analyses with the ASTM method for foam analysis. For more information, check out the published method with the ASTM.

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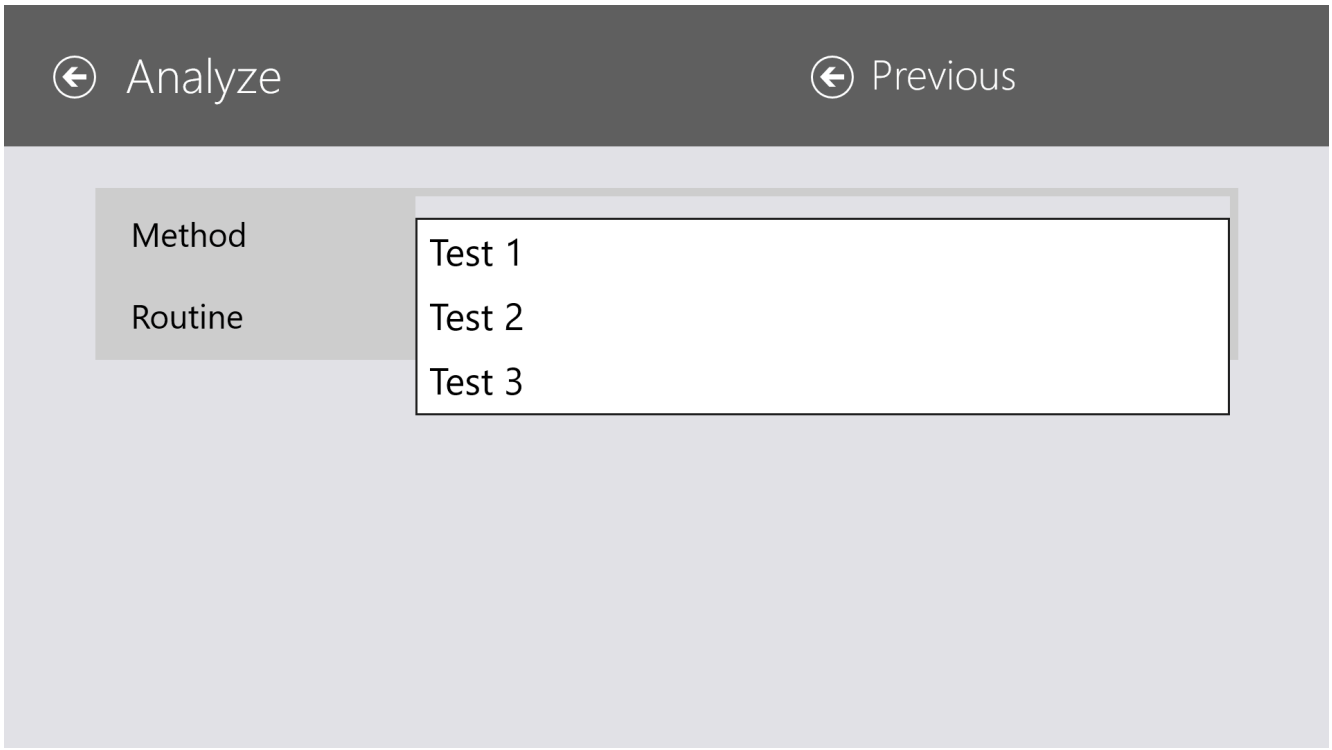
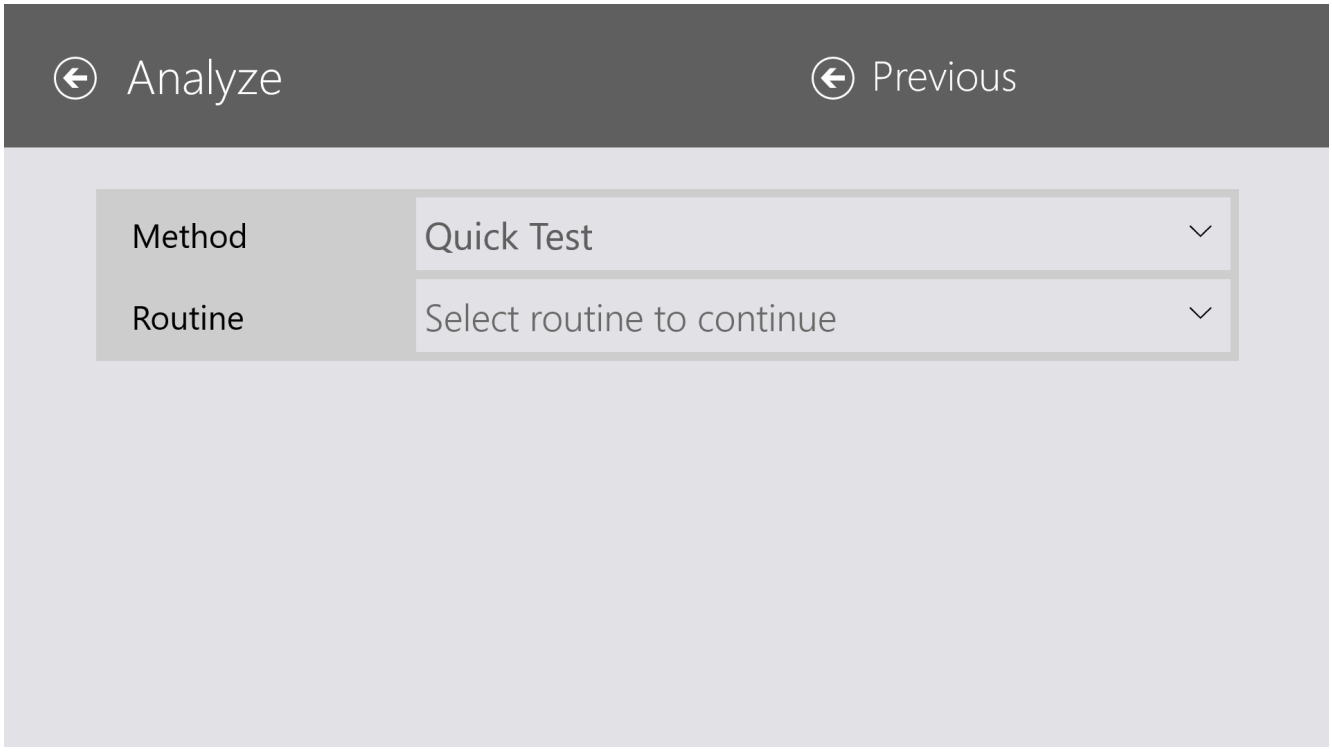


The operator will also have the choice to select a Quick Test. This option is a feature that conducts a room temperature demo analysis.



If a Quick Test is selected, the operator will need to select Test 1 as the routine.

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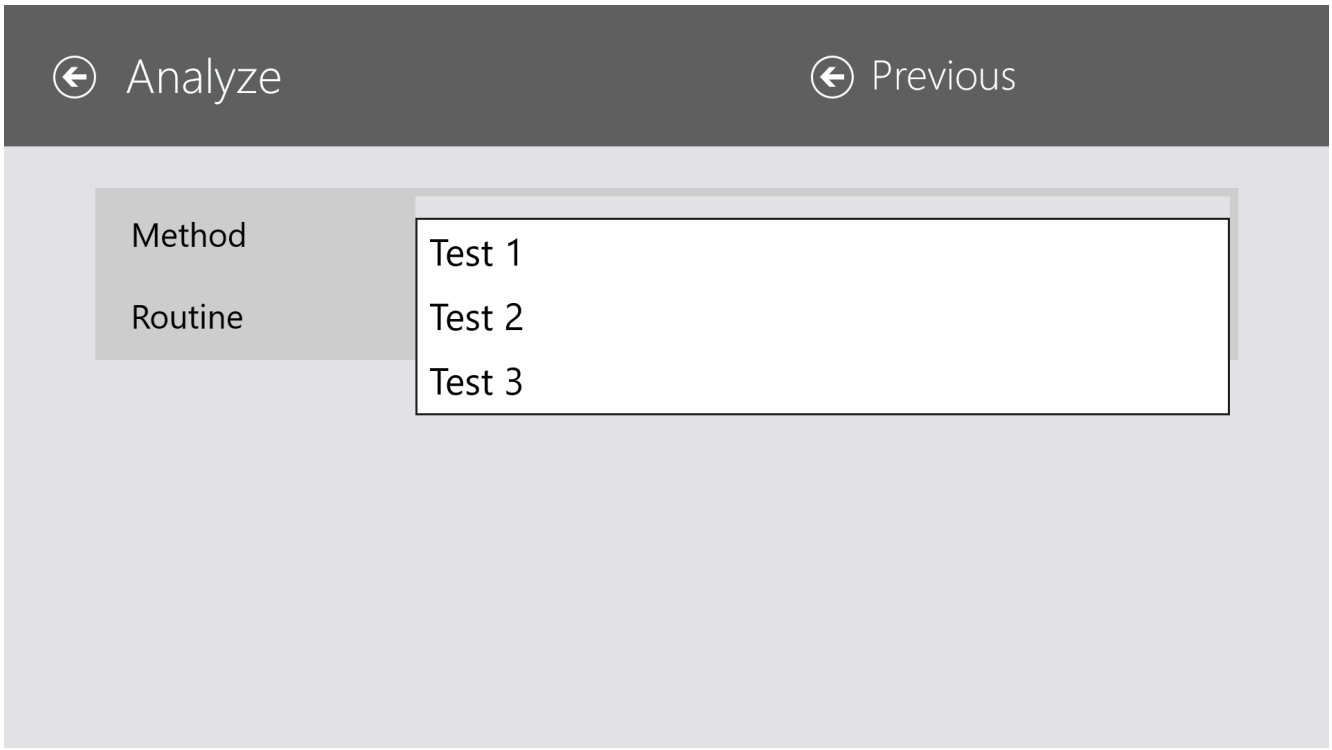

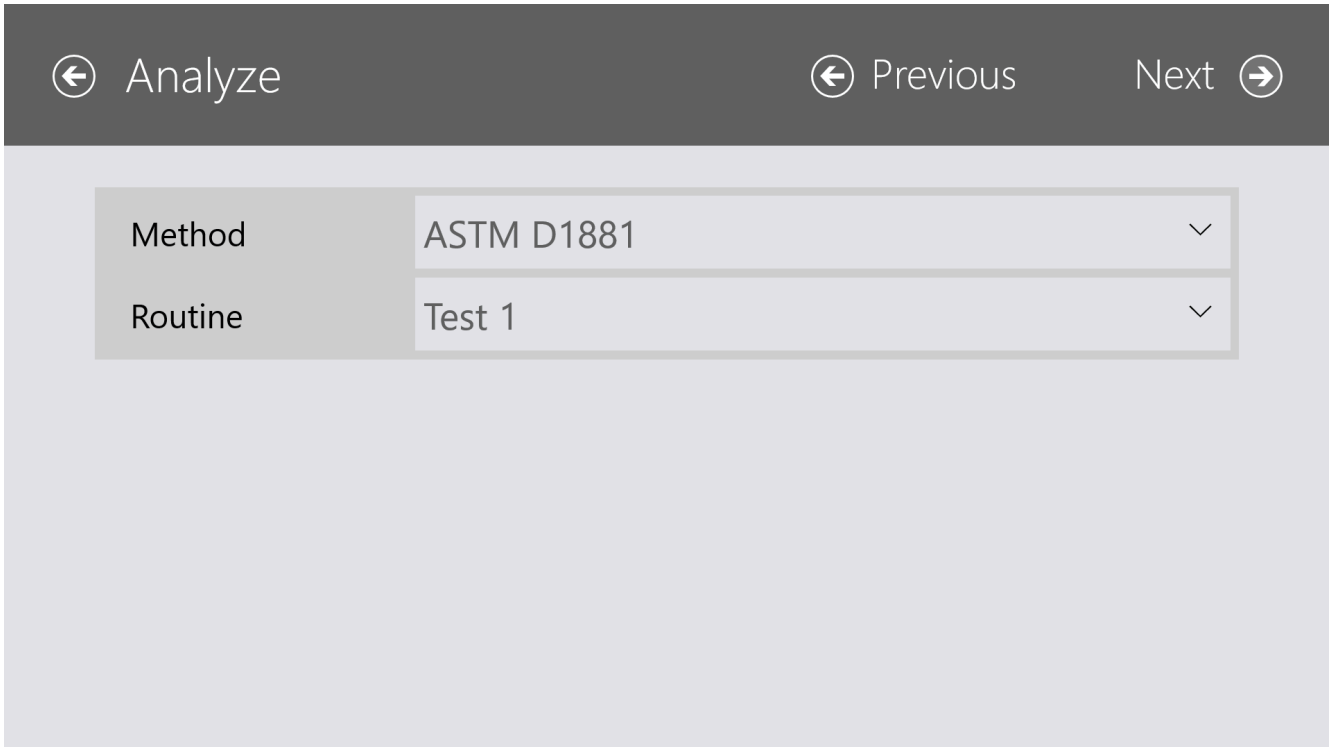

Once the method and routine are selected, Next will appear at the top-right hand corner. Select Next.

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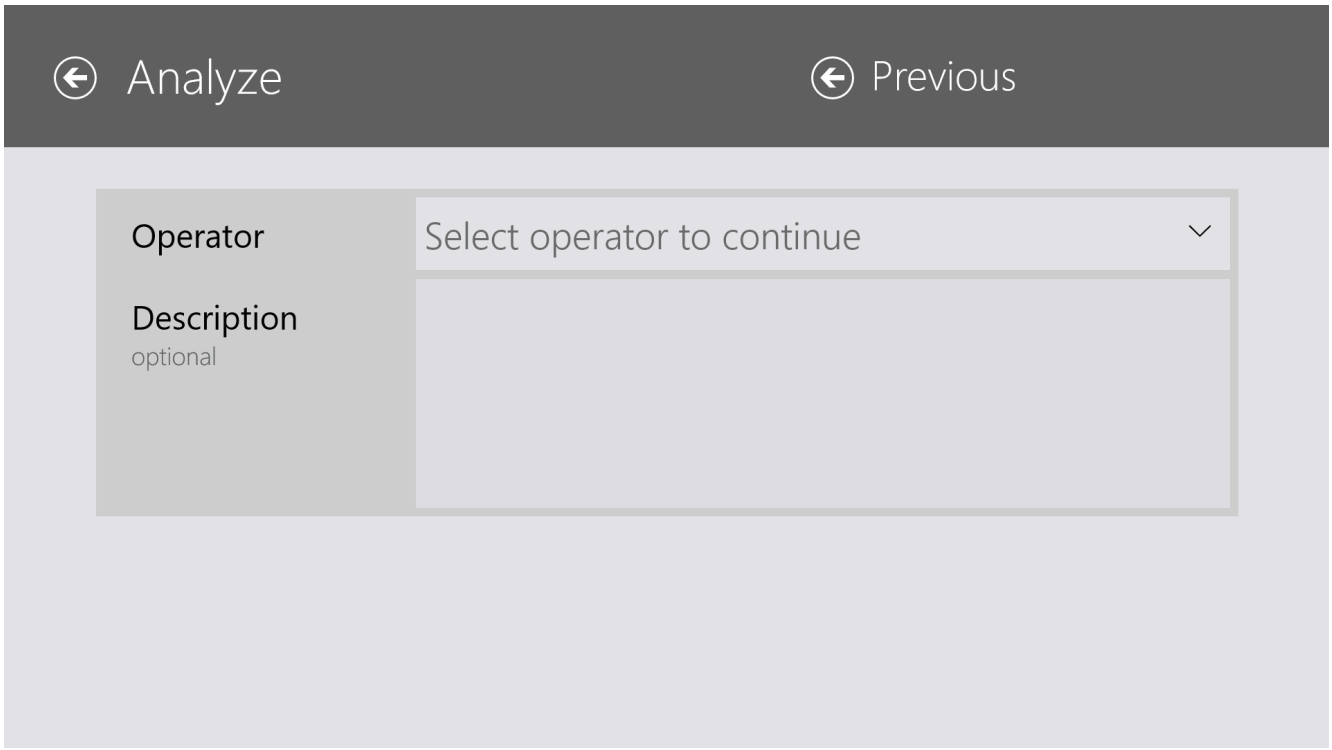
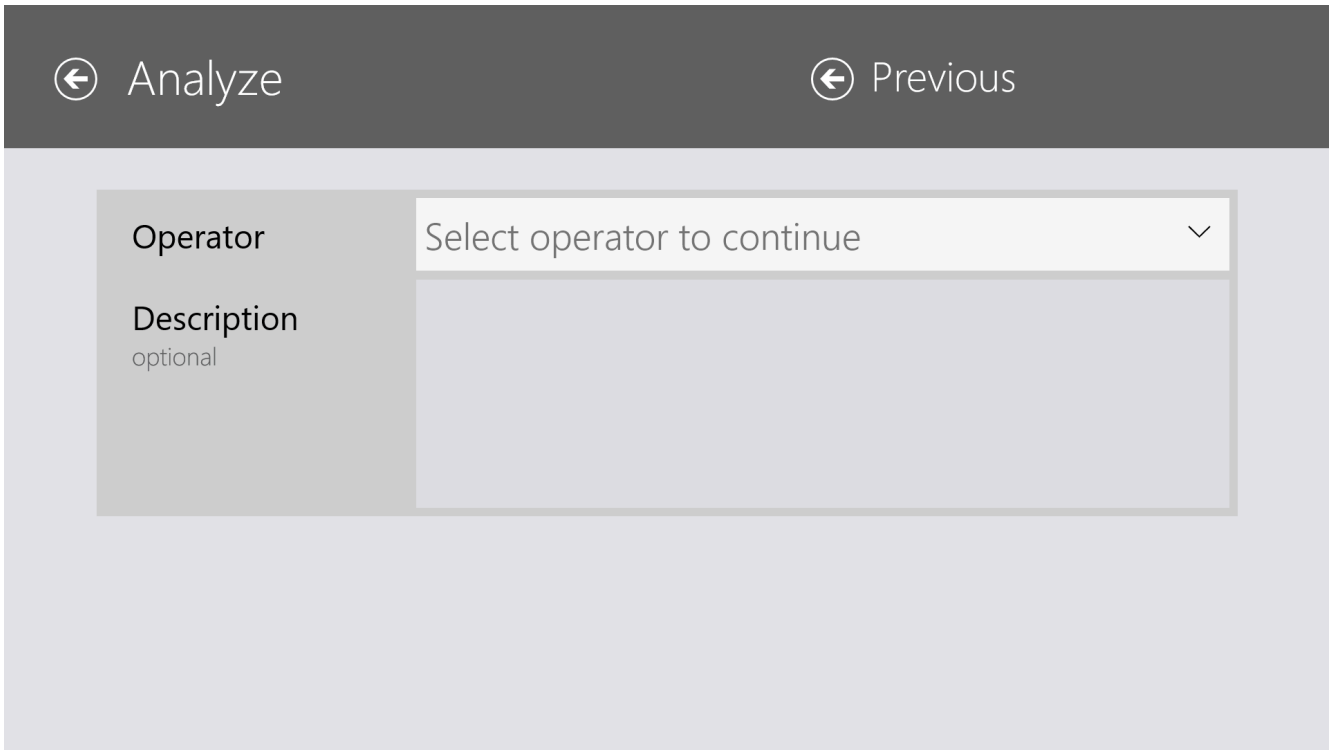
The next selection is the Operator. Operator names are fixed and selected using a dropdown box. To add new operator names, please consult the Settings section of this manual (**Section 5.4.2**).





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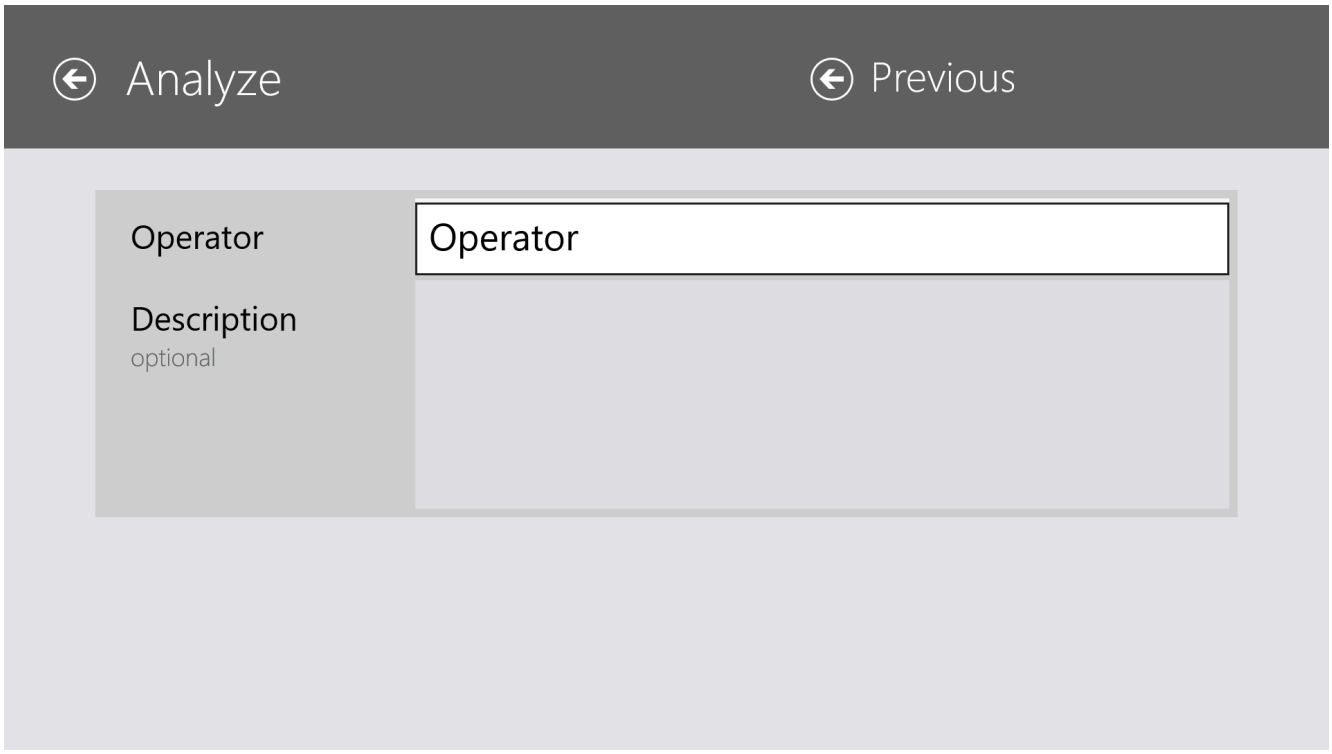
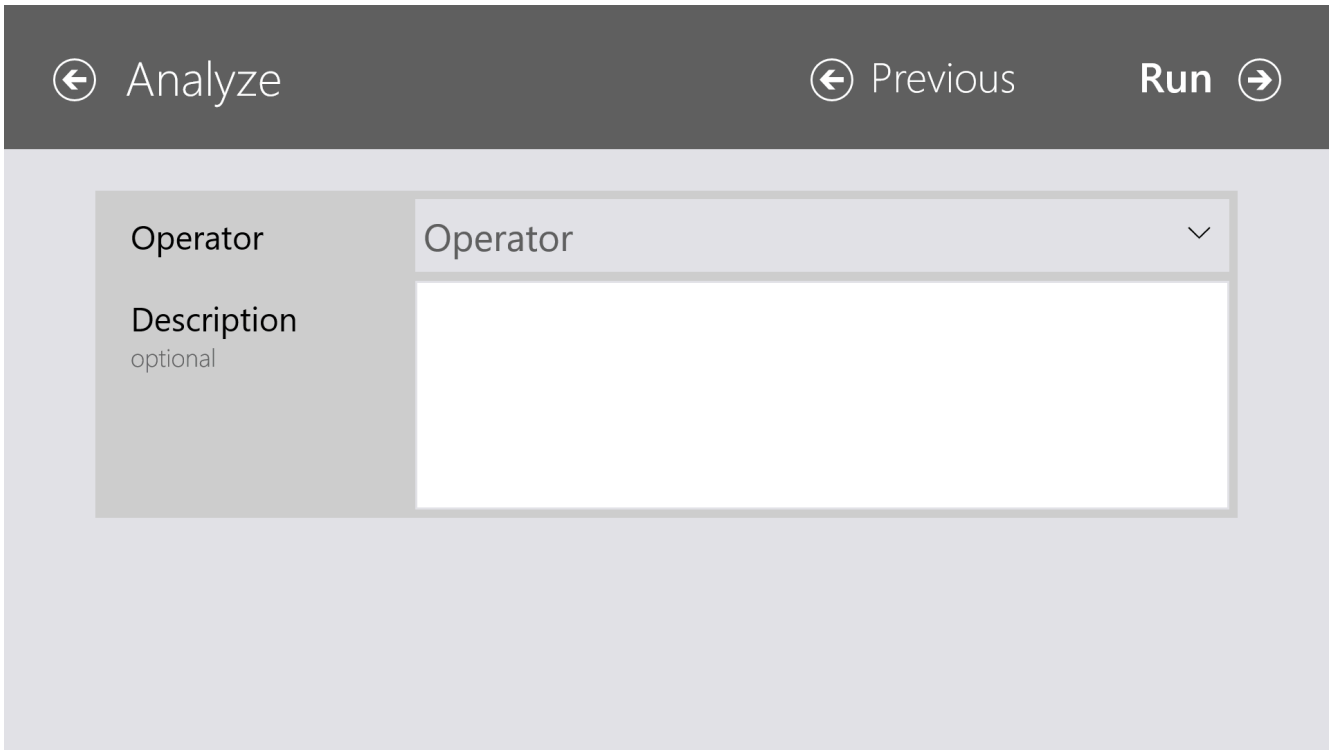


Below the Operator selection there is an optional Description or comments of a sample inspection. This field is used to enter other information, such as notes, batch ID, etc. To enter notes, simply tap or click on the white entry box to display an onscreen keyboard.



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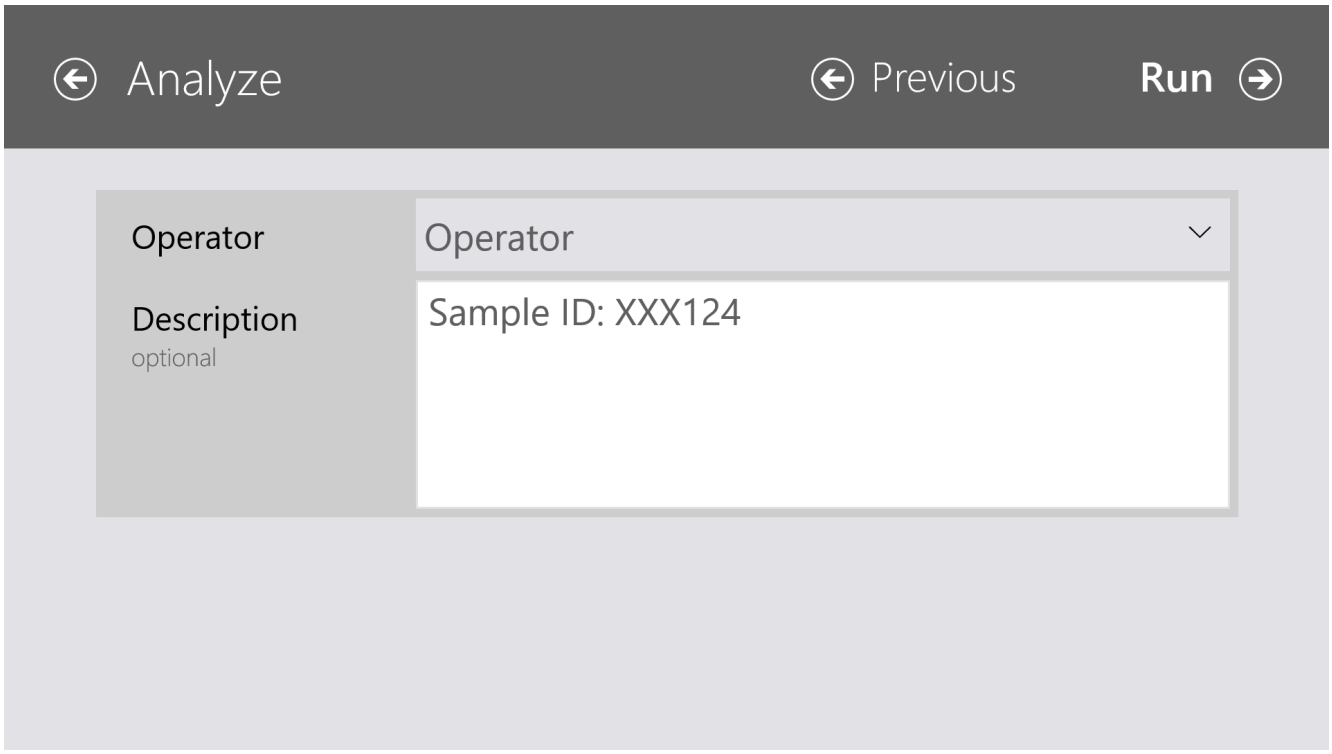
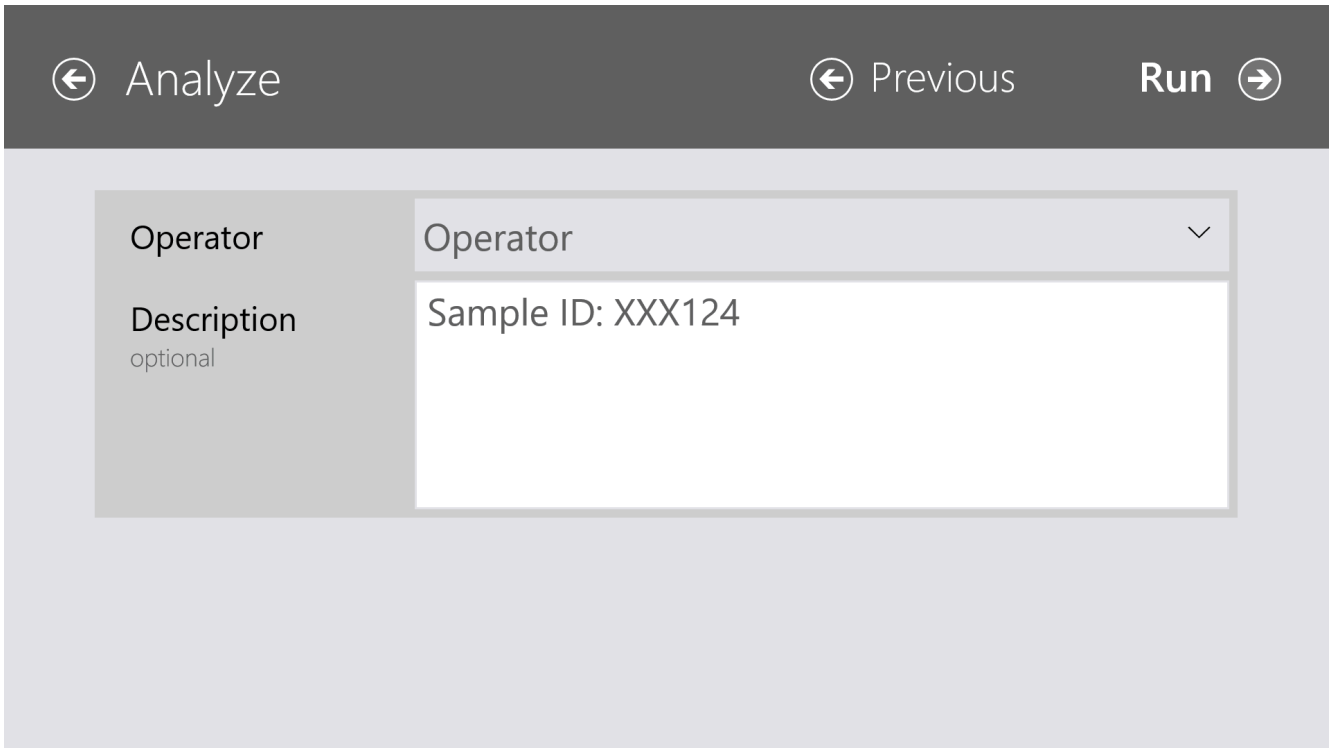


Once complete, press the Run button to proceed with your inspection.



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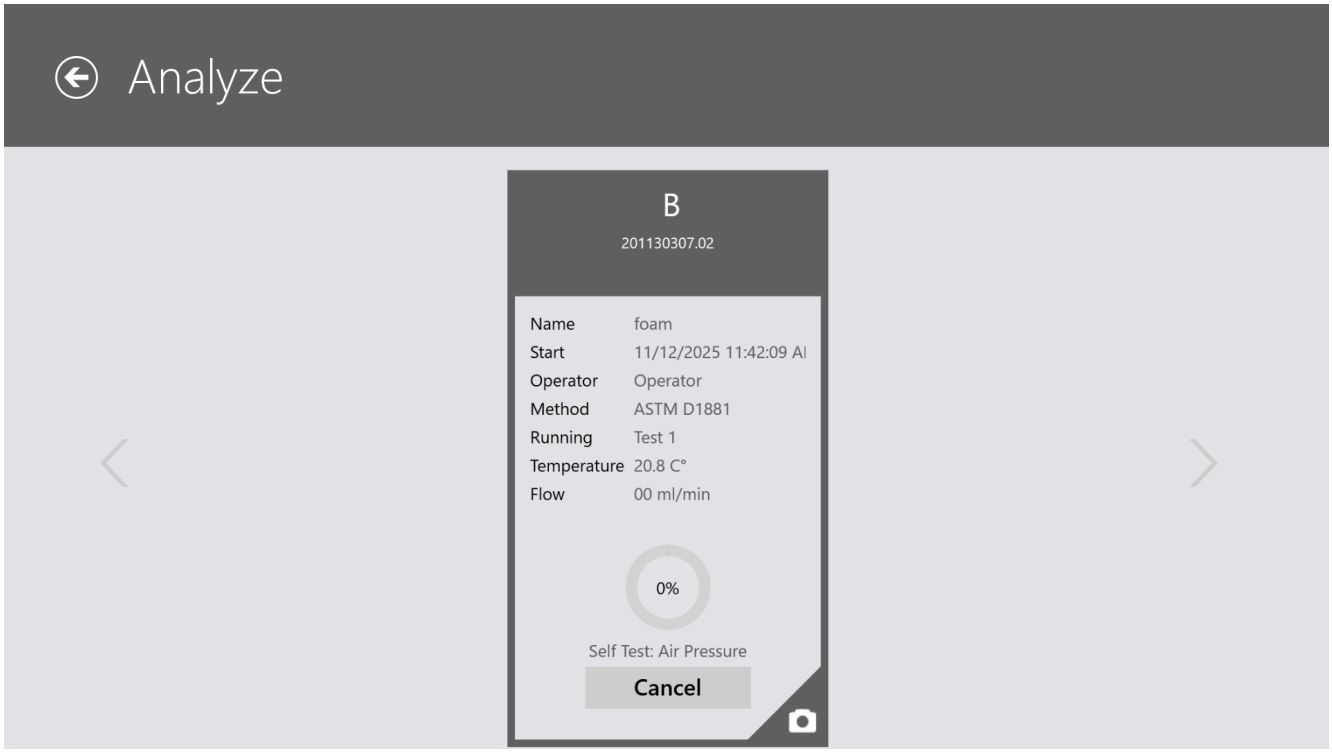
After completing all the necessary sample identifiers (sample ID, operator name, test method, and optional description) and clicking Run, the automated visual analysis starts.

A progress and information bar will appear and display the status of the analysis.



A live feed showing the sample can be accessed by clicking the camera icon on the bottom right. To return to the identification screen, simply click the information icon.

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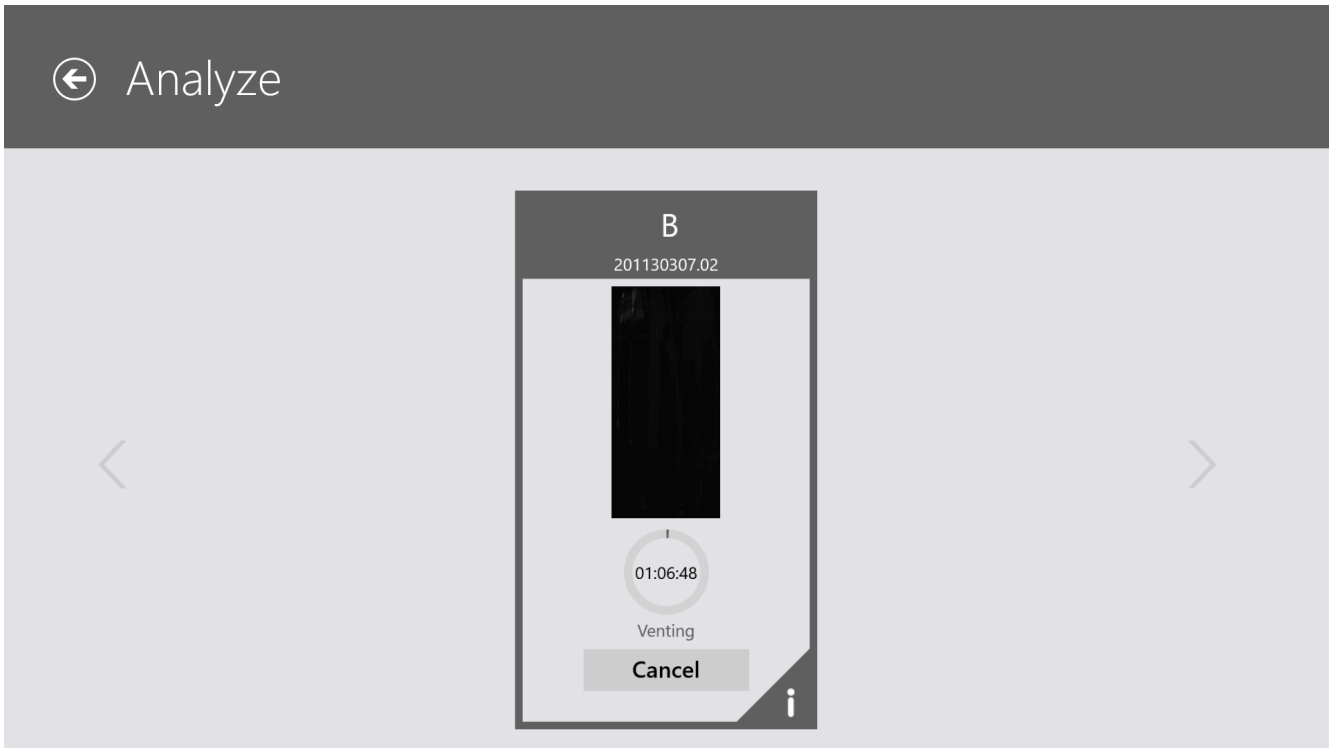
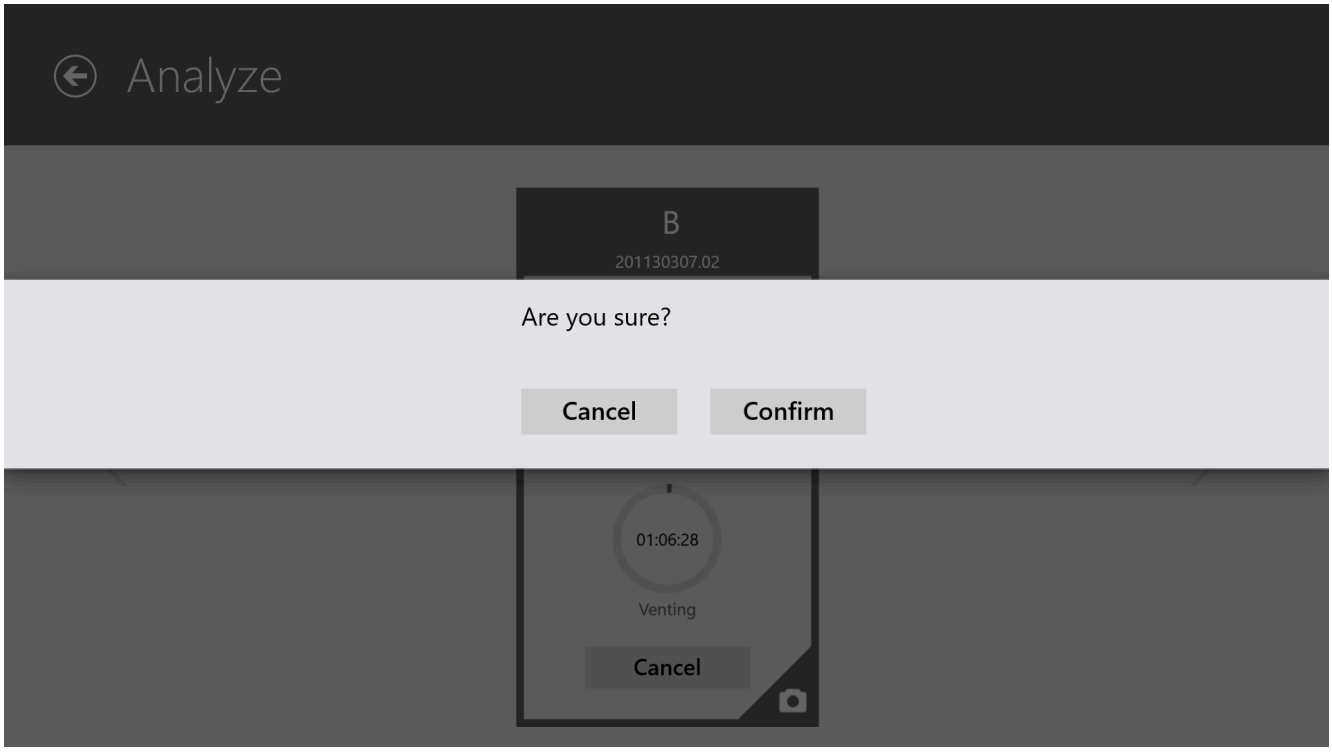


At this point, the operator may select a second module to start a second asynchronous analysis following the same procedures as described above.

An analysis can be cancelled at any time by Cancel and selecting confirm.



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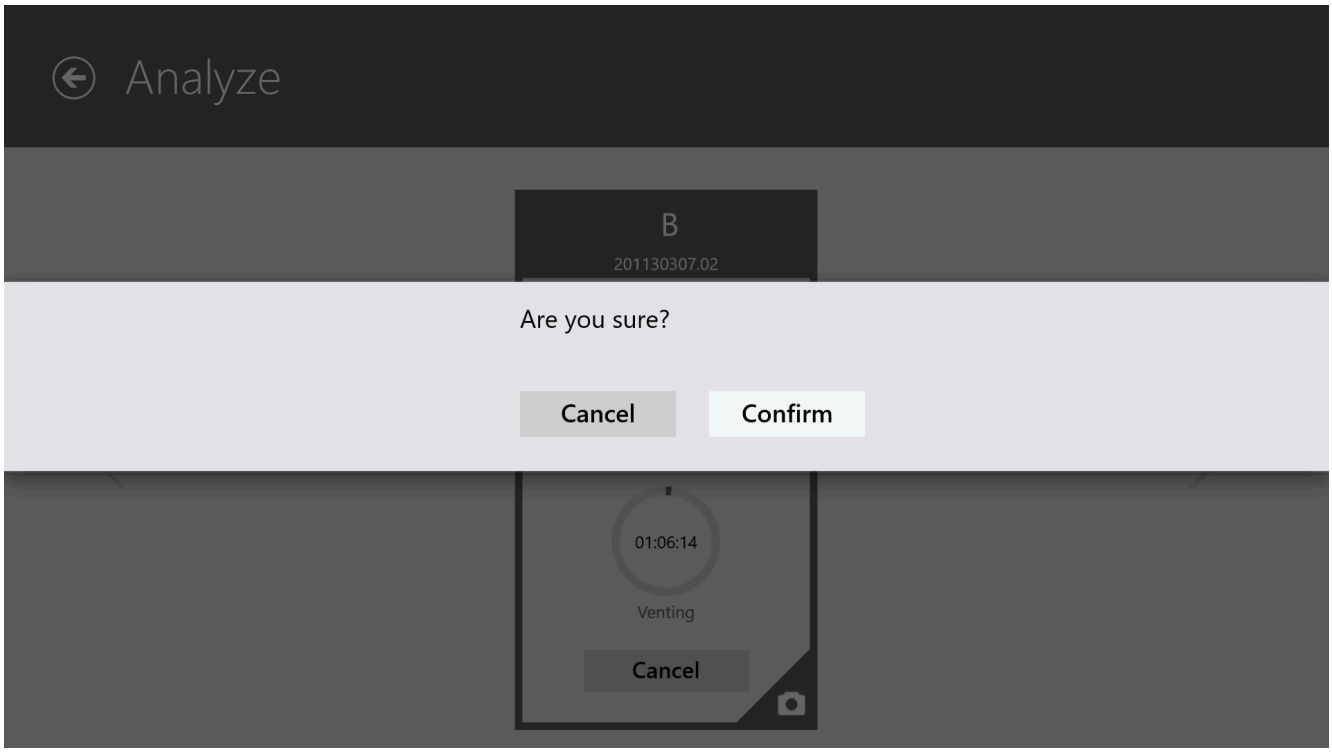


5.2.3. Amines

This section details the protocols and procedures to test to foamability of lean amines.

5.2.3.1. Sample and Cylinder Preparation

Prepare 150 mL of the lean amine under the conditions desired to be tested.

Do not use rich amine or amine with sulfur species. The corrosive nature of sulfur species will deteriorate the system and operators will be exposed to potentially dangerous sulfur species during the foaming procedure.

Attach the gas diffuser to the stopper assembly. Insert the stopper firmly into the cylinder to ensure an airtight seal.

Insert the prepared cylinder into the FoamDDI chamber with the white alignment mark facing forward. Connect the probe and air tubes.

Adjust the cylinder so that the graduations face to the side for the vision system. If connections cause the cylinder to rotate, re-align by carefully adjusting the stopper. Ensure the cylinder sits centrally within the module.

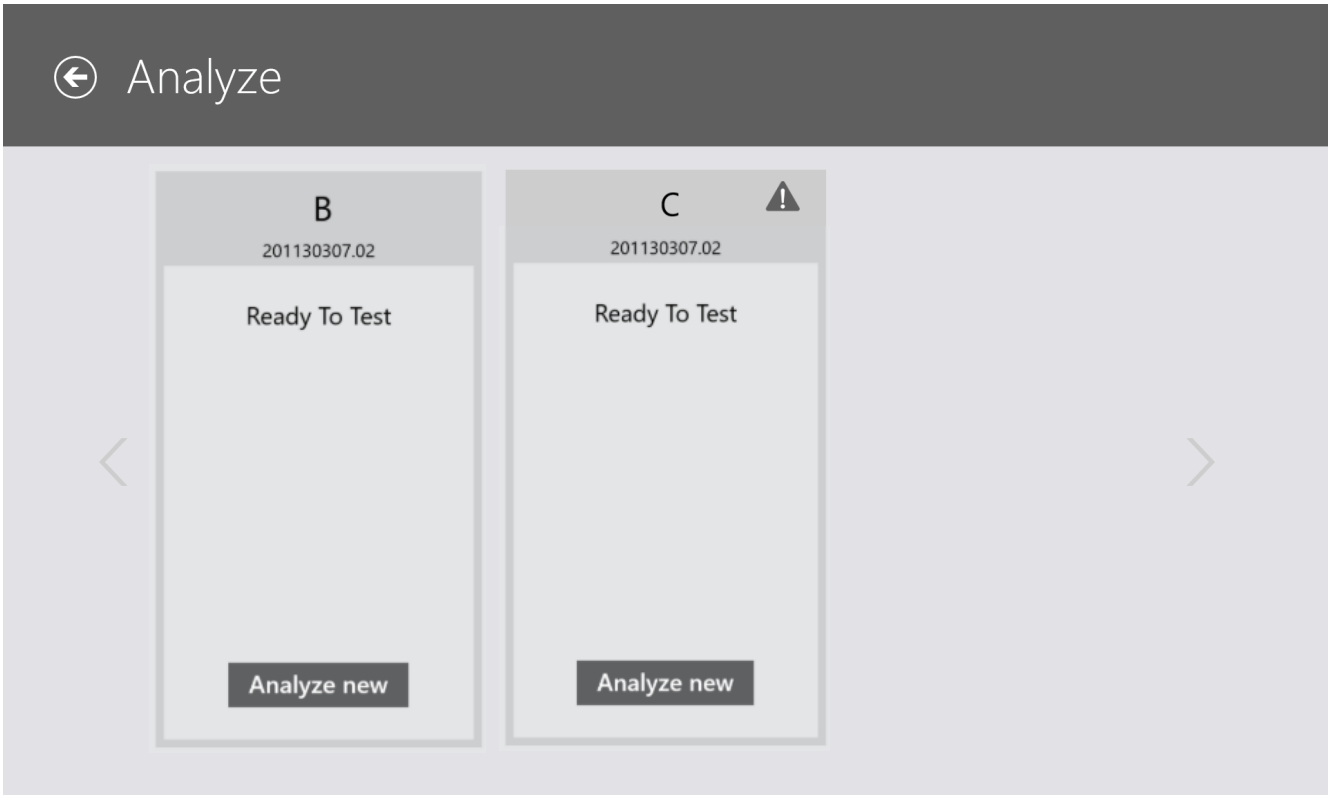
From the Main Screen, tap Inspect.

5.2.3.2. Running an Aminic Analysis

Running an inspection is achieved using a simple Wizard-driven menu. Upon entering the Inspect menu, the user is presented with a box showing connected FoamDDI module(s).

Choose whichever module is to be used (if multiple are active) and select Analyze New. Checkout **Section 5.4.8** to connect multiple modules.

If two modules are connected, they will both appear side by side. Any warnings or errors will be indicated by the triangle containing an exclamation mark. Click on the warning icon or the Detail icon to find out more information. **Section 6** has more information on warnings and errors.



Once selecting Analyze New, two options will propagate: New or Continue. Choose New to run a completely new sample. The Continue option allows you to append to an existing sample ID, if one has started within the past 3 hours.

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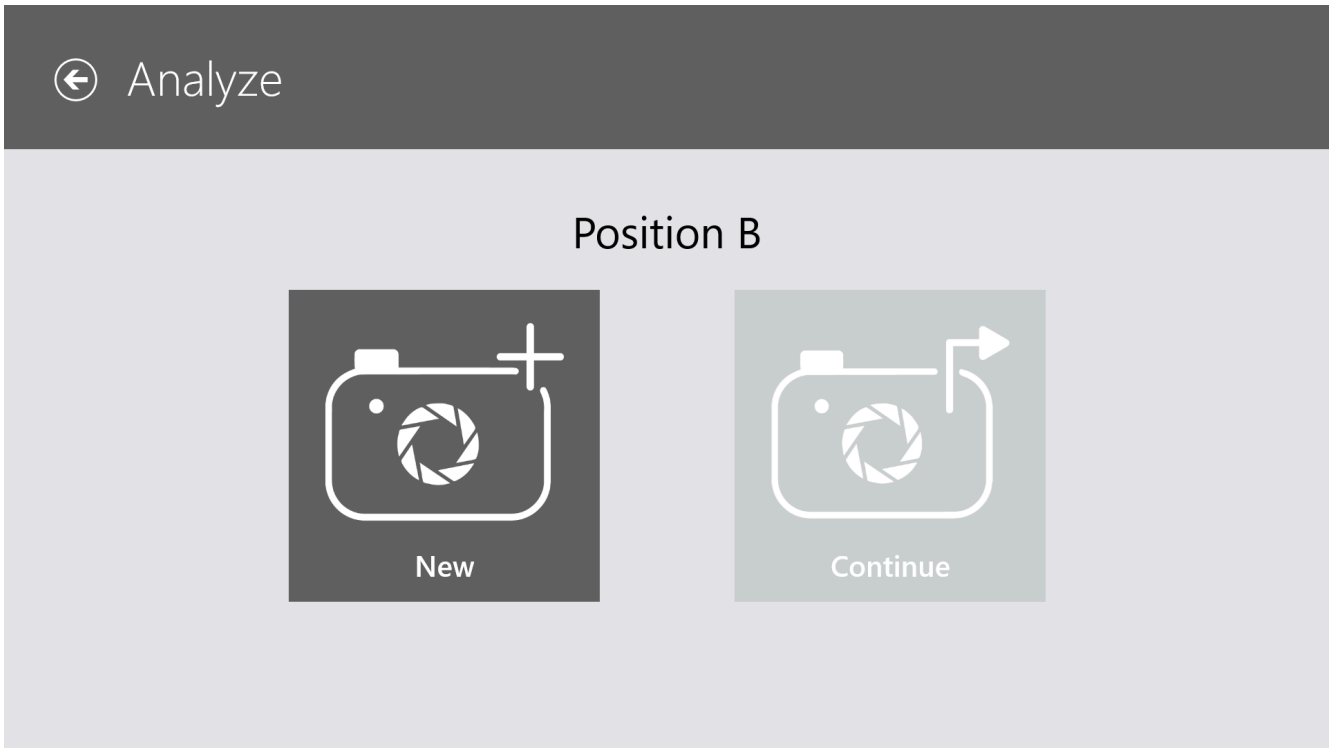

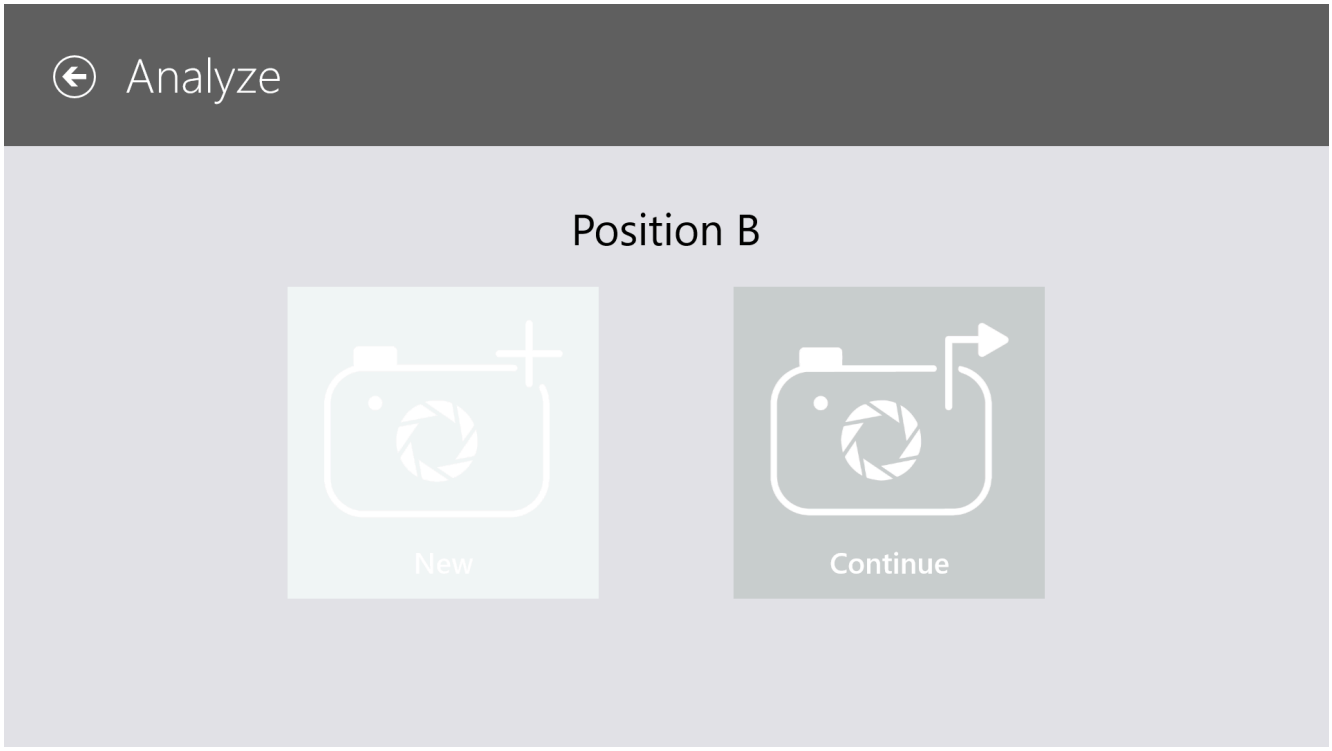

When starting a New sample, you will be asked to enter a sample name. Using the Continue option will propagate this information for the operator automatically. The user

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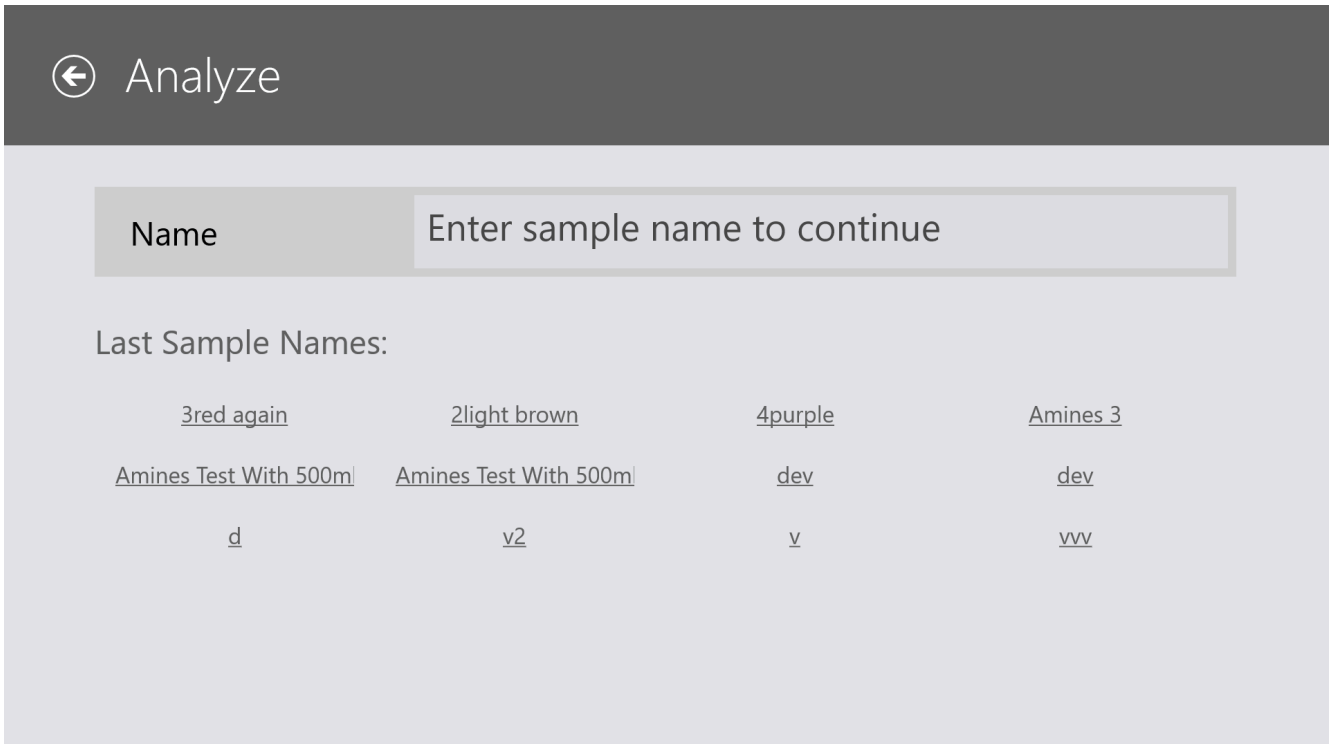
can click or tap in the sample name section to display an onscreen keyboard for entry of a new sample ID or select a sample name from the last twelve samples tested.

The keyboard allows the entry of up to 255 characters and accepts standard ASCII characters. Changes and deletions of text can also be made using the keyboard.



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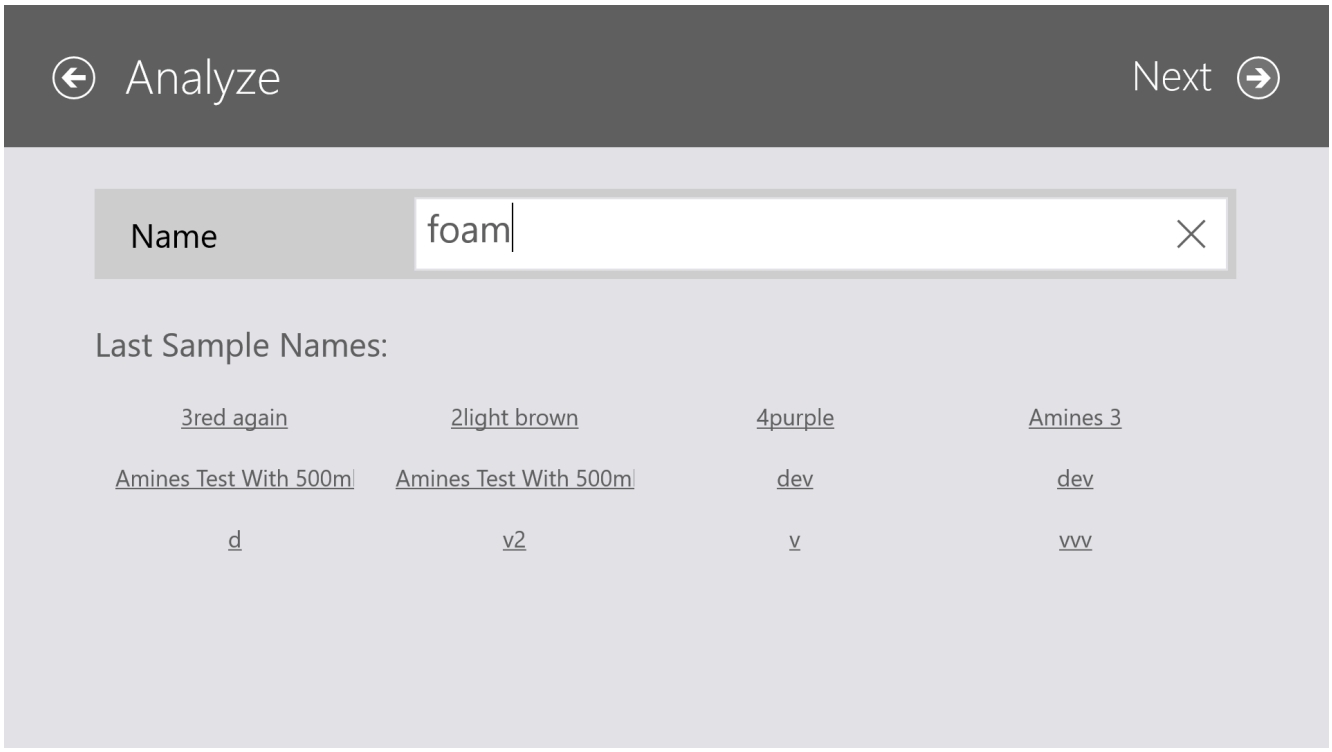
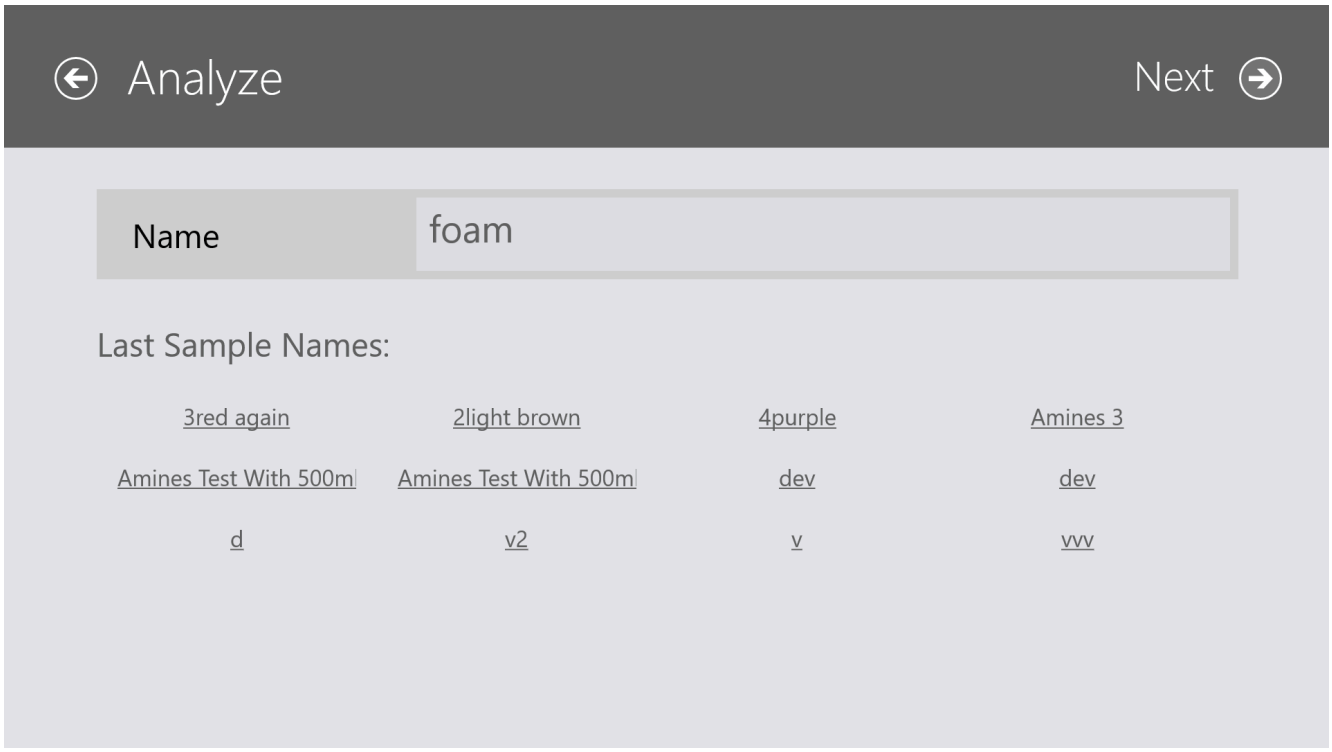


After successful entry of a valid sample ID, the Next button will become activated. The sample ID is confirmed using the Next button located at the top-right hand corner. Select Next.



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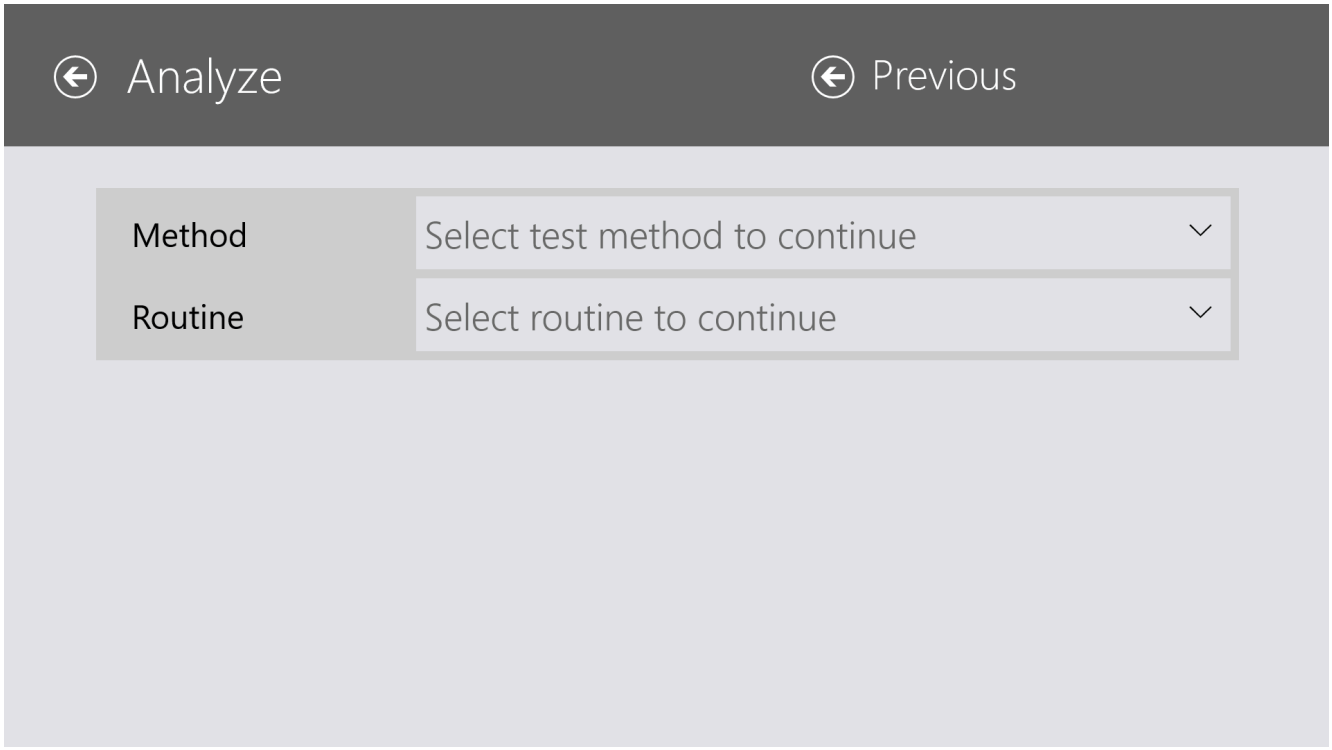
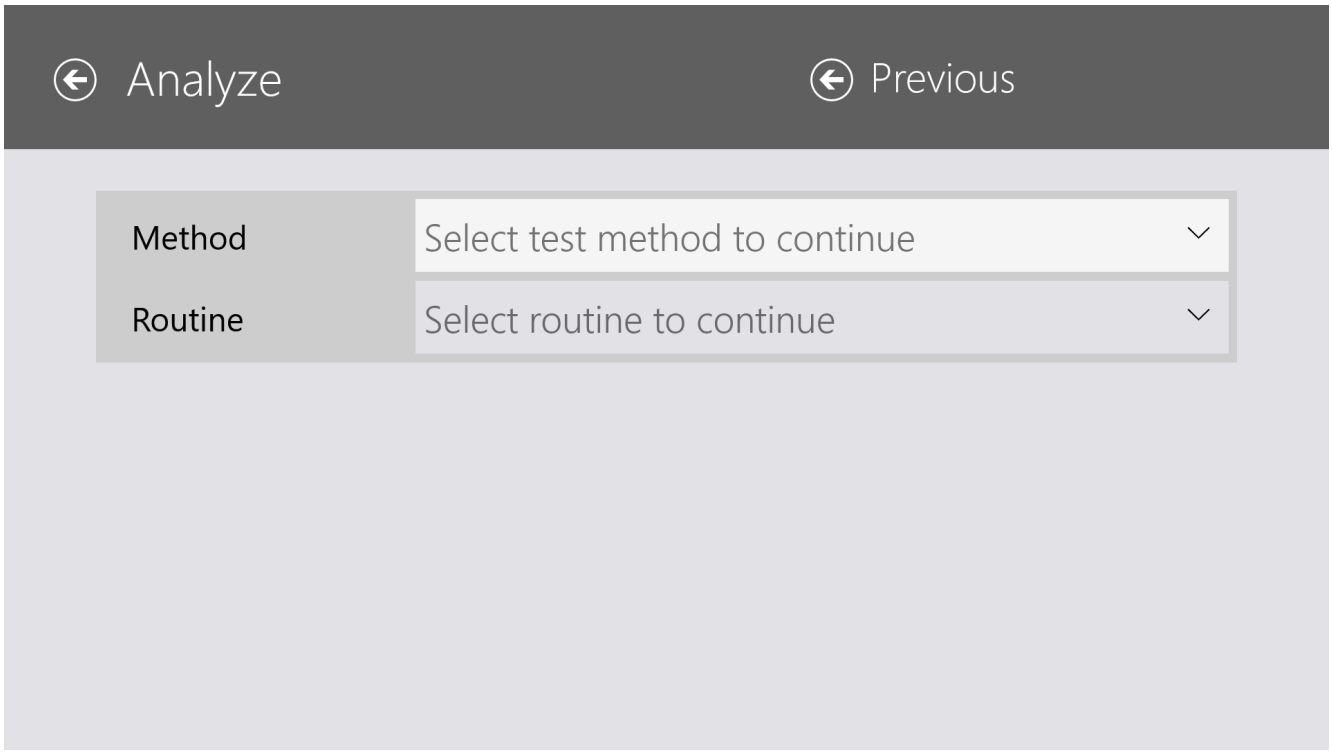
After pressing the Next button, the operator is presented with options to enter a Test Method and Routine.

Methods and Routines are fixed and selected using a dropdown box.





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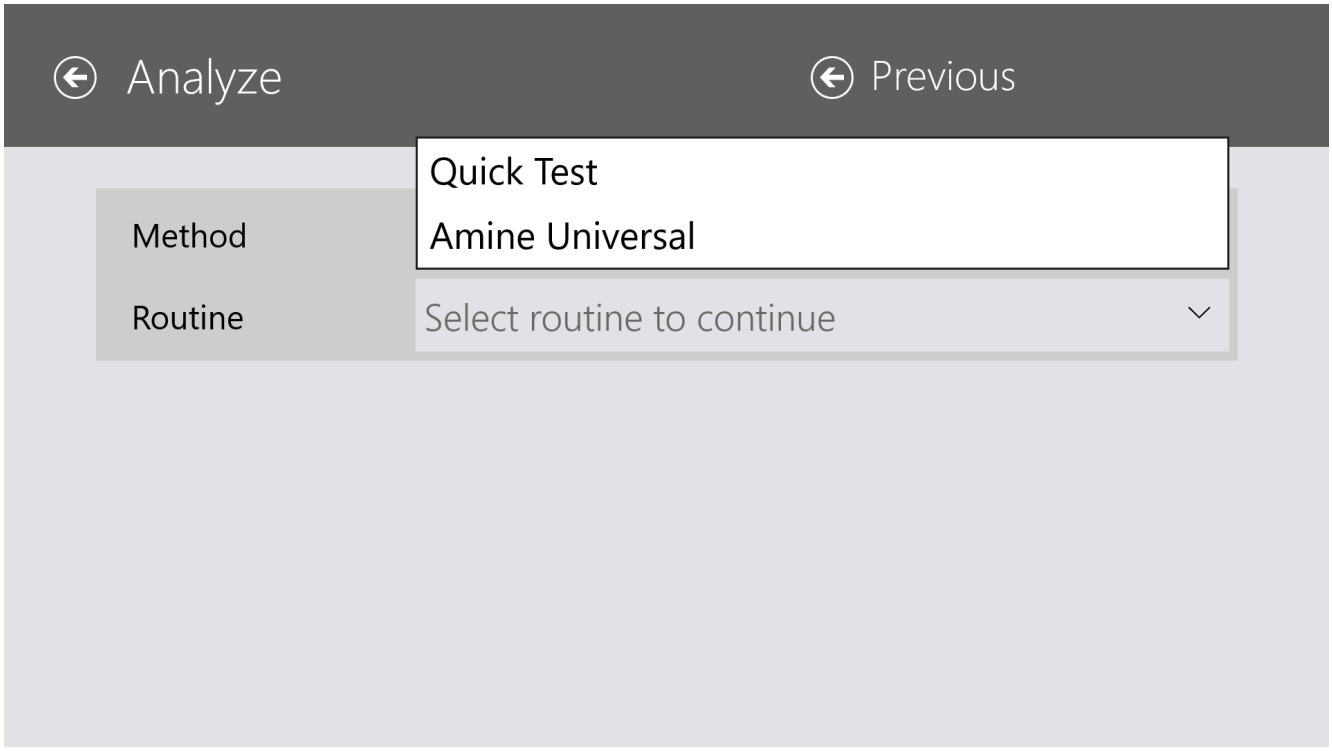
The operator will choose whether they want to run an Aminic analysis or a Quick Test.



Then they will select the type of test to run from the Amine Universal method or run a Quick Test:

Test	Parameters	Flow
Test 1	24℃	100 mL / min
Test 2	35℃	500 mL / min
Test 3	95℃	1000 mL / min
Quick Test	24℃	1000 mL / min

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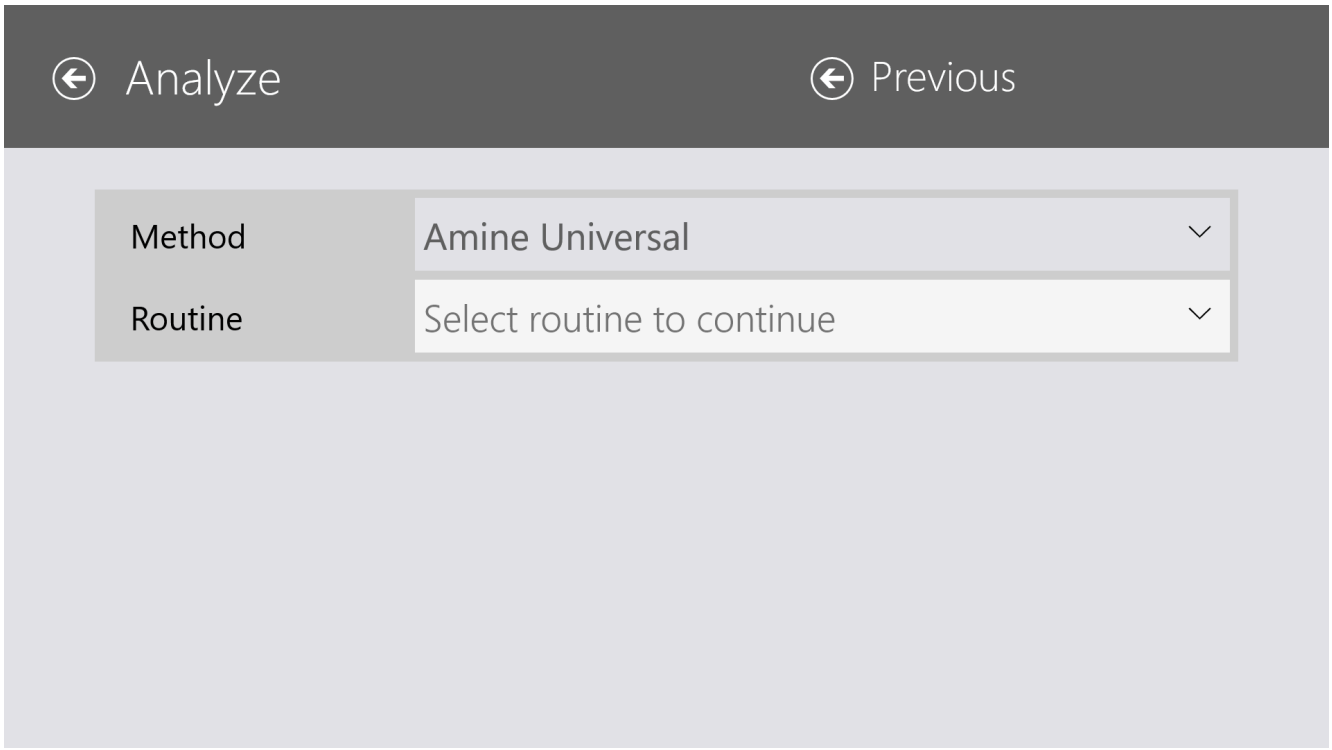
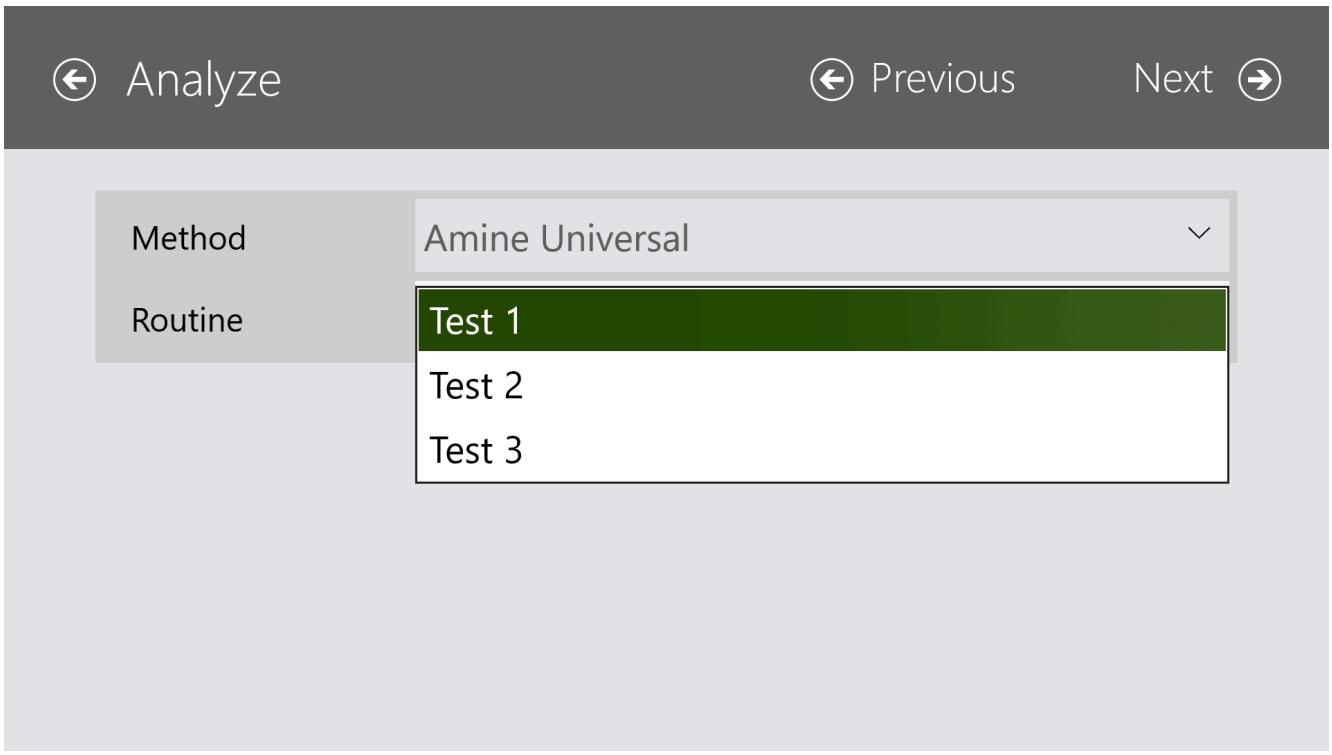
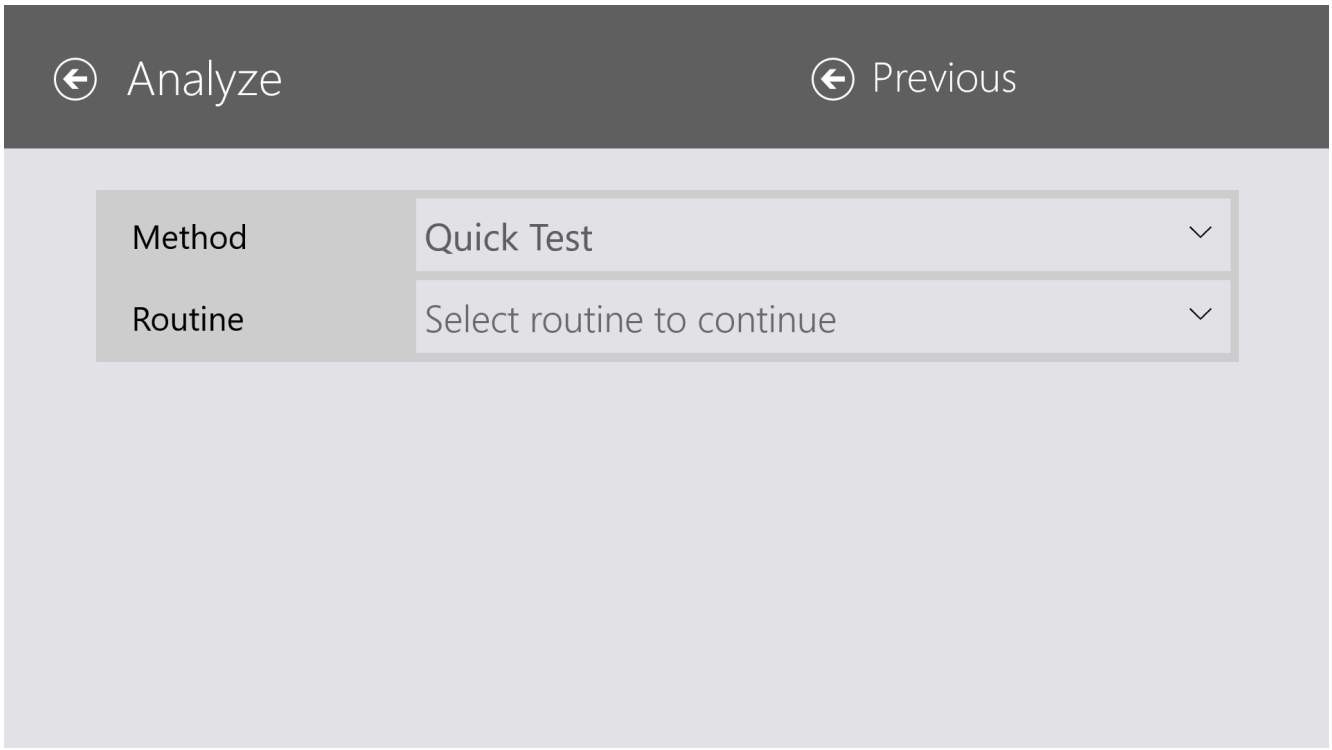
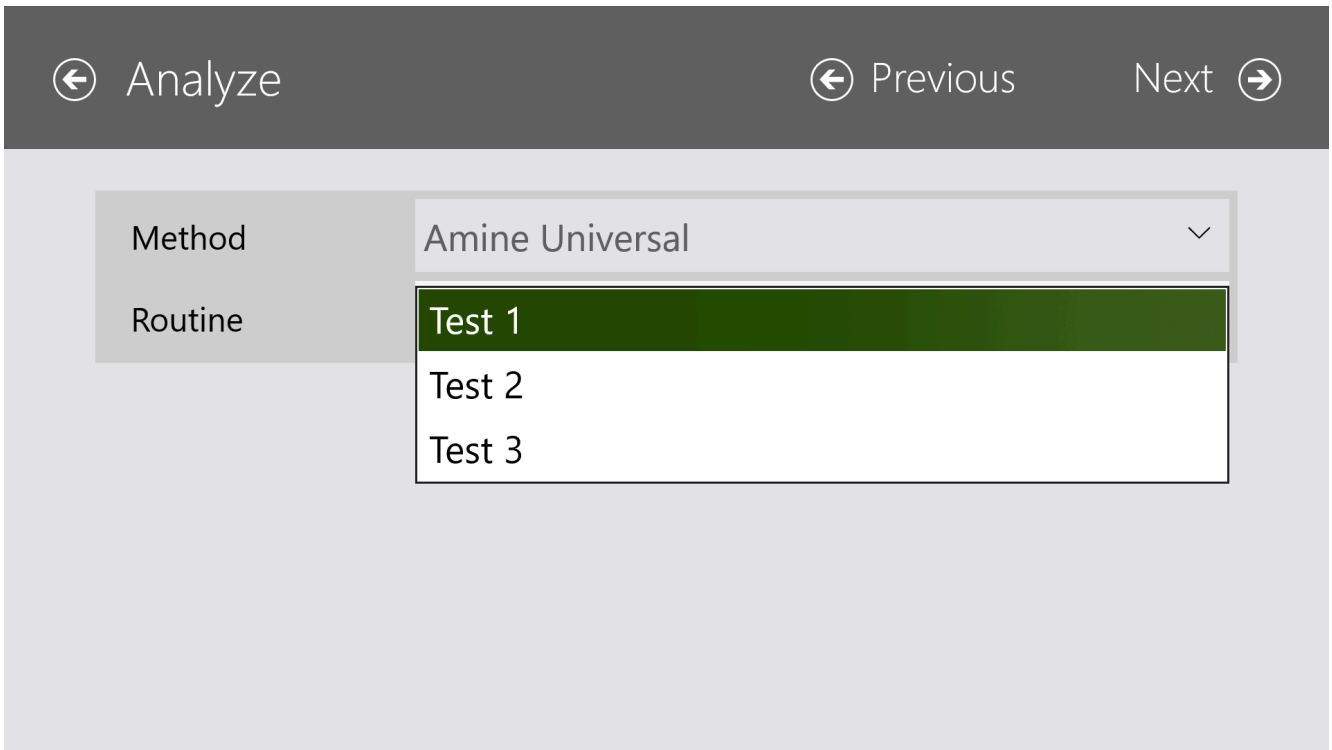




The operator will also have the choice to select a Quick Test. This option is a feature that conducts a room temperature demo analysis.



If a Quick Test is selected, the operator will need to select Test 1 as the routine.



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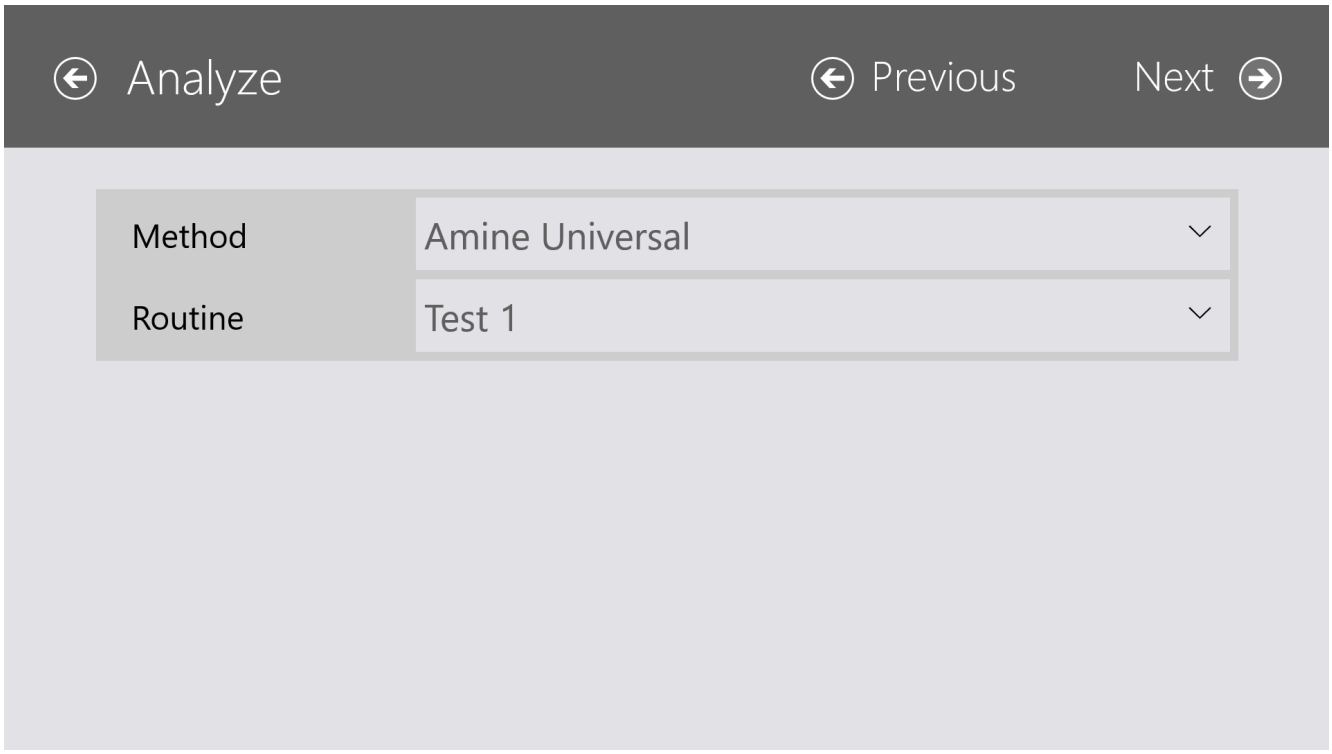

Once the method and routine are selected, Next will appear at the top-right hand corner. Select Next.

The next selection is the Operator. Operator names are fixed and selected using a dropdown box. To add new operator names, please consult the Settings section of this manual (**Section 5.4.2**).

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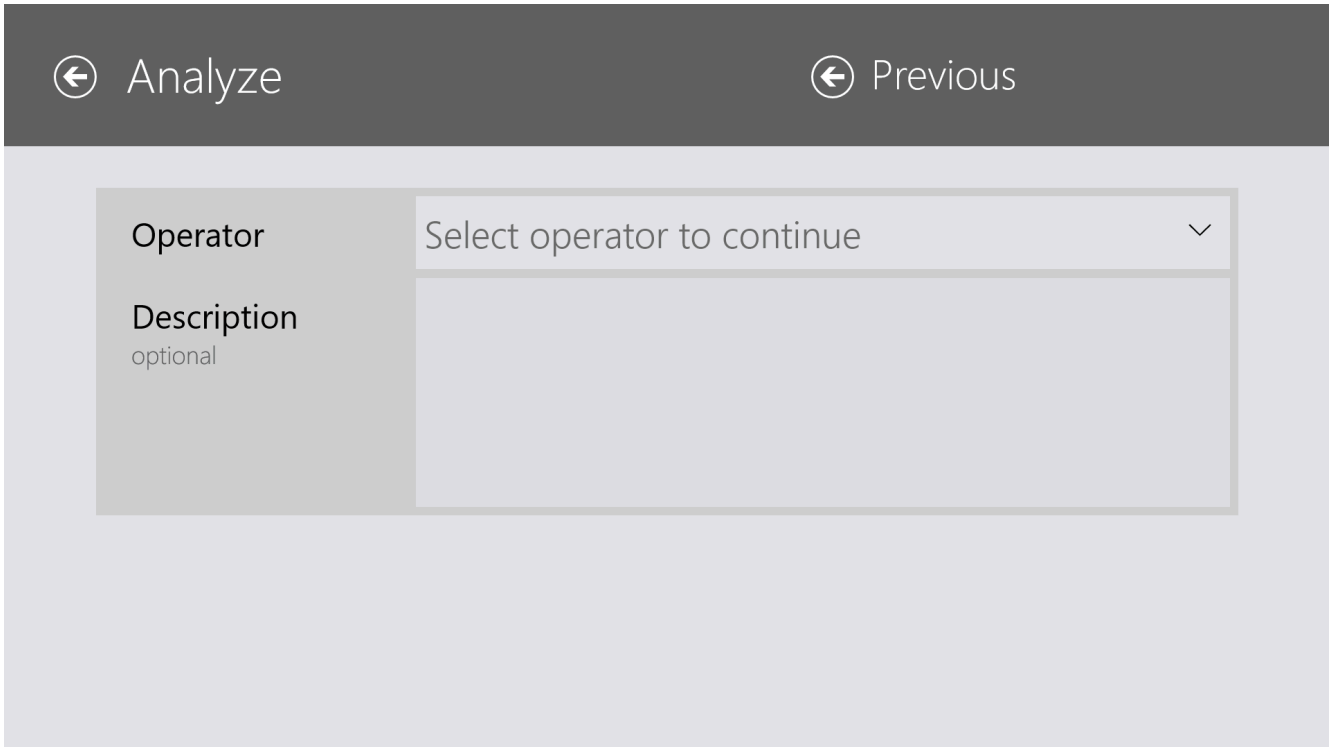

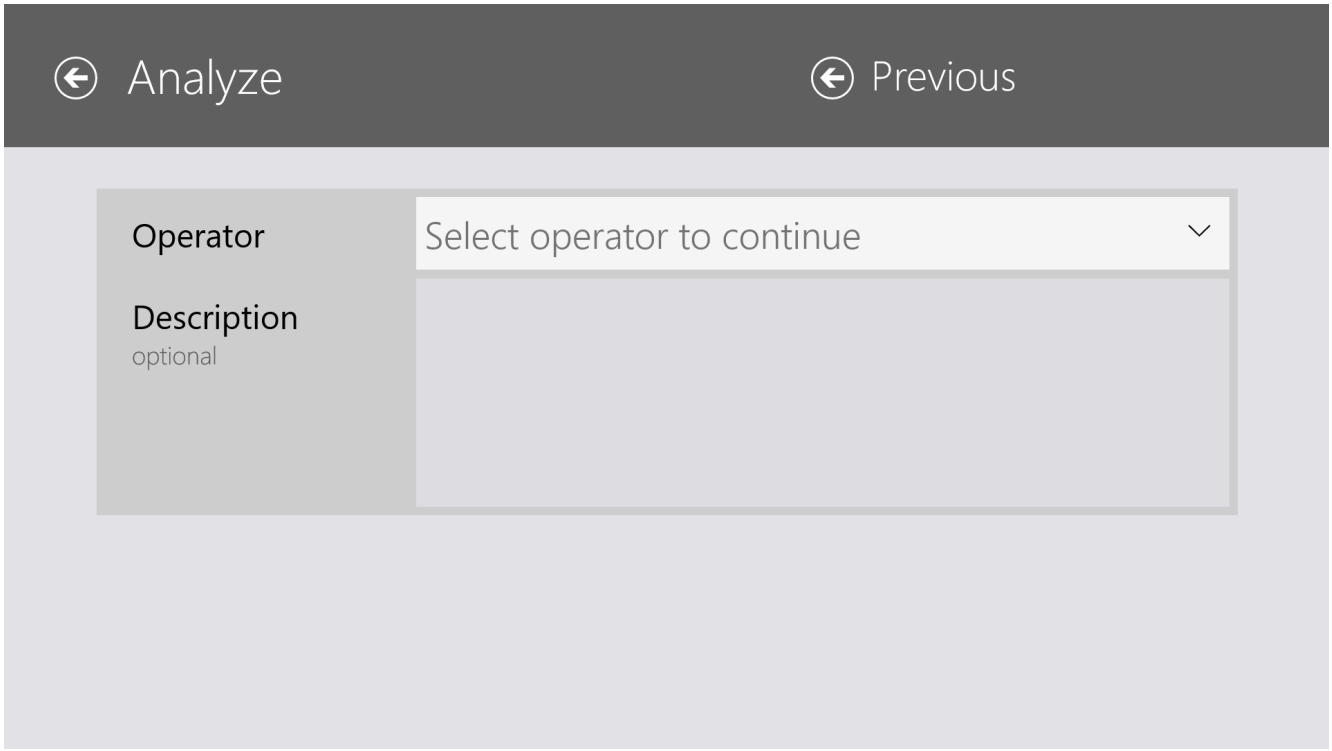

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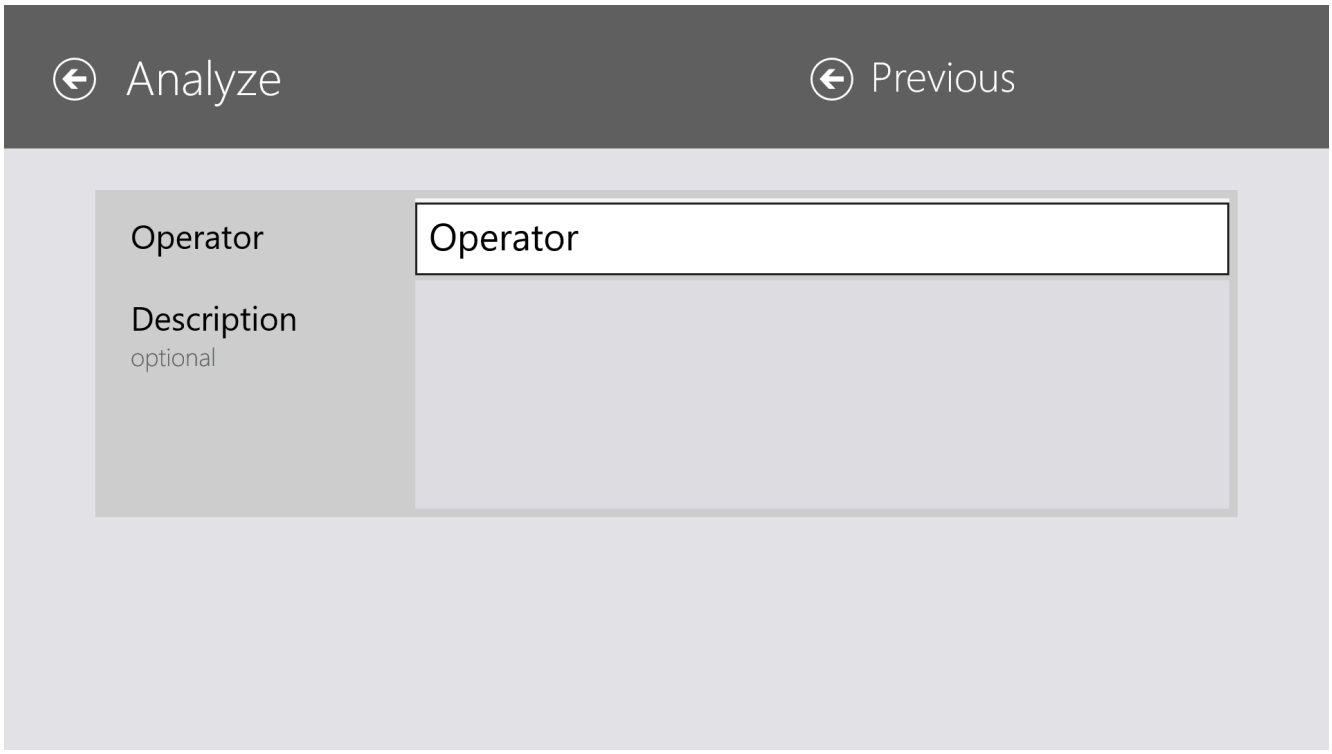
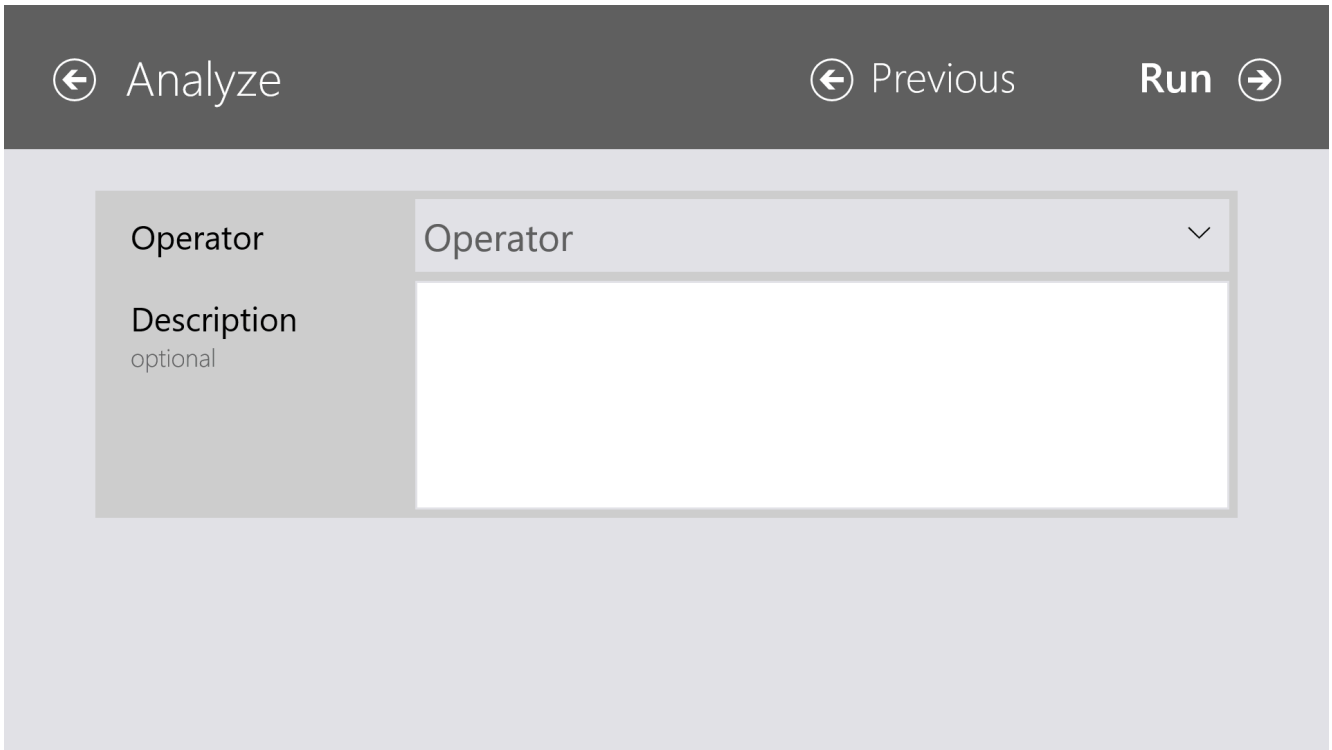


Below the Operator selection there is an optional Description or comments of a sample inspection. This field is used to enter other information, such as notes, batch ID, etc. To enter notes, simply tap or click on the white entry box to display an onscreen keyboard.



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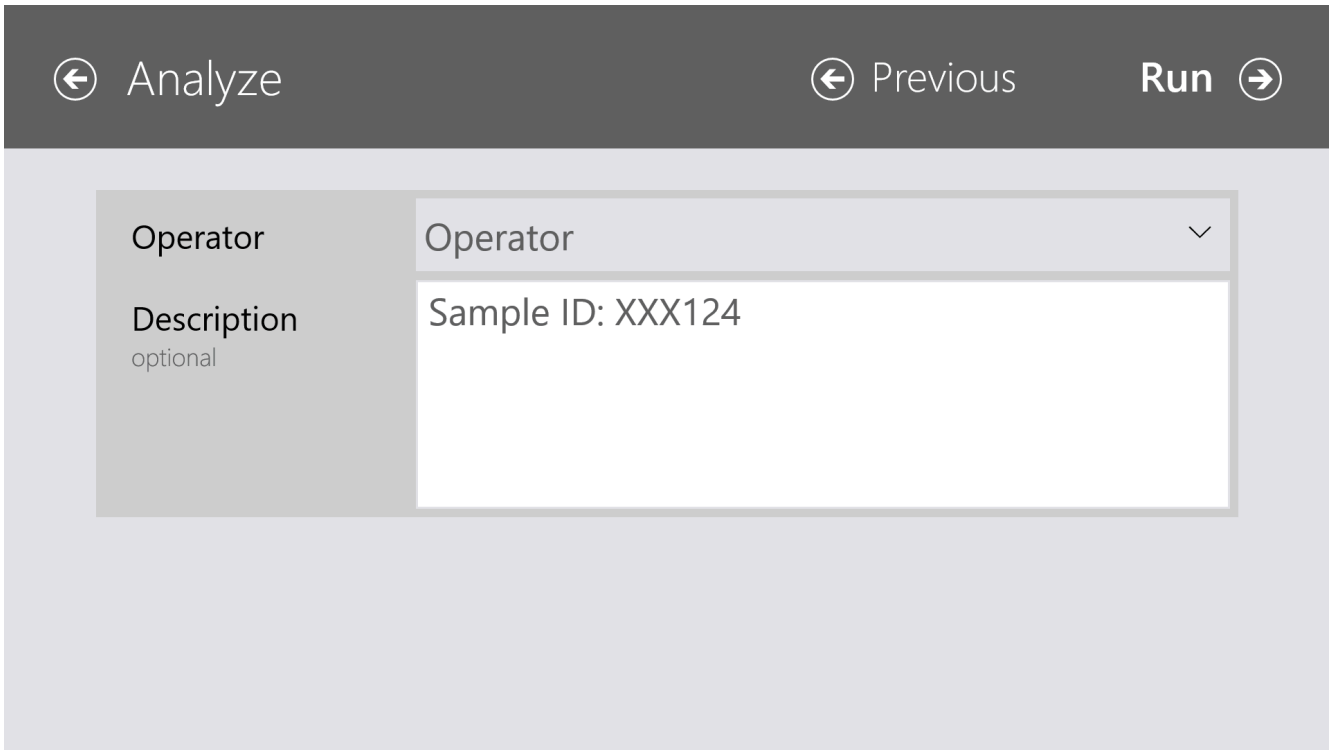
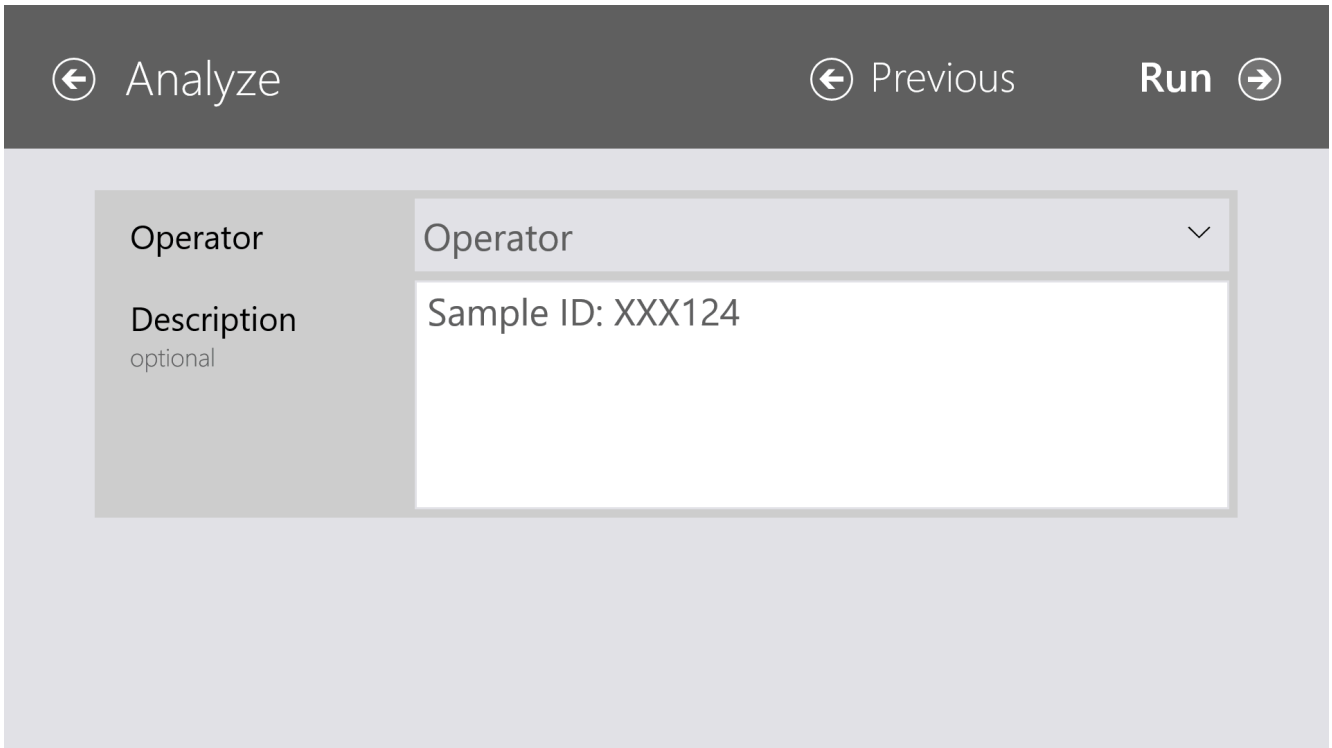


Once complete, press the Run button to proceed with your inspection.



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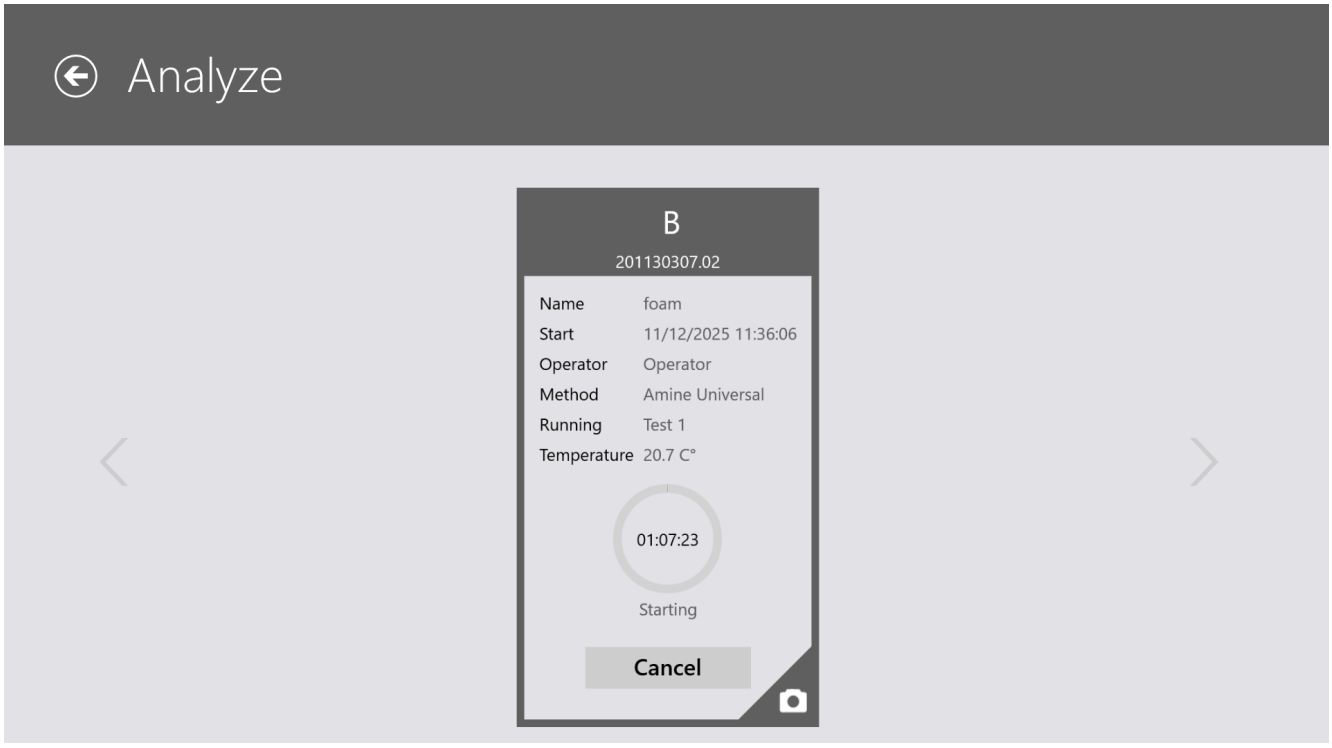
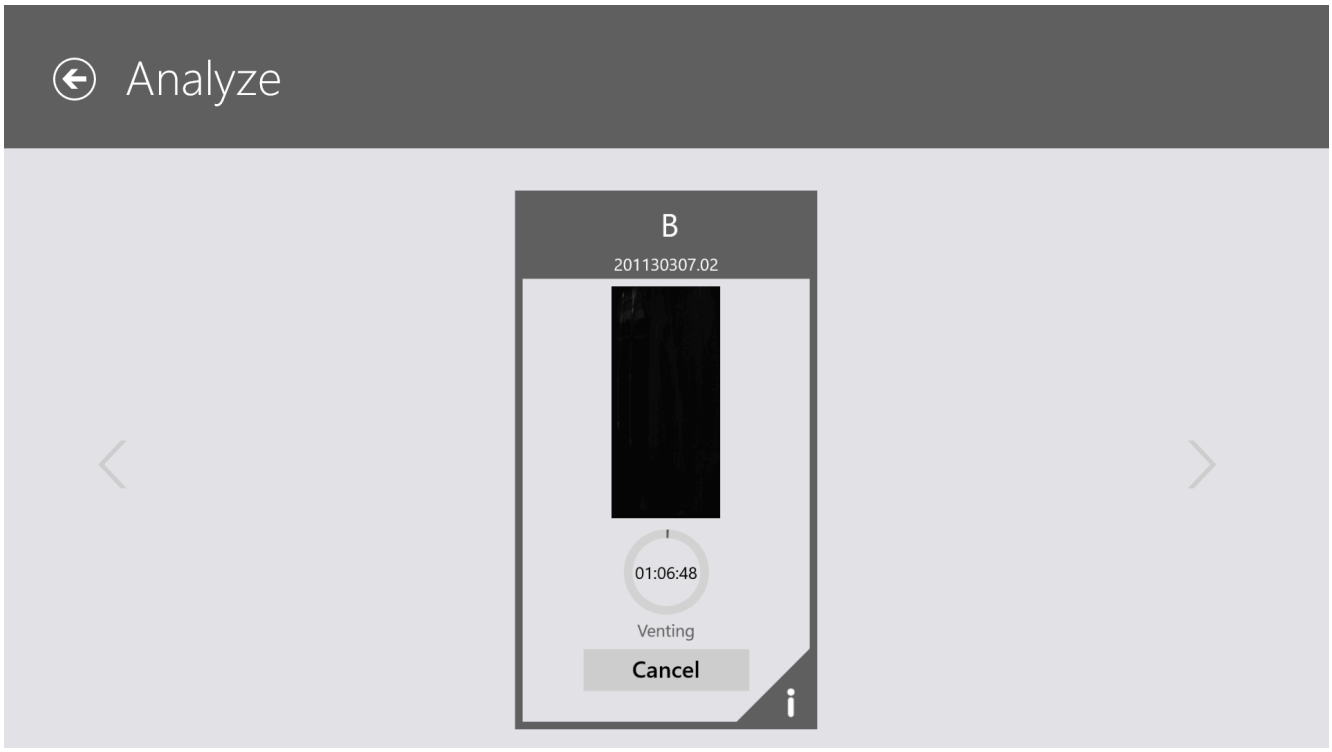
After completing all the necessary sample identifiers (sample ID, operator name, test method, and optional description) and clicking Run, the automated visual analysis starts.

A progress and information bar will appear and display the status of the analysis.

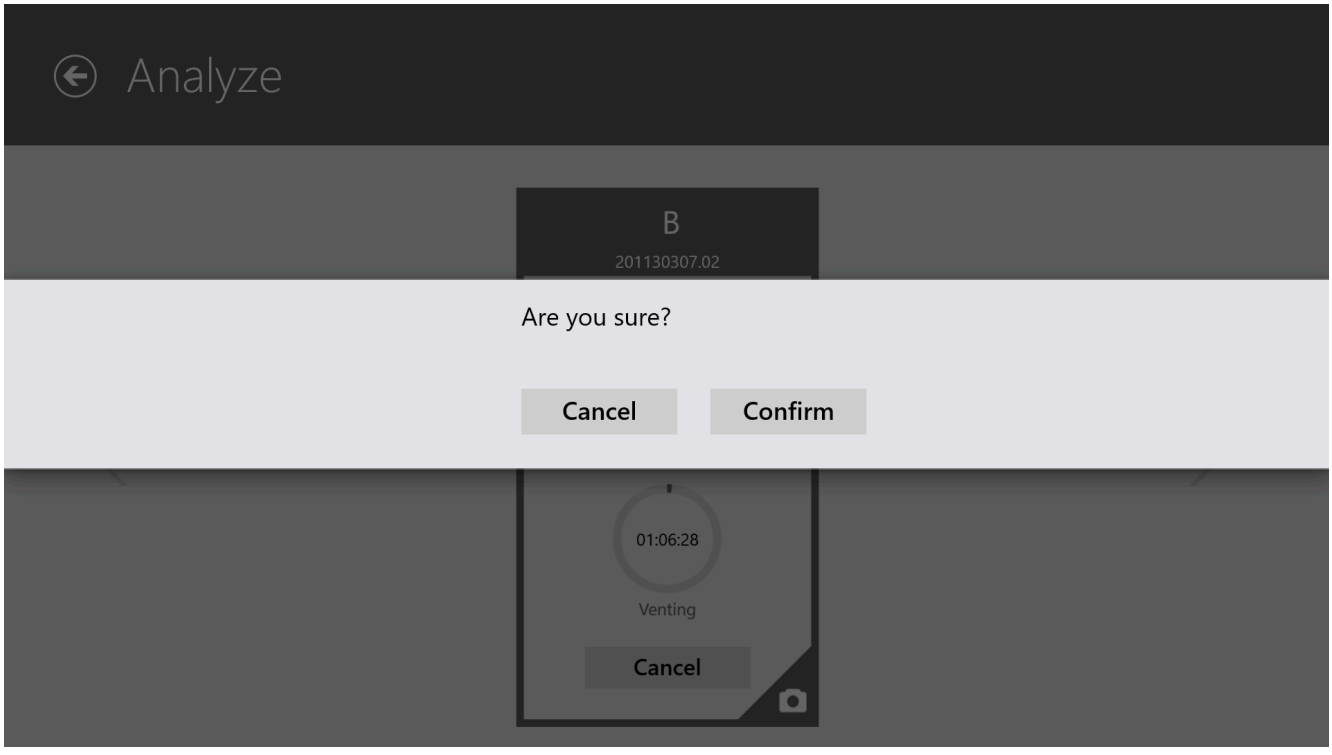


A live feed showing the sample can be accessed by clicking the camera icon on the bottom right. To return to the identification screen, simply click the information icon.



At this point, the operator may select a second module to start a second asynchronous analysis following the same procedures as described above.

An analysis can be cancelled at any time by Cancel and selecting confirm.



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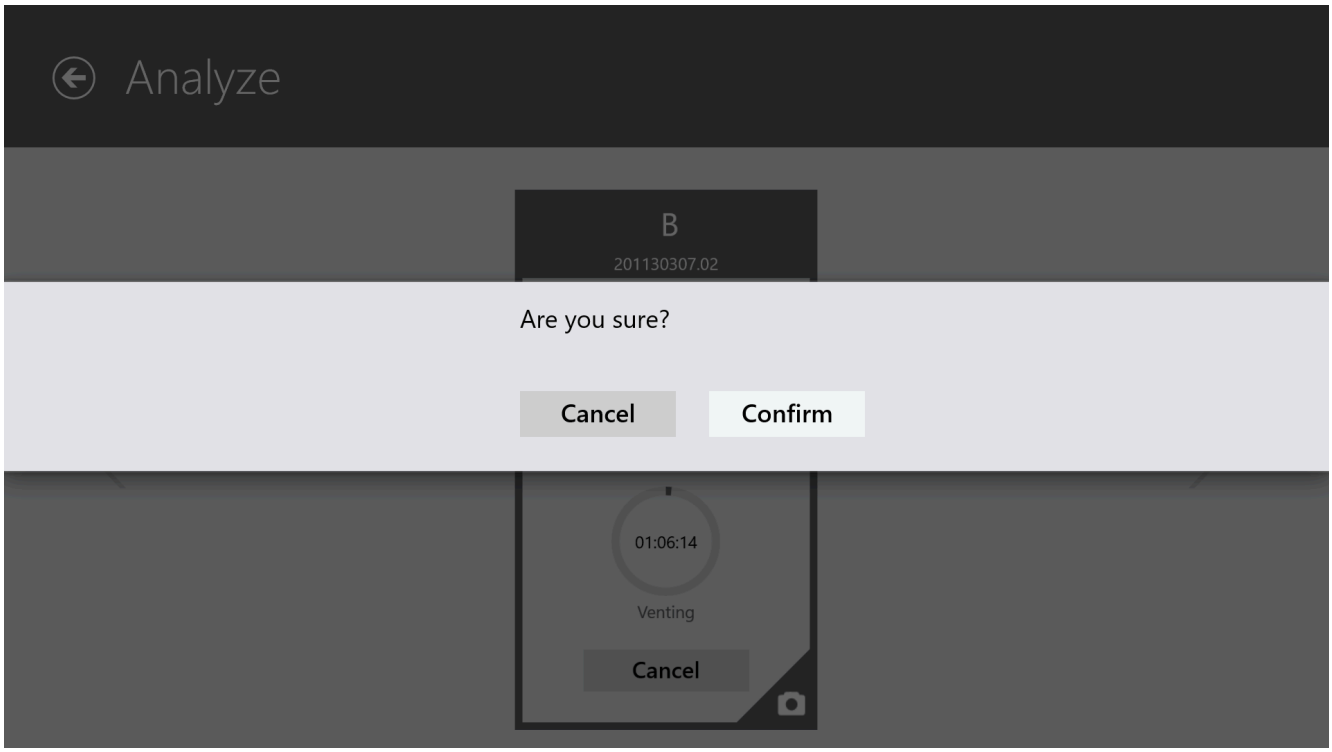


5.2.4. Analysis Procedure

Once the Test starts to run, the Testing Screen will appear. The underlying line, shown below, will change throughout time, depending on what the unit is performing as follows:

- Venting - Venting the inlet tube for 5 minutes while heating/cooling
- Heating or Cooling - Waiting until the target temperature is reached. May take between 20 - 30 minutes
- Flow - Air is released to the diffuser/sample at a controlled rate.
- Measuring Foam Stability - Vision is monitoring foam until volume is either zero or a total of 10 minutes has elapsed
- Finalizing - Disabling modules, temperature control, and preparing test results (storing data)

The system automatically:

Opens the air valve and initiates foam generation.

Activates LED illumination for optimal imaging.

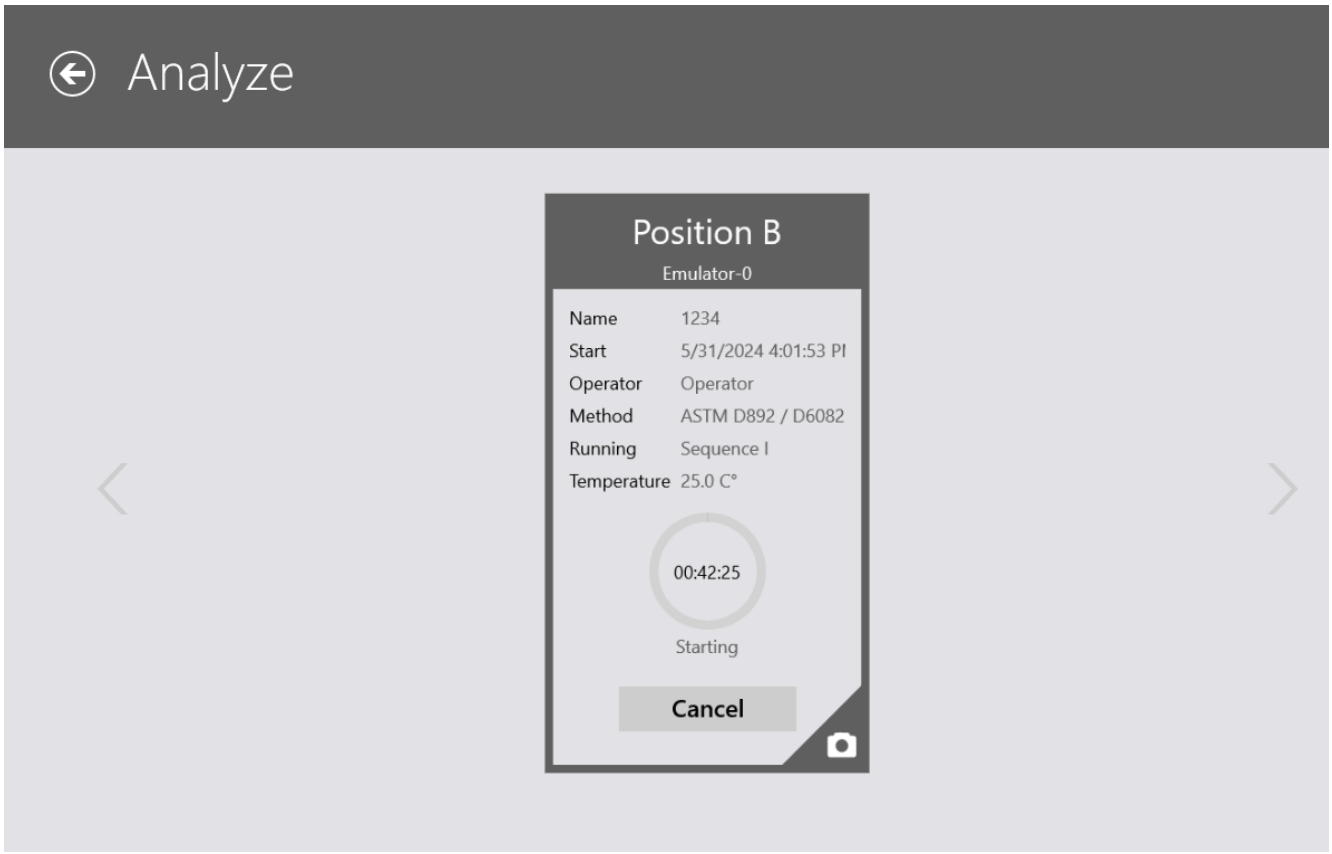
The FoamDDI will control air flow, temperature, and timing automatically according to the method.

Monitors foam growth and collapse in real time.

Captures images and video for playback.

Foam tendency (maximum height) and stability (residual foam after a set period) are calculated automatically.

The vision algorithm records foam growth and collapse in real time.



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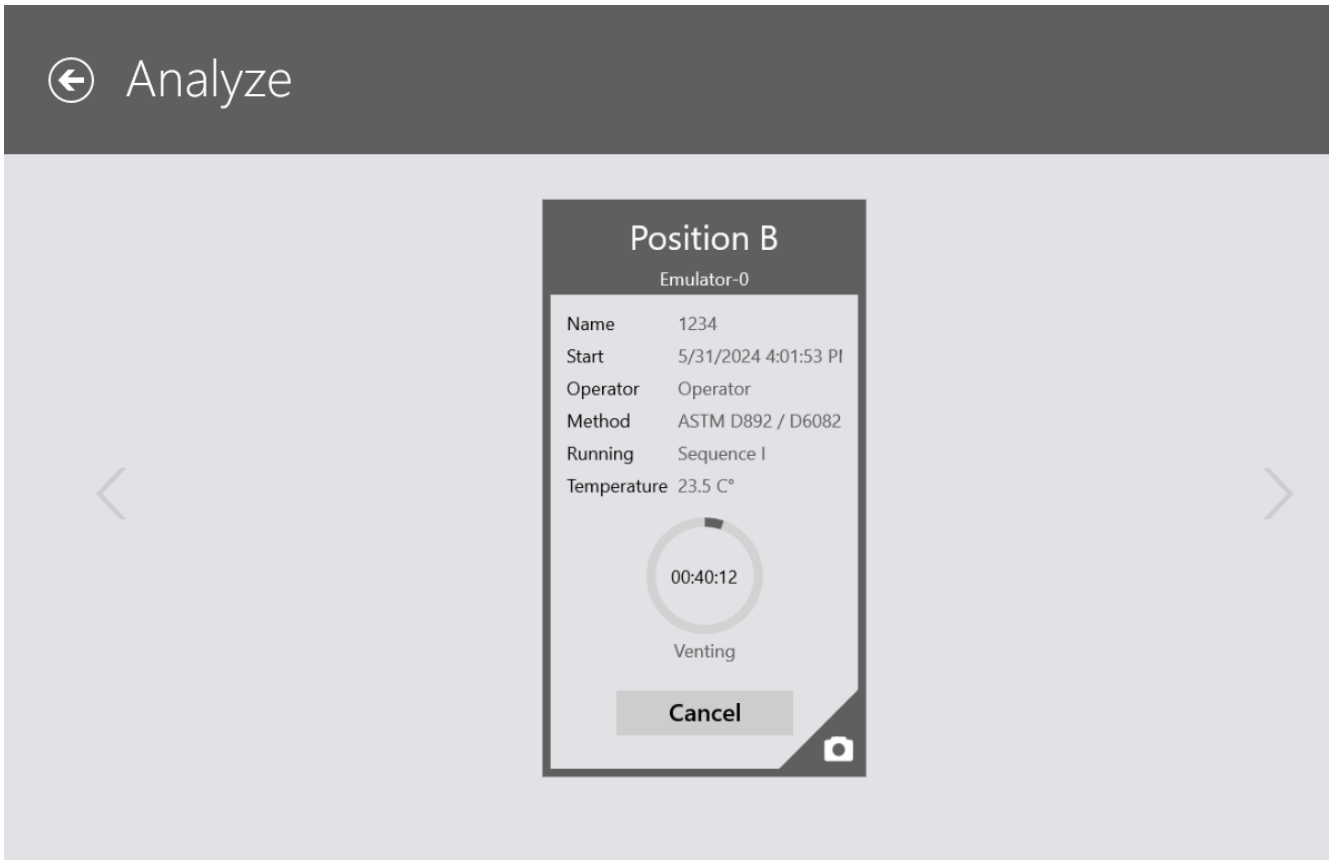
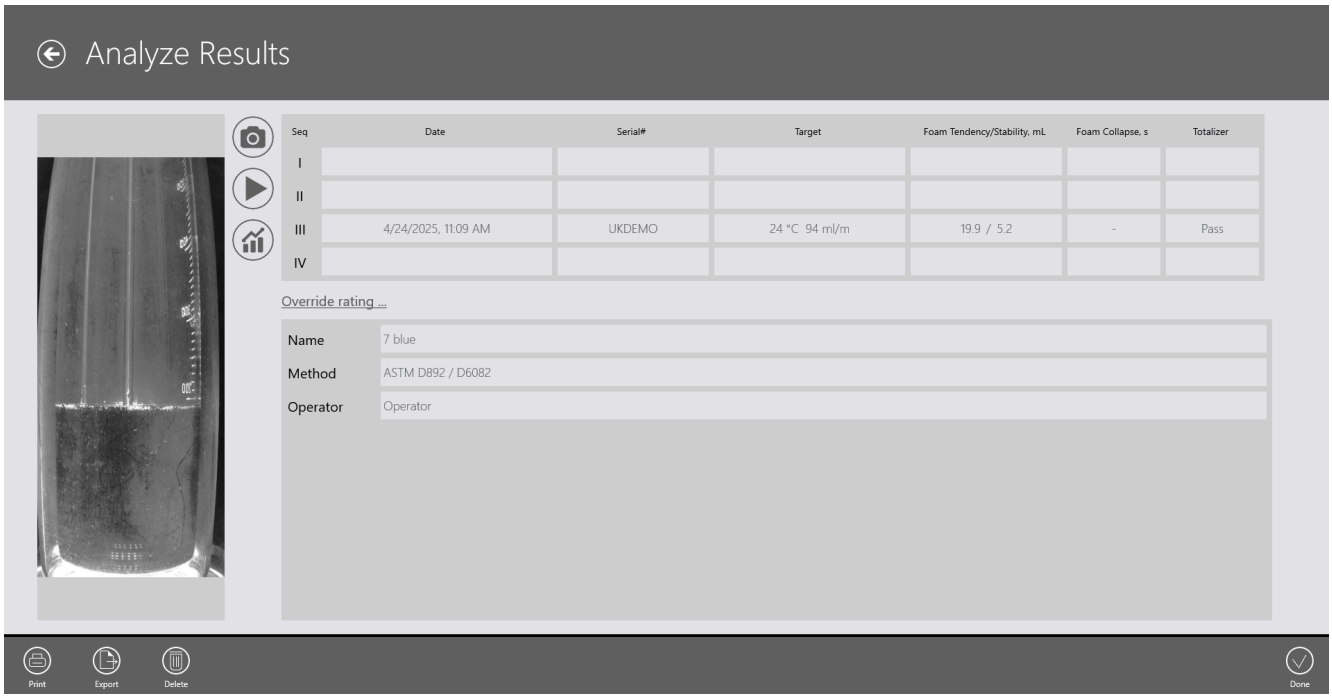


5.2.5. Interpreting the Results

At the completion of the test, results will automatically propagate. The results displayed at the end of a test, and the results in the History database, are the exact same. For more comprehensive instructions on navigating the results screens, navigate to **Section 5.3.2**. For more information on what the results mean, read below.

After successfully running a test sequence, the FoamDDI presents a comprehensive Analyze Results screen. This screen combines raw data, calculated results, visual evidence, and system status information.



The results screen has all of the information from the analysis: foam height vs. time graph, foam tendency (mL), foam stability (mL at 10 min), collapse rate graph, digital images and time-lapse video, and any identifiers like date and sample name.

Data Field	Description and Source
Name, Method, Operator	Sample identifiers entered during the test setup are displayed in the center of the screen.
Sequence Parameters	Details the specific parameters used for each test (e.g., Target temperature and air flow rate).

Foam Tendency / Stability (mL)	This is the primary result, displayed in milliliters (mL). Foam Tendency is the amount of foam determined immediately after the cessation of air flow. Foam Stability is the amount of foam remaining at the end of the specified settling period (typically 10 minutes for D892). If the foam collapses to 0 *mL* before the 10-minute settling time, the test may be terminated, and the result recorded as 0 *mL*.
Foam Collapse (s)	Records the time, in seconds, required for the foam to collapse to 0 after the 5-minute air blowing period (applicable mainly to D6082/Sequence IV).
Totalizer	Displays the air flow totalizer data, which measures the accumulated air flow *out* and the air moving *back in* (Flow In) via the Mass Flow Monitor.
Pass/Fail Status	The result screen indicates whether the test sequence successfully ran, generally displaying Pass. If abnormal foam is detected, a manual override may be suggested.
Override Data (O)	If a value has been manually adjusted by the operator, an (O) will denote that the data has been overridden, while the original values are preserved and saved.

Operator notes can be amended or added before saving the result as well.

Playback view allows the physical review of photographic and videographic results showing the entire test in a x10 interval.

Operators will also have the option to override the results at this stage. To override, manually adjust the edge contrast on the resultant photos. The results will automatically be tabulated. An overridden result is denoted with an ‘O’.

Results can be exported in multiple formats: PDF, CSV, TXT, or XML. To configure export settings, go to **Section 5.4** of this manual.

Note: Due to possible minor discrepancies from the glass blowing procedure to make the cylinders, it is recommended for repeatability, to use the same cylinder for sequential runs.

5.2.5.1. Visual and Graphical Data Validation

The FoamDDI is an optical device designed to provide extensive data for validation, allowing operators to visually and graphically audit the automated results.

Image View:

By clicking the camera button or the cylinder photo, the operator accesses the Image View.

This view displays still images captured during the test for the Foam Tendency and Foam Stability for each sequence.

A green/yellow/red line is superimposed on the image to indicate the foam's height as determined by the vision algorithm for the given sequence.

The operator can zoom in on the image to get a clear view of the measurements.

Foam Resolution: The resolution of the foam level (e.g., 0.1 mL, 1.0 mL, 5.0 mL, or 10.0 mL) can be changed in the settings, affecting how the automatic analysis and manual override function.

Video Playback:

Clicking the playback button (play icon) reveals the Playback View.

This feature plays the recording from the internal camera, allowing the operator to replay the sequence over again.

The Playback View is crucial as it allows the operator to observe at what exact time and temperature foaming occurs, which is helpful in cases where the Tendency sample occurs quickly.

Graph View:

Clicking the graph button (graph icon) displays the Graph View.

Graphs show continuous data plotting temperature, air flow, and foam height data captured throughout the duration of each sequence.

5.2.6. Override Options Following An Analysis

If the user does not agree with the automated foam measurement, they have the option to manually adjust the results.

The results displayed at the end of a test, and the results in the History database, are the exact same. For more comprehensive instructions on navigating the results screens, navigate to **Section 5.3.2**.

Override Rating Button: Clicking this button allows the operator to manually select the Foam Tendency and Foam Stability values.

Data Preservation: Both the original automated data and the manually overridden data are saved under this tab.

Authorized users (Admin Mode) can override automatic classifications in specific cases, such as:

- Re-analyzing with modified parameters.
- Applying manual notes to test runs.
- Enabling the “Wait for Operator” or “Audible Alert” features during inspection.

5.2.7. Saving Results

Data is stored with time stamps, sample ID, operator ID, and method parameters for traceability.

Results are stored in the internal database, accessible through the History tab. From the results screen, the operator has options to handle the data:

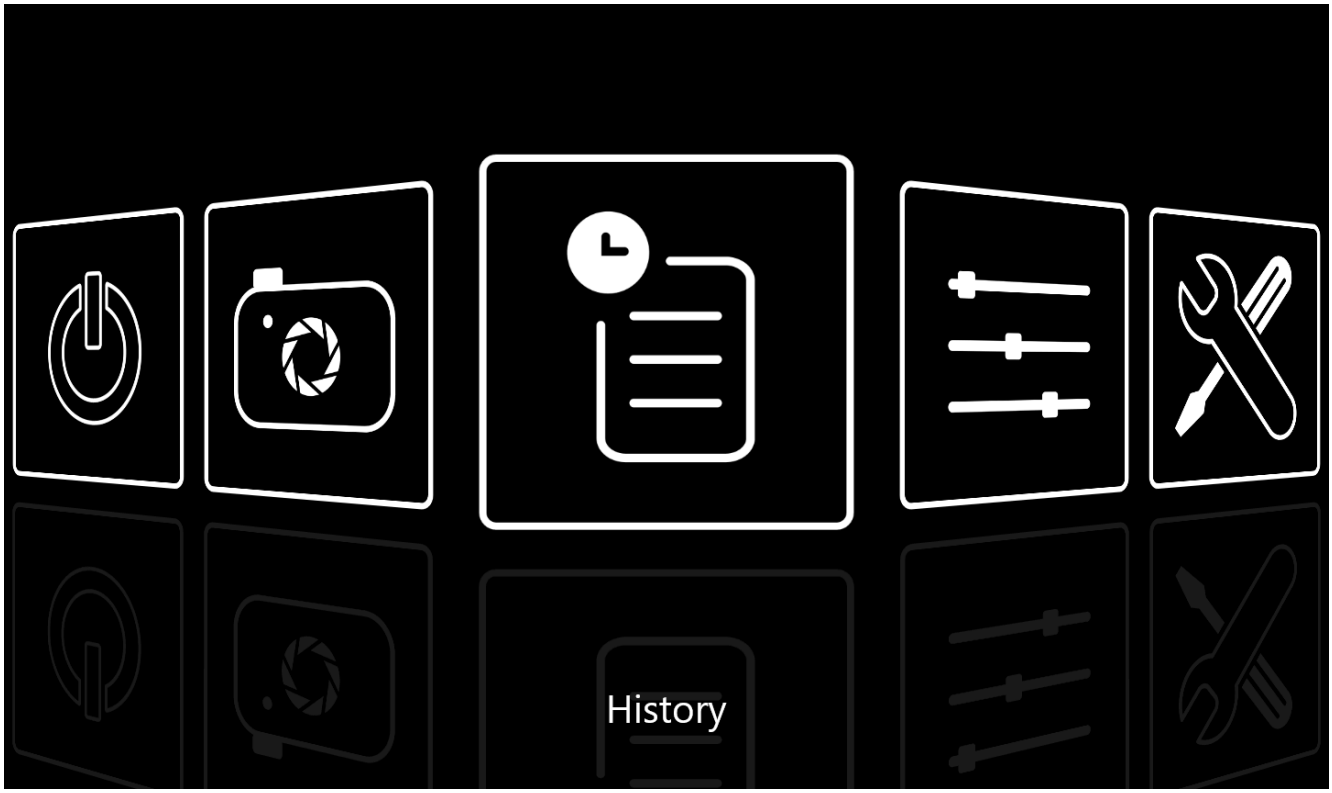
- Done: Saves results to the local IPC database.
- Export: Allows results to be exported to LIMS, USB, or email.
- Print: Sends the results page and images to a predefined connected printer.
- Delete: Permanently removes the result from the database (admin restrictions may apply).

5.2.8. Asynchronous Second Test

If there is a second module available, the FoamDDI supports performing a secondary analysis asynchronously while a primary run is logged. This allows operators to prepare the next sample without losing previous data continuity.

Follow the same procedures above for the second module. The modules are independent of each other, so you can run any method and sequence independently.

5.3. History



The History menu provides access to all stored results.

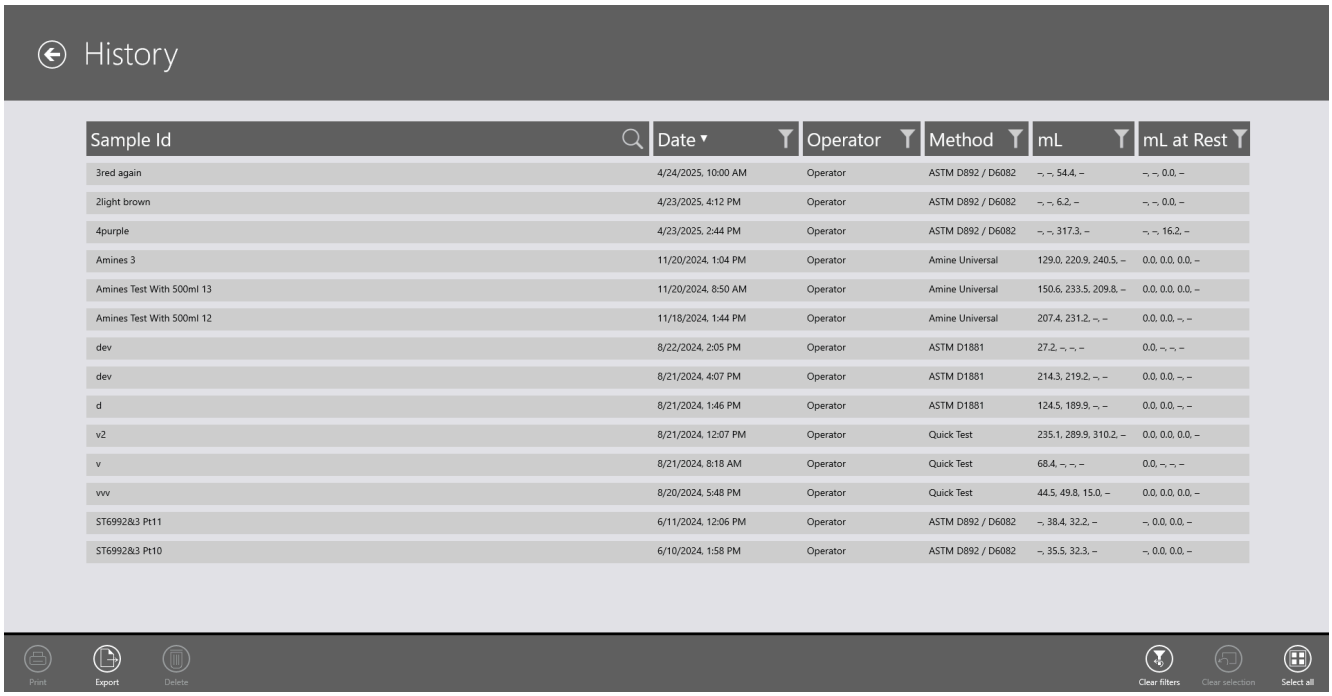
5.3.1. Database

To access stored results on the logic box, navigate to the History tab on the main screen.

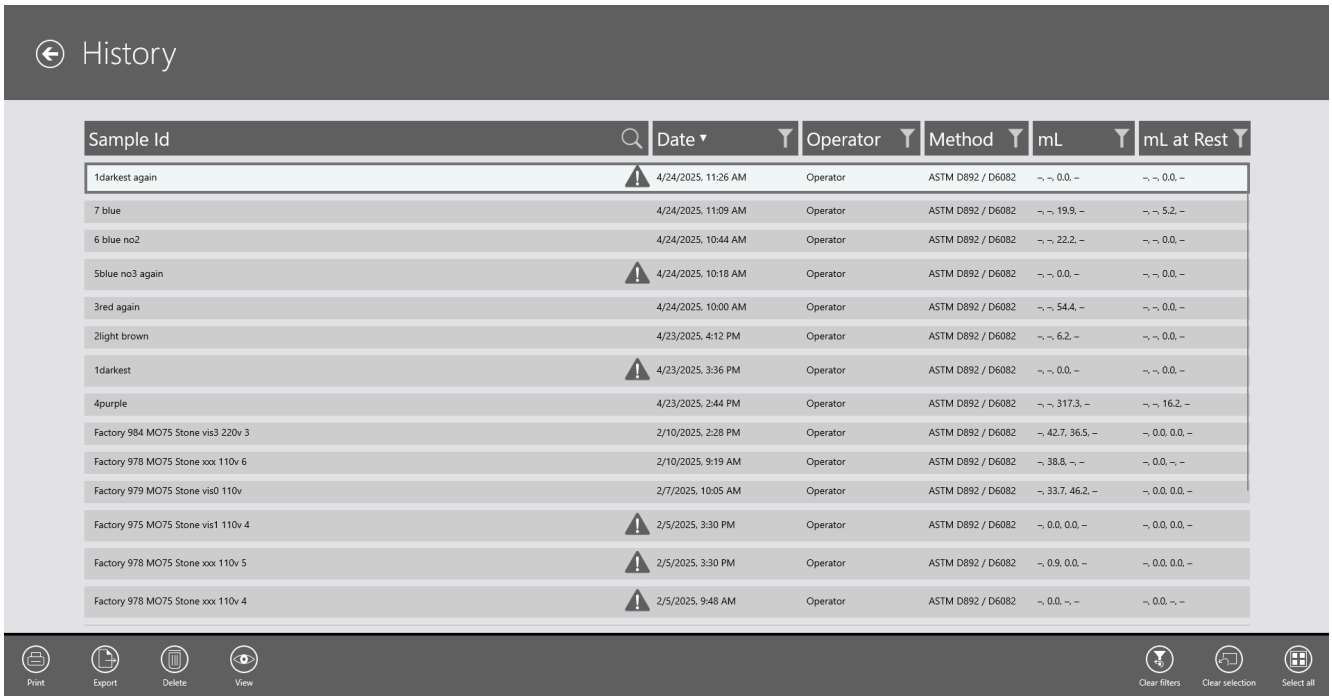
Once open, short form results of the most recent tests will be displayed. The parameters displayed on this screen include the Sample Id, Date of the test, Operator, Method used, Foam Tendency in mL, and Foam Stability in mL.

Any tests that could not be analyzed properly are indicated with a warning triangle with an exclamation mark.

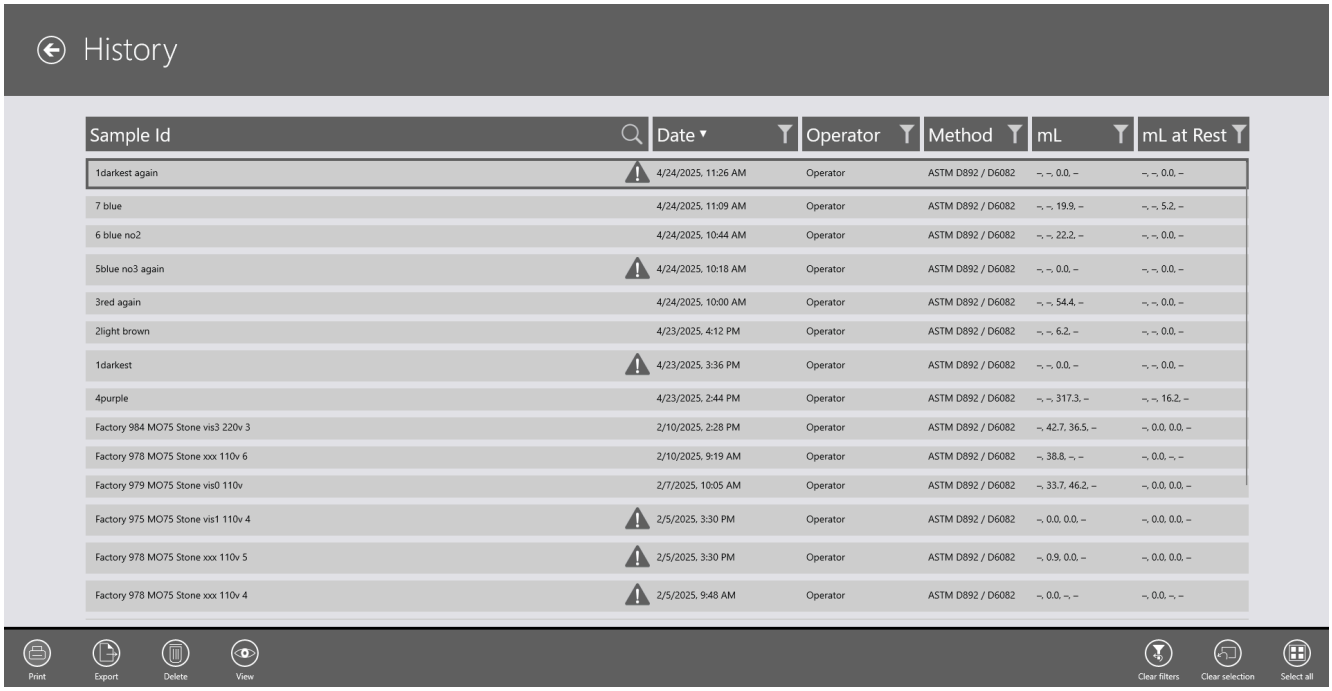
To see earlier results, simply scroll down to peruse through all of the results in the database.



To select a result, simply click on the row of the test. Once selected, a myriad of options are available to the user. These include export options, detailed view, and option to delete.



Export options from this screen are located in the bottom left hand corner of the screen. To print the result, simply click the Print button on the farthest left and, if there is a printer connected, the result will print immediately. To configure print settings, go to **Section 5.4.7** of this manual.



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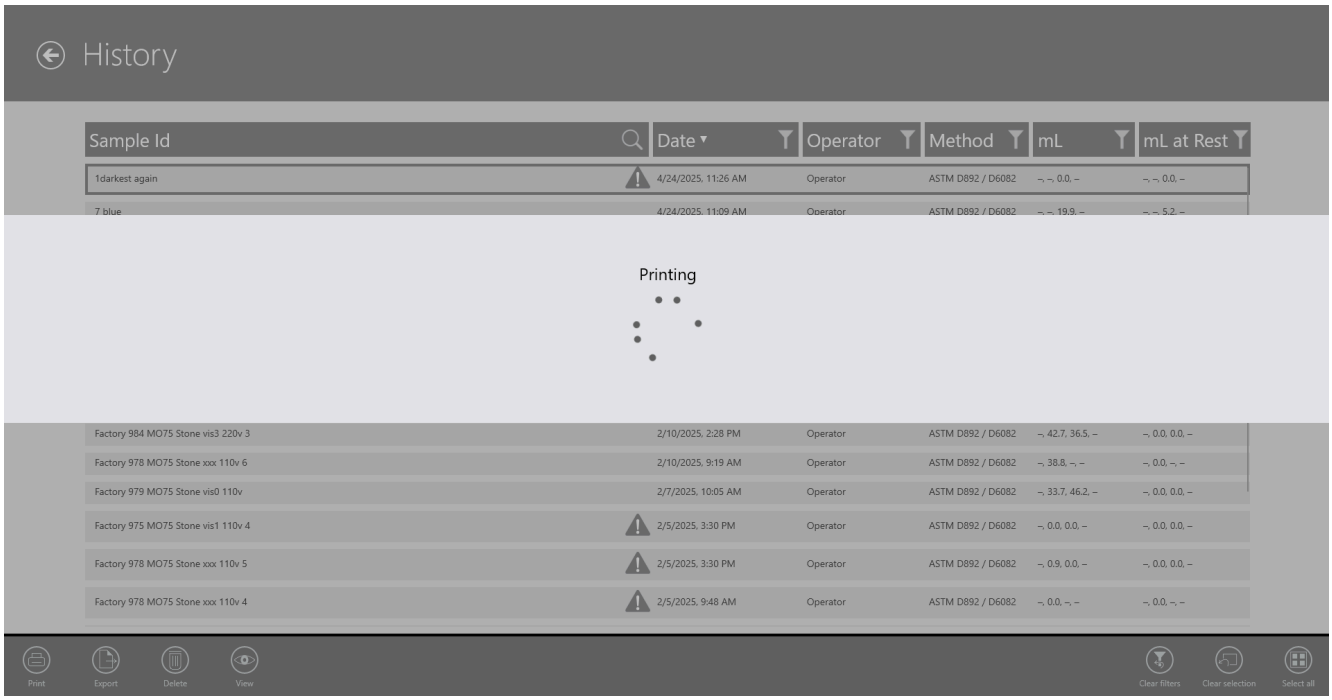
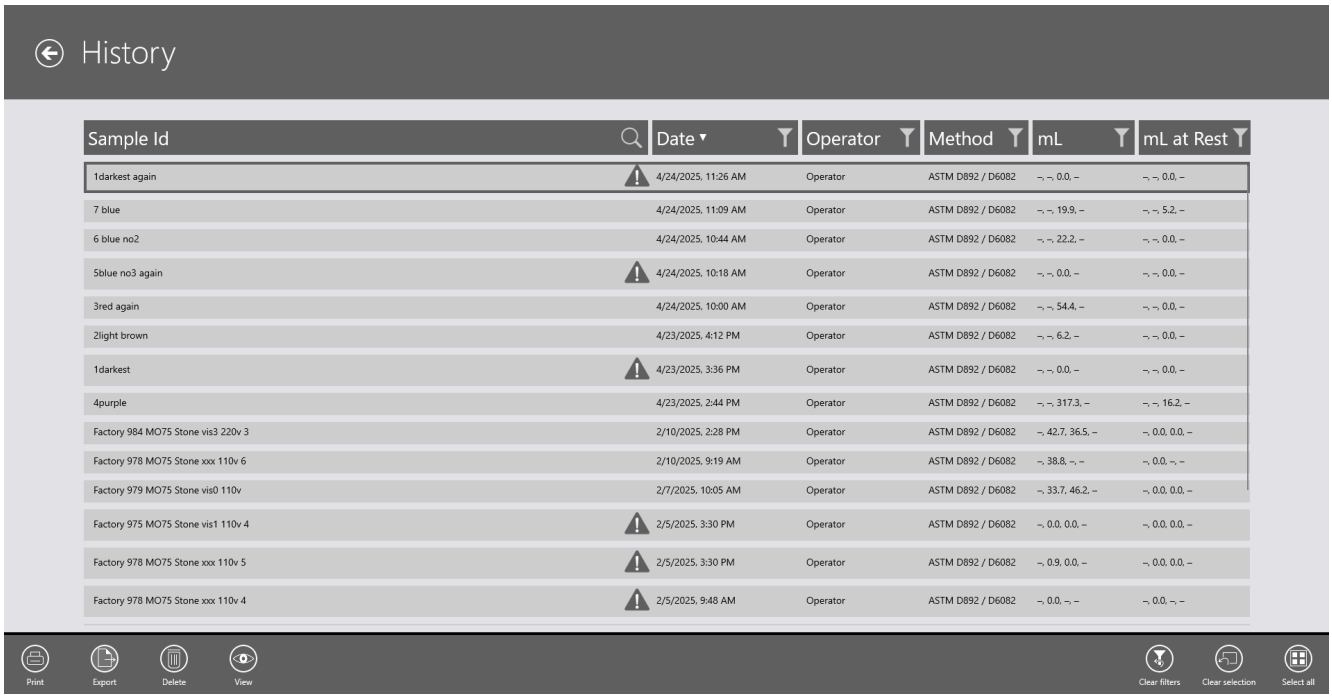


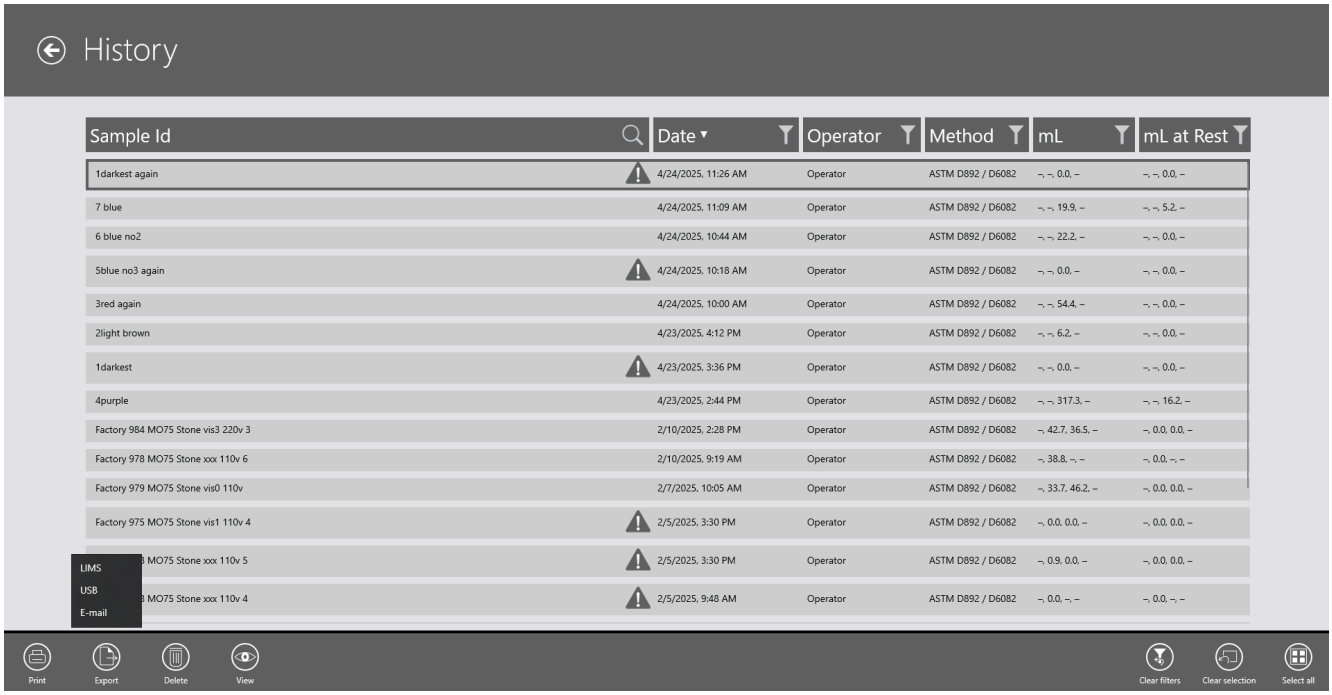
All other export options are located one button to the right of the print button. Click Export and three options will be presented. To export the result to a LIMS system, click LIMS and the result will immediately transfer over to that system. To configure a LIMS system with the device, navigate to **Section 5.4.6** of this manual.

Results can also be exported to a USB if one is plugged into the device. Simply click USB and the result will immediately be transferred to the USB.

To configure the format of the exported result for both LIMS and USB, navigate to **Section 5.4.6** of this manual.

Results can also be emailed directly from the device. This requires an email system to be set up. To set up an email and format the email along with the results, navigate to **Section 5.4.4** of this manual.





The third icon in the bottom left of the screen can be used to delete test results. Simply click the Delete button and a prompt will appear asking if you are sure that you want to delete the result.

Note: Once deleted, data can not be recovered.

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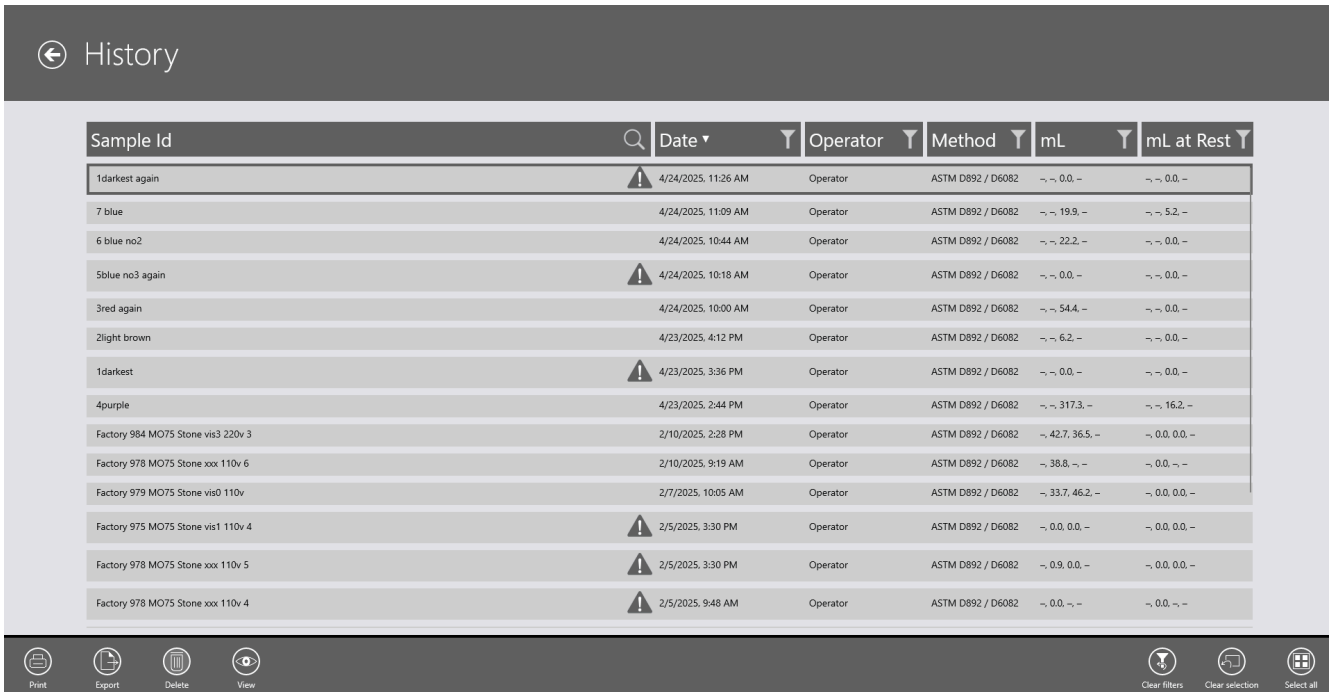

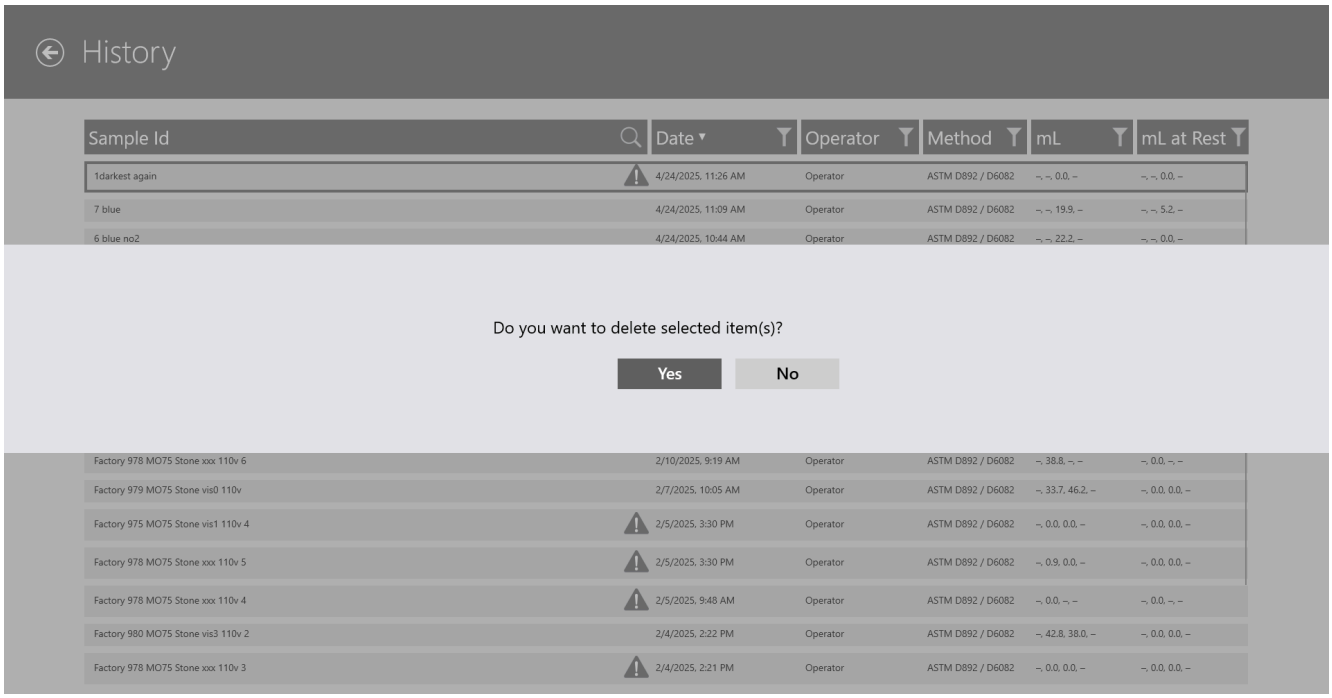

All of the above options—Print, Export, and Delete—are also available when multiple selections are made. To Print, Export, or Delete multiple results at once, simply click all items desired and follow the instructions above.

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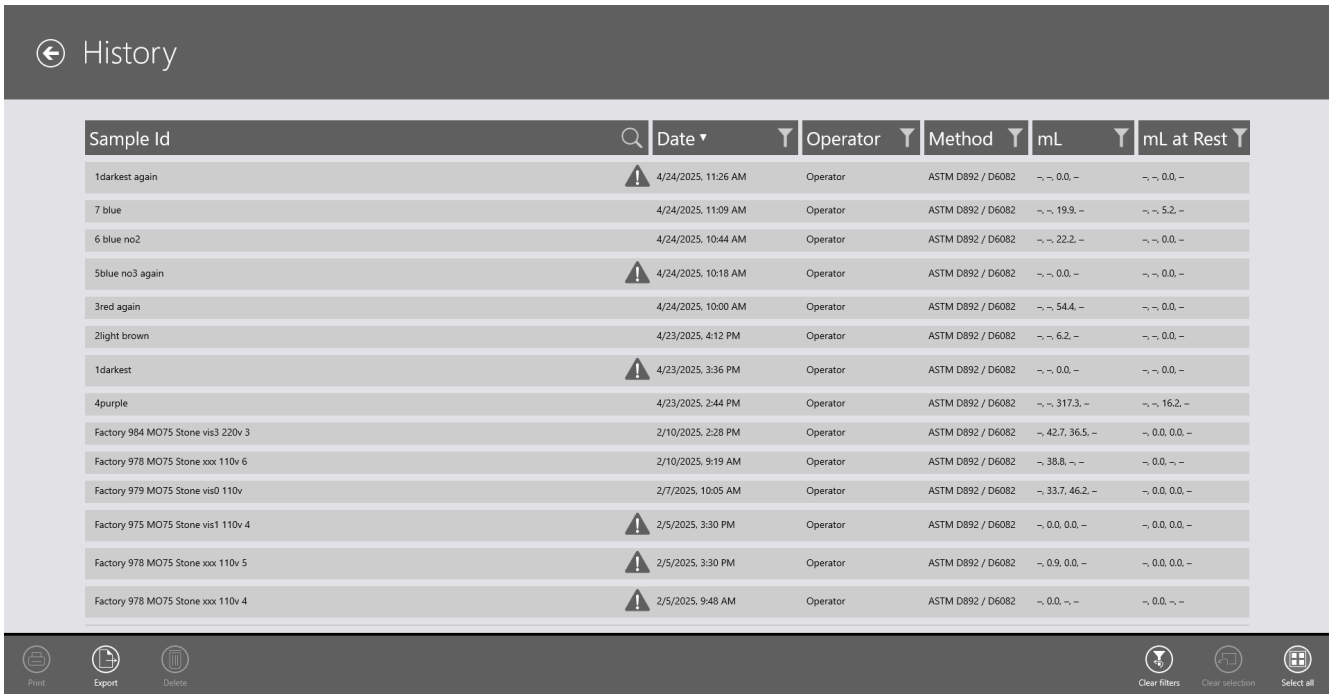
In the bottom right hand corner, there is a Select All option to print, export, or delete the entire database at once. To clear the all selection or multiple selection, click Clear Selection just to the left of the Select All icon.

To deselect a singular sample, either click the selection again or click Clear Selection.



5.3.2. Detailed Result View

To see the full and comprehensive results of an analysis, select one test and View icon will appear next to the export and delete options on the bottom left of the screen.

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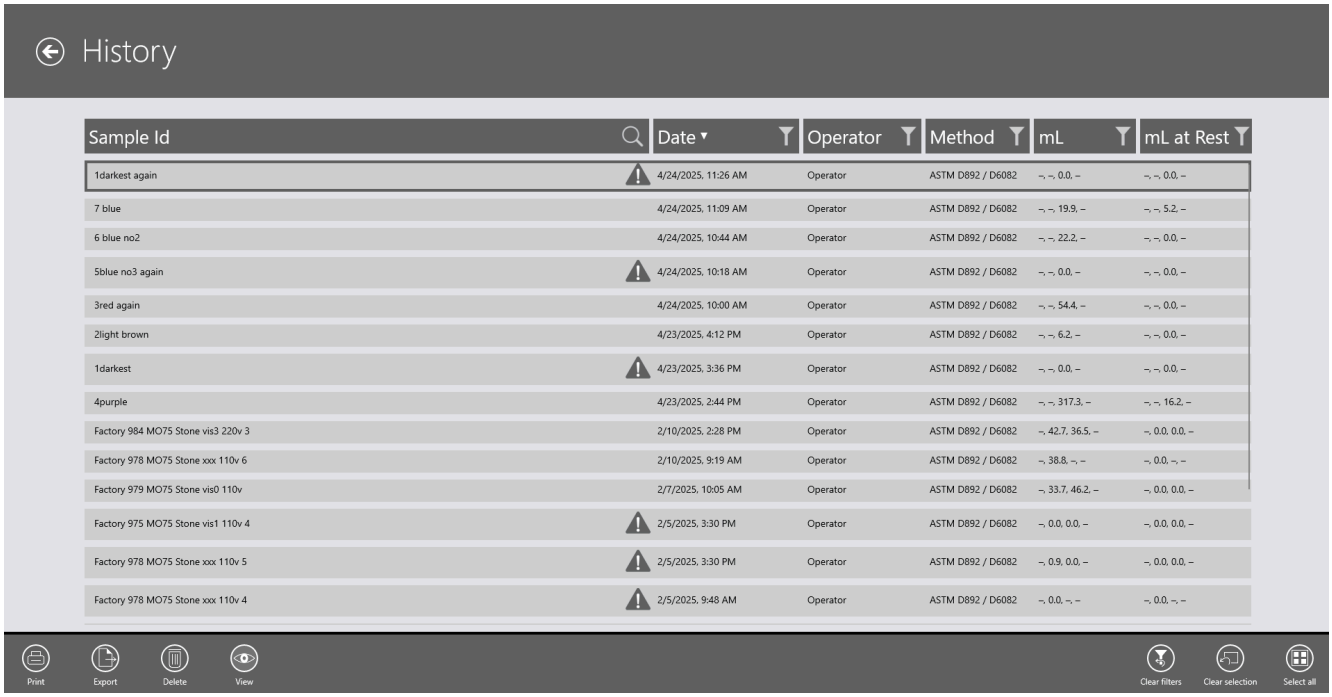
Note: When multiple tests are selected, the view icon will not be available. Only one test can be opened in the detailed result view at a time.

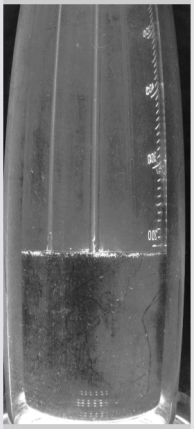
Click View and the primary detailed result screen for that test will appear.



The detailed result screen displays a photo of the Foam Stability, all of the identifiers such as the sample Name, Method, and Operator, the Date, Serial Number of the module, the parameters of the sequence, the Foam Tendency and Stability in mL, the Foam Collapse results in seconds, and a Totalizer pass/fail result. If any details or notes were added when the sample was tested, then they will also appear here.

This is also the screen in which results can be overridden if desired.

Analyze Results



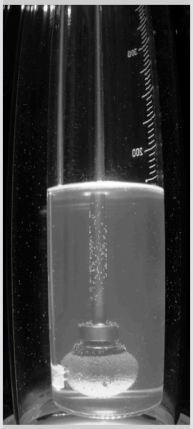
Seq	Date	Serial#	Target	Foam Tendency/Stability, mL	Foam Collapse, s	Totalizer
I						
II						
III	4/24/2025, 11:09 AM	UKDEMO	24 °C 94 ml/m	19.9 / 5.2	-	Pass
IV						

Override rating ...

Name	7 blue
Method	ASTM D892 / D6082
Operator	Operator

Print Export Delete Done

Analyze Results



Test	Date	Serial#	Target	Foam Tendency/Stability, mL	Foam Collapse, s
1	8/21/2024, 4:07 PM	220922495-02	88 °C 1000 ml/m	214.3 / 0.0	139.8
2	8/21/2024, 4:35 PM	220922495-02	88 °C 1000 ml/m	219.2 / 0.0	141.4
3					
Avg				216.8 / 0.0	140.6

Override rating ...

Name	dev
Method	ASTM D1881
Operator	Operator

Print Export Delete Done

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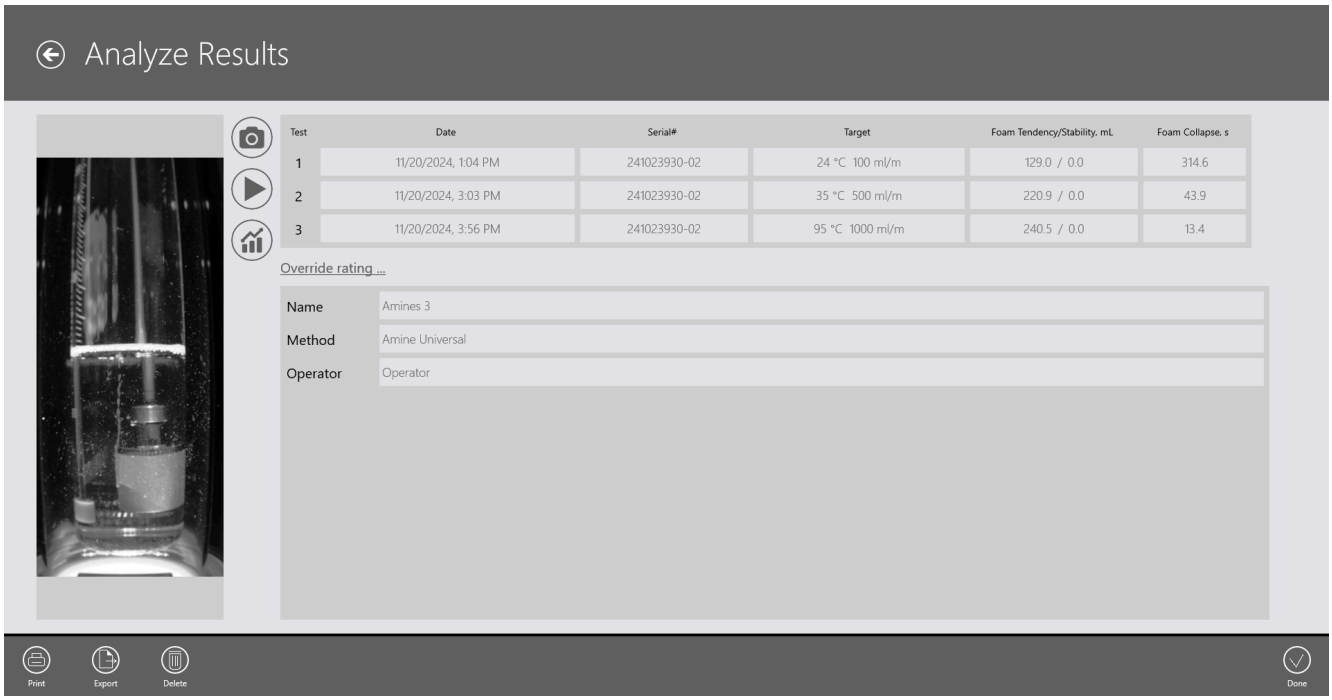
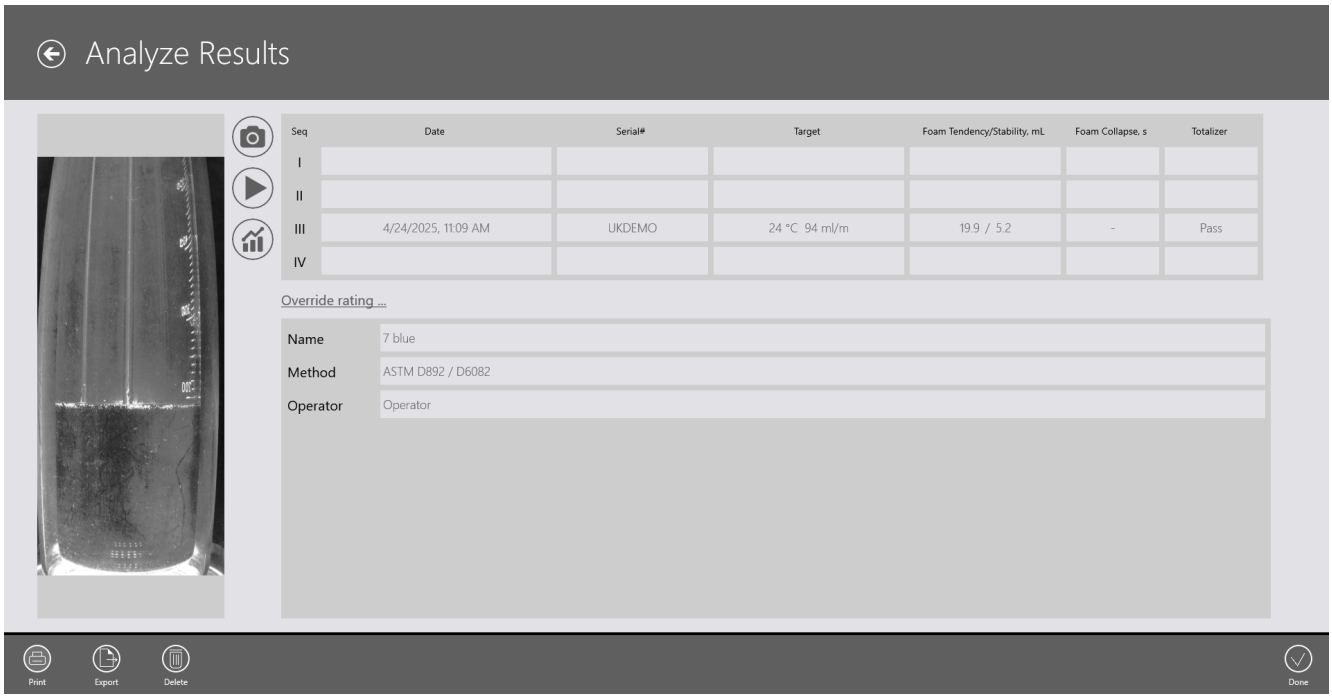


The result can be printed, exported, or deleted in the same manner as described in the Database section. These icons are located at the bottom left of the detailed result screen.



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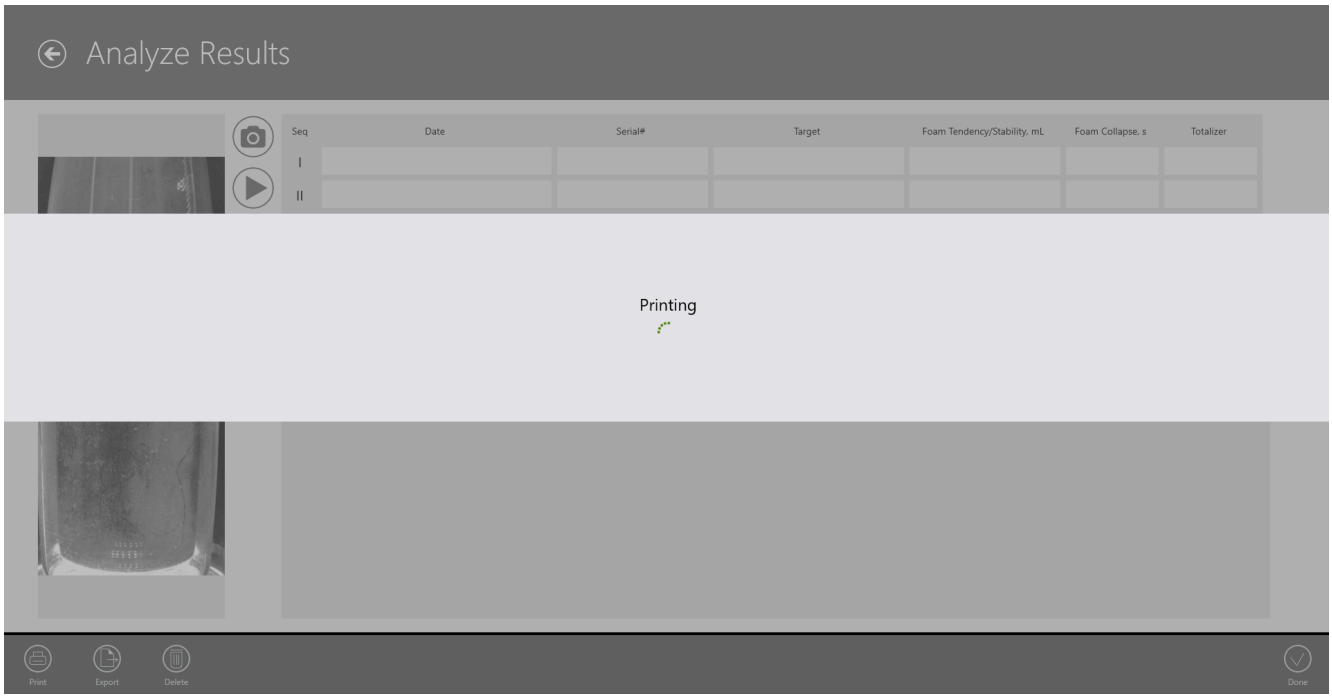

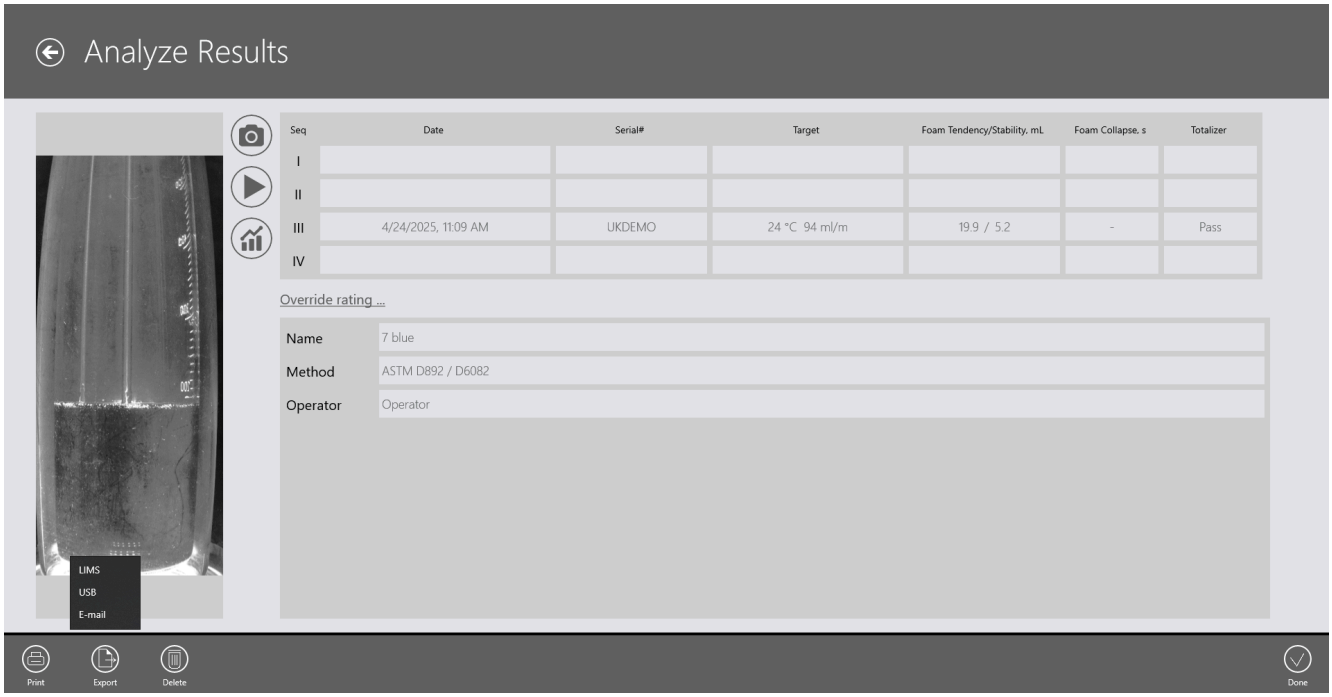

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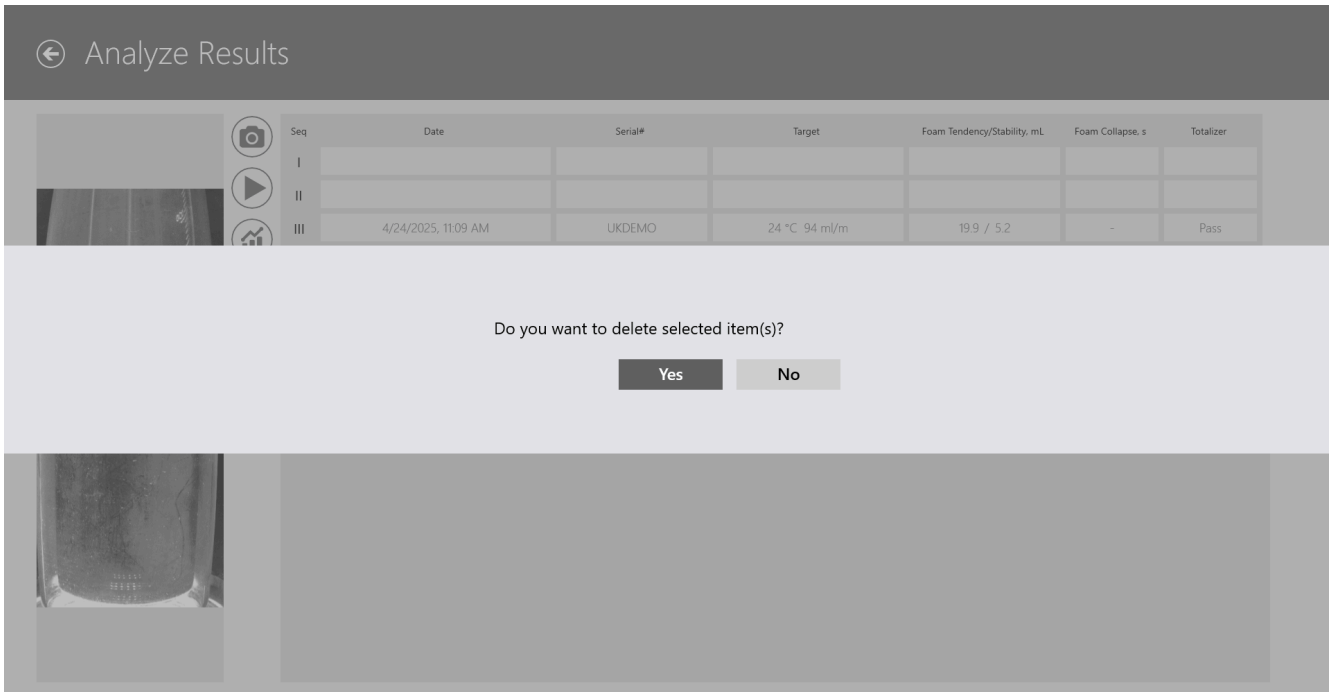
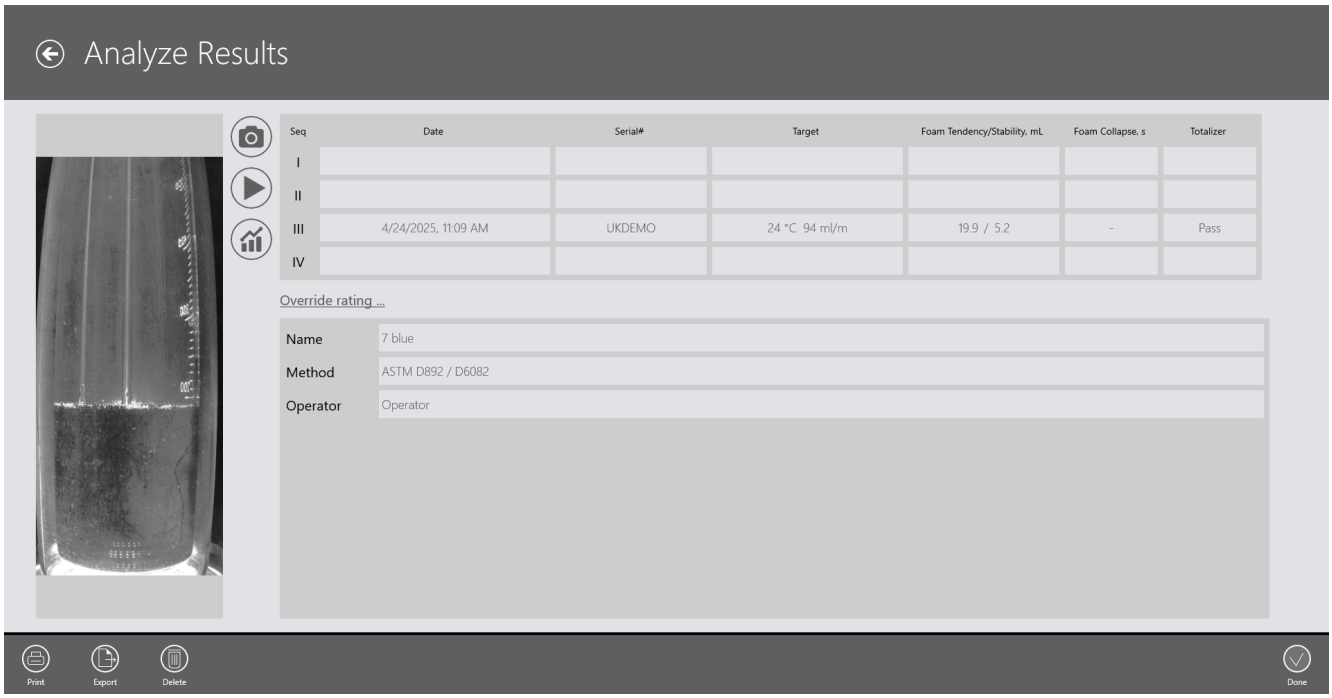
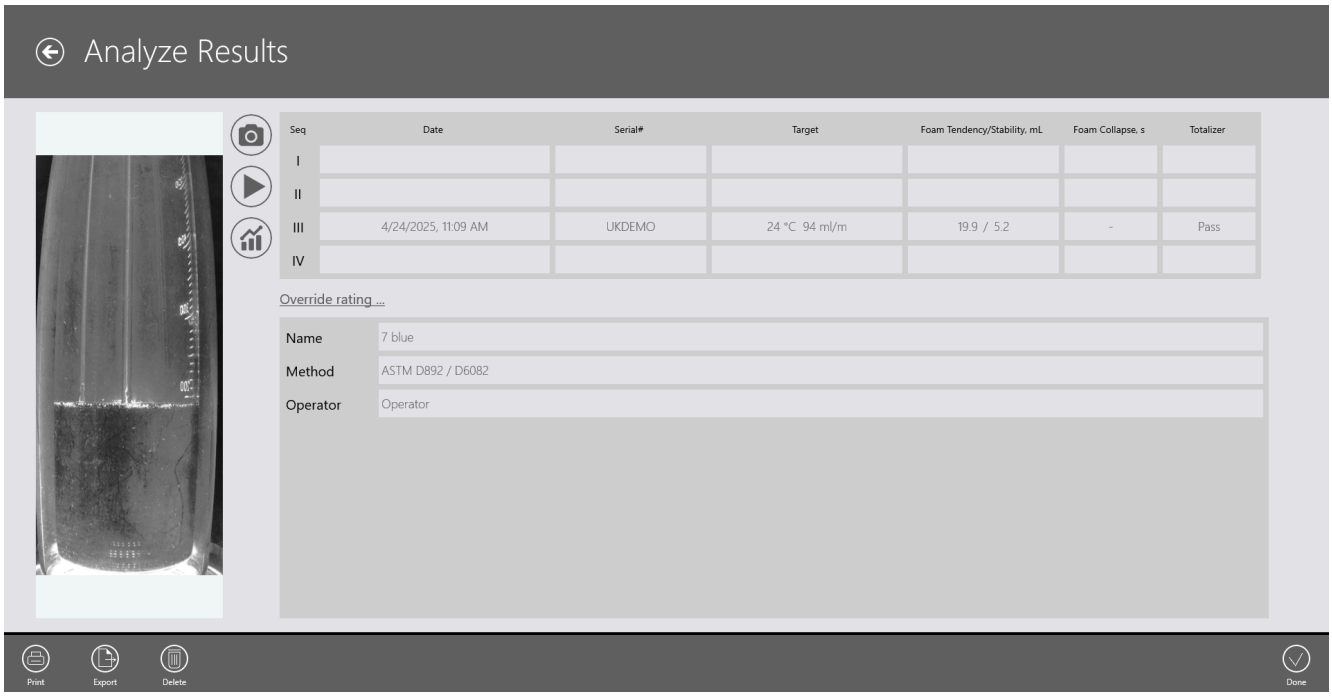


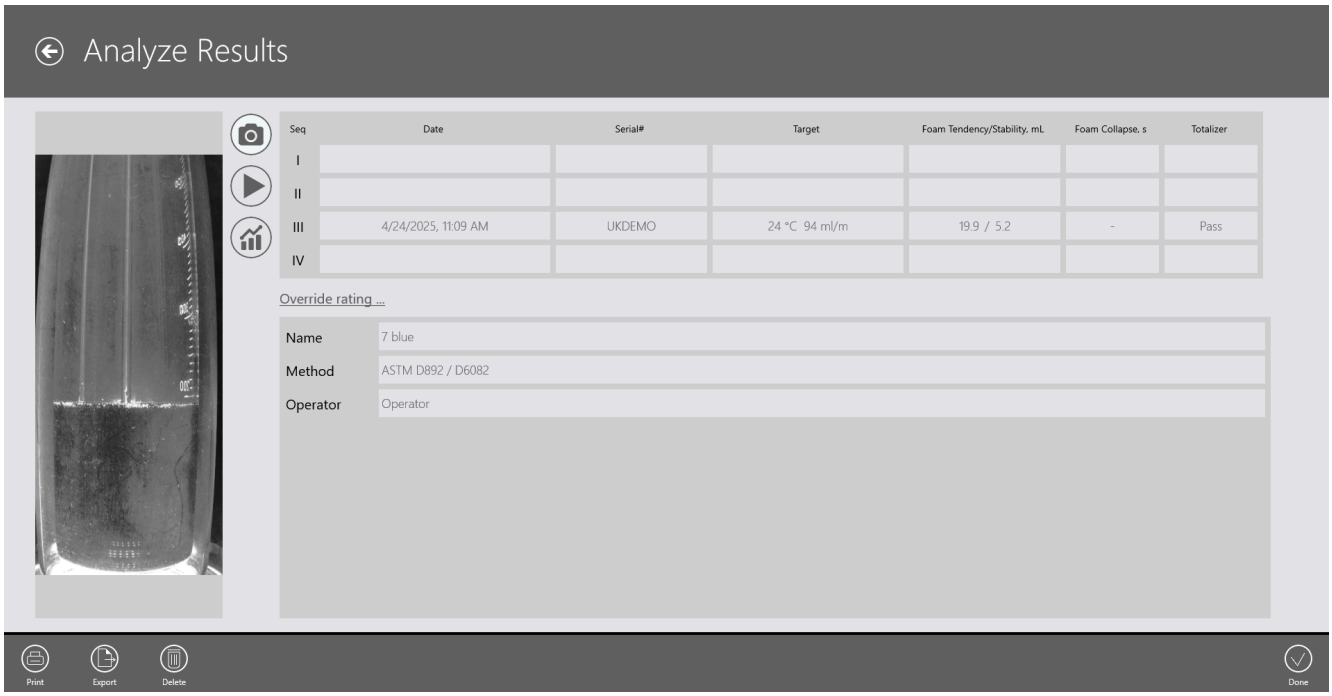
To exit the detailed result screen and reenter the database, click Done at the bottom right hand corner of the primary screen.



5.3.2.1. Secondary Screen - Image View

To see the detailed photographic results, click on either the photo displayed itself or the photo icon.





Both of these options will show the Image View. The images show the visual results of the test and the edges of the foam for both tendency and stability results.

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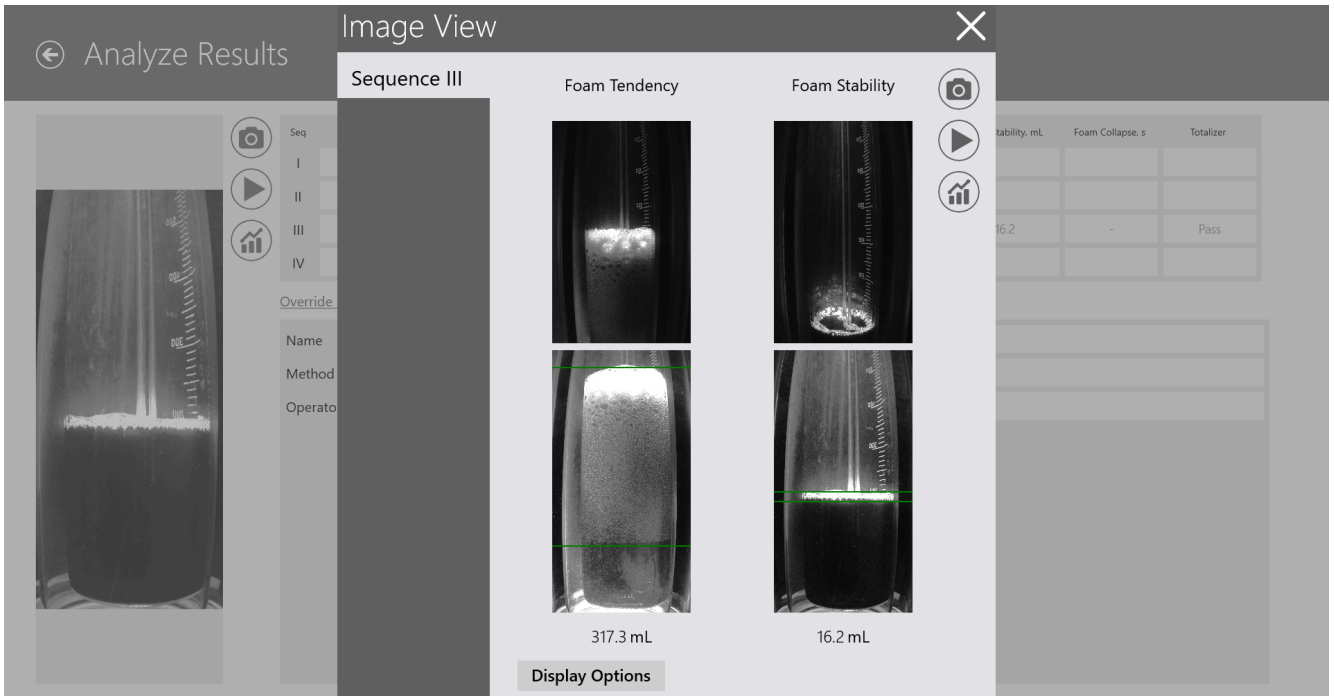

Graduations can be displayed in the Amine and Coolant software by checking Show Graduations under Image View.

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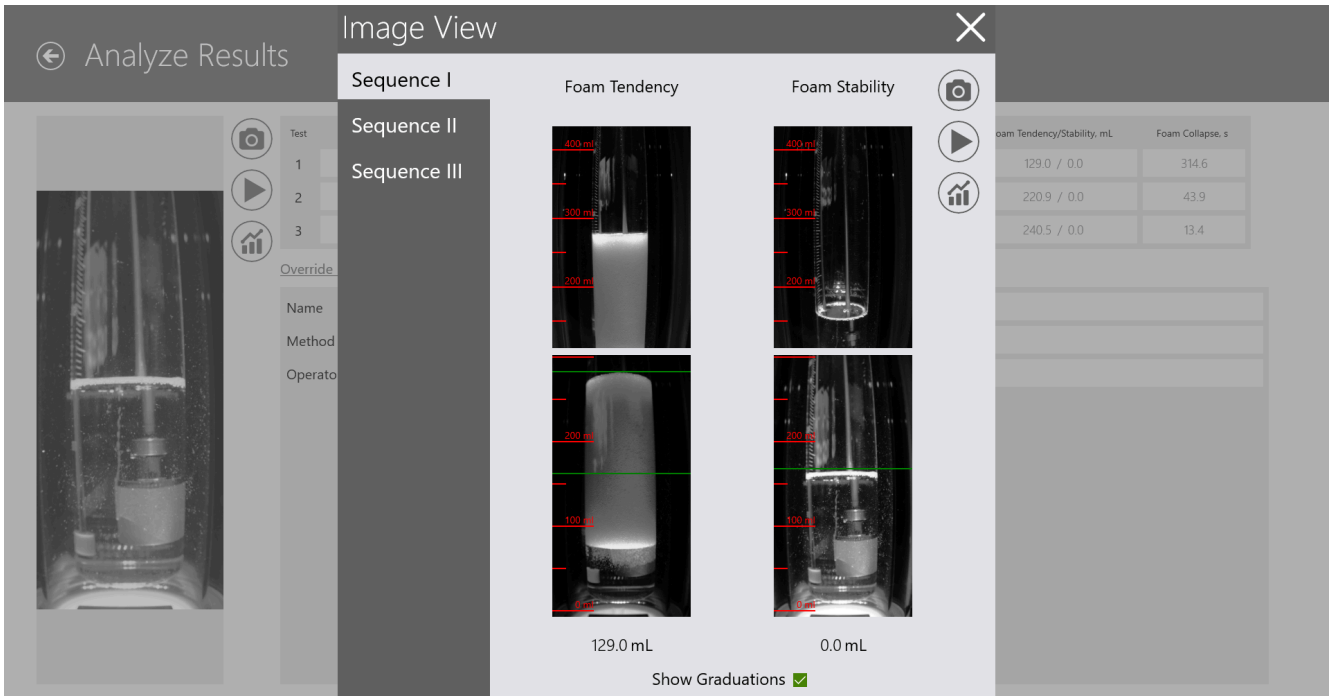
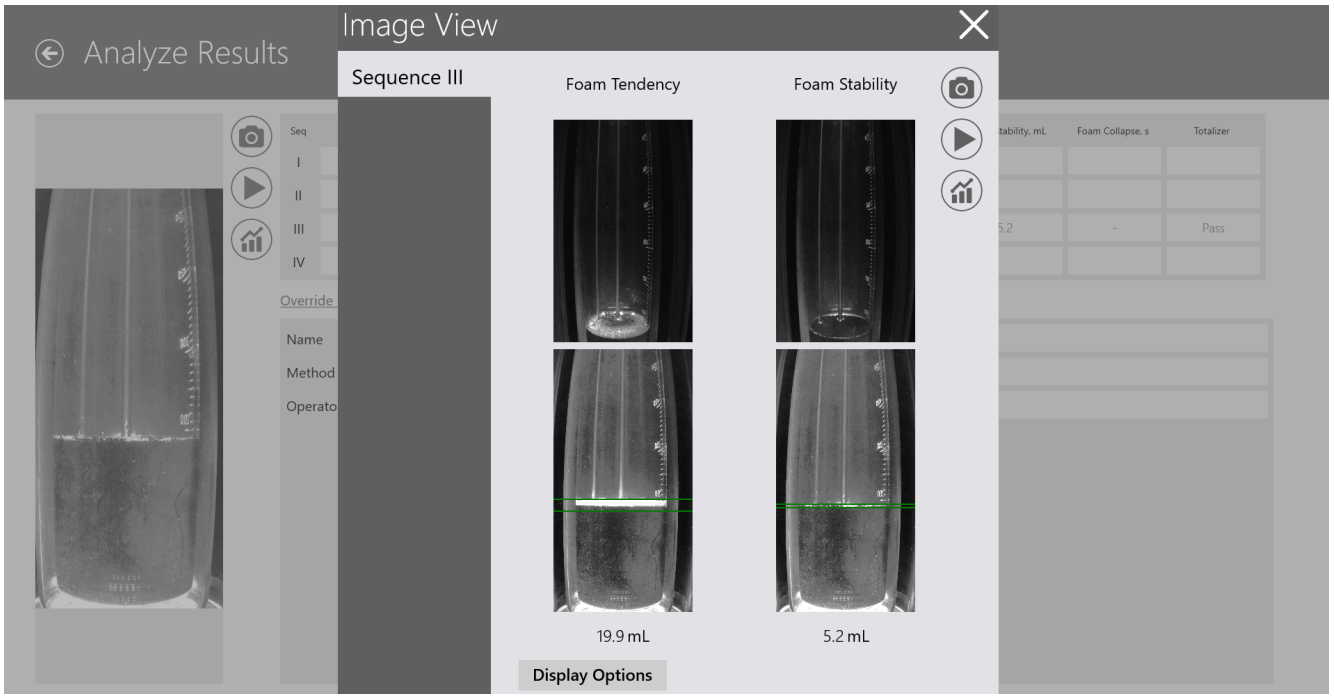


There is an option to turn up the brightness or show graduations under Display Options in the Lubricant software.



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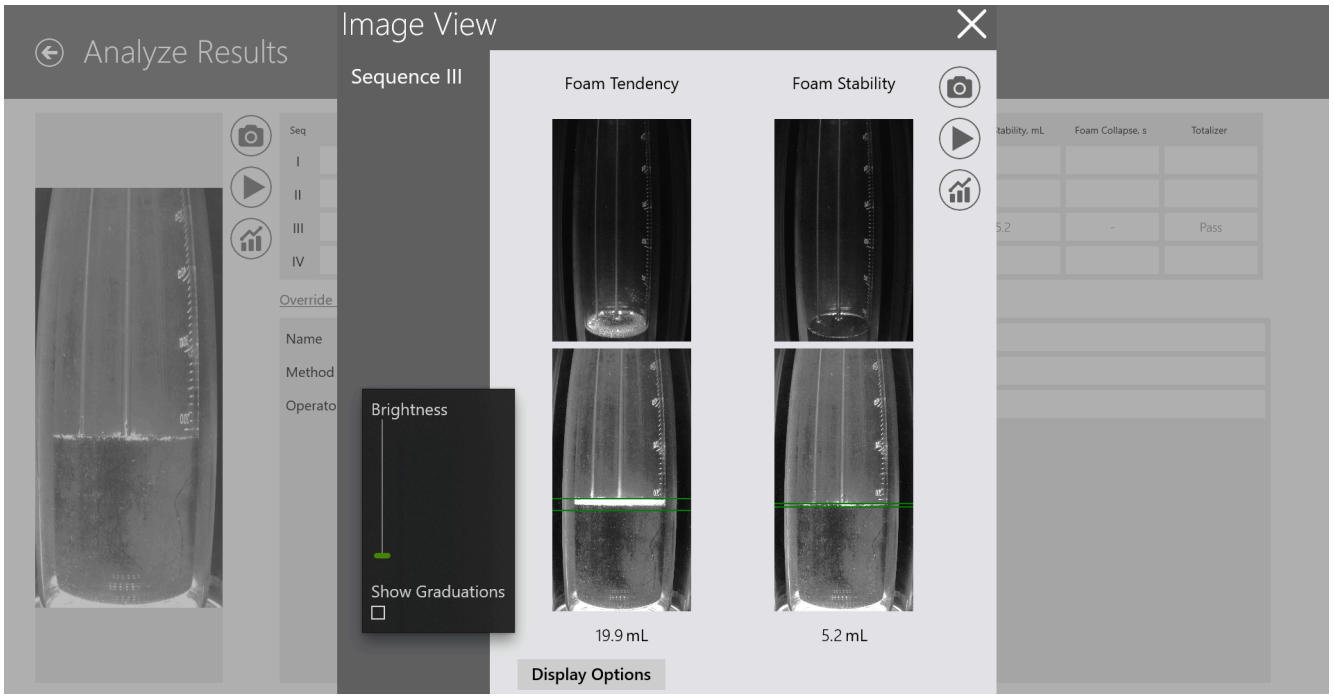
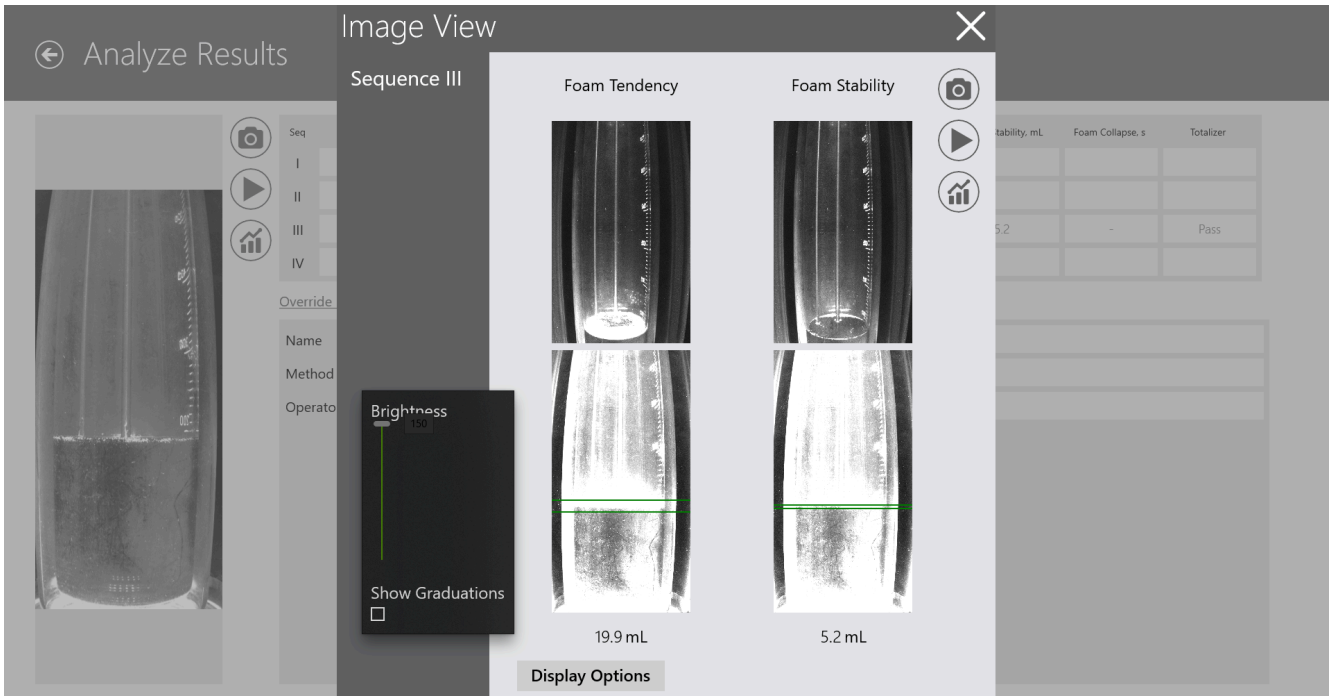


Use the slider to increase or decrease the brightness of the results and click the check box to show thye graduations used to calculate the heights of the foam.



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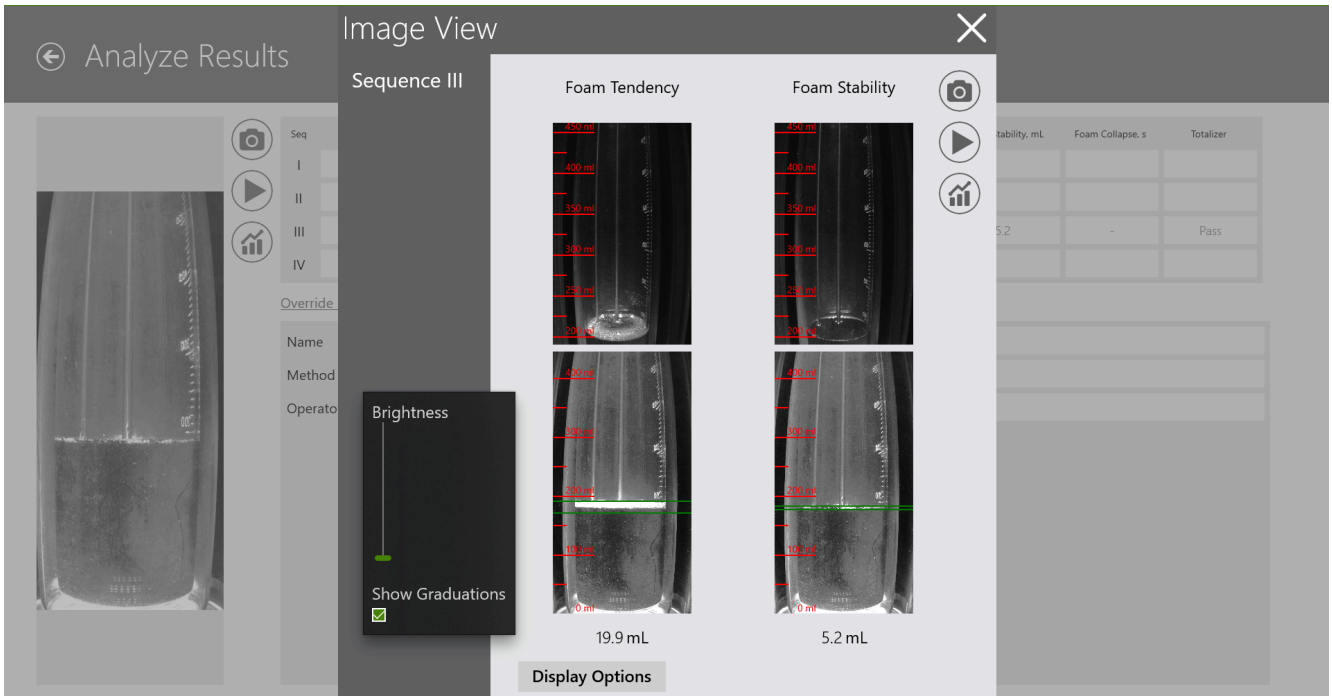
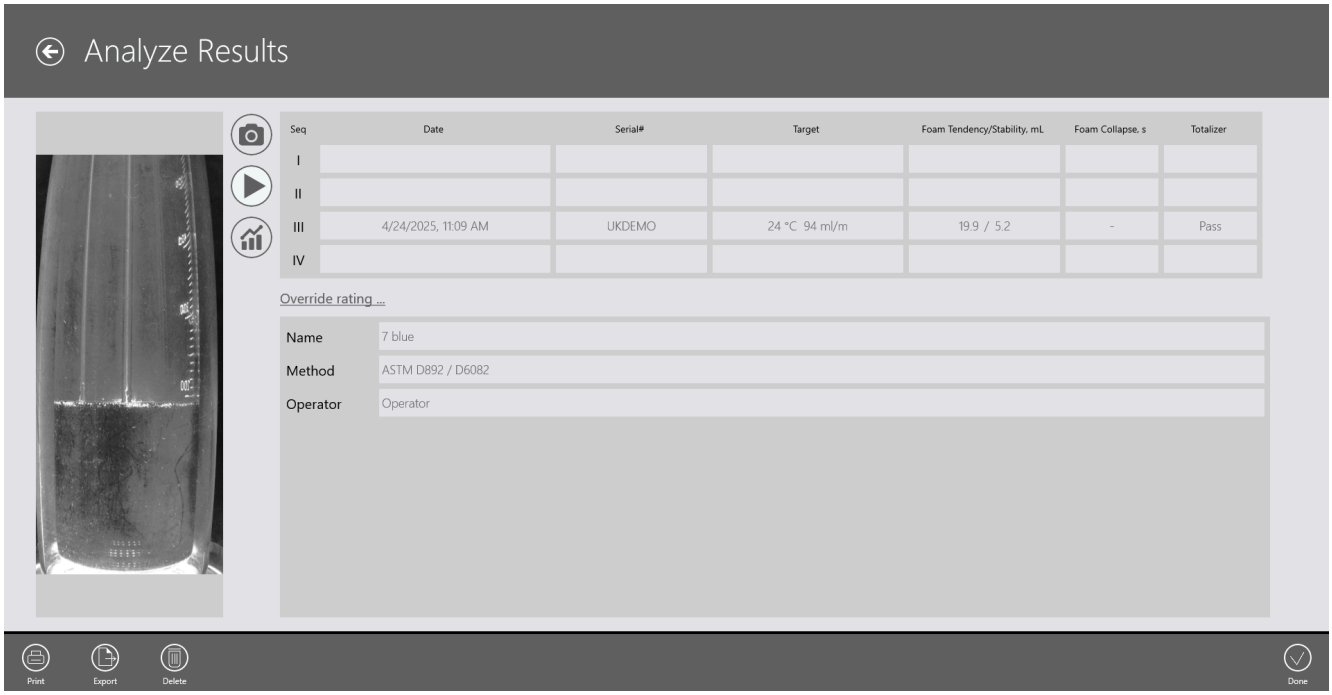


5.3.2.2. Secondary Screen - Playback View

To navigate to the videographic results, select the Play icon from the initial detailed results screen or in the Image/Graph View screen.



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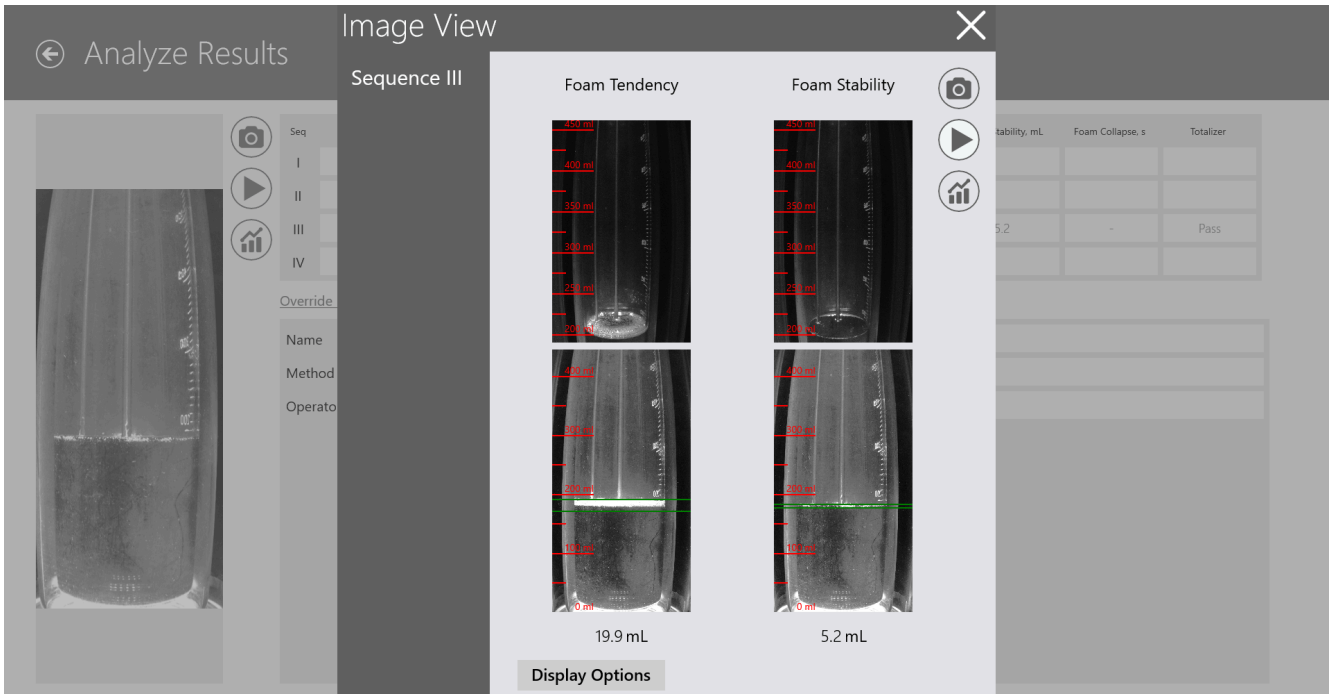
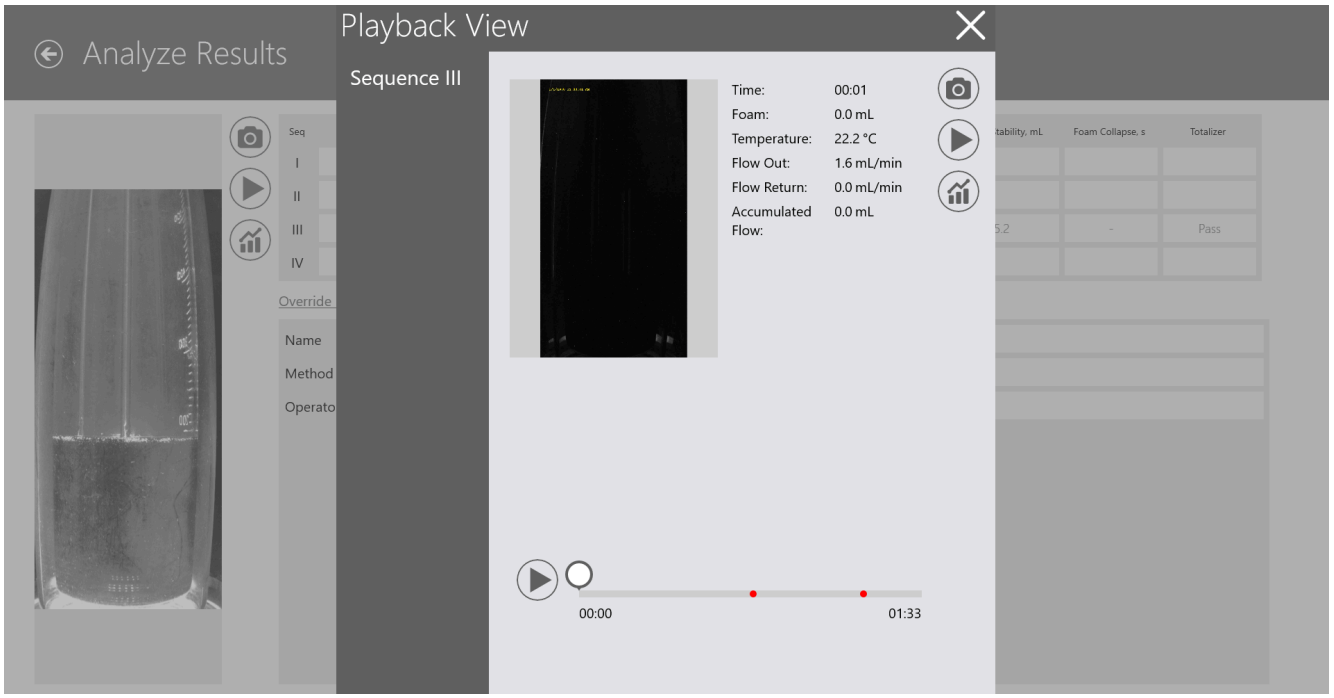


The Playback View opens with a full recording of the entirety of the test. The readings of Time, Foam, Temperature, and Flow throughout the analysis are also displayed as the video plays back.

To start the recording, click the Play icon on the bottom of the screen. The timeline is displayed to the right of the play icon.



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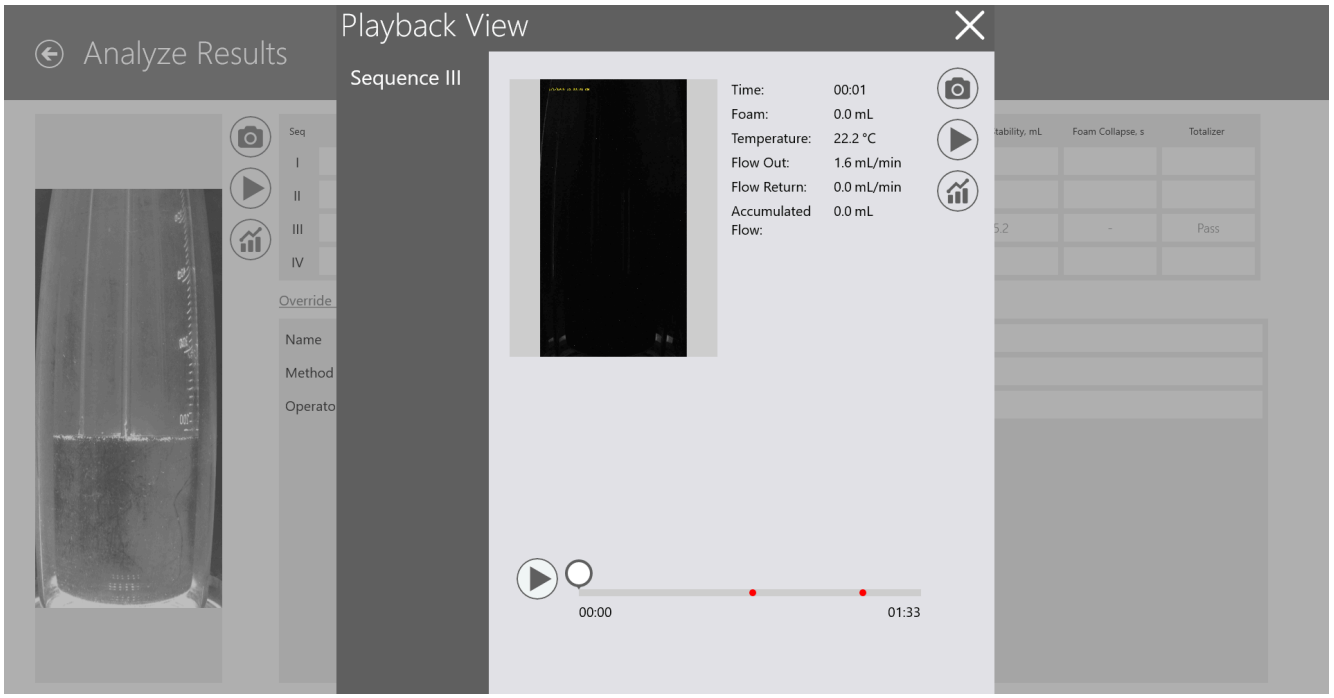

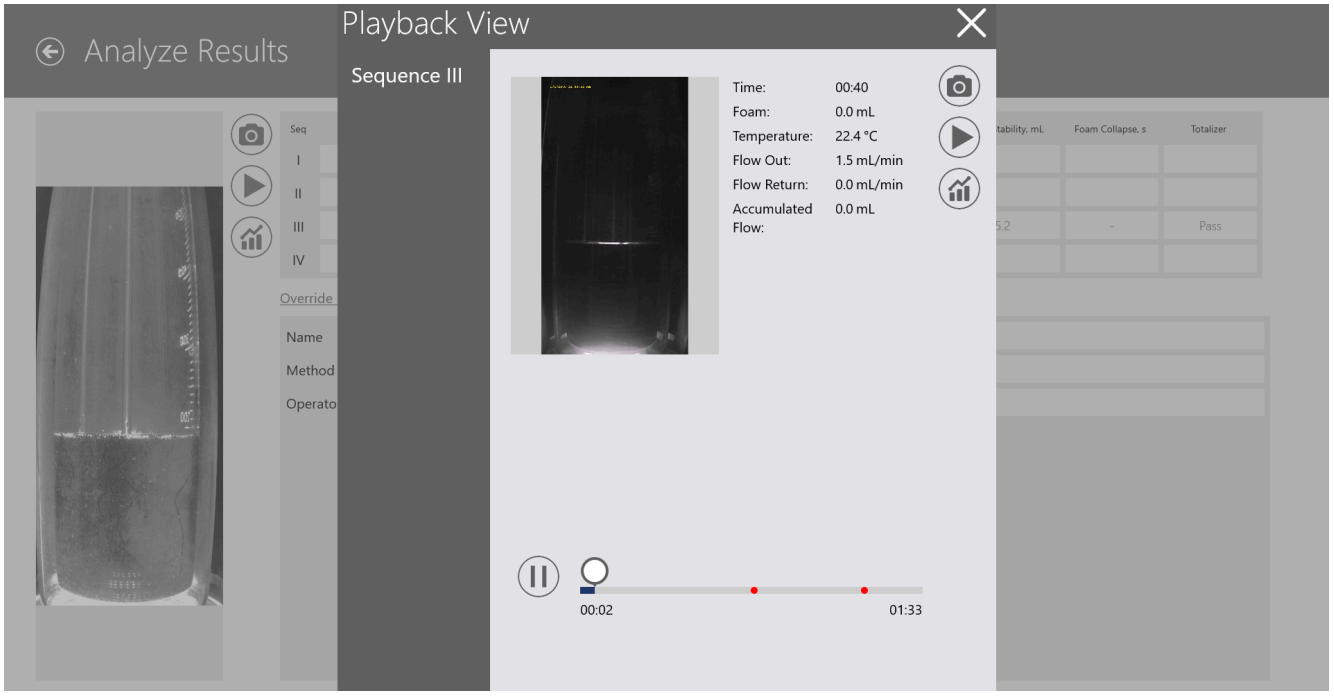

The Red markers show the location within the test the airflow begins and ends respectively. Drag the slider to fast track to specific parts of the video. These markers are currently only available in the Lubricant software.

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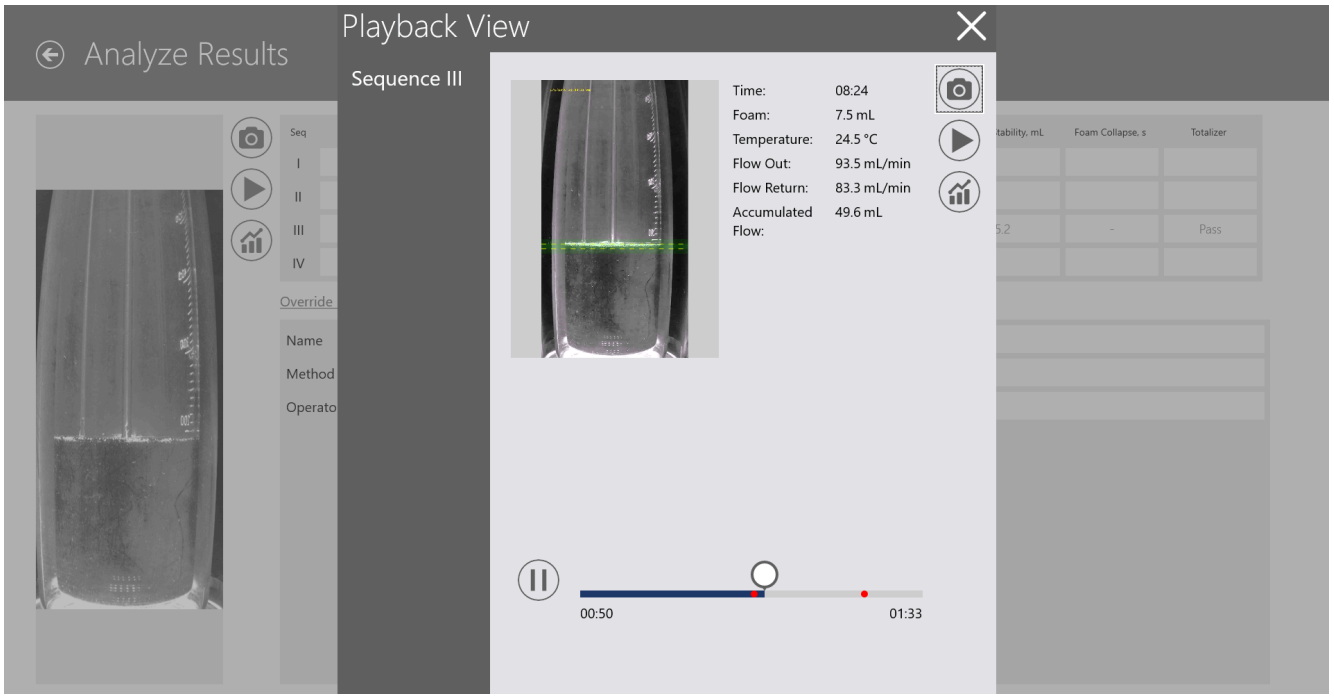

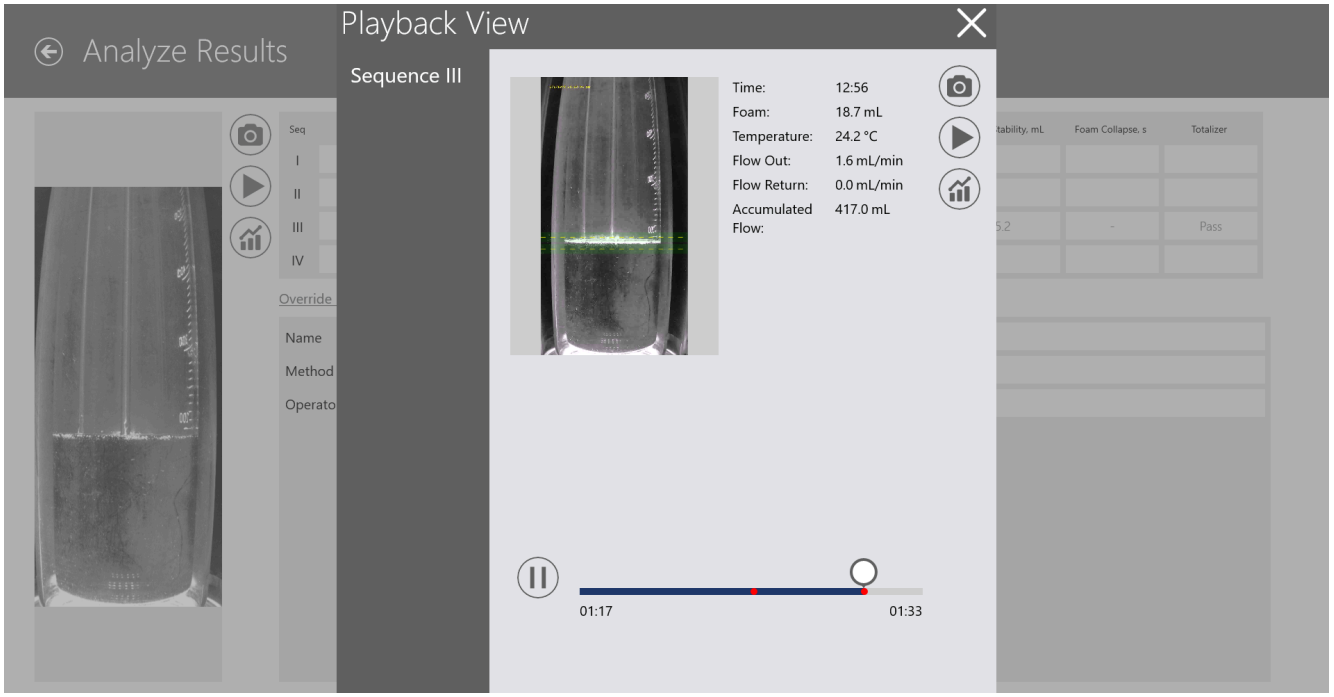

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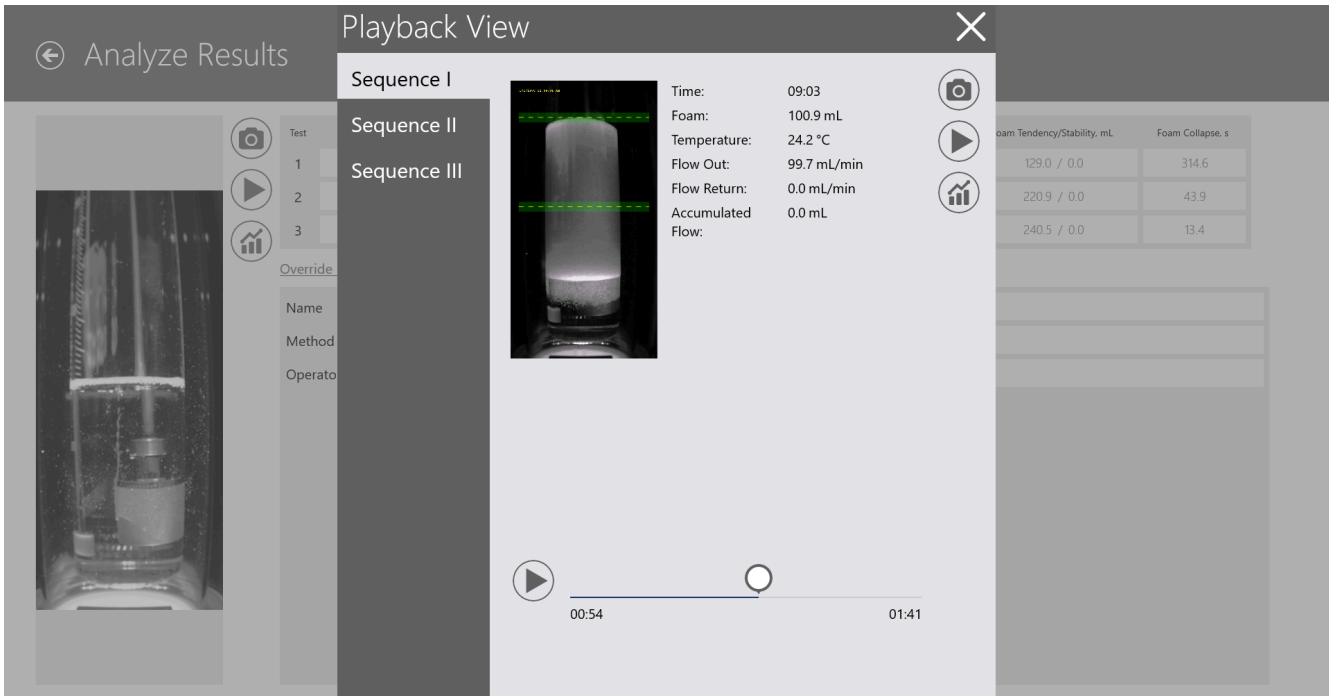
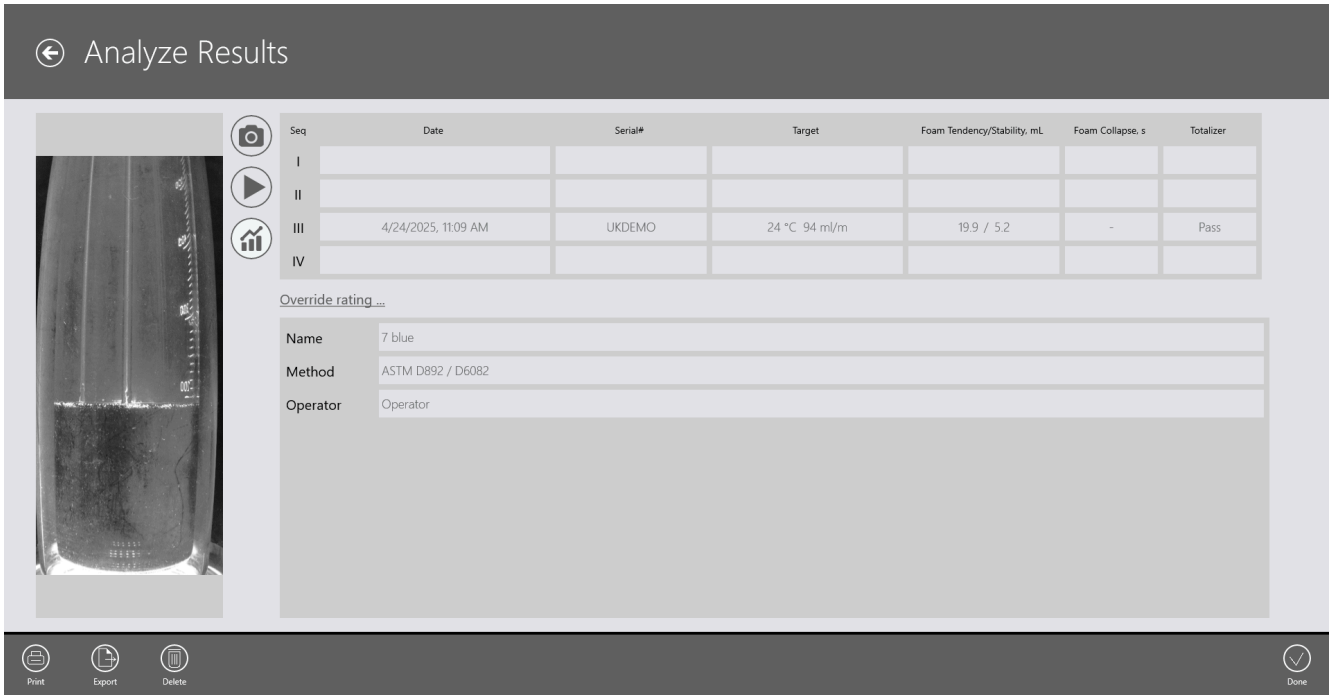


5.3.2.3. Secondary Screen - Graph View

To navigate to the graphic results, select the Graph icon from the initial detailed results screen or in the Image/Playback View screen.



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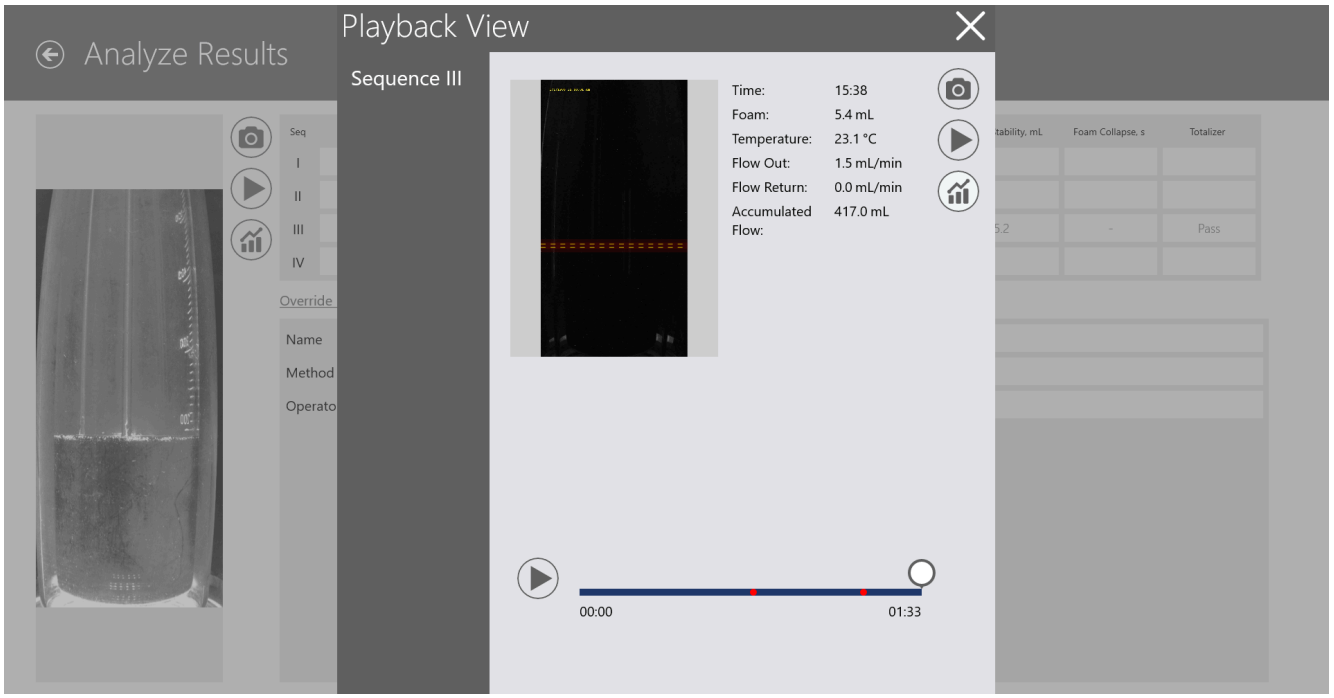
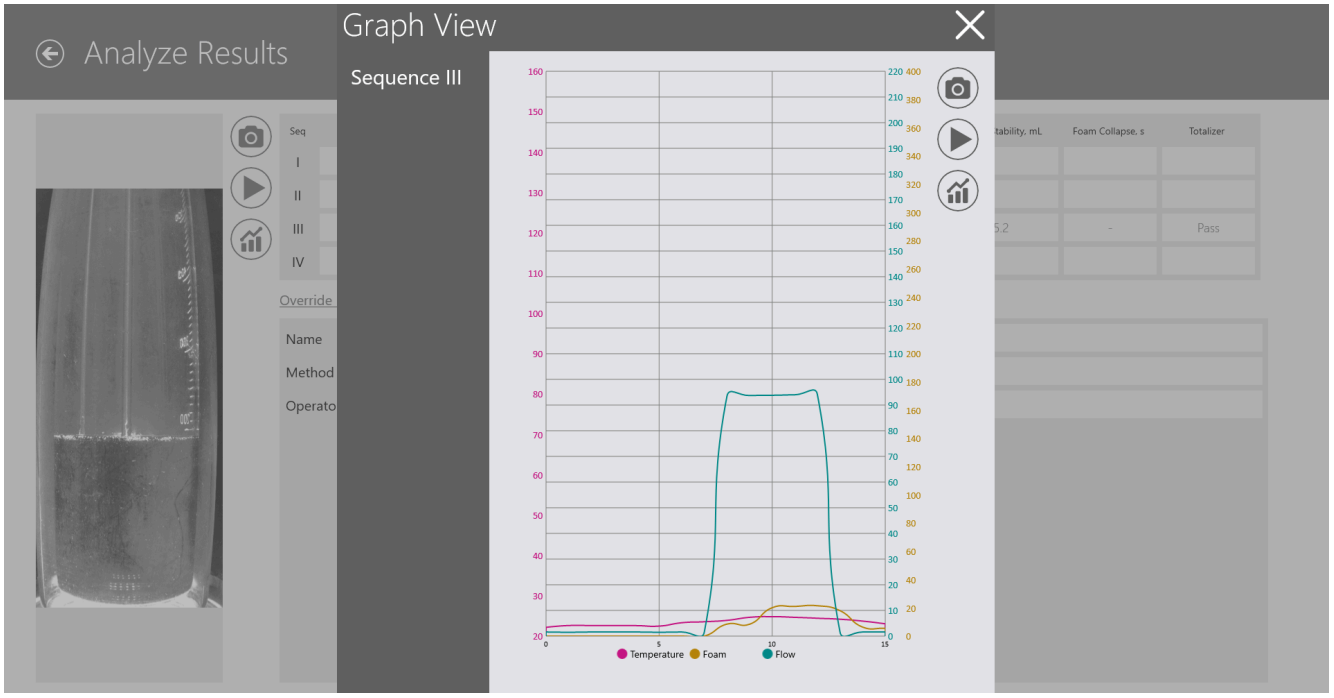


The Graph View screen shows the Temperature, Foam, and Flow throughout the test.



Hovering over specific points on the graph displays the actual readings associated.

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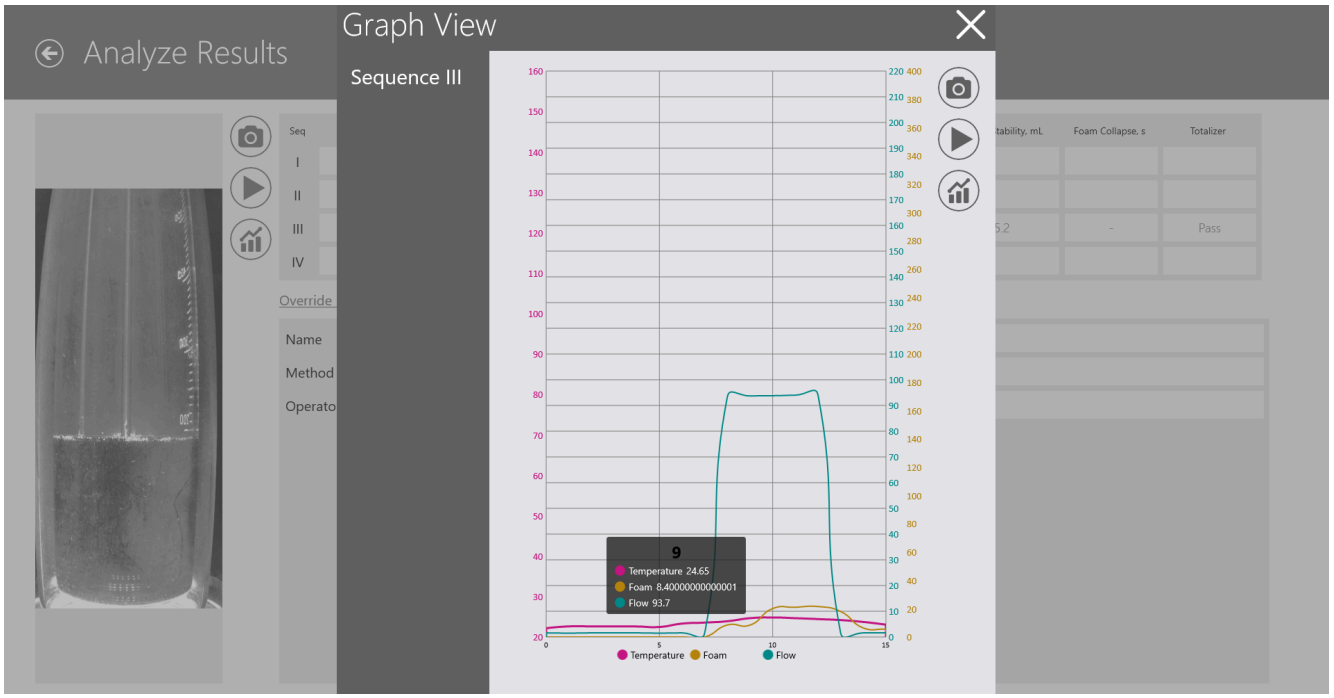
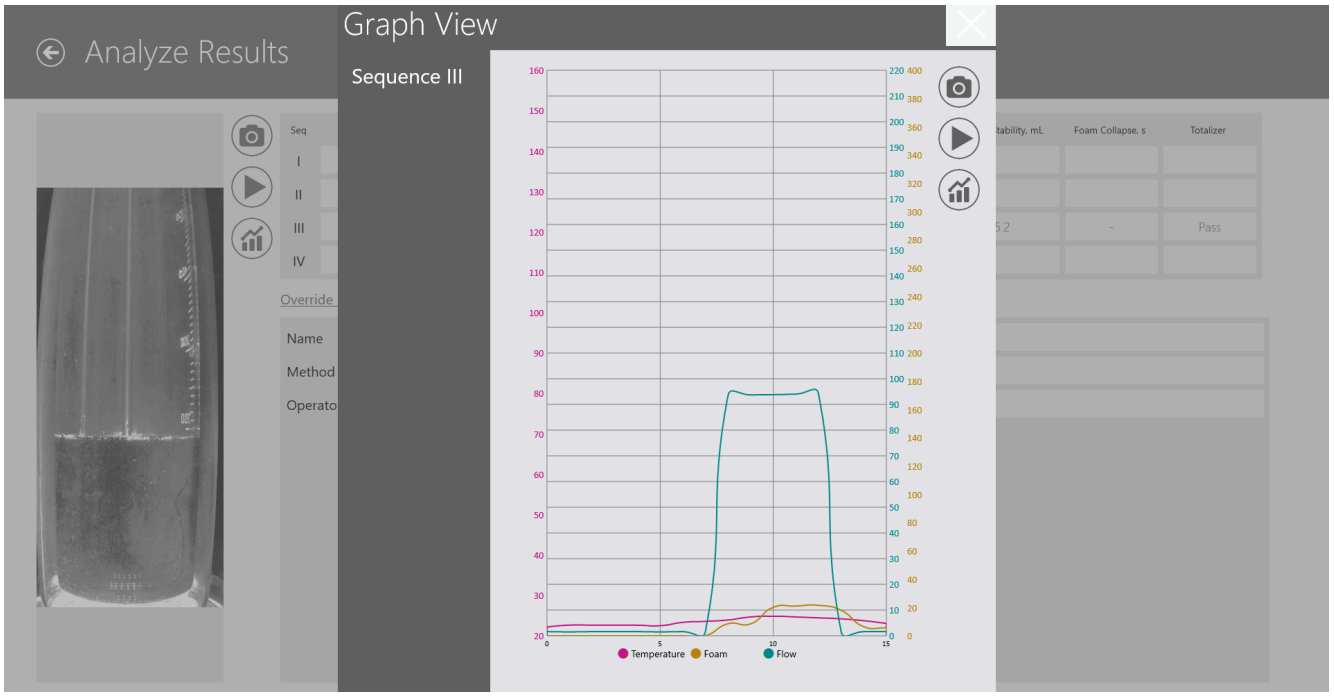


To navigate back to the initial detailed results screen from the Image, Playback, or Graph View screens, simply click the *X* button at the top right of the secondary screen.



5.3.3. Override Results from History

To override the rating, click the underlined option.

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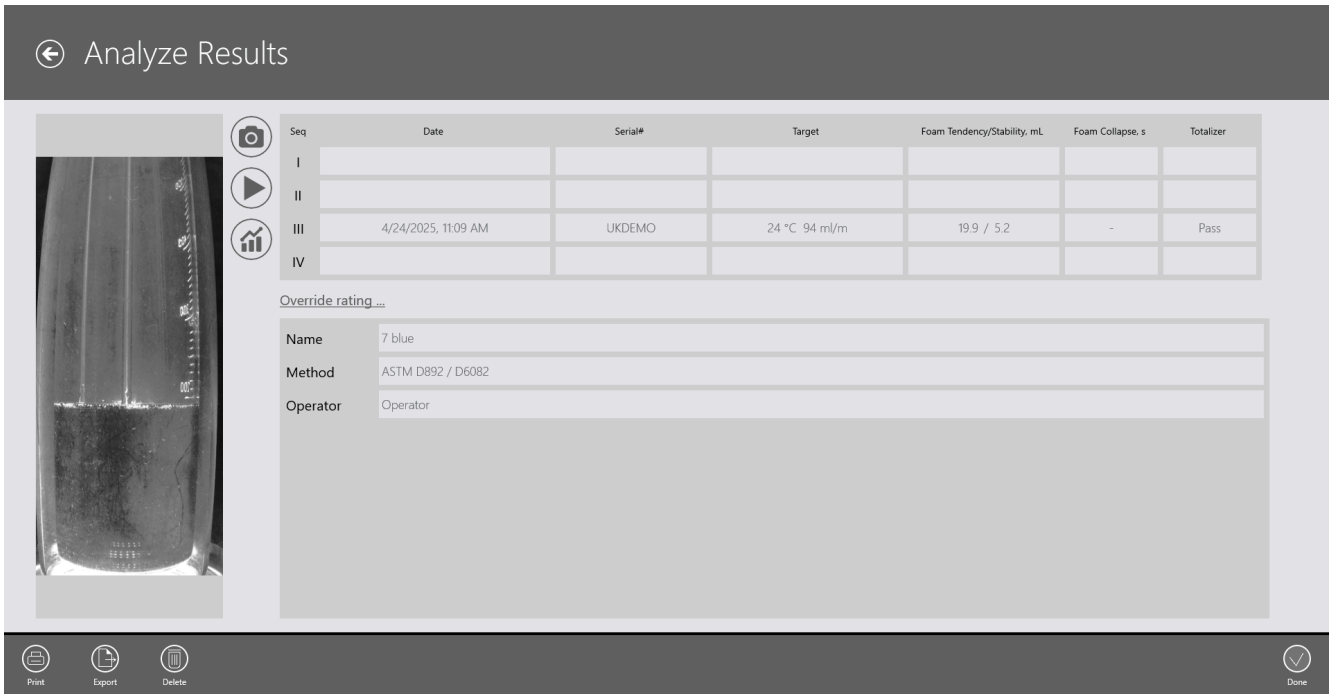
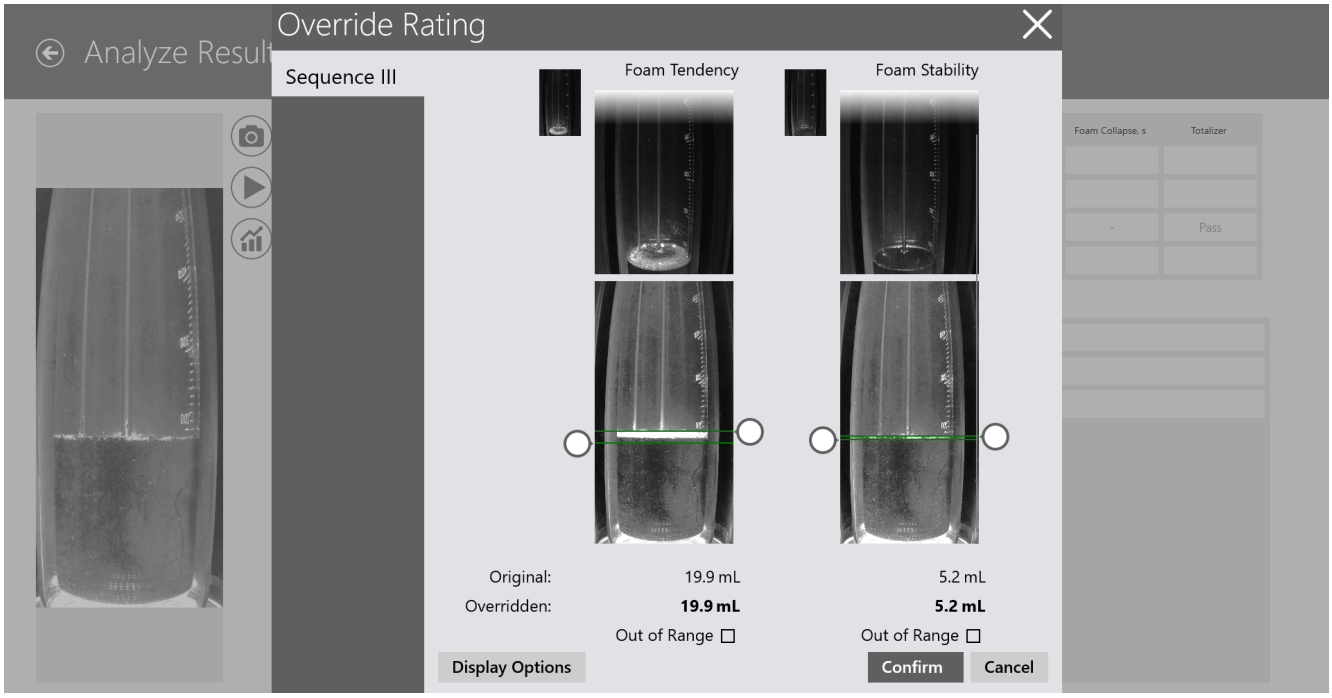
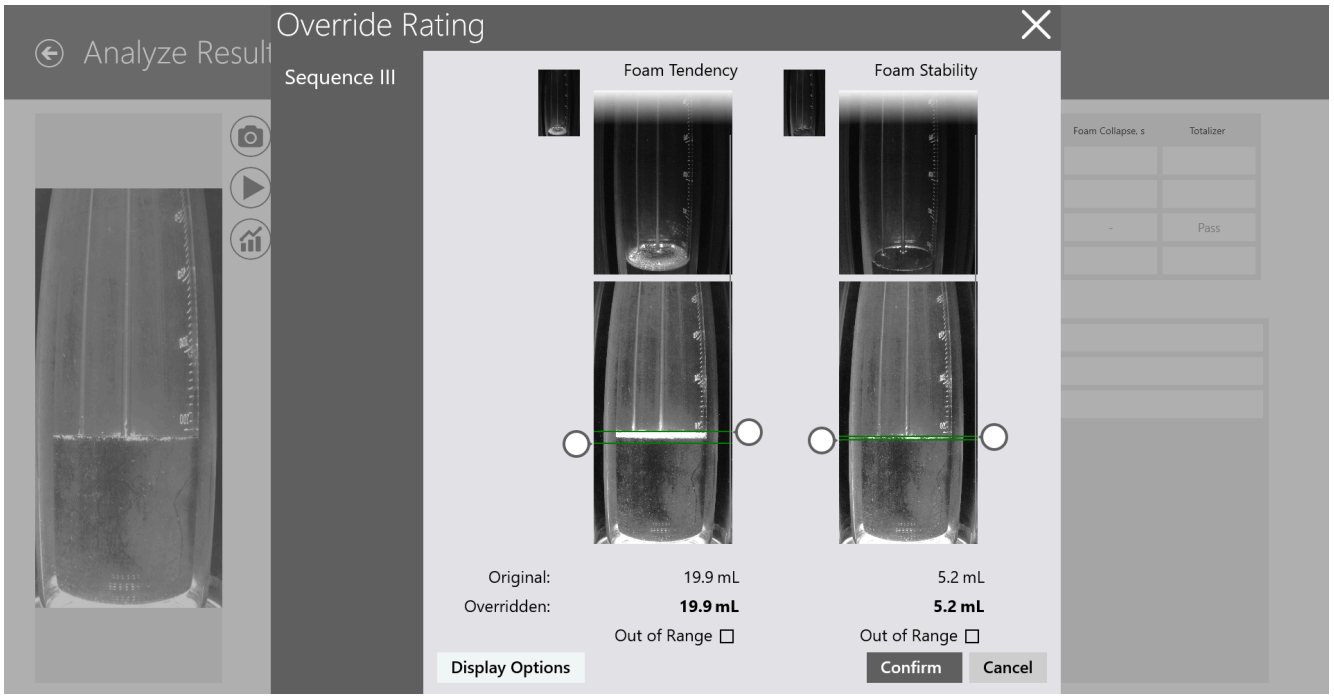
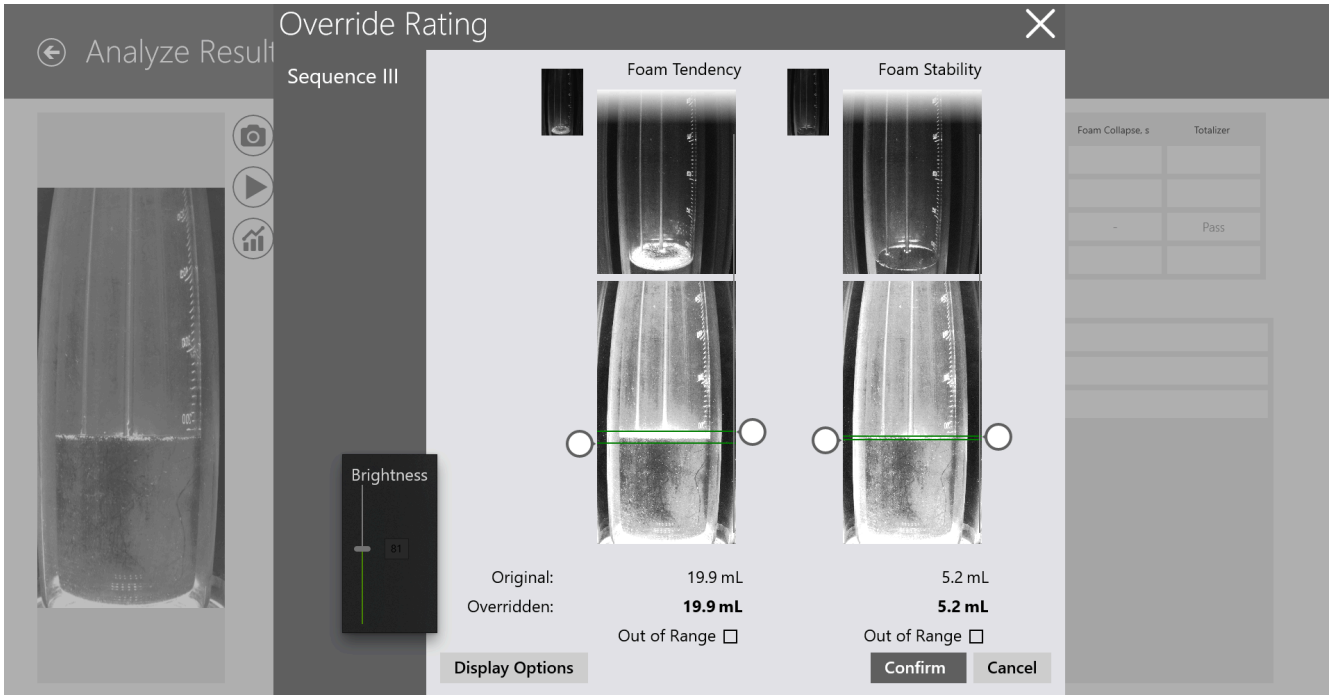


A screen similar to the Image View will be displayed.



Similar to in Image View, the Display Options allow the user to turn up or down the brightness.





To override the results, manually drag the slider on either the tendency or stability or both photos to the more accurate edge of the foam. The results will automatically be calculated and displayed as the Overridden result. The original result will be retained.

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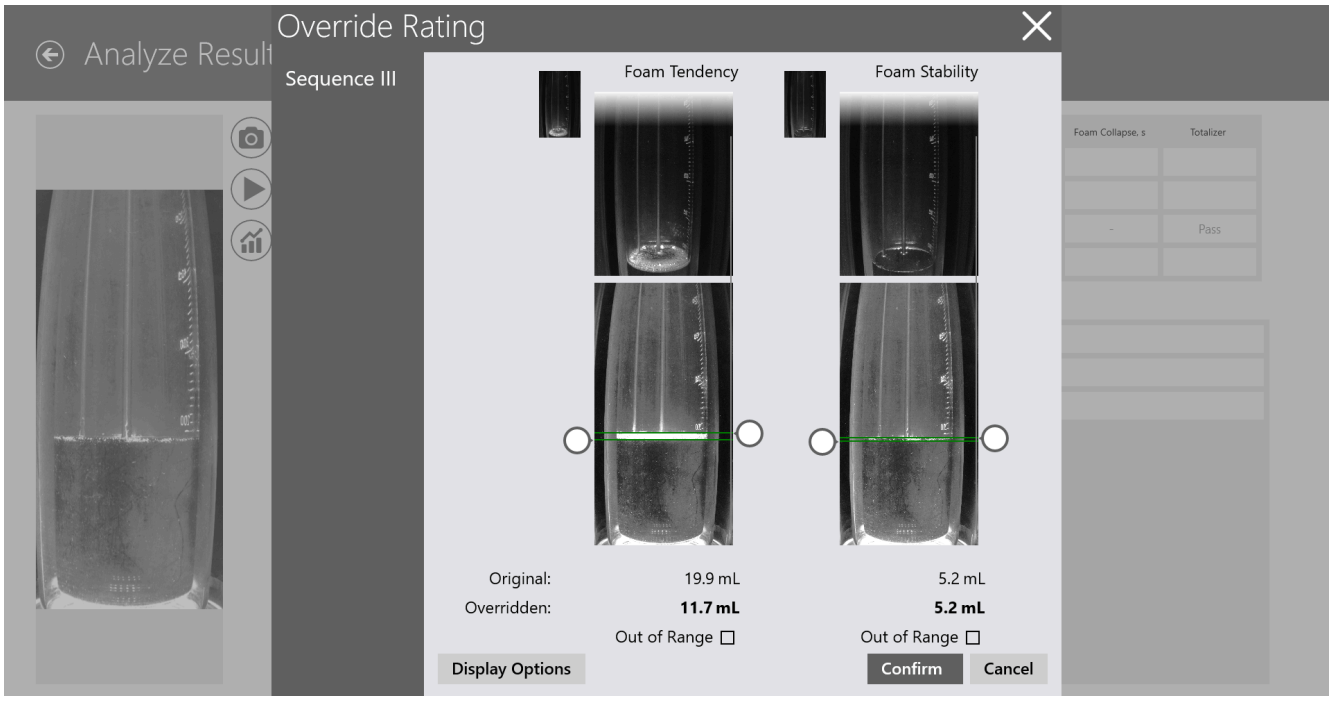
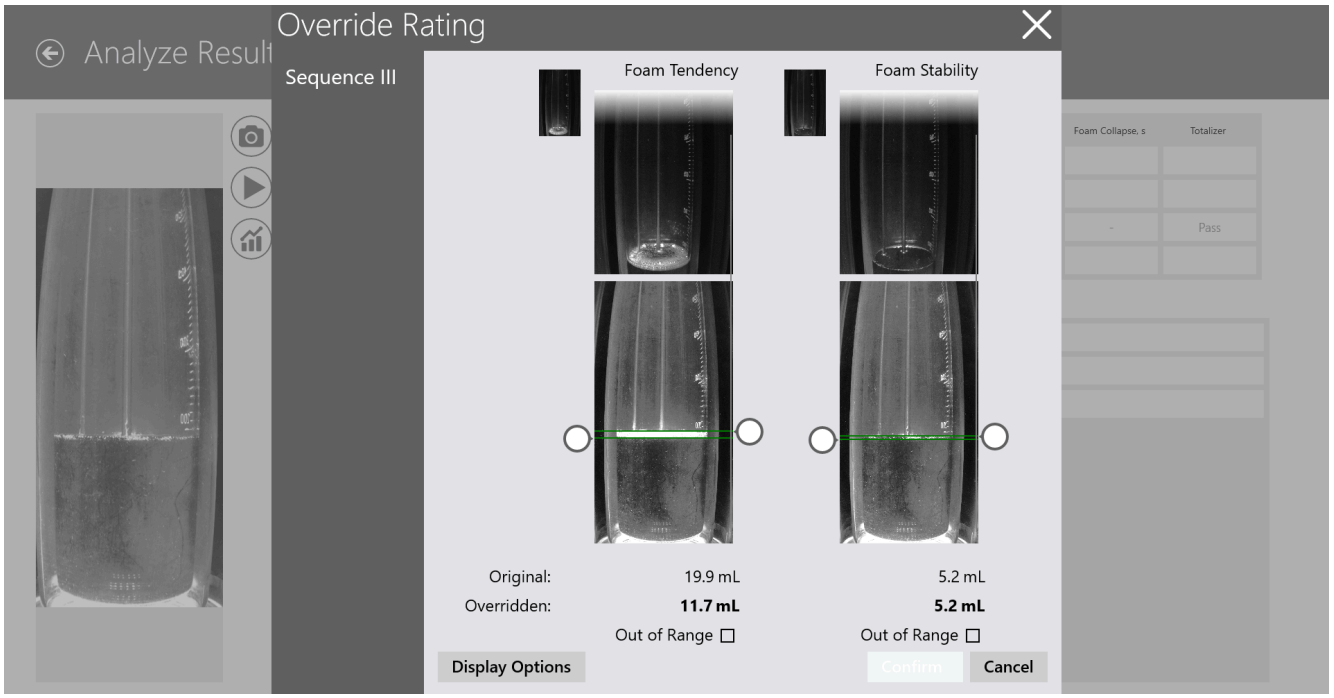


To save the overridden result, click Confirm at the bottom right of the secondary screen. To cancel the override, click Cancel at the bottom right next to Confirm.



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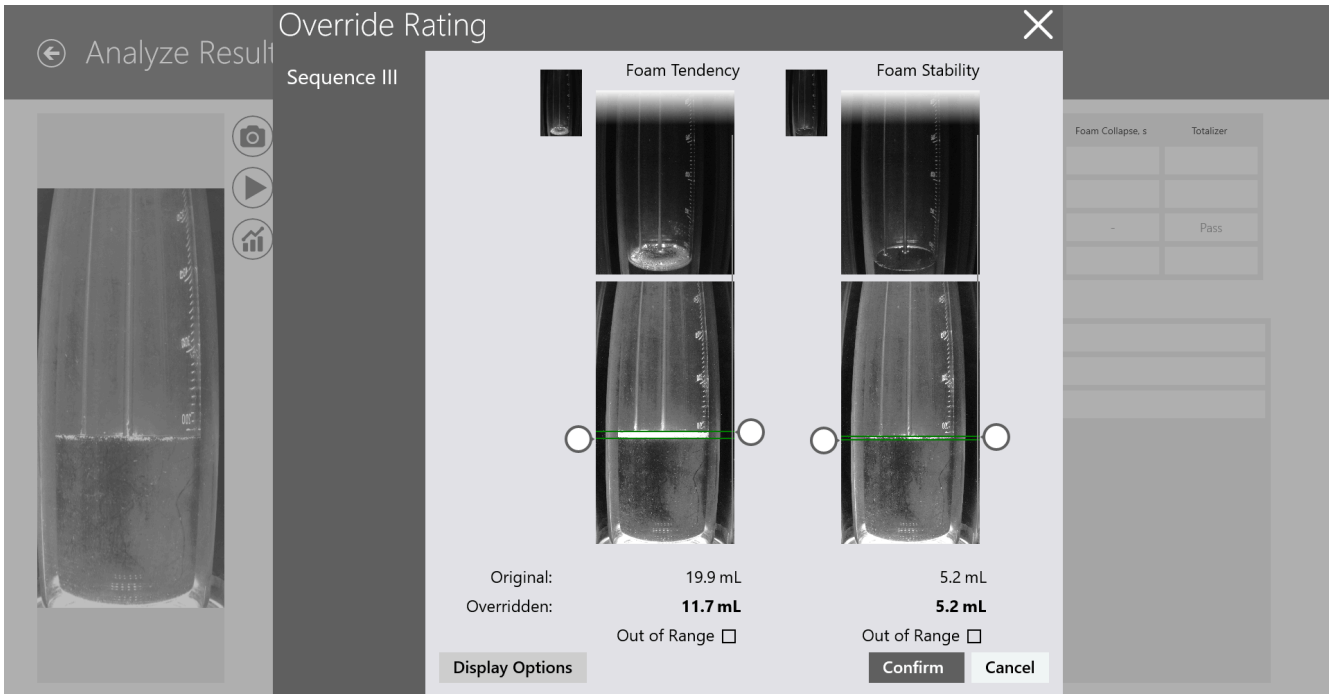
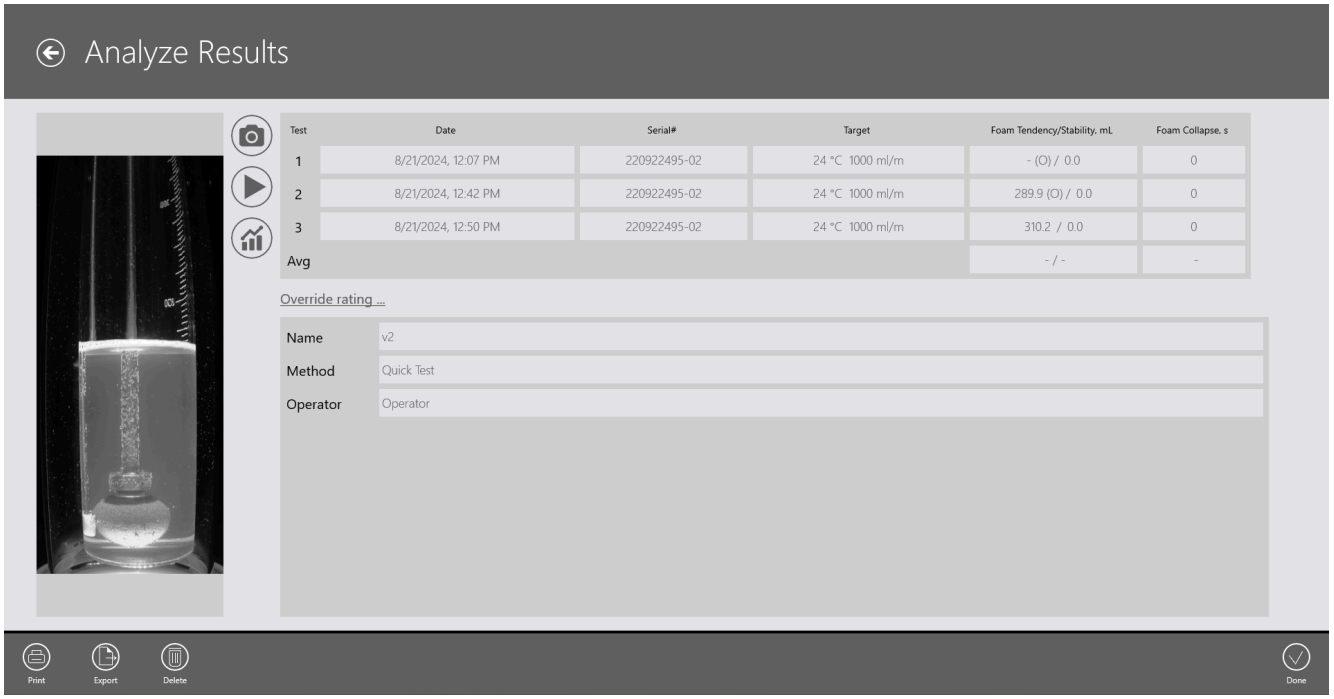
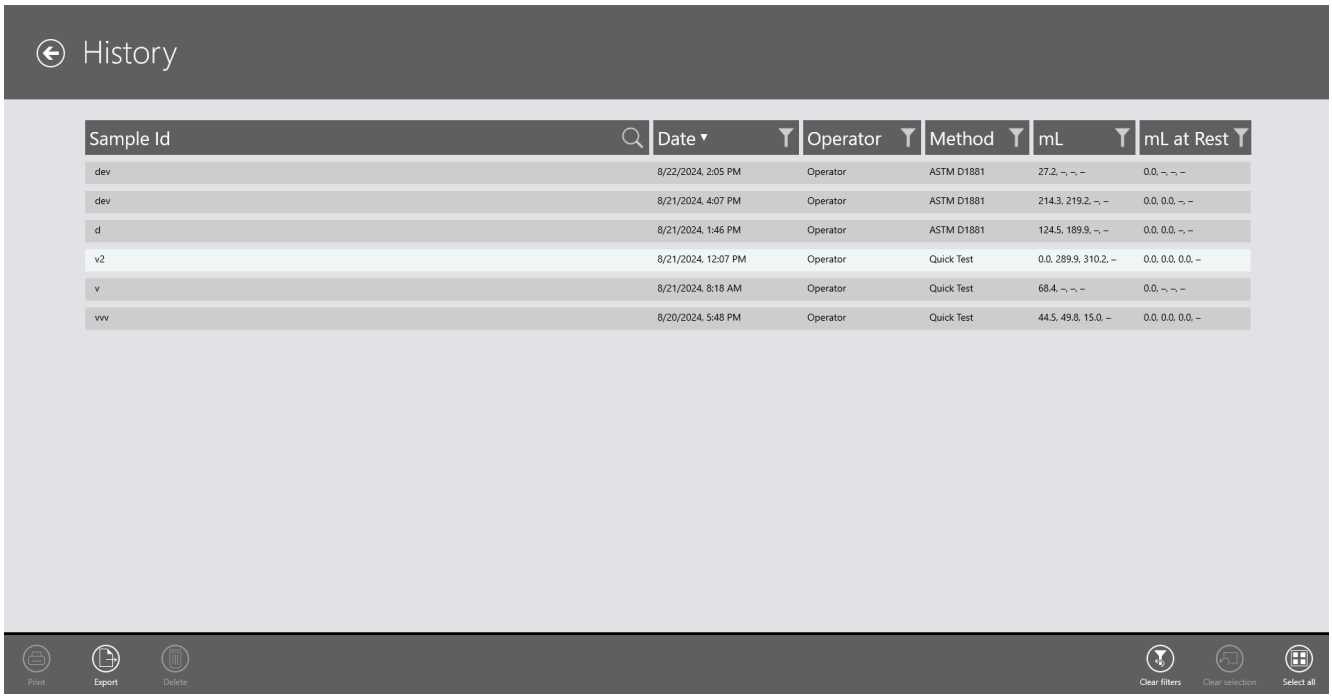


Overridden results are indicated by an '(O)' in the detailed results screen.

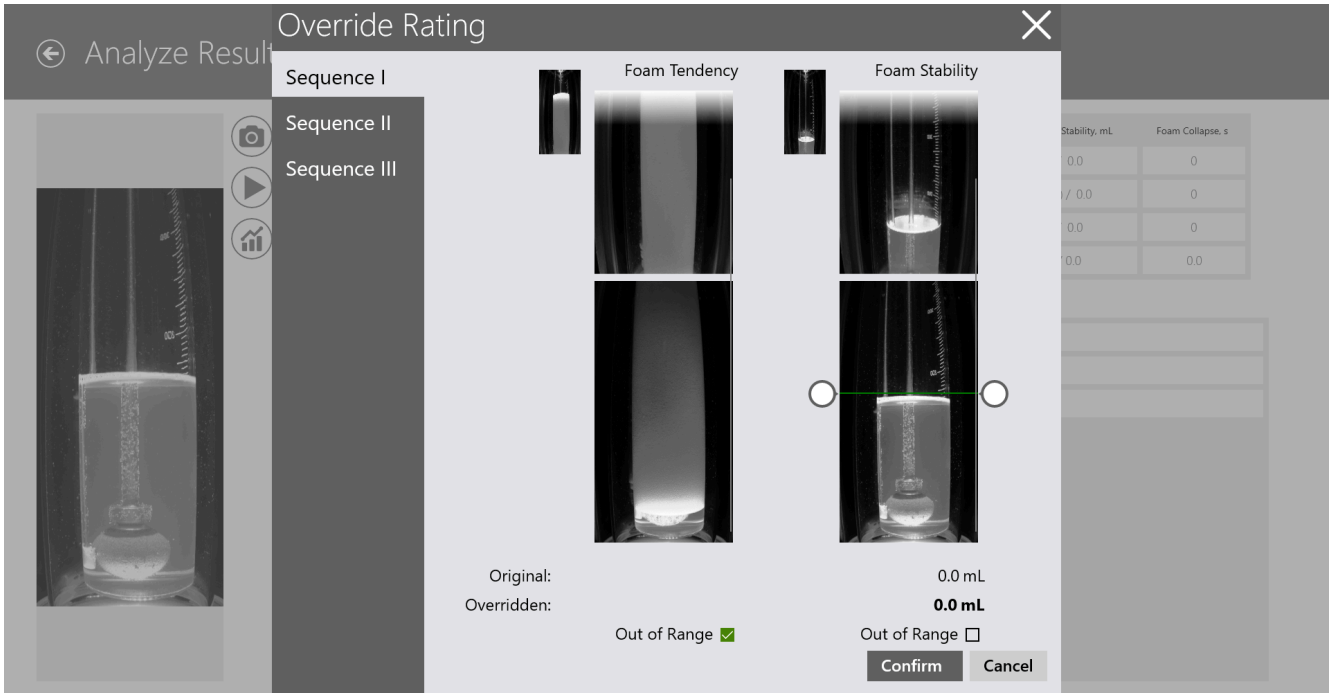


Note: Overridden results are not displayed on the history screen. Only noted on the detailed results screen.

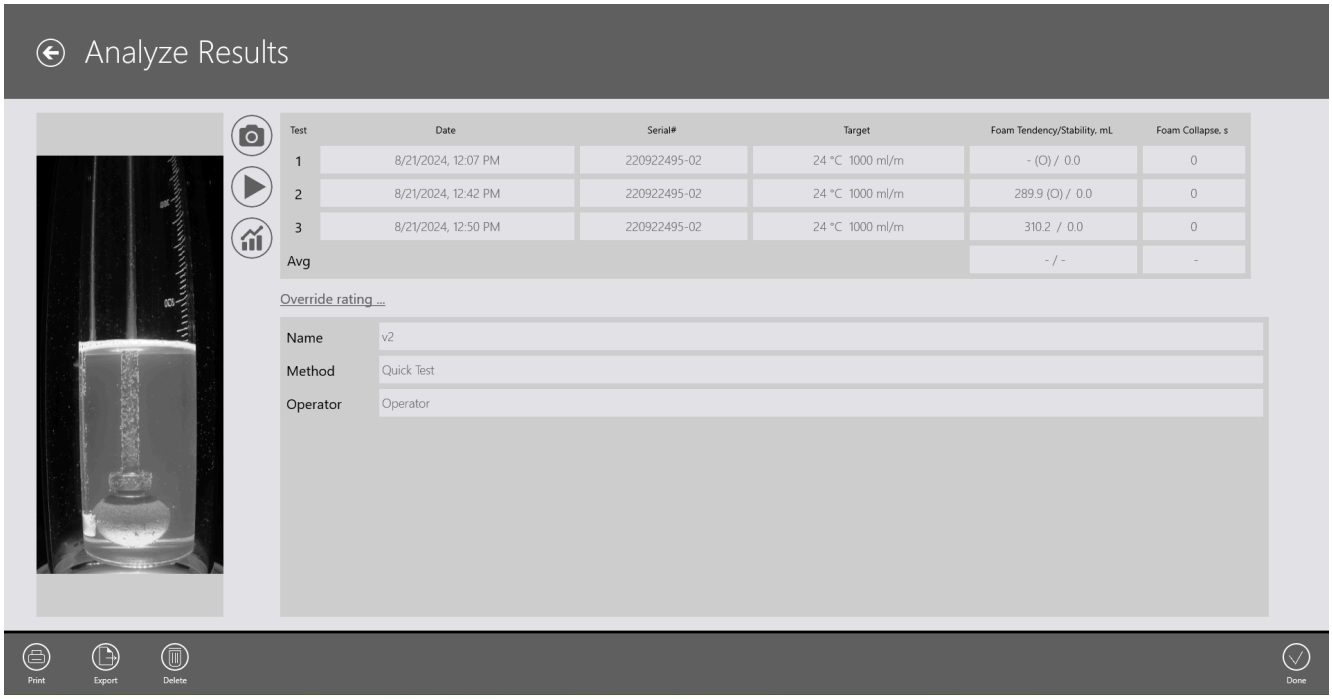


5.3.4. Out-Of-Range Override

In high foam samples, sometimes the camera will not be able to identify edges of the foam. Checking the Out-of-Range tab, will display a too high to analyze result.

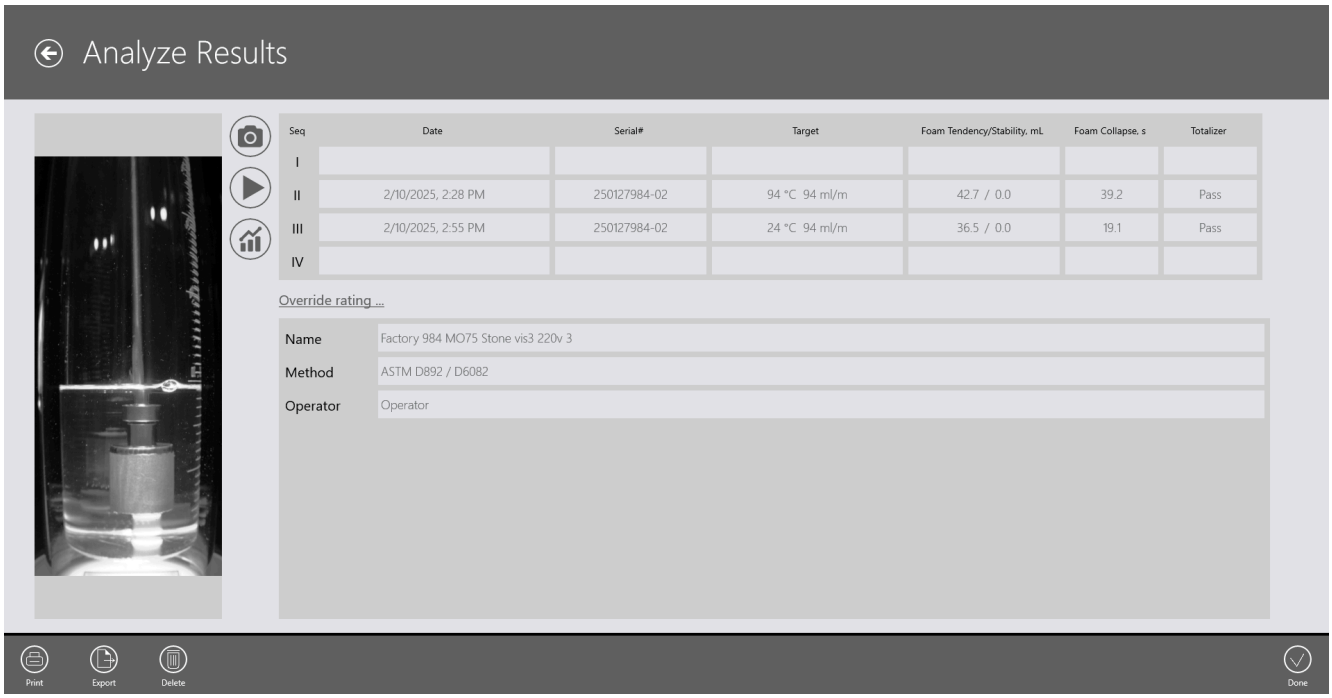


Under the primary detailed result screen, an out of range override will display as a dash (-) indicating the value could not be determined due to being out of range.

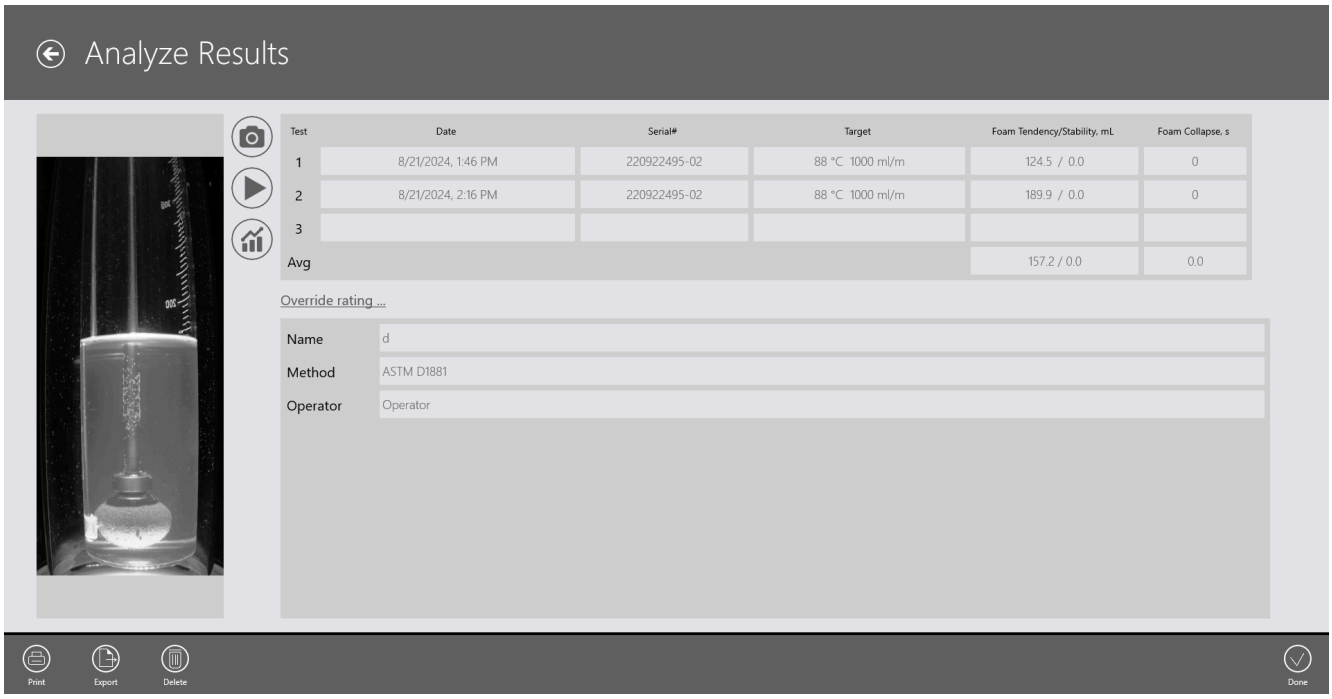


5.3.5. Multi-Sequence/Test Runs Results

With samples that undergo multiple sequence runs. All sequence results or multiple runs associated with the sample will be displayed together.



Within the Coolant software, an average result is calculated in the case multiple tests are conducted on the same sample.



When navigating to secondary screens, there will be tabs with all sequences ran to analyze the detailed results.

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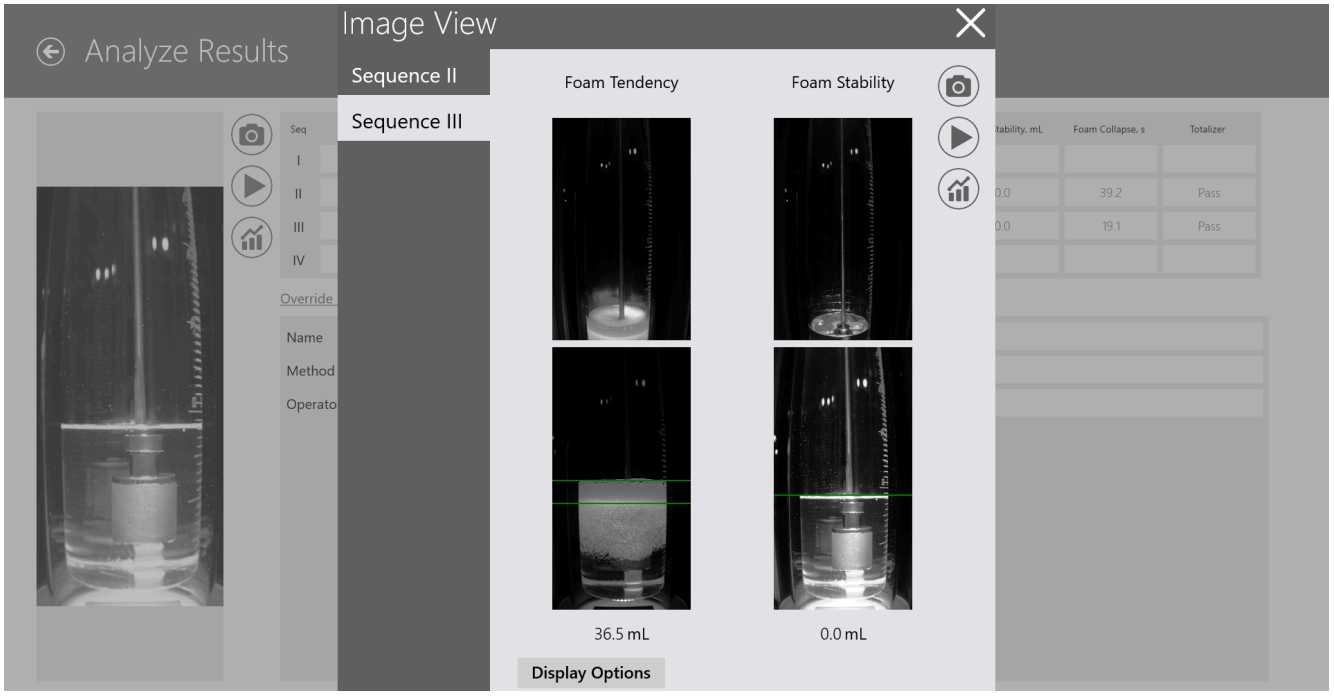
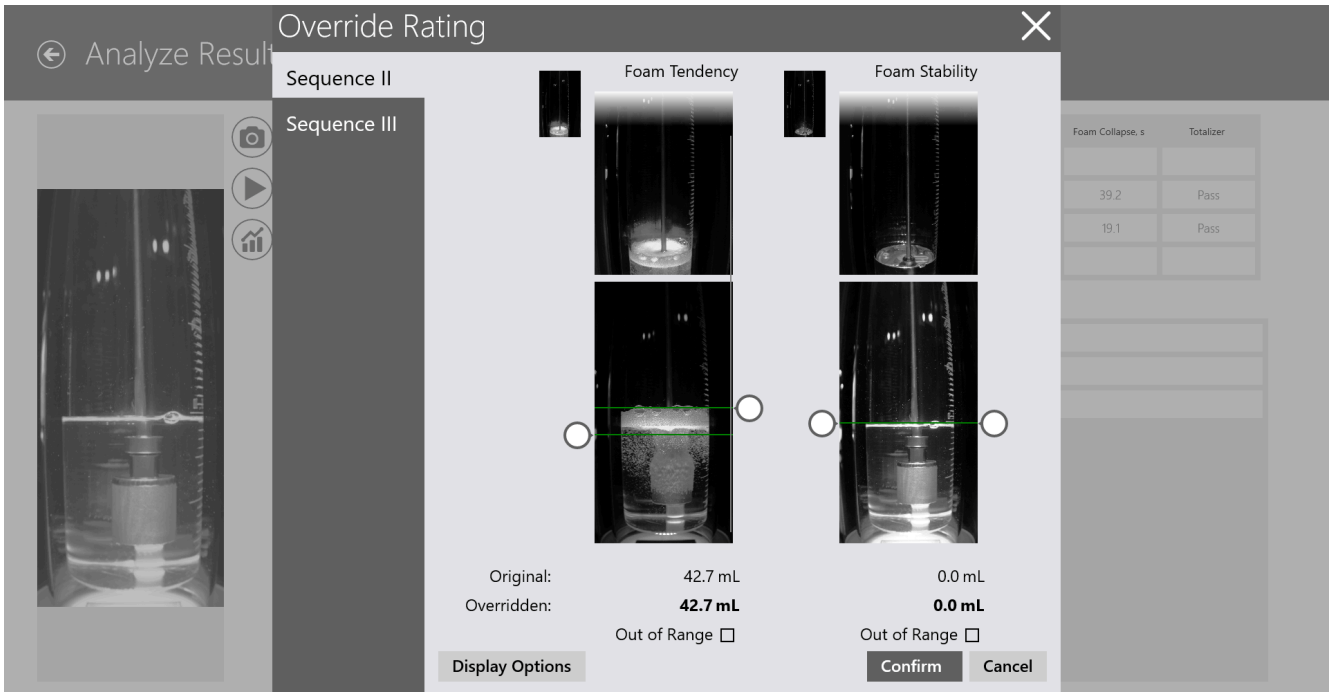


To override these results choose which sequence to override and follow the same procedures as above. One or more sequences can be overridden in these cases.



5.3.6. Failed to Analyze Result

If a result cannot be properly analyzed, a warning message will appear next to the result. This can occur for a variety of reasons. Some of these reasons may be that the sample is too dark, the sample is behaving in an unpredictable manner, lighting issues, flow issues, too high or too low of foam, or other miscellaneous issues.

In majority of these cases, simply rerunning the sample corrects the issue.

If issues persist, check out the troubleshooting and error sections of this manual.

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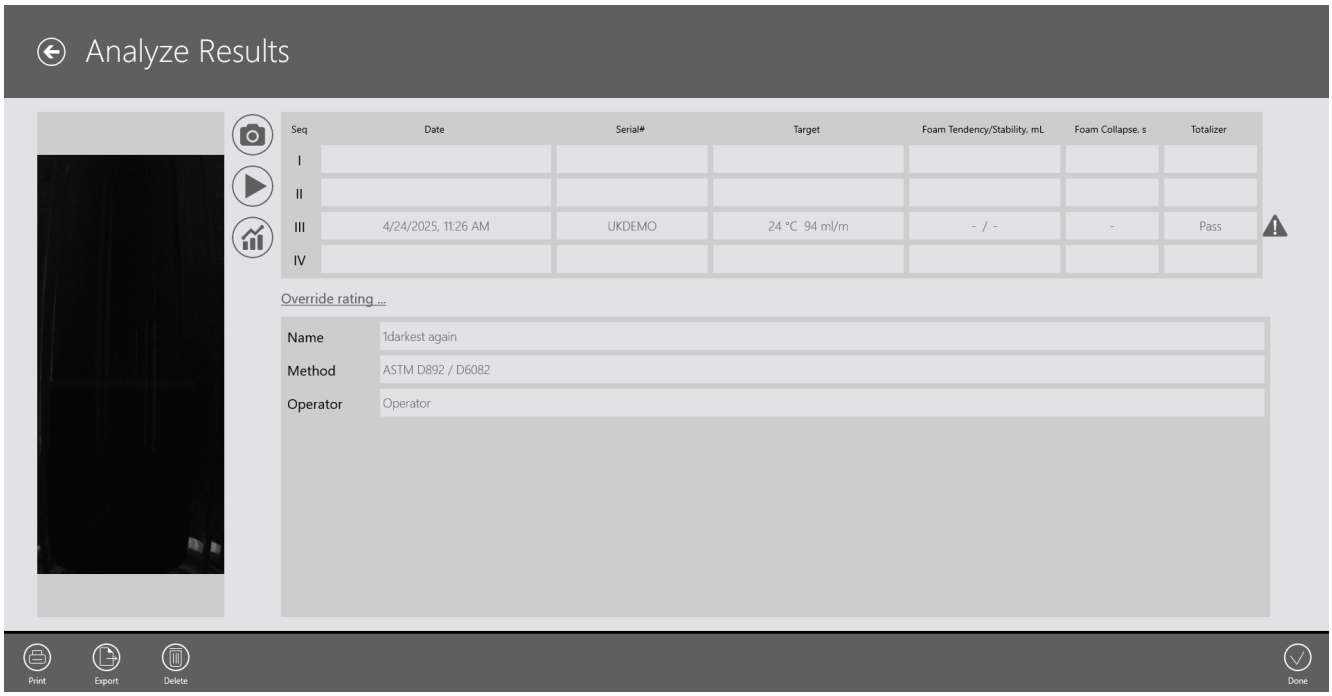
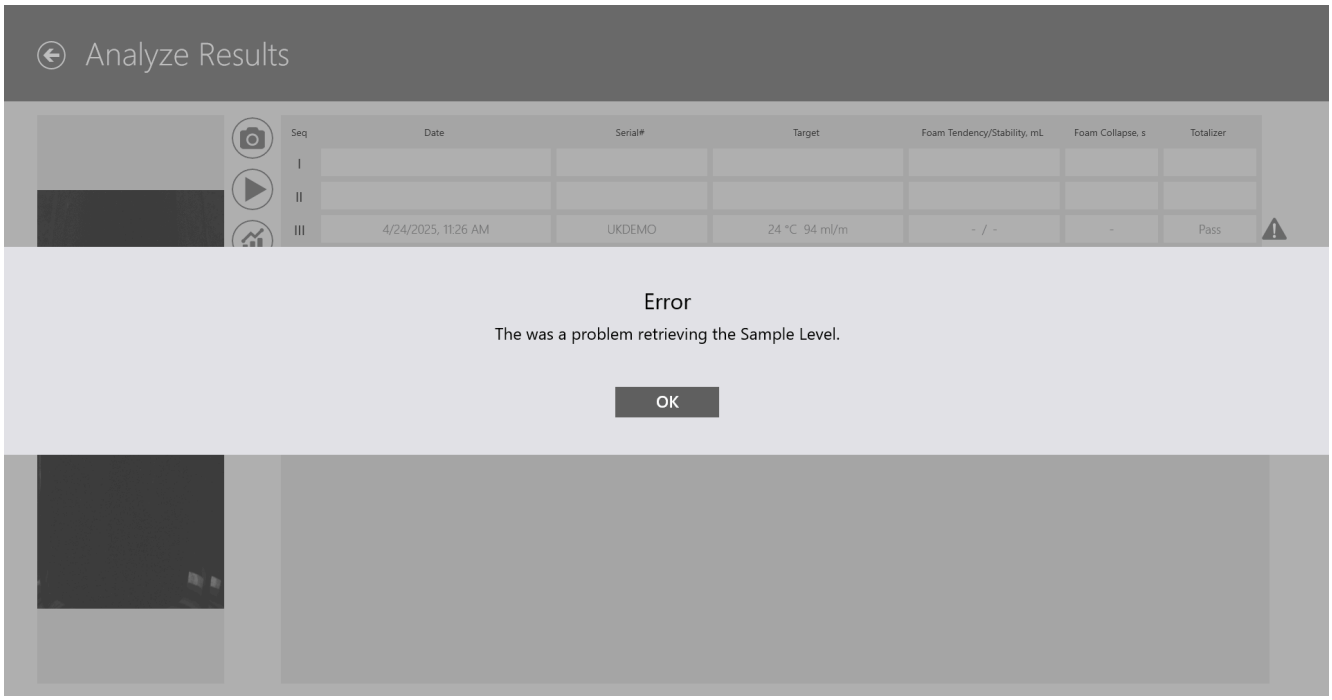


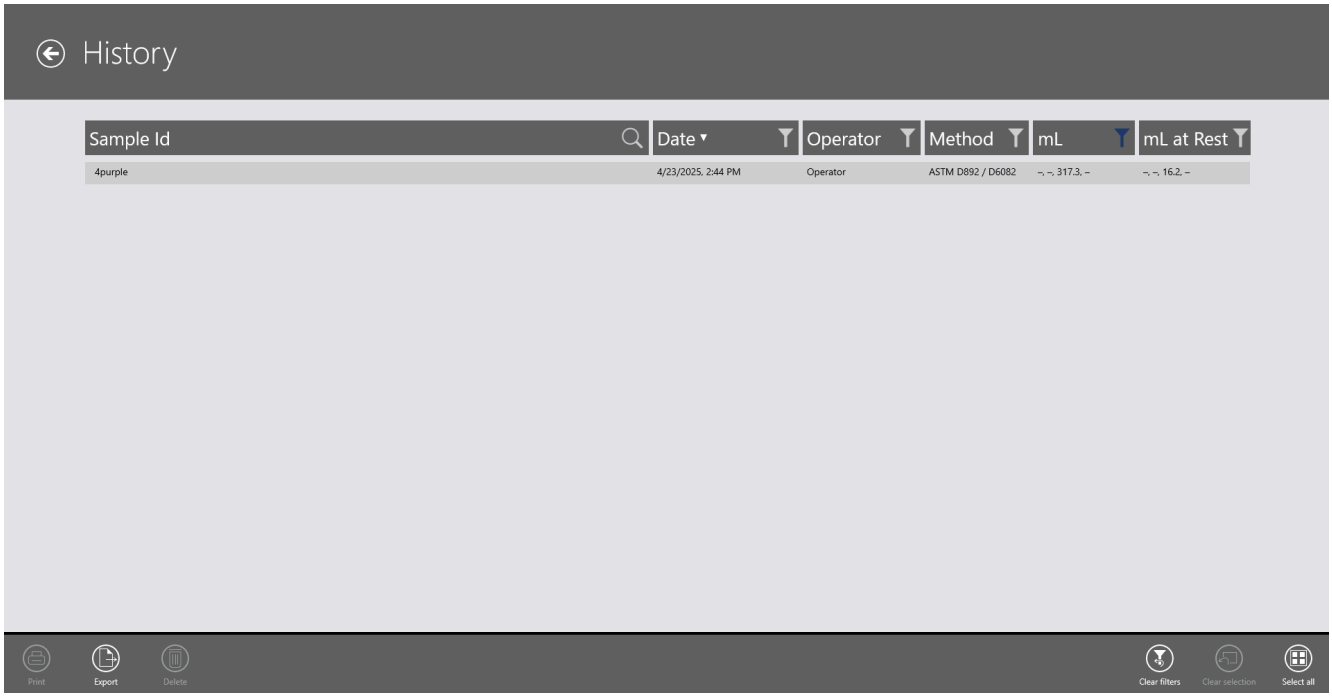


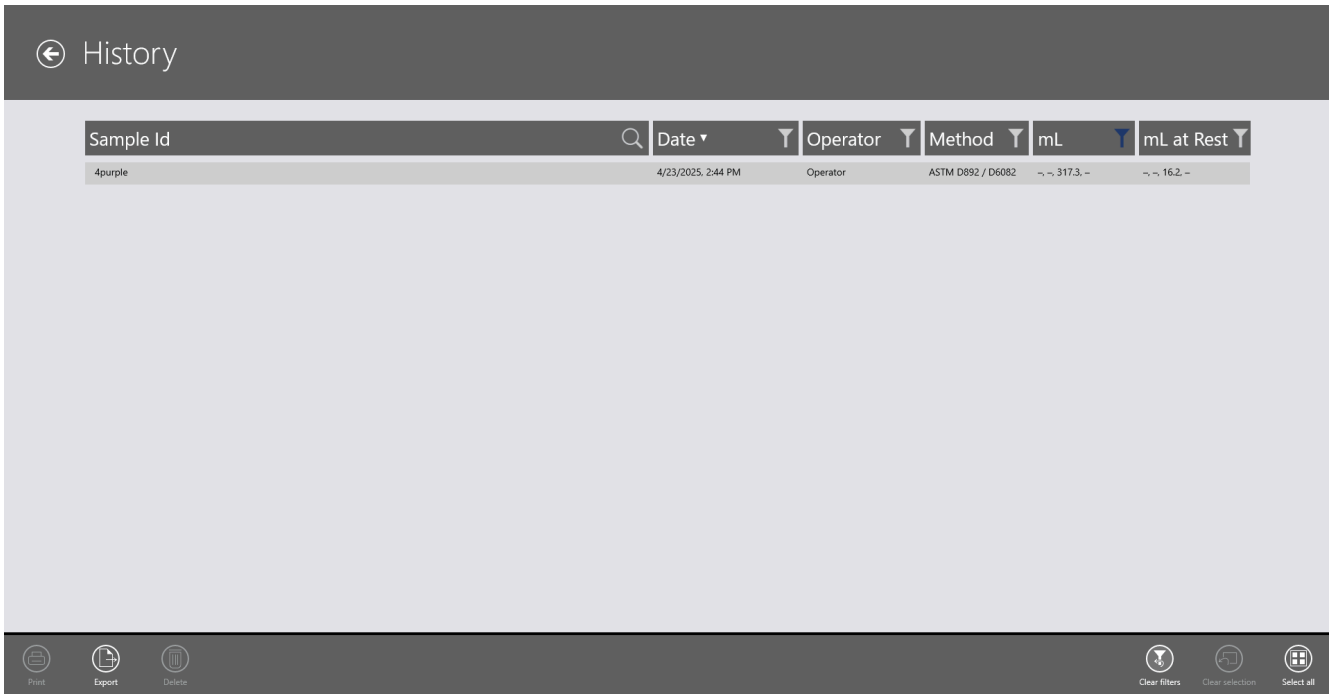
5.3.7. Filter Results

Results are archived in a secure database, organized by date, operator, and sample ID.

There are a variety of ways to navigate or filter results in the database. Multiple filters can be applied at once.

When filter are applied, the Filter icon will turn blue. To return to the unfiltered database, click Clear Filters at the bottom right hand corner of the screen.

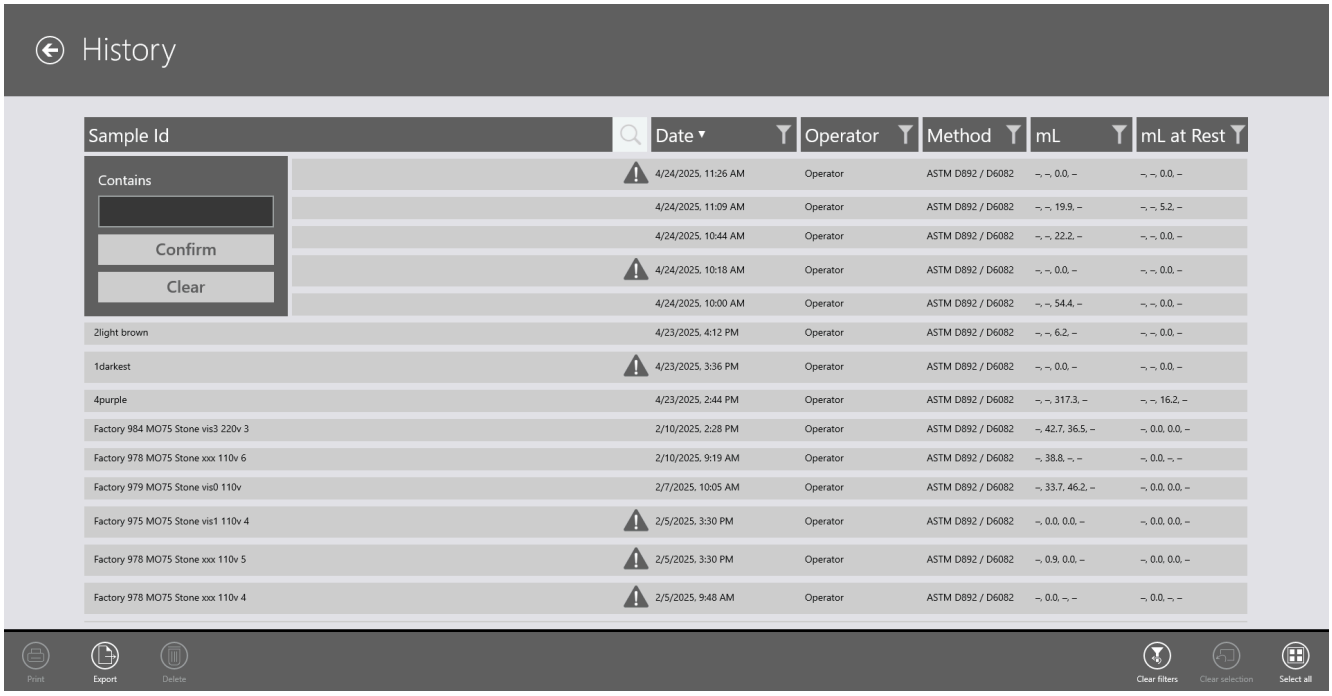


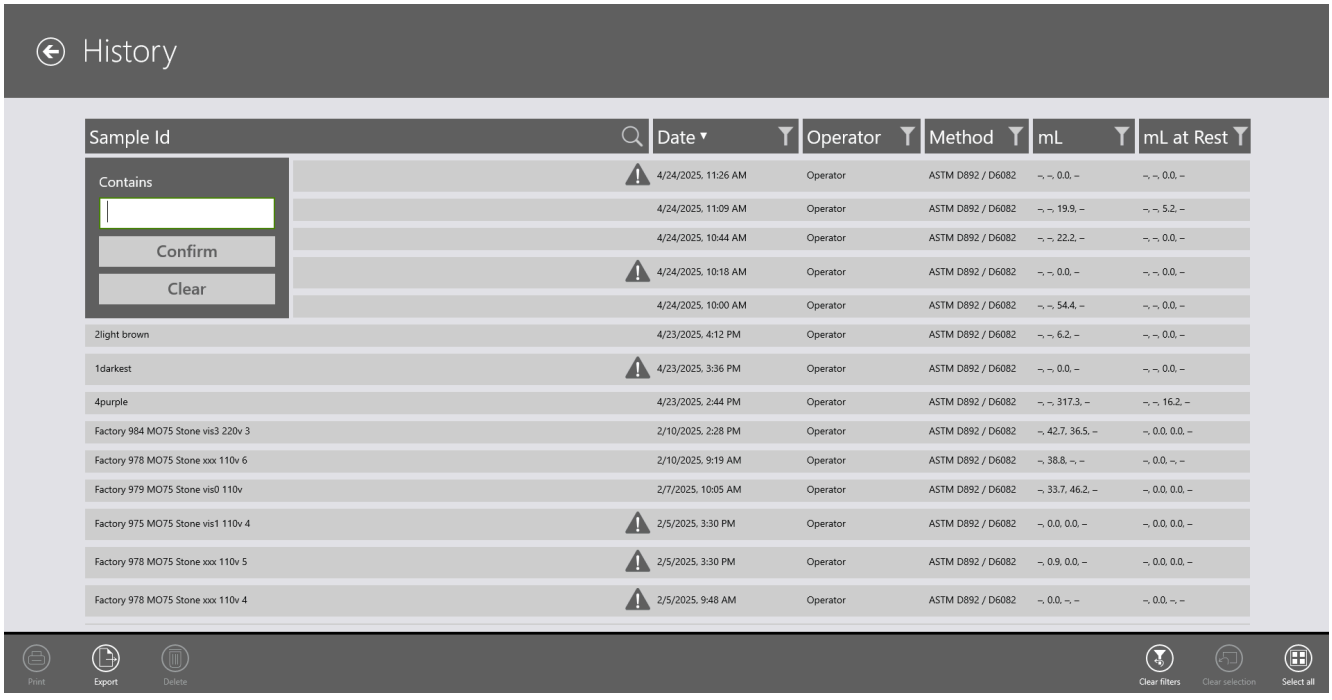


5.3.7.1. Filter by Sample Id

Clicking the Magnifying Glass icon in the Sample Id column will allow the user to search for specific parts of the sample name.

Any ASCII characters can be sued to search the database.





Click Confirm to search the database for the specific chgaracters and click Clear to return to the unfiltered database.

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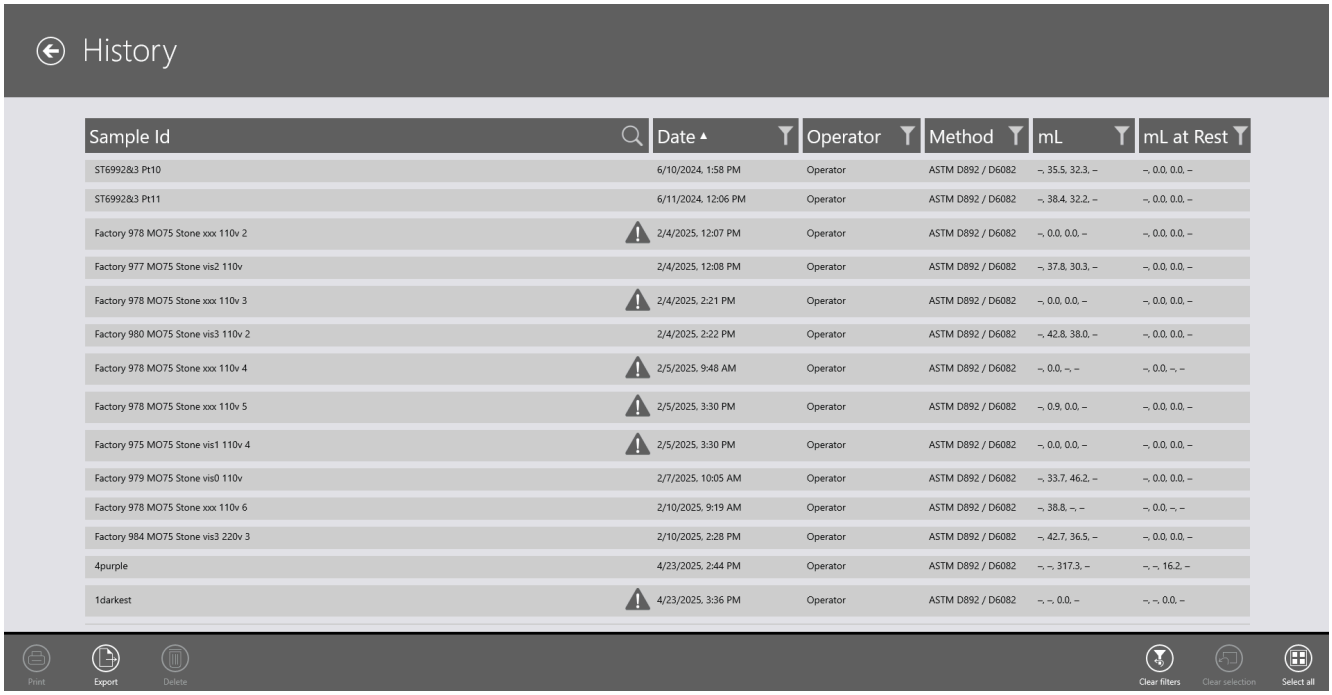
5.3.7.2. Filter by Date

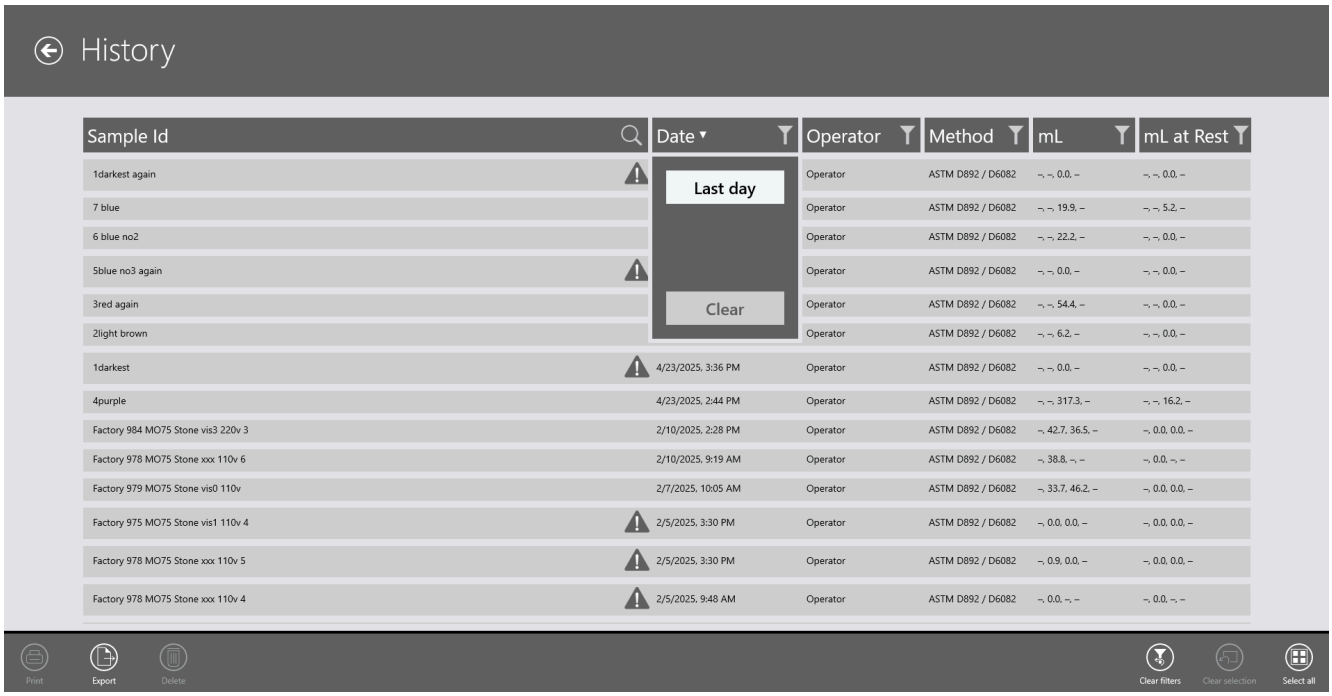
By default, the results are listed from most recent to oldest test. To reverse the order to oldest at the top, click the arrow directly to the right of Date.

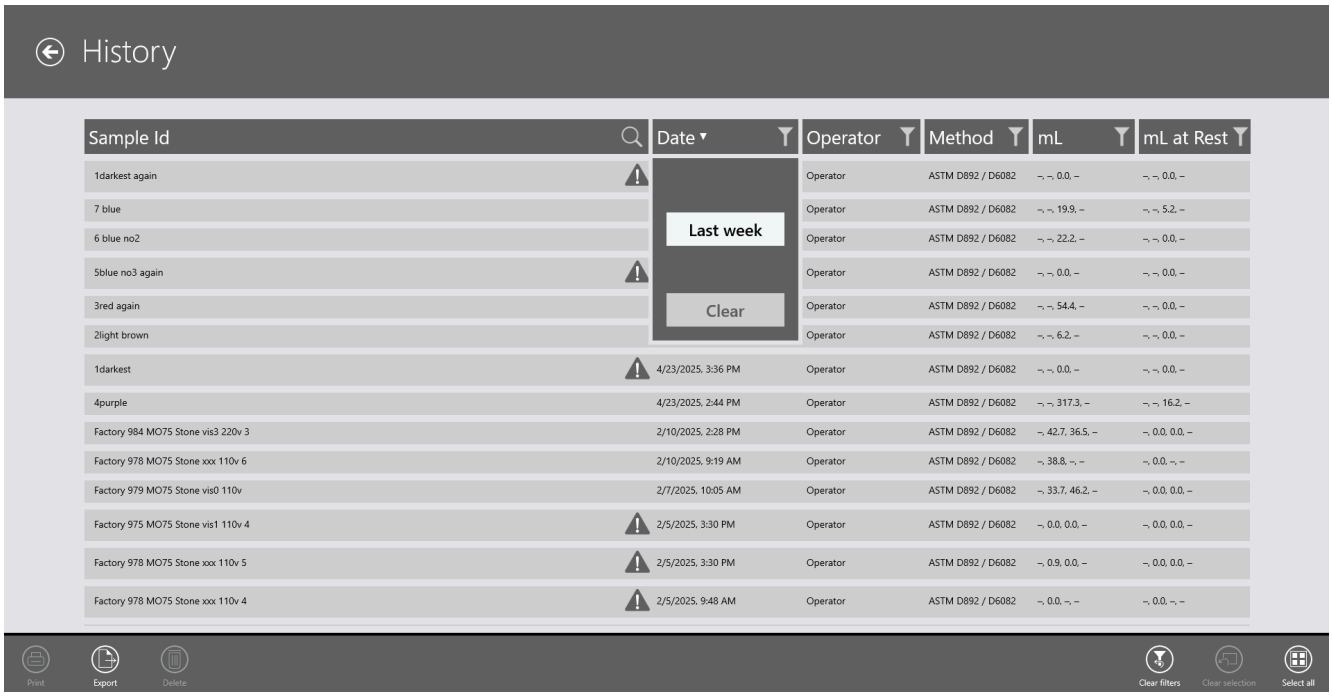
To filter more specifically by date, click the Filter icon on the right side of the Date column.

There are three options to choose from: Last Day, Last Week, or Specific Date.



The Last Day option will show all tests completed in the previous 24 hours. The Last Week option will display tests completed in the last 7 days. The specific Date option allows the user to search for results from a specific time period.





The results will automatically filter by the selection. Click Clear to return to the full unfiltered database.

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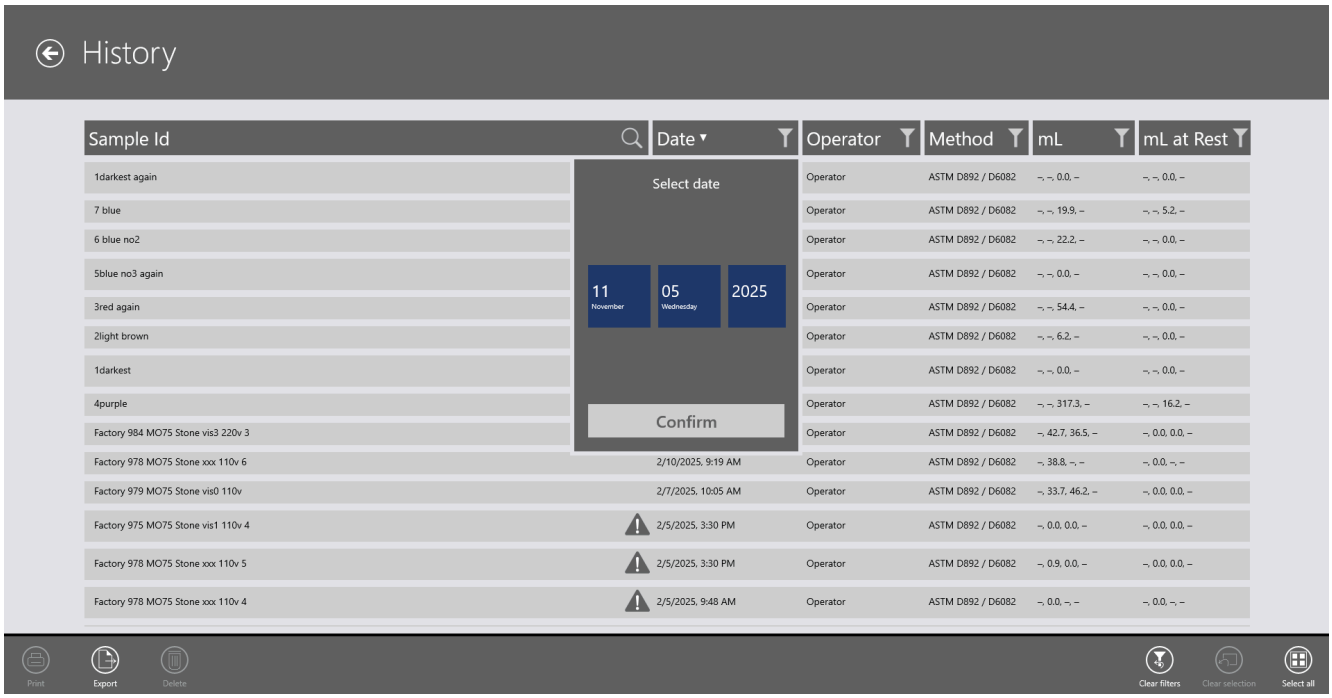
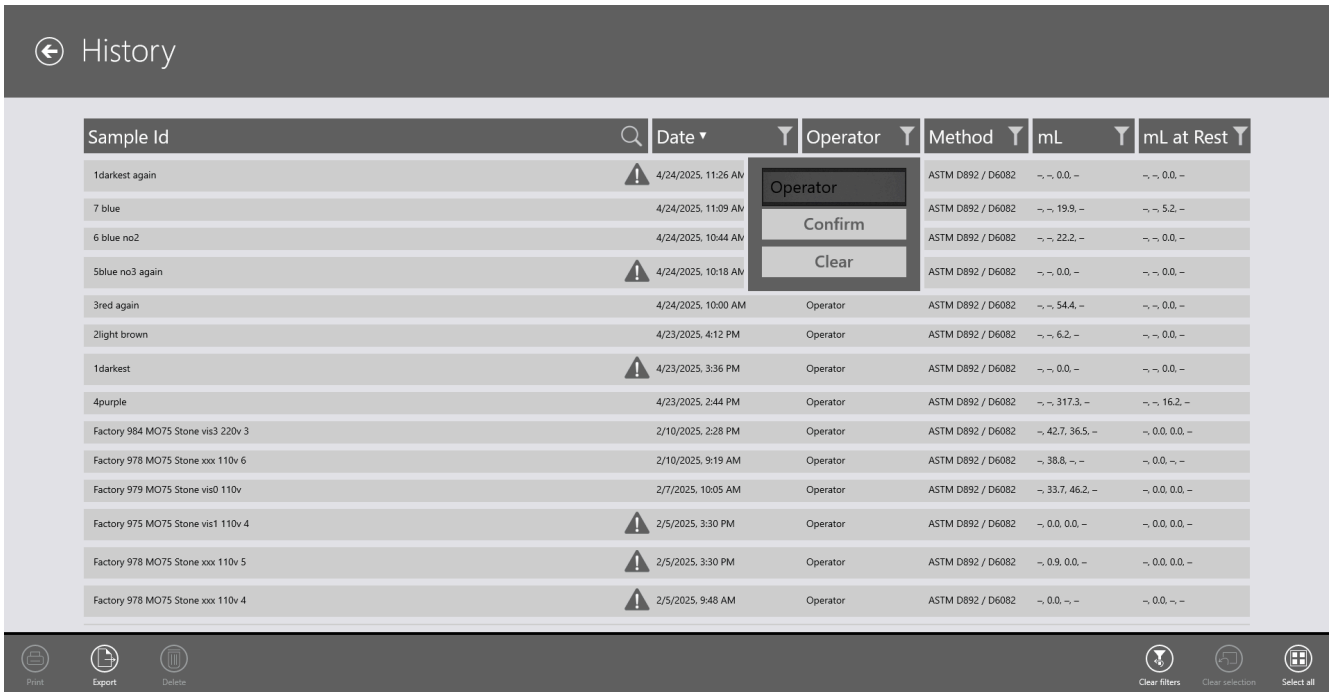


5.3.7.3. Filter by Operator

To filter by who conducted the test, click the Filter option. A dropdown menu will appear listing the different operators set up in settings. Choose the operator to filter by and click Confirm. To return to the unfiltered database, click Clear.



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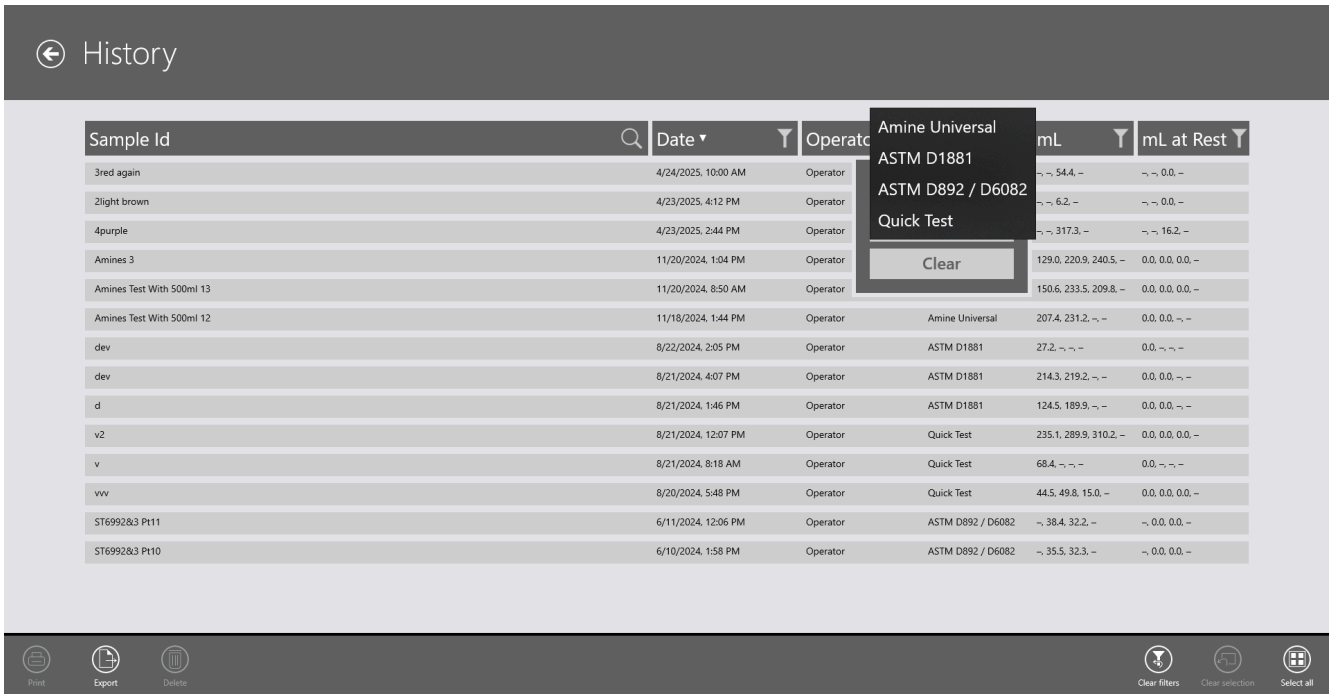
5.3.7.4. Filter by Method

To filter by the method, click the Filter icon in the Method column. Choose the method to filter by and click Confirm. To return to the unfiltered database, click Clear.



5.3.7.5. Filter by Results

There is also the option to filter by the results.

To filter by the Foam Tendency results, select the Filter icon in the mL column. The sequence options will display along with a slider for each. Drag the slider to the range in which the user would like to filter under the sequence(s) they would like to filter and click Confirm. To return to the unfiltered results, click Cancel.

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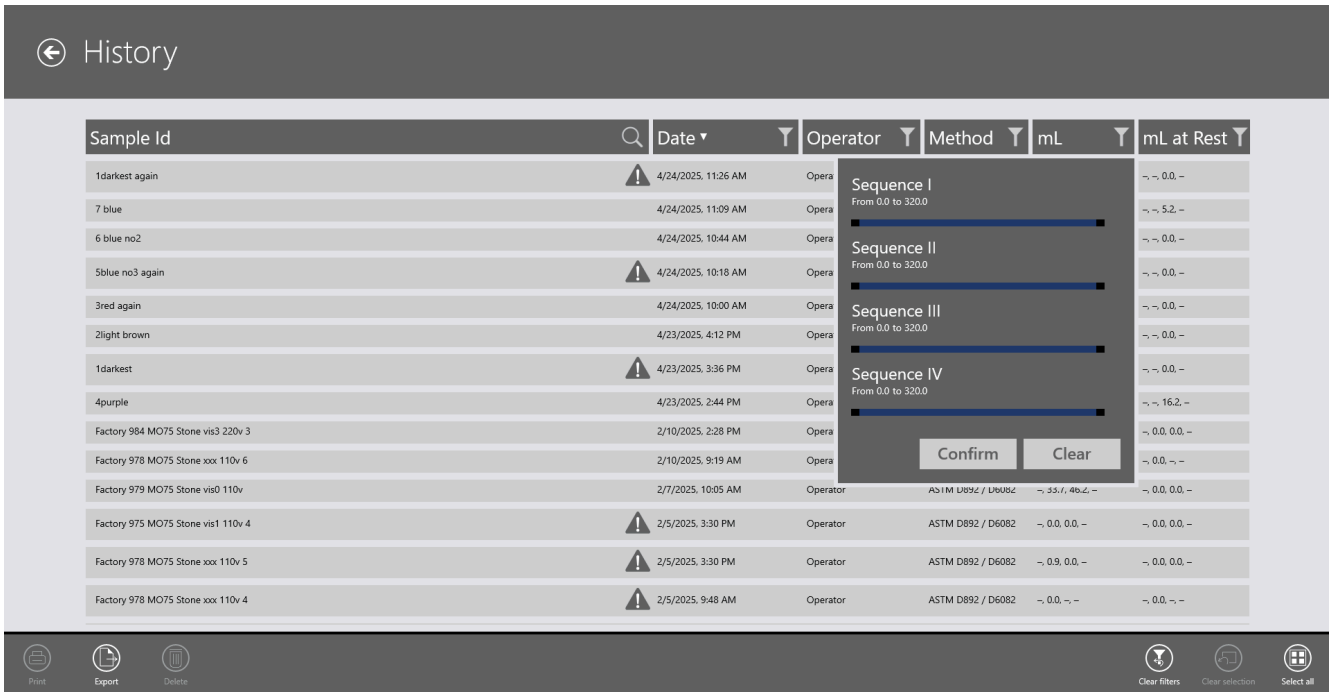

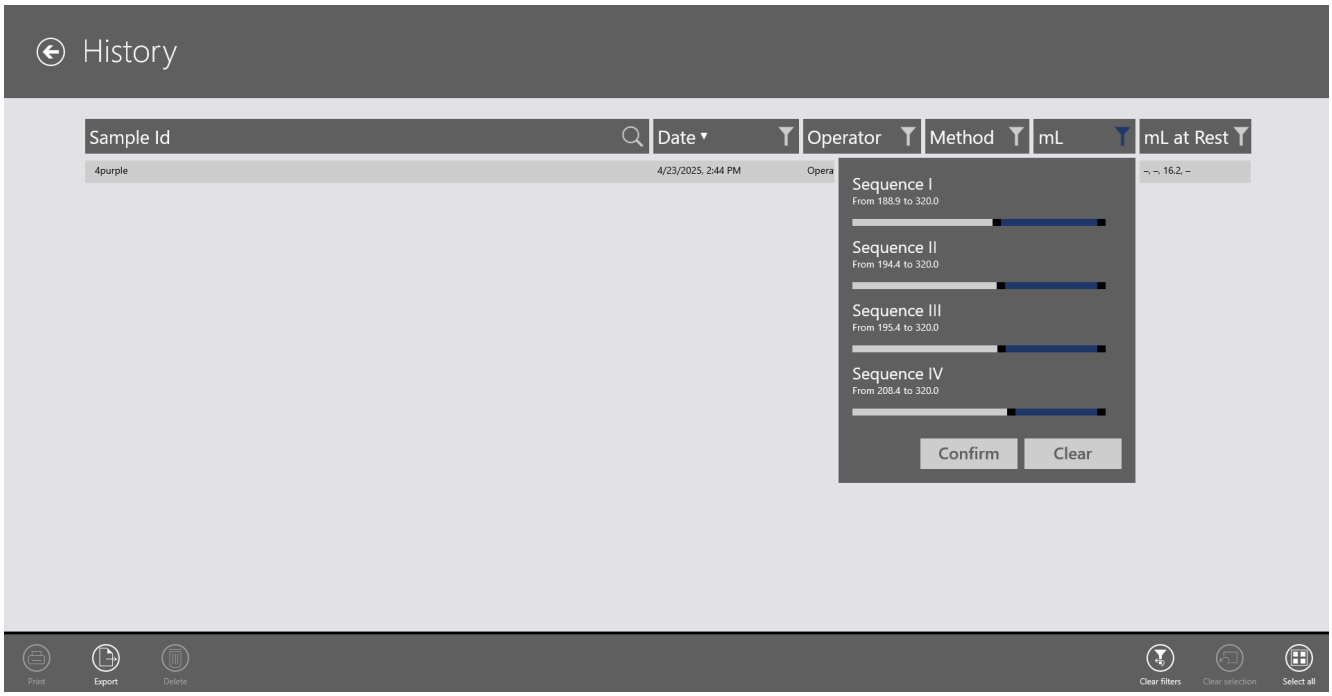

To filter by the Foam Stability results, select the Filter icon in the mL at Rest column. The sequence options will display along with a slider for each. Drag the slider to the range in which the user would like to filter under the sequence(s) they would like to filter and click Confirm. To return to the unfiltered results, click Cancel.

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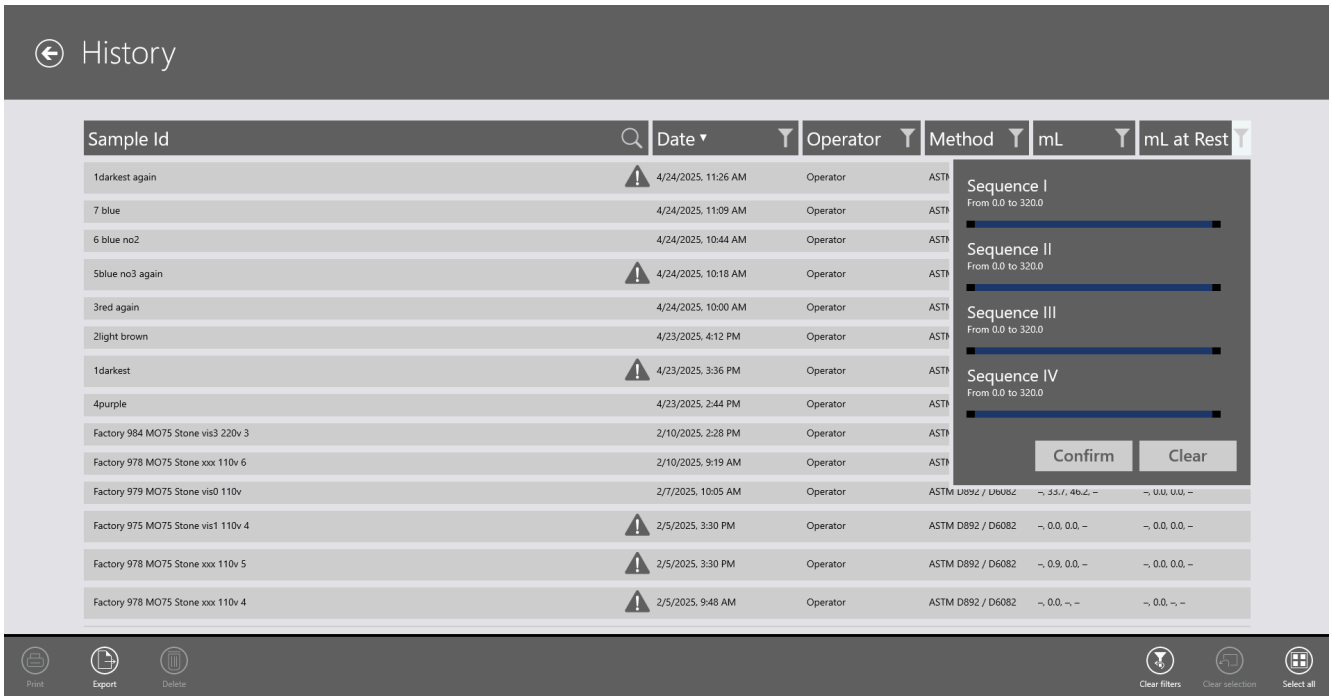
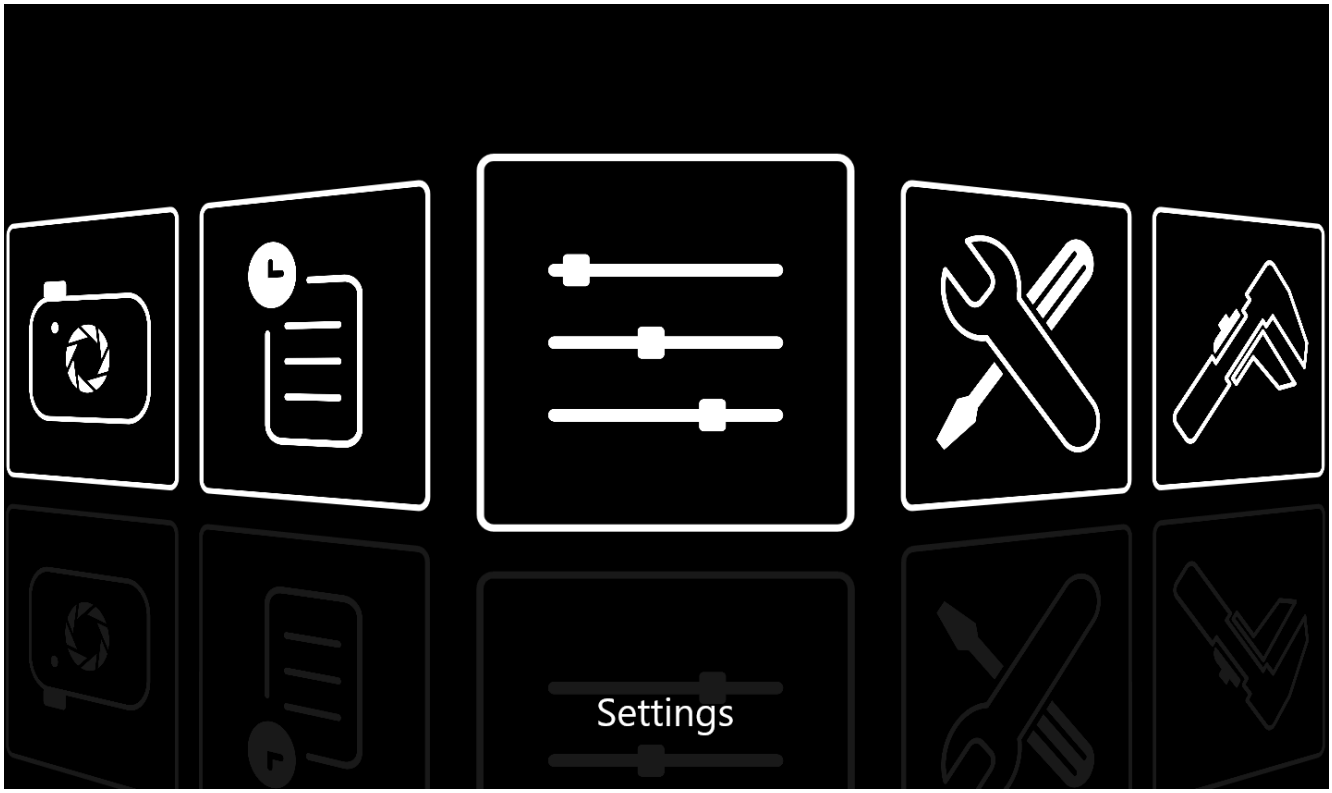


5.4. Settings

The Settings menu contains submenus for system customization.



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5.4.1. About

Displays software & firmware version, hardware information, and licensing details.

The About screen displays basic information about the FoamDDI analyzer. This includes:

Model Number: Displays model number and hardware version of the analyzer

Serial Number: Displays serial number, which also appears on the back of the analyzer

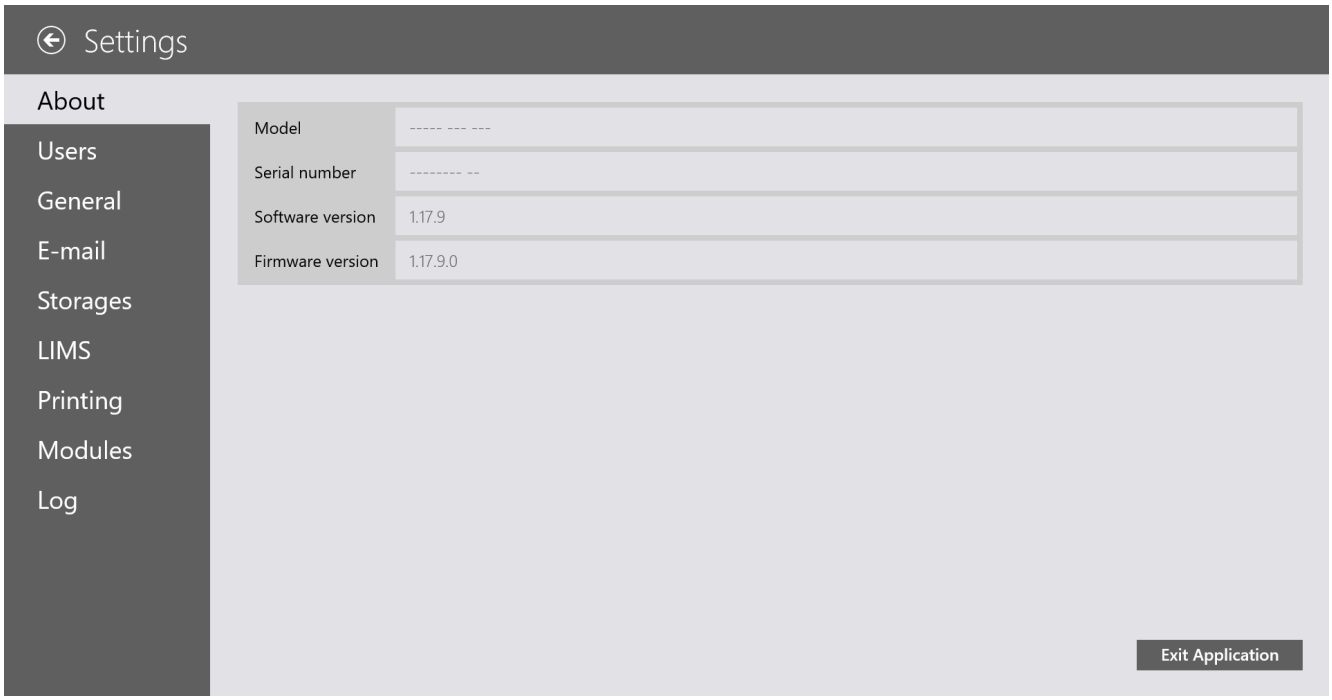
Software Version: Software version of GUII or Interface

Firmware Version: Hardware control software

The FoamDDI is operated in a protected mode called KIOSK or Assigned Access mode. This mode is critical to protect against changes to the software and operating system, as well as to prevent the ability to run malicious programs on the FoamDDI. This operating mode allows only the approved FoamDDI software to run.

It may be necessary to exit this mode to allow access to the administrator account to perform functions such as adding to the Domain, defining Network Printers, and IP Address functions. To exit the FoamDDI's operating mode, press the Exit Application button and allow the administrator to access a FoamDDI Admin account. Please contact VISAYA for more information on making changes and the password for the VISAYA Admin account.

Note: The Exit Application Mode button is only available when operating when no Admin mode is activated.



5.4.2. Users

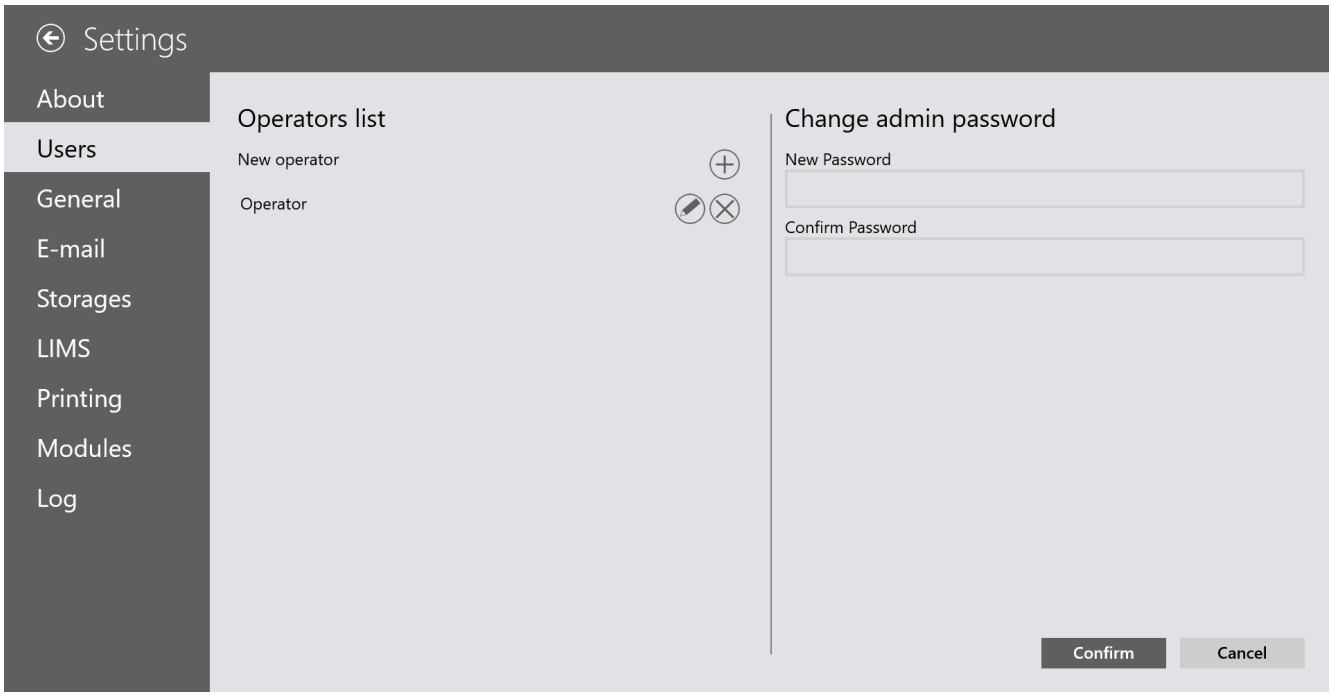
User accounts can be created with different access levels (Operator, Admin).

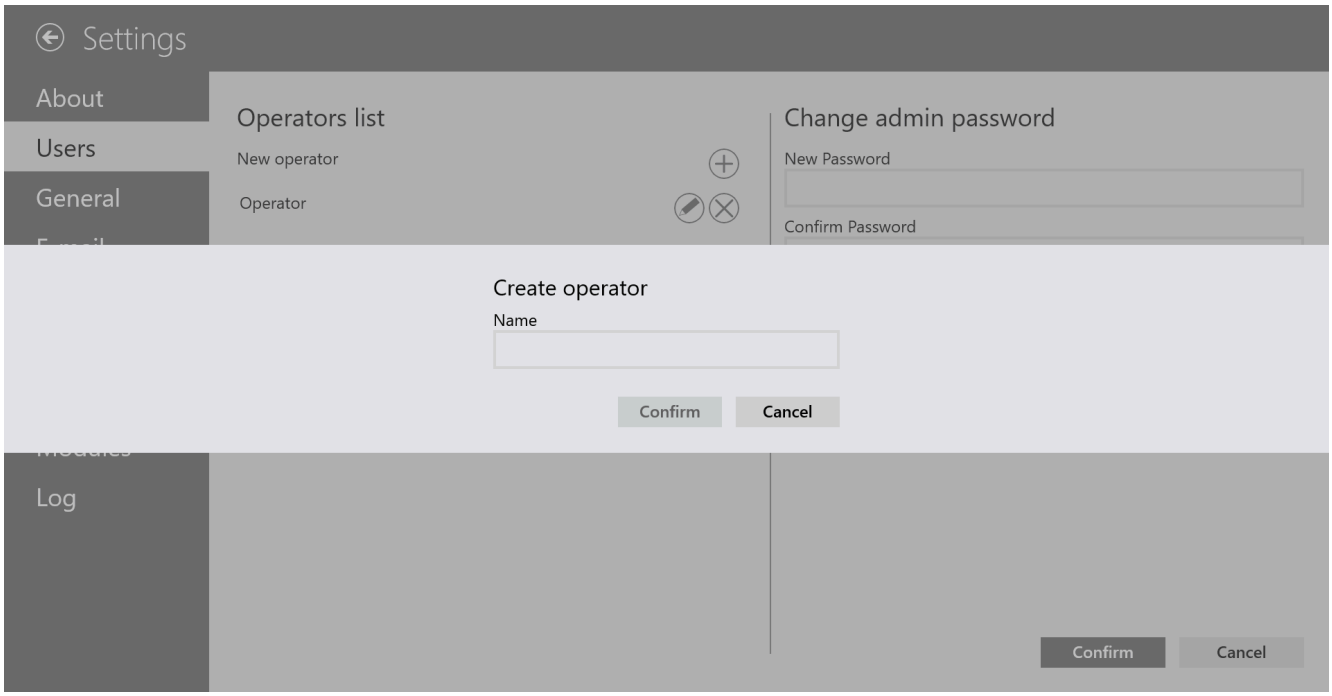
Admin Mode: Enables access to override options, calibration parameters, and advanced data export.

Operators, for use within the Inspect analysis screen and filtering options detailed in History, are defined in this section. To create a new operator click the + button next to New Operator. The user will be prompted to enter a new Name

Note: Name must be unique from other operators and can be alpha-numeric

Click Confirm to confirm a new entry and the new user name will be displayed in the Operators List.





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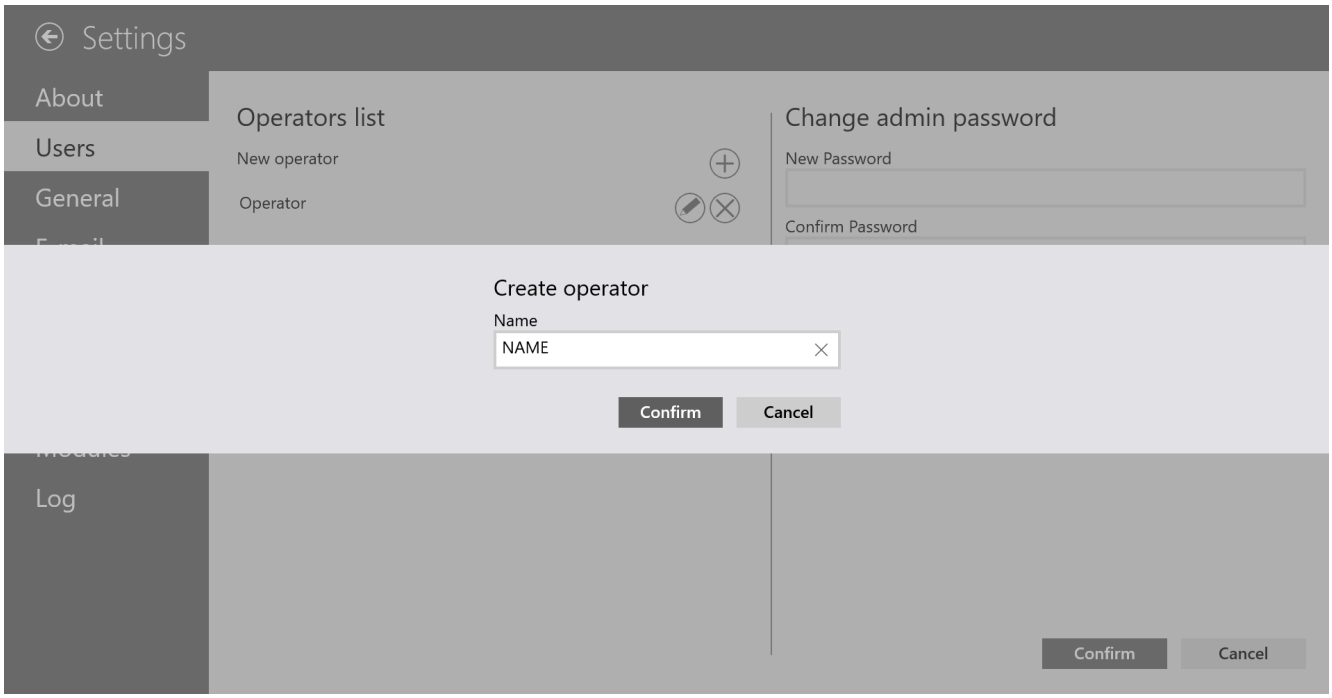

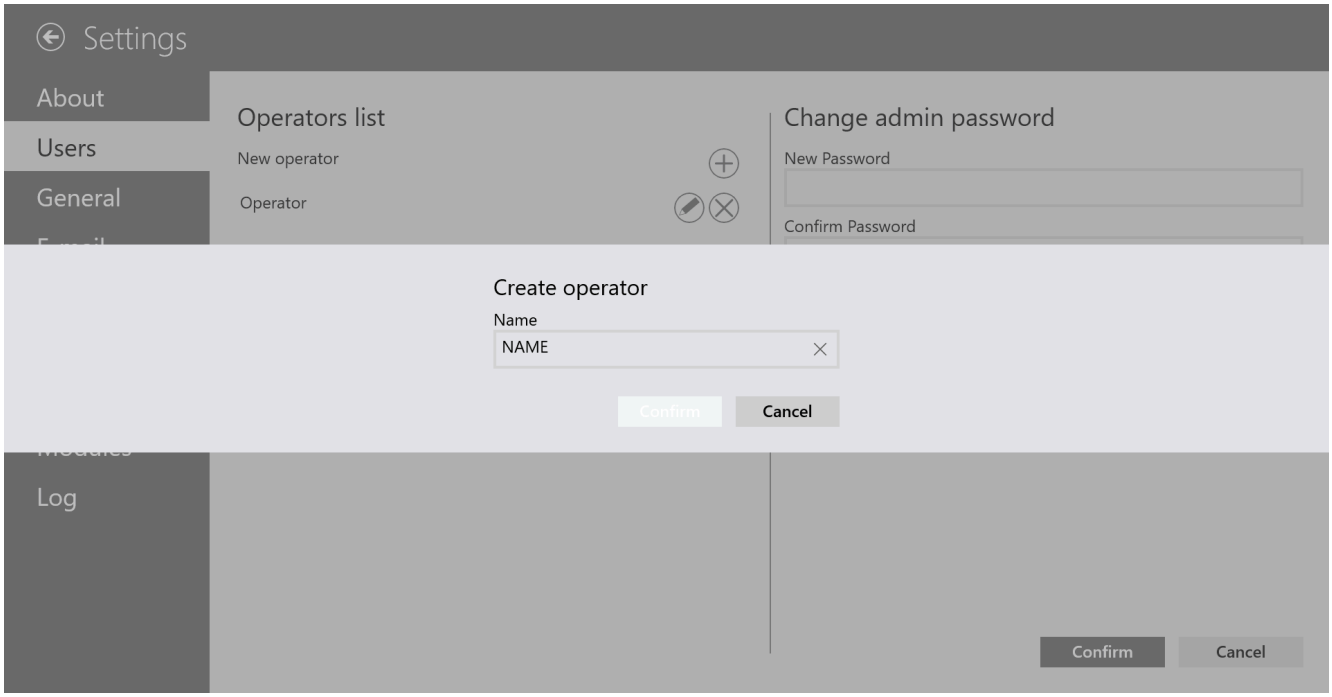

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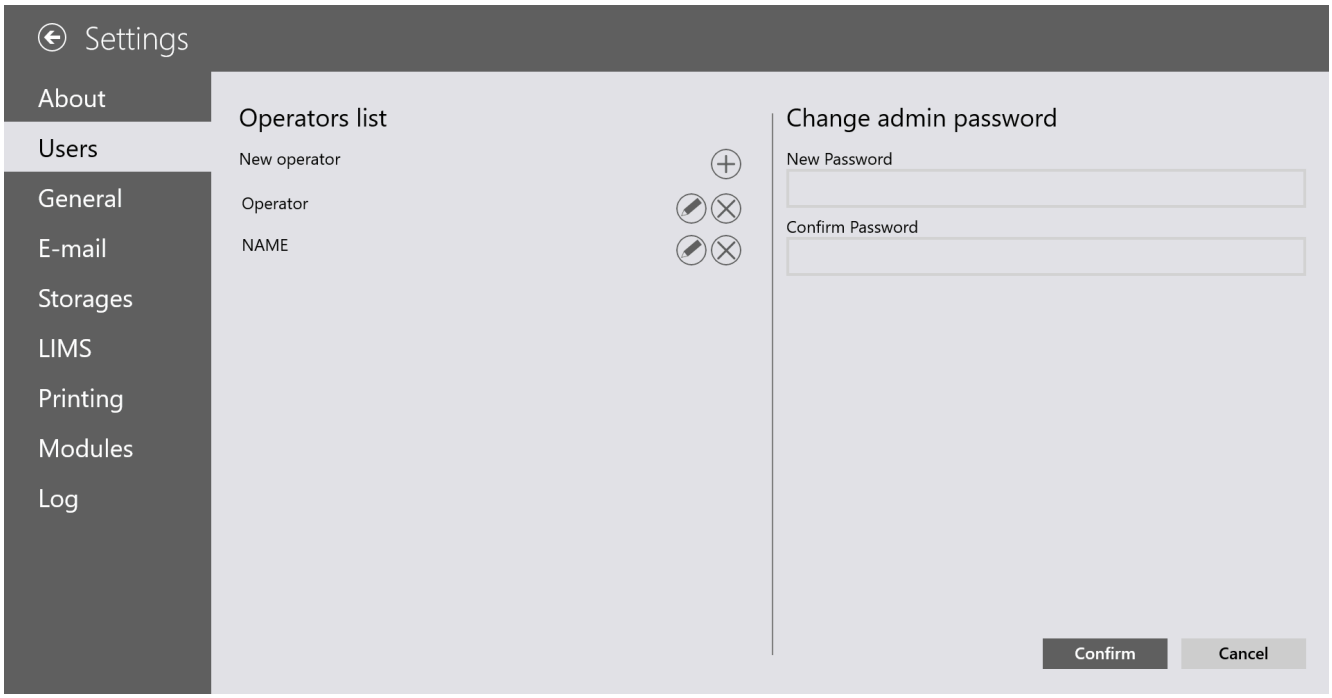
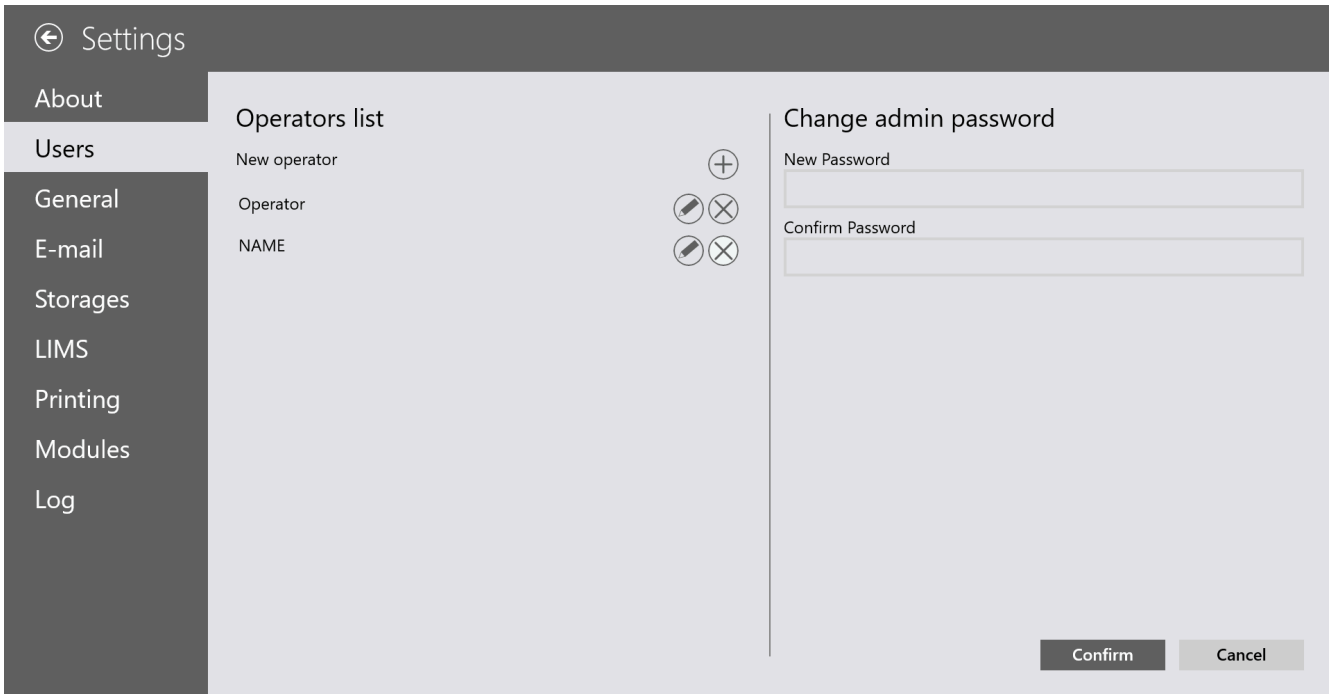


Once created, an Operator Name can be edited and deleted. To edit and delete a user, click on the Edit Icon next to the Operator name to edit or click on the Delete Icon next to the Operator name to delete.

Note: It is not necessary to confirm these changes using the Confirm button. All edits are permanent.



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5.4.2.1. Admin Mode

The FoamDDI's software provides an easy-to-use Admin mode to prevent access to critical parts of the software. Any critical functions can be locked with a prompt for a password to allow access and/or changes. To activate this function, simply type a password in the Change Admin fields. Password must be entered a second time to confirm entry. To activate this function, it is necessary to confirm using the Confirm button at the bottom of the screen. Alternately, press cancel to start all over with a new password.

Note: Users versus admin will still have the ability to perform a Calibration Verification procedure.

Once this feature is activated, any attempt to enter critical parts of the software will be prompted with a request to enter a password. The operator can enter the Admin password to enter, or cancel to exit the screen. Admin Mode prevents access to the following screens:

Entering the Settings screen.

Advanced Calibration screen.

Admin Mode provides an easy-to-use protection mode without the need for complex usernames and passwords, with various user rights and access levels.

To remove Admin Mode, simply:

Enter Settings Page

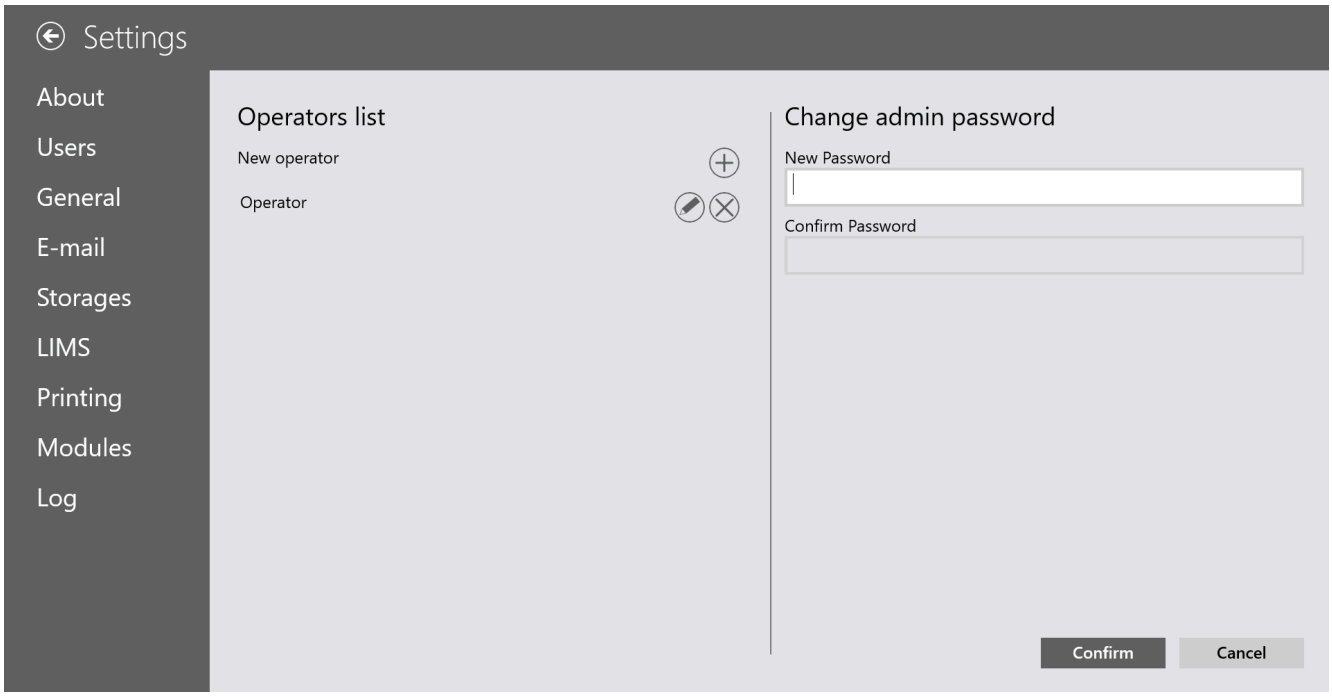
Enter Admin Password when prompted

Enter the Users section and

Erase password from both "New Password" & "Confirm Password" fields

Press the "Confirm" button to accept the removal of admin mode

Note: If the Admin password is lost, please contact VISAYA Service for the master password to regain access to the system.



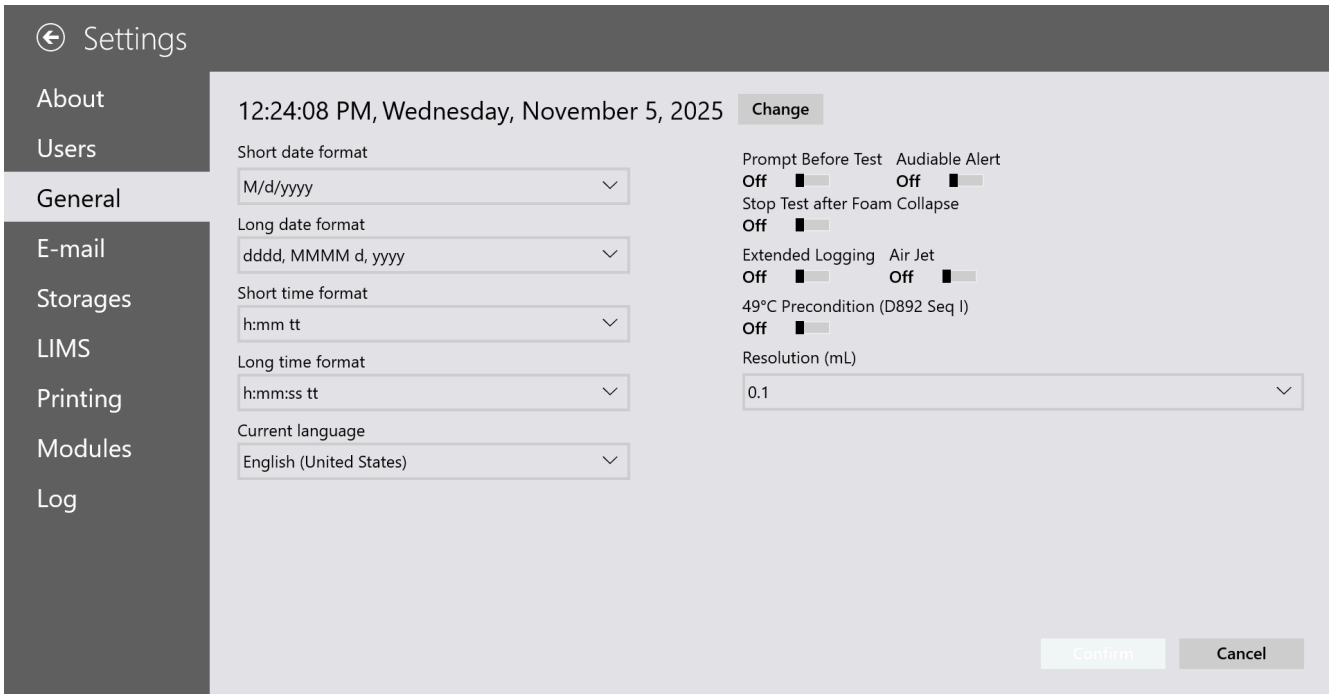
5.4.3. General

The General section of the *Settings* page is used to view and adjust key system parameters such as the current time, date format, and display language.

Additional options allow the user to enable or disable certain features that influence test behavior.

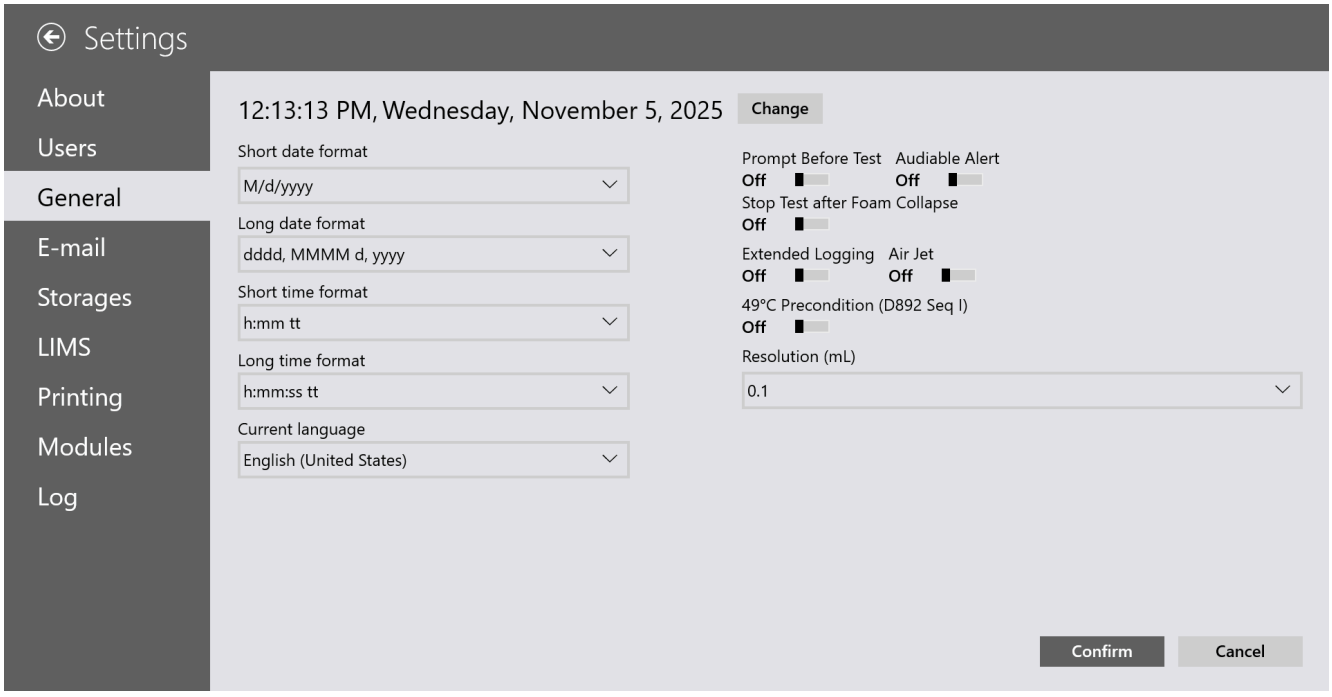
Changes made within this section are only applied after the user confirms the settings. These adjustments affect future tests only — historical results found under *History* remain unchanged.

To save changes, click Confirm at the bottom right of the screen before navigating elsewhere. To ensure changes are NOT saved, click Cancel before navigating elsewhere.

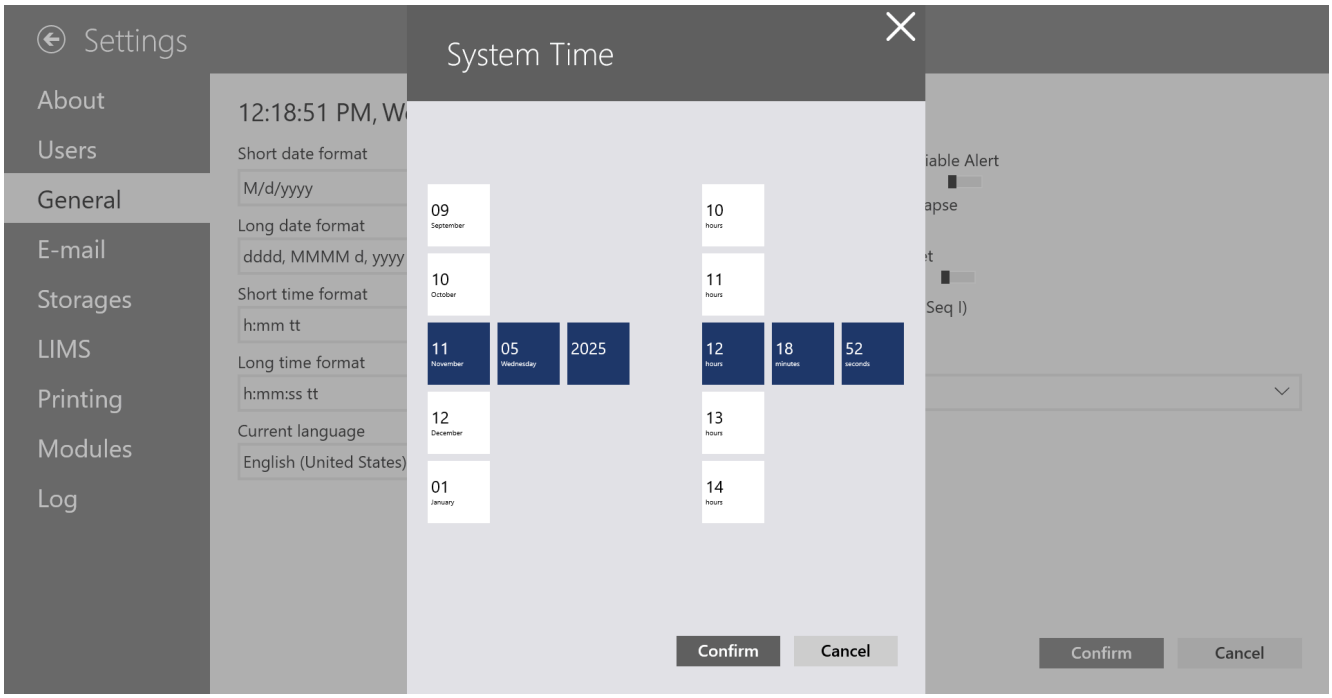


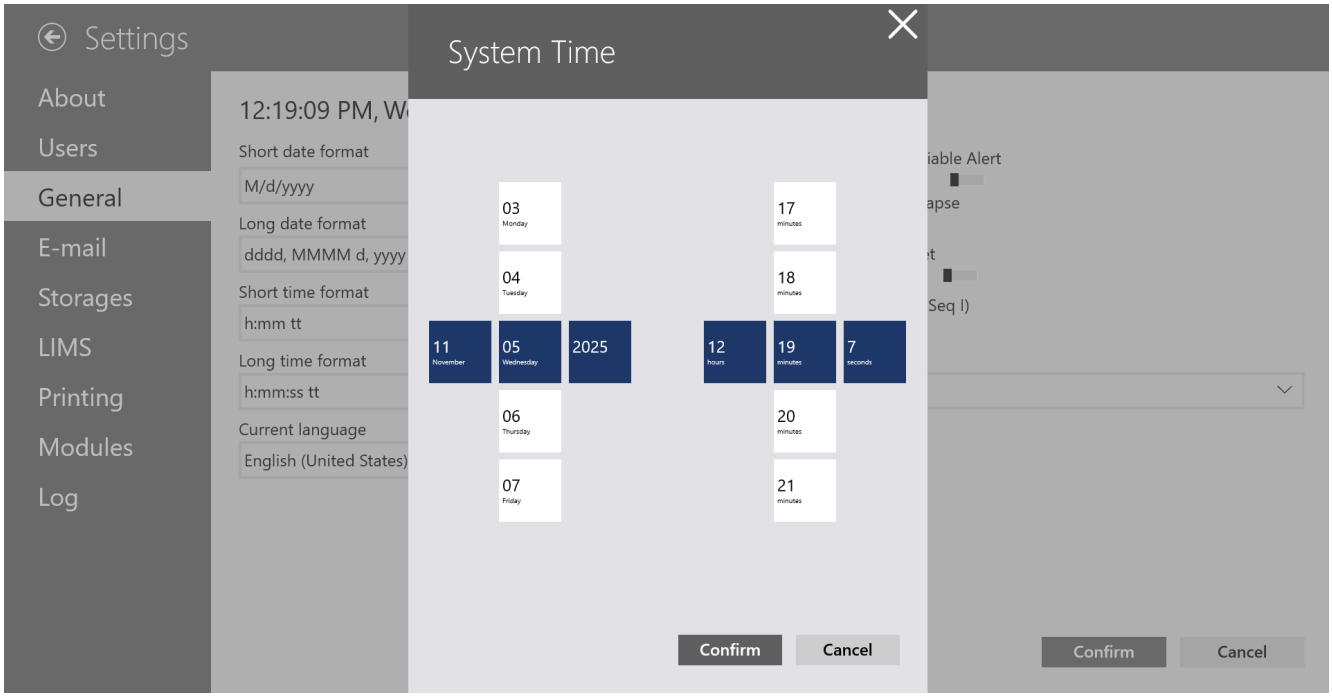
5.4.3.1. System Date and Time Setup

The date and time can easily be changed to best accommodate any lab anywhere.



To change the analyzer's Date and Time, click Change at the top of the screen, next to the current listed day and time. Follow the prompts to choose the correct day, time, and year. This is the date and time the unit will use to record when the analyses take place.





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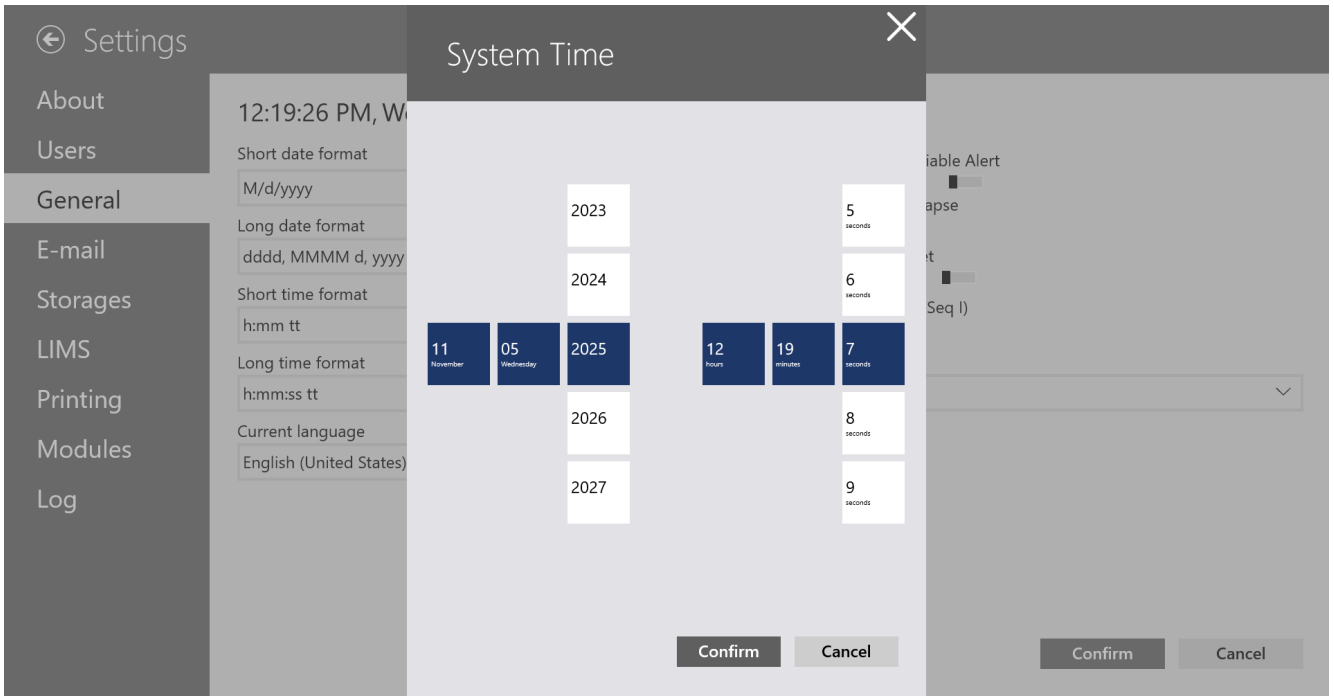

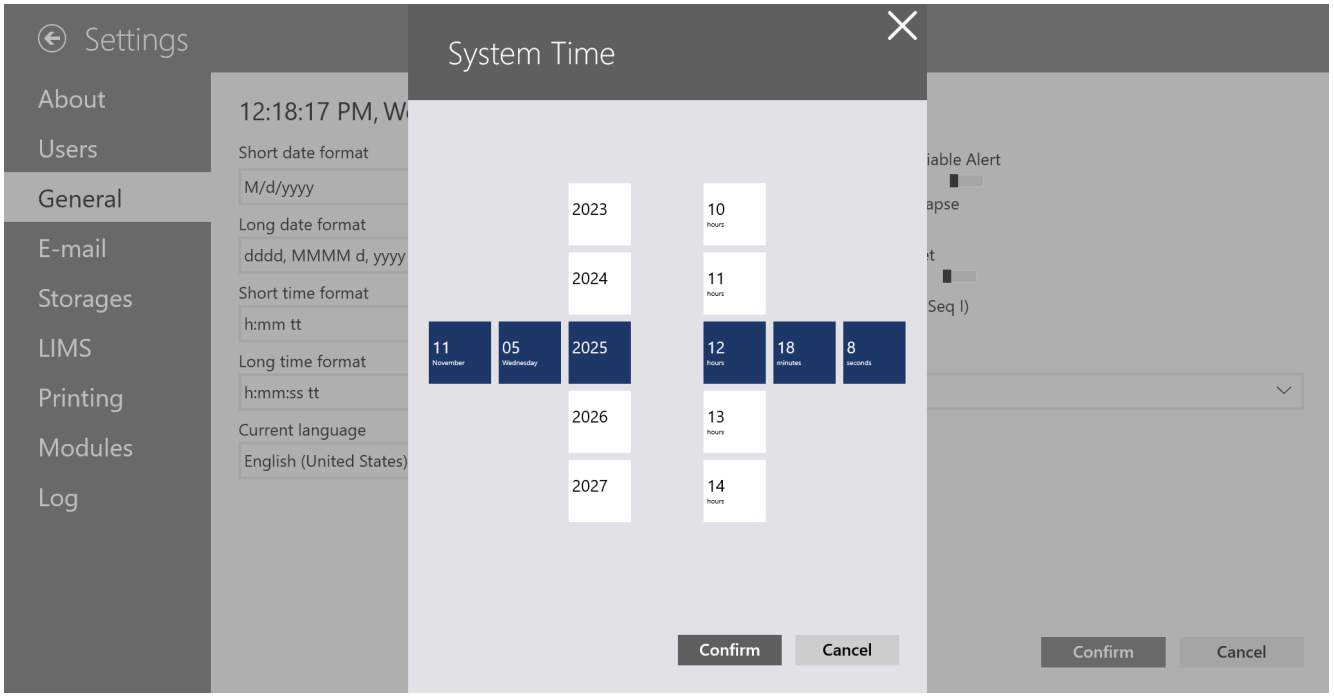

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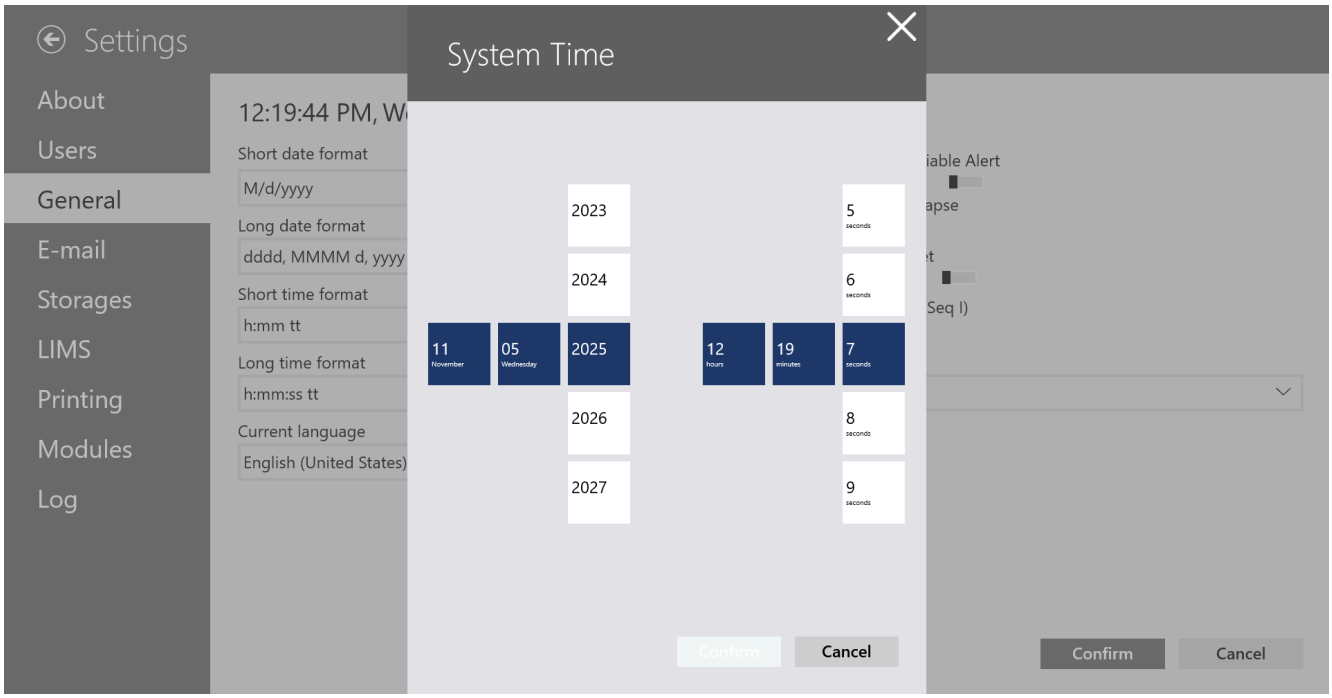

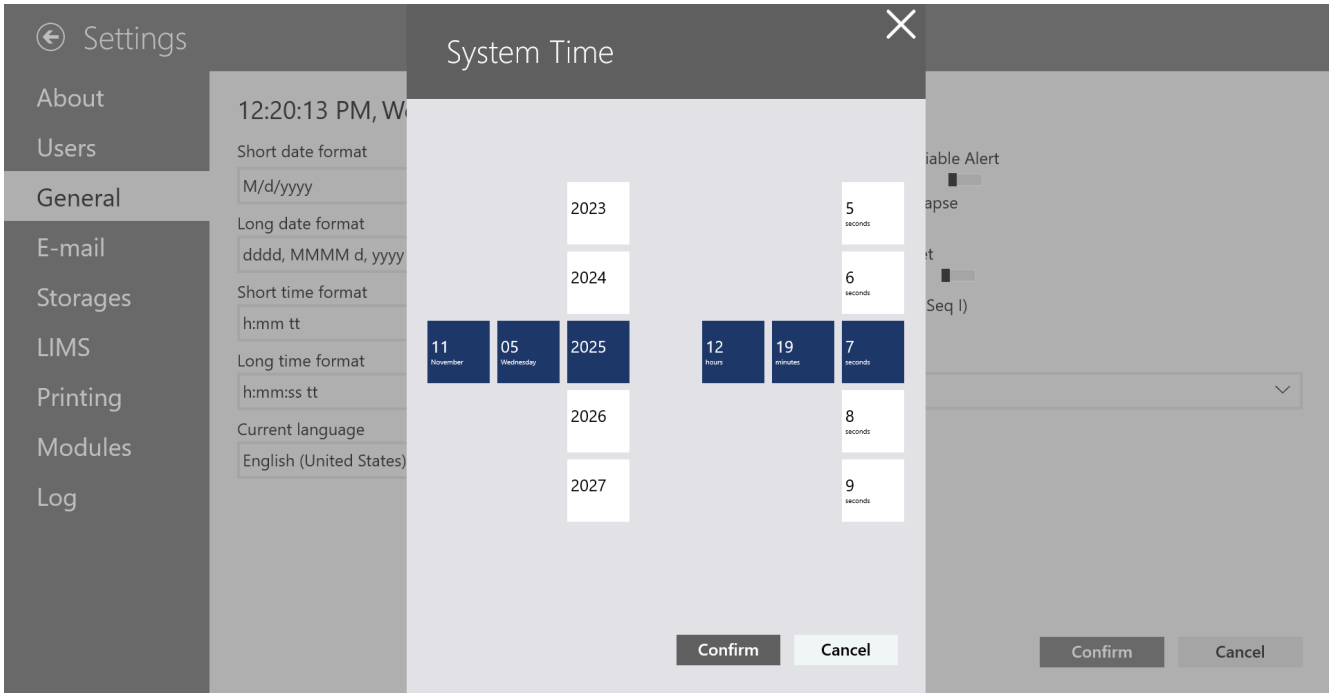

5.4.3.2. Date and Time Reporting Options

The Date and Time reporting options are categorized into two subsets:

Short Date/Time Format – Defines how date and time appear in reports and list views within *History*.

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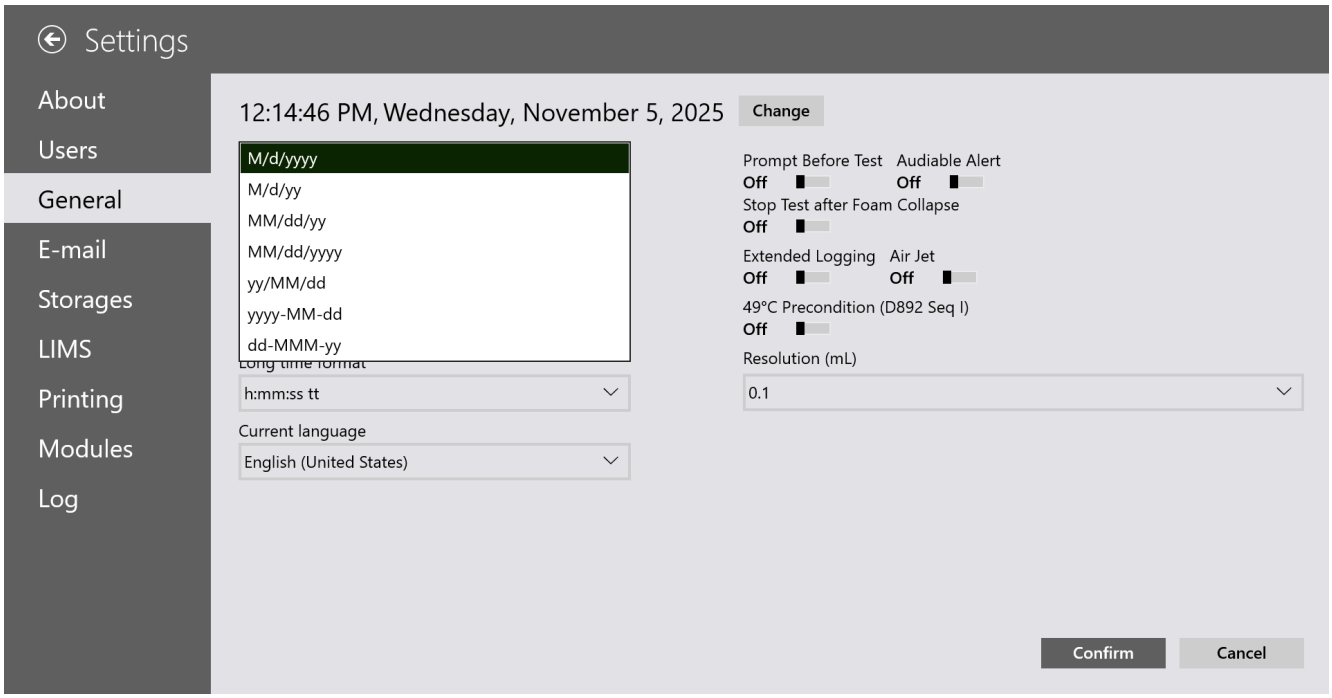
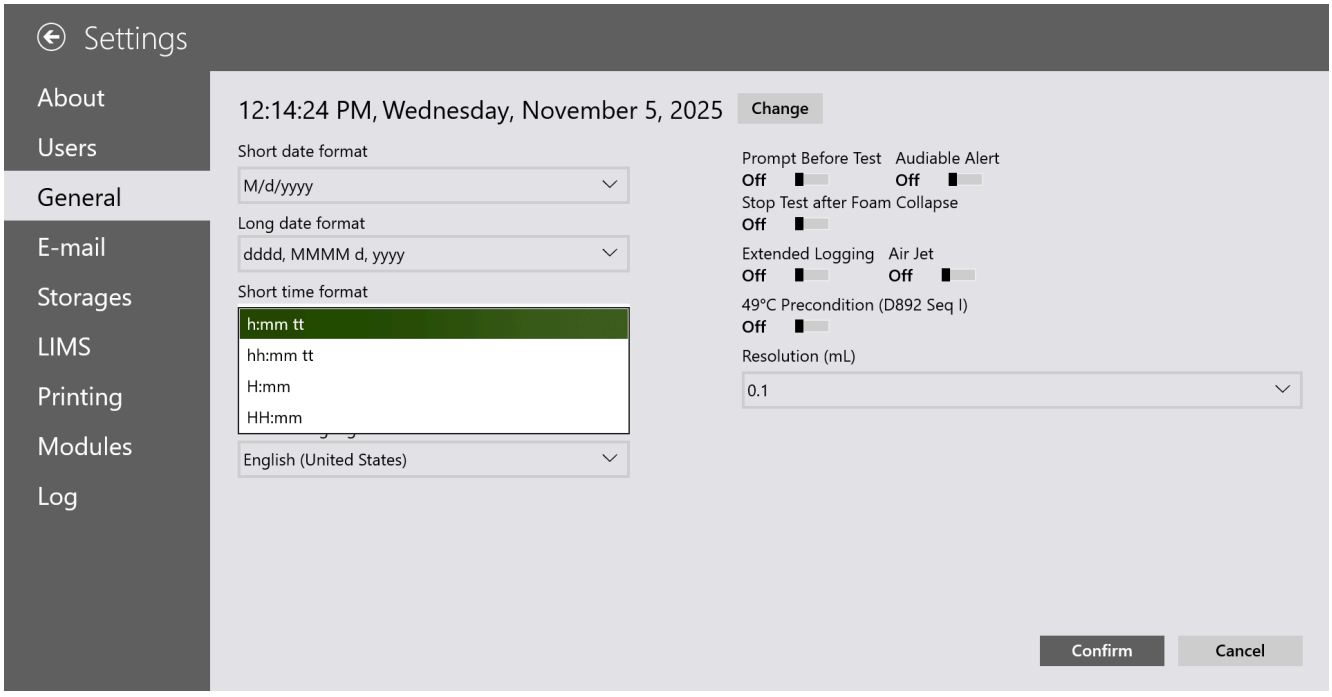
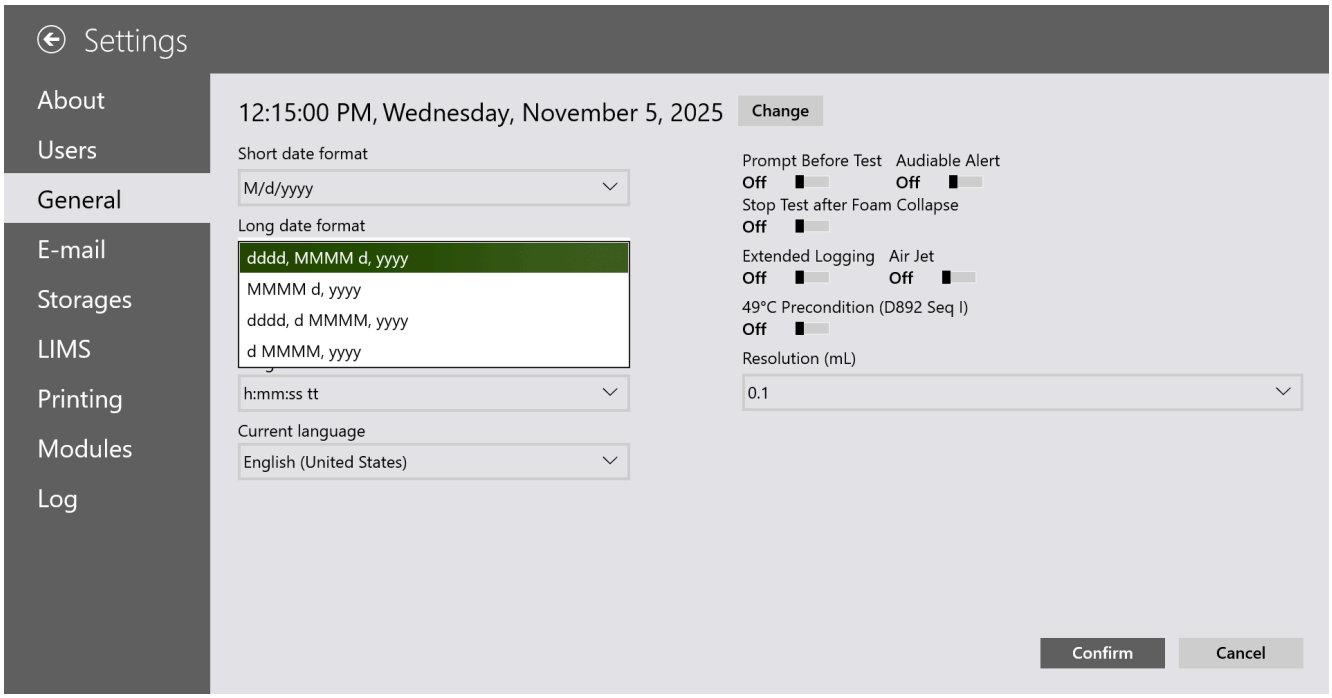
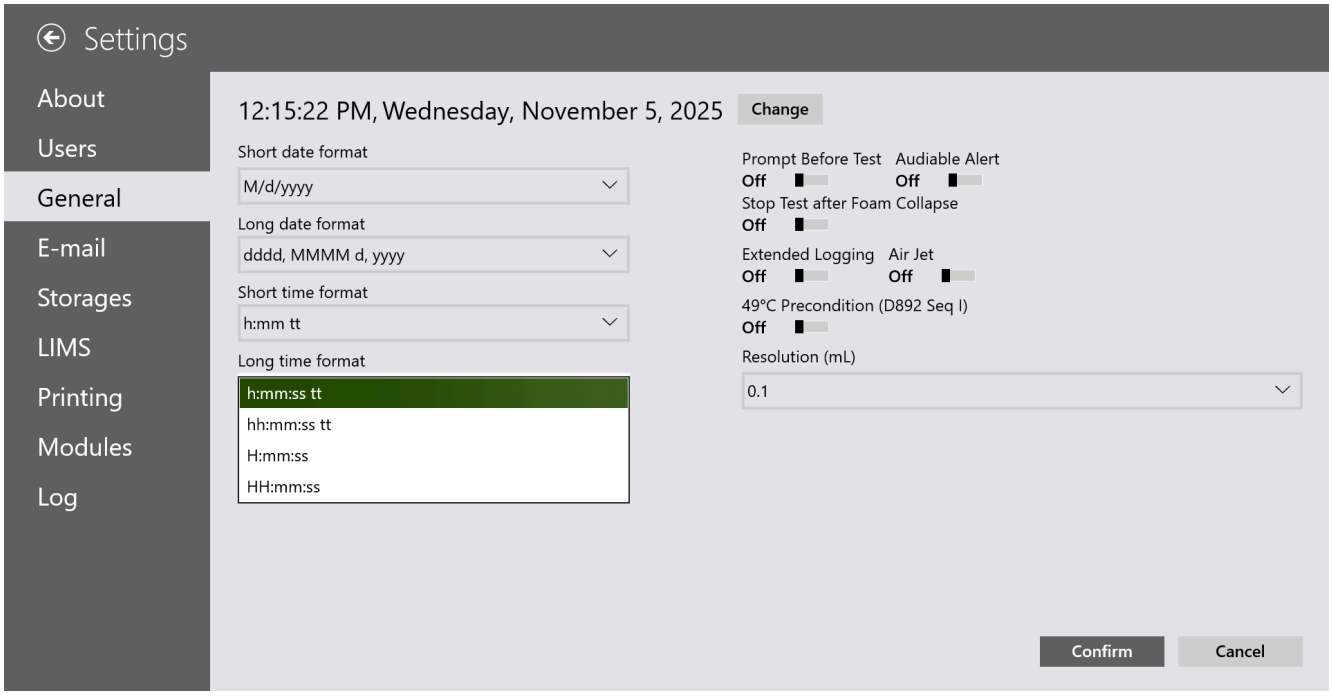




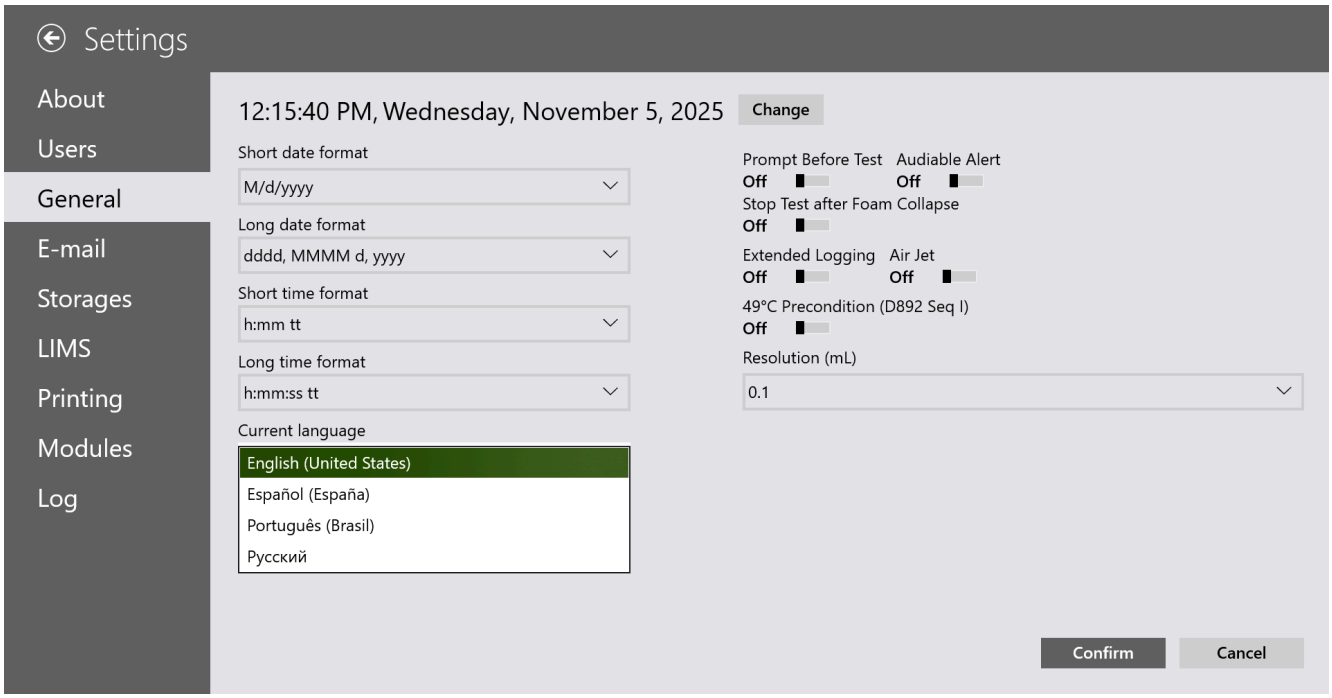
Long Date/Time Format – Defines how date and time appear in LIMS and database entries.





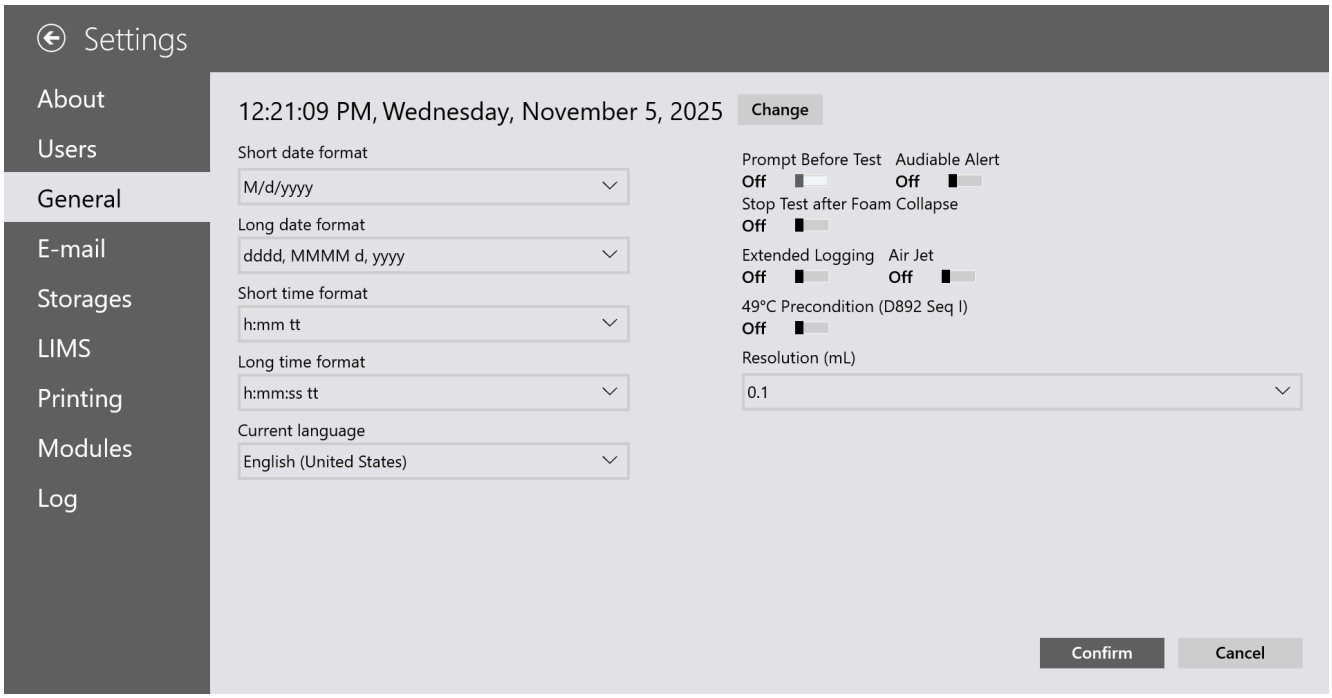
5.4.3.3. Current Language

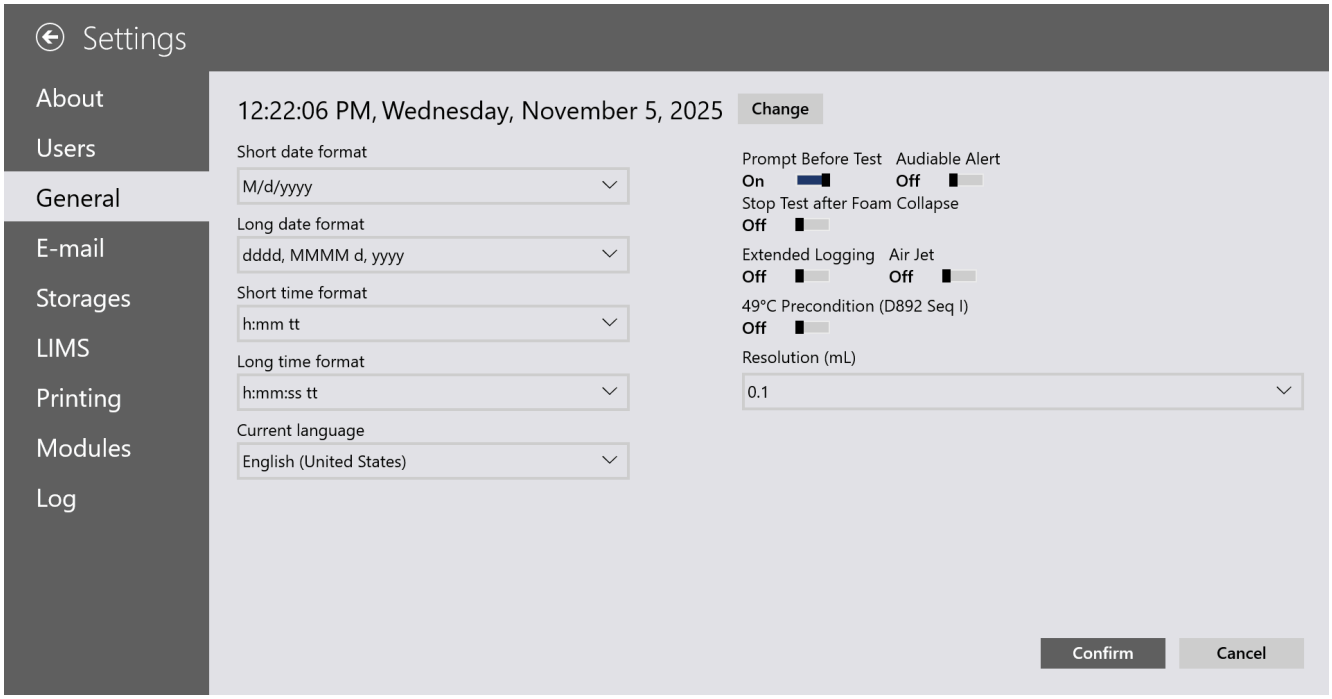
The FoamDDI interface supports multiple display languages. To change the language, select the desired option from the dropdown menu. The software will automatically restart within a few seconds, reloading in the newly selected language.



5.4.3.4. Semi Automatic Mode: Prompt Before Test

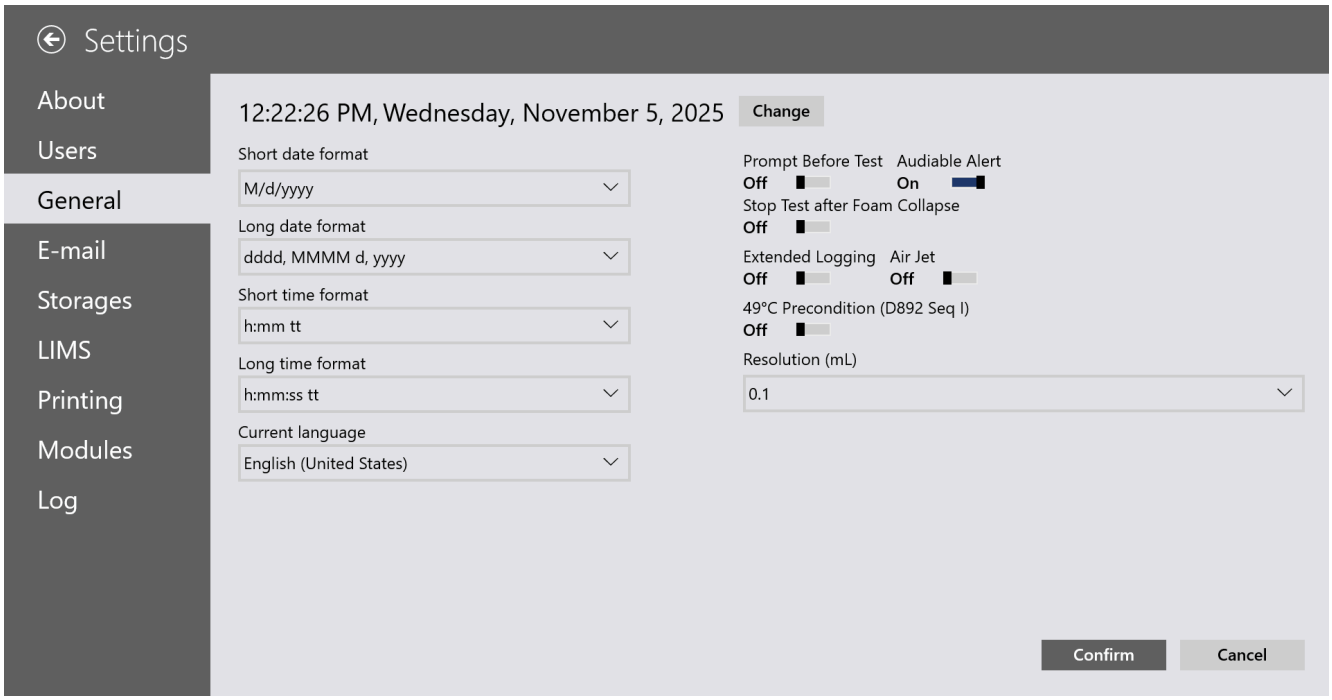
When enabled, the FoamDDI automatically prepares the sample temperature (24 °C or 93.5 °C) and then pauses before proceeding to the analysis. This feature ensures that an operator is present to observe and record *Foam Tendency*. An audible alert signals when the system is ready, while temperature is maintained at the target value.





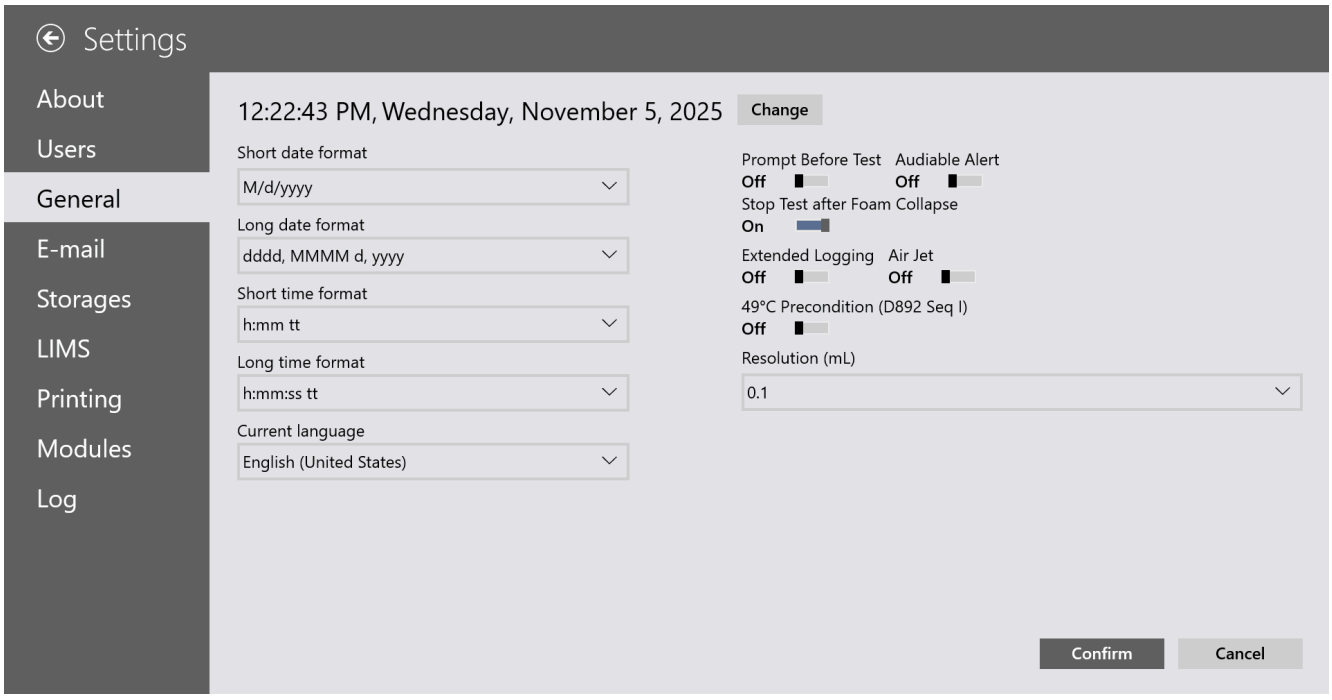
5.4.3.5. Audible Alert

Activates an audible signal from the module to notify the operator of key stages during Foam Analysis.



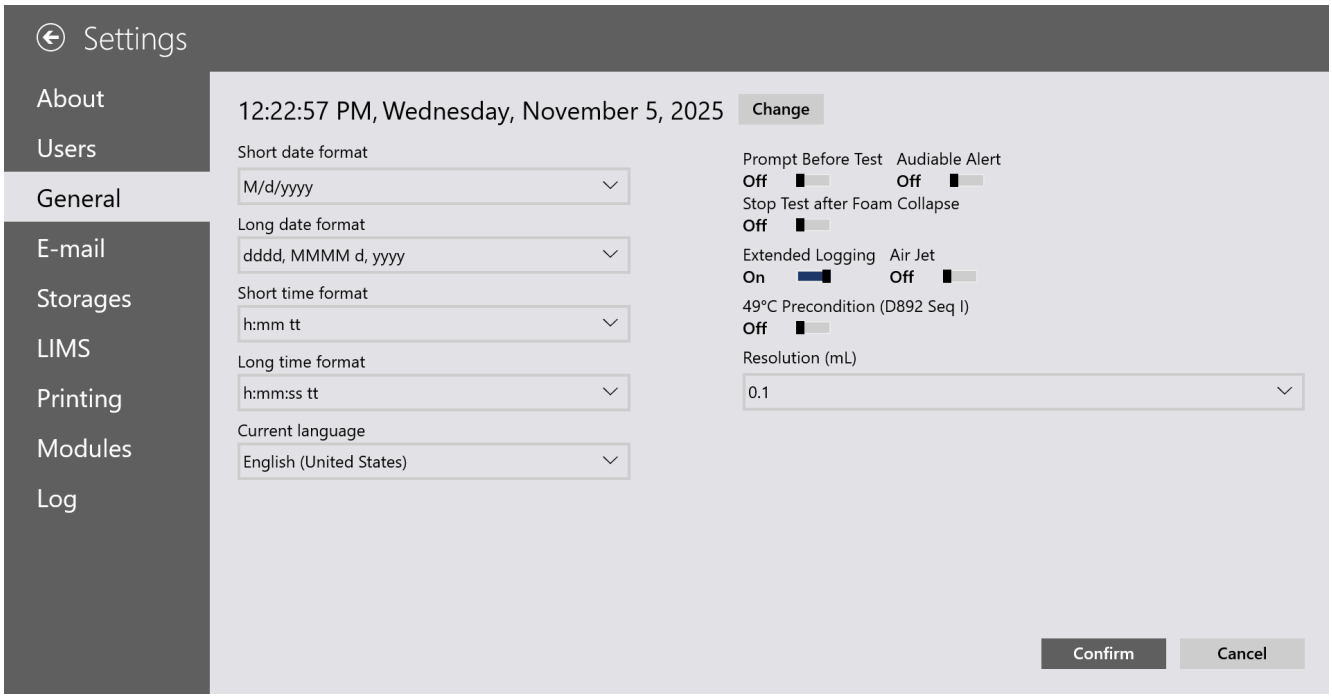
5.4.3.6. Stop Test After Foam Collapse

Terminates the test automatically if the foam collapses completely to a value of ‘0’ before reaching the standard 10-minute duration.



5.4.3.7. Enable Extended Logging

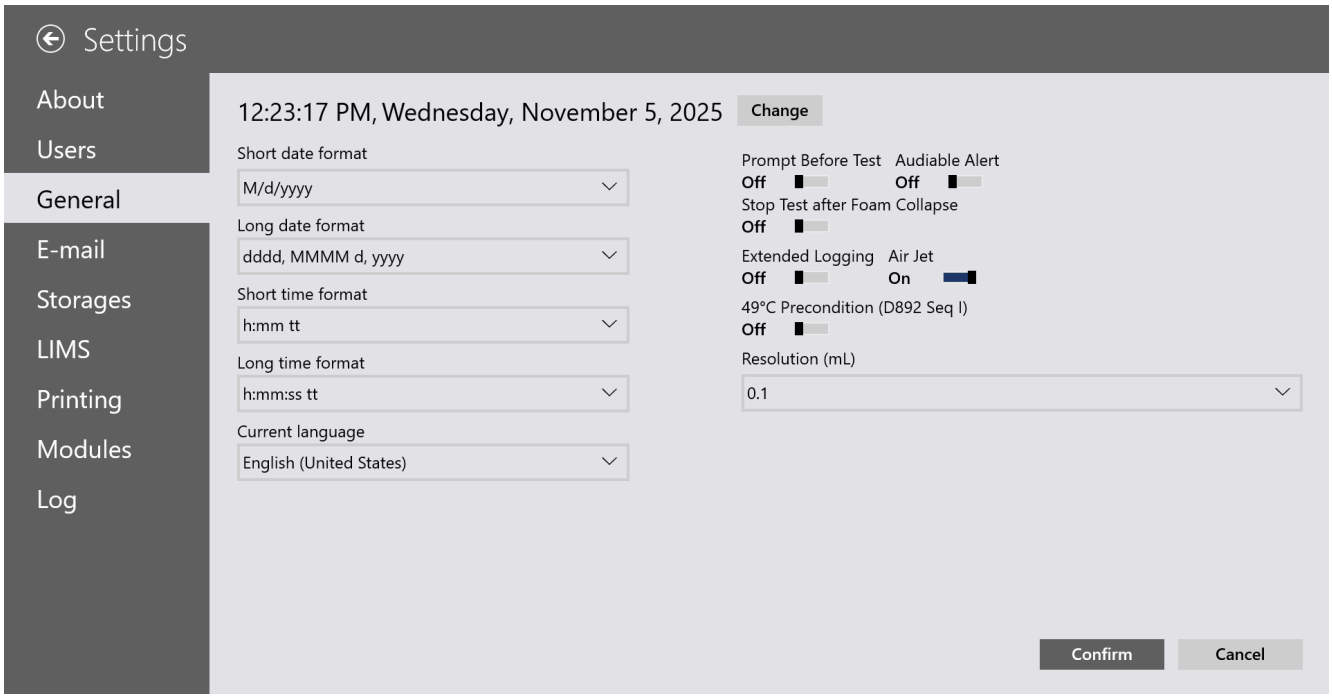
Enables detailed log output for diagnostic purposes. This option is primarily used by Visaya Engineering when troubleshooting.



5.4.3.8. Air Jet

When enabled, compressed air is directed onto the hot sample during *Sequence 3* to accelerate cooling from the 93.5 °C test.

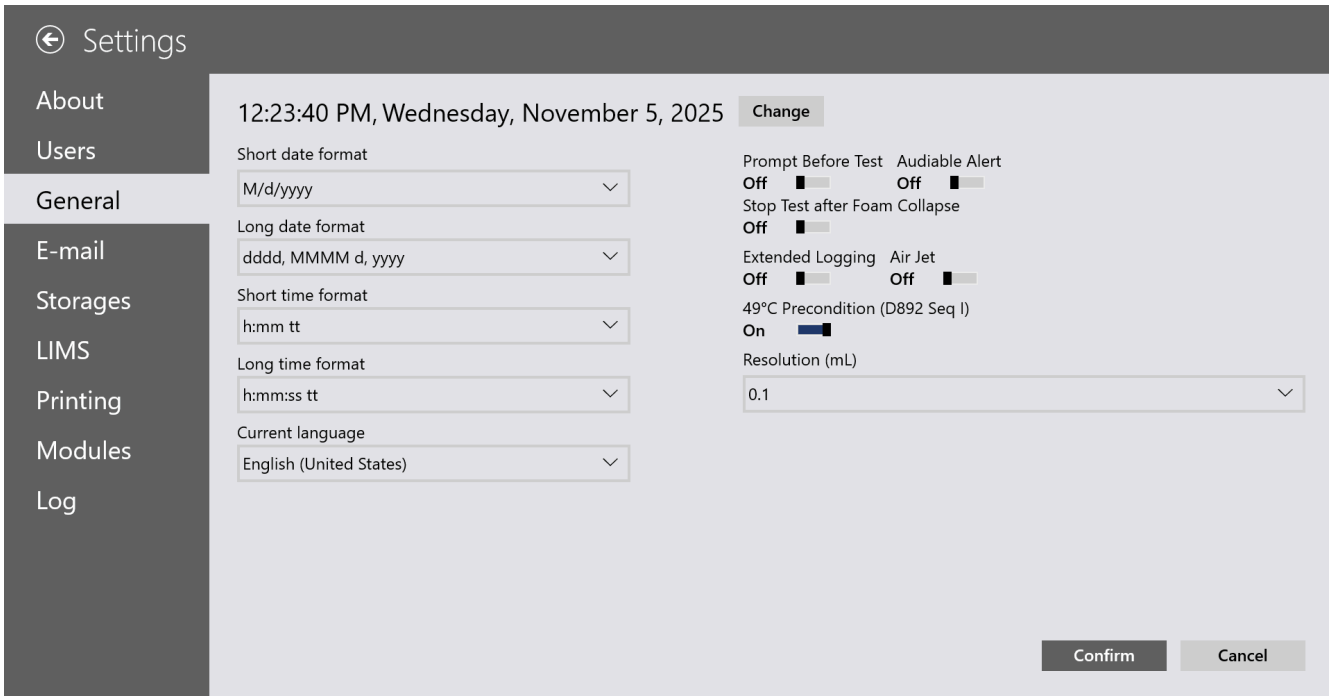
Warning: This feature increases air consumption. Contact a Visaya representative to confirm your installation has sufficient air supply capacity.



5.4.3.9. 49 °C Precondition (D892 Seq 1)

When enabled, the sample is first heated to 49 °C, then cooled to 24 °C during *Sequence 1*. This optional step, defined in ASTM D892, helps eliminate *thermal memory* from the sample.

Refer to ASTM D892 section 10.1 for more information.

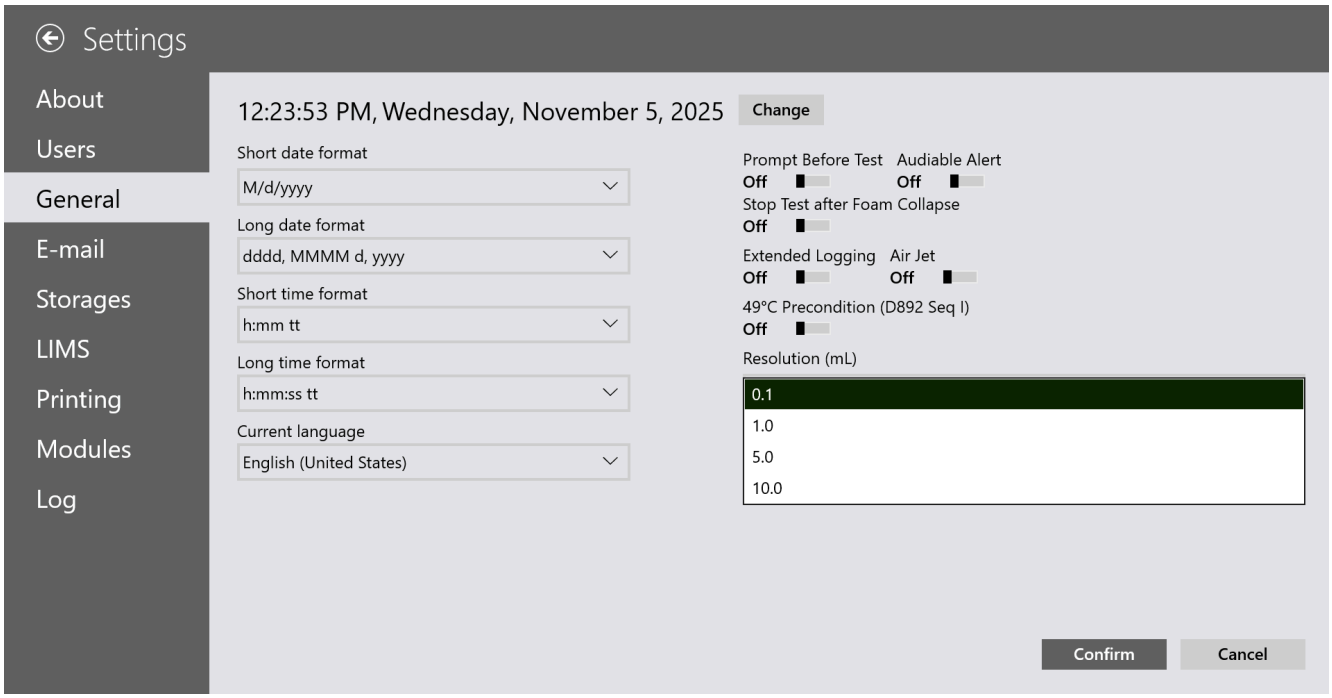


5.4.3.10. Foam Resolution (mL)

Specifies the measurement resolution for foam level detection. Available options: 0.1 mL, 1.0 mL, 5.0 mL, and 10.0 mL.

This setting applies to both *Automatic Foam Analysis* and *Override / Semi-Automatic Mode*.

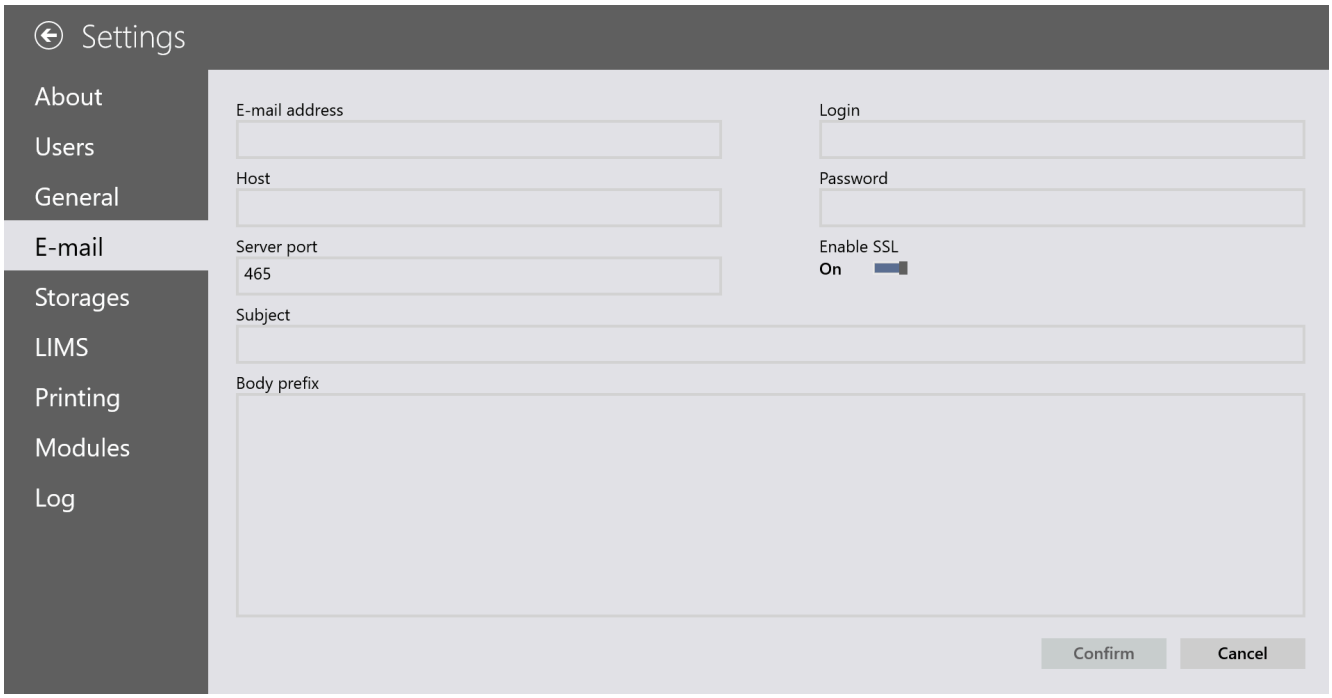
For example, a 0.1 mL resolution provides finer measurement detail than a 10.0 mL setting.



5.4.4. Email

The FoamDDI can automatically send test results by email upon request from the Results View, History, or Log Files screens.

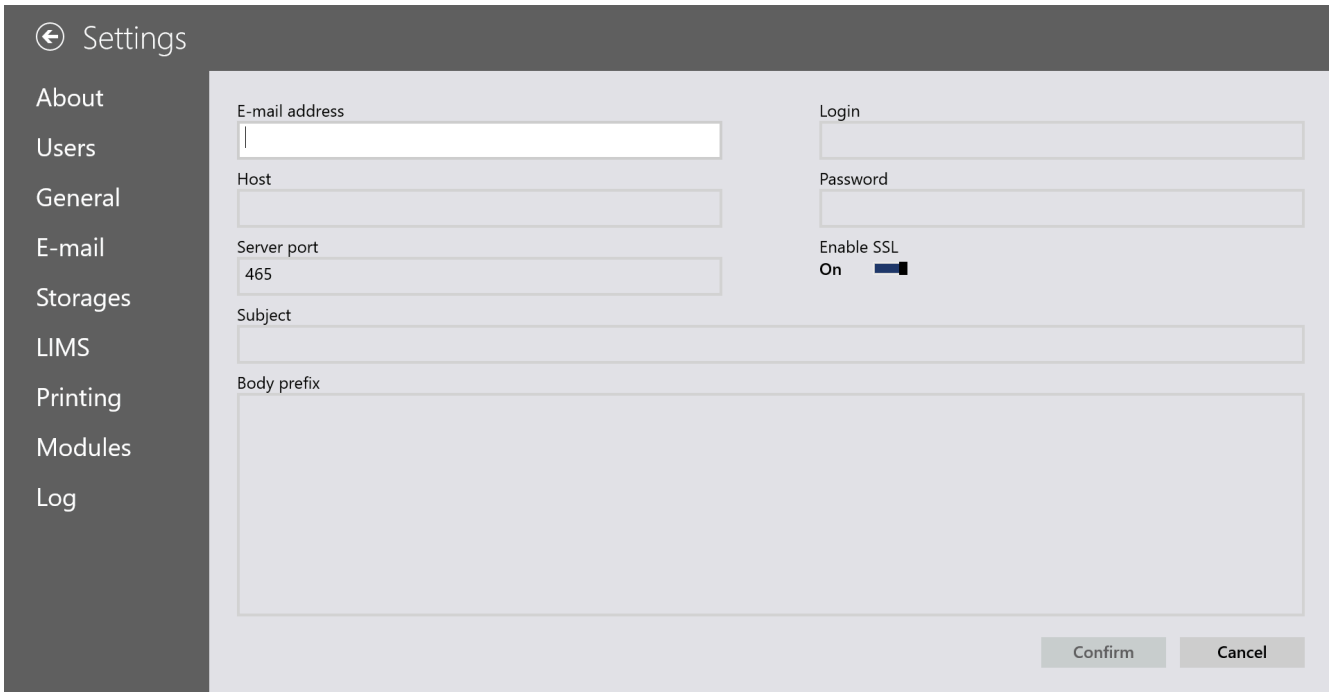
Before using the email feature, the FoamDDI must be configured with valid outgoing mail server credentials. These settings define how the system connects to your organization's email service to send messages and attachments.



The following parameters must be defined for email transmission:

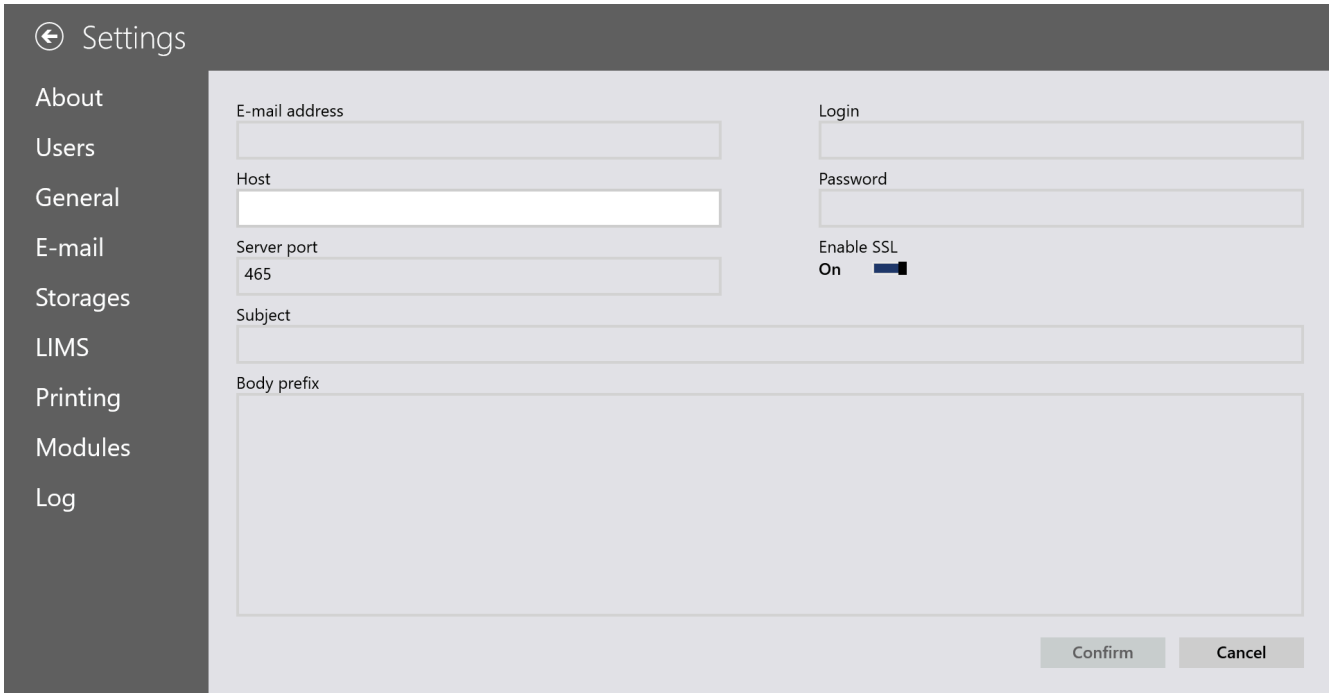
5.4.4.1. Email Address

The sender's unique email address used to transmit results. This address will appear in the "From" field of outgoing messages. It must correspond to the account credentials entered below.



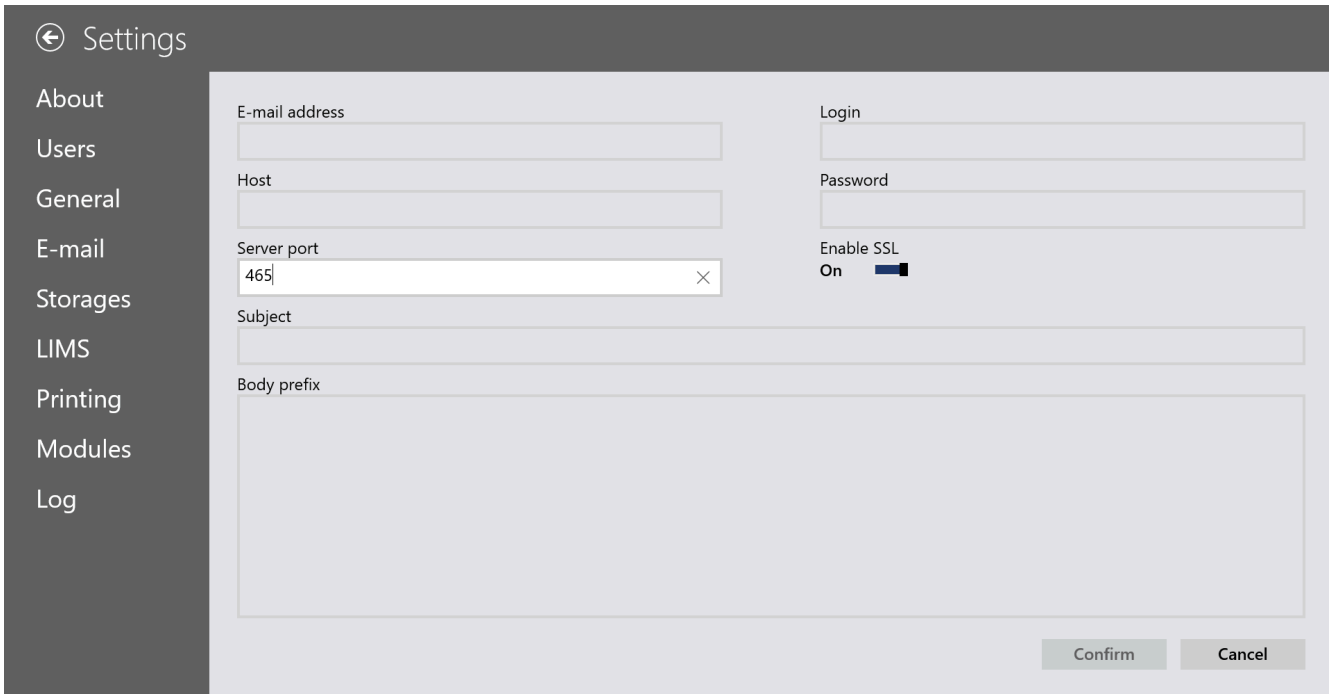
5.4.4.2. SMTP or IMAP Host

The outgoing mail server (SMTP) address, such as *smtp.office365.com* or *smtp.gmail.com*. In some installations, an IMAP host may also be required for authentication.



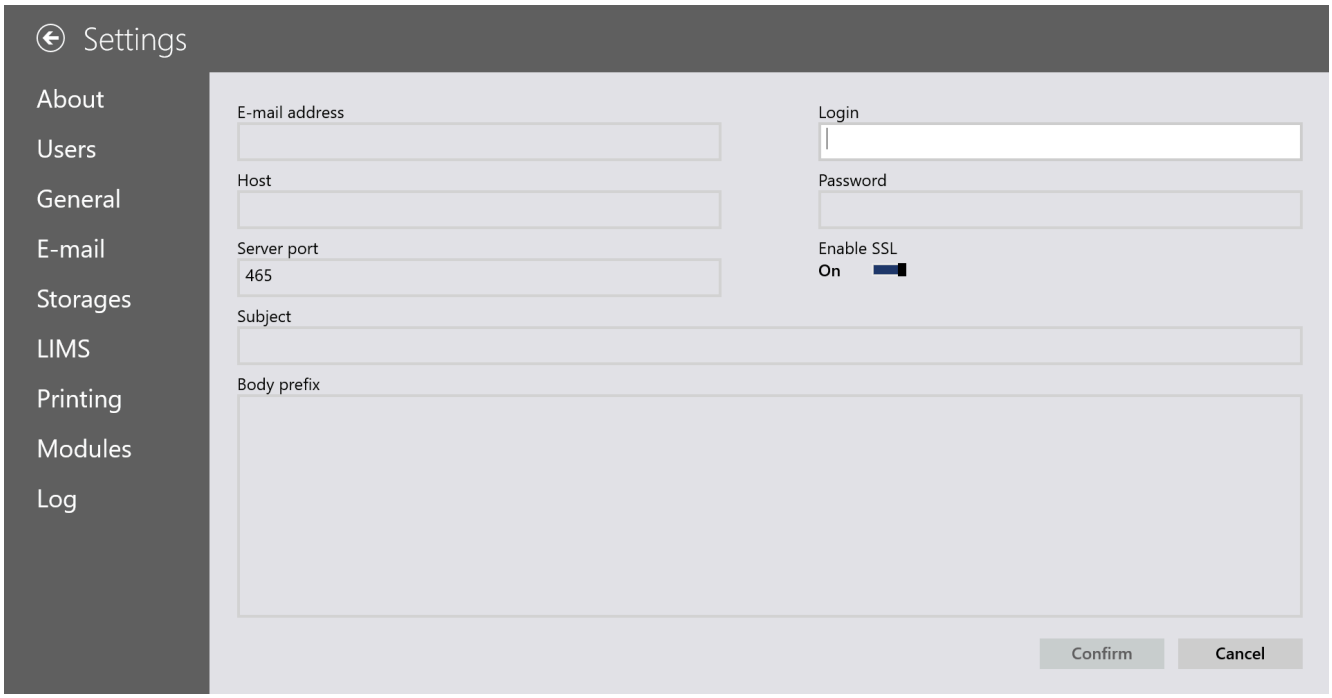
5.4.4.3. Server Port

The port number used by the mail server for outgoing mail traffic. Common ports include 25, 465, or 587, depending on whether encryption (SSL/TLS) is enabled.



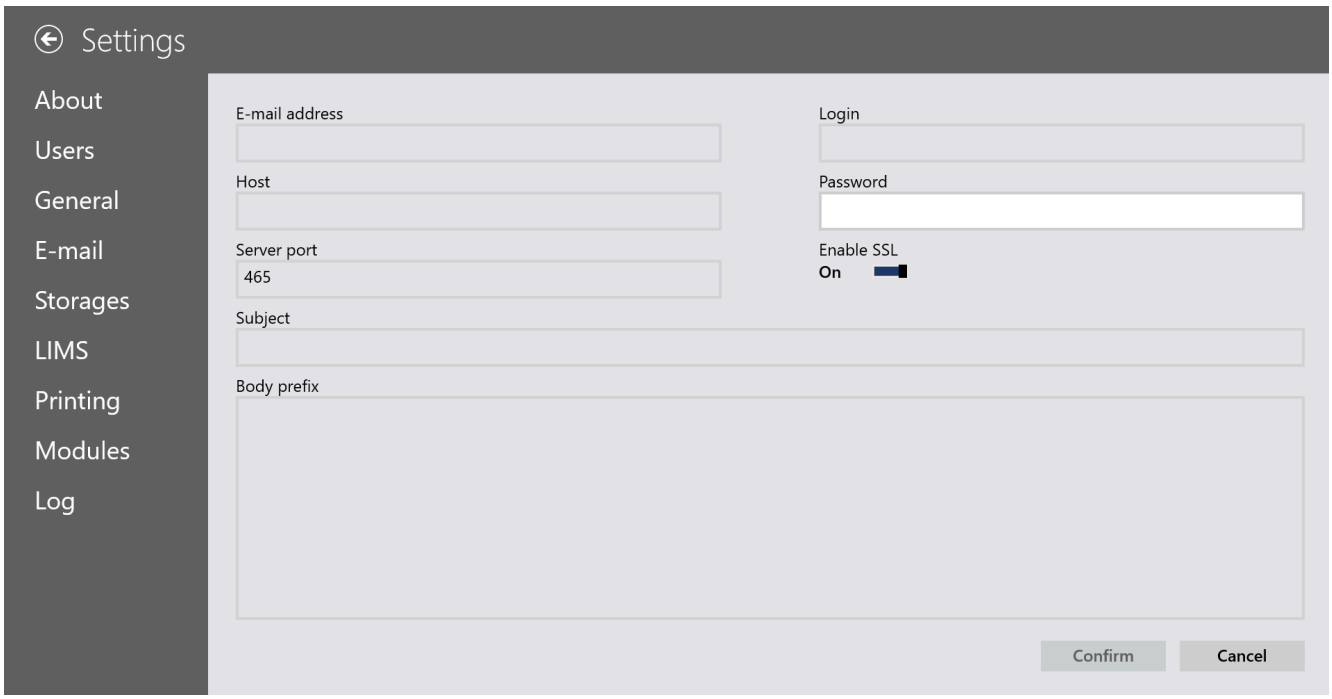
5.4.4.4. Login

The username associated with the email account. In most cases, this is the same as the email address (e.g., *user@company.com*), though some servers may require a domain-based or short-form username.



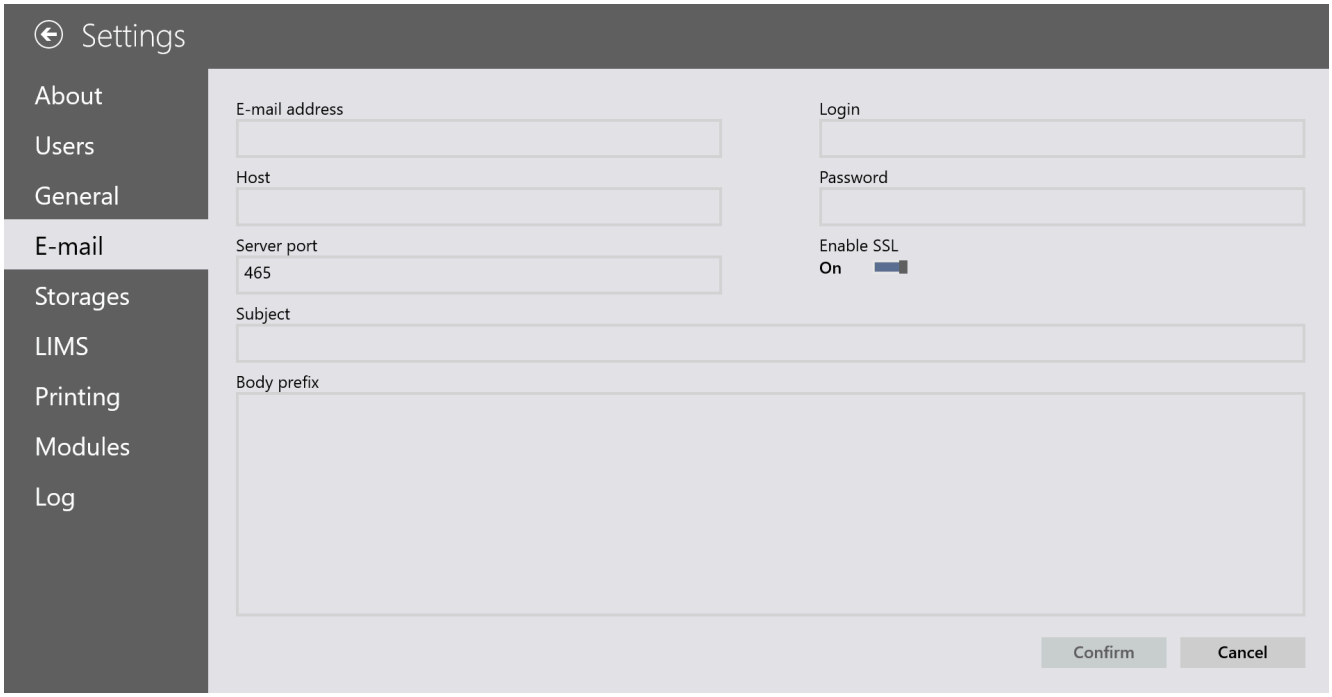
5.4.4.5. Password

The password corresponding to the account defined in the *Login* field. This is used by the FoamDDI to authenticate with the mail server during transmission.



5.4.4.6. SSL Encryption

Enables or disables Secure Socket Layer (SSL) encryption for the connection. SSL/TLS is recommended for secure communication. Most modern mail servers require this option to be turned On.



5.4.4.7. Email Subject and Body

If the Subject and Body Text fields are left blank, the system will use the following default message:

Default Subject:
FoamDDI Test Results / SN #XXXXXXXX-XX

Default Body:

Dear Lab Manager,

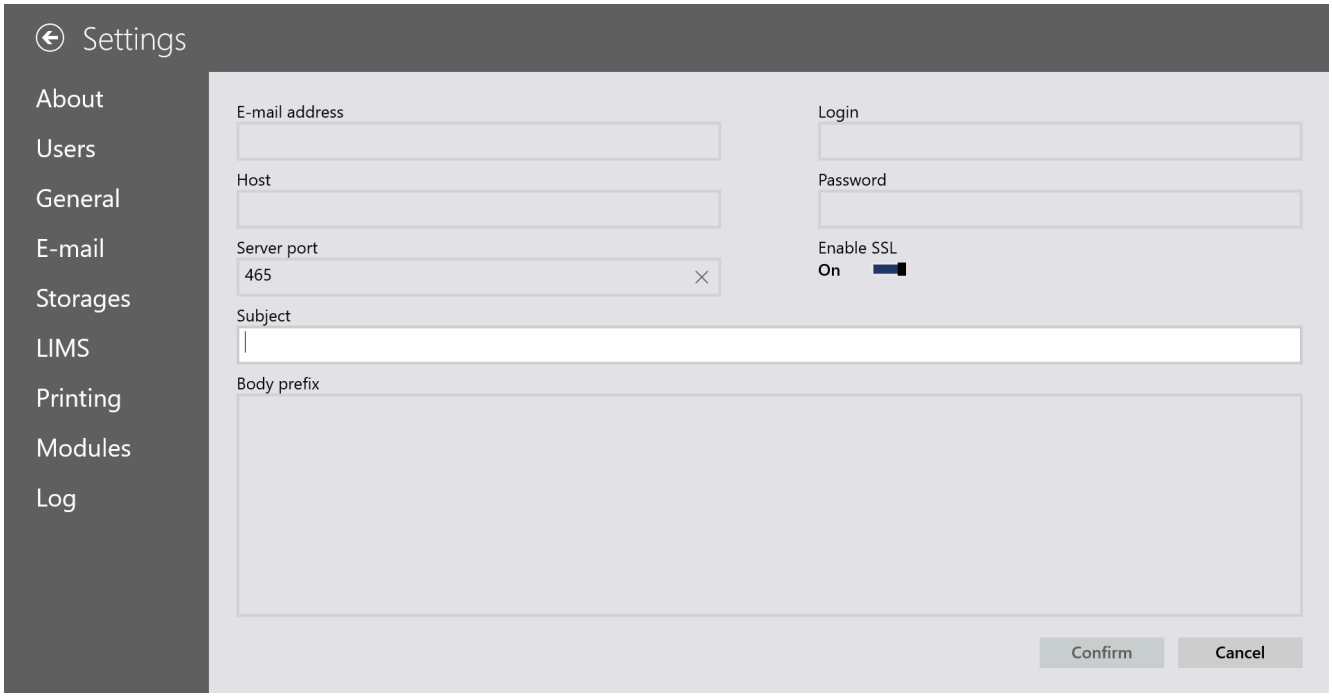
Please find attached the test results.

Sincerely,

FoamDDI Test Results / SN #XXXXXXXX-XX

Sample ID - Filename ID

To use a custom subject line or message, simply enter your preferred text into the corresponding fields.



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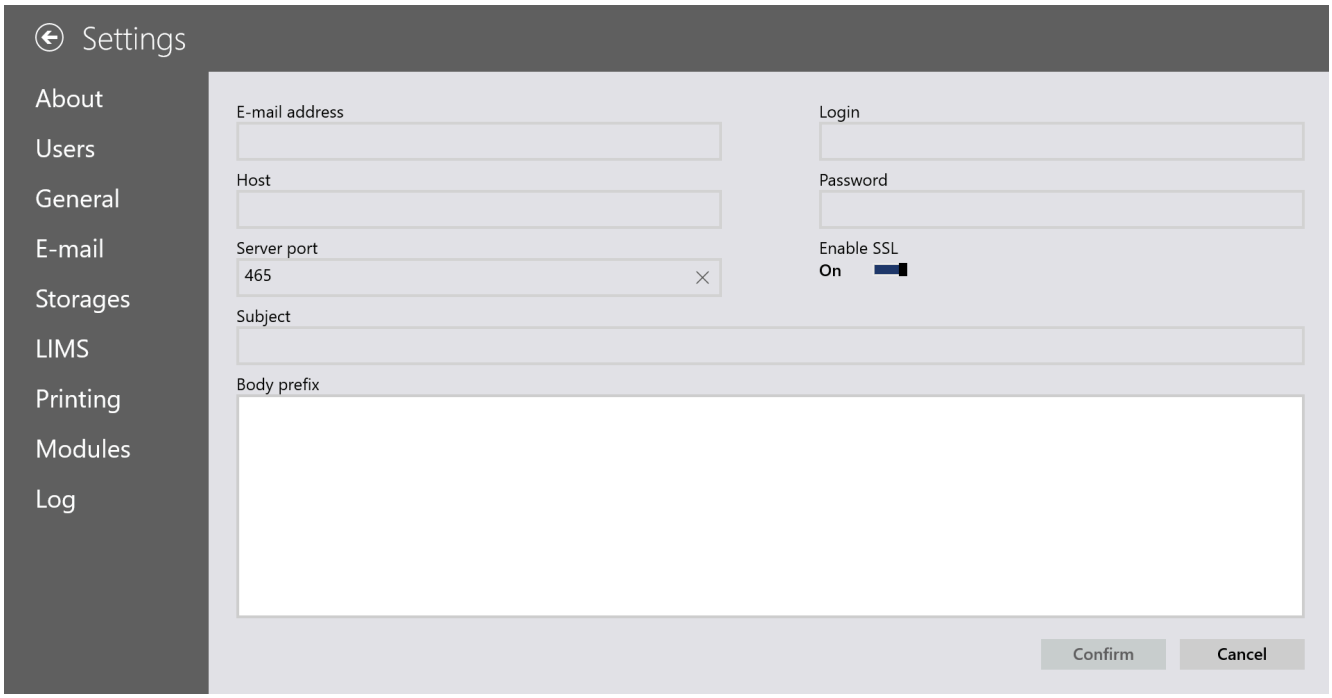
E: info@ayalytical.com



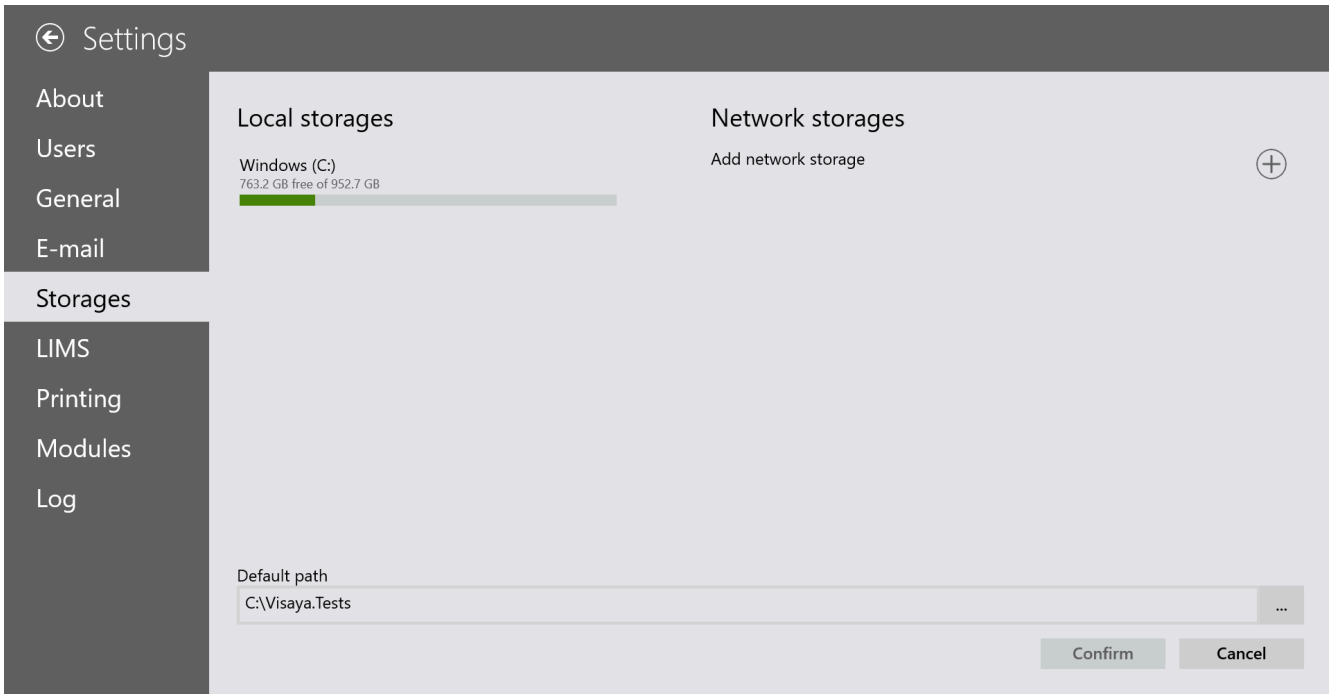
5.4.5. Storages

The FoamDDI operates within a specialized industrial environment running a Windows-Embedded operating system.

All test results and related data are stored on an internal solid-state drive (SSD).

The green progress bars indicate the remaining storage space on the internal SSD, typically designated as Drive C:.

Any external drives or USB storage devices connected to the system will also appear in this list.

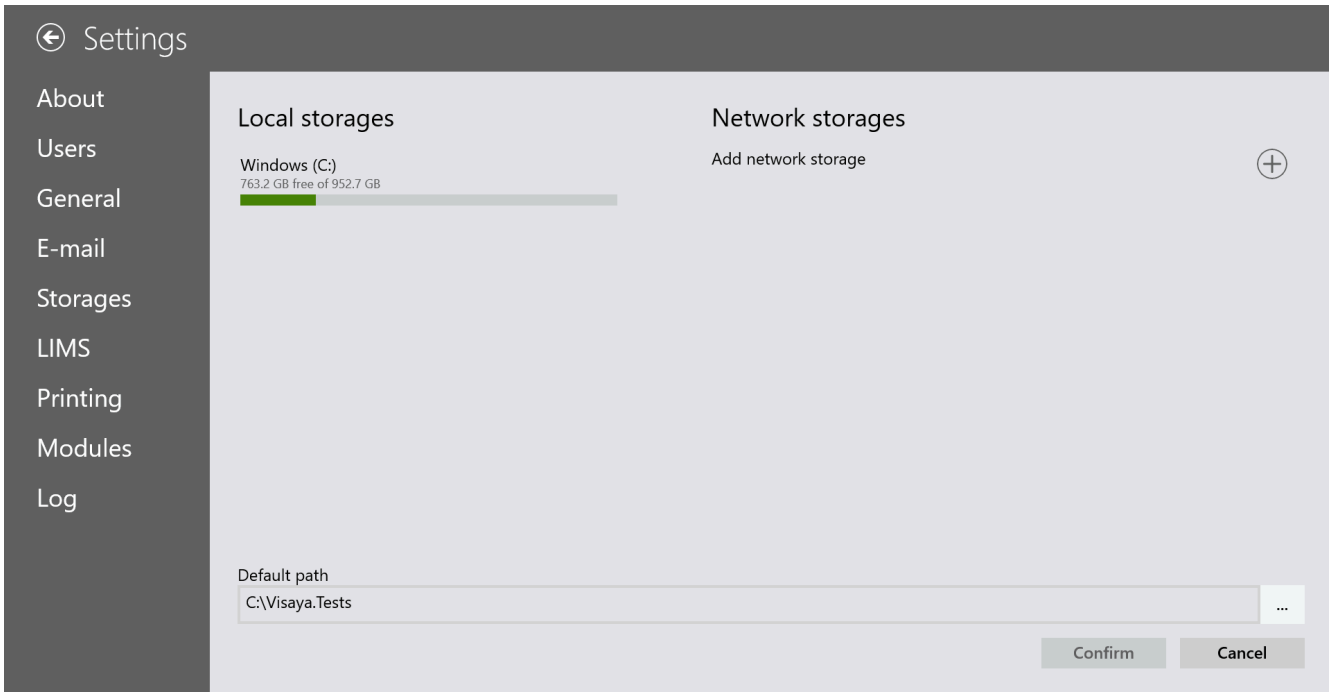


5.4.5.1. Default Path

The Default Path shows the current directory used for storing test results.

To change the storage location, click the browse button (···) at the end of the file-path field and select a new folder.

Under normal operation, the internal SSD provides sufficient capacity for all testing and history data.



5.4.5.2. Add Network Storage

If needed, administrators can configure a network storage location such as a shared network drive.

Selecting Add Network Storage opens a dialog window that allows entry of the network path and credentials such as Letter, Path, Login, and Password.

When defining network storage (mapping a network drive) in Windows, you explicitly choose an available drive letter from a dropdown list during the setup process.

How to know which letter to choose:

> View available letters: The "Map Network Drive" window in File Explorer presents a dropdown menu showing only the letters that are not currently in use by local drives or other mapped network resources.
>
> Check existing assignments: You can view all currently assigned letters in the Disk Management tool or by opening File Explorer and selecting "This PC" (or "Computer" in older Windows versions). This will show which letters are already taken by your C: drive, optical drives, external hard drives, and other network mappings.

Use best practices:

Reserve local letters: It is a best practice to leave letters like C: (system drive) and those immediately following (D:, E:, etc.) free for internal and physical devices like hard drive partitions or DVD drives.

Use letters from the end of the alphabet: Letters from the end of the alphabet (Z:, Y:, X:, W:, etc.) are typically used for network drives to minimize the chance of conflicts if you add more local drives later.

Standardize within an organization: In a business environment, IT administrators often standardize specific letters for certain network shares (e.g., H: for home directories, P: for public files) for consistency across all users' computers.

During the mapping process:

Open File Explorer and go to This PC.

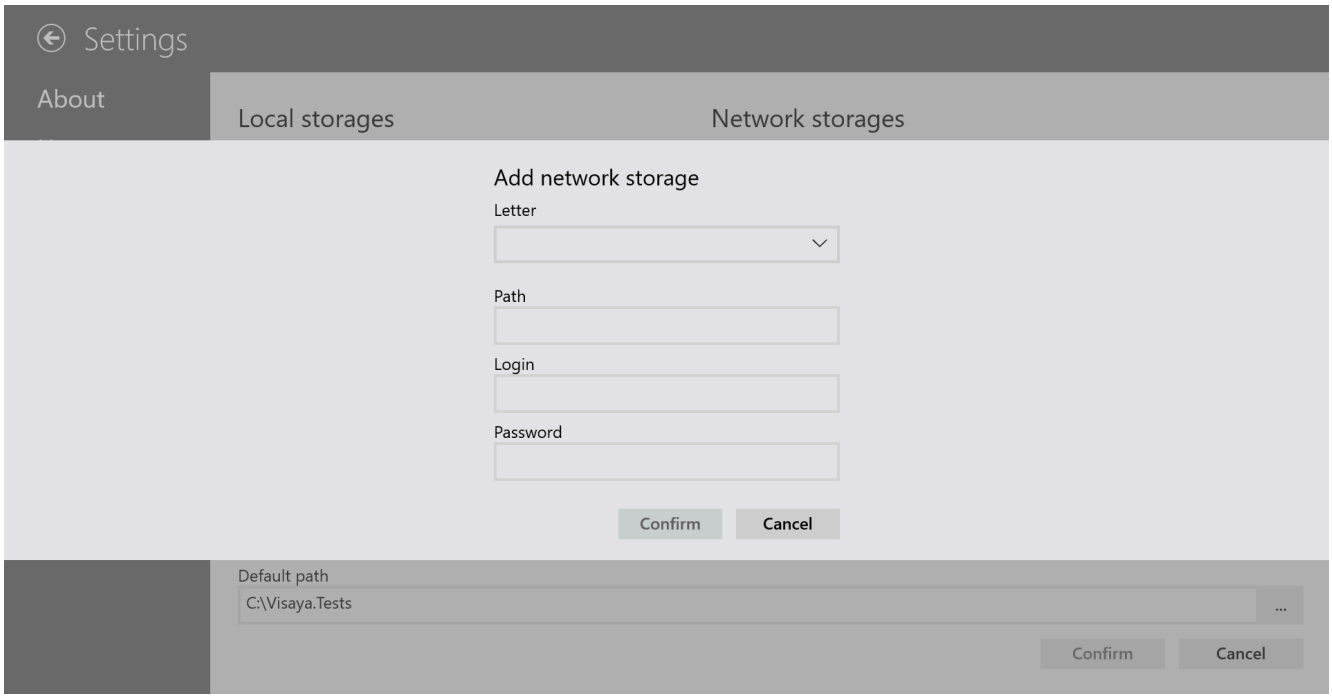
Click Map network drive from the ribbon menu.

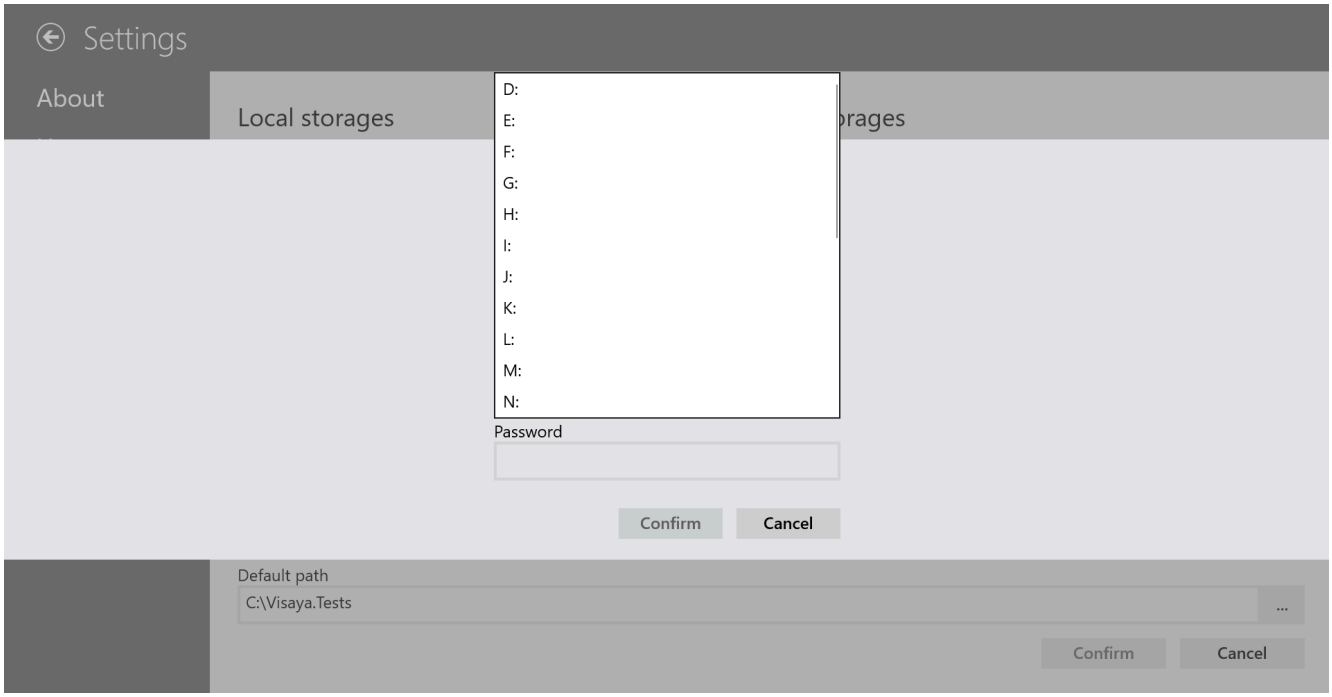
In the "Drive" dropdown list, select your desired available letter.

Enter the network path in the "Folder" field.

Click Finish to complete the mapping.

The letter you select is how you will then access the network storage.





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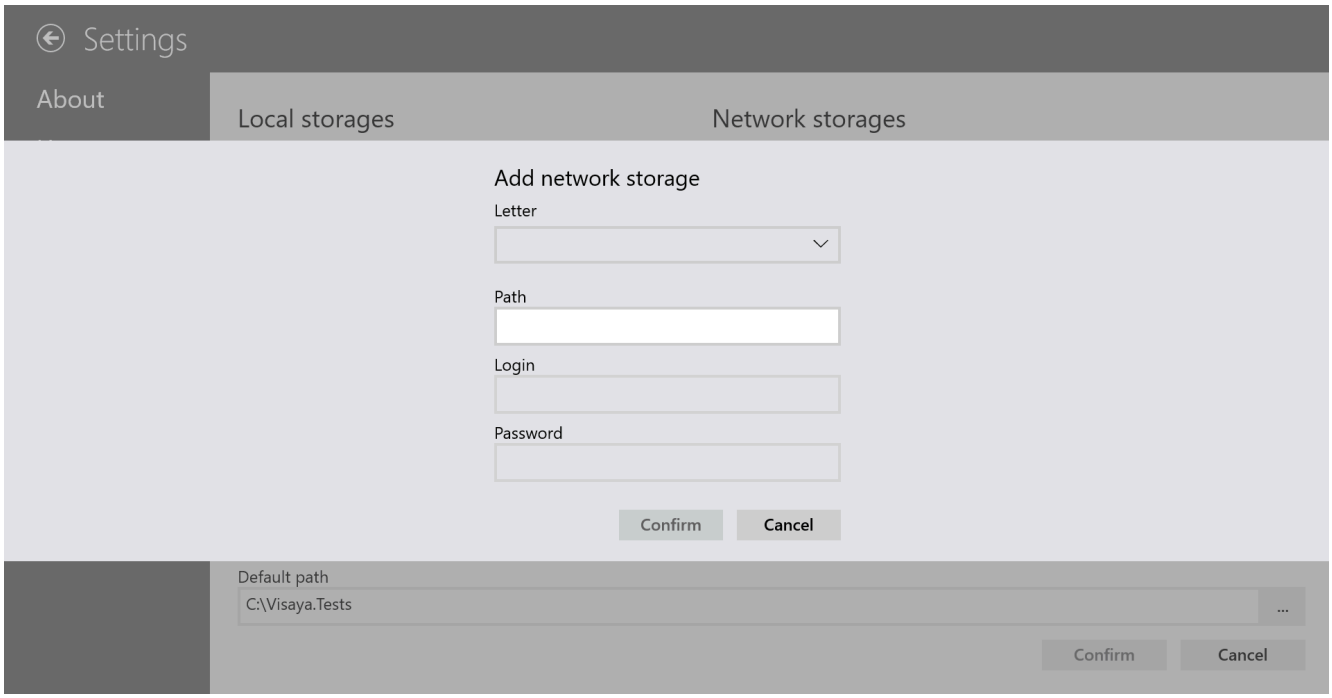
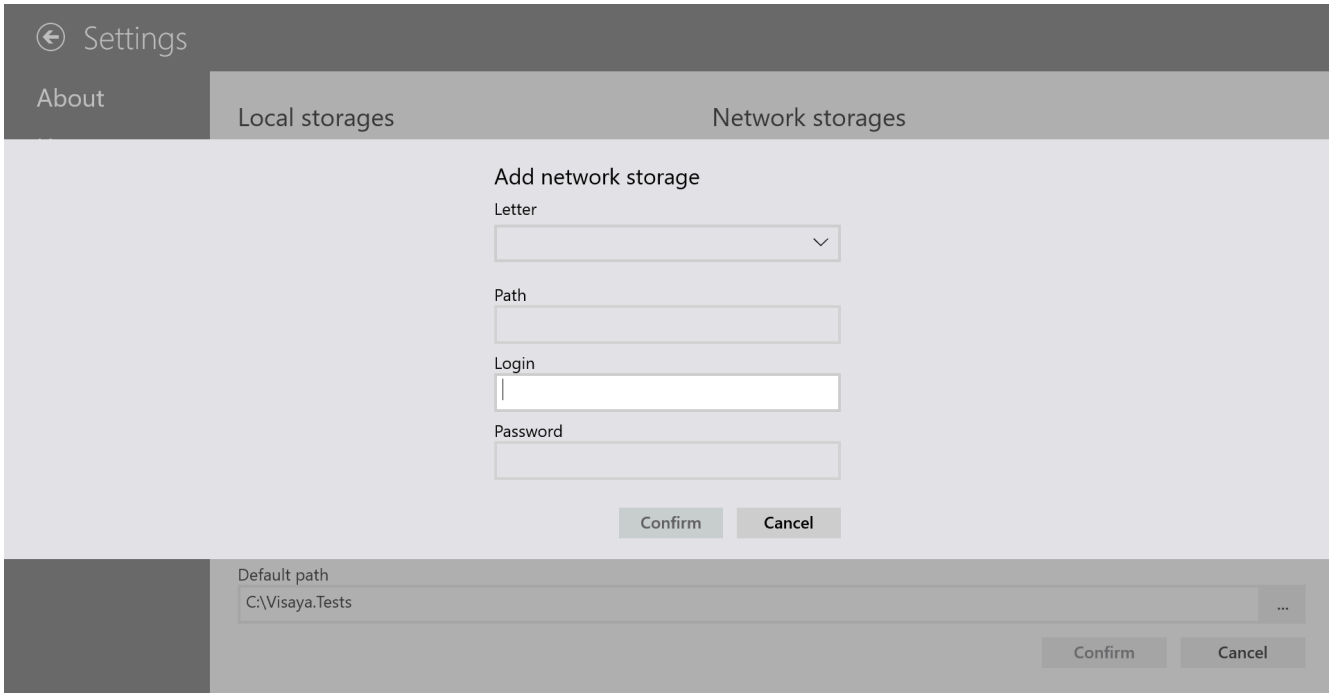


Use the credentials required for the path desired as the newtork storage.



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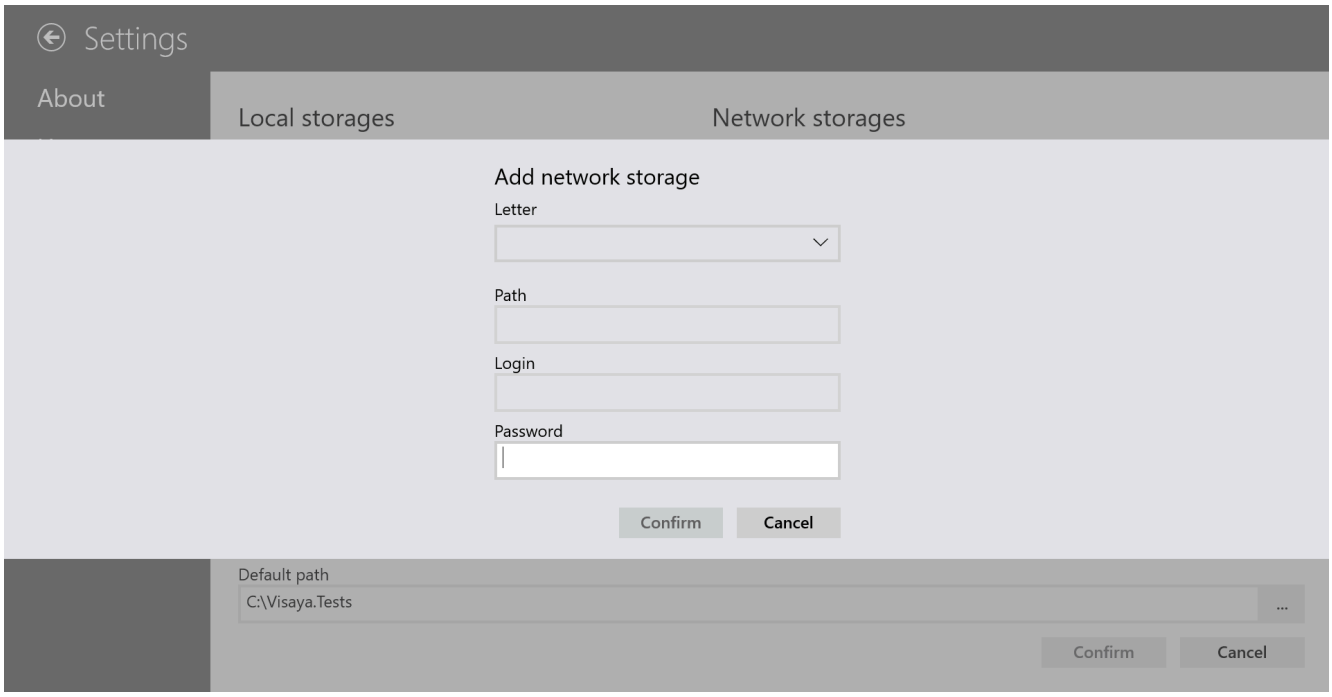


Once added, the Default Path can be reassigned to this network location for future data storage.

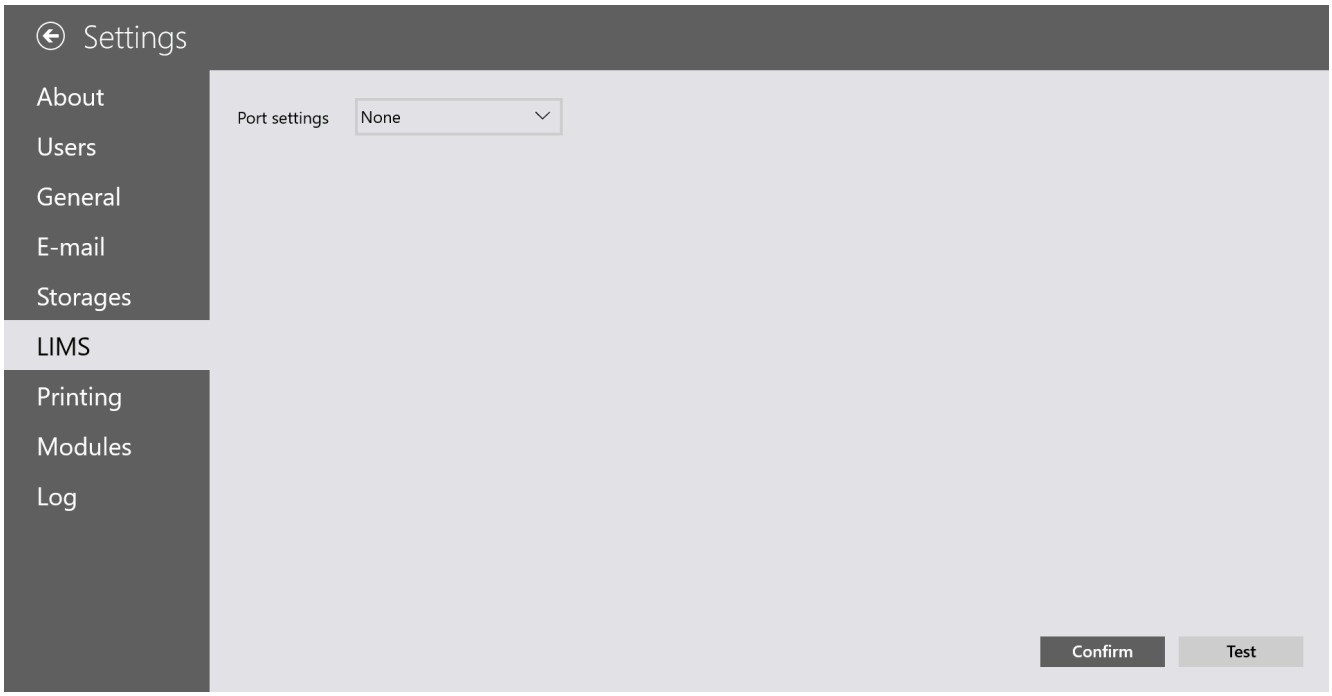
5.4.6. LIMS

The FoamDDI supports full integration with Laboratory Information Management Systems (LIMS) through multiple communication methods. These options provide flexibility to accommodate virtually any LIMS environment.

FoamDDI can communicate with a LIMS system using any of the following methods:

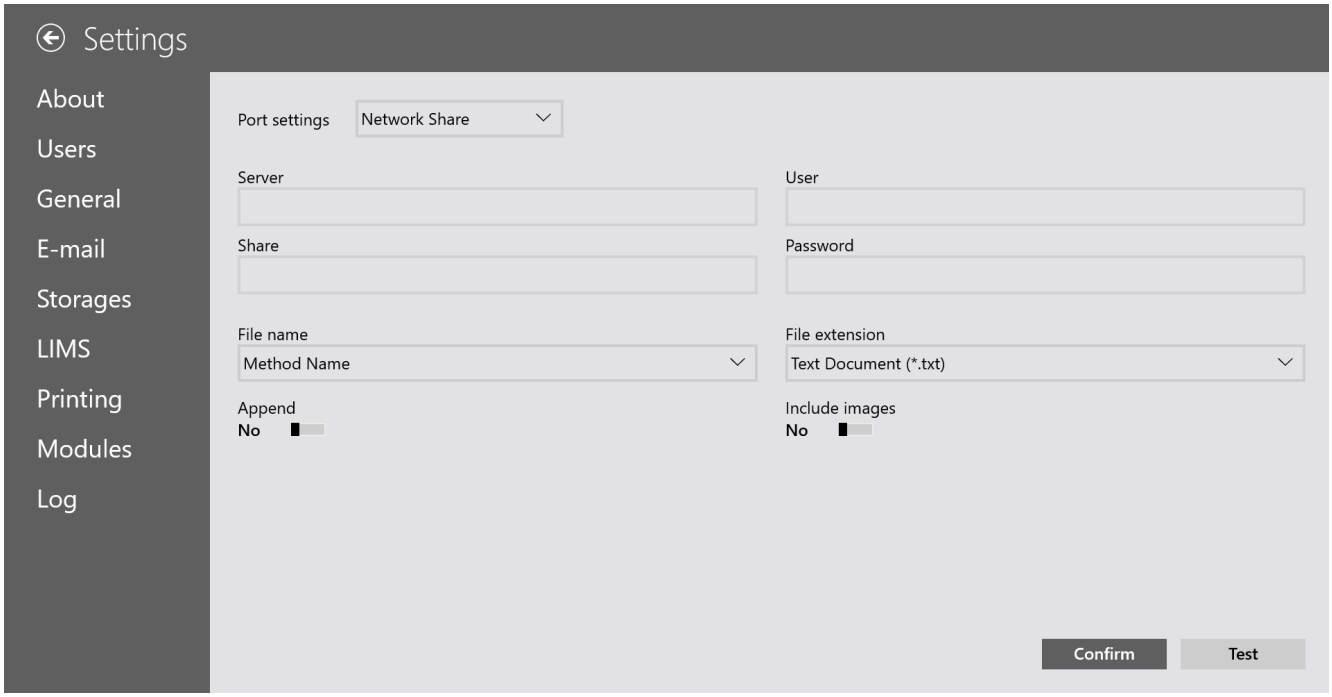
- Network Share – Uses a mapped network location or shared folder on a remote server.
- USB Save – Treats a connected USB drive as a shared storage location.
- USB COM – Uses the USB port for serial communication.
- COM 1 – Uses the RS-232 serial port for direct serial communication.

Select the desired communication method from the Mode dropdown menu to access the configuration options specific to that mode.



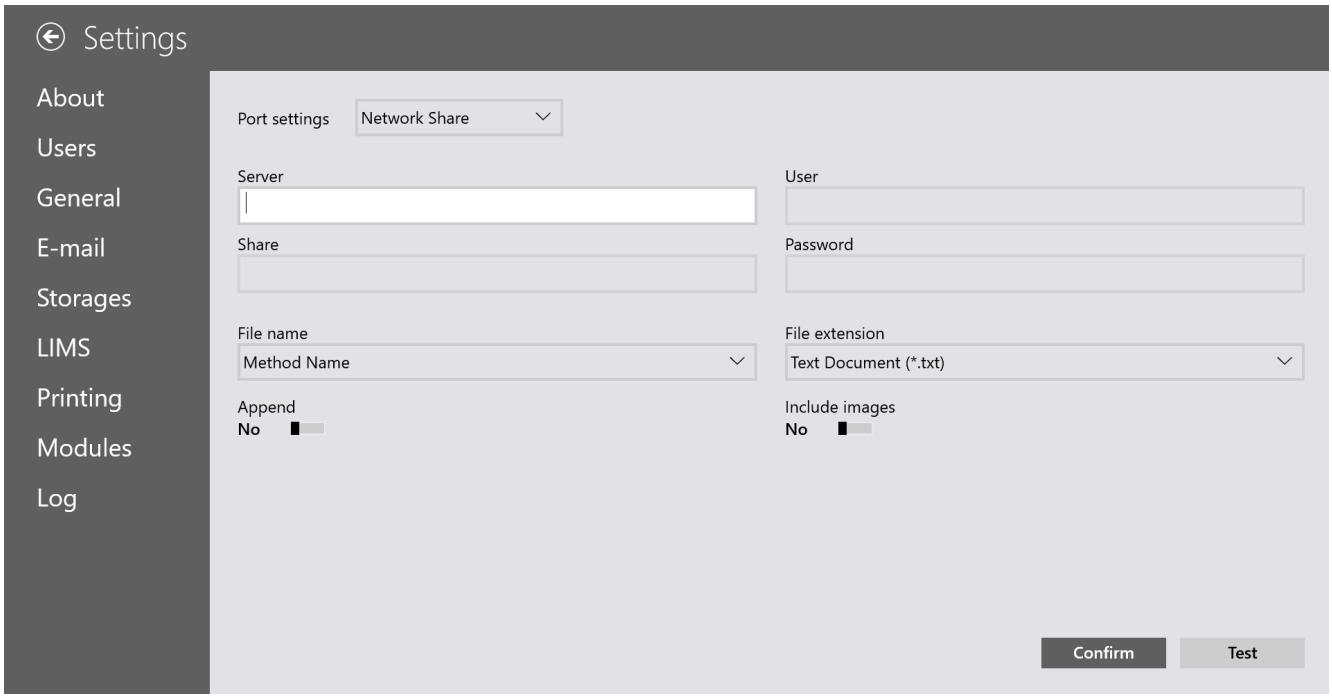
5.4.6.1. Network Share

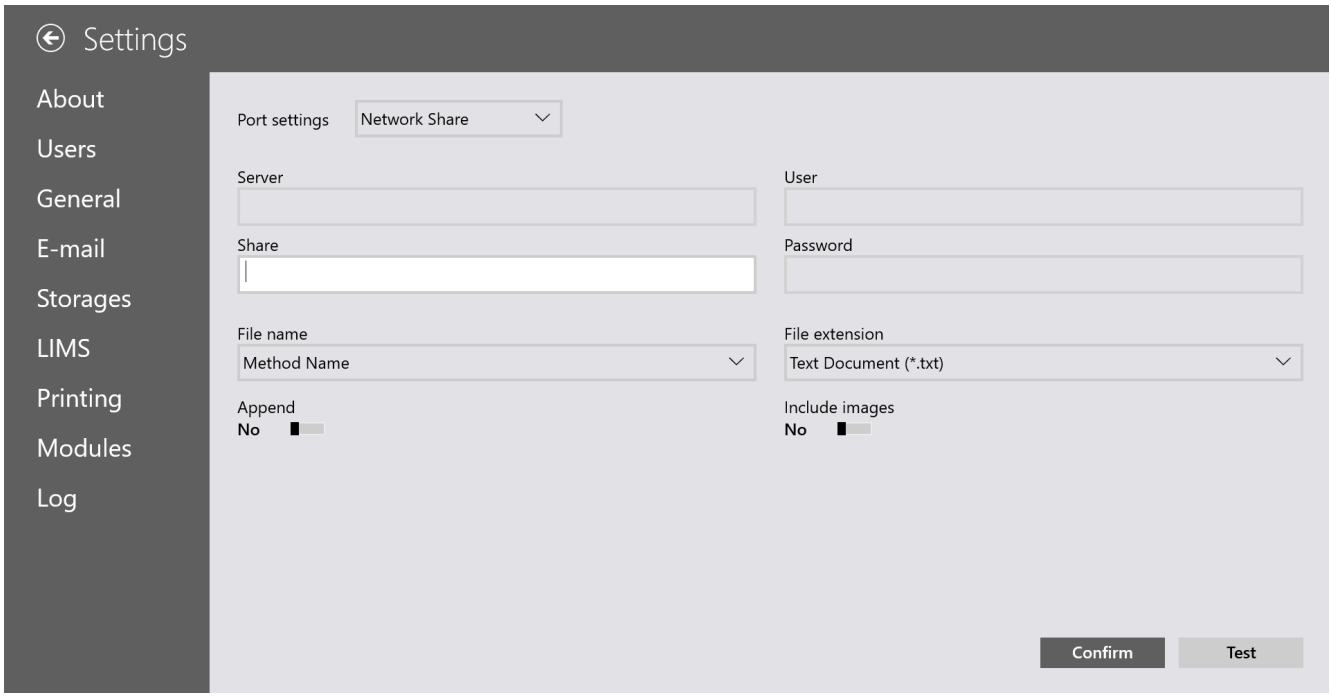
This mode allows the FoamDDI to export test results directly to a shared network location.

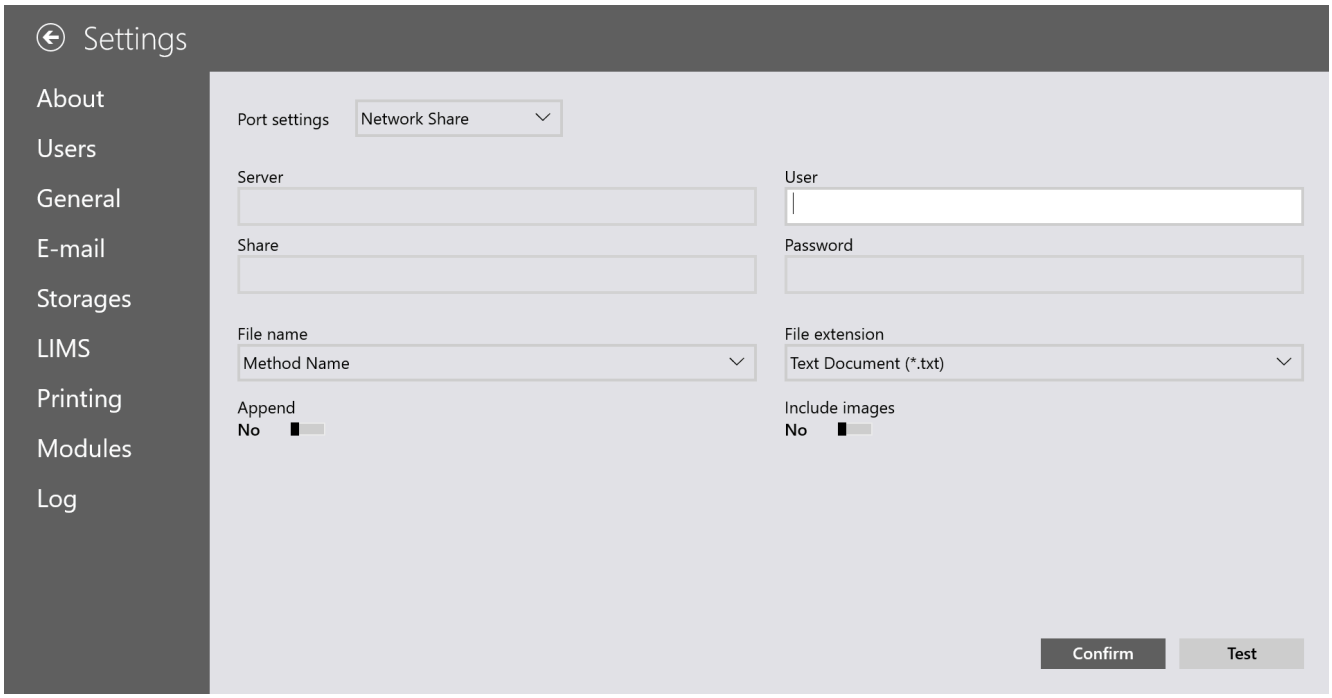


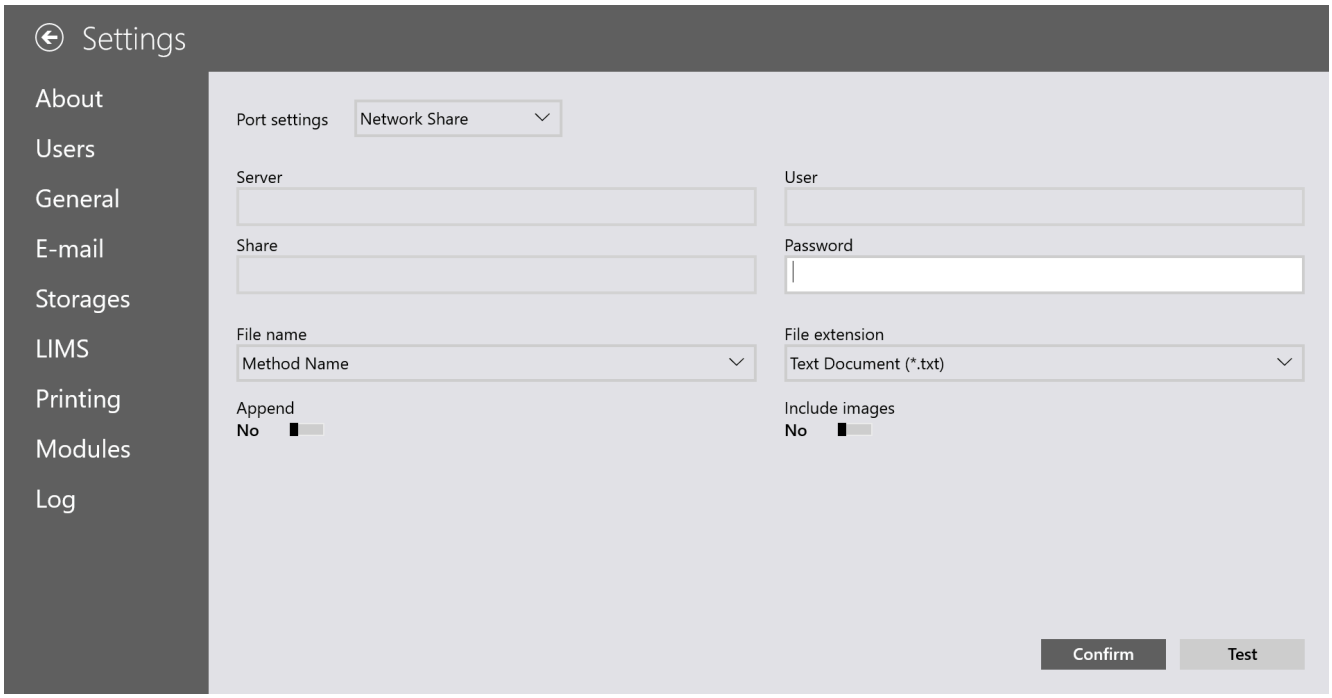
Before enabling communication, the following parameters must be defined:

- Server – Address or hostname of the target server.
- Share – File path or share name where the test results will be written.
- User – Authorized username with write access to the target location.
- Password – Password corresponding to the above credentials.







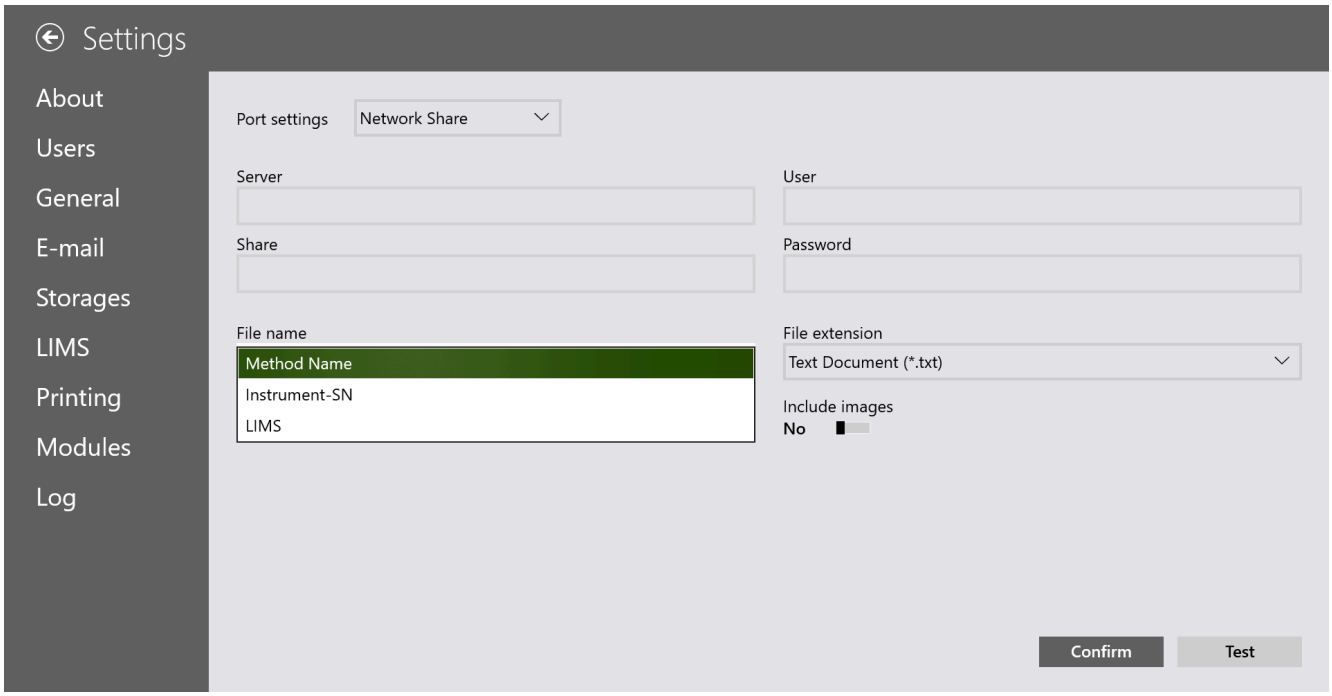


FoamDDI allows the output filename to be customized to suit your laboratory's LIMS format requirements. The following file-naming templates are available:

- Method Name – Uses the test method name (e.g., *ASTM_D892.csv*).
- Instrument-SN – Uses the instrument's serial number as the file name.
- LIMS – Automatically assigns *LIMS* as the file name.

To ensure compatibility with different LIMS systems, FoamDDI supports several export formats selectable via dropdown menu:

- Text Document (.txt) – Simple plain-text file with comma-delimited fields.
- CSV Format (.csv) – Standard comma-separated values format for spreadsheets or databases.
- XML Format (.xml) – Structured XML file with field identifiers and markup, ideal for automated parsing or custom PDF generation with images.



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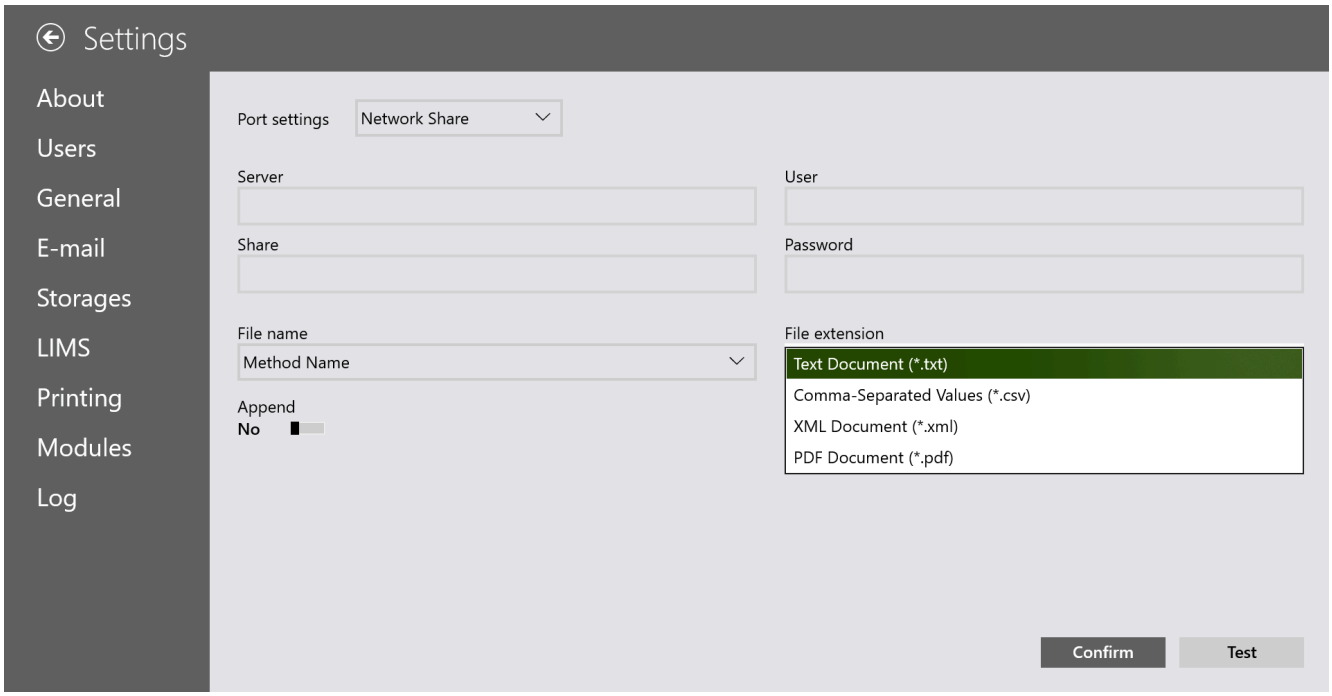
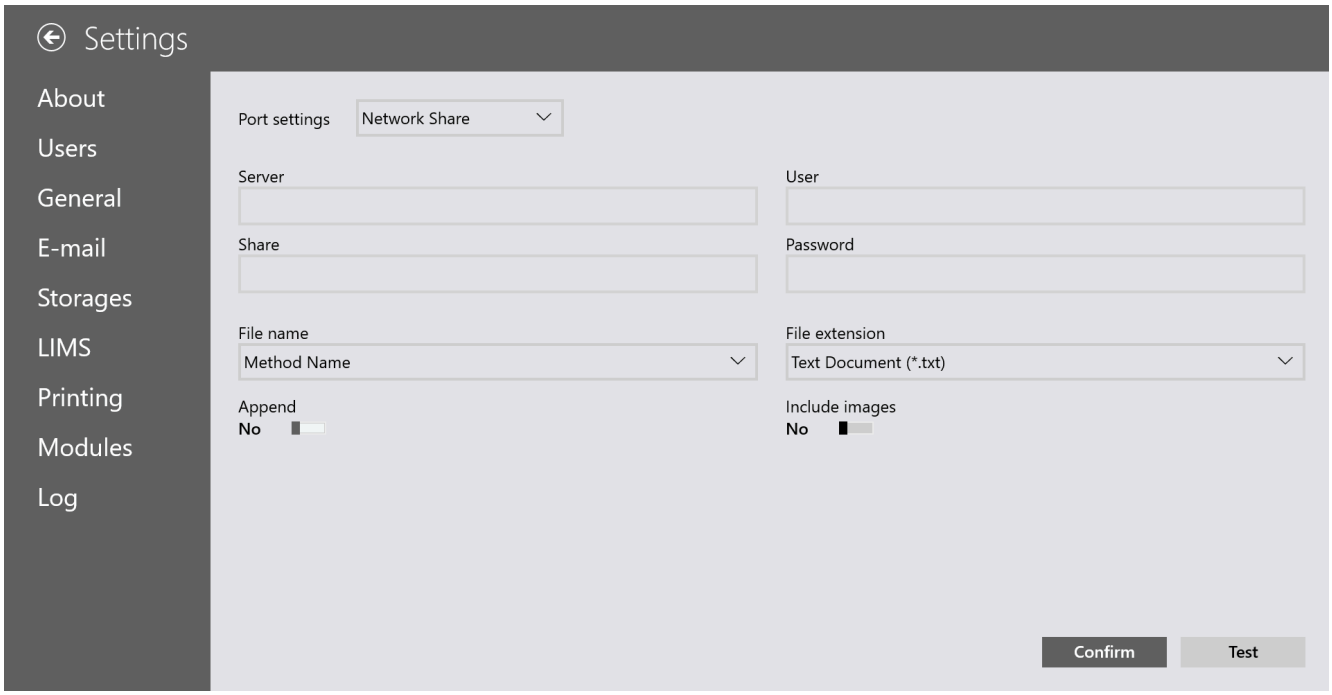


Append Options:

- Append = Yes – Adds each new test as a new data line to an existing file.
- Append = No – Creates a new file for each test result, overwriting any previous file with the same name.



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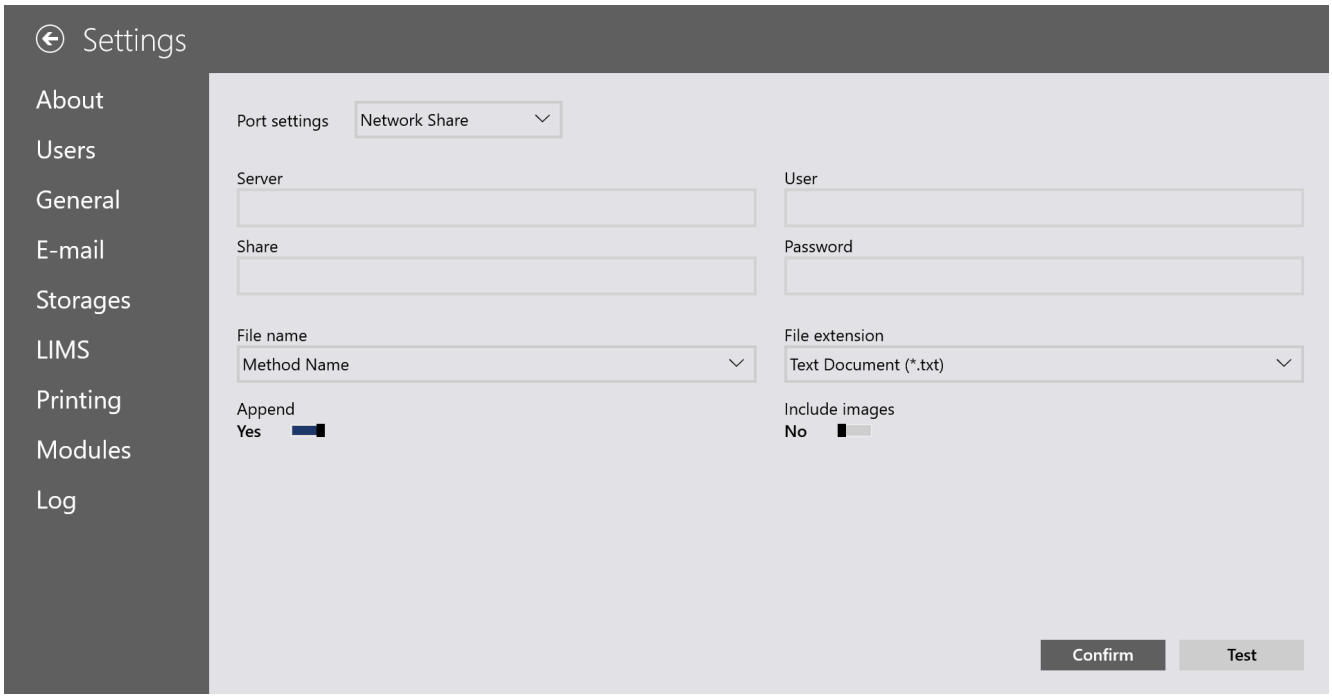
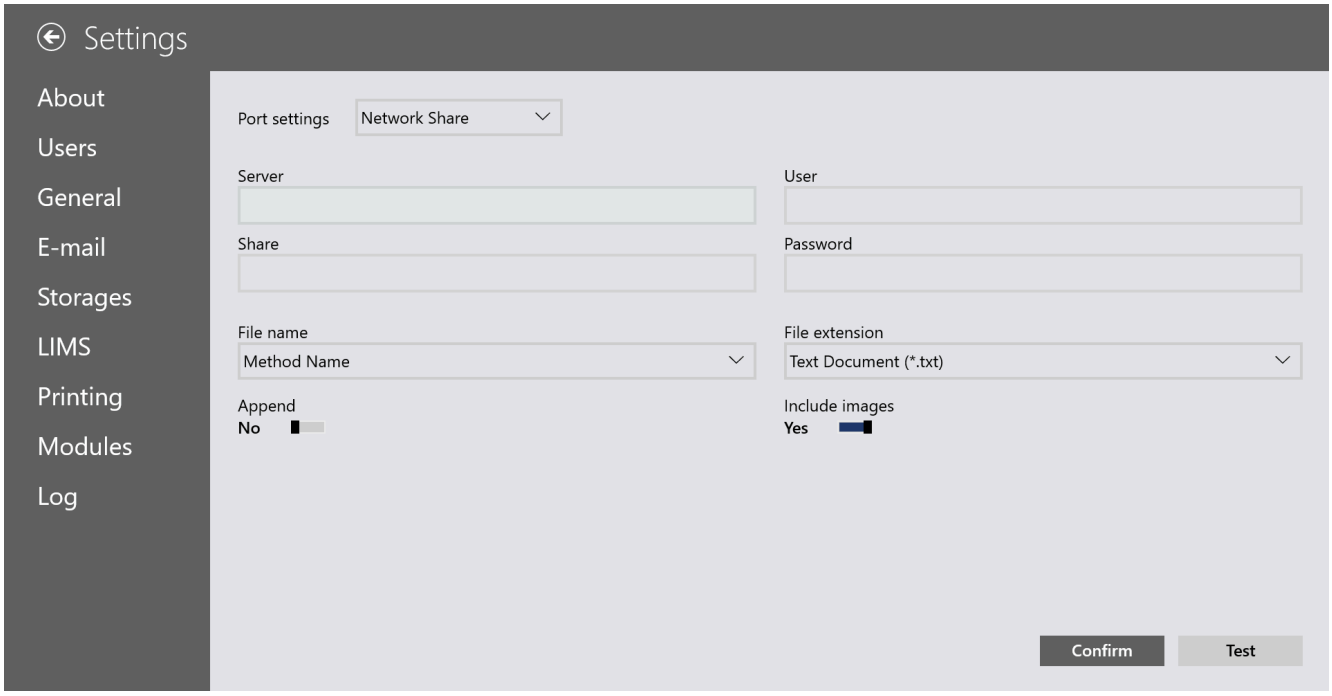


Include Images:

- When set to Yes, FoamDDI will export test images to a separate *Images* folder using a serialized ID.



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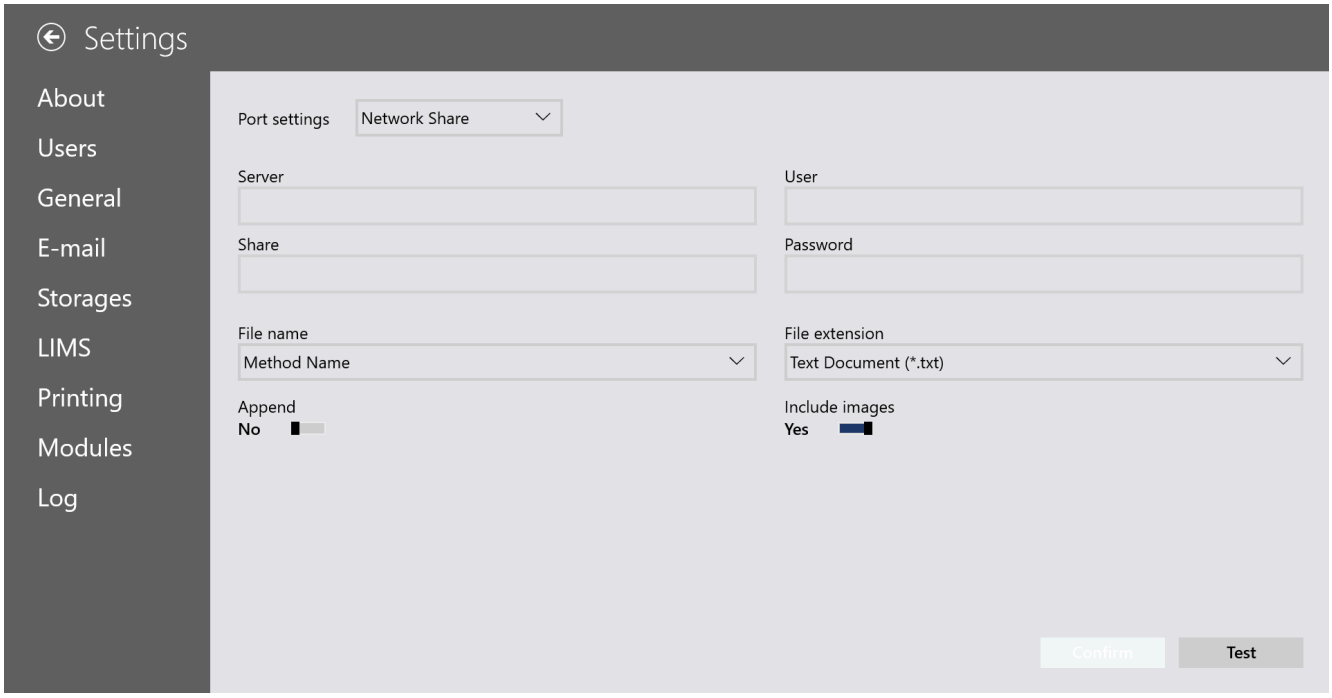
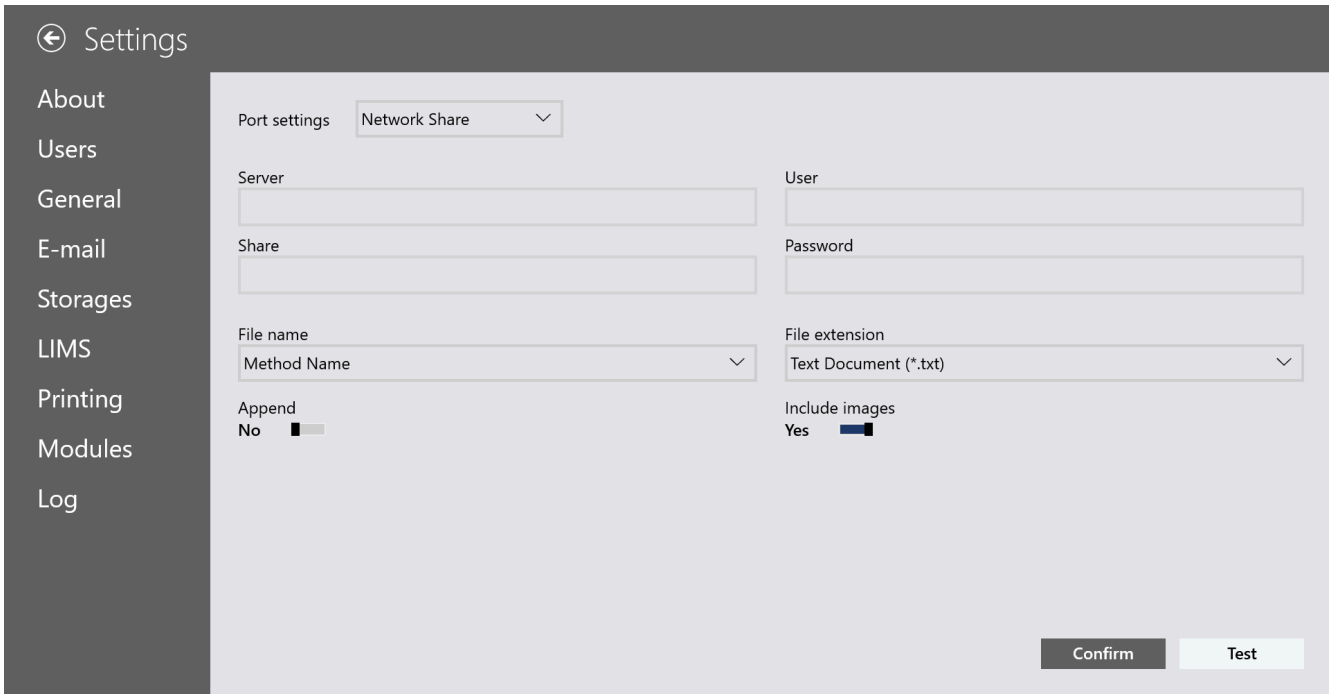
To save changes, click Confirm. To check the configurable settings were done correctly, click Test and check to make sure the test save was saved correctly.





5.4.6.2. USB Save

This mode allows the FoamDDI to export test results directly to a USB storage device.

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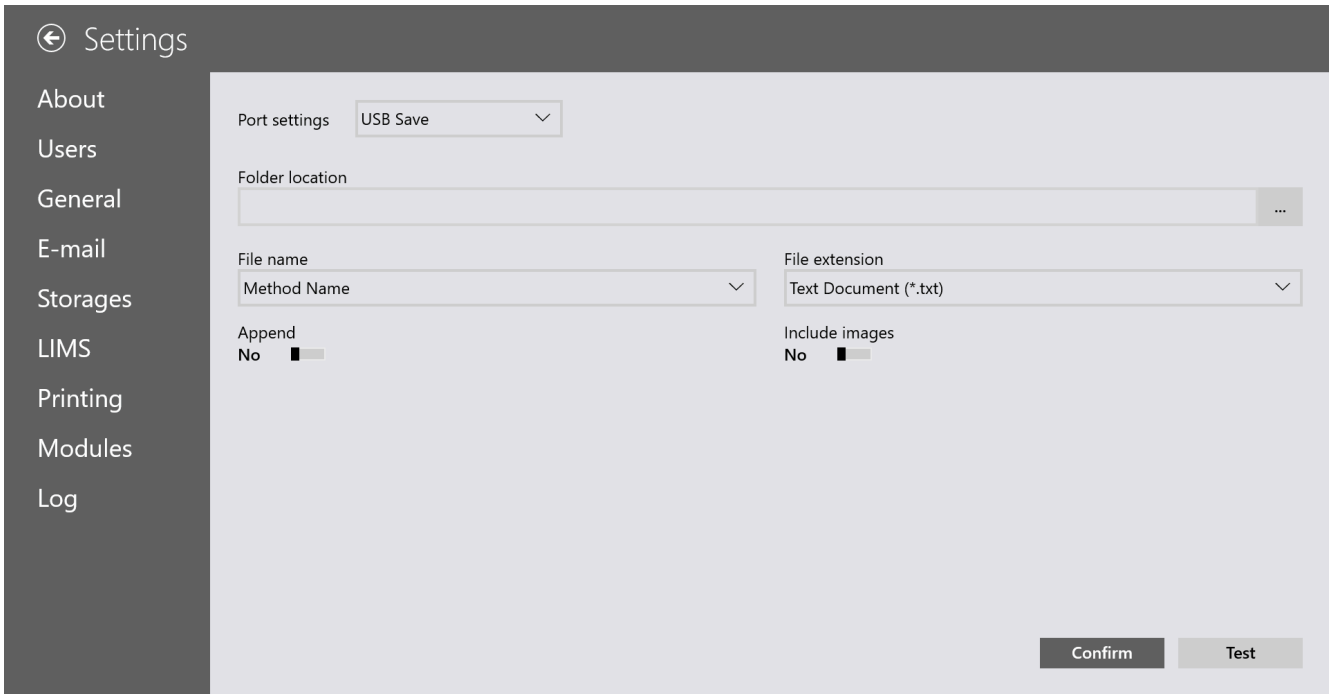
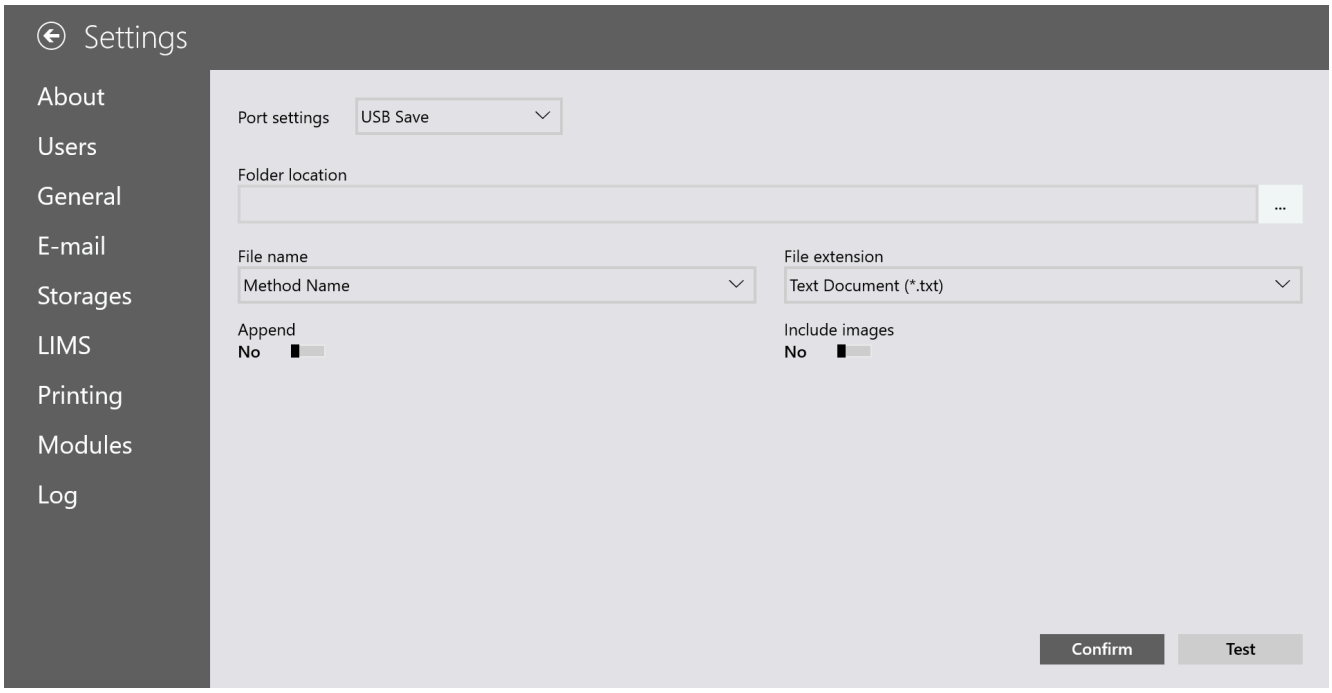


Before enabling communication, the following parameters must be defined:

- Folder Location – Path for the results to be saved.

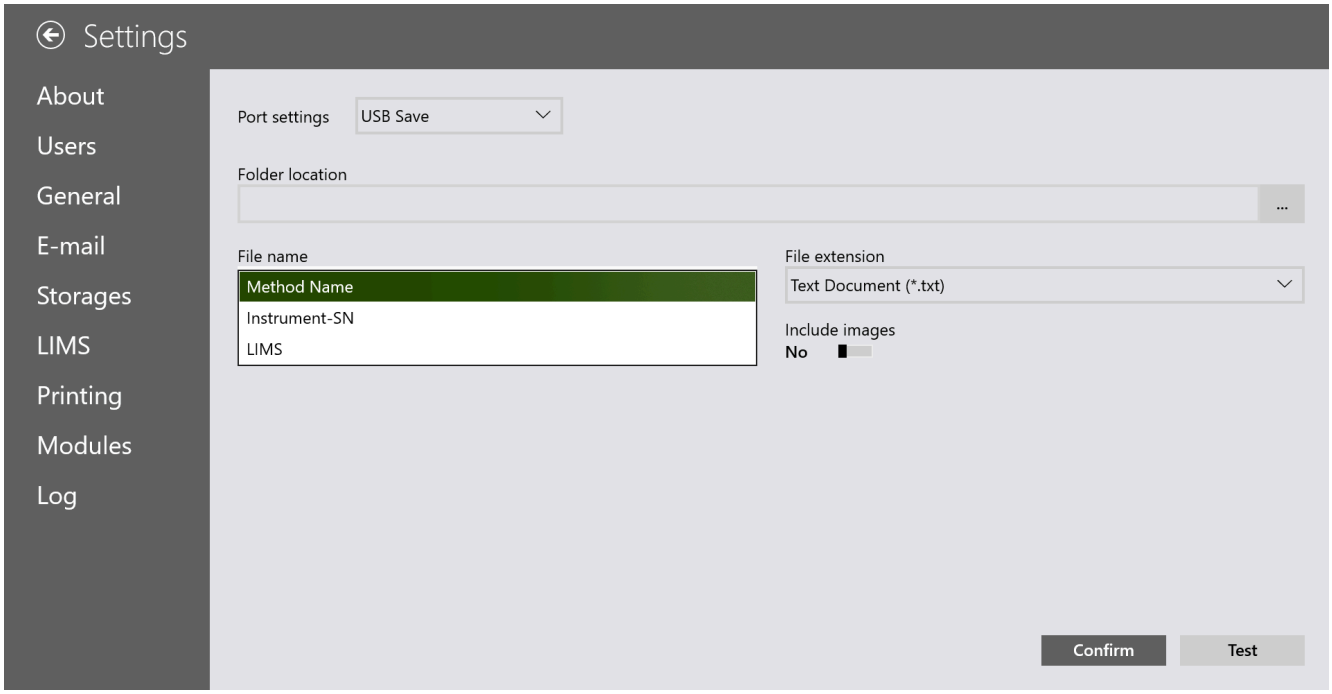


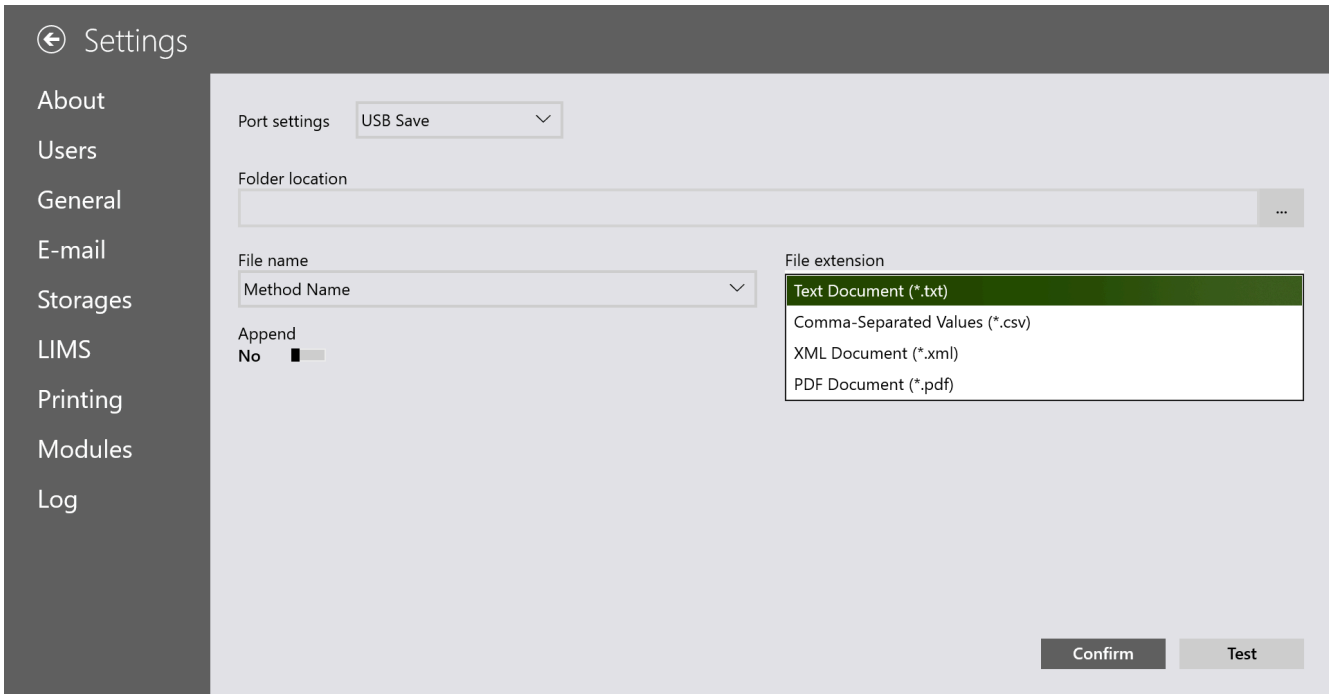
FoamDDI allows the output filename to be customized to suit your laboratory's LIMS format requirements. The following file-naming templates are available:

- Method Name – Uses the test method name (e.g., *ASTM_D892.csv*).
- Instrument-SN – Uses the instrument's serial number as the file name.
- LIMS – Automatically assigns *LIMS* as the file name.

To ensure compatibility with different LIMS systems, FoamDDI supports several export formats selectable via dropdown menu:

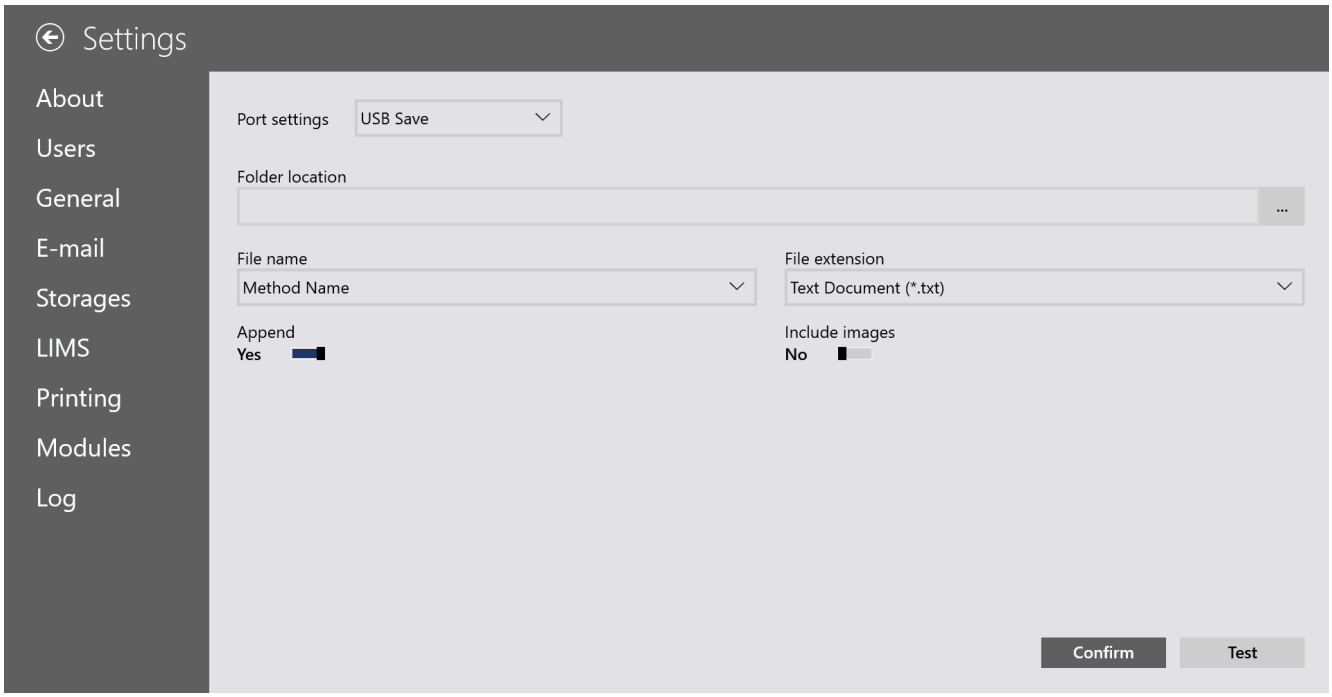
- Text Document (.txt) – Simple plain-text file with comma-delimited fields.
- CSV Format (.csv) – Standard comma-separated values format for spreadsheets or databases.
- XML Format (.xml) – Structured XML file with field identifiers and markup, ideal for automated parsing or custom PDF generation with images.





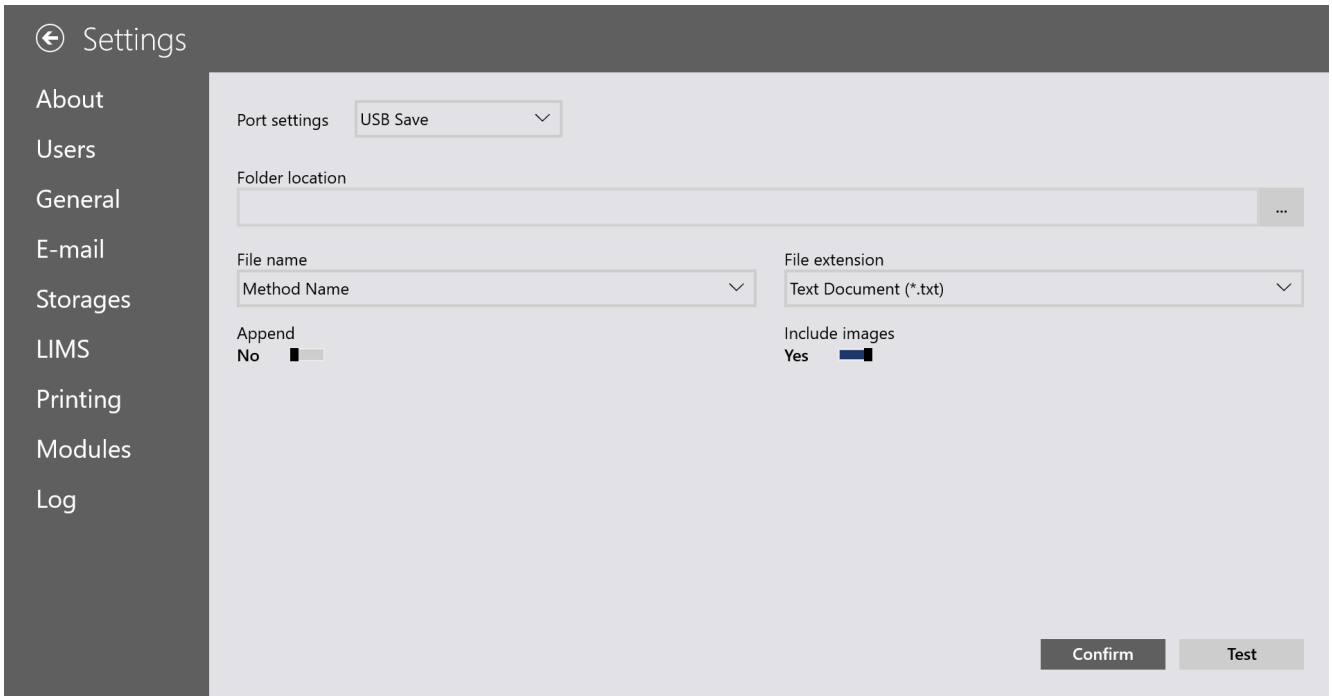
Append Options:

- Append = Yes – Adds each new test as a new data line to an existing file.
- Append = No – Creates a new file for each test result, overwriting any previous file with the same name.

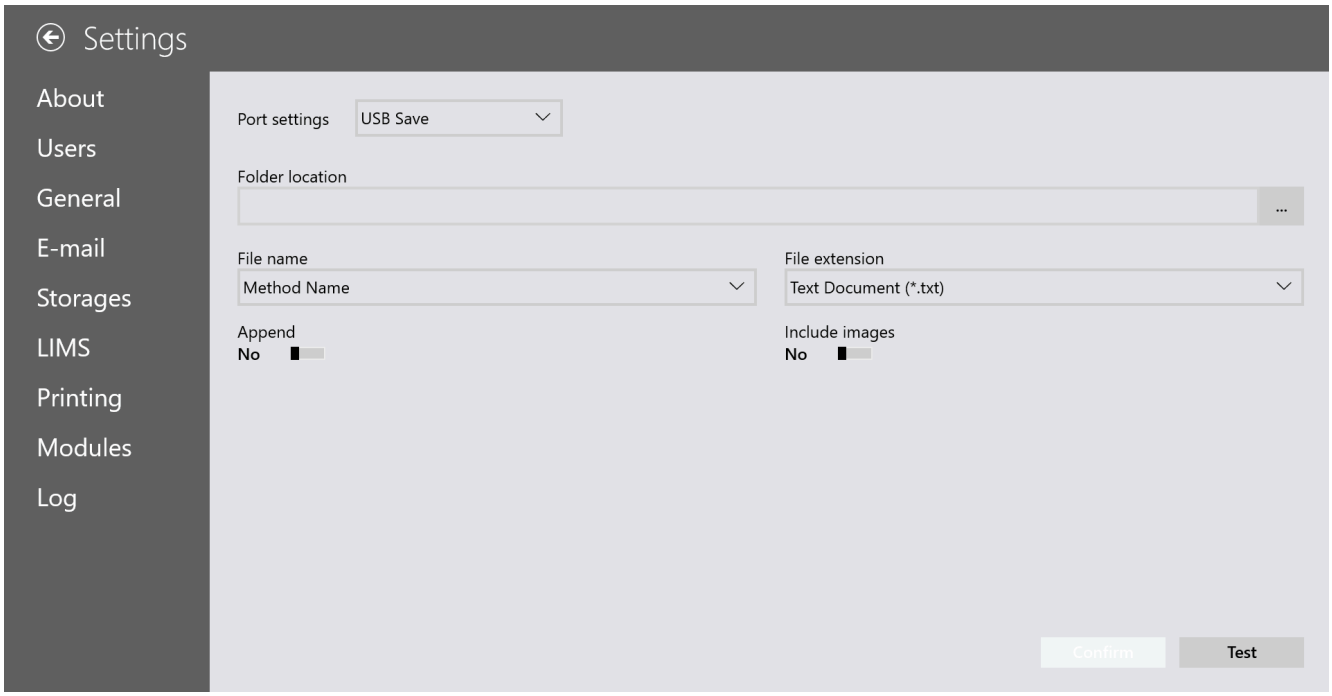


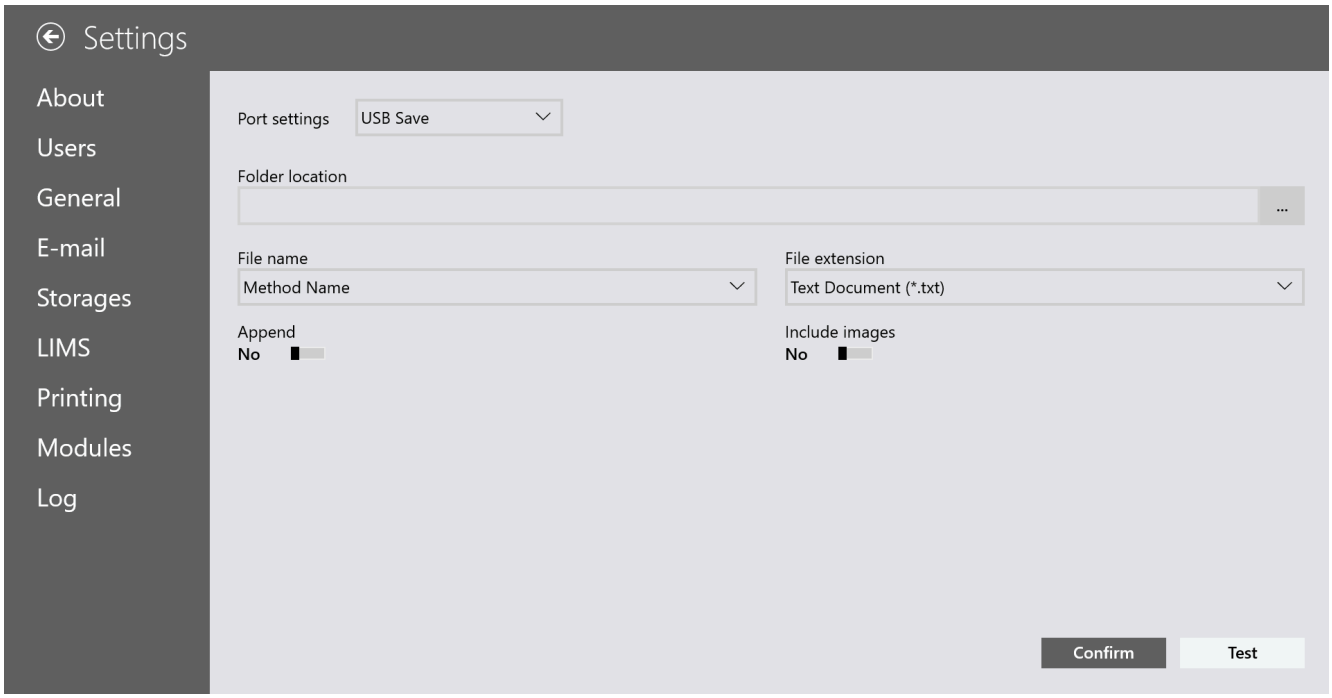
Include Images:

- When set to Yes, FoamDDI will export test images to a separate *Images* folder using a serialized ID.



To save changes, click Confirm. To check the configurable settings were done correctly, click Test and check to make sure the test save was saved correctly.



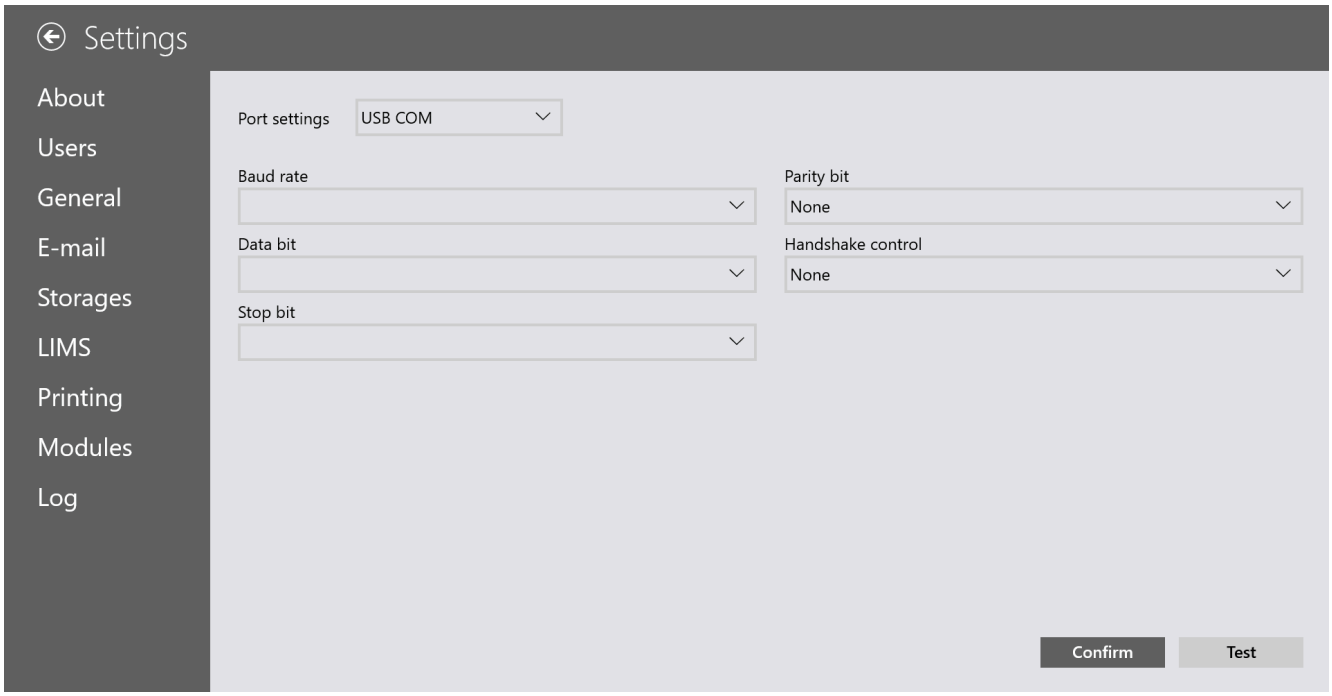


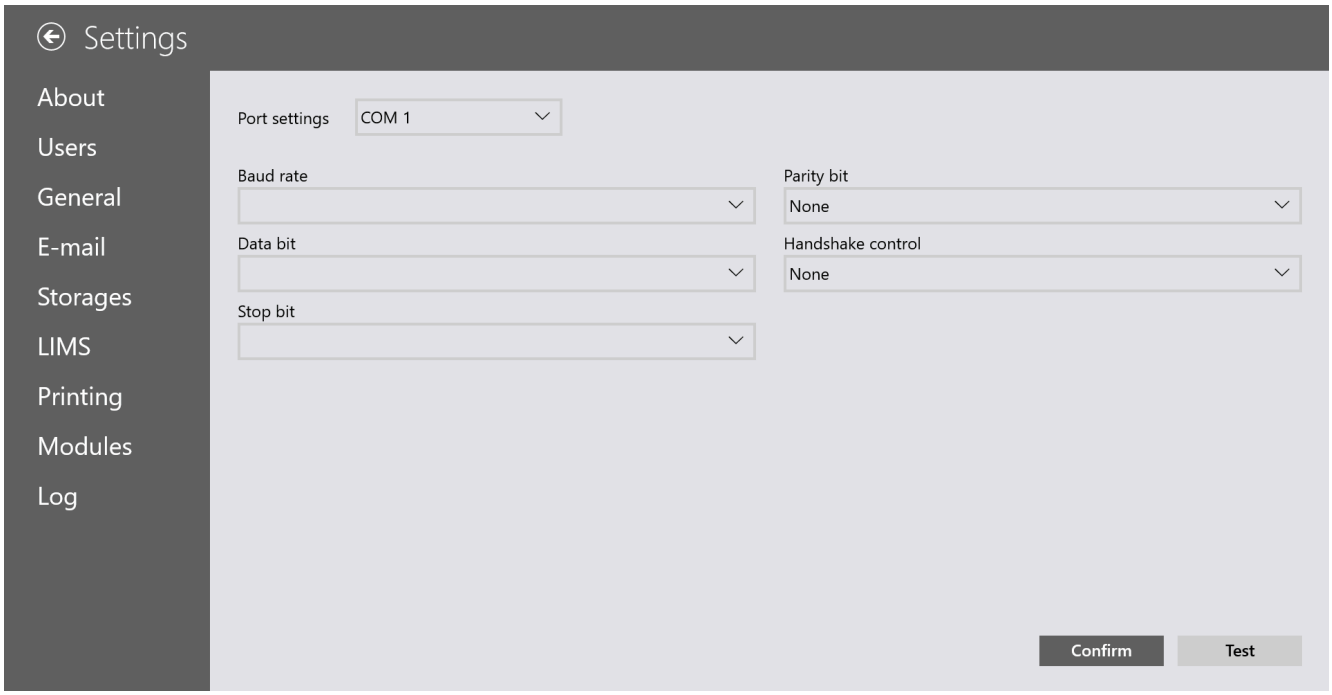
5.4.6.3. USB COM and COM 1

The other two options for LIMS connectivity is direct connection to USB COM or COM 1. Both options require the same information:

- Baud Rate

- Data Bit
- Stop Bit
- Parity Bit
- Handshake Control





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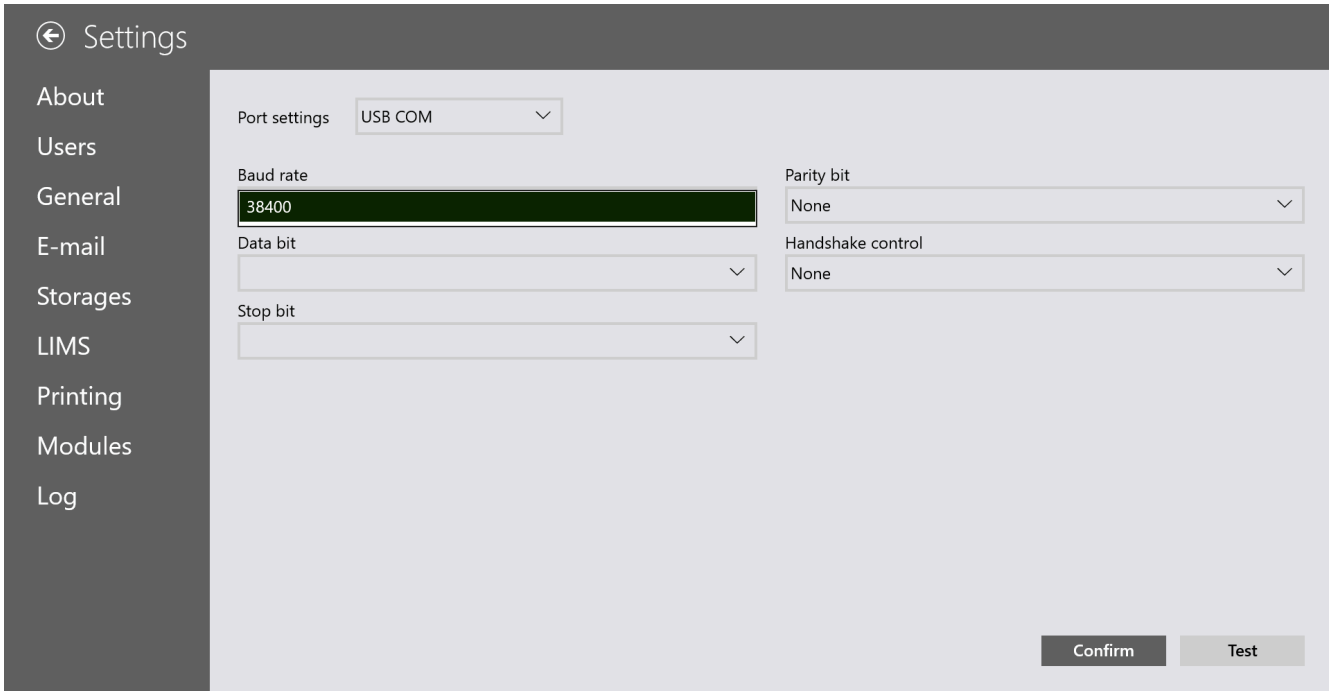
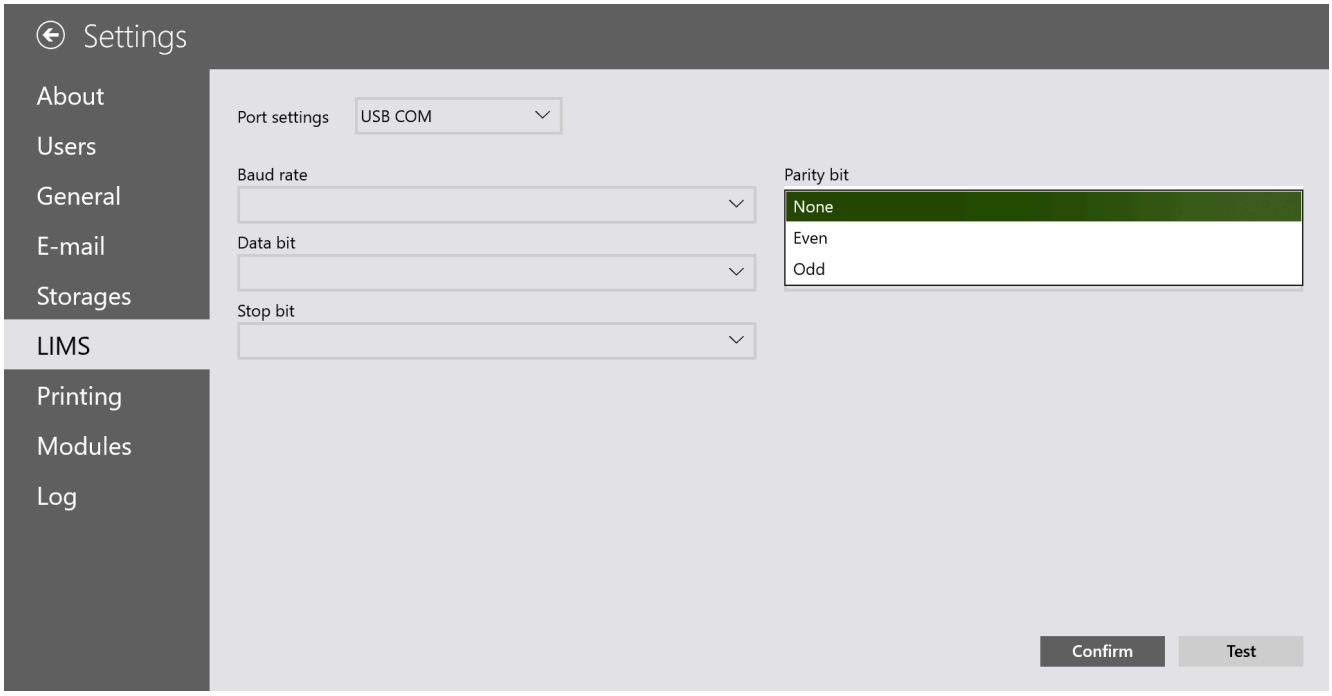
The Baud Rate for the FoamDDI is 38400. Ensure the system is compatible with this Baud Rate.



The Parity Bit options are None, Even or Odd.



The Data Bit options are 7 or 8.

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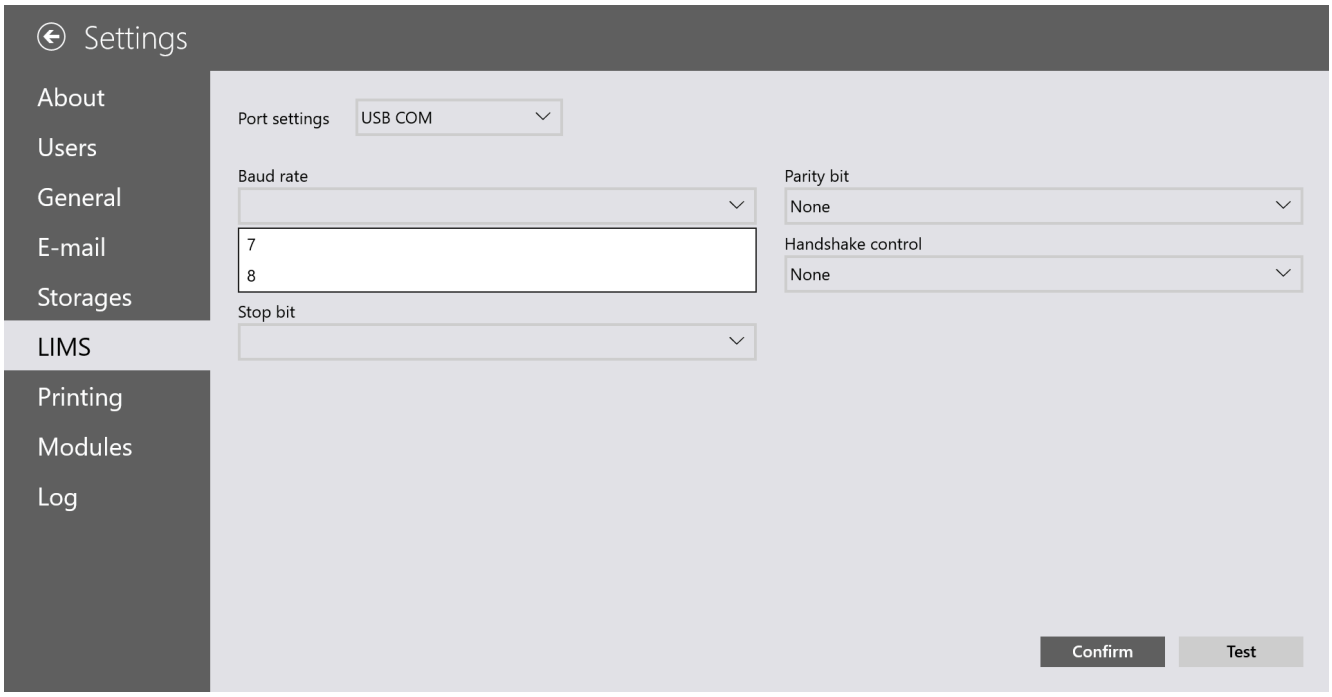
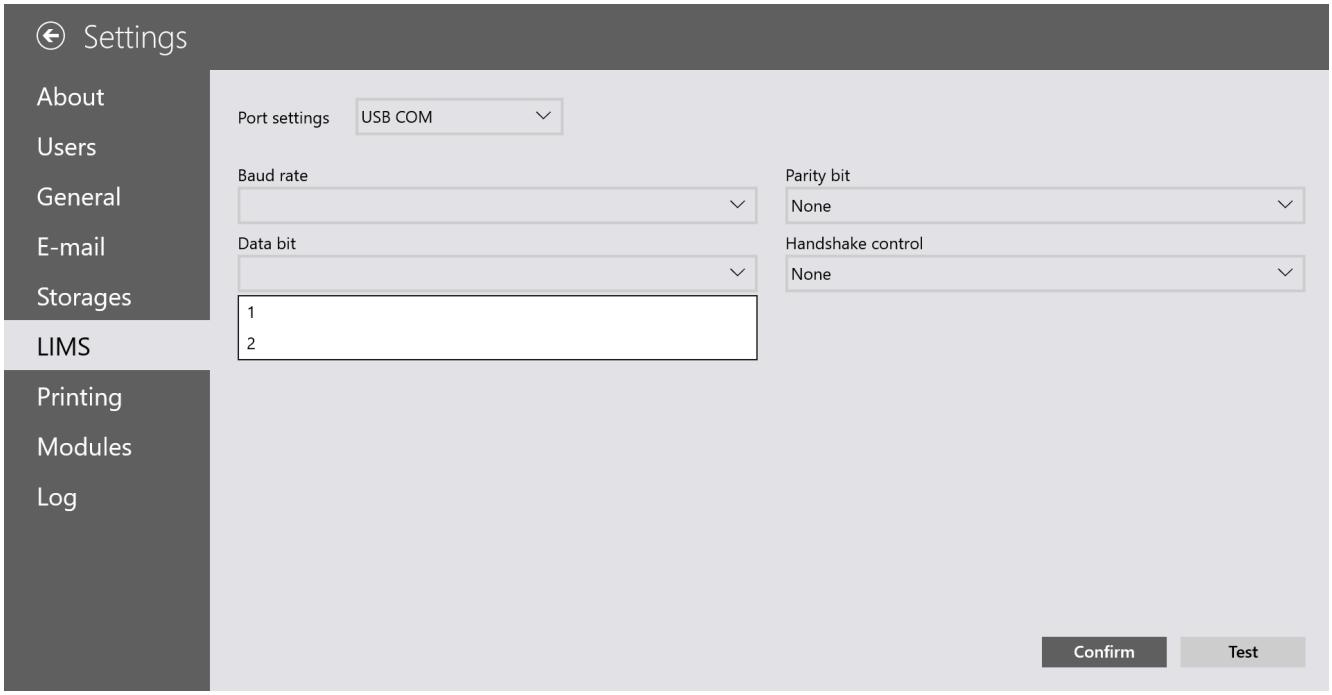
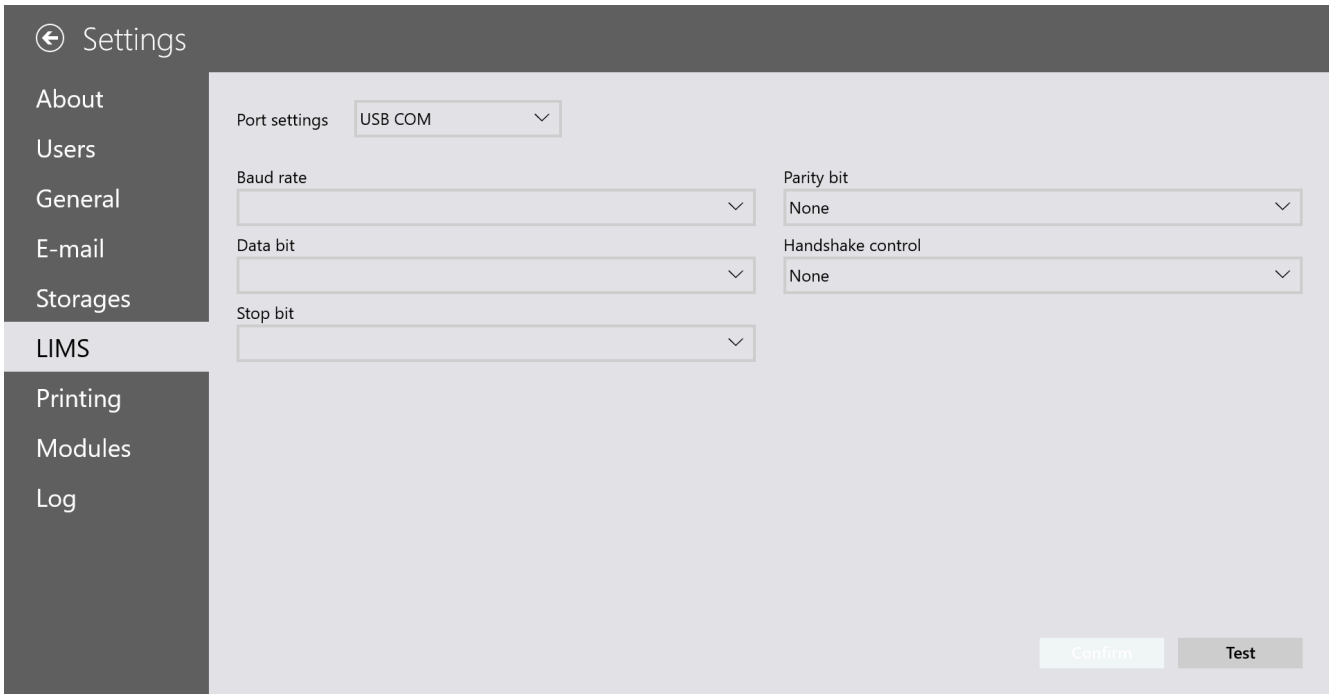
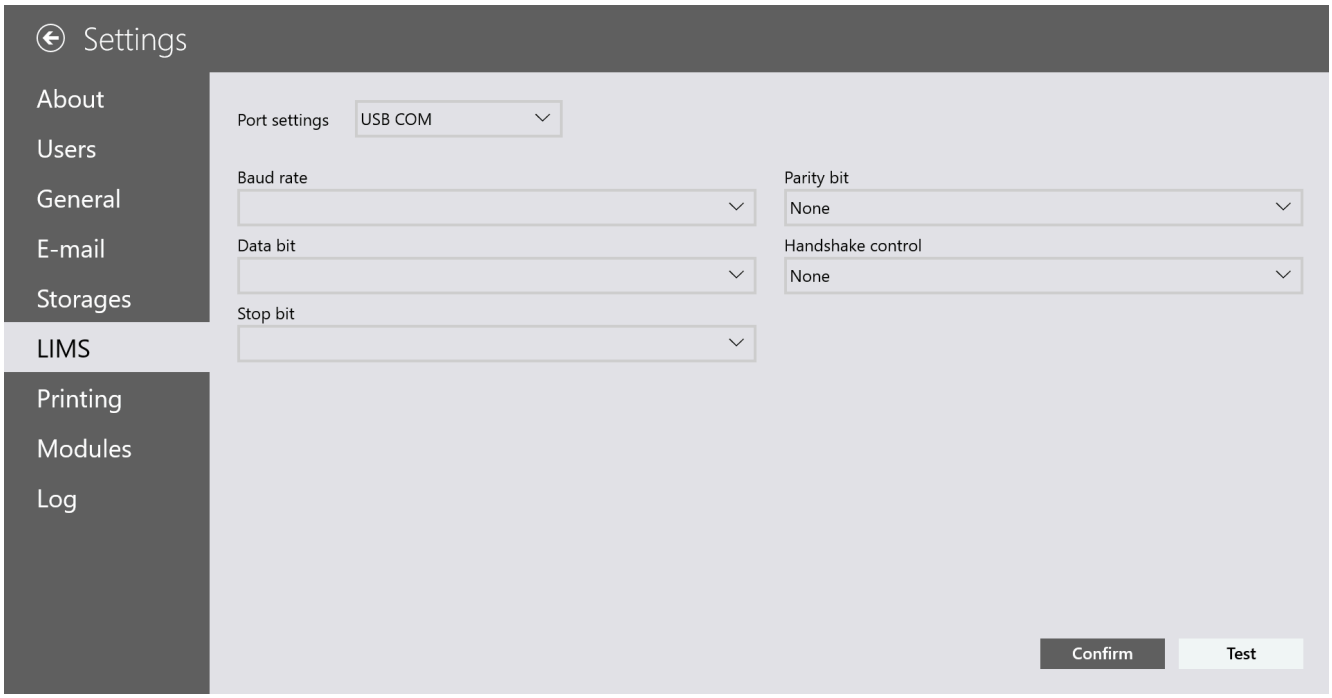


The Stop Bit options are 1 or 2.



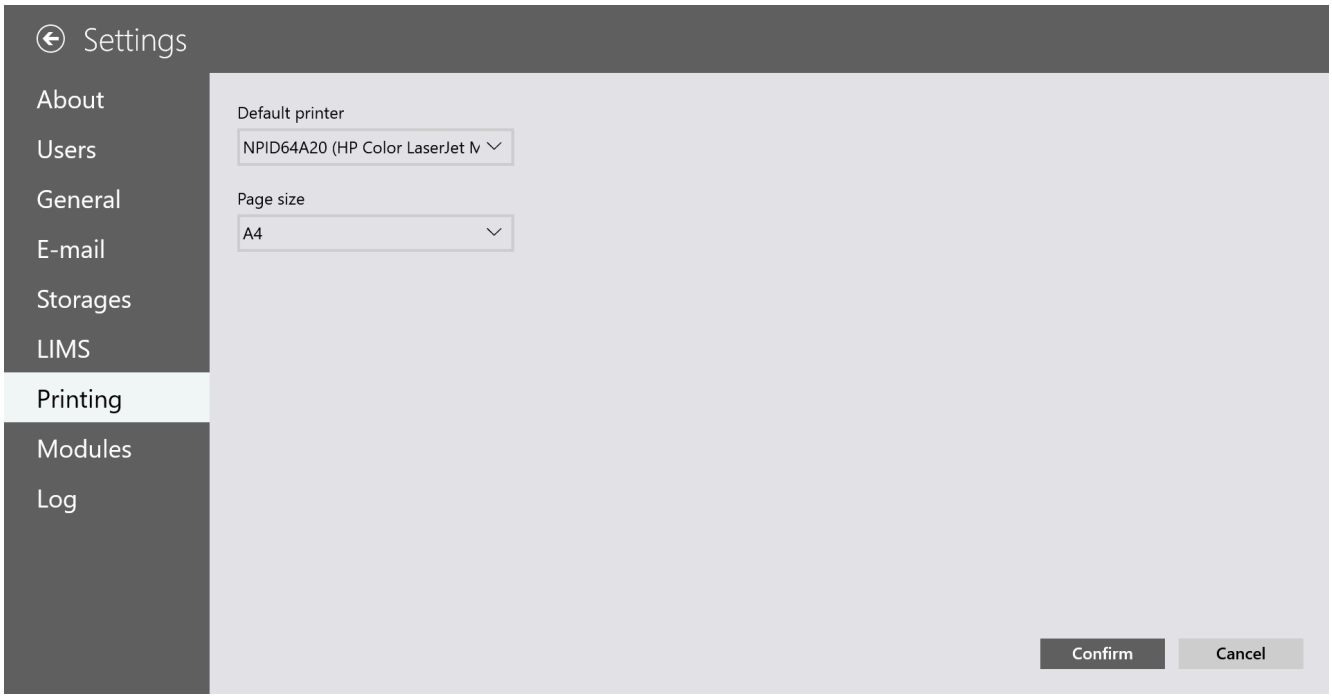
To save changes, click Confirm. To check the configurable settings were done correctly, click Test and check to make sure the test save was saved correctly.





5.4.7. Printing

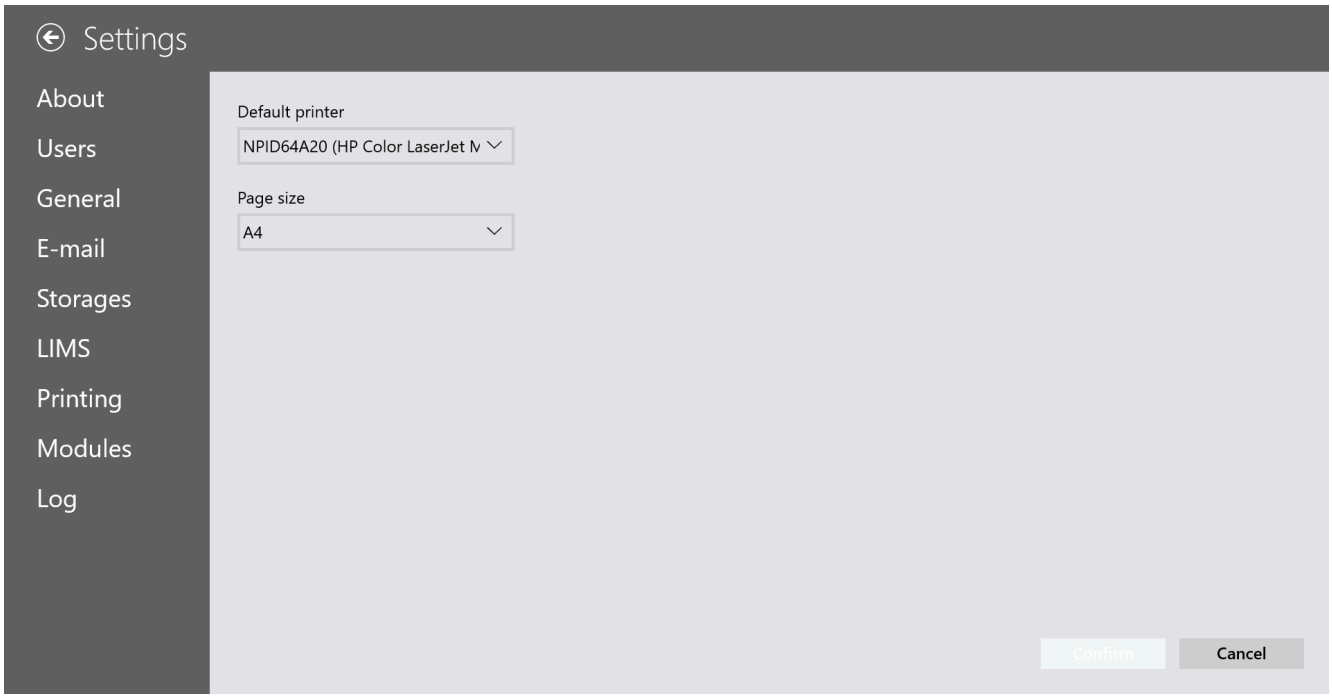
The FoamDDI allows test results to be printed directly after an analysis or at any time through the History view.



Printing can be performed to any network or USB-connected printer that is properly installed and configured.

Note: Some printer setup tasks — such as driver installation or network printer configuration — may require administrator assistance.

Make sure to save any changes by clicking Confirm or erase changes by clicking Cancel.



5.4.7.1. Default Printer

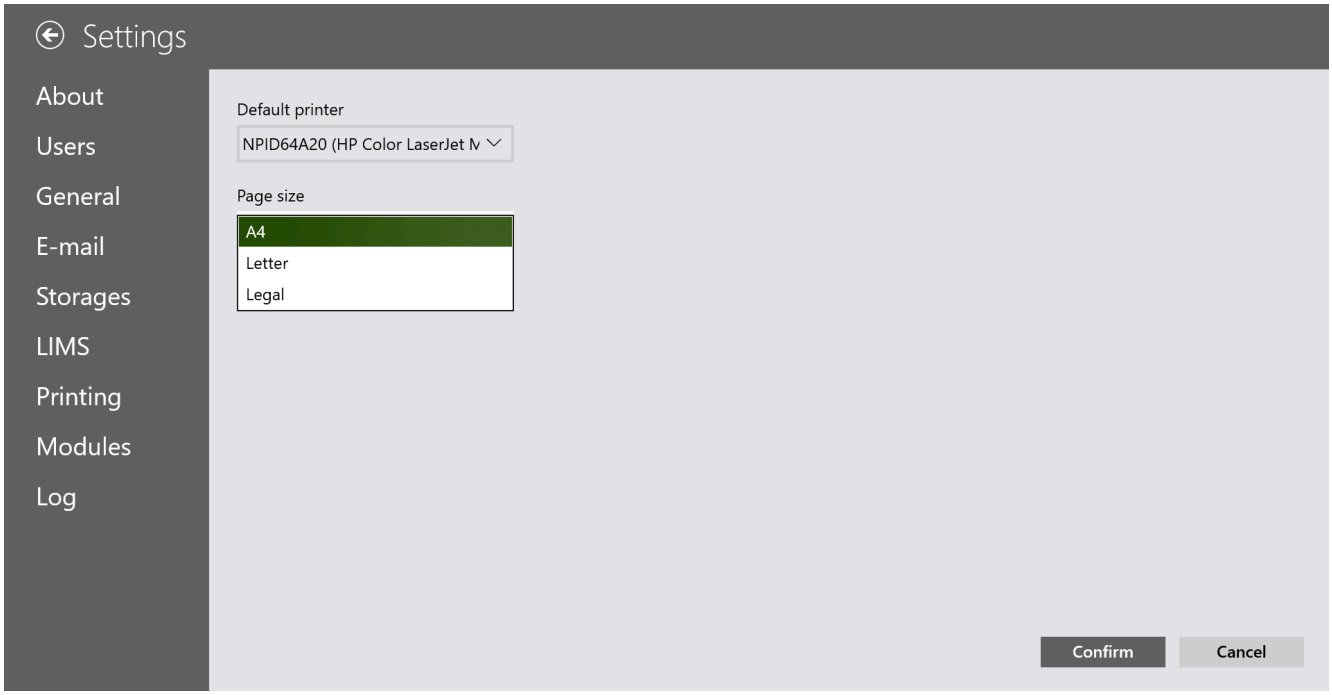
Displays a list of all available printers connected to the system, including both network-shared and locally installed devices.

The operator can select a preferred printer to be used as the default output device.

5.4.7.2. Page Size

Specifies the document size used for printed and PDF reports.

Available options include Letter (8.5" × 11"), Legal (8.5" × 14") and A4 (210 × 297 mm) formats.



5.4.8. Modules

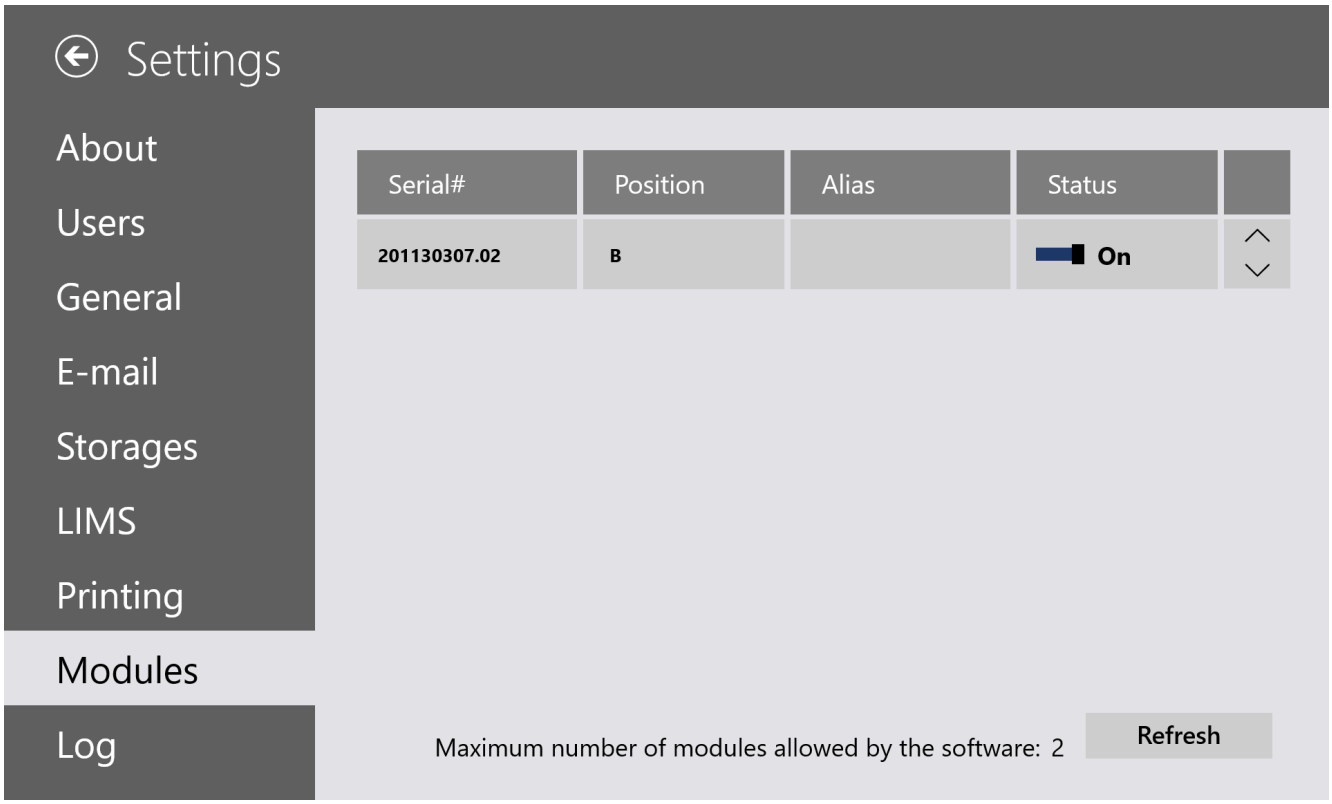
5.4.8.1. Connecting Multiple Modules

The FoamDDI allows the connection of up to two asynchronous modules by default.

To connect a second module, simply use the USB connection built in to the operating system. Connect the module to the back of the logic box and click Refresh in the module tab under Settings.

Morei information on the module tab in Settings can be found below.

The Modules tab is used to view and manage the FoamDDI units currently connected to the system's logic box.



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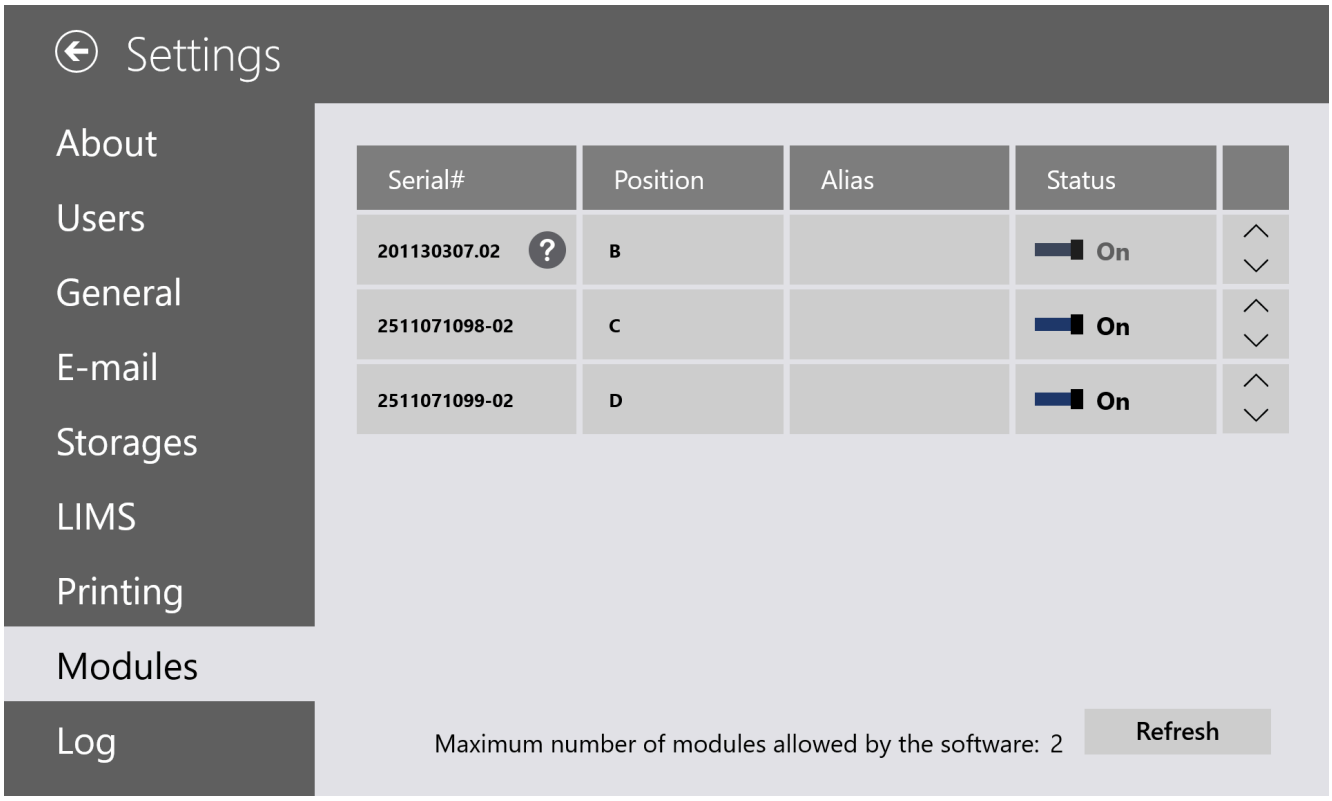

Each connected module is displayed with identifying information and control options.

- Serial # – Displays the serial number of the installed FoamDDI module.
- Position – Indicates the assigned position of the module as recognized by the operating system. The position can be changed using the ↕ (Up/Down) keys.

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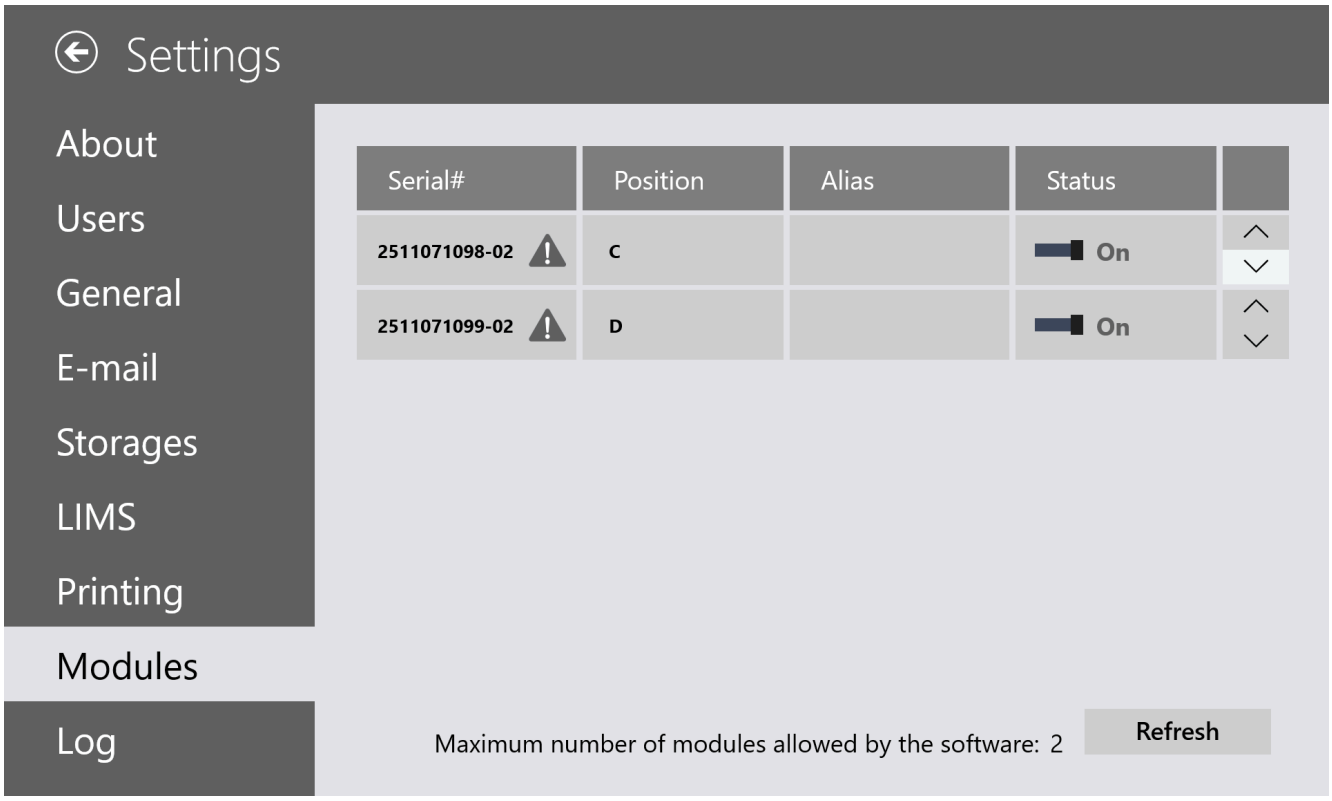

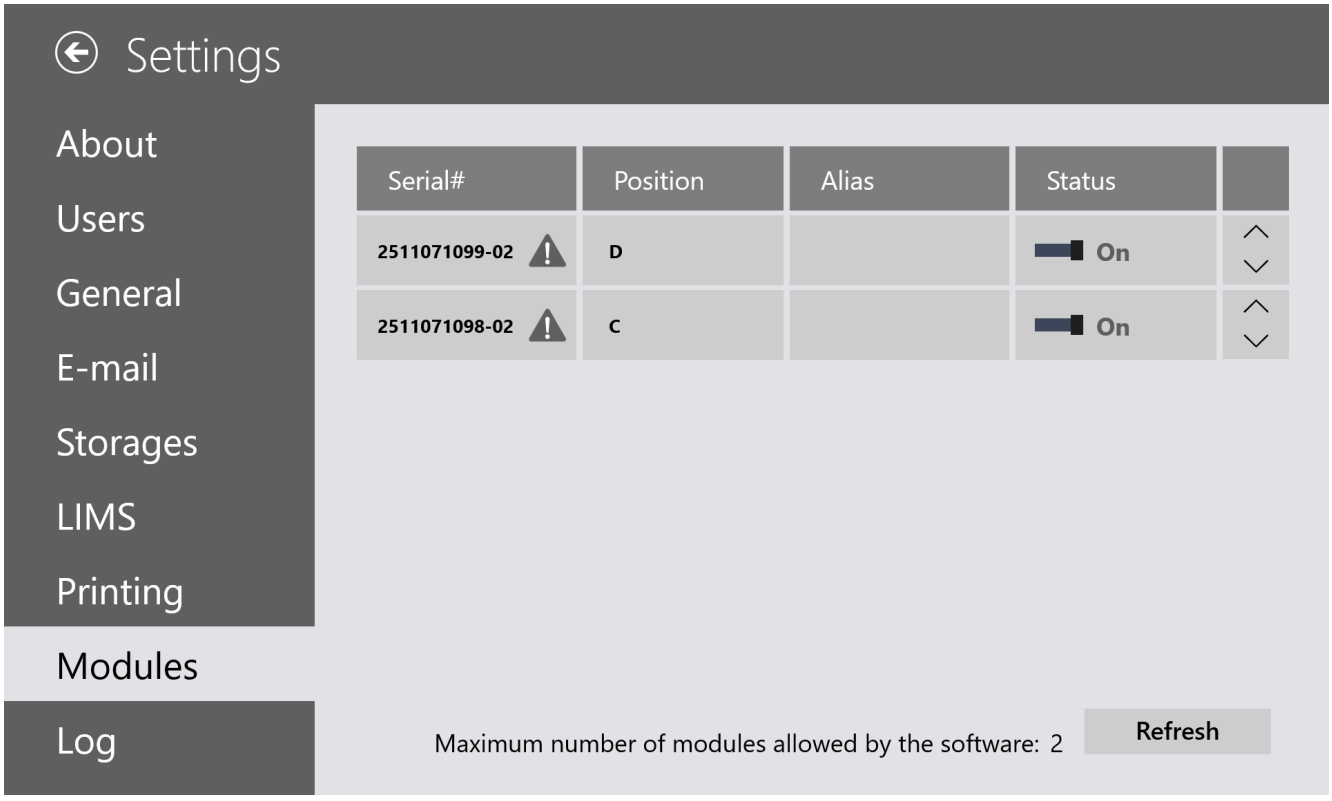

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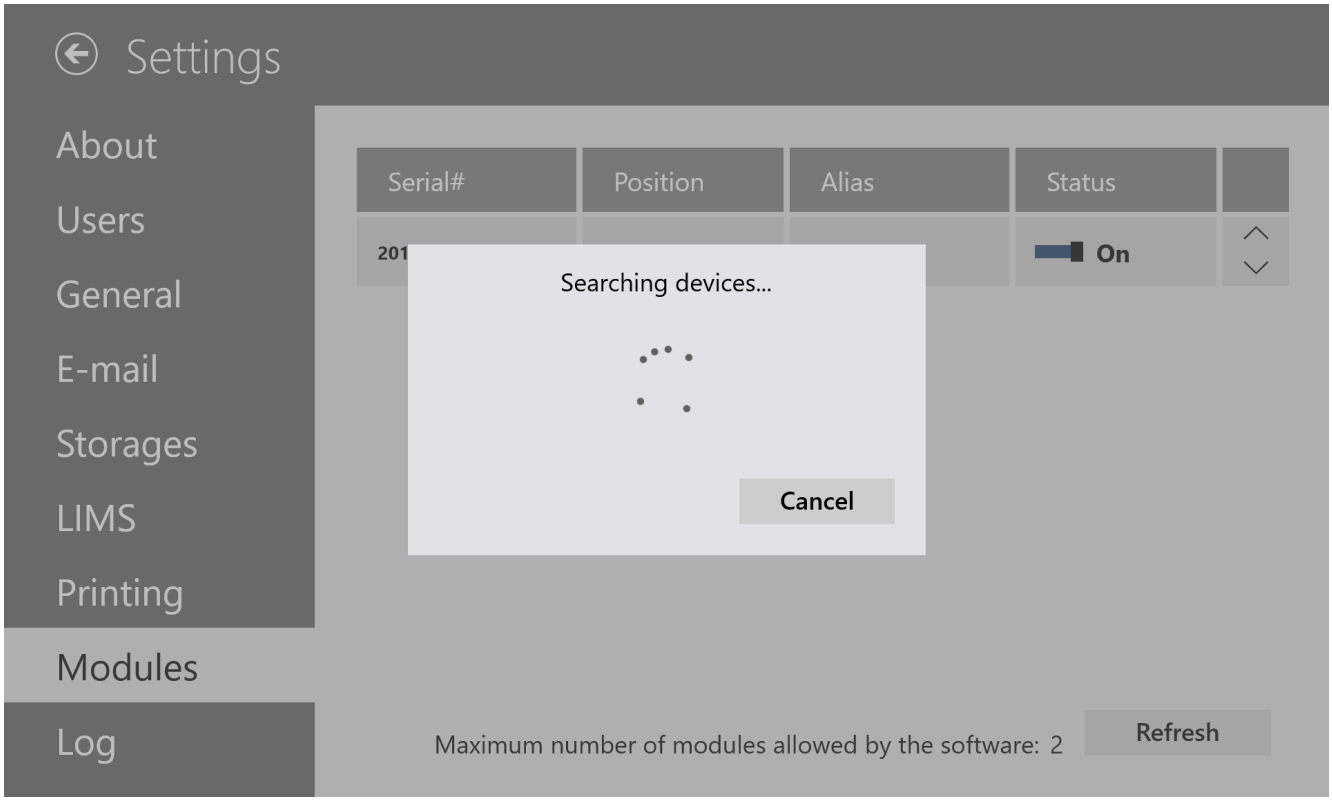
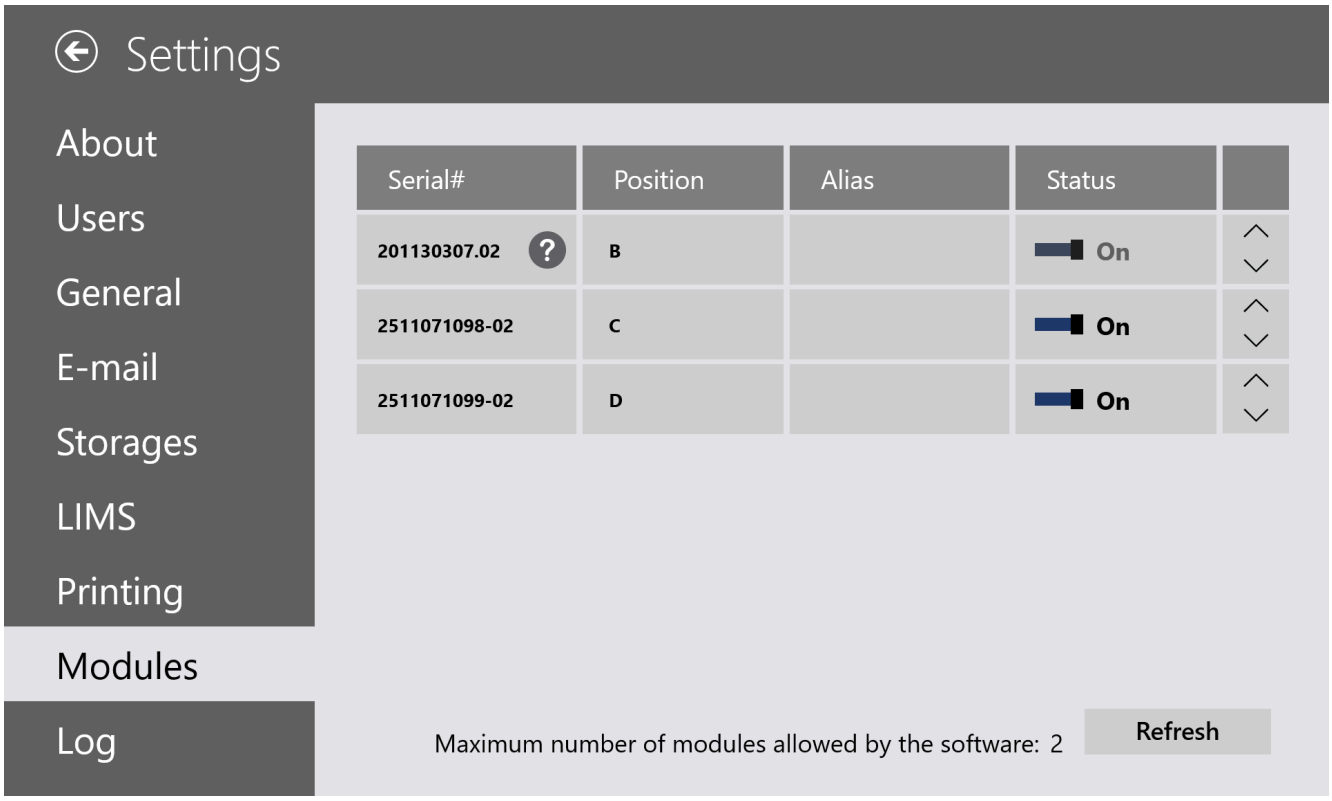
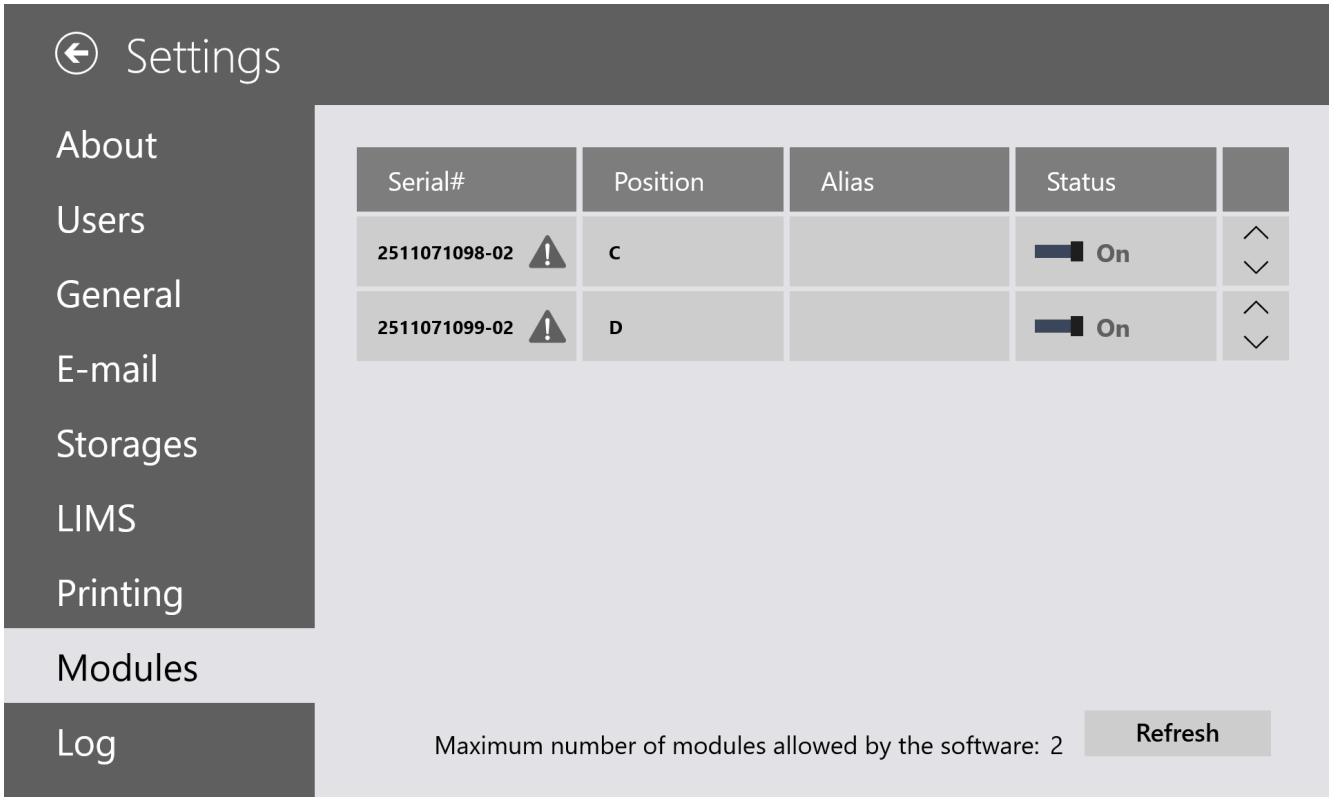
- Alias – Allows the user to assign a custom name to a specific module for easier identification. Click the Alias field to edit or rename the module.
- Status – Displays the current operating status of the module and allows each FoamDDI to be individually turned On or Off.
- Refresh – Initiates a search for active modules and updates the list with any newly detected or reconnected devices.



Any warnings or error messages will also be displayed utilizng a question mark or exclamation icon.





Clicking on the error icon will display more information.

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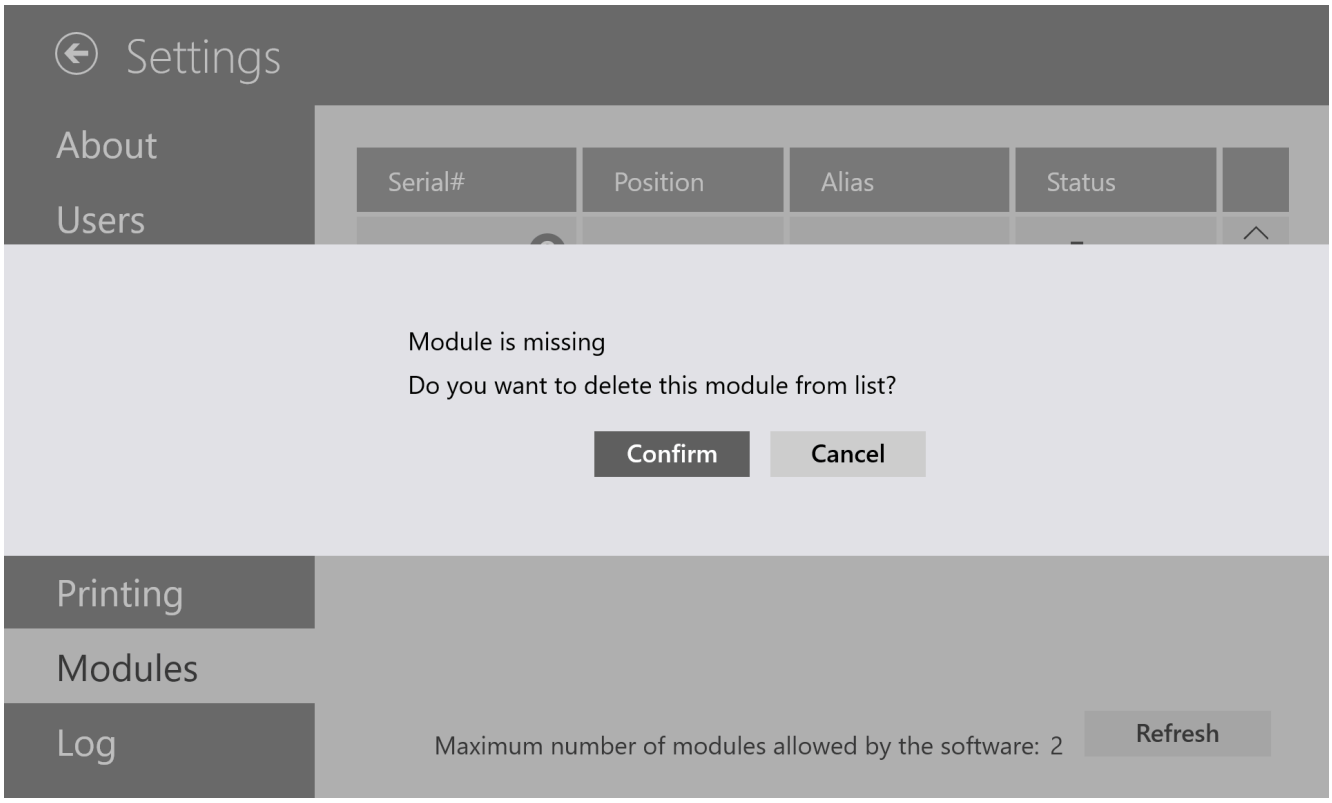

5.4.9. Log

The FoamDDI maintains a system log file that records all communication between the instrument's firmware and software. This log is useful for diagnostics and troubleshooting, and can be viewed, saved, refreshed, or emailed directly from the interface.

The log is useful especially for Visaya engineering. Any communication with Visaya should include the relevant log files.

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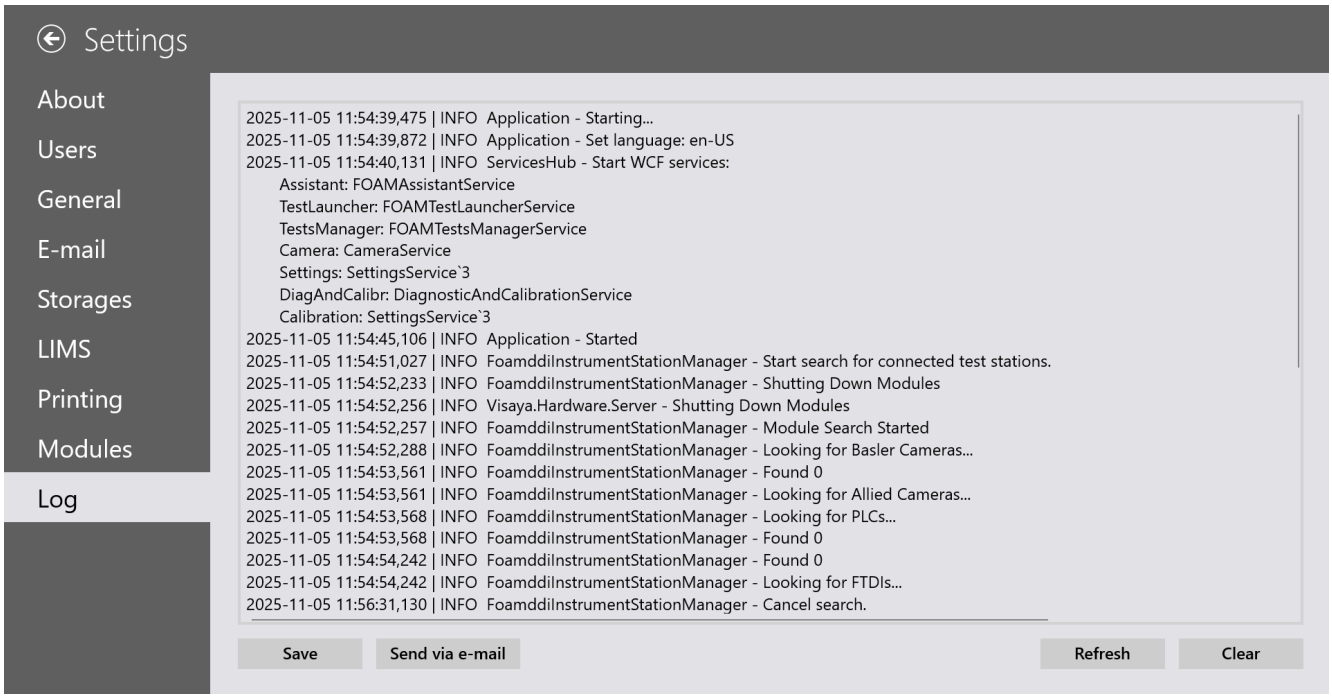
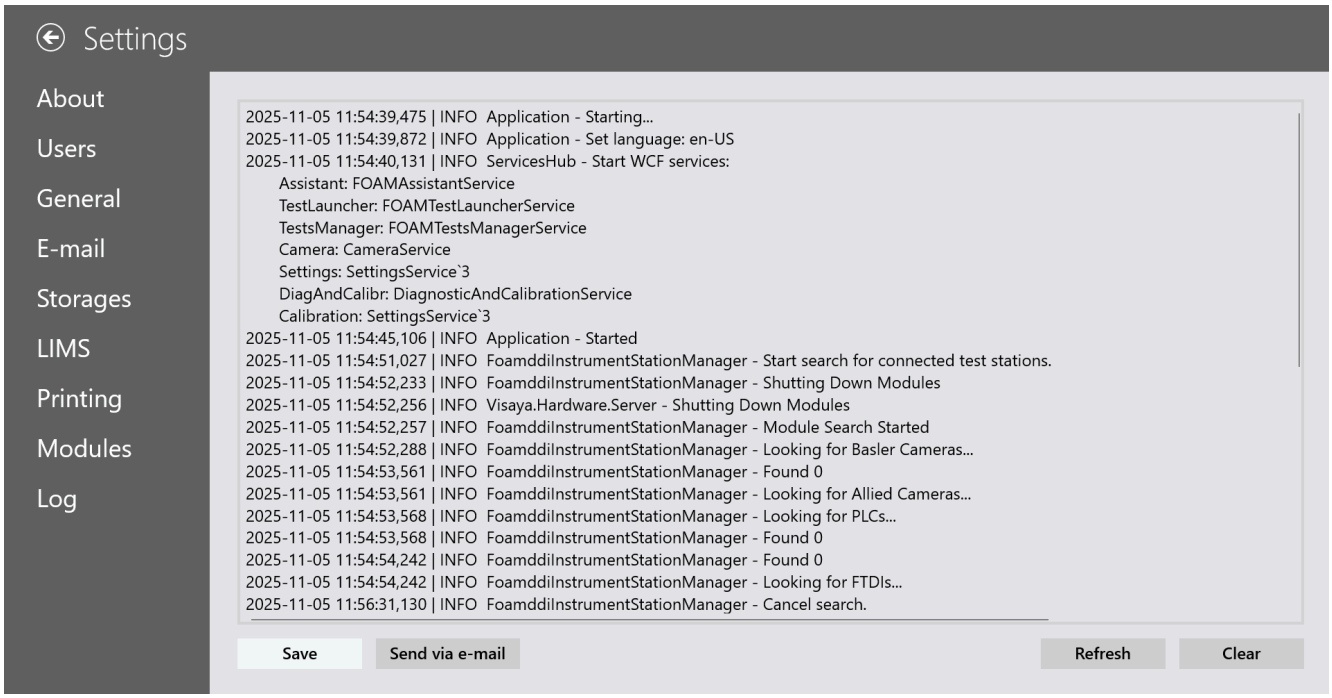


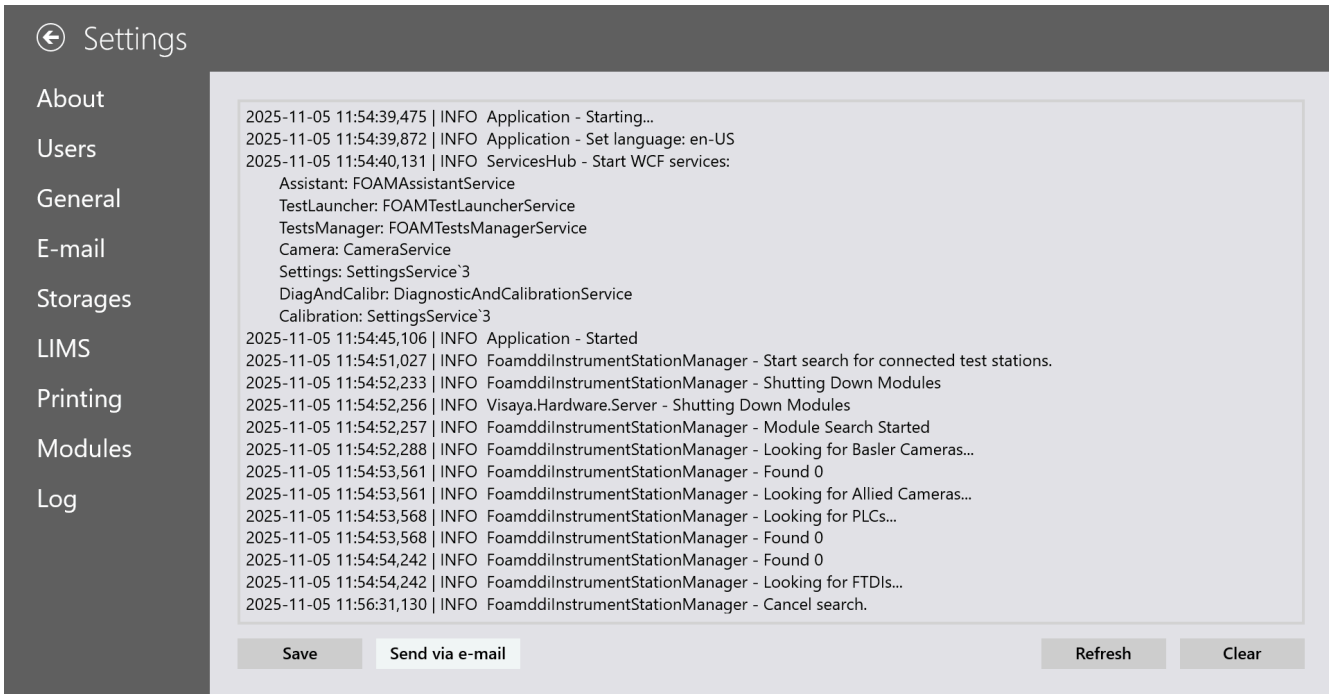
5.4.9.1. Save Log

Saves the current log file to the internal hard drive or to an external USB storage device.



5.4.9.2. Send Log via Email

Opens a dialog box where the operator can enter an email address to send the log file as an attachment.



Note: To use this feature, the Email Settings must be configured. Refer to Email Configuration for setup details.

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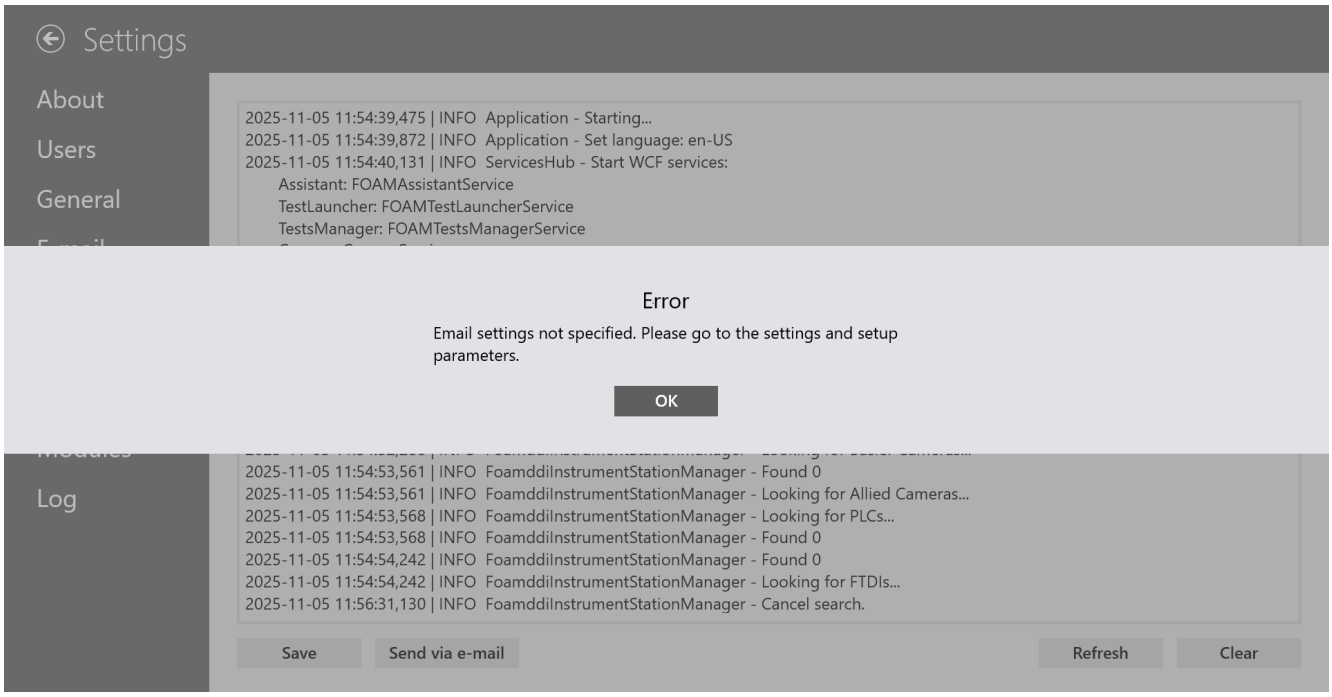
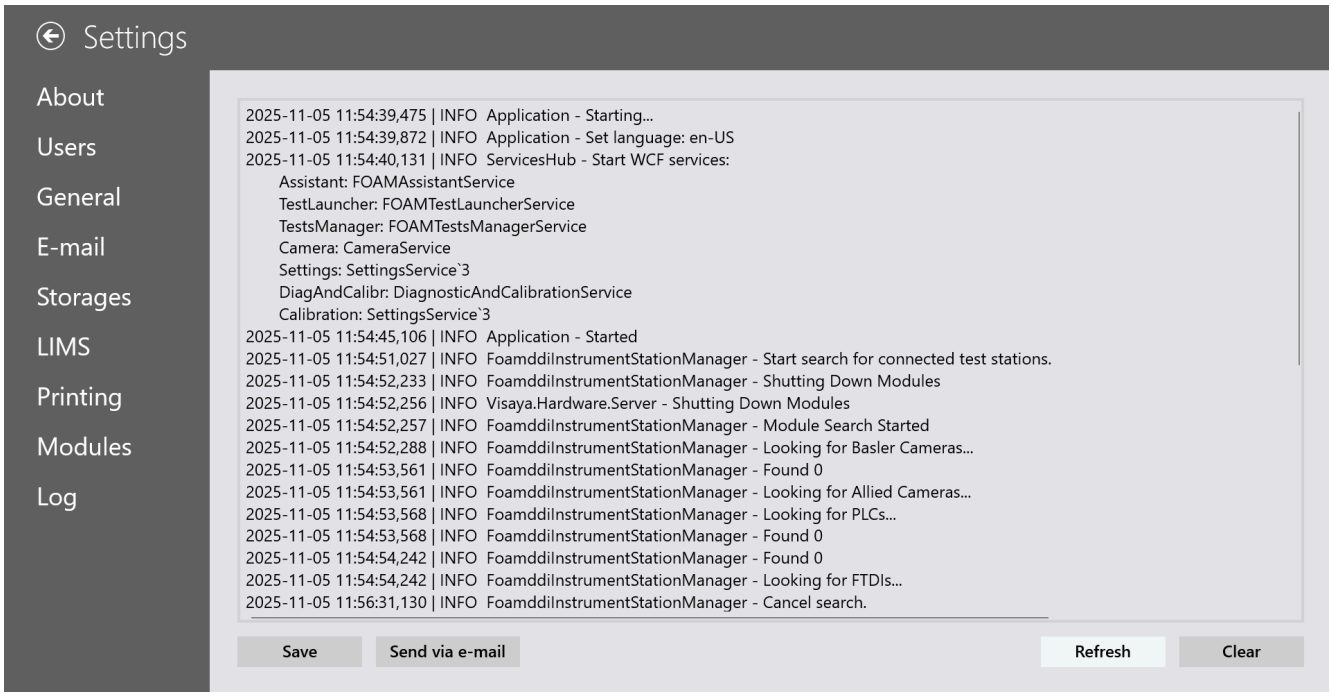


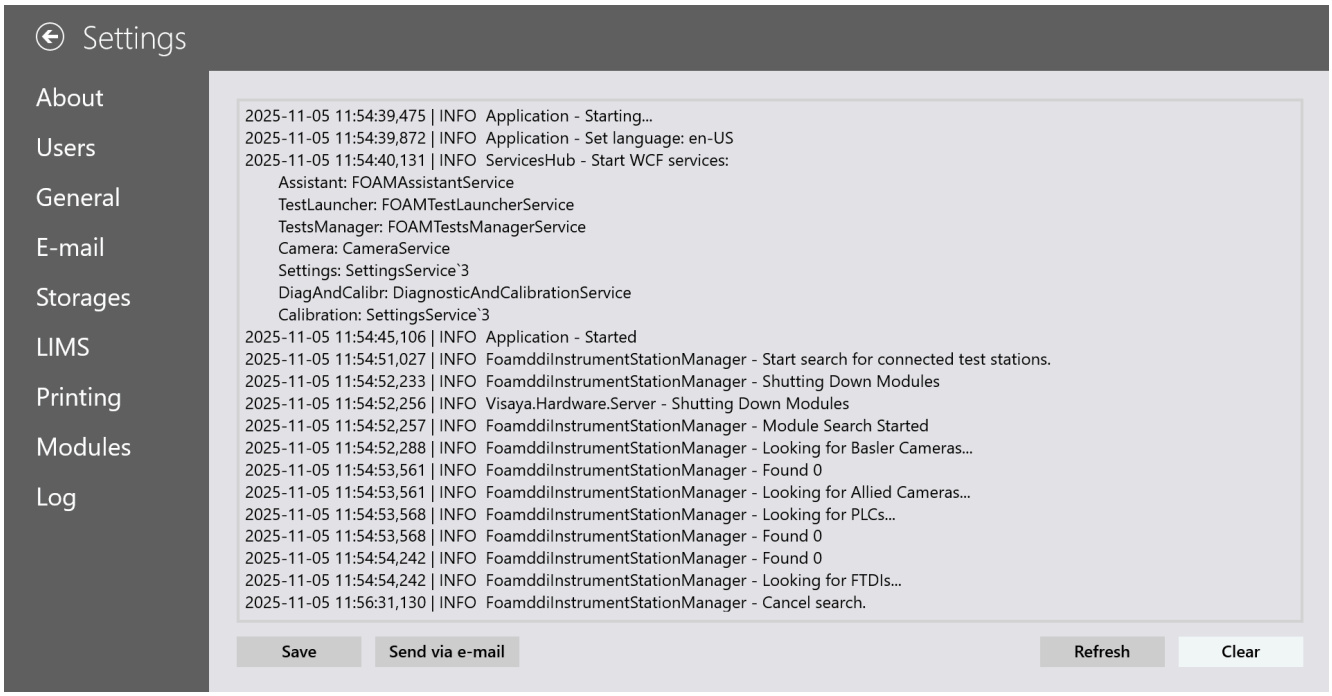
5.4.9.3. Refresh Log

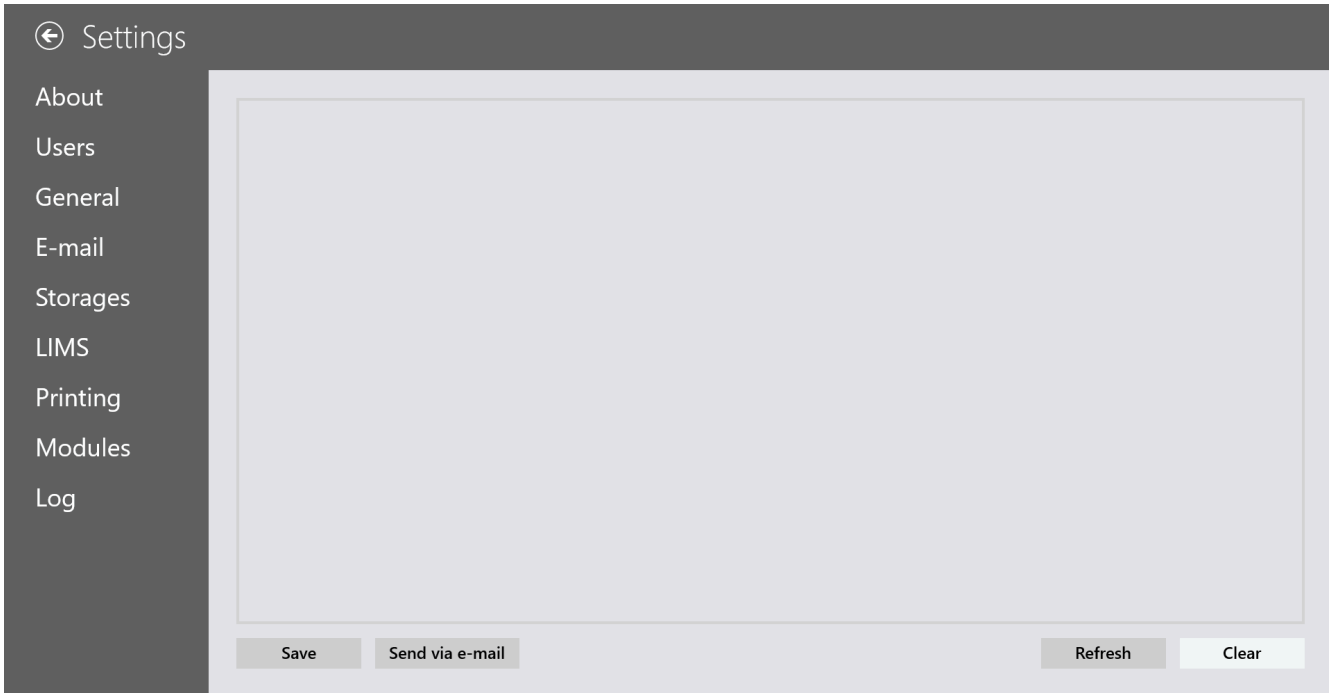
Updates the display to include the most recent log entries generated by the system.



5.4.9.4. Clear Log

Clears the on-screen log display and resets the view to show only the newest entries as they are created.

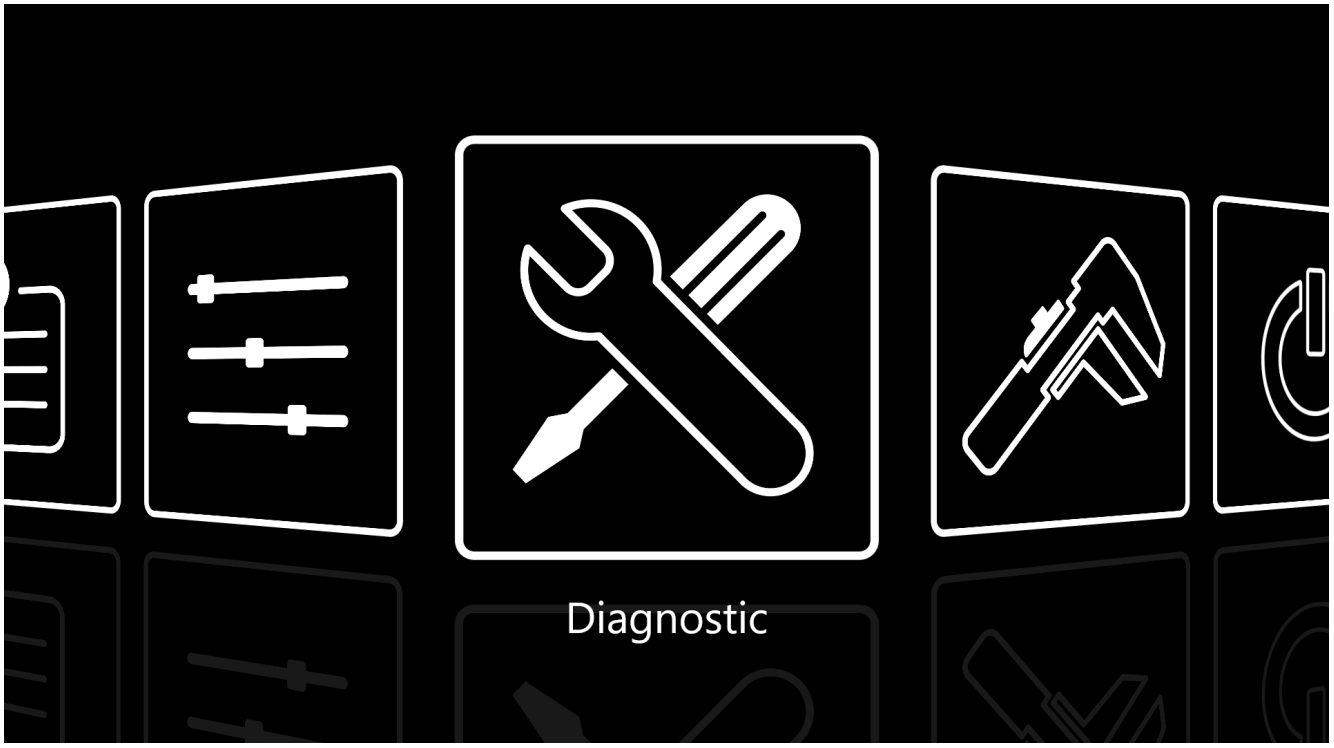




5.5. Diagnostics

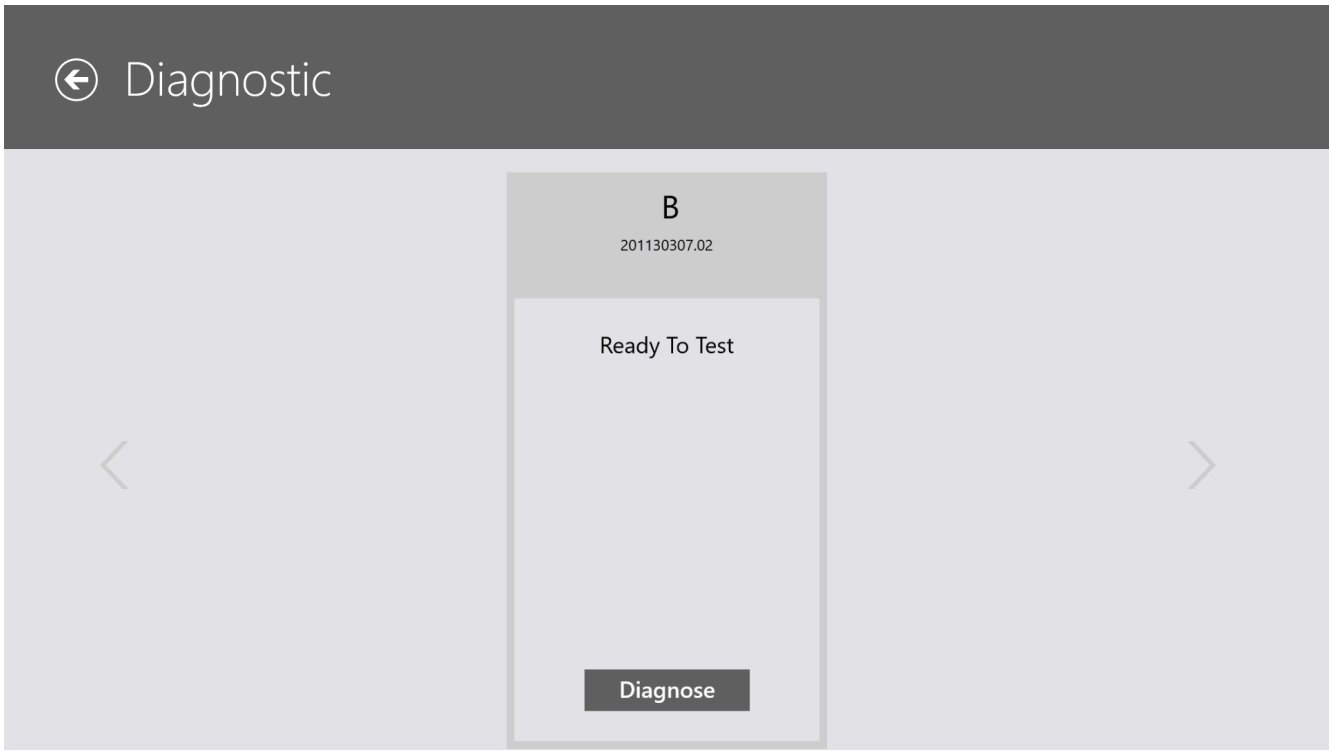
The FoamDDI software includes an easy-to-use diagnostics screen designed for real-time system health monitoring, troubleshooting, and manual verification of the instrument's electro-mechanical components.

This screen is used to identify the source of a fault, manually operate key devices, and run individual component tests to confirm proper system performance.

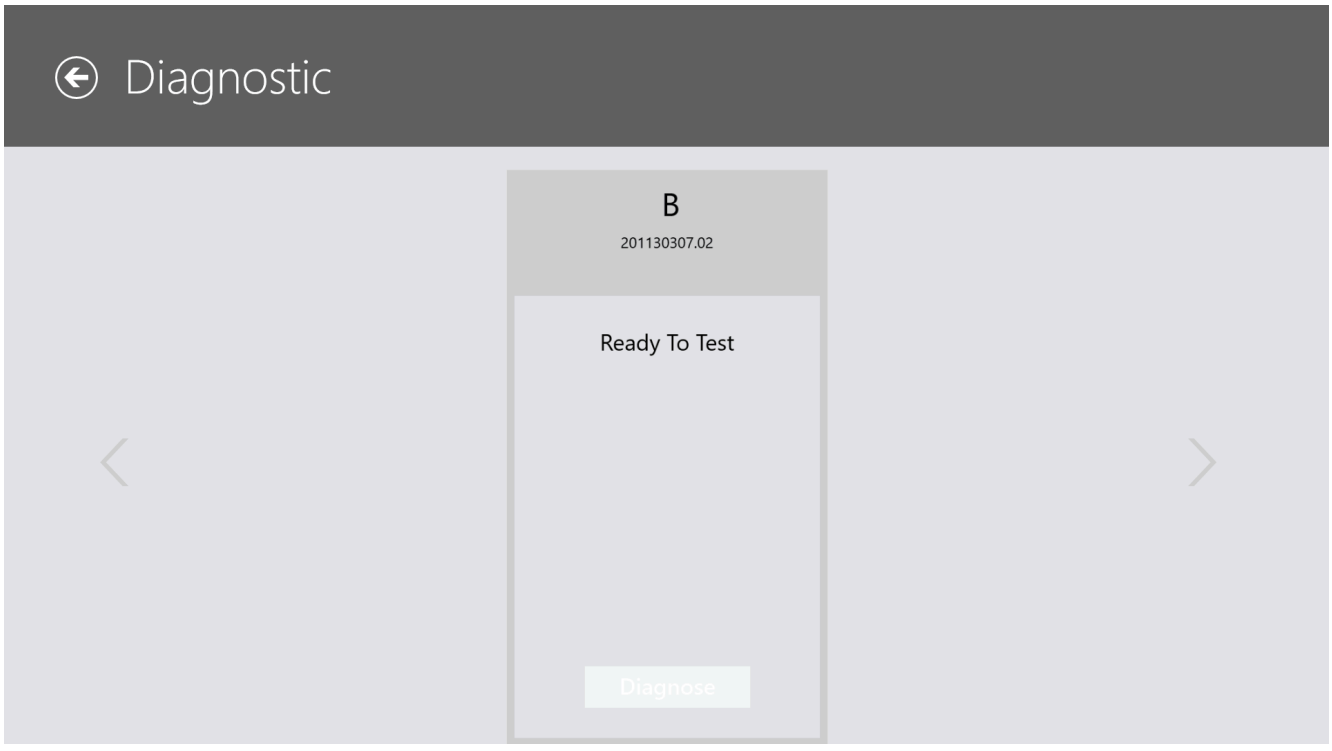


The diagnostics screen displays similarly to the inspection and calibration screens.

The connected modules will be displayed. If multiple modules are connected, they will both be displayed.



Click Diagnose on the module desired (if multiple are connected).



The Reset All Devices turns off any of the activated settings below.

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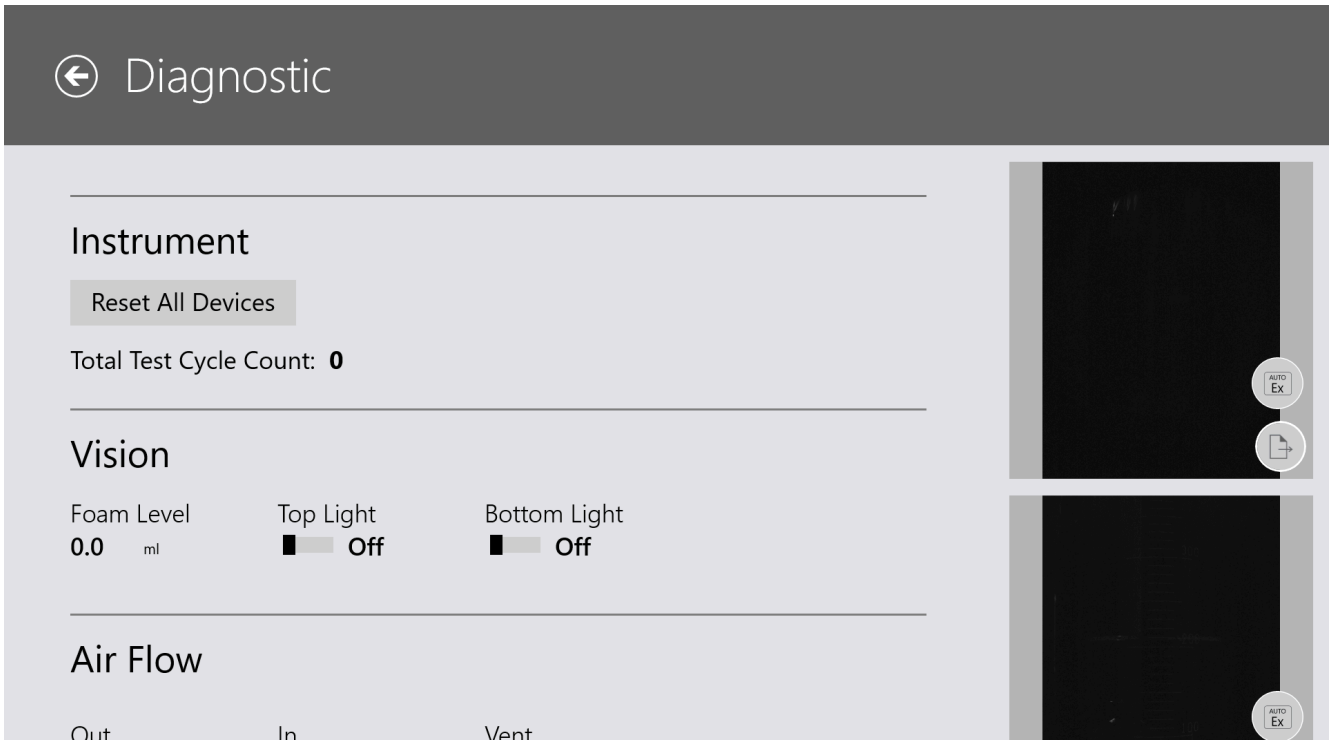

5.5.1. Vision

Displays a live camera feed from both cameras on the FoamDDI, updating several times per second.

This section is used to verify camera communication, image alignment, and lighting conditions.

Use the Top Light and Bottom Light buttons to manually toggle illumination.

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Diagnostic

Instrument

Reset All Devices

Total Test Cycle Count: **0**

Vision

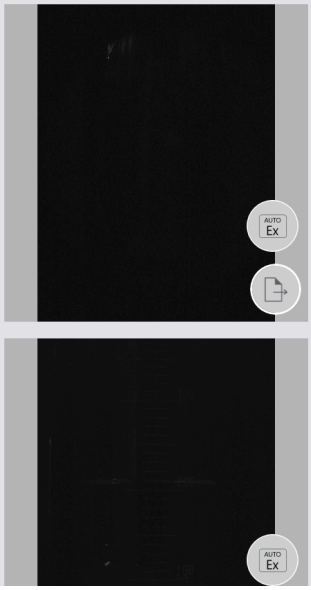
Foam Level
0.0 ml

Top Light
Off

Bottom Light
Off

Air Flow

Out In Vent



Diagnostic

Instrument

Reset All Devices

Total Test Cycle Count: **0**

Vision

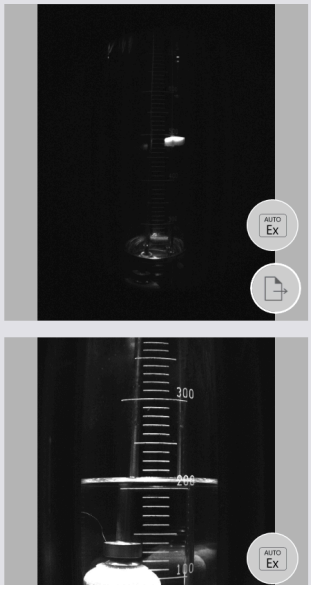
Foam Level
0.0 ml

Top Light
On

Bottom Light
Off

Air Flow

Out In Vent



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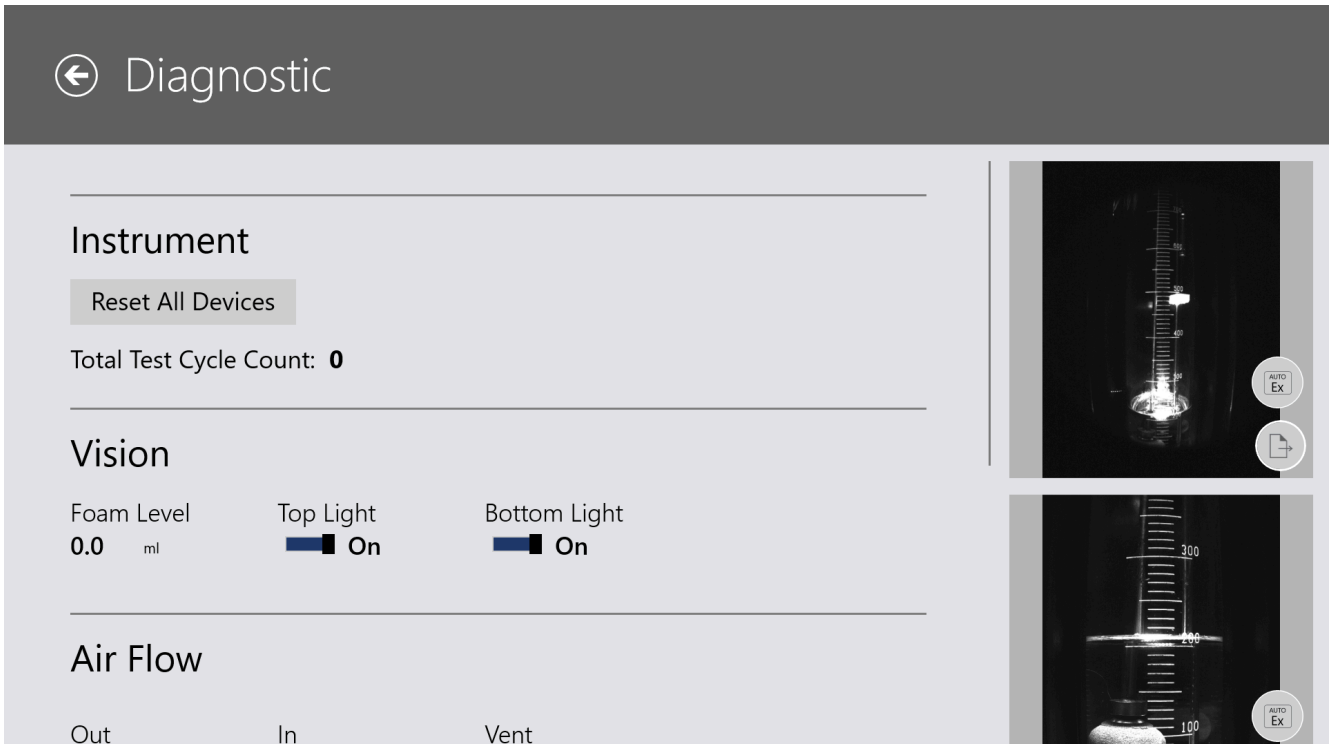
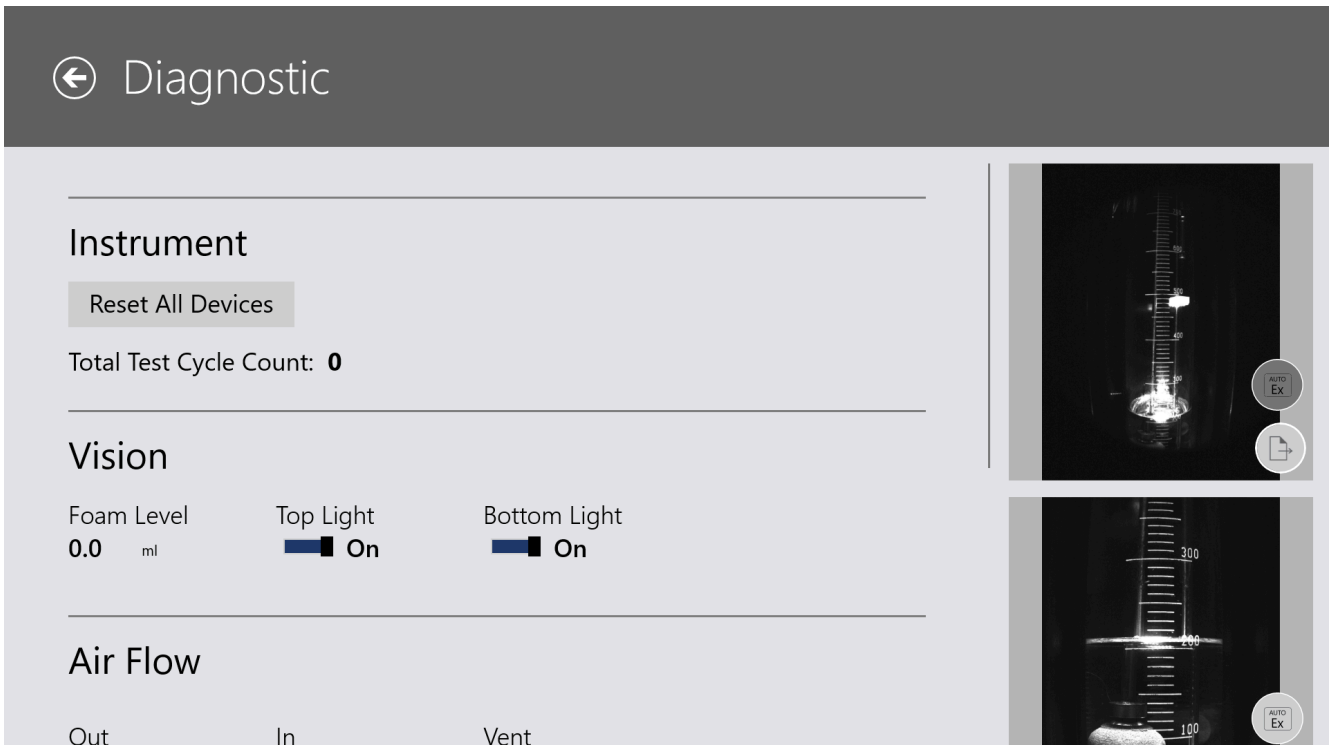


The Auto Exposure button automatically adjusts brightness (enabled by default).



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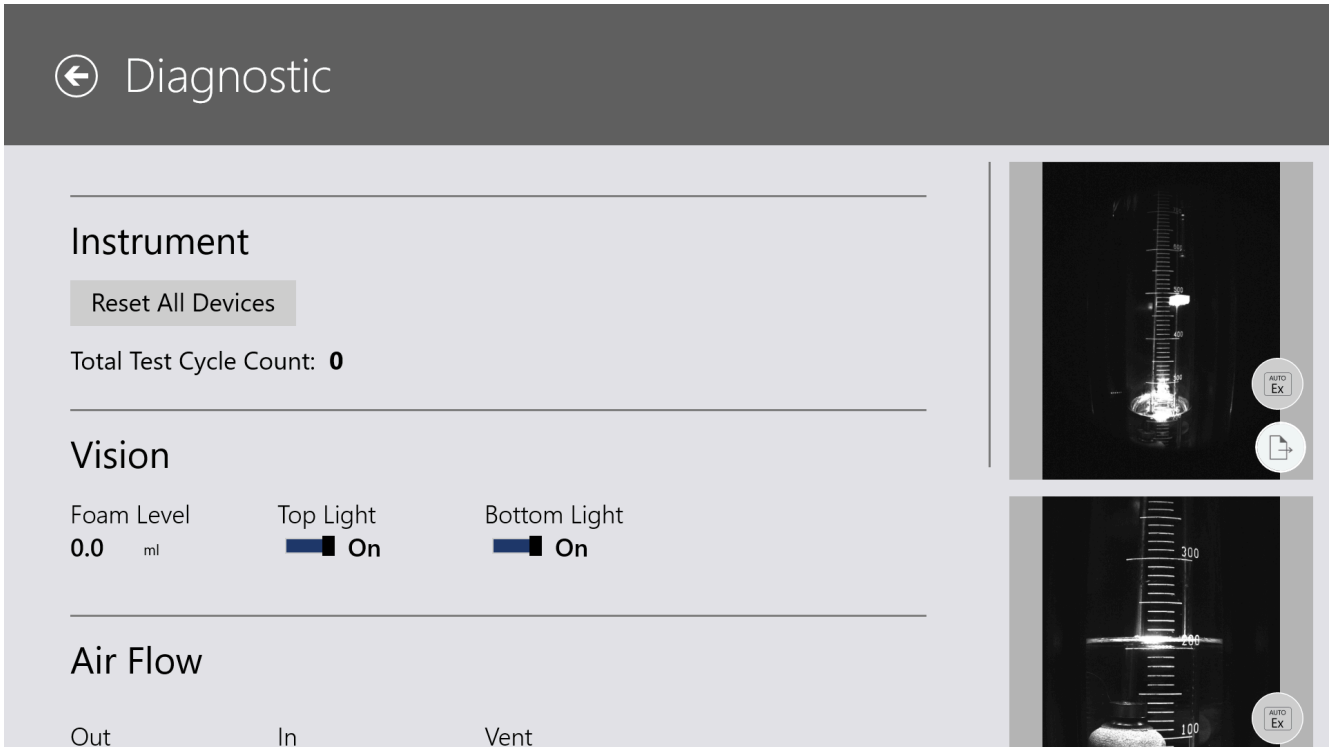
The Export button saves the current image to a selected folder or network share for review or troubleshooting.



The live camera feeds can also be zoomed in or out by pinching the screens and scrolling.

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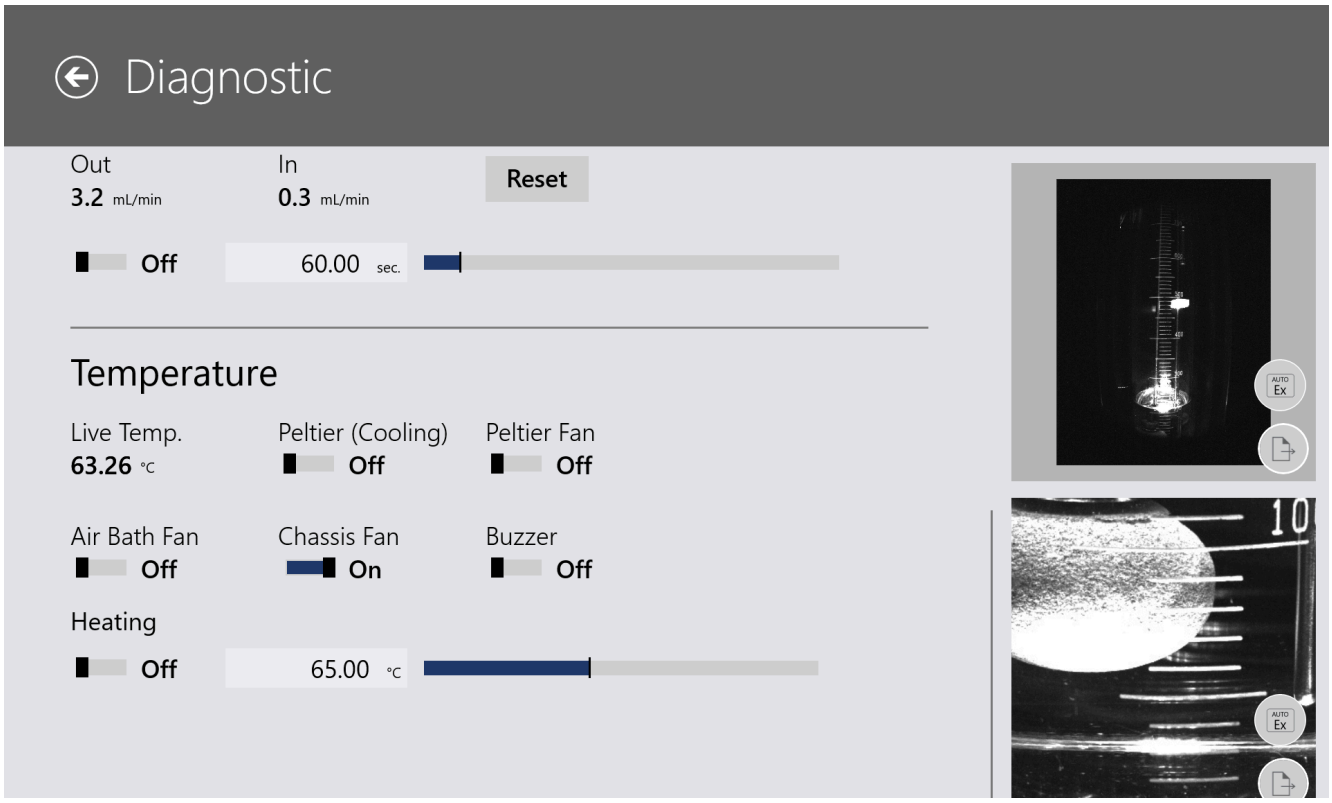

5.5.2. Air Flow

Scroll down to see Air Flow. This shows live airflow readings through the diffuser and associated valves.

Adjust the Flow Rate slider or enter a mL/min set-point, then press On/Off to enable flow.

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- **Out:** Displays real-time flow from the Mass Flow Controller.
- **In:** Displays return flow measured by the Mass Flow Monitor (Totalizer).
- **Vent:** Opens or closes the vent valve to release air from the circuit.

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5.5.2.1. Totalizer

Tracks the cumulative Flow Out and Flow In values for diagnostic purposes.
Set a duration in seconds, press On/Off to begin totalizing, and use Reset to clear values when needed.

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5.5.3. Temperature

Scroll down to see Temperature. This monitors and controls the sample's thermal system.

Adjust the Temperature Set-Point (°C) or move the slider, then press On/Off to enable heating.

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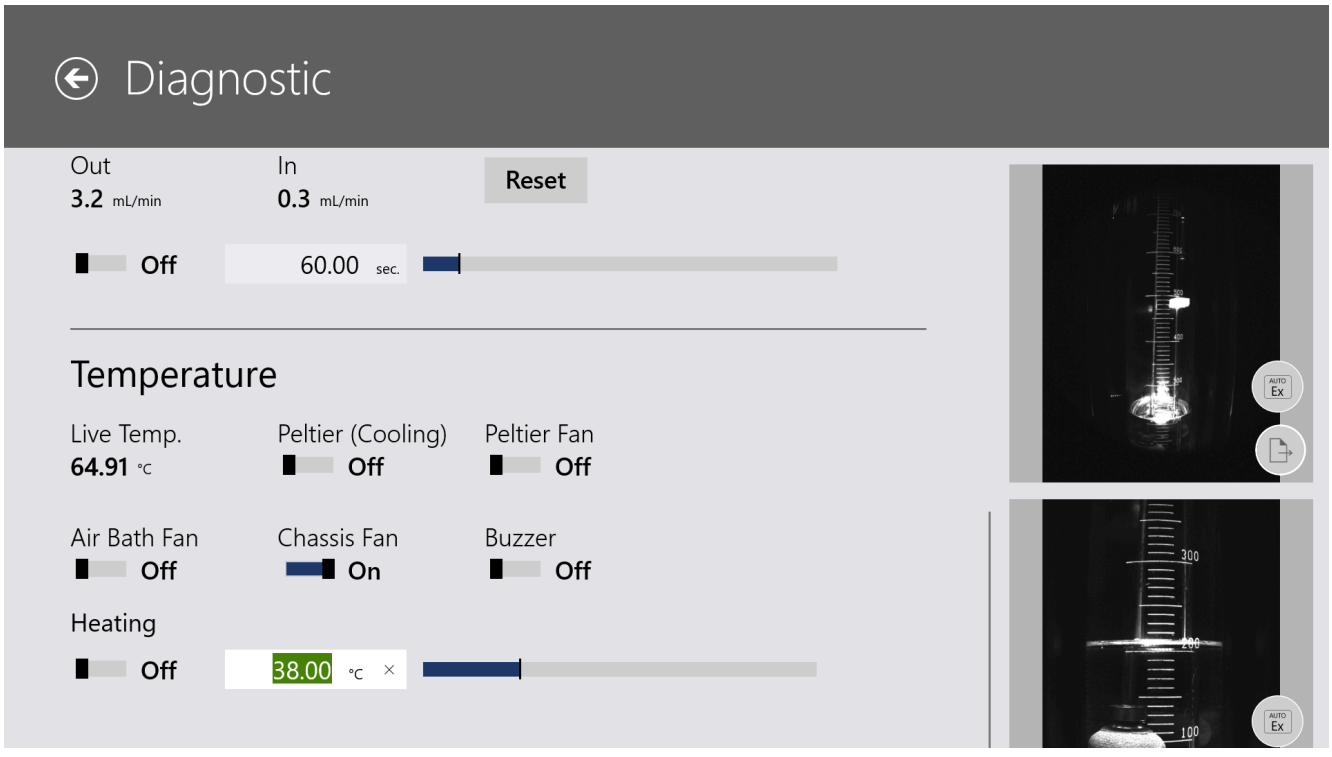

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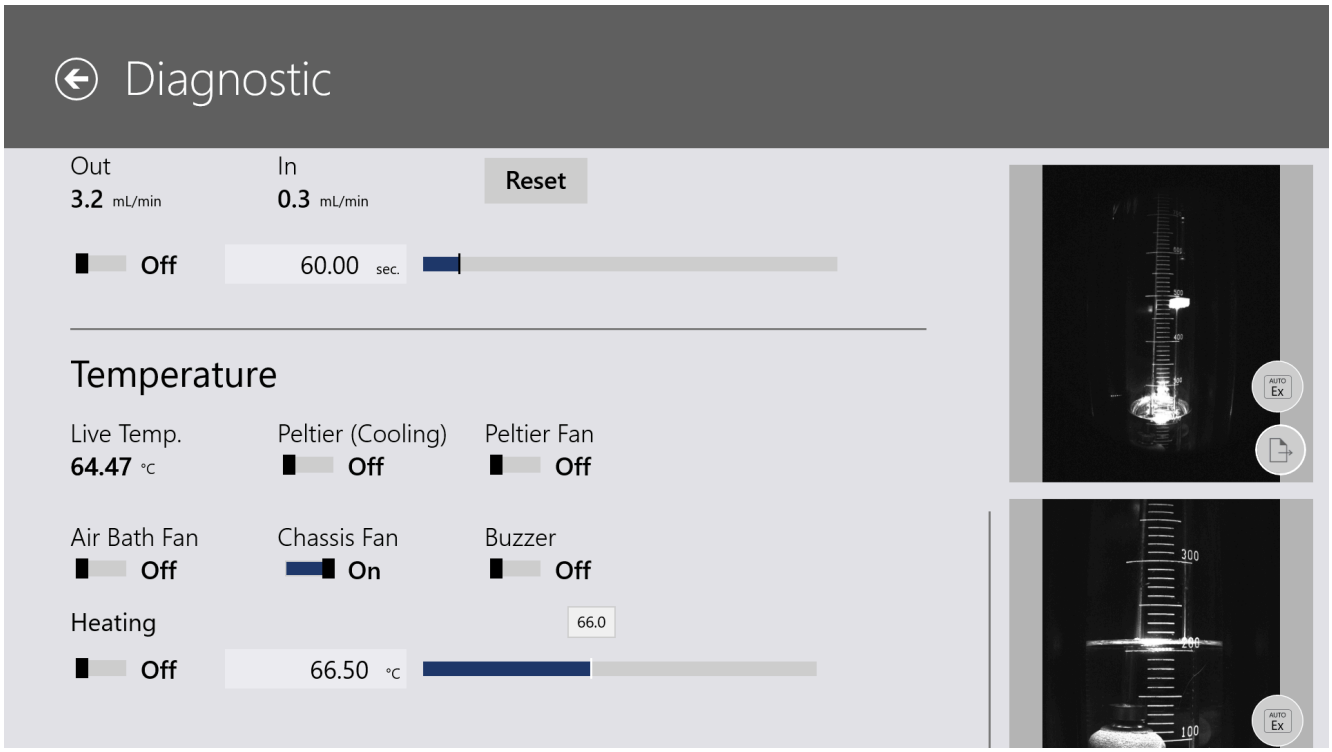

Enable the Peltier for active cooling.

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Note: The Peltier Fan turns on automatically when Peltier cooling is activated.

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The Peltier Fan cools the hot side of the Peltier element. The active cooling using the Peltier can be turned off, while the fan continues operating.

The Air Bath Fan circulates air across the sample cylinder.

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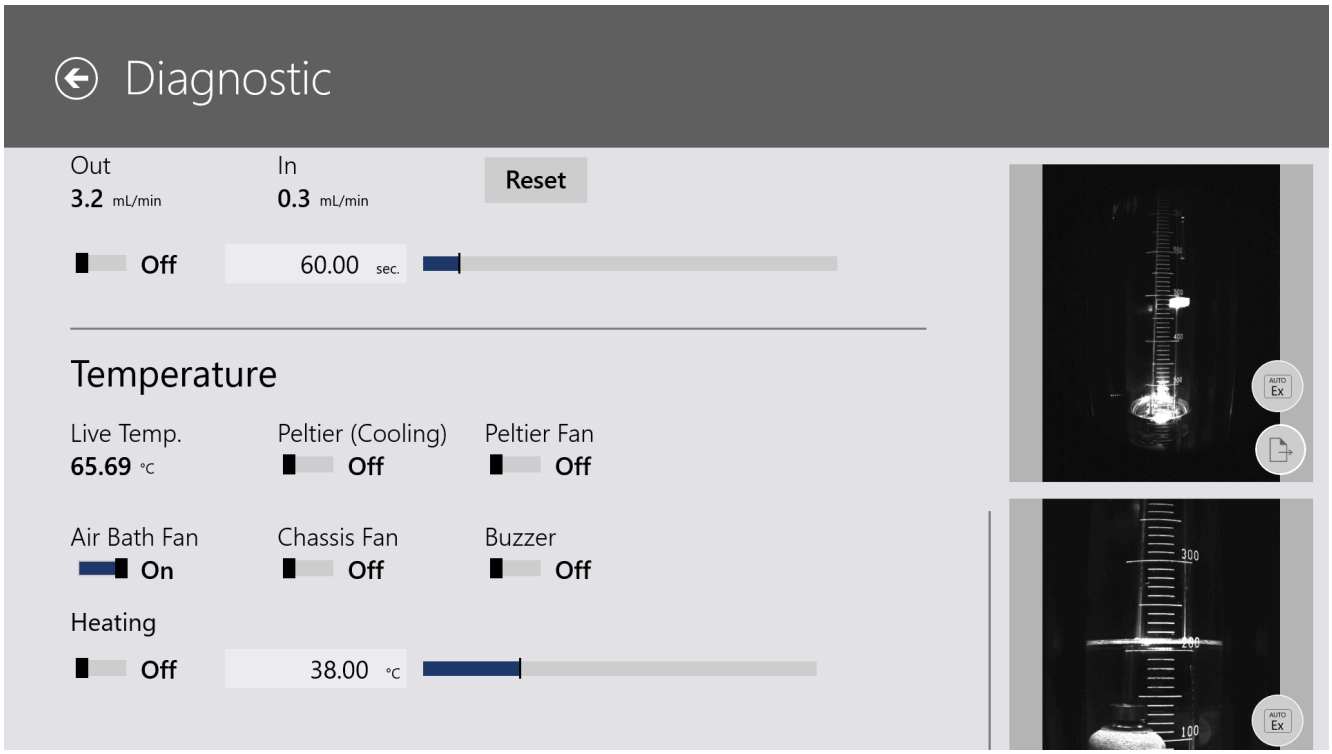

The Chassis Fan prevents internal overheating during heating operations.

The Buzzer activates the sound system used in semi-automatic/audible alert mode.

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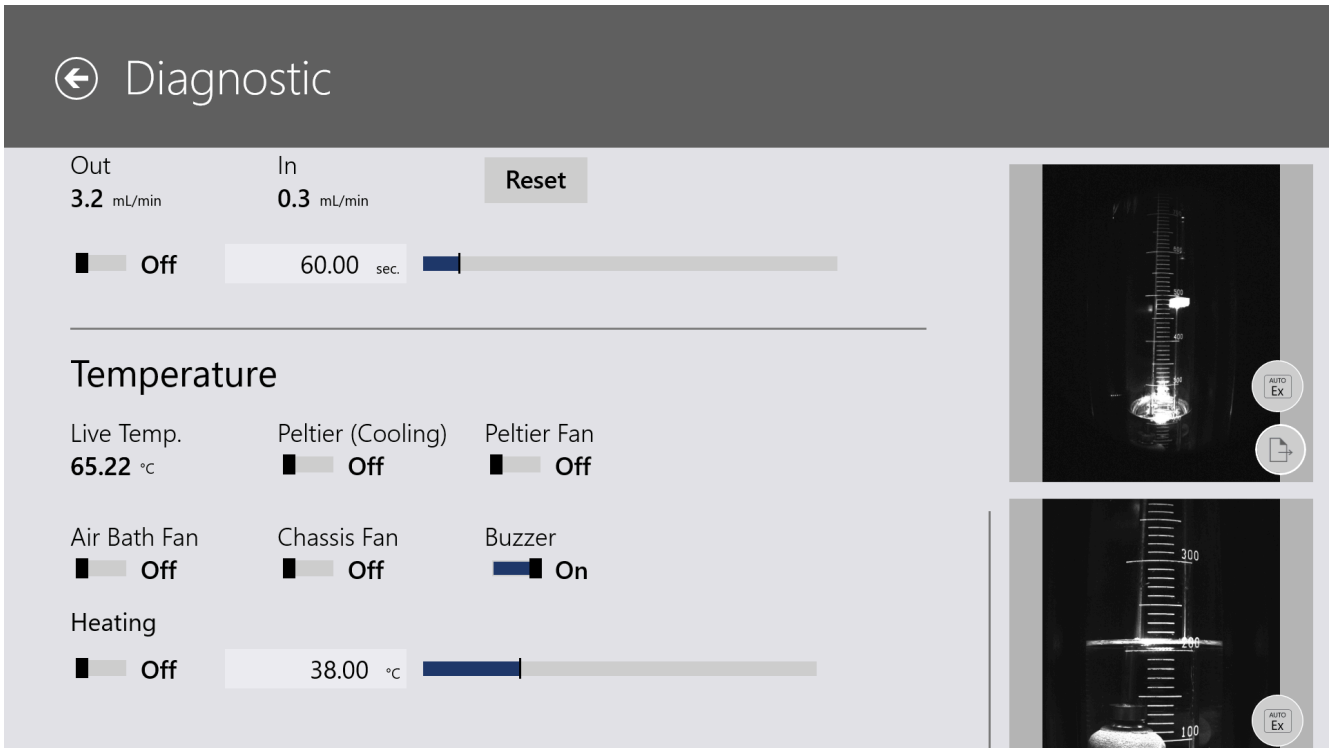
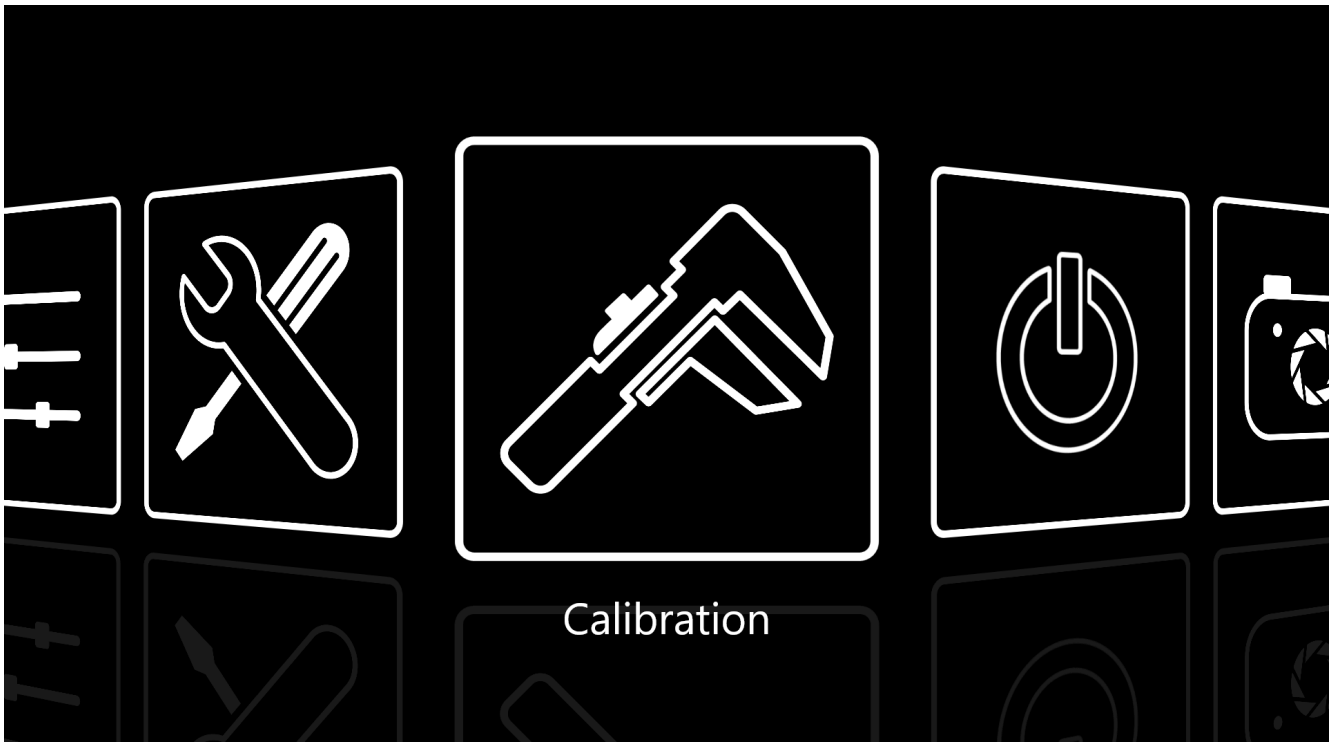


5.6. Calibration

Calibration ensures accuracy and reproducibility.



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5.6.1. Verification

A verification procedure for temperature, airflow, and camera focus is conducted automatically when the modules are booting up.

If a verification fails, diagnostic messages will specify whether it is a temperature probe, camera alignment, airflow imbalance, or general calibration.

In a majority of cases, simply refreshing the modules under the settings screen or powering off then back on again will correct the issue.

If the issue persists, check the connections and ensure all polarized glassware is correctly inserted.

5.6.2. Calibration Features

If calibration fails, diagnostic messages will specify whether it is a temperature probe, camera alignment, or airflow imbalance.

Calibrations should be conducted at least once a year.

It is also recommended to perform a calibration when receiving new equipment (sample cylinders - vision calibration, diffusers - flow calibration, and temperature probe or stopper - temperature calibration, etc.).

Equipment Required:

- The Testo 735 and Testo 0609-7072 probe — Temperature Calibration
- Alicat Meter MB-55LPM-D/5M — Flow Control Calibration

5.6.2.1. Import/Export Calibration

There is an import or export calibration option in the calibration screen.

To import or export a calibration, please reach out to the local distributor or Visaya personnel.

5.6.3. Advanced Calibration

The calibration screen displays similarly to the inspection and diagnostic screens.

The connected modules will be displayed. If multiple modules are connected, they will both be displayed.

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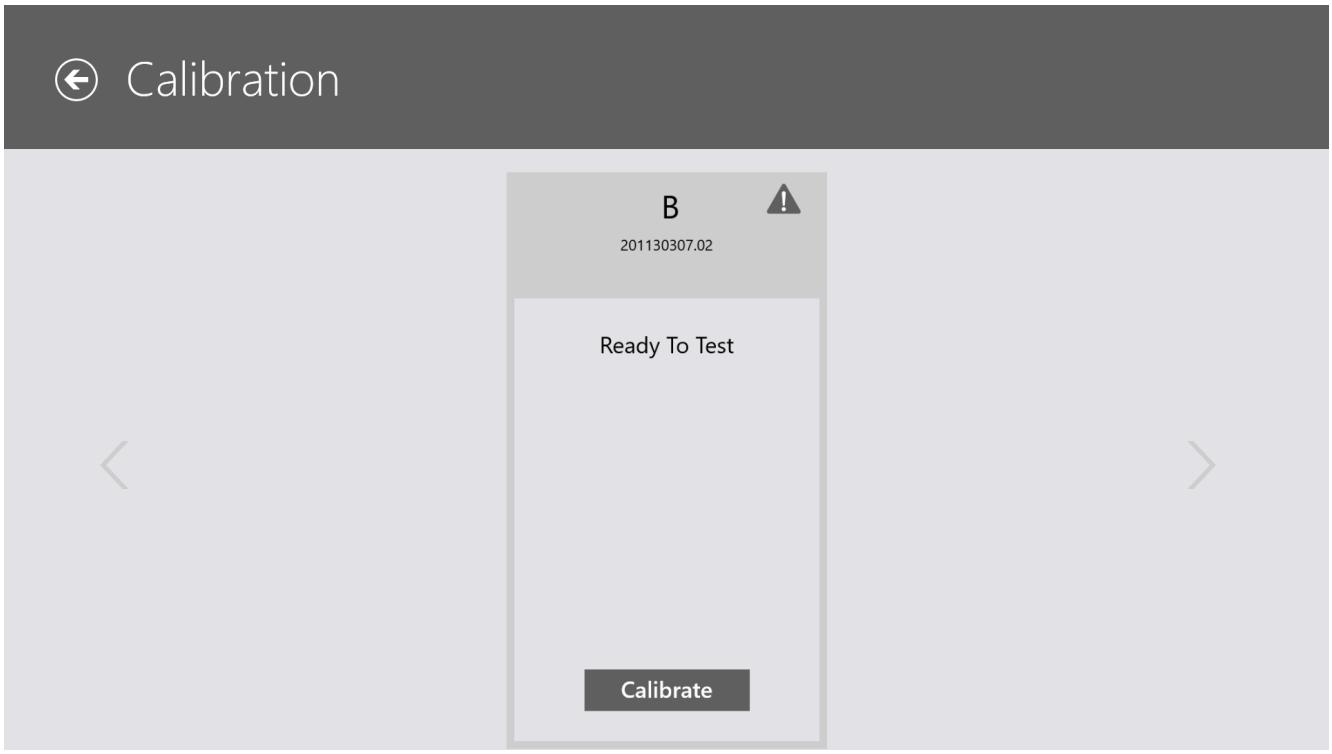
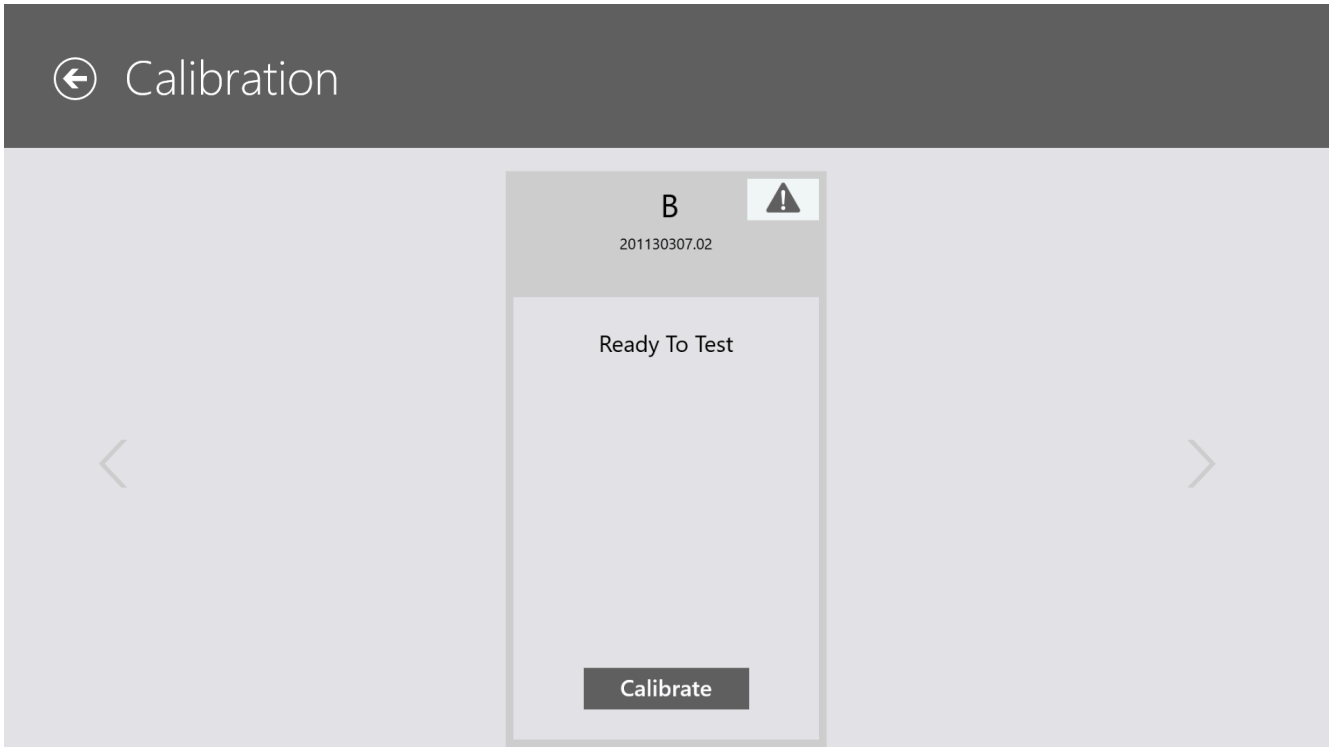


Any errors and warnings will be displayed at the top right corner of the module column. Click on the warning icon for more detailed information.



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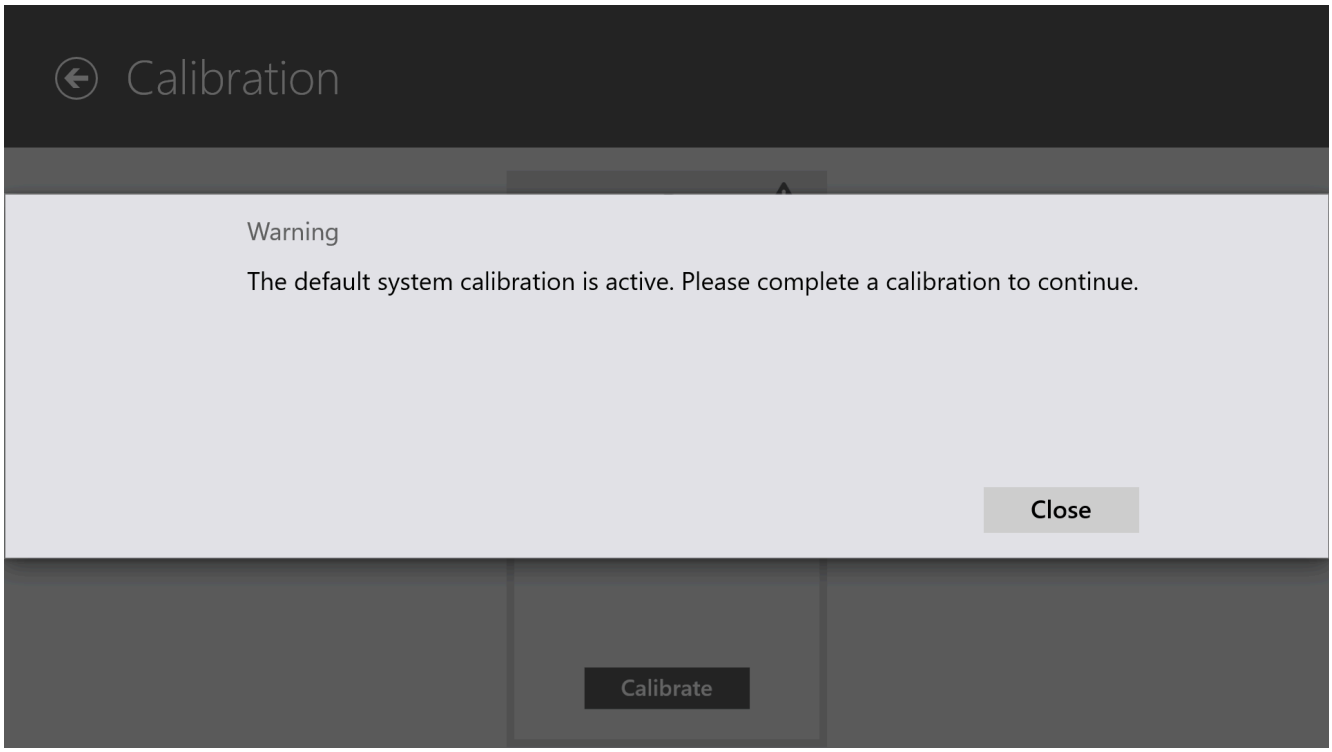


If necessary, click Calibrate to start a new calibration for any or all of the features: flow, temperature, and vision.

Note: Calibrations should be done thoroughly and by trained professionals. Check with a distributor or Visaya representative to see if your FoamDDI requires a new calibration.

5.6.3.1. Flow Control Calibration

To calibrate the flow control, an Alicat Meter (gas flow device) is required. It is recommended to use an Alicat Meter MB-55LPM-D/5M. Others devices can be used, but must match or exceed the specifications defined by the Alicat.

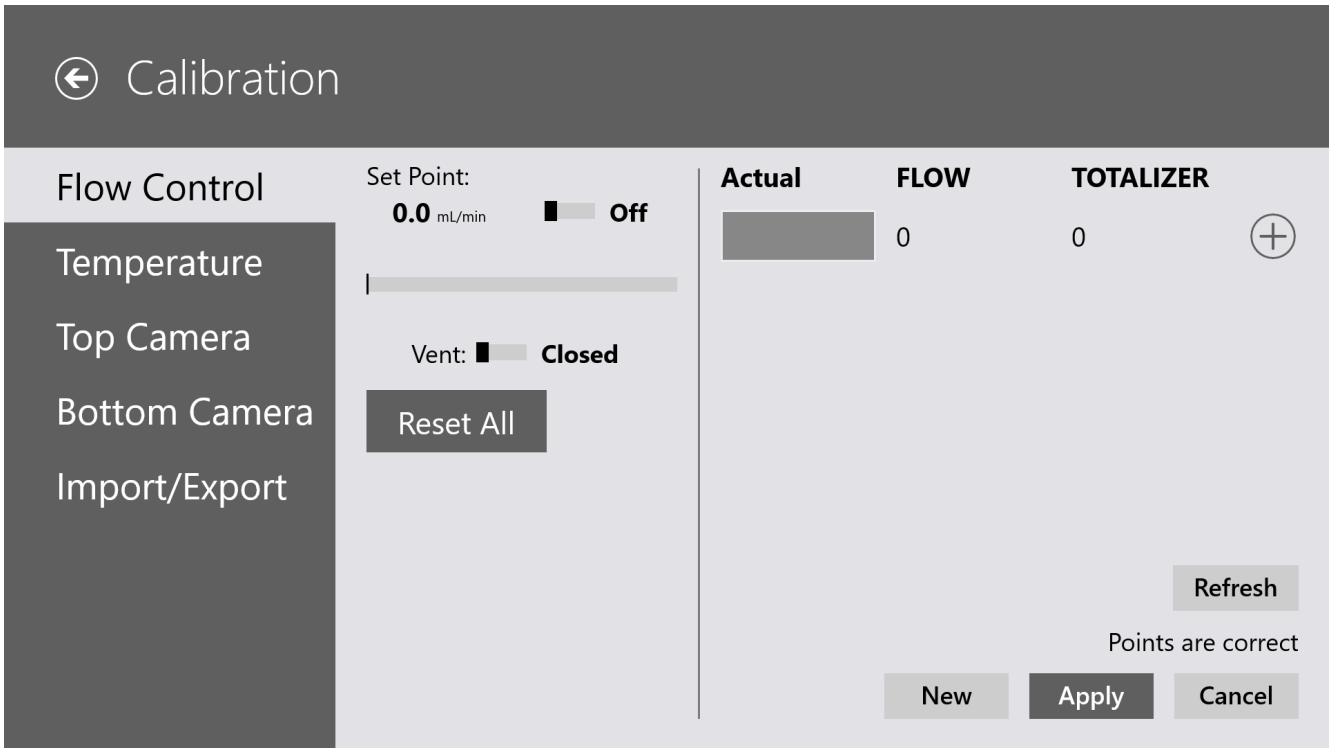
> The Alicat MB-55LPM-D/5M is a portable mass flow meter with a 5 SLPM (Standard Liters Per Minute) full-scale flow rate, a measurement accuracy of ±0.8% of reading + 0.2% of full scale, and a repeatability of ±0.2% of full scale.

The airflow calibration procedure works by adjusting the diffuser airflow using a reference standard and confirmation via diagnostic output.

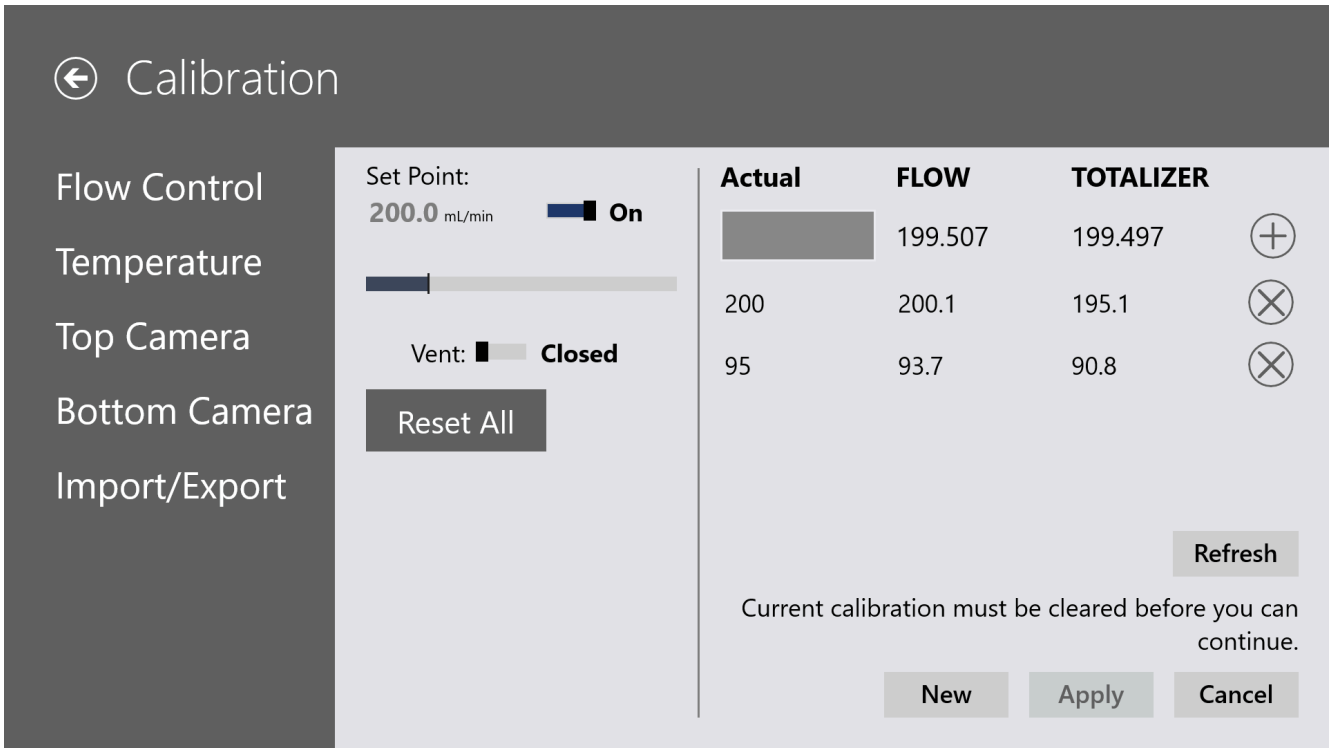
Procedure:

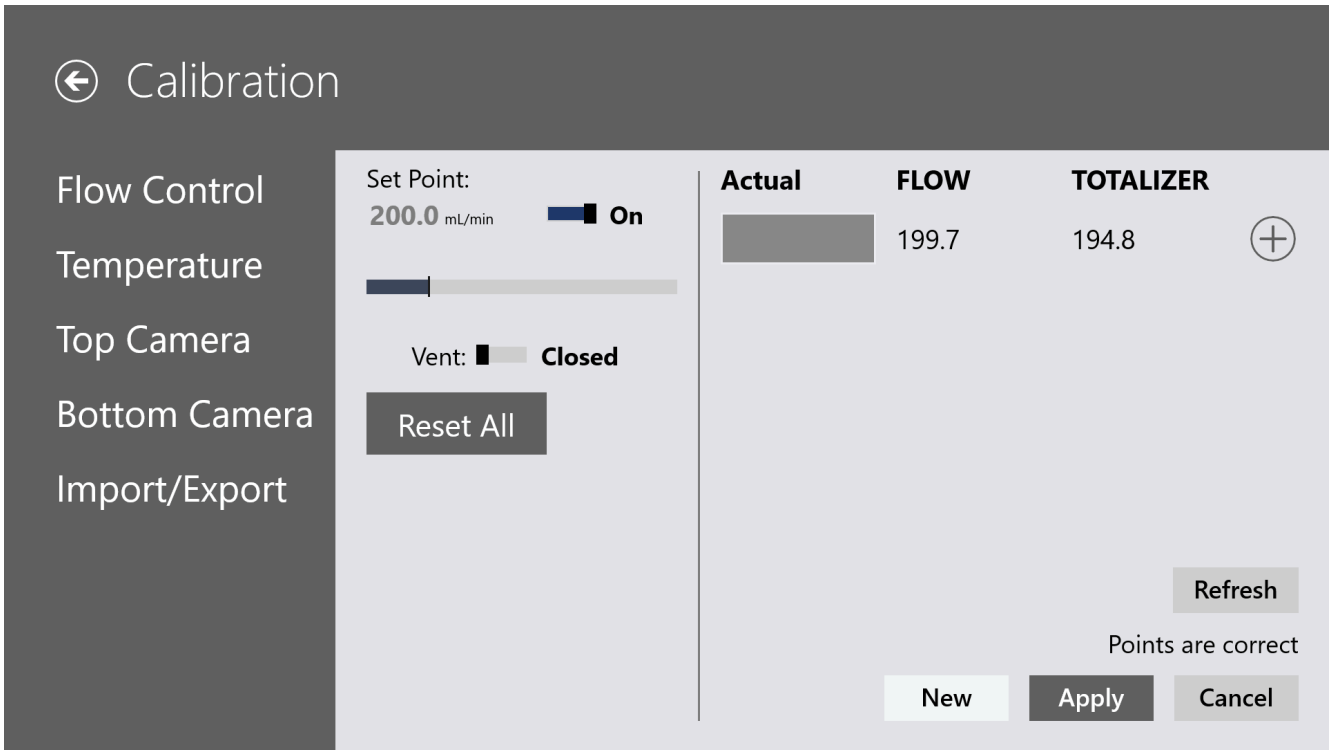
Start by attaching the inlet and outlet hoses to the Alicat Meter at the appropriate connectors.

Once Calibrate is selected for a module under the initial calibration screen, the flow control tab should open by default.



Click New to start a new calibration. If there is an existing calibration outside of the factory calibration, it will be displayed on the right side of the screen. Clicking New will erase the calibration.





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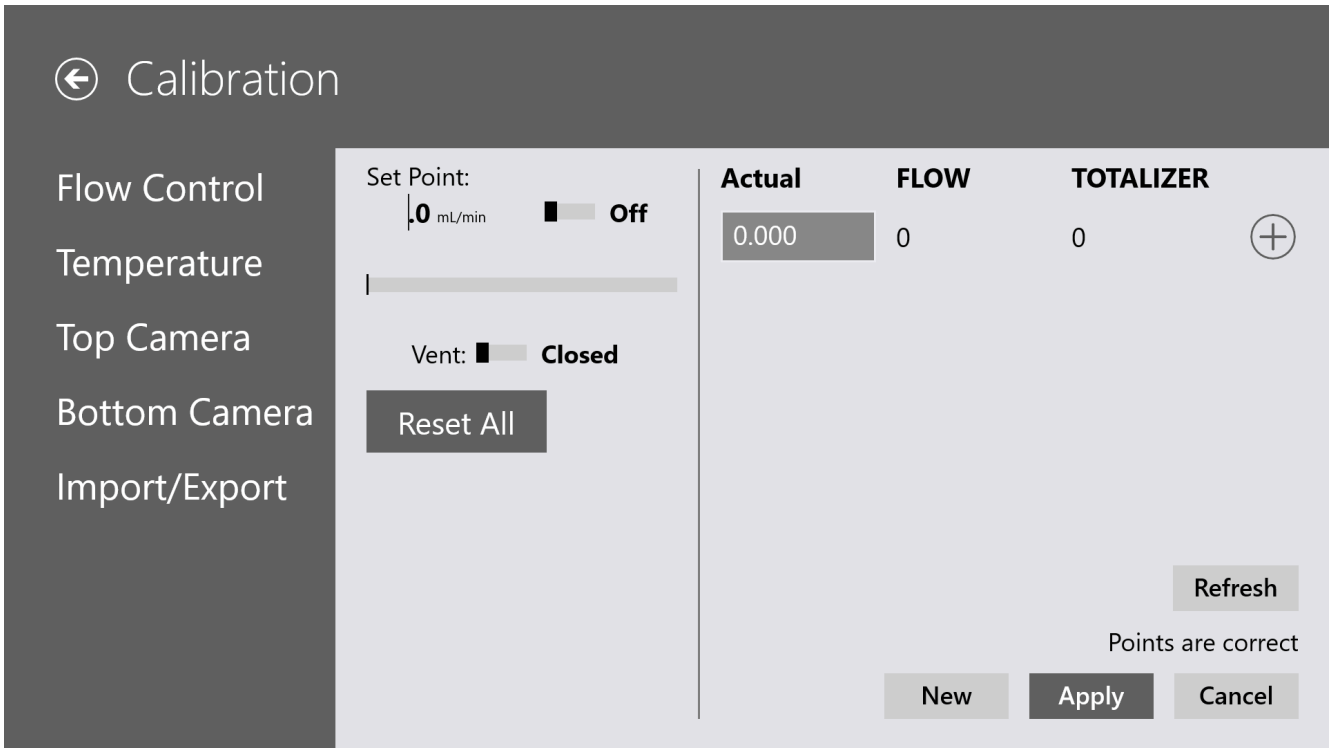
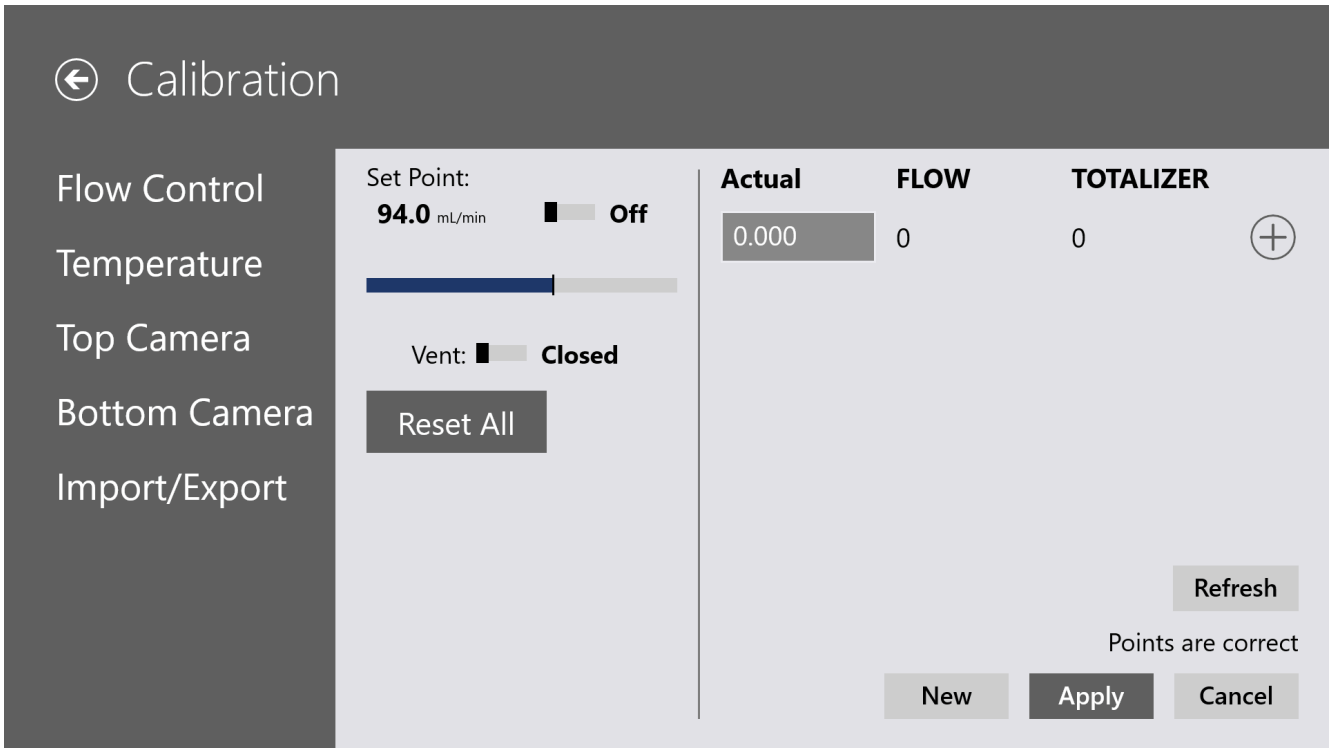
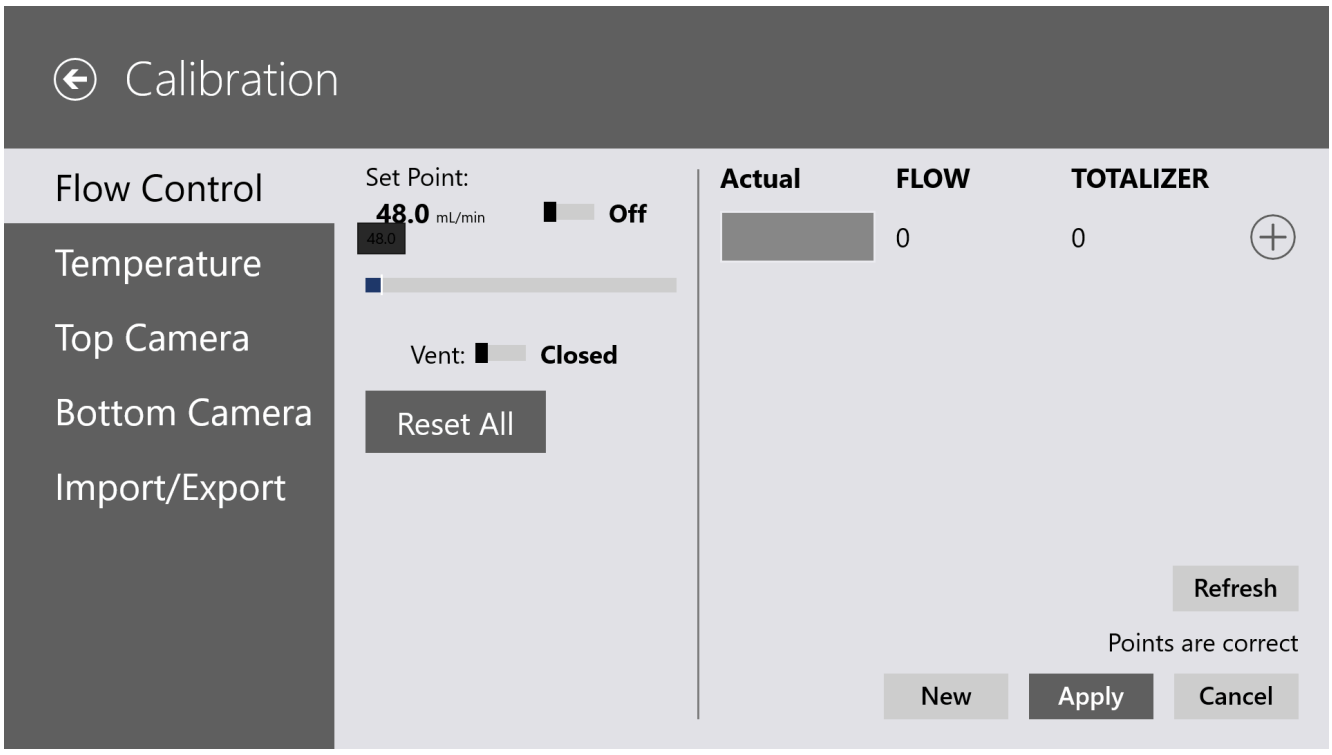
Note: Any airflow parameters can be used for a calibration. The values below are the suggested specifications.

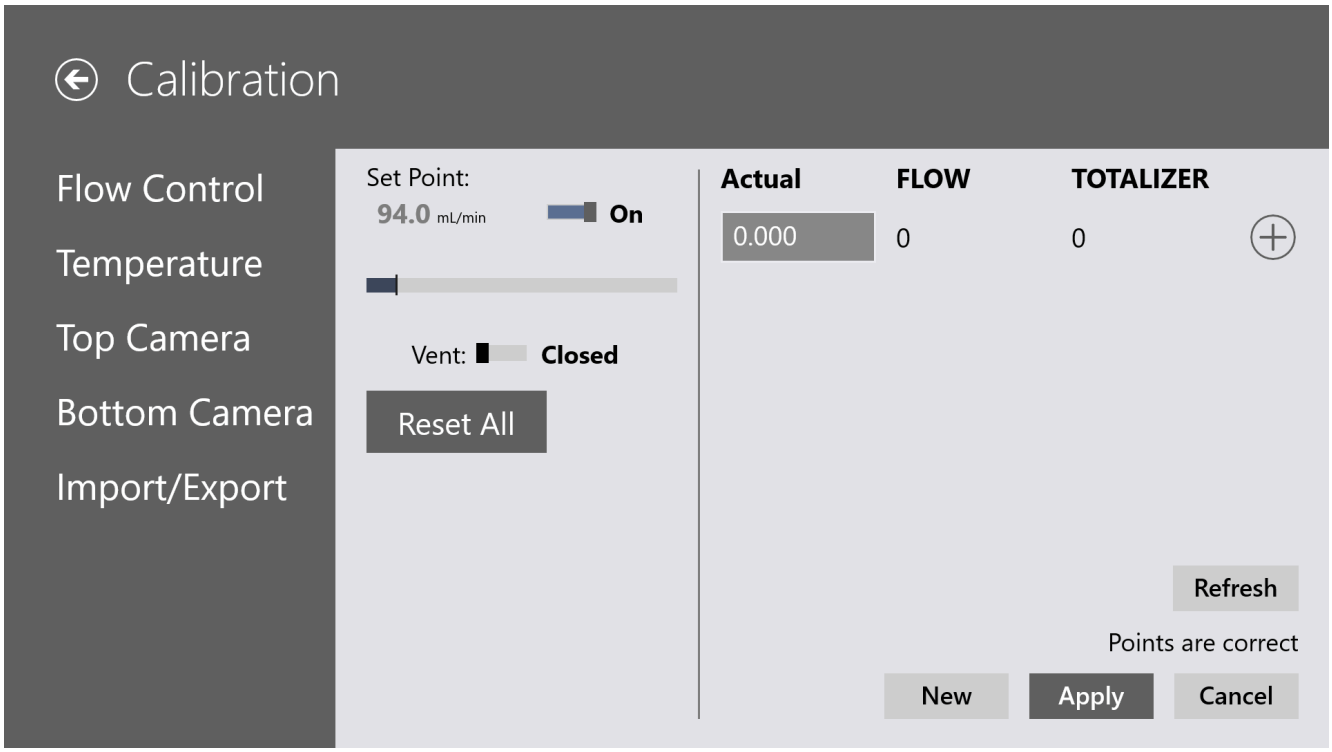
Under Set Point type 94.0 to set the flow rate. The slider can also be used to adjust the Set Point.





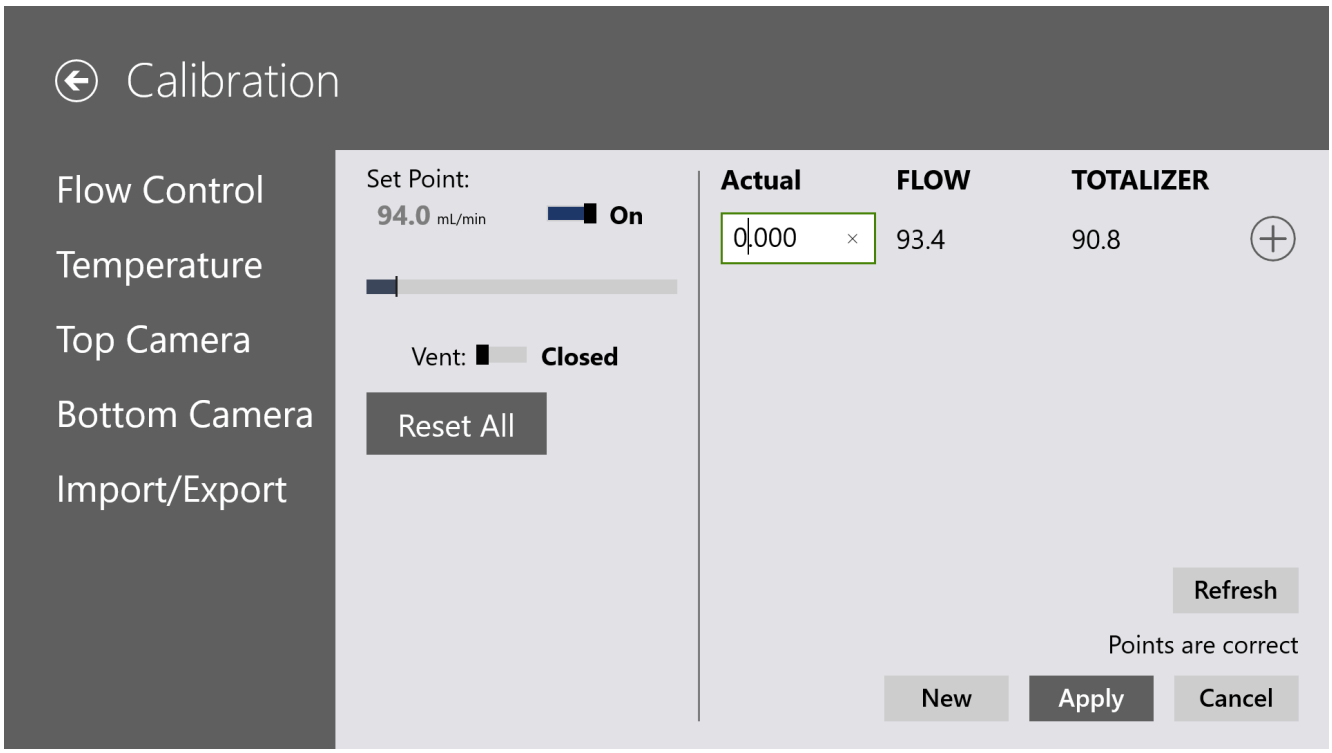


Once set, use the On/Off toggle on the Set Point's right and set it to On.



Wait until both the Flow column and the Alicat Meter have stabilized and the number does not fluctuate. Less than a +/- 2 point fluctuation is recommended.

Once stabilized, type the Alicat reading of flow, into the Actual column.



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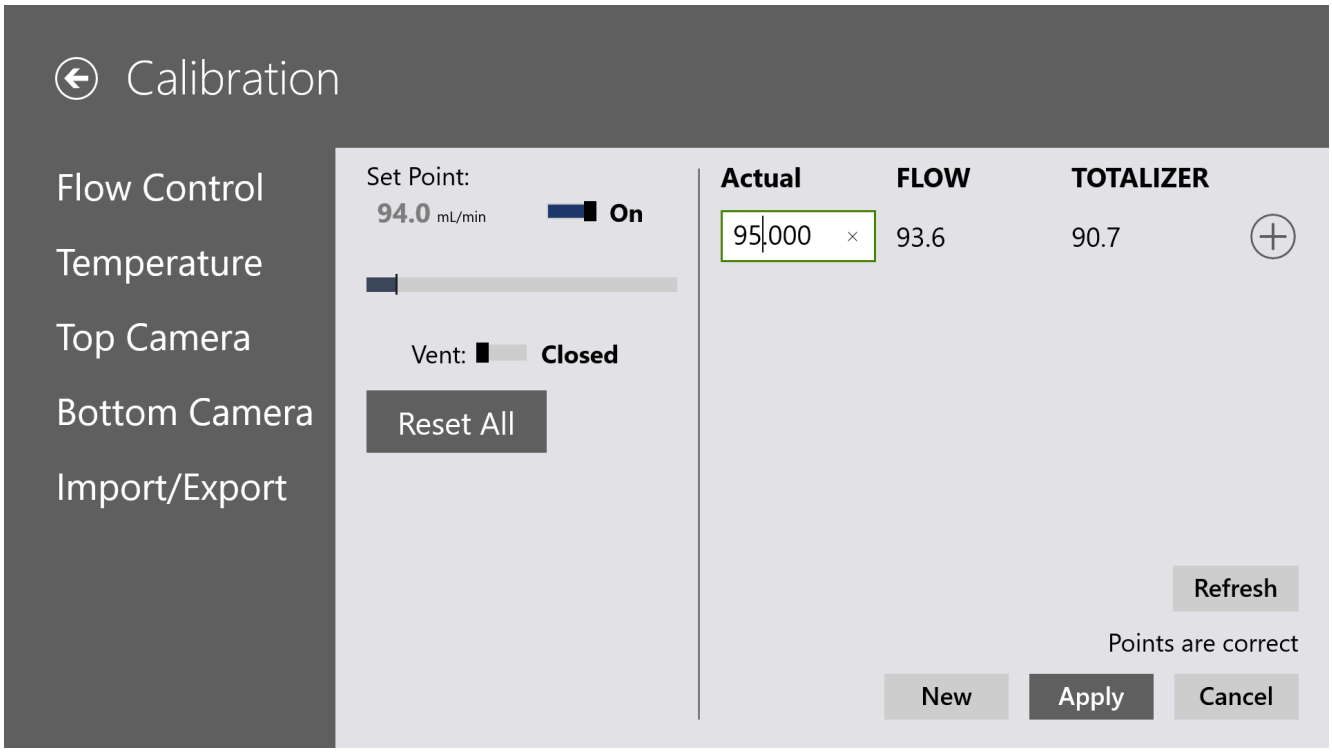

Use the plus (+) icon on the right to add the set point to a new calibration.

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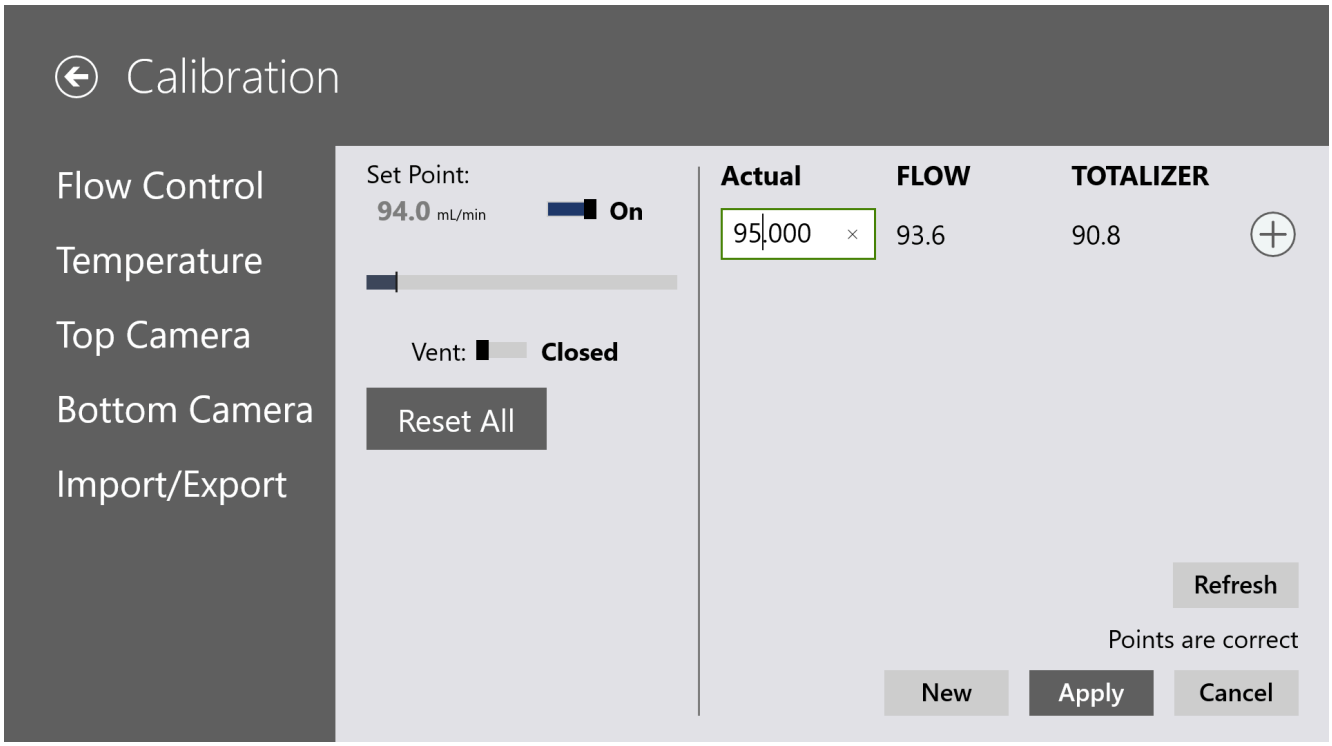

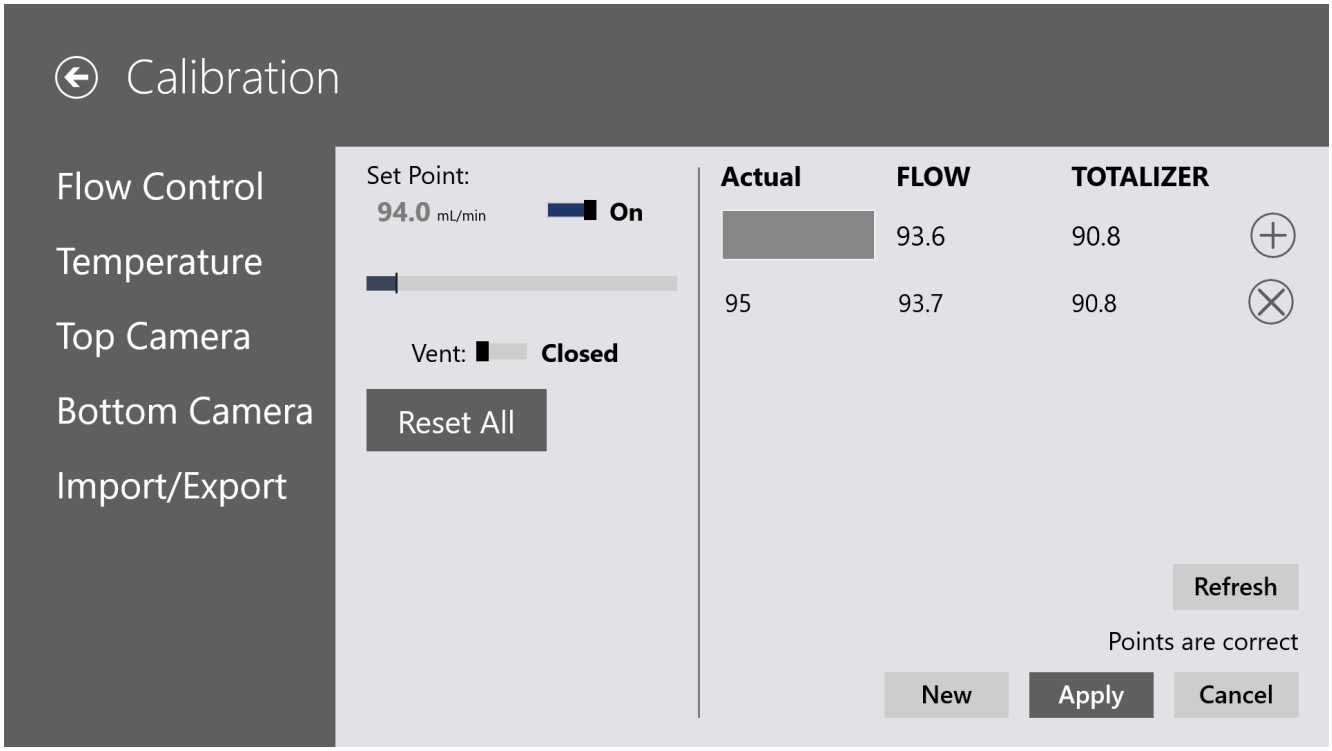

Once added, turn off the set air flow using the toggle and set a new flow rate of 200 mL / min.

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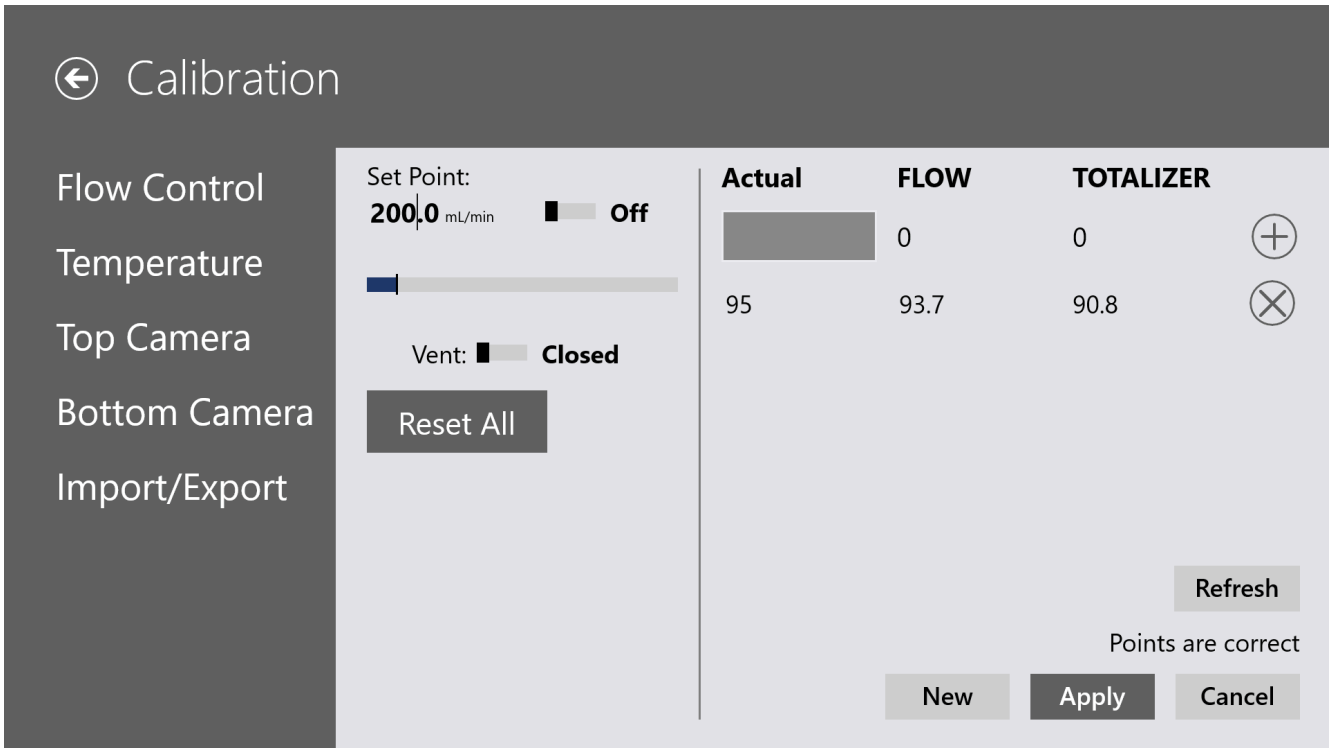
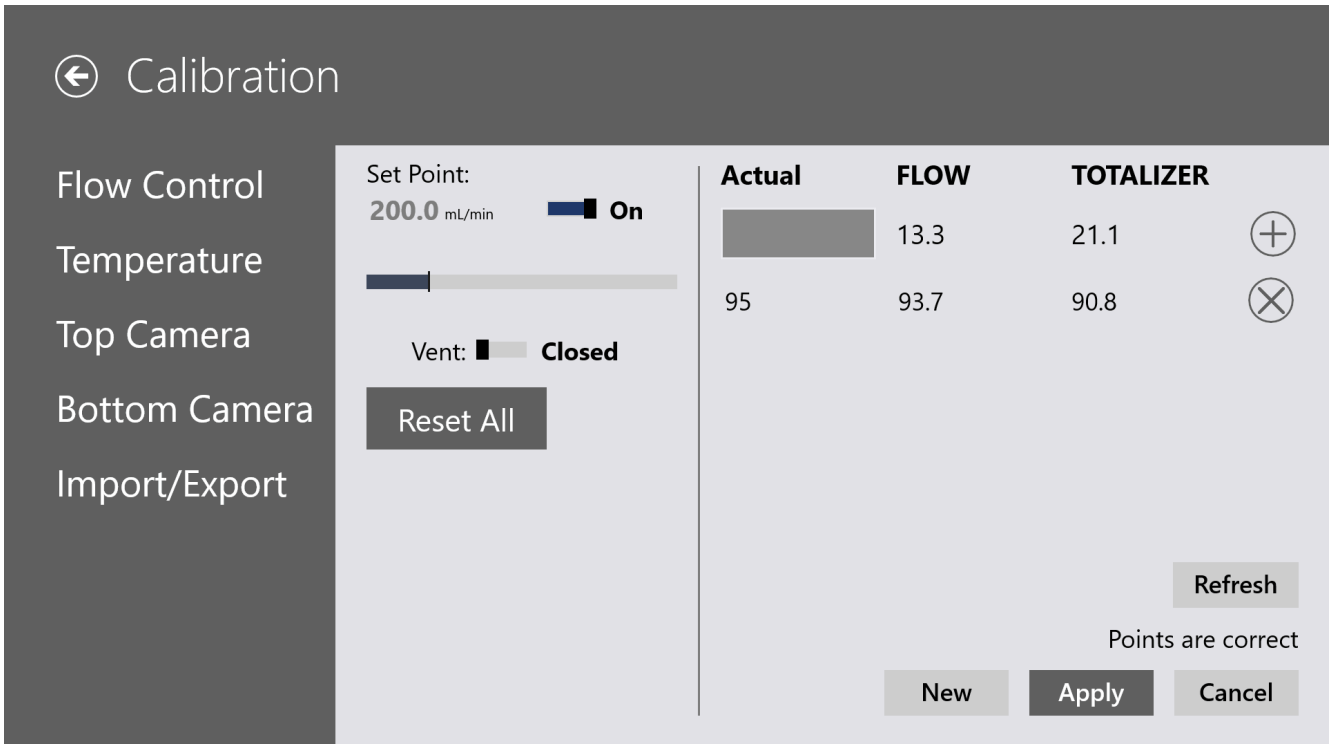


Once the new flowrate is set, toggle the airflow to On.



Wait until the Alicat Meter and Flow stabilize.

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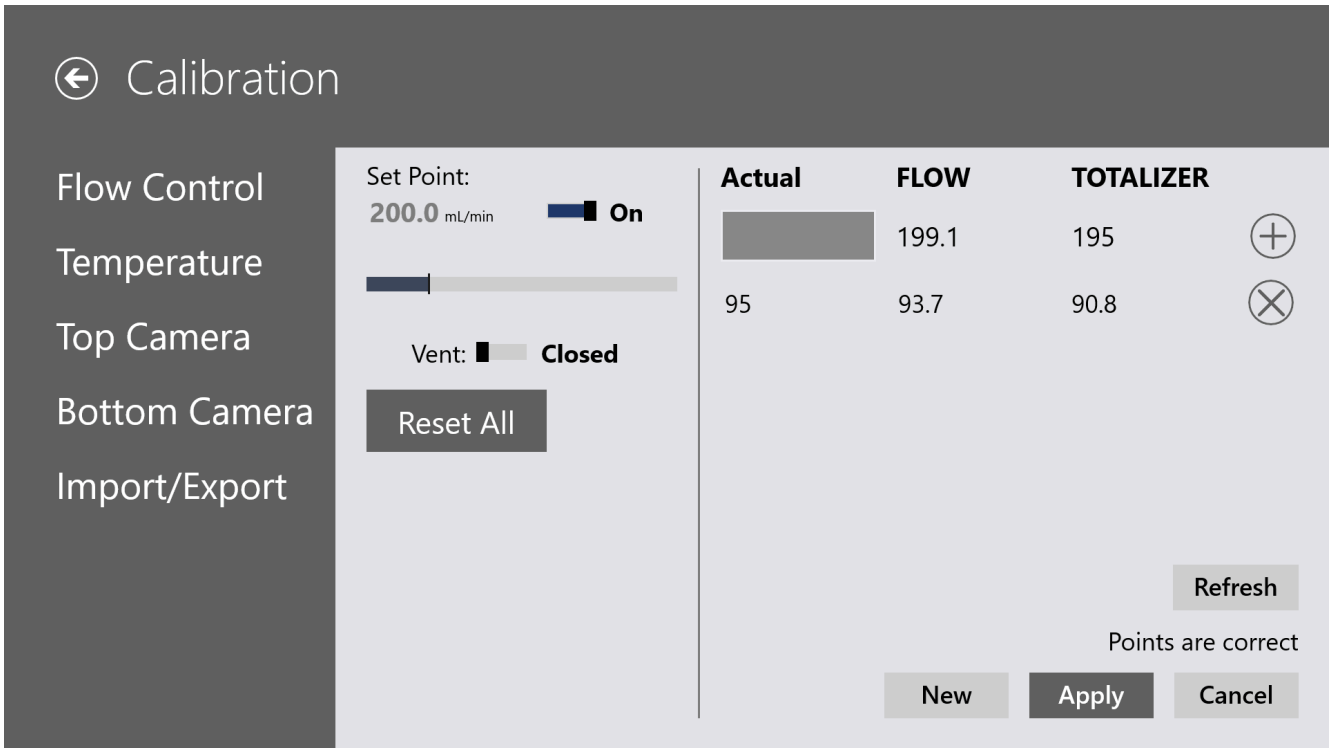
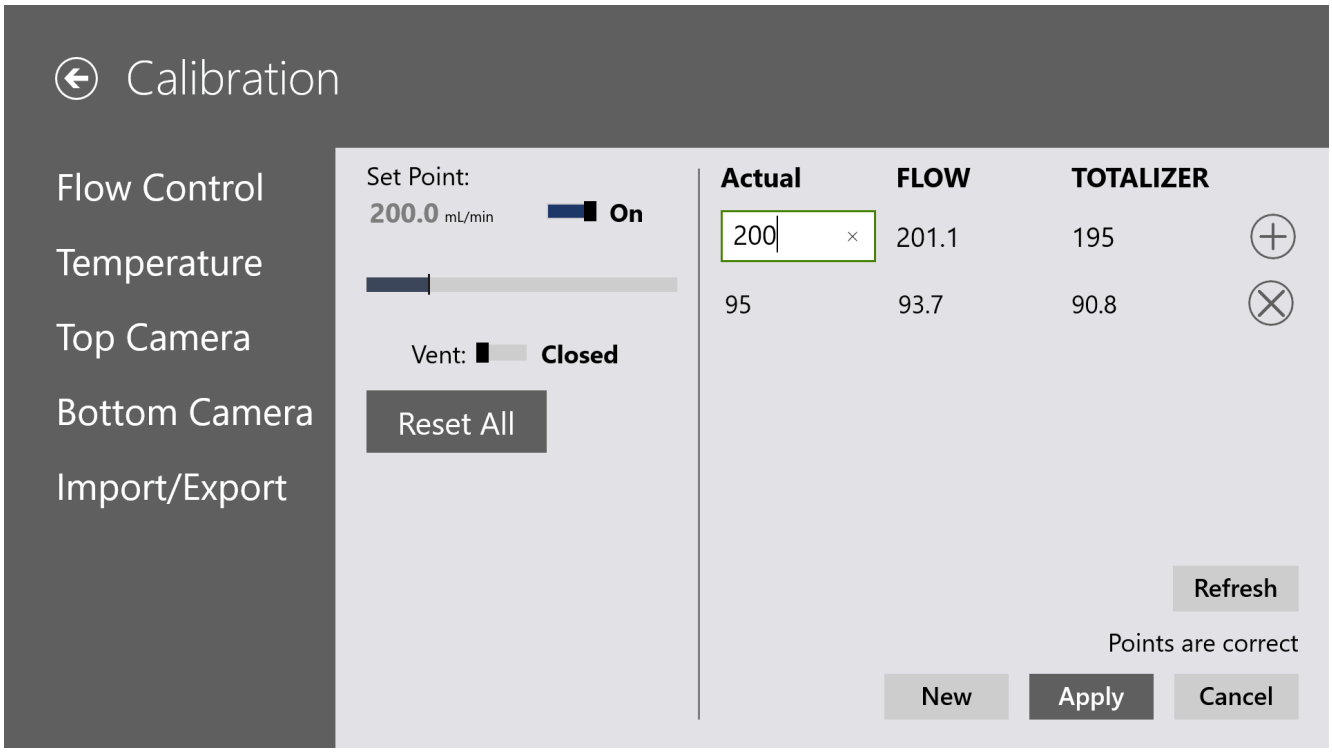


Once stable, record the Alicat reading in the Actual column.



Click the plus (+) icon to add the new set point to the calibration.

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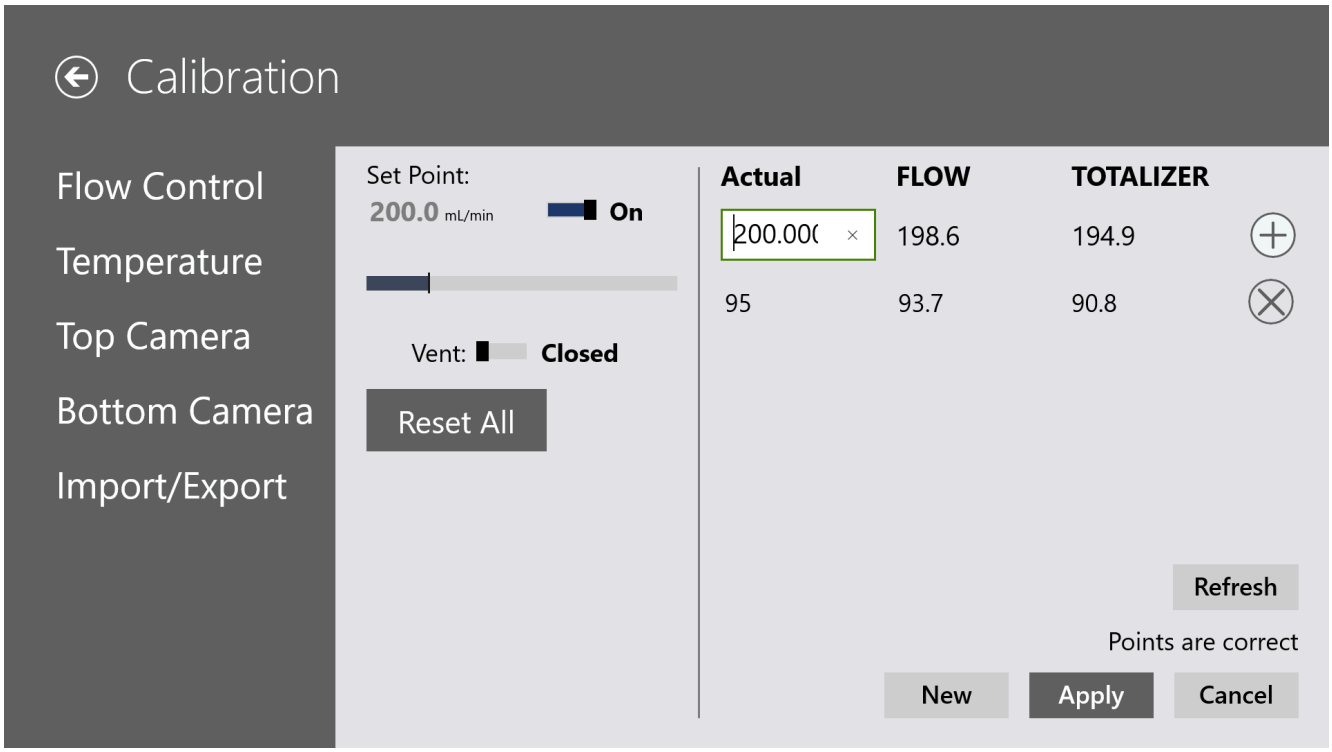

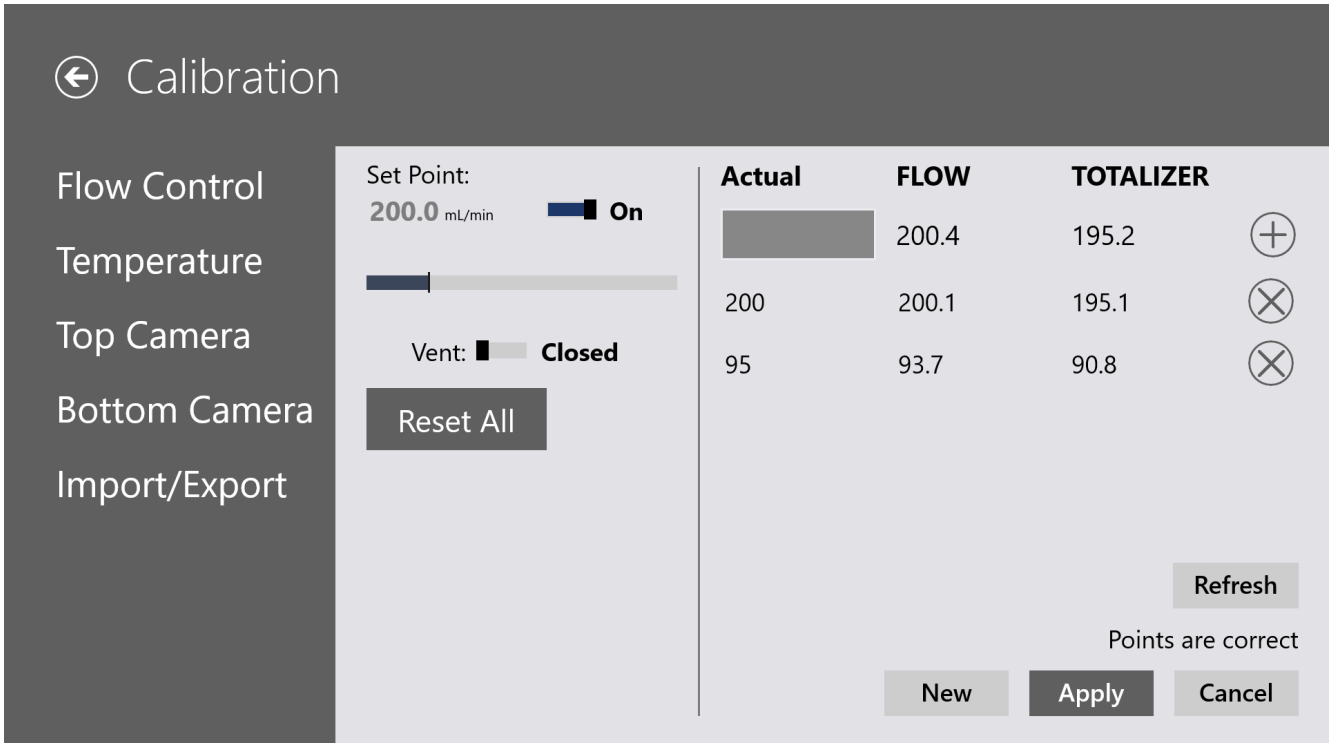

Once both set points are added, click Apply to confirm this new calibration.

Note: Once applied and confirmed, the old calibration is disregarded.

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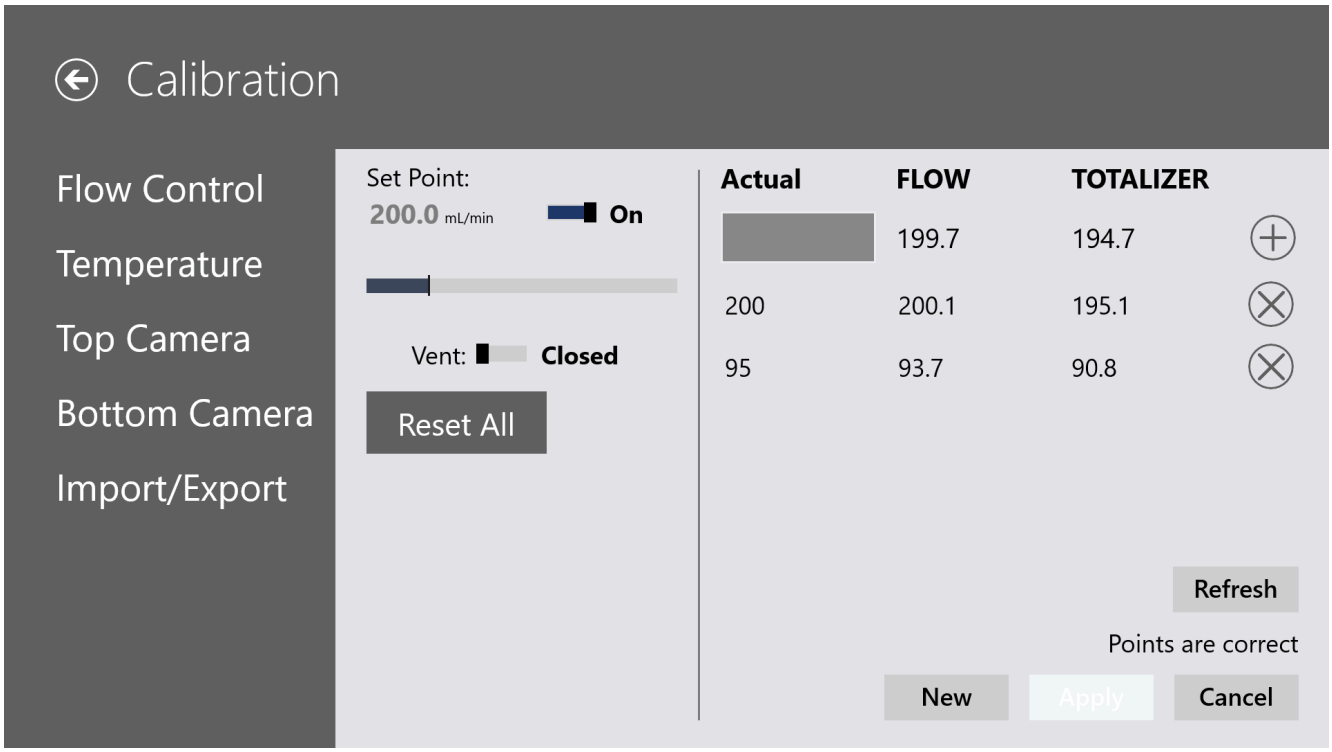
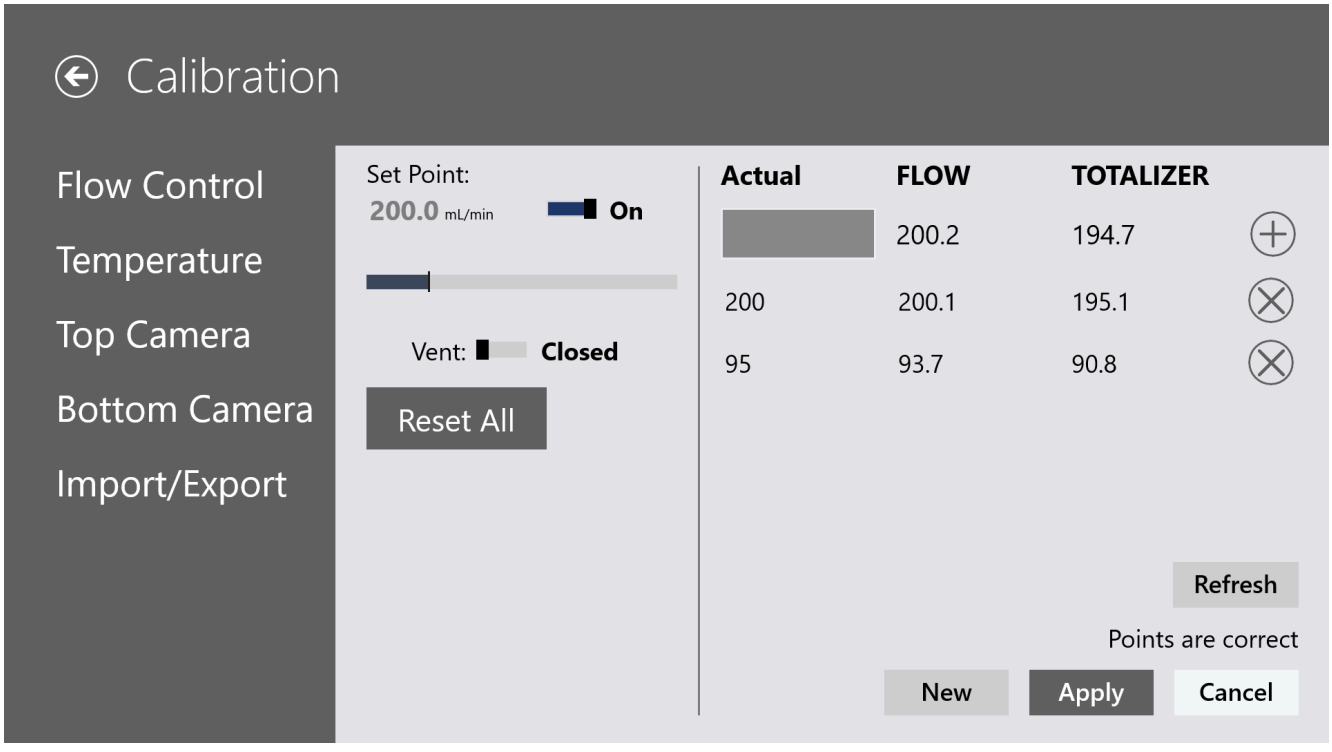


If the operator wants to return to the prior calibration, they can simply click cancel at any time *BEFORE* confirming the calibration with the Apply button.



To delete only part of the calibration, select the X icon on the right side of the screen.

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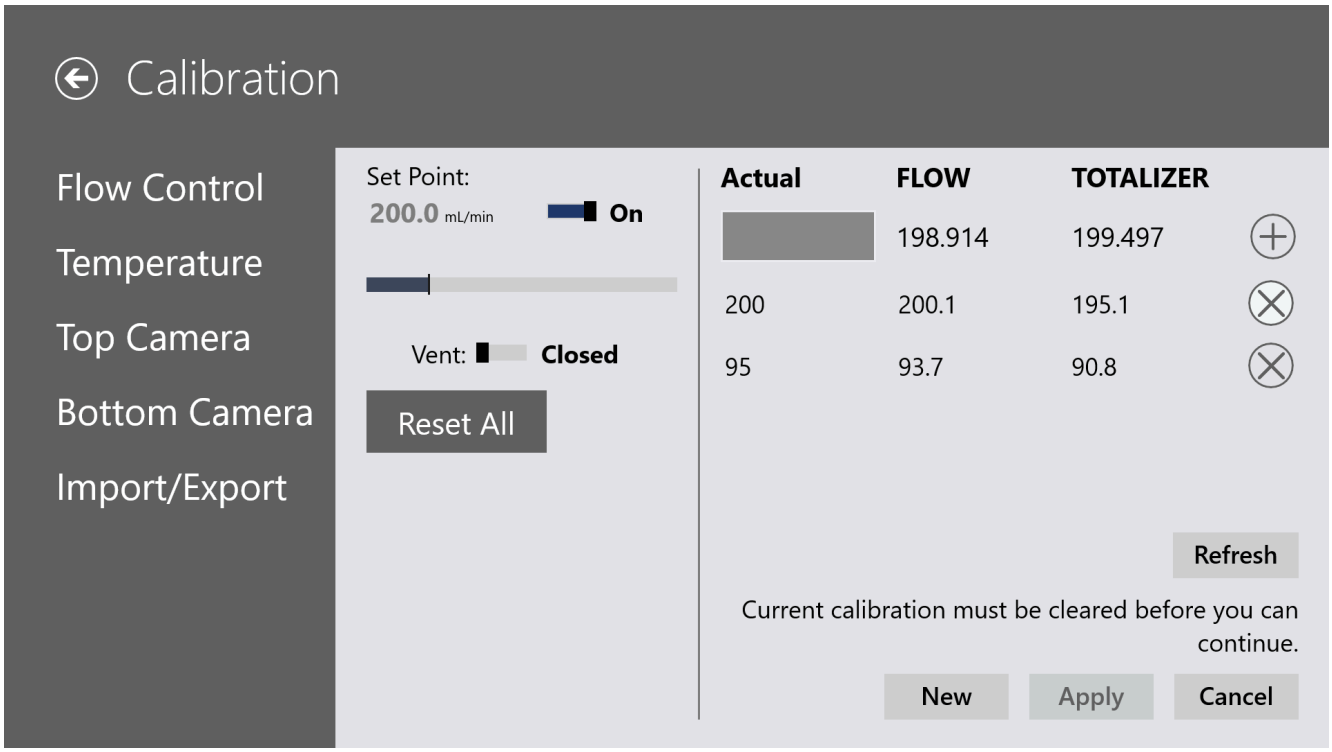
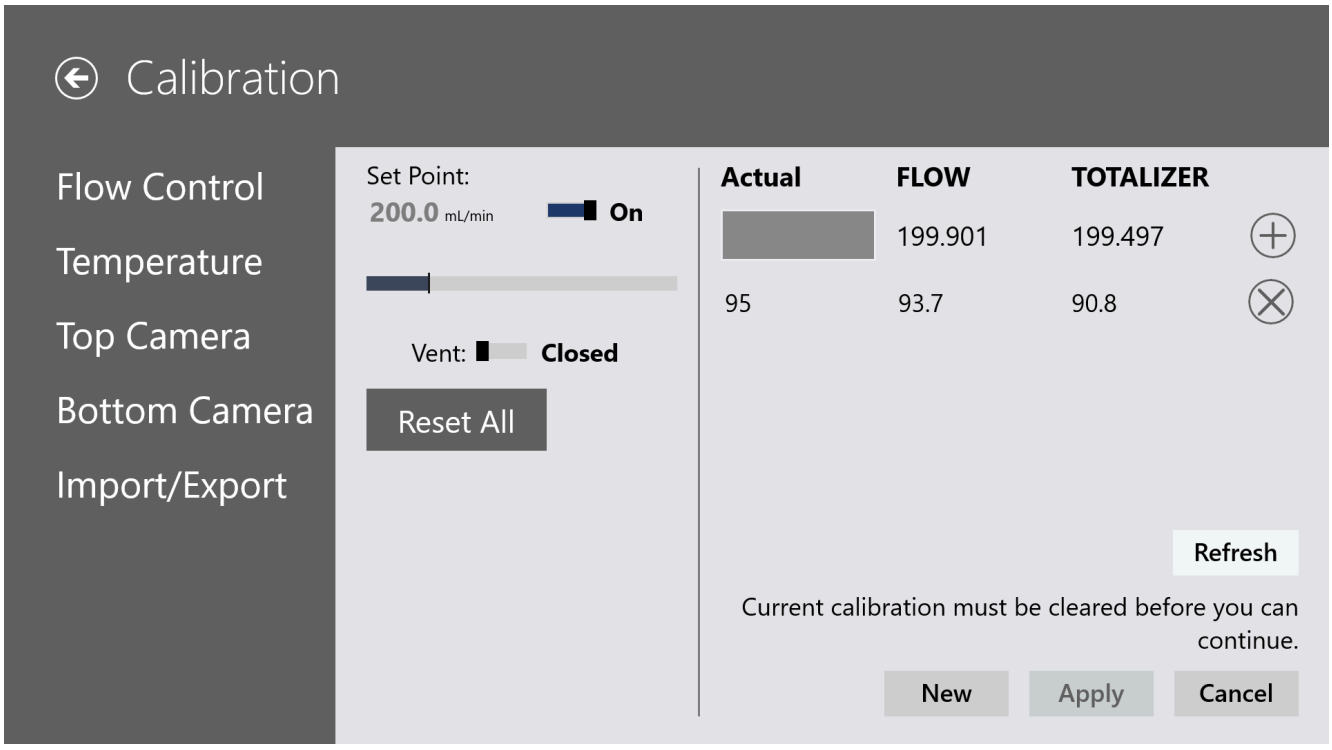
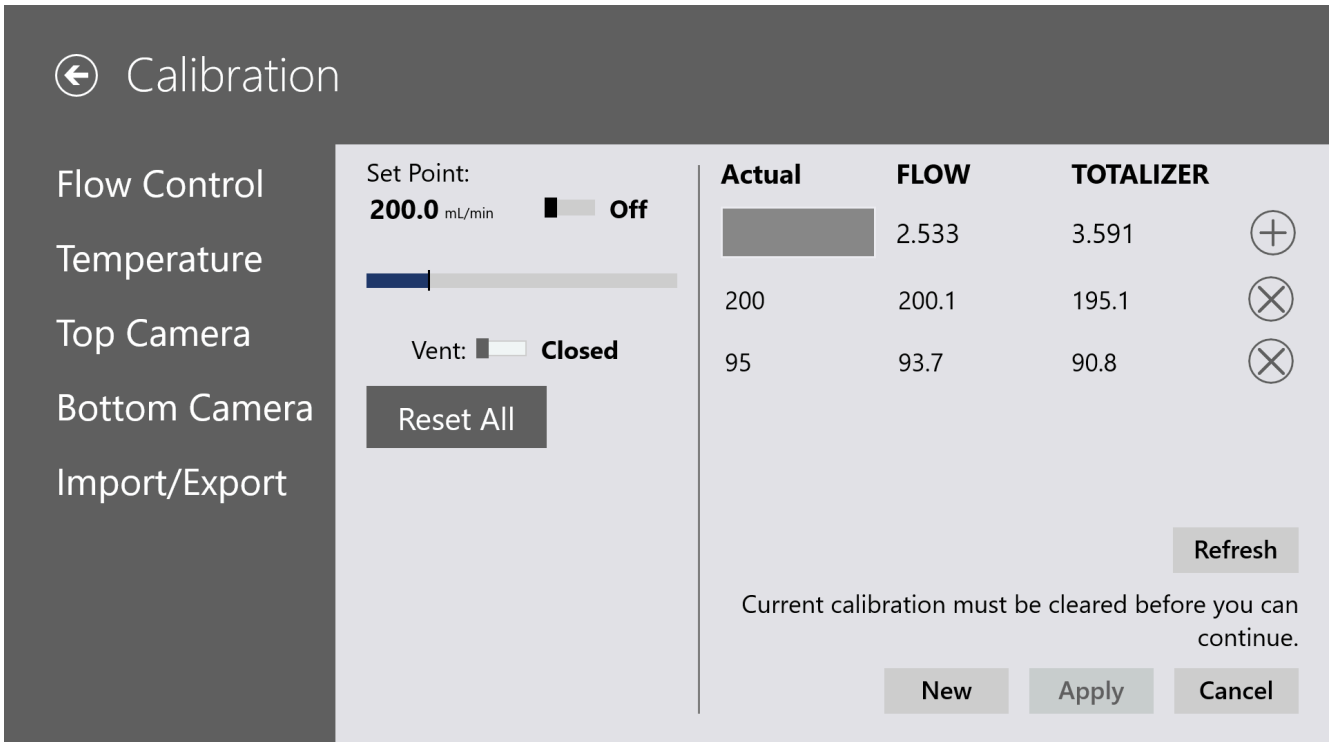
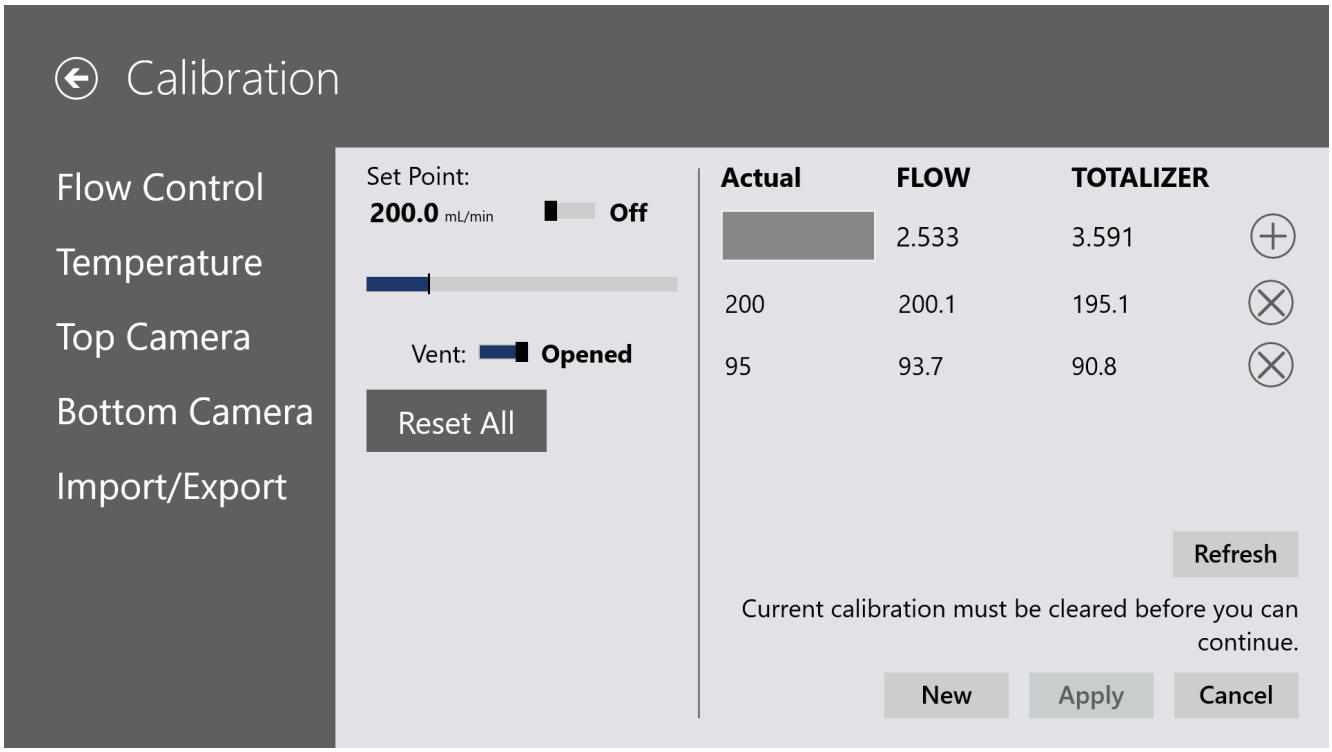


To undo a deletion, click Refresh.





The Vent toggle on the left is used to release air trapped in the tubing. This allows the user to quickly release air between calibrations.



The Reset All option will turn off the airflow allowing for the reset of parameters.

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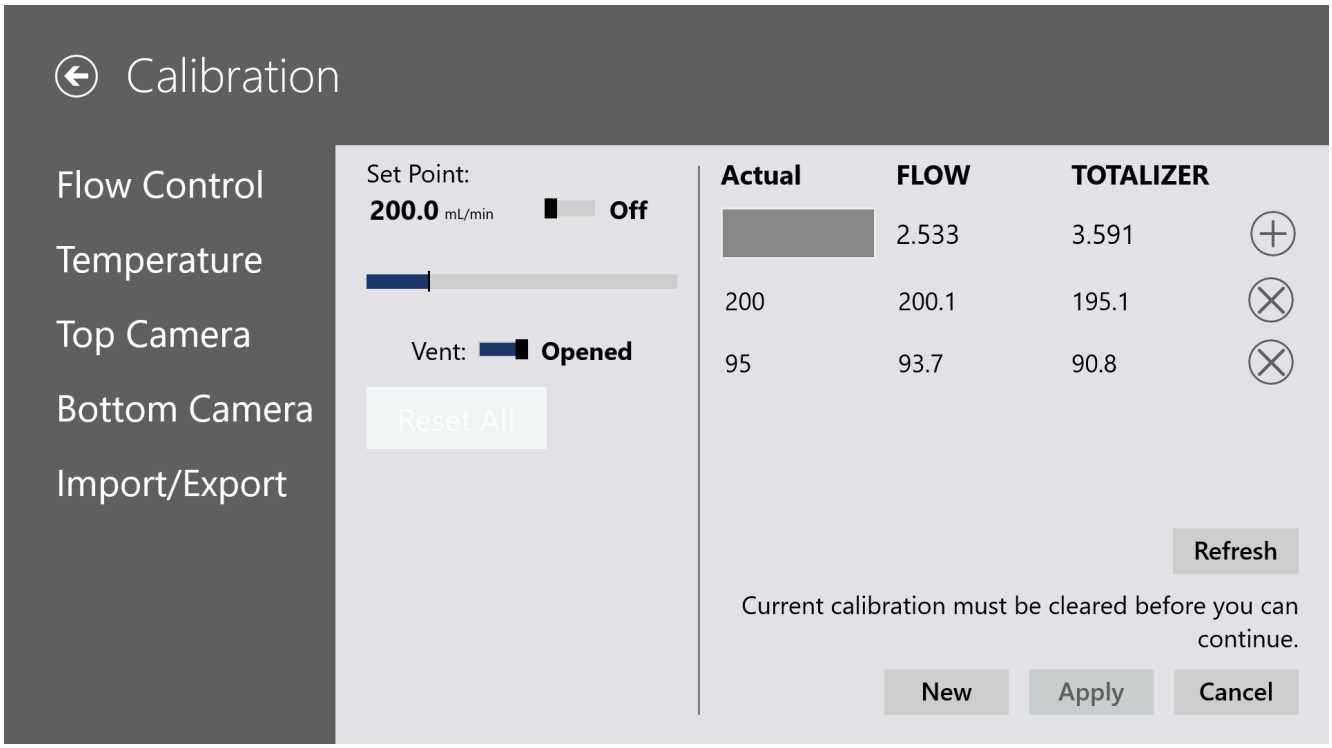

5.6.3.2. Temperature Calibration

To calibrate the temperature, a secondary thermometer and probe are required. It is recommended to use a Testo 735 and Testo 0609-7072 probe. Others devices can be used, but must match or exceed the specifications defined by the Testo.

> The Testo 735 is a multi-channel temperature measuring instrument with a large temperature range. It includes general specifications such as a -20°C to 50°C operating temperature and IP65 protection class.
>
> The Testo 0609-7072 is a glass-coated Pt100 laboratory probe designed for measuring corrosive media, with a temperature range of -50°C to 400°C. Key specifications include a fixed cable, a replaceable glass pipe made of Duran 50, and an accuracy of Class A for -50°C to 300°C and Class B for the remainder of the range. Its dimensions include a probe tip with a 5 mm diameter and 30 mm length, and a 6 mm diameter by 200 mm length shaft.

The temperature calibration procedure works by adjusting the offset by running reference test fluid.

Procedure:

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To start, make sure 180 mL of a reference sample (food grade mineral oil, etc.) is in the sample cylinder and a diffuser is connected to the stopper.

To calibrate the temperature, a second probe is required. Depending on the stopper generation of the module to be calibrated, there are two methods to inserting the probe into the sample.

Newer generation stoppers can simply detach the airflow tubes and feed the secondary probe through the unobstructed airflow exit connection.

Older generation stoppers have an obstructured connection.

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In this case, simply feed the probe along the side of the stopper and do not fully insert the stopper into the cylinder.

Ensure both probes are roughly the same height in the sample to make sure consistent temperature readings.

Note: Make sure the two probes are NOT touching eachother which is known to affect readings.

Use a small rubber band or connector to hold the probes in place apart from one another if available.

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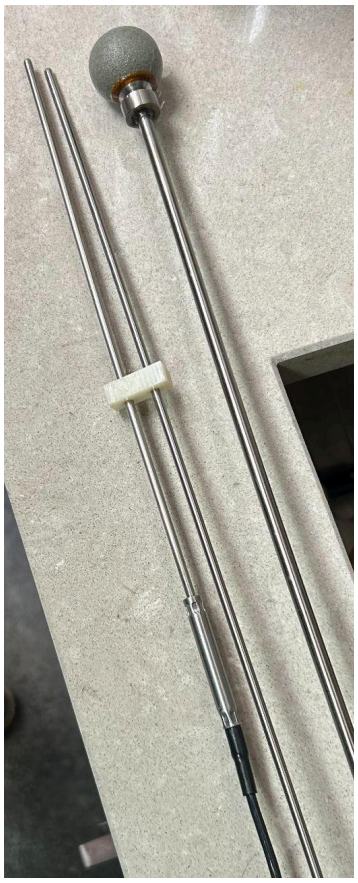

Ensure both probes are not touching the glass or diffuser as well.

Note: Operators may also attach the airflow to the In connector and set the flow to 94 mL / min in Diagostics to ensure consistent and ubiquitous temperature throughout the sample.

Before proceeding, ensure that the temperature probe is connected to the top of the stopper and both probes are at the same height in the sample.

Once Calibrate is selected for a module under the initial calibration screen, navigate to the Temperature tab.

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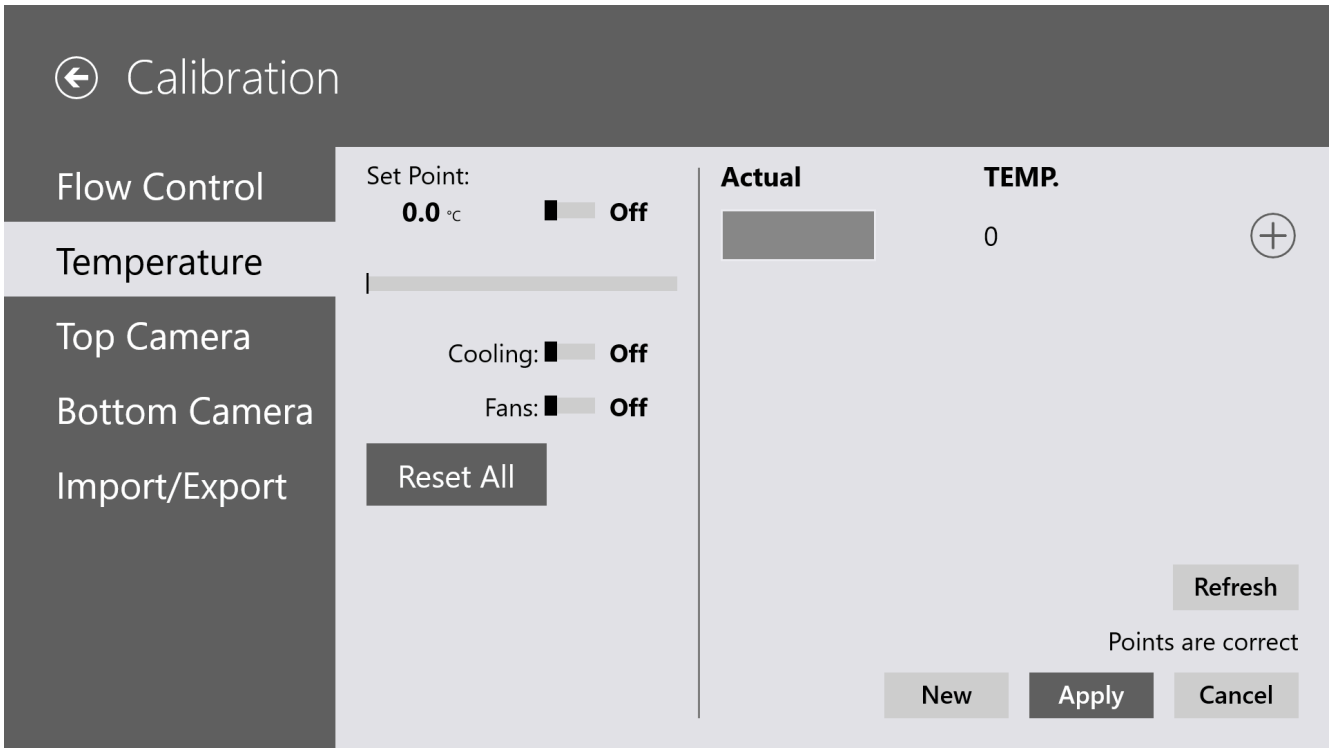
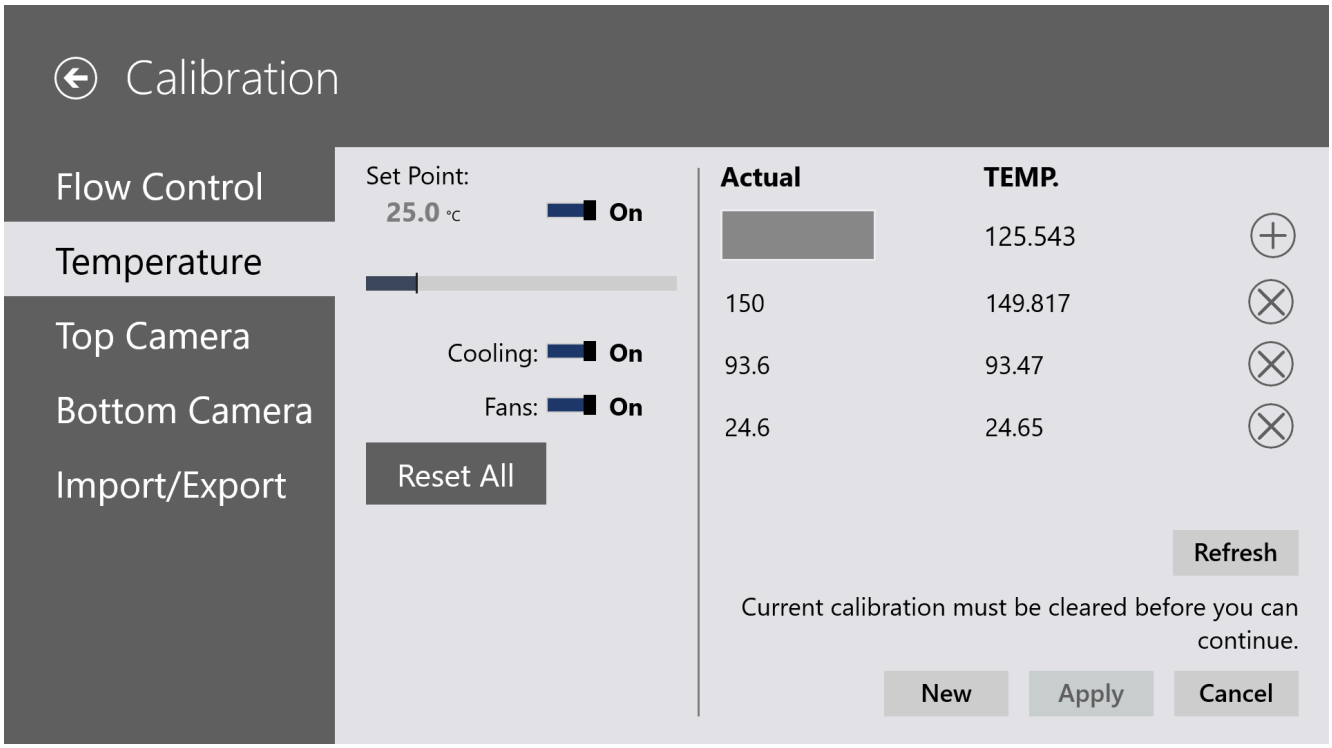


Click New to start a new calibration. If there is an existing calibration outside of the factory calibration, it will be displayed on the right side of the screen. Clicking New will erase the calibration.



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If there is not a calibration displayed on the right side of the screen, clicking New is not necessary and the operator can proceed.

The temperature set points used for the calibration should be representative of the software variant of the FoamDDI:

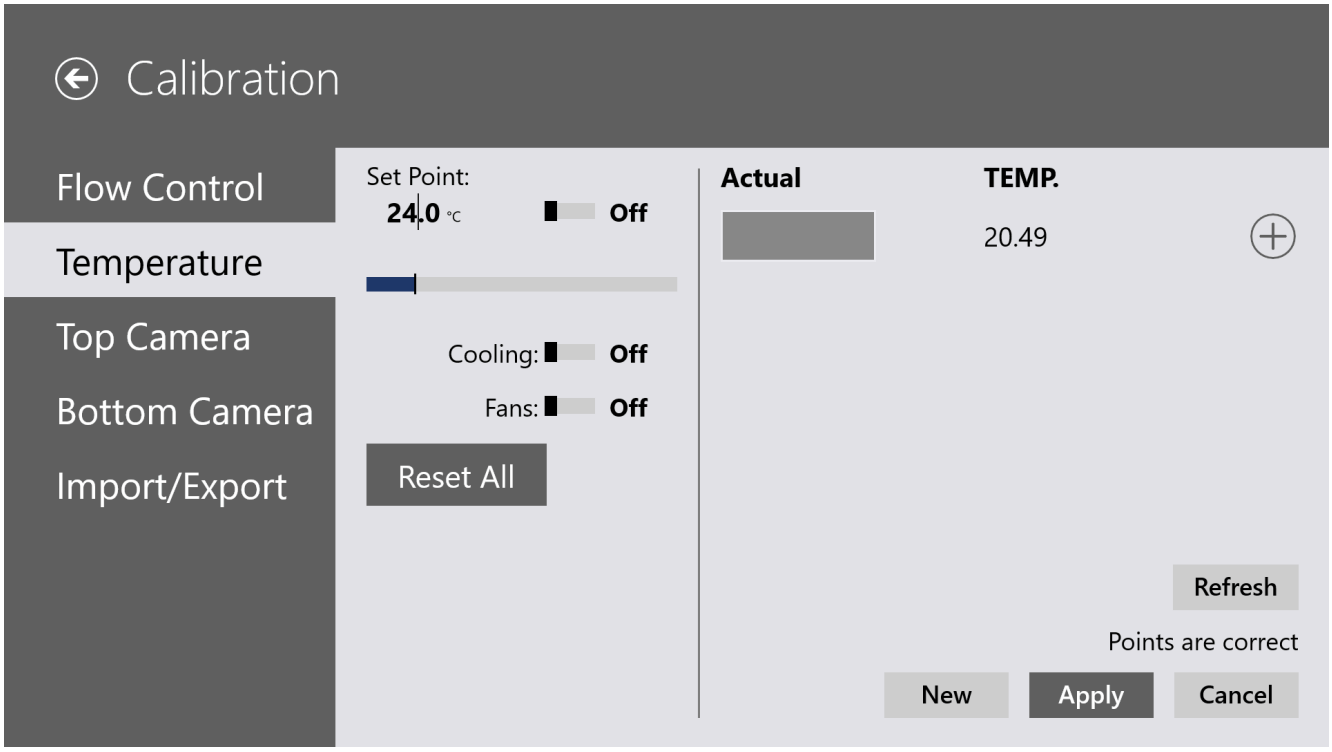
Lubricant	Coolant	Aminic
24°C	88°C	24°C
93.5°C	–	35°C
150°C	–	95°C

Note: For the coolant version, only the 88°C set point is required since it is the only temperature tested under ASTM D1881.

Note: Other and/or more temperature set points can be used for calibration. These are only the suggested/bare minimum suggested set points.

Note: For the purpose of consolidation, the below instructions will follow the set points suggested for the lubricant variant of the FoamDDI.

Click below the Set Point or use the slider to set the first temperature set point.



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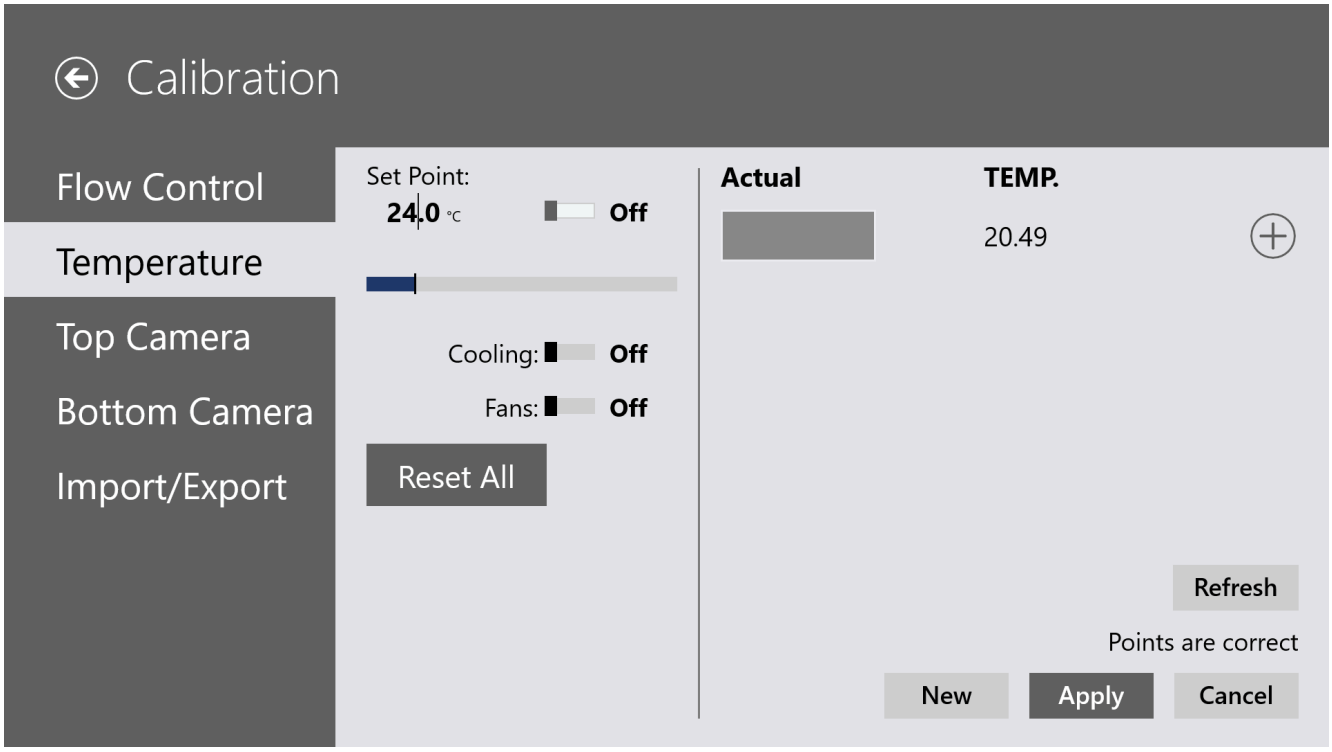
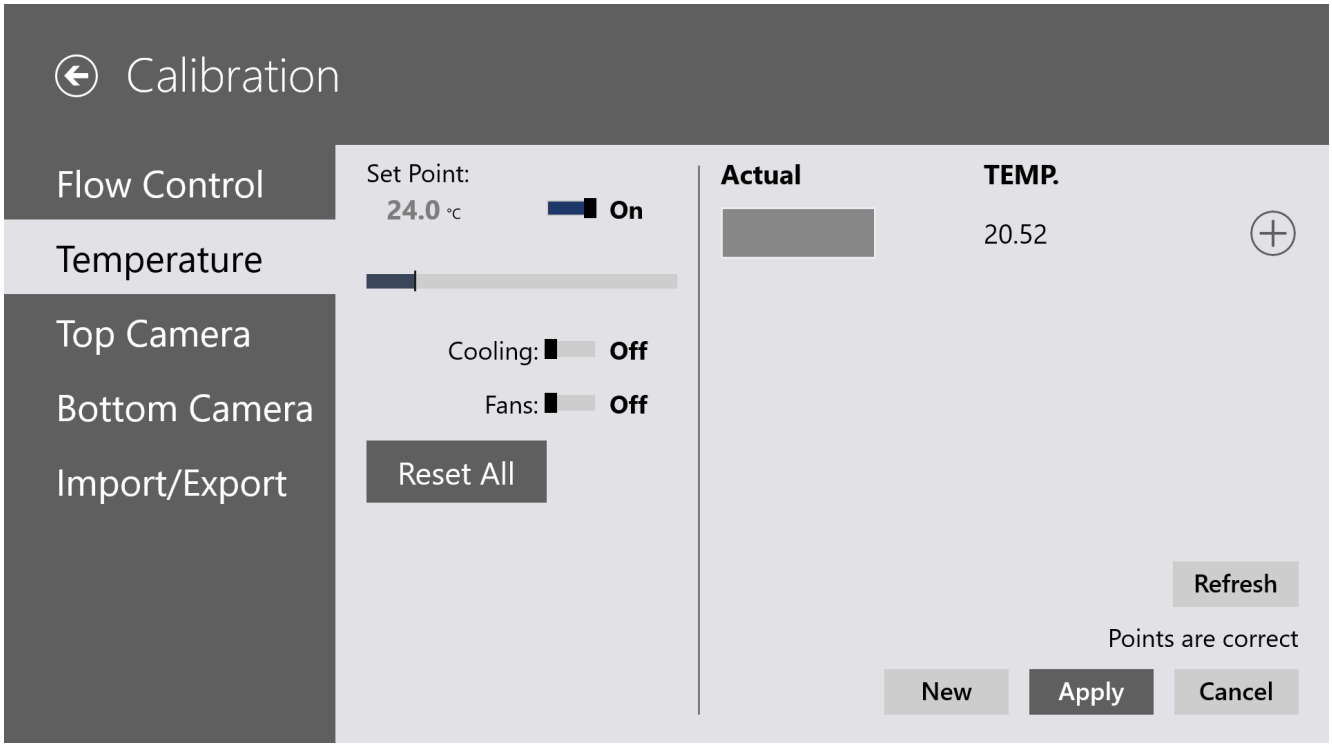
Use the On/Off toggle beside the set point to turn on the temperature regulation elements. Click the toggle to On.





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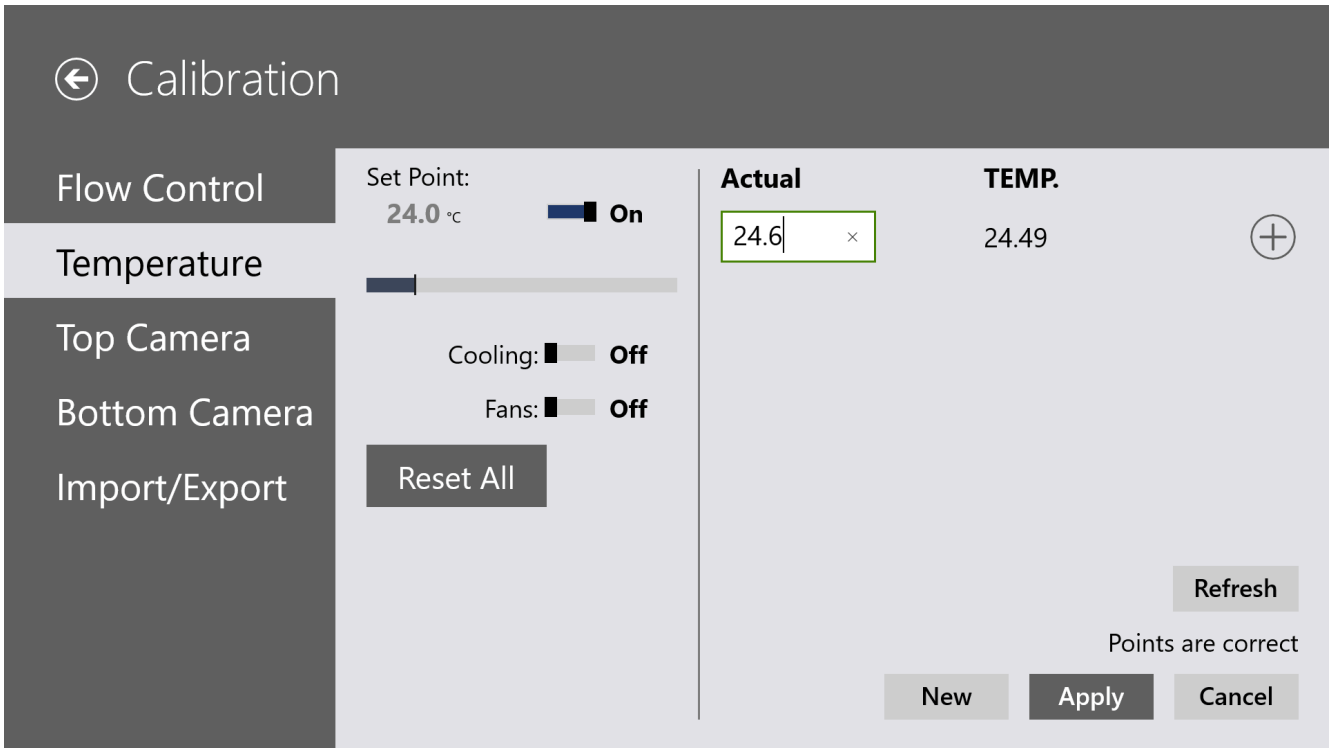
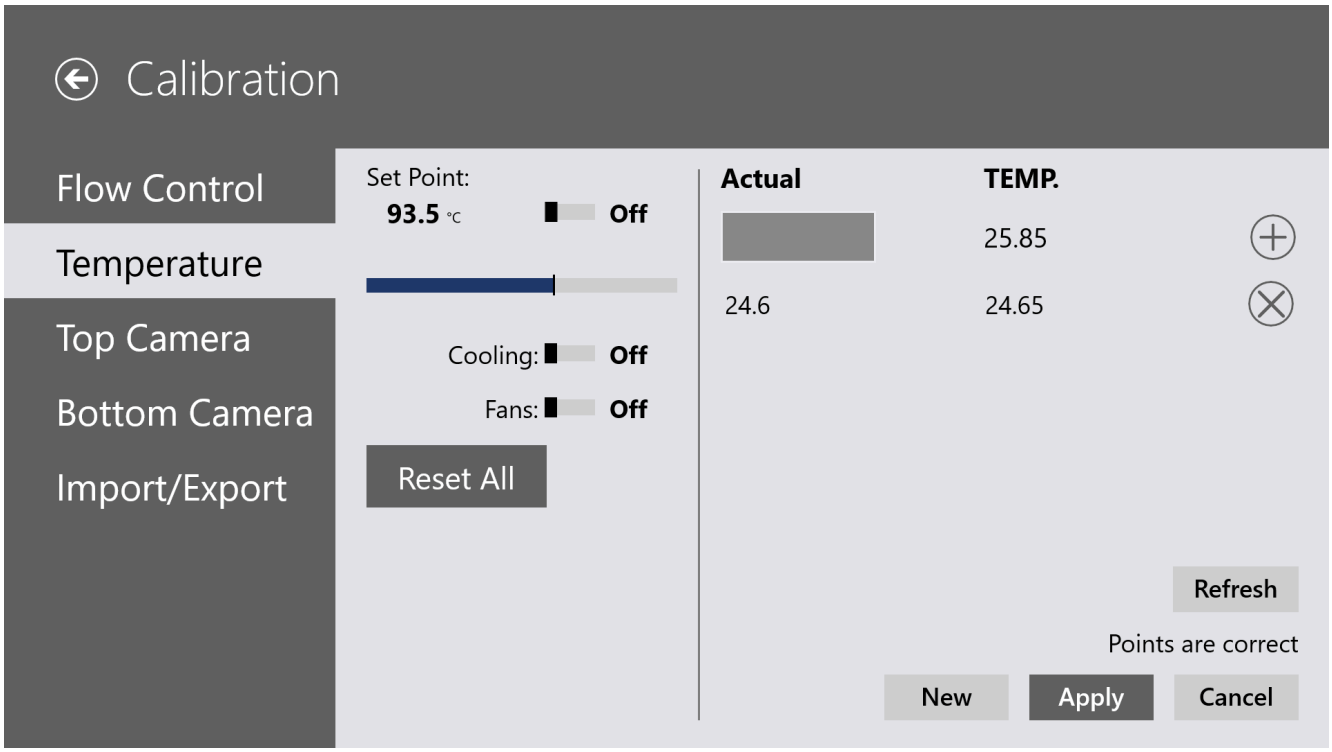
Wait until bot temperature probes have stabilized before proceeding. Less than a +/- 0.4°C fluctuation is recommended.

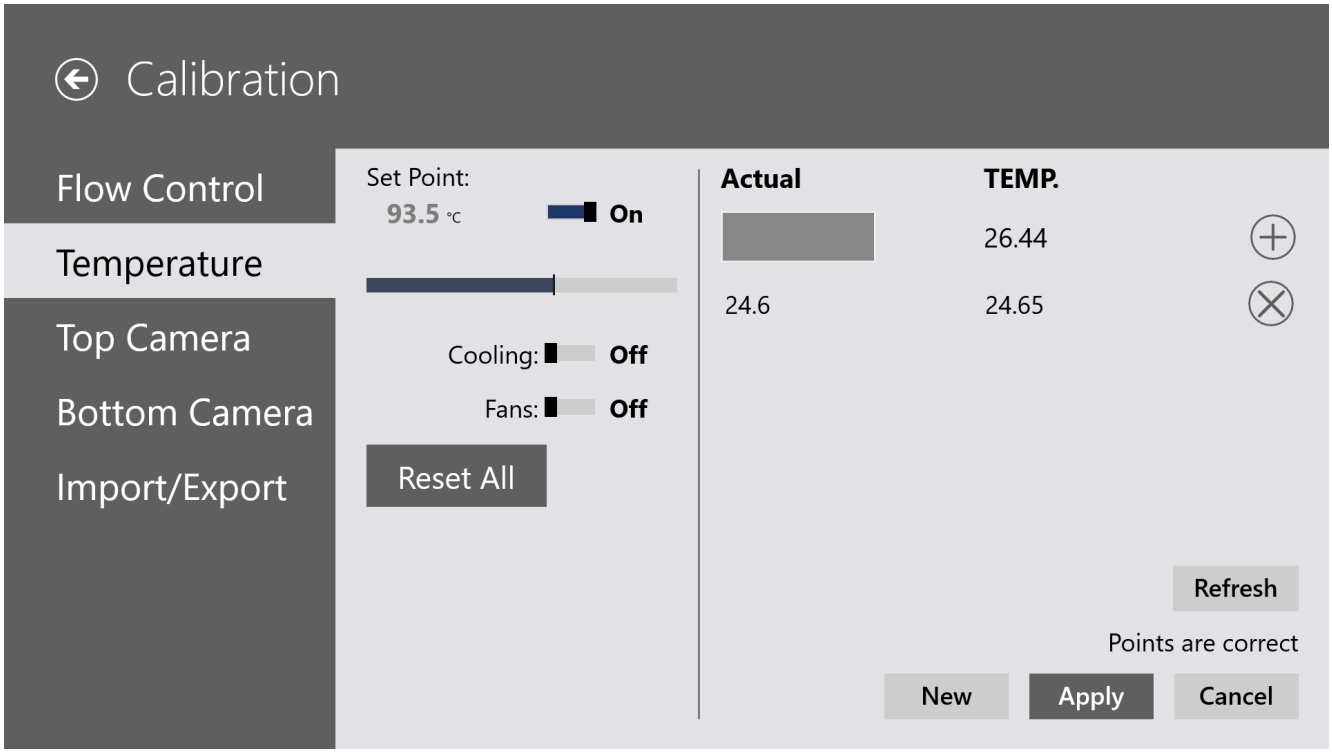
Note: The temperature calibration takes the longest due to the waiting periods allowing the sample to stabilze its temperature.

Once stable, type the secondary temperature probe reading into the Actual column.



Click the plus (+) icon on the right to add this set point to the calibration. Once set, proceed with the next (if applicable) Set Point to be added by following the same instructions as above.





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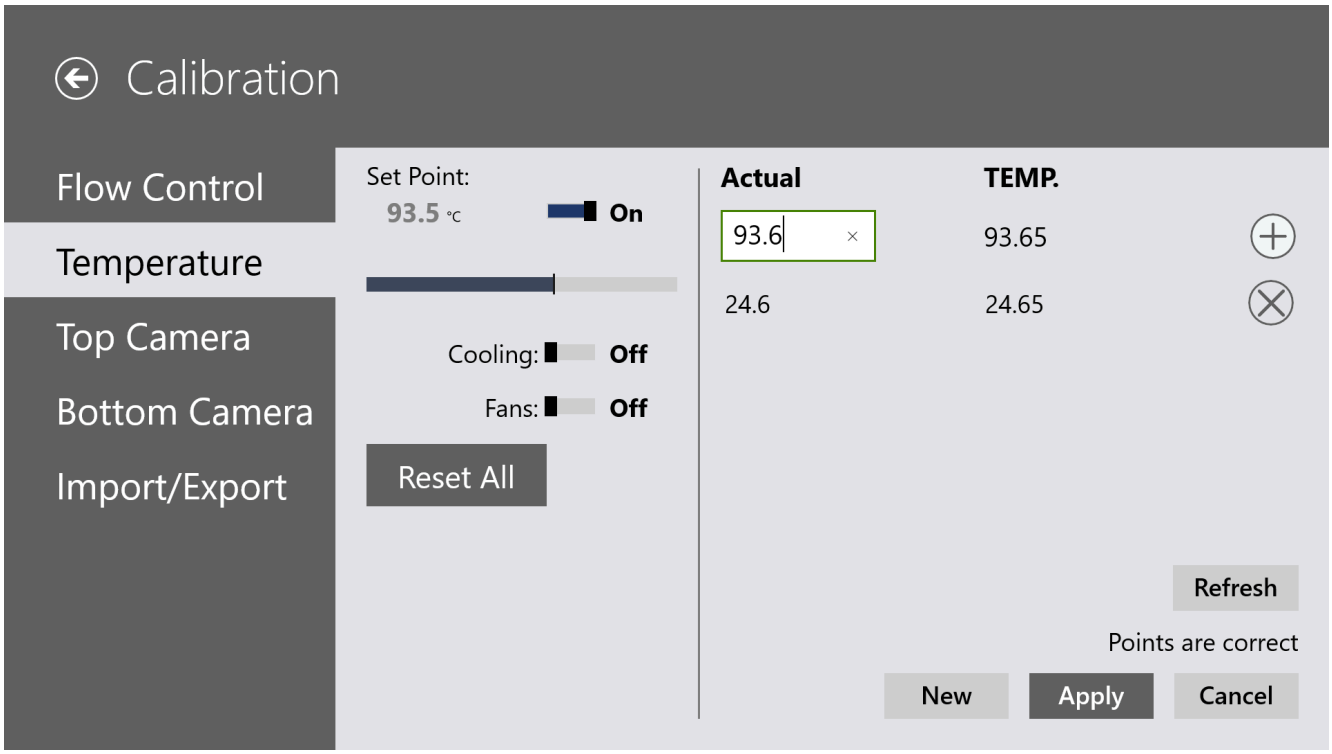

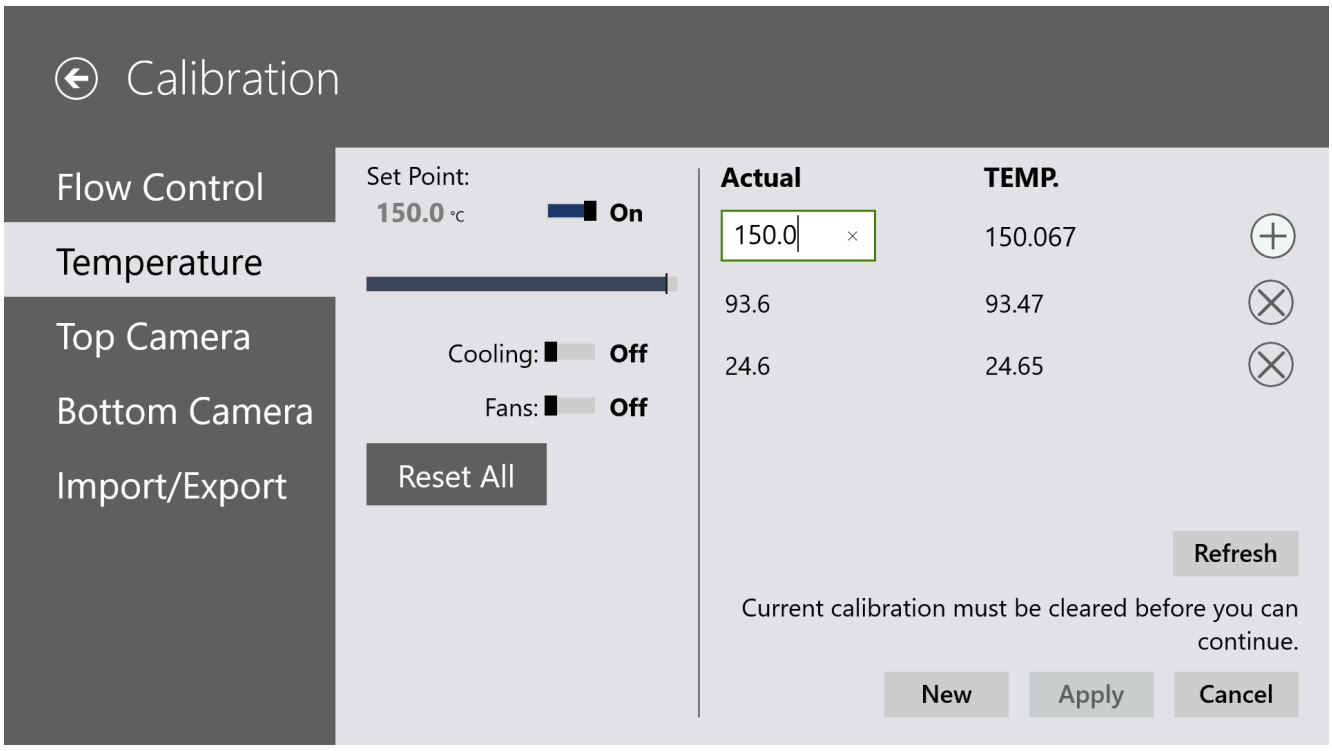

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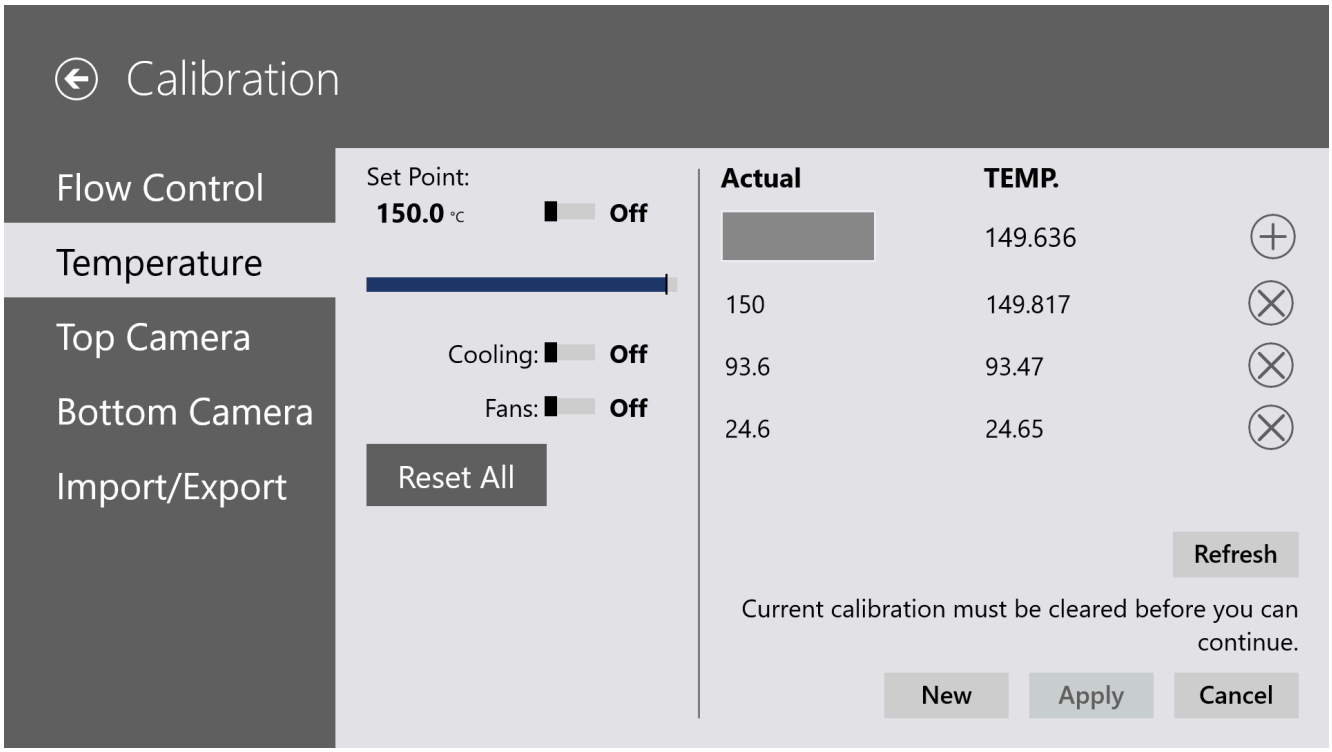


Once all set points have been added, click Apply to confirm and save the calibration.

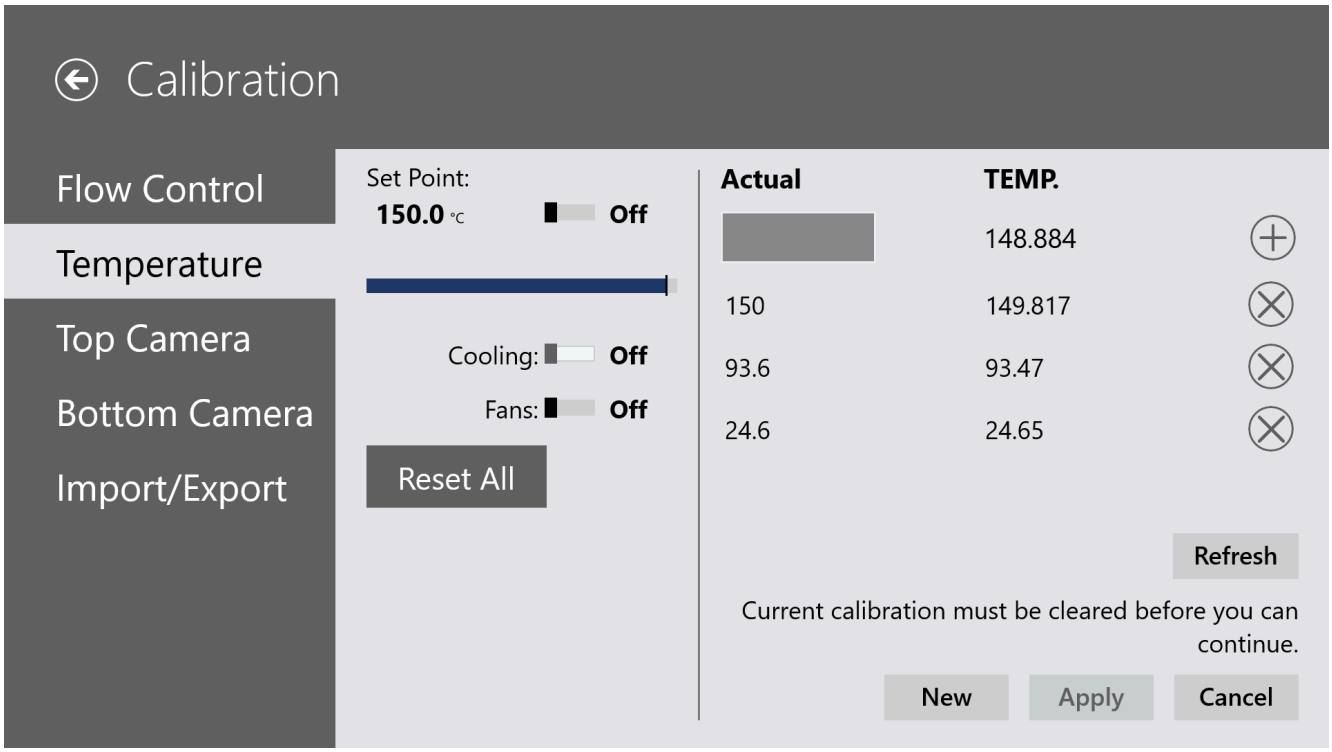
If the previous calibration is preferred, the operator can delete the current calibration by clicking Cancel or Refresh BEFORE saving the current calibration by clicking Apply.

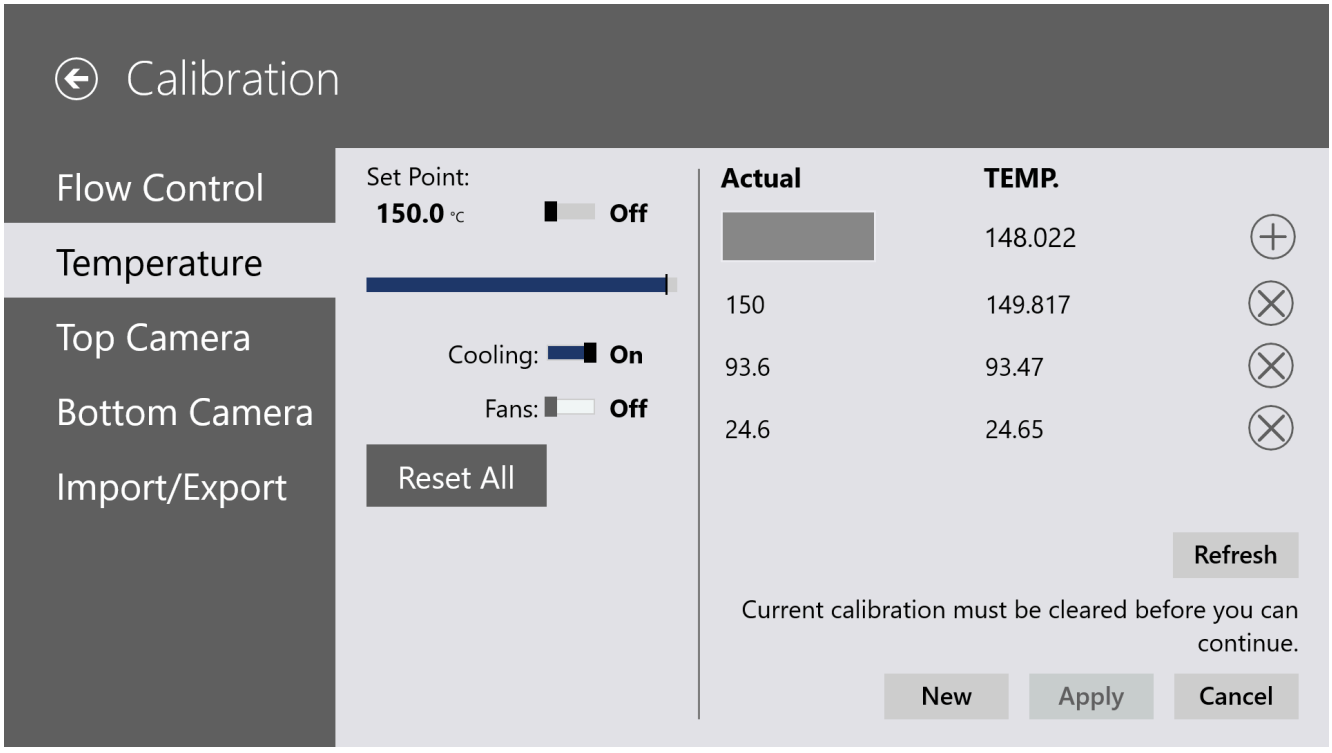
Note: Once applied and confirmed, the old calibration is disregarded.

To delete only part of the calibration, select the X icon on the right side of the screen.

To undo a deletion, click Refresh.

To speed up the cooling process, the toggles for Cooling and Fans can be turned on by switching the toggles to On.





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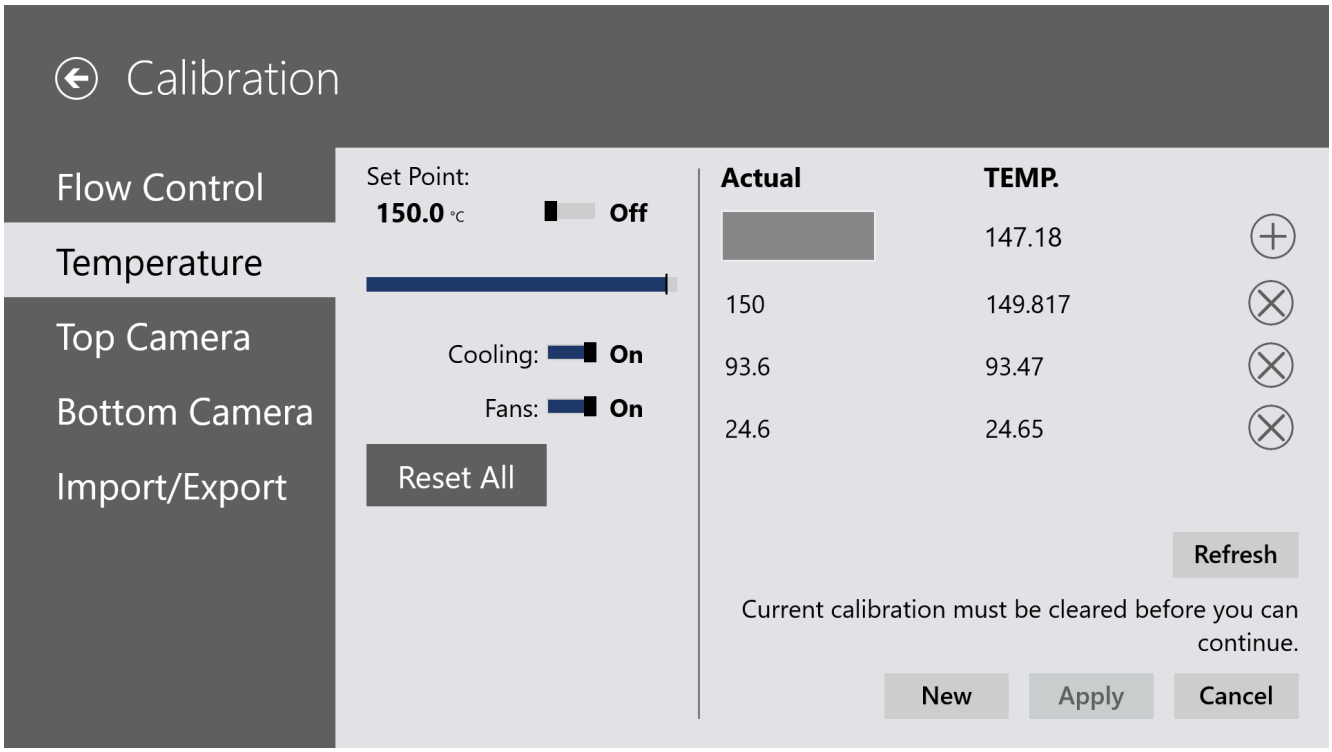
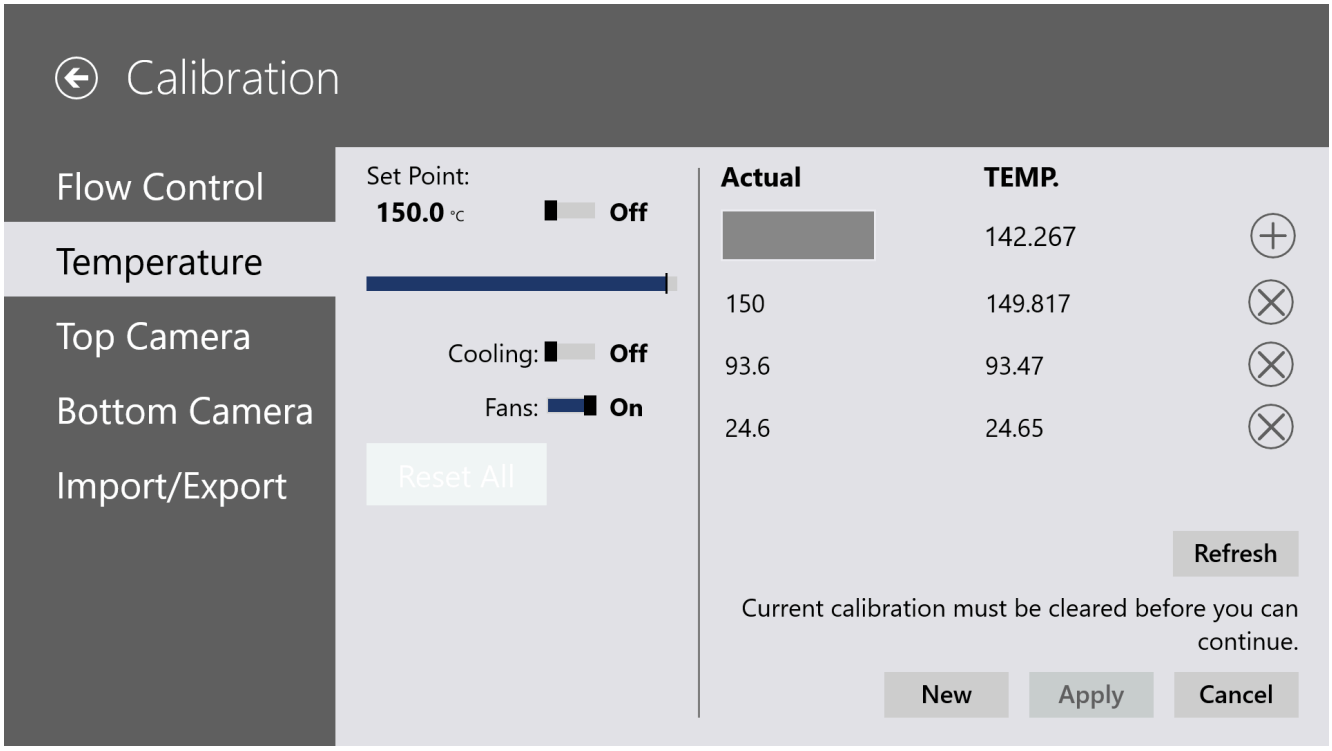


The Reset All button, turns off all temperature regulators (heating and cooling).



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5.6.3.3. Top Camera Calibration

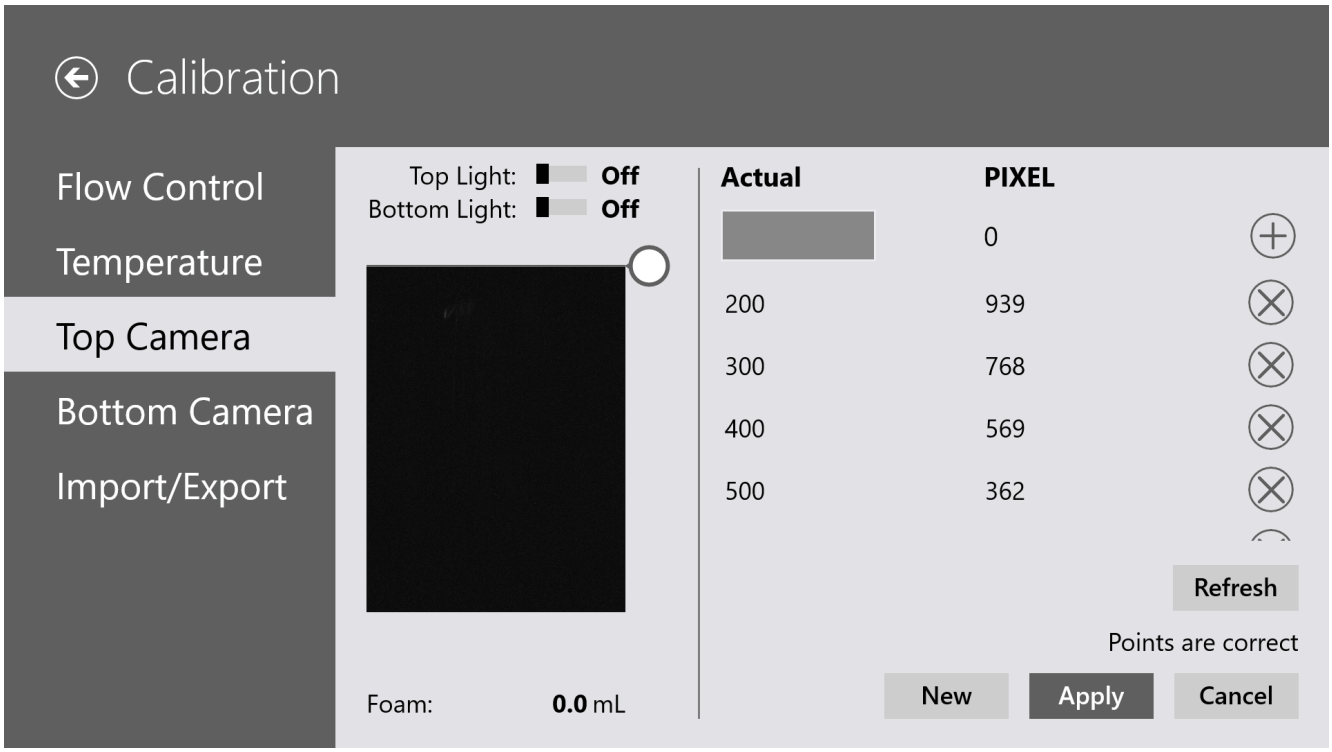
To calibrate the vision system, an empty sample cylinder is suggested.

The vision calibration procedure works by adjusting the algorithm graduations using the calibration sample cylinder.

Procedure:

Ensure that the graduations on the sample cylinder are rotated in a way that they are facing the camera (backwards).

Once Calibrate is selected for a module under the initial calibration screen, navigate to the Camera tab.



Turn on both the top and bottom light by using the On/Off toggles at the top left of the screen.

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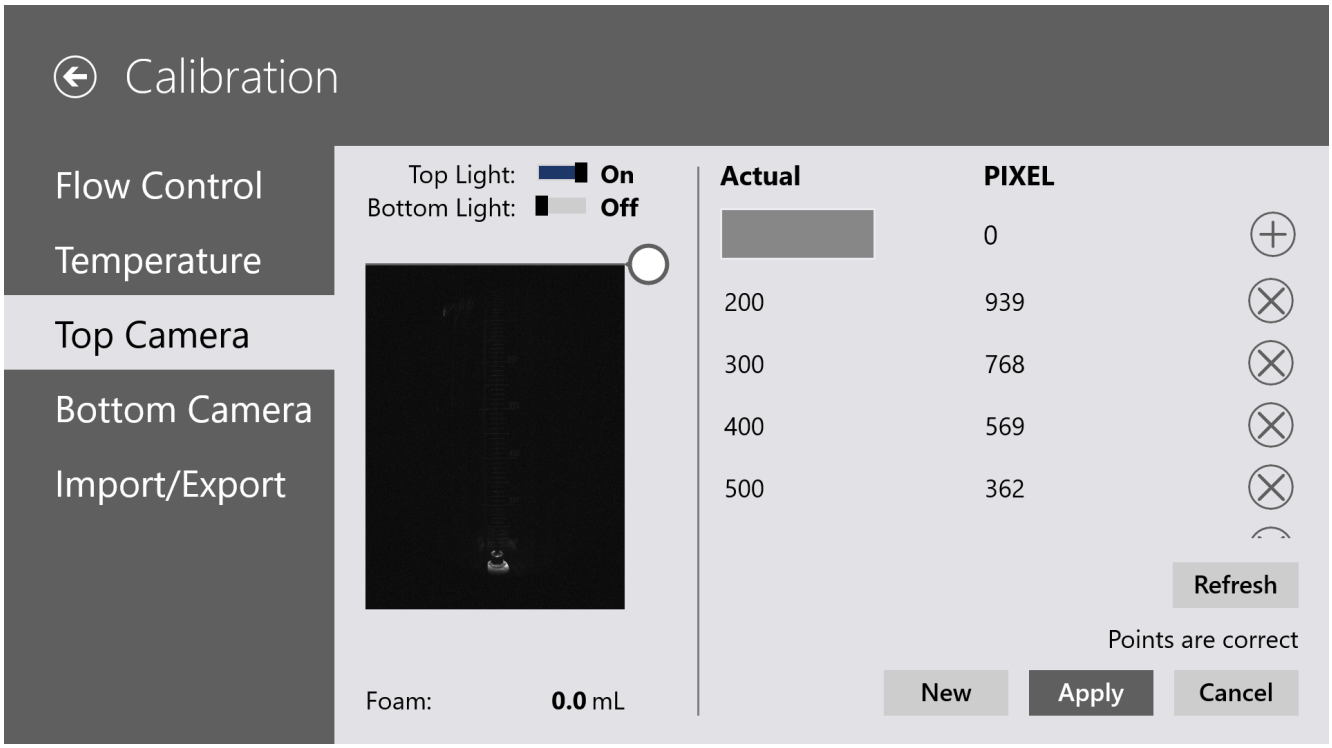

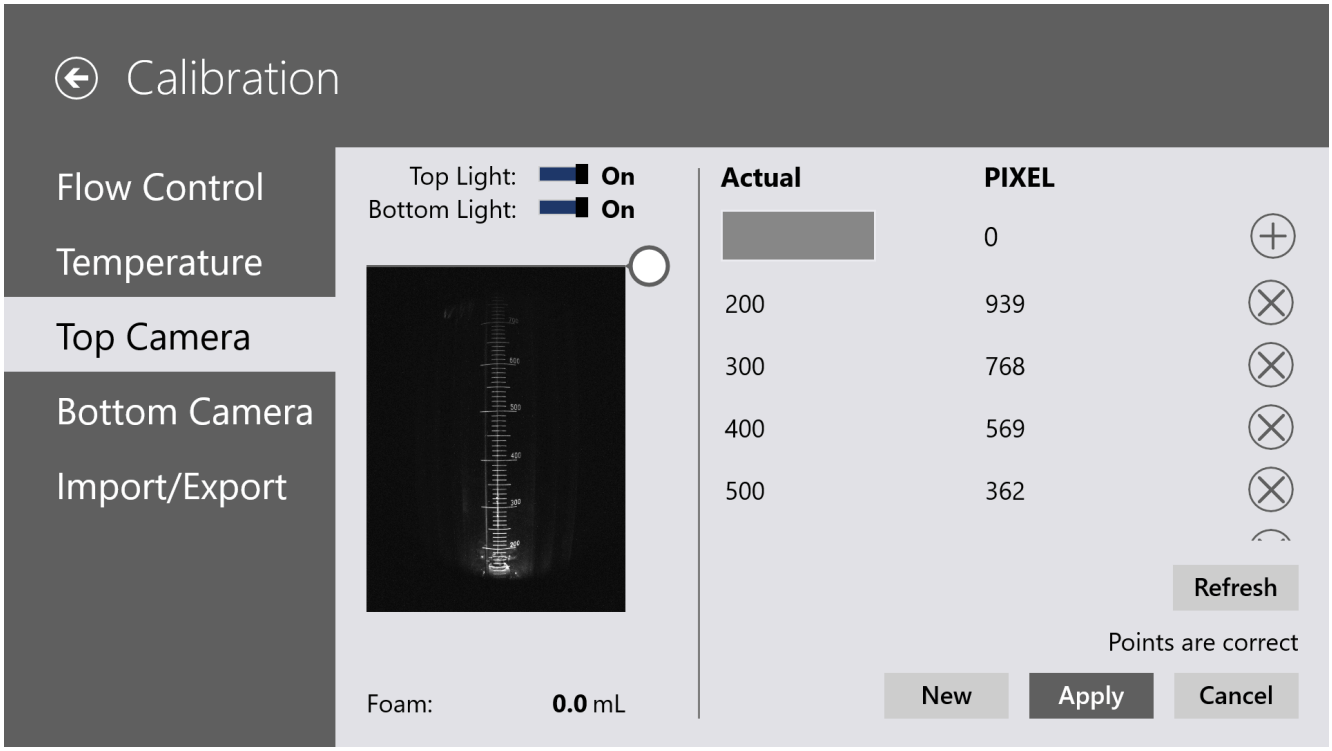

Click New to start a new calibration and disregard the previous calibration.

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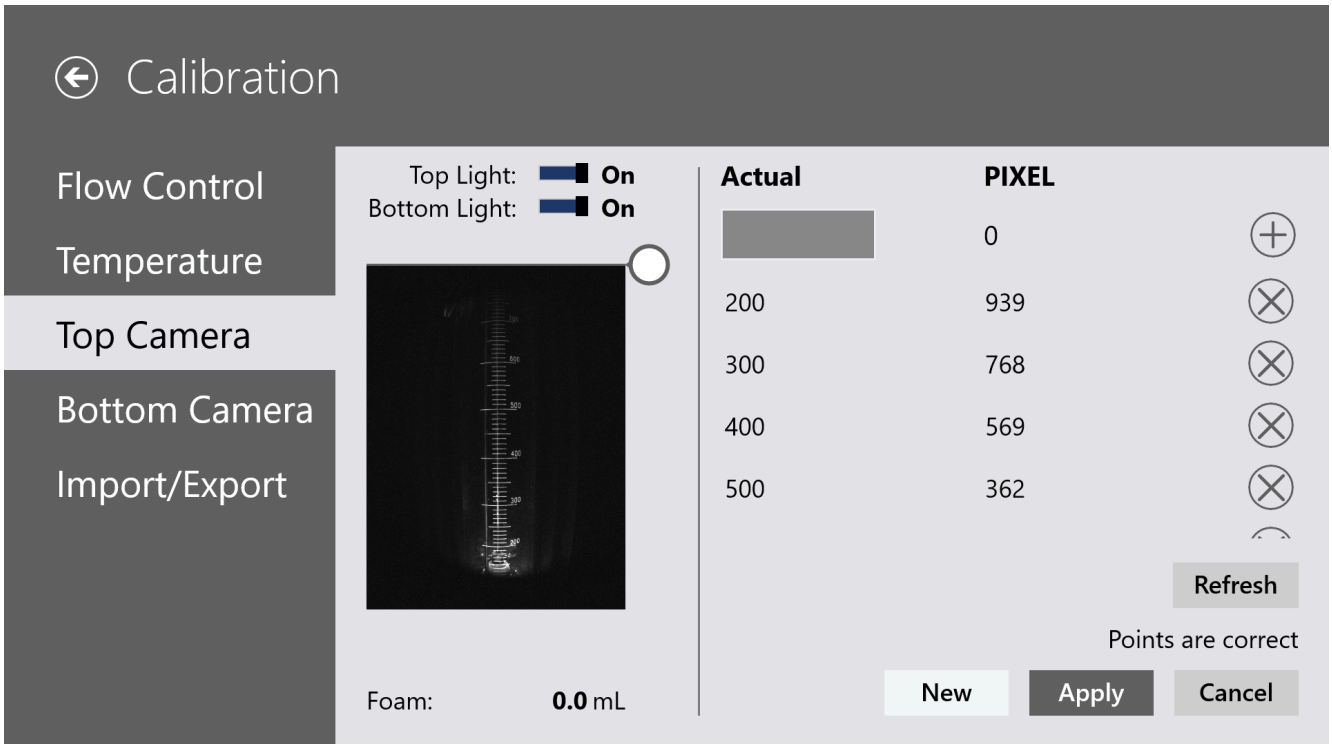
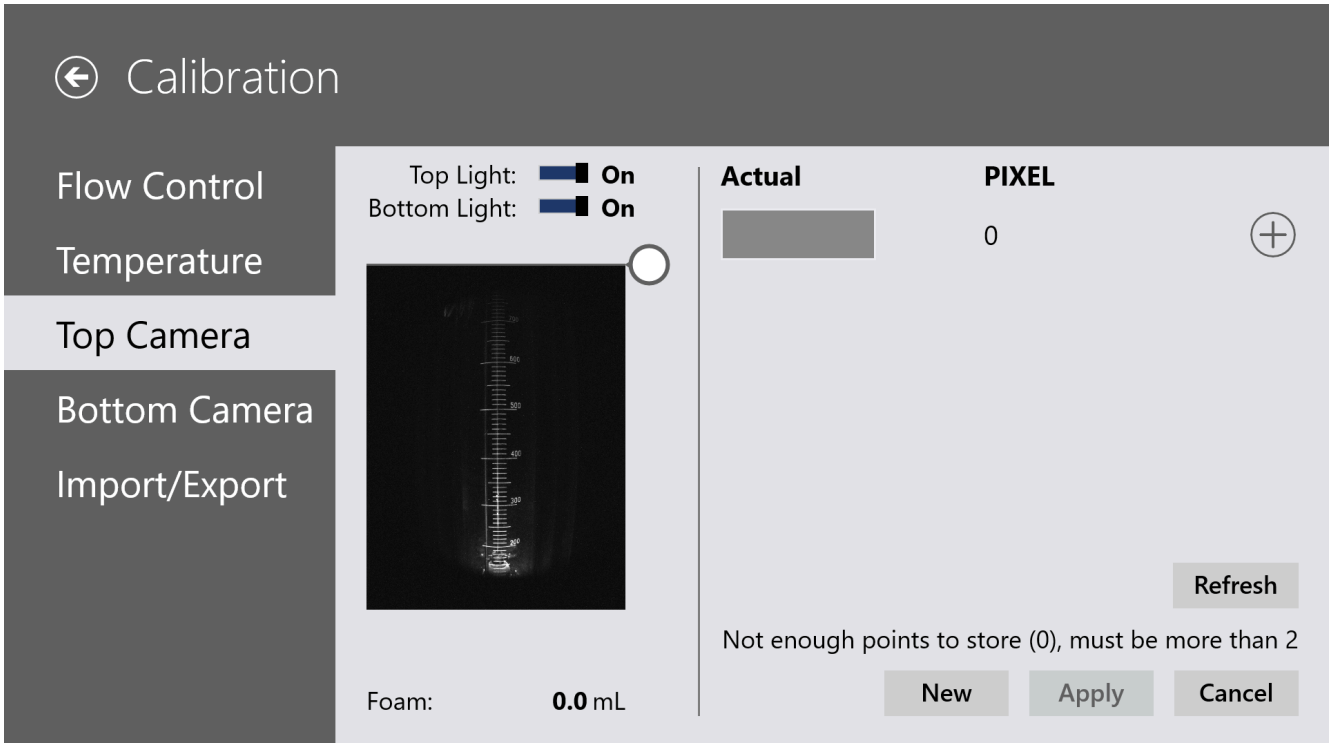




Determine which graduations to use for the set points.

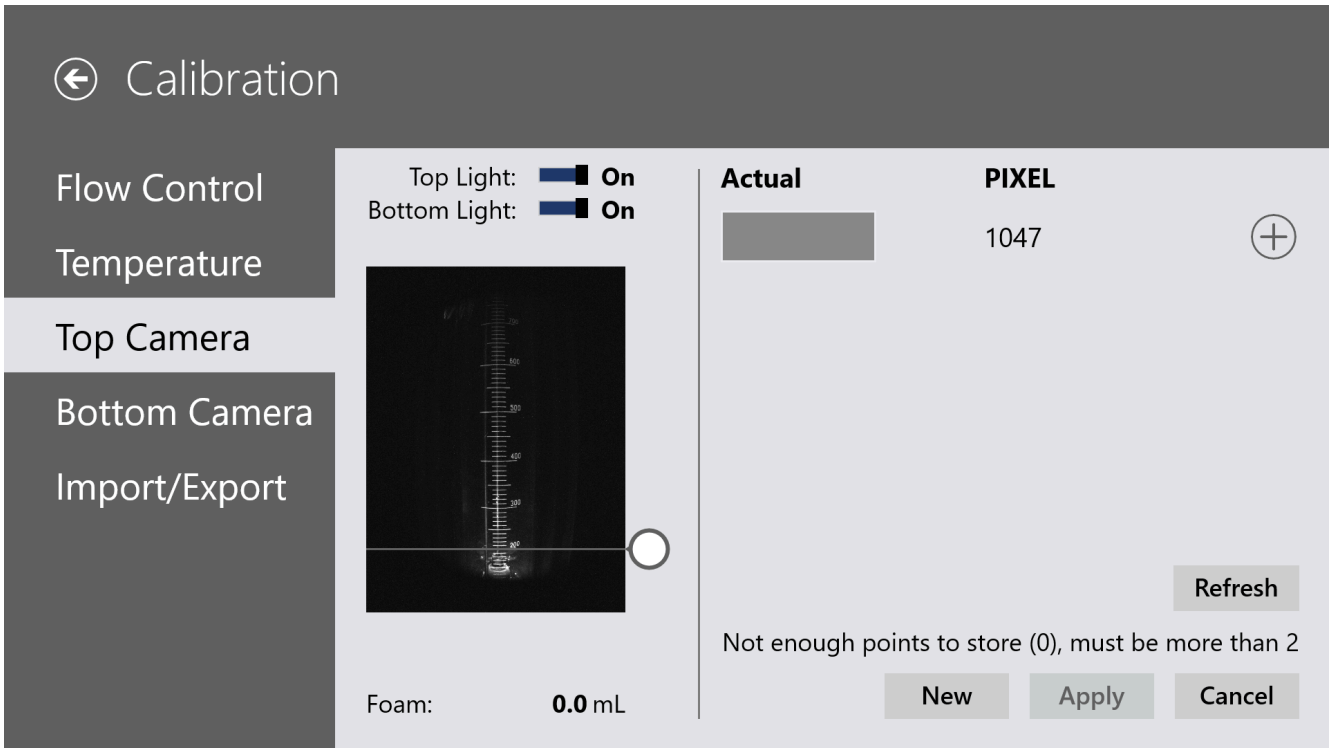
For a 1000 mL sample cylinder (lubricant variant), the following are recommended:

- 200 mL
- 300 mL
- 400 mL
- 500 mL
- 600 mL

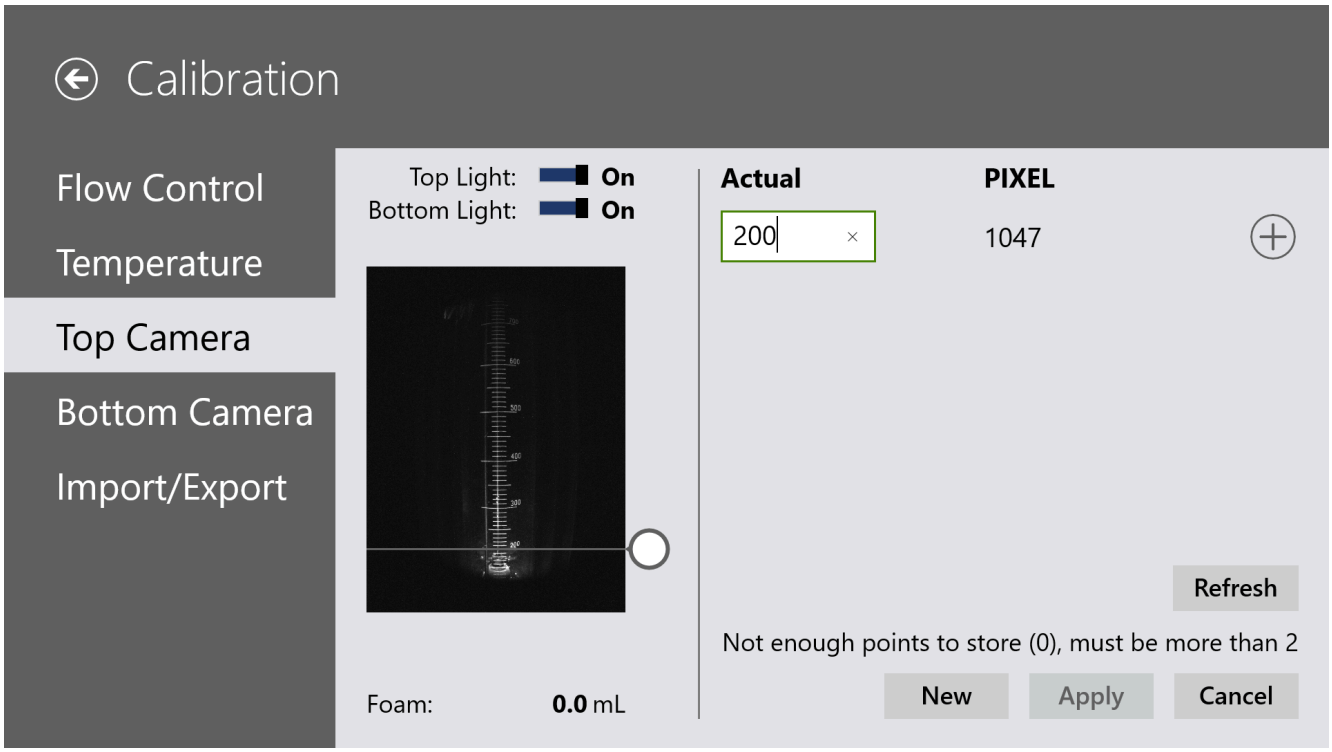
For a 500 mL sample cylinder (coolant and aminic variant), the following are recommended:

- 100 mL
- 200 mL
- 300 mL
- 400 mL
- 500 mL

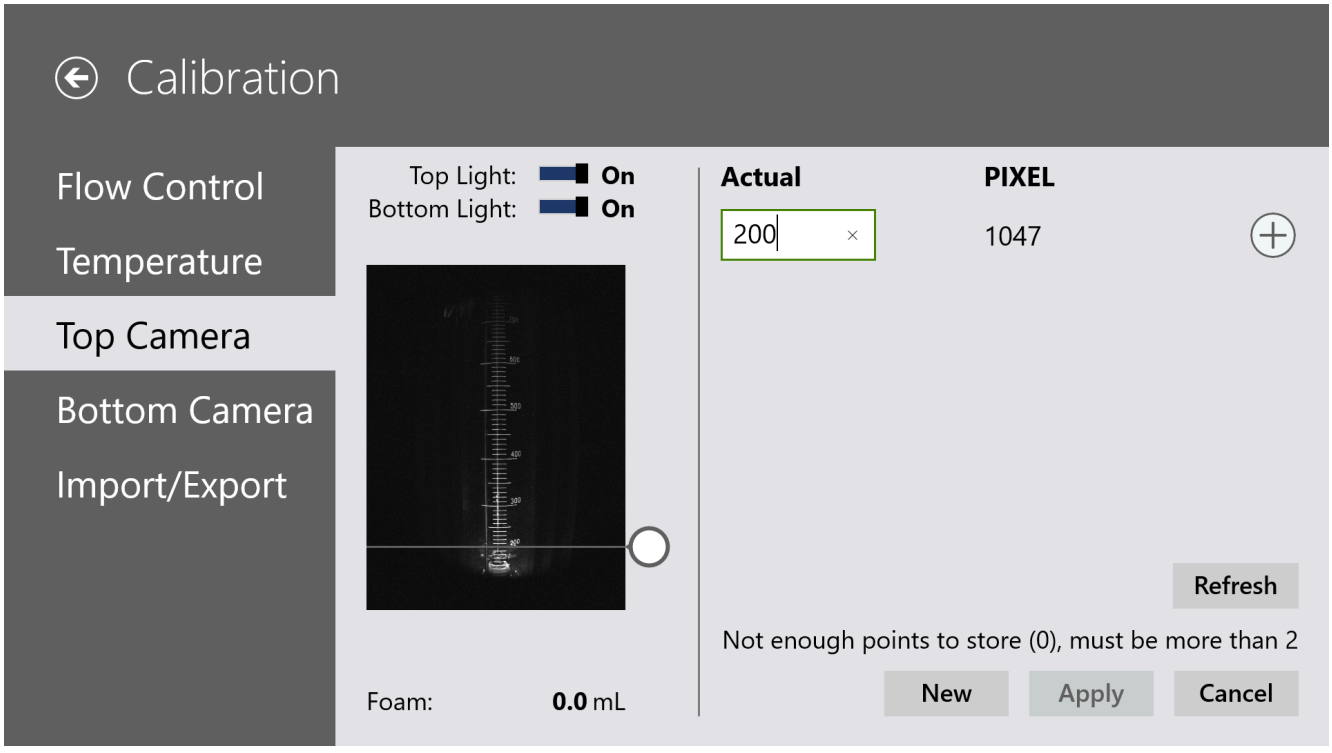
Drag the slider to the graduation mark in the live photo. Due to the angle of the camera, ensure the meniscus of the sample would be the measurement level.



Once the level is accurately marked with the slider, type the graduation in the Actual column.



Click the plus (+) icon on the right to add the graduation set point to the new calibration.



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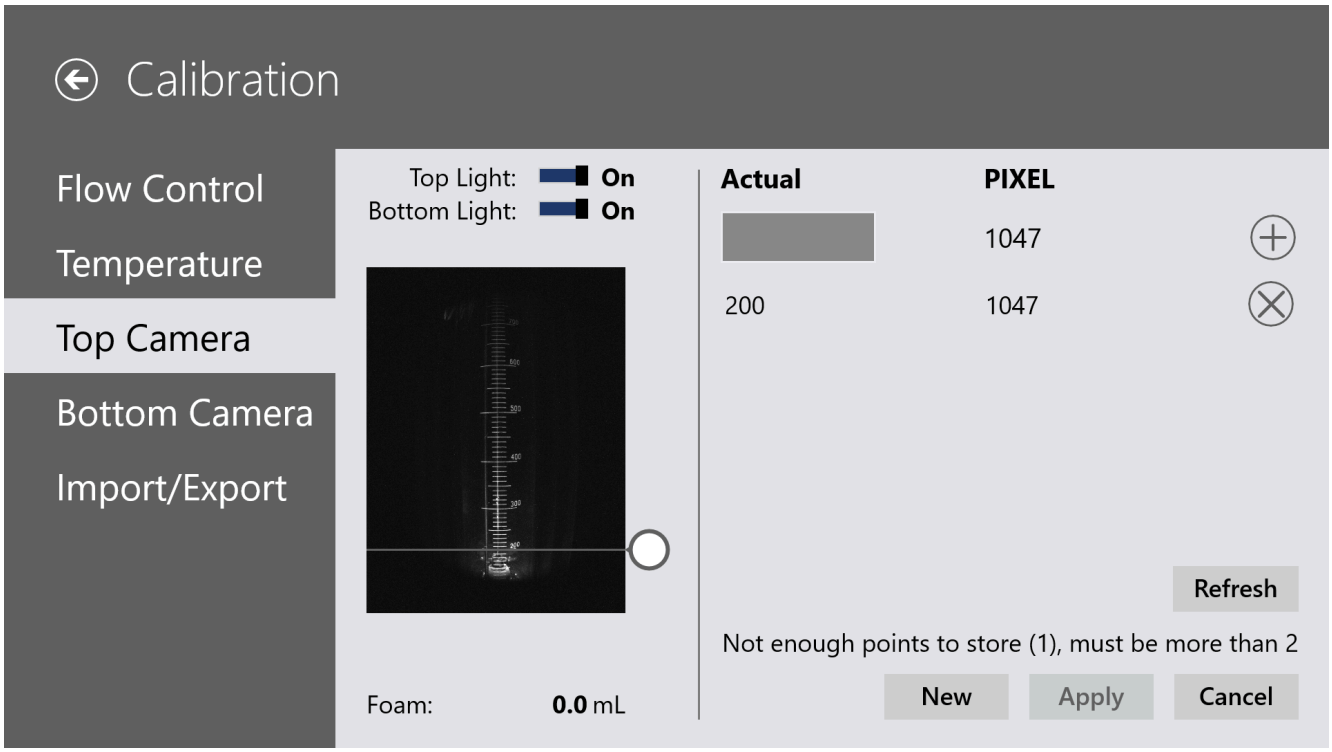
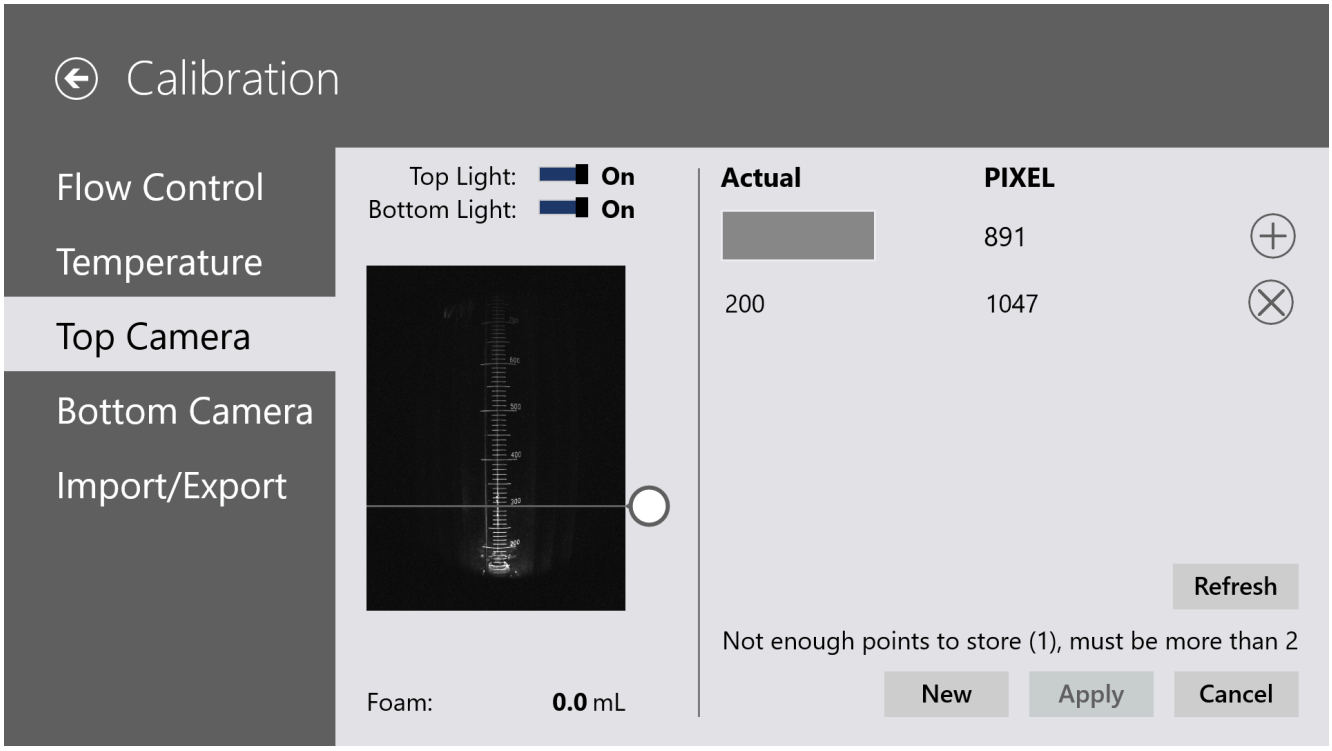


Repeat this procedure for at least one more graduation.



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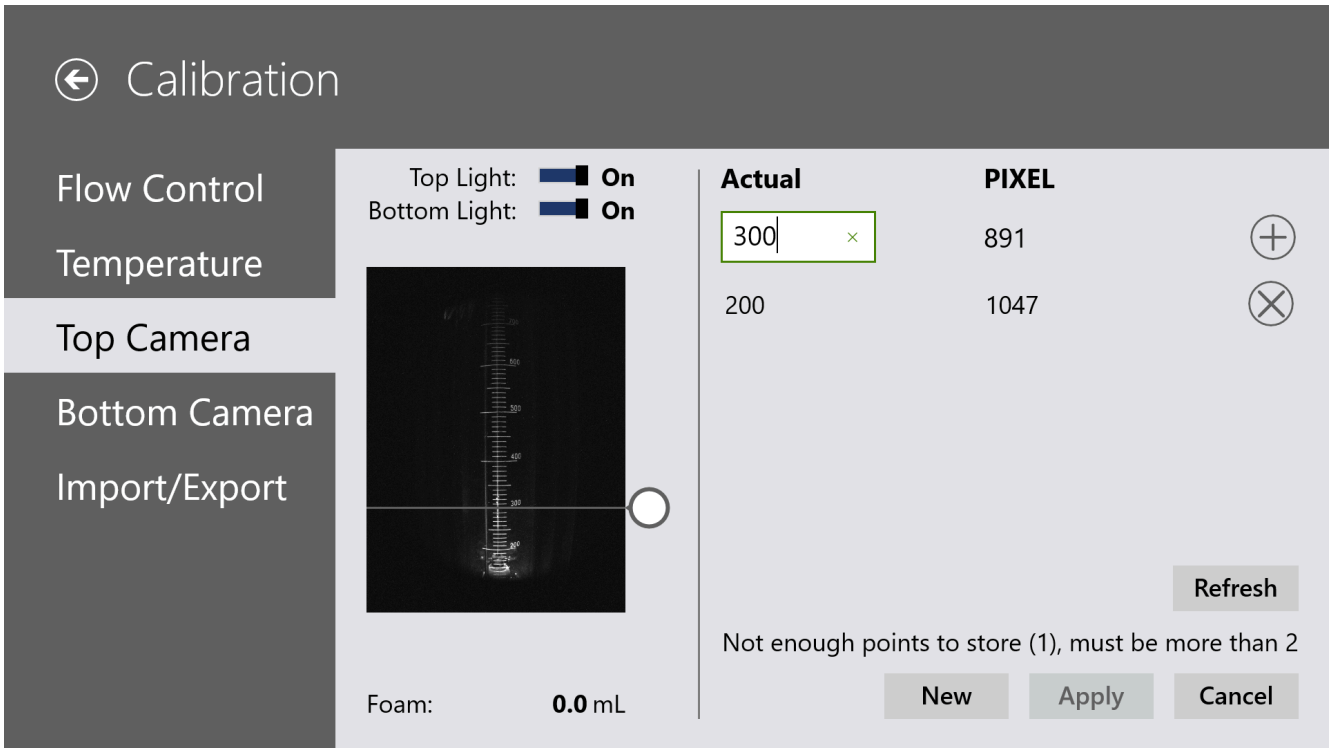

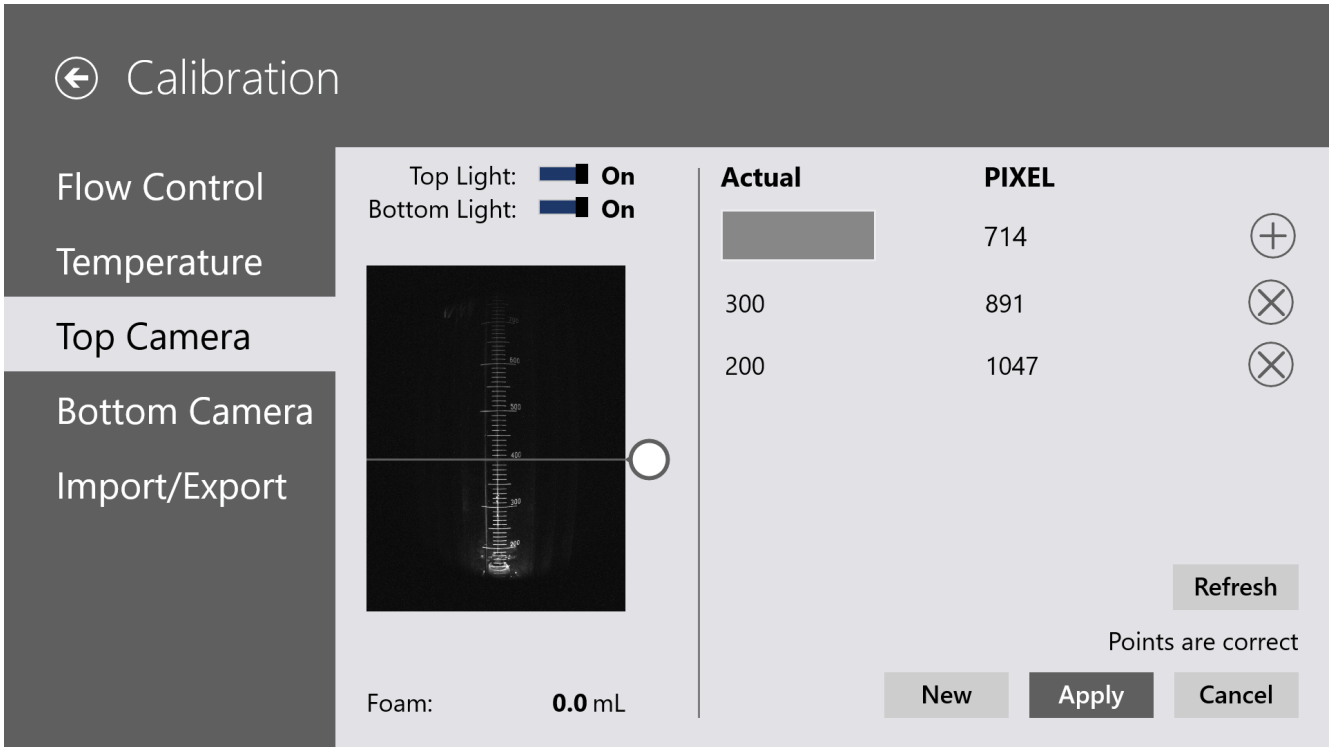

Continue until all desired set points have been added to the calibration on the right.

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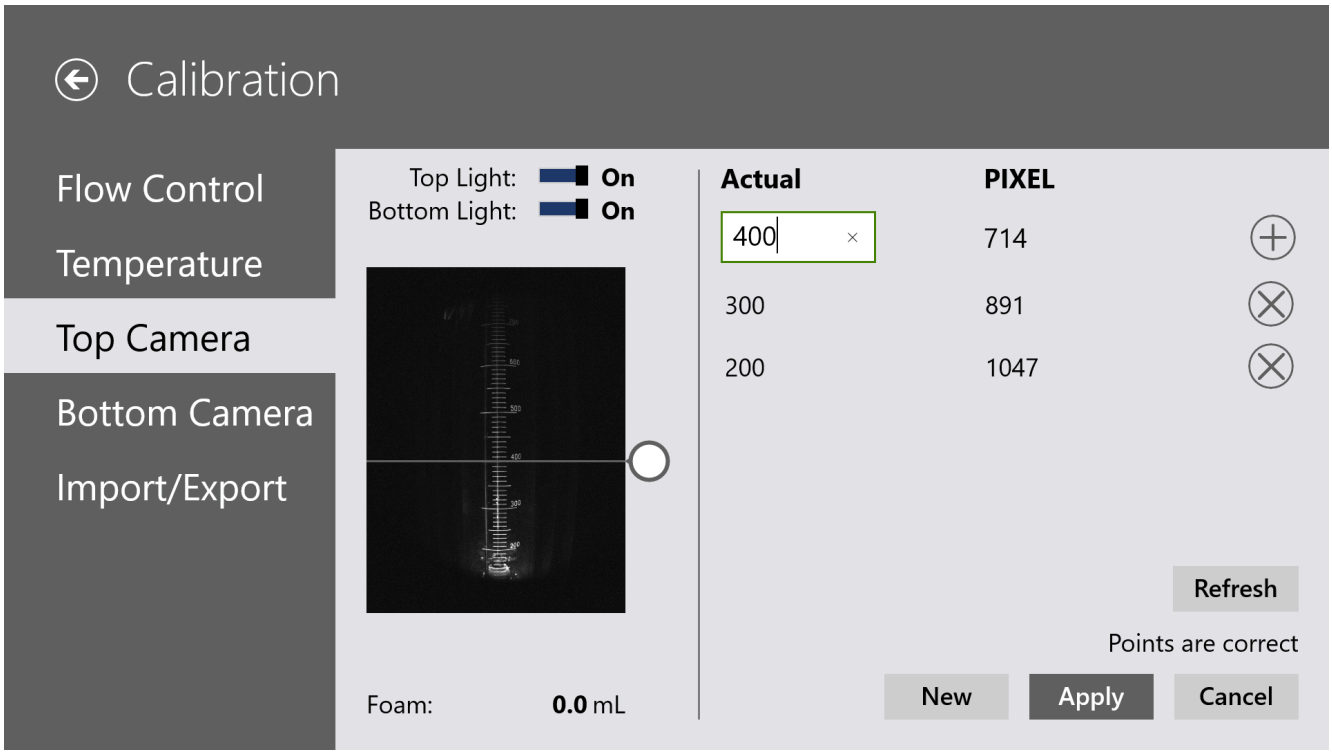

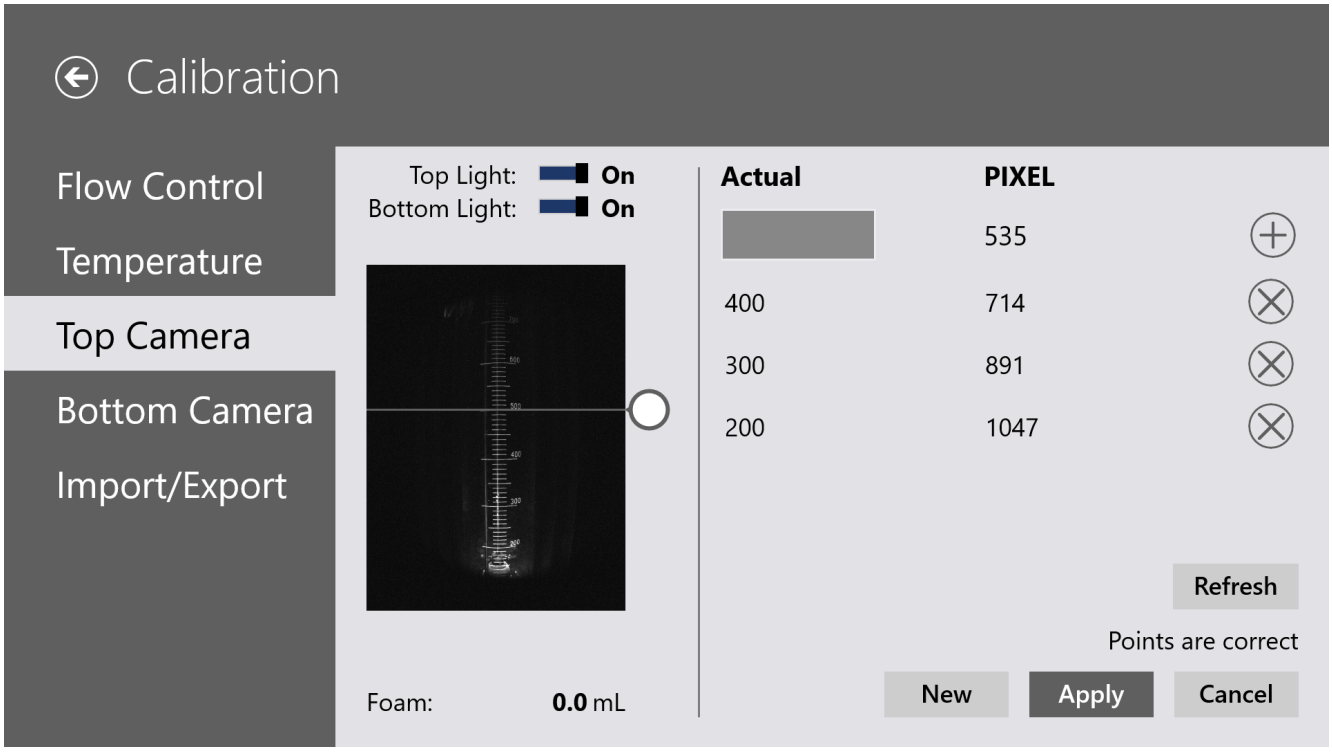

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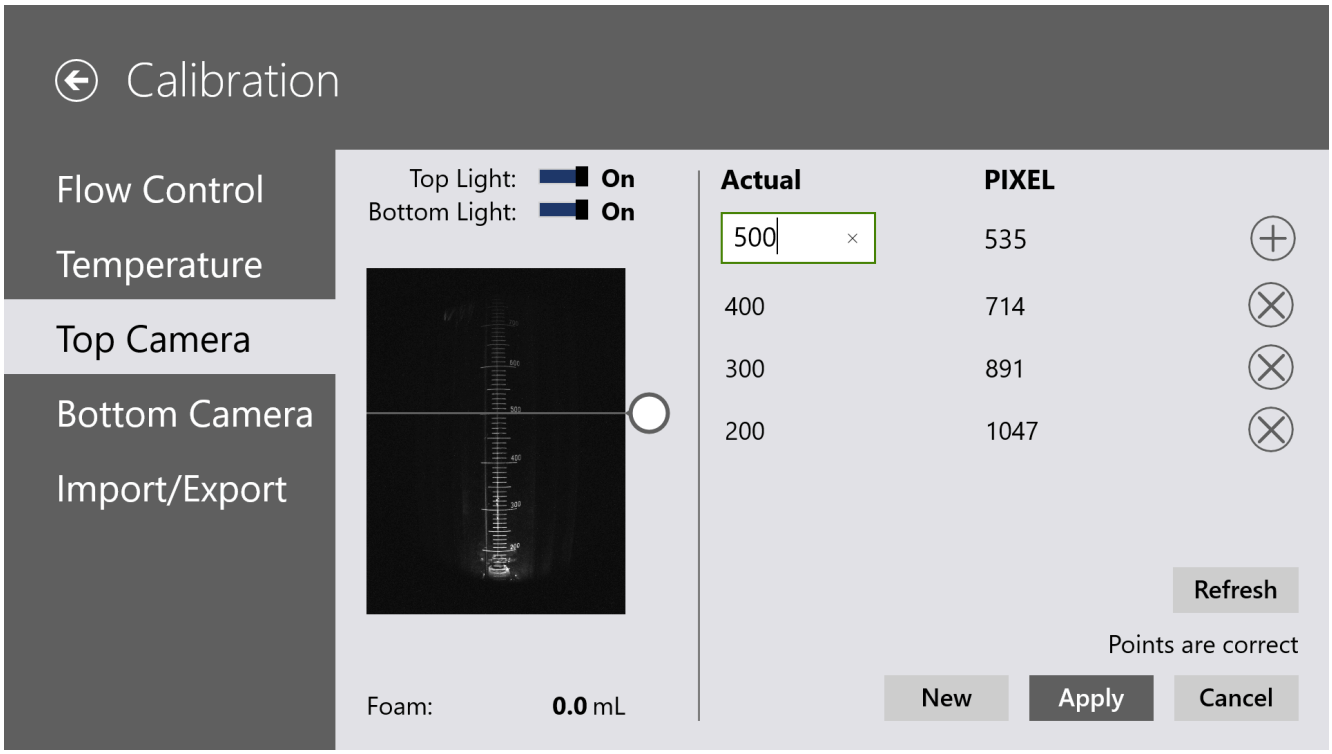

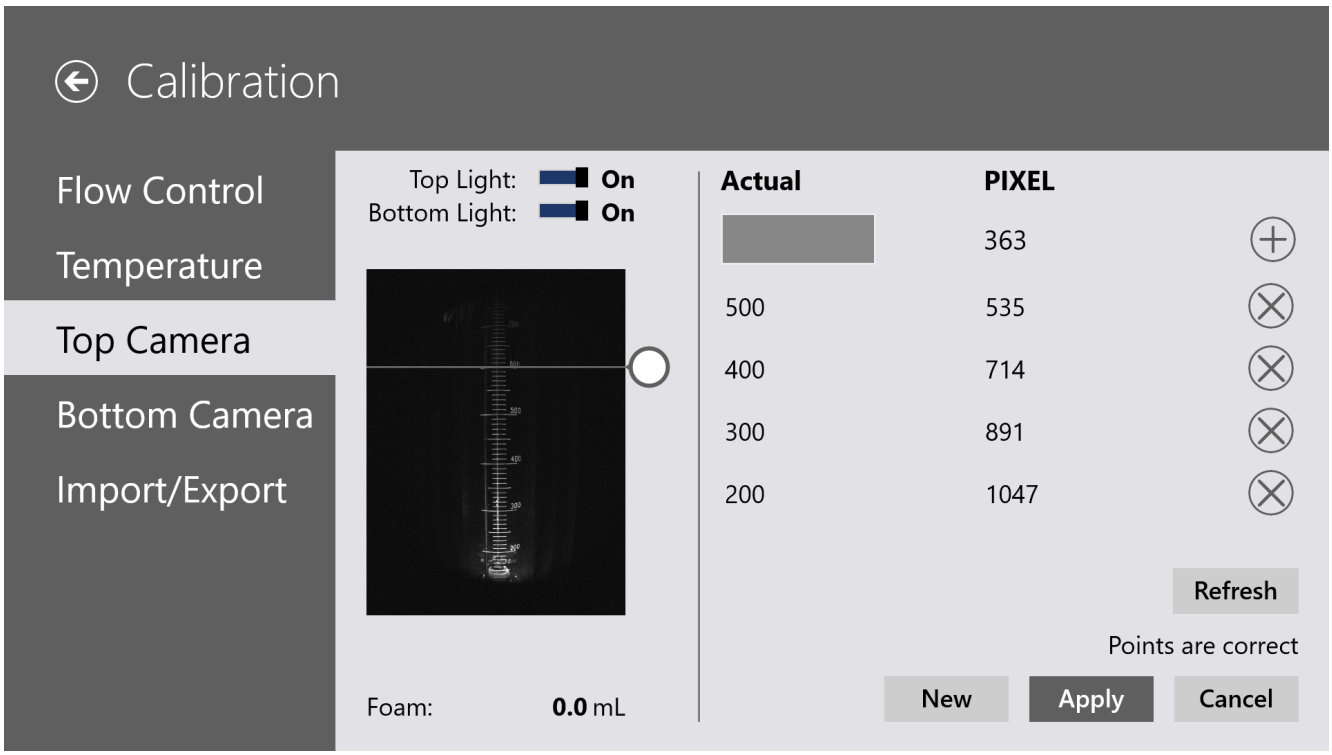

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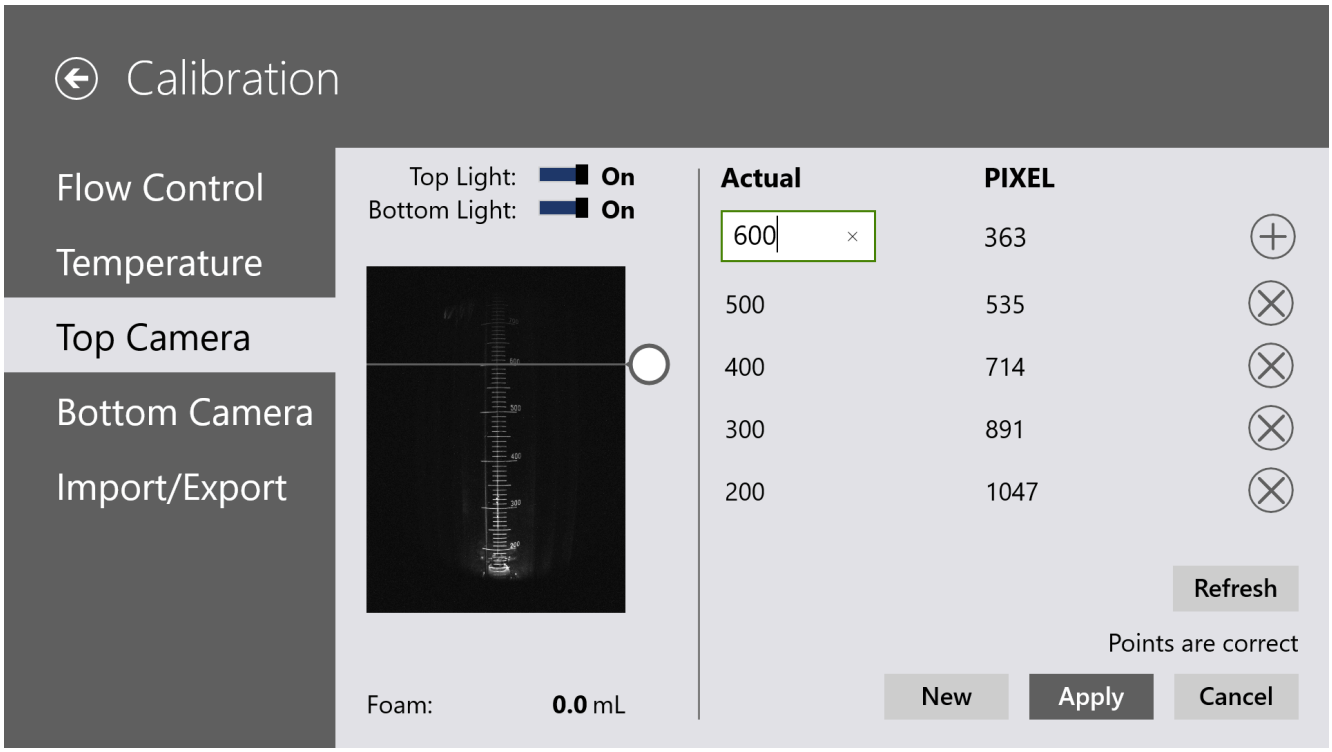
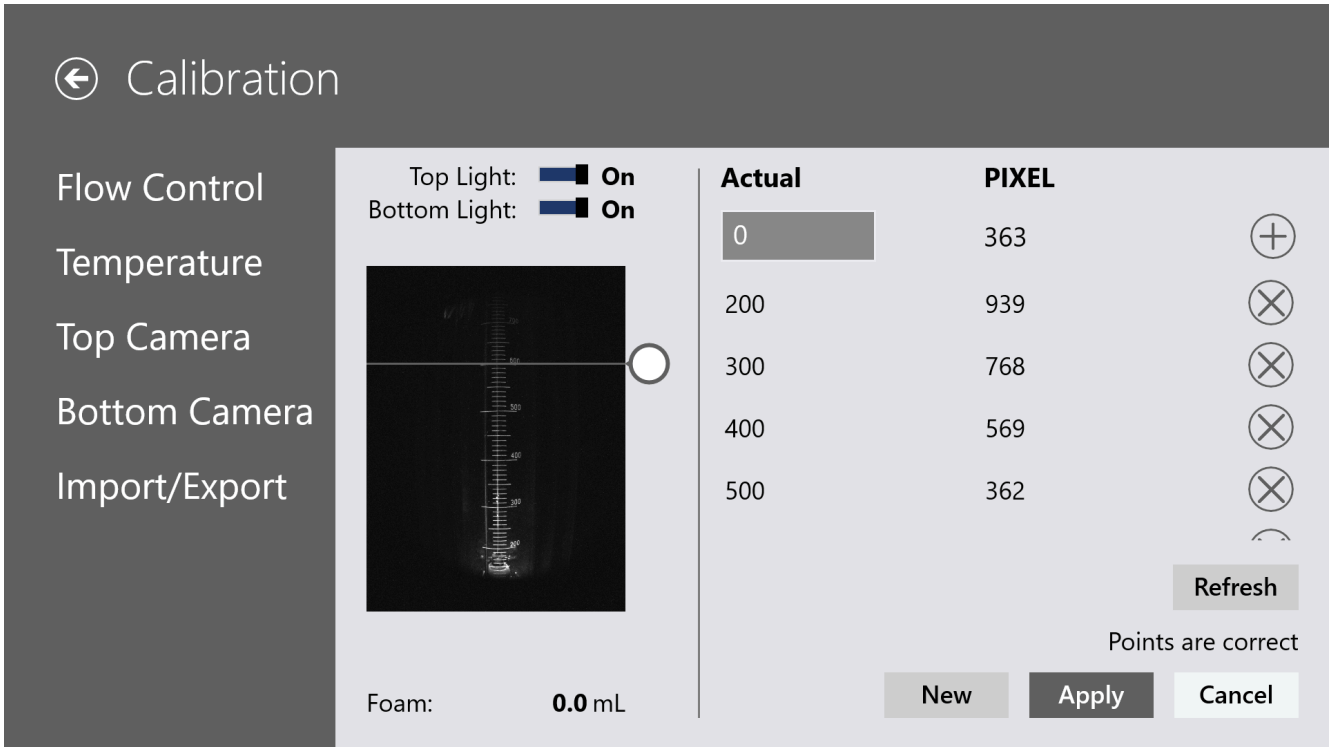


Once all graduations have been added, click Apply to confirm and save the calibration.

Note: Once applied and confirmed, the old calibration is disregarded.



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If the operator wants to return to the prior calibration, they can simply click cancel at any time *BEFORE* confirming the calibration with the Apply button.

To delete only part of the calibration, select the X icon on the right side of the screen.

To undo a deletion, click Refresh.

5.6.3.4. Bottom Camera Calibration

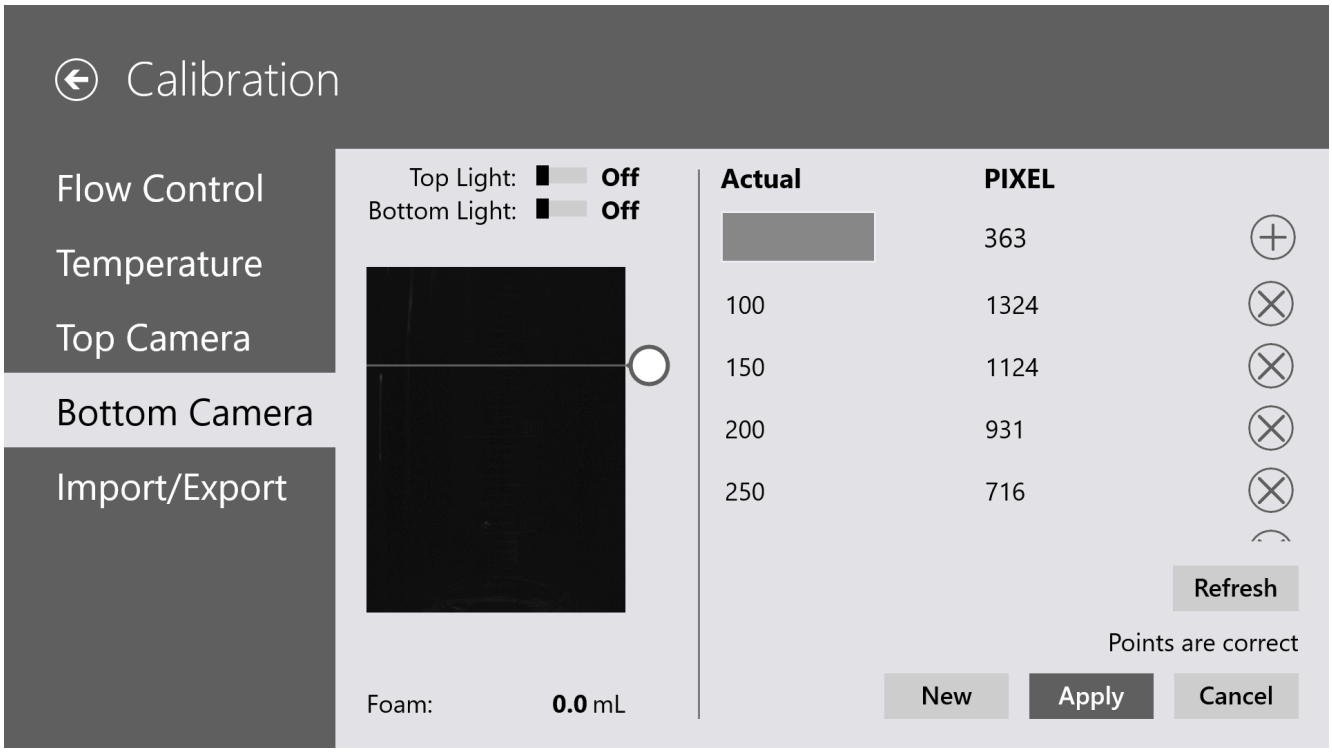
To calibrate the vision system, an empty sample cylinder is suggested.

The vision calibration procedure works by adjusting the algorithm graduations using the calibration sample cylinder.

Procedure:

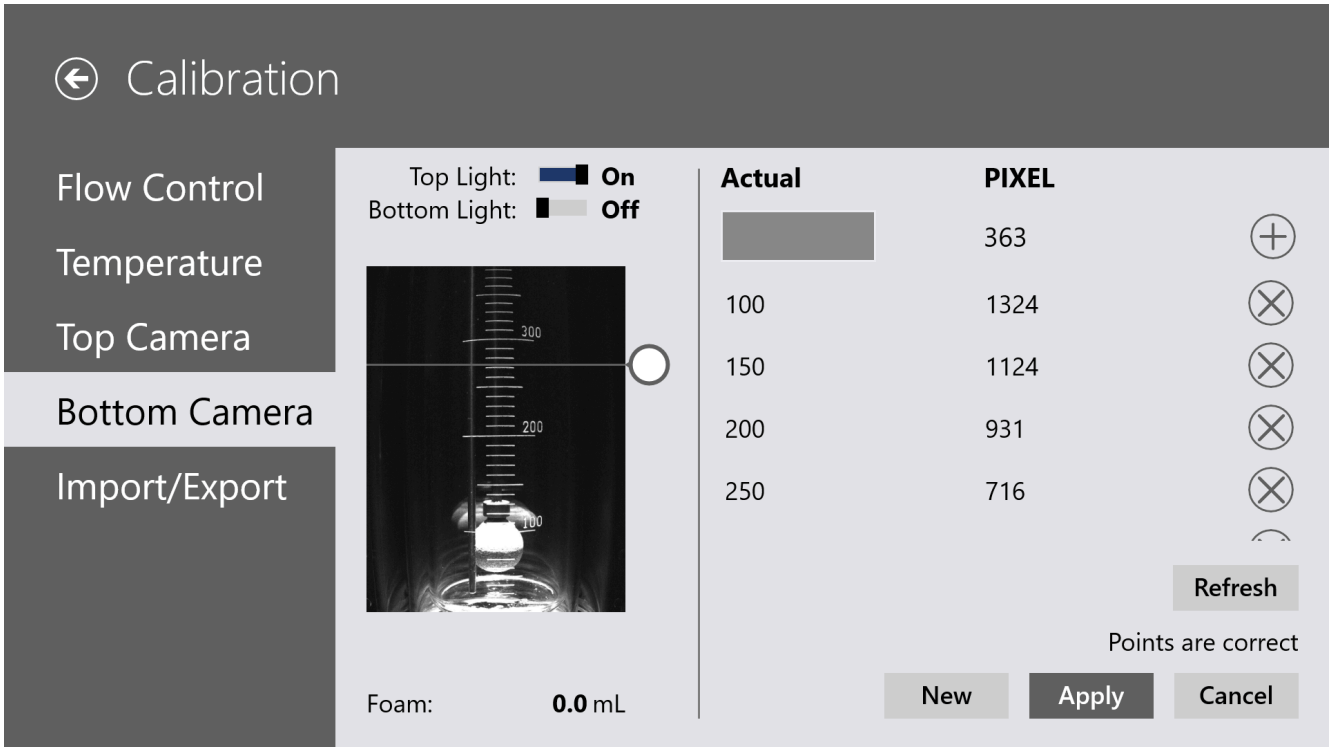
Ensure that the graduations on the sample cylinder are rotated in a way that they are facing the camera (backwards).

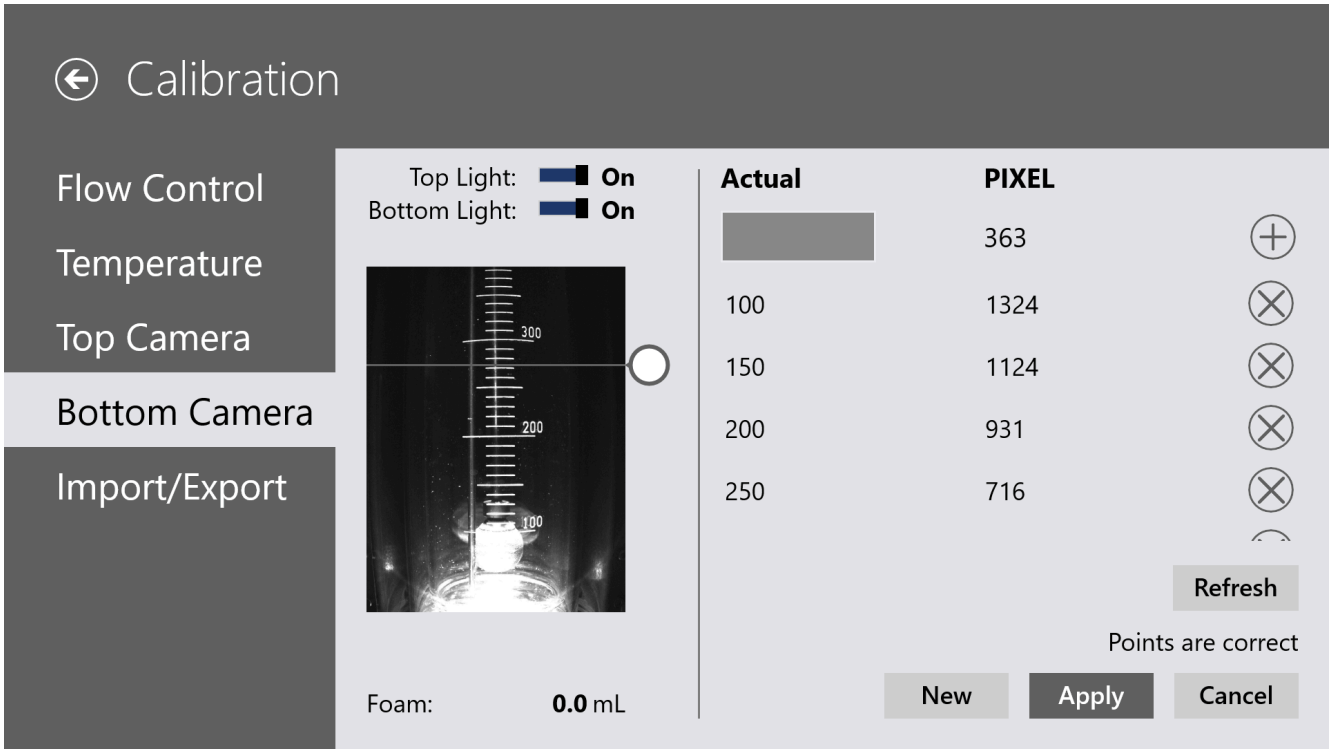
Once Calibrate is selected for a module under the initial calibration screen, navigate to the Camera tab.



The procedures for the bottom camera are exactly the same as the top camera, but the graduation set points are more zoomed in.

Turn on both the top and bottom light by using the On/Off toggles at the top left of the screen.





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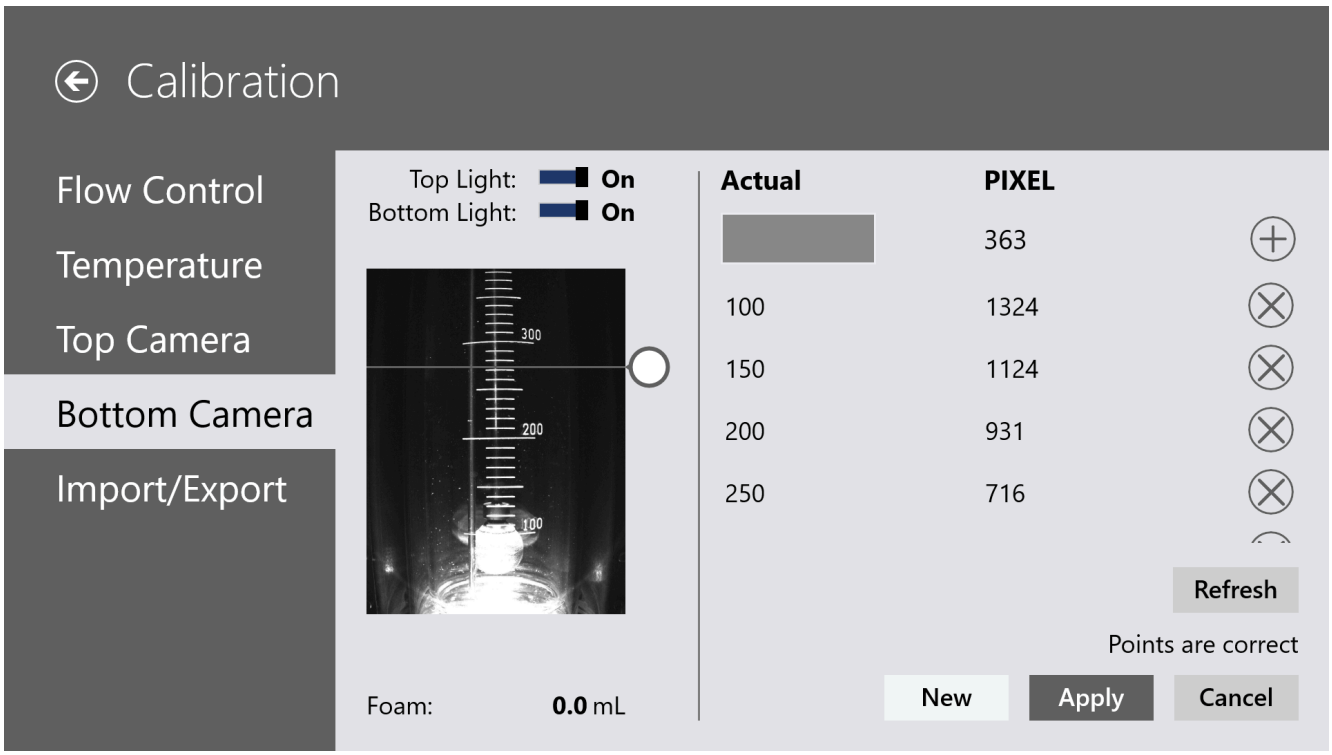
Click New to start a new calibration and disregard the previous calibration.



Determine which graduations to use for the set points.

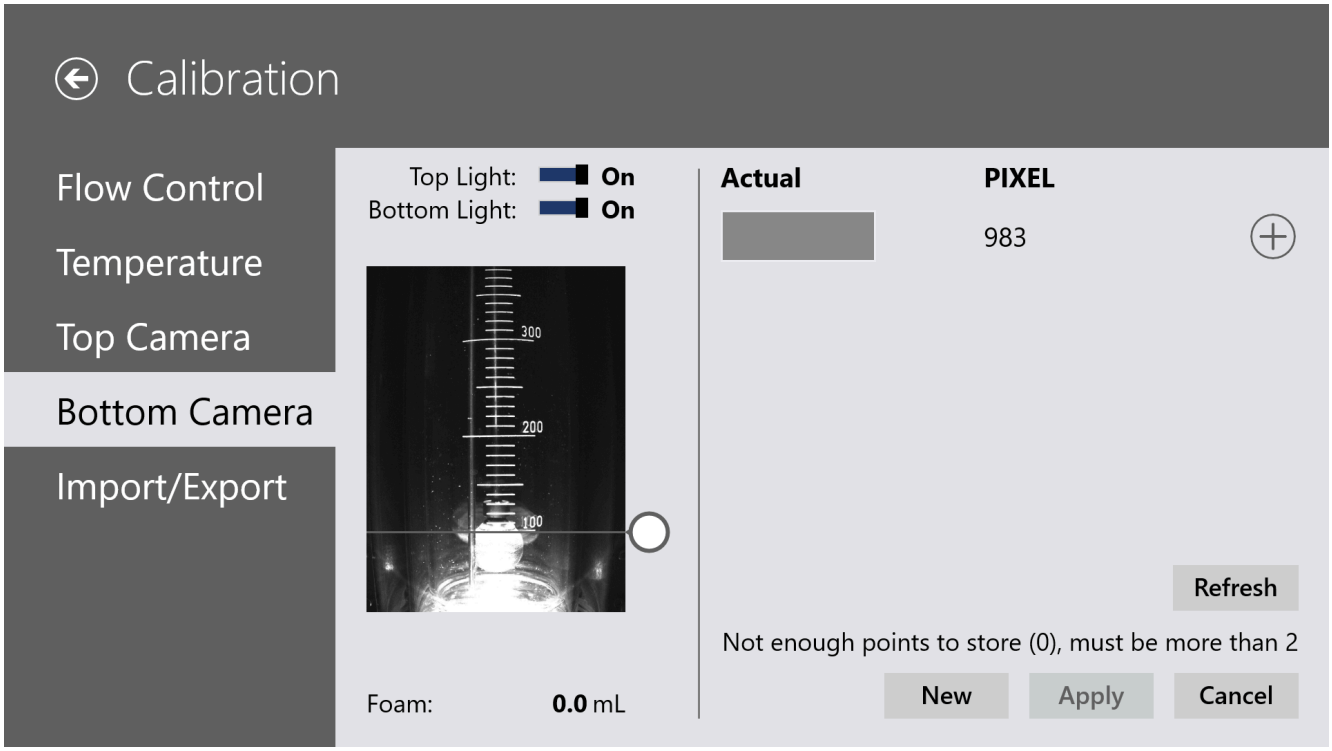
For a 1000 mL sample cylinder (lubricant variant), the following are recommended:

- 100 mL
- 150 mL
- 200 mL
- 250 mL
- 300 mL
- 350 mL

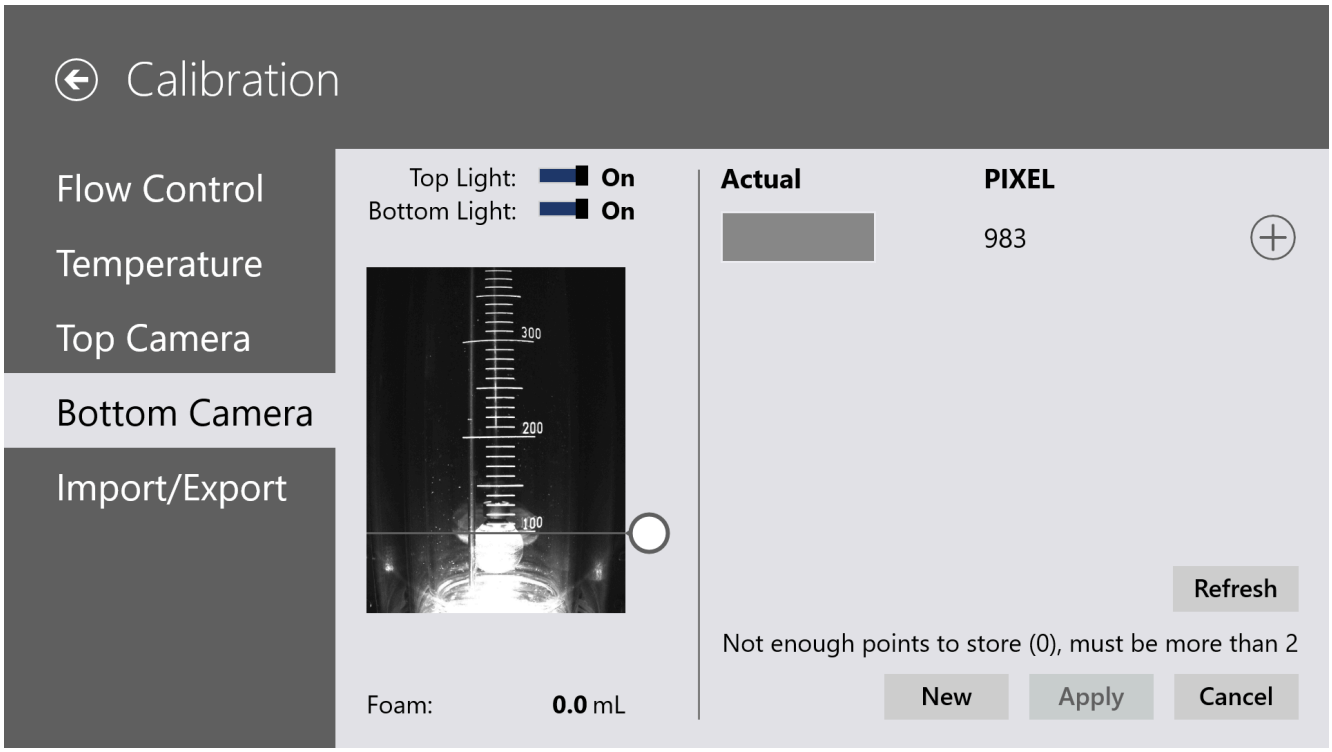
For a 500 mL sample cylinder (coolant and aminic variant), the following are recommended:

- 50 mL
- 100 mL
- 150 mL
- 200 mL
- 250 mL
- 300 mL

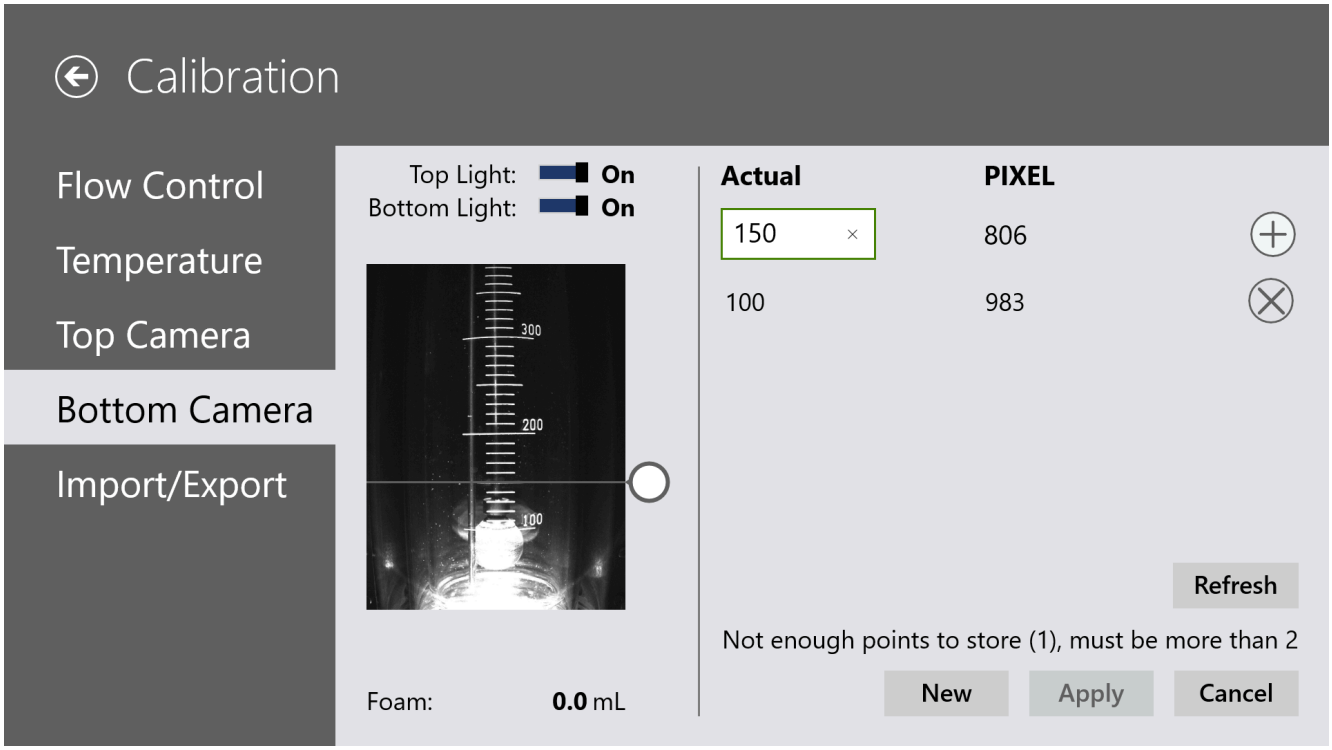
Drag the slider to the graduation mark in the live photo. Due to the angle of the camera, ensure the meniscus of the sample would be the measurement level.



Once the level is accurately marked with the slider, type the graduation in the Actual column.



Click the plus (+) icon on the right to add the graduation set point to the new calibration.



Repeat this procedure for at least one more graduation.

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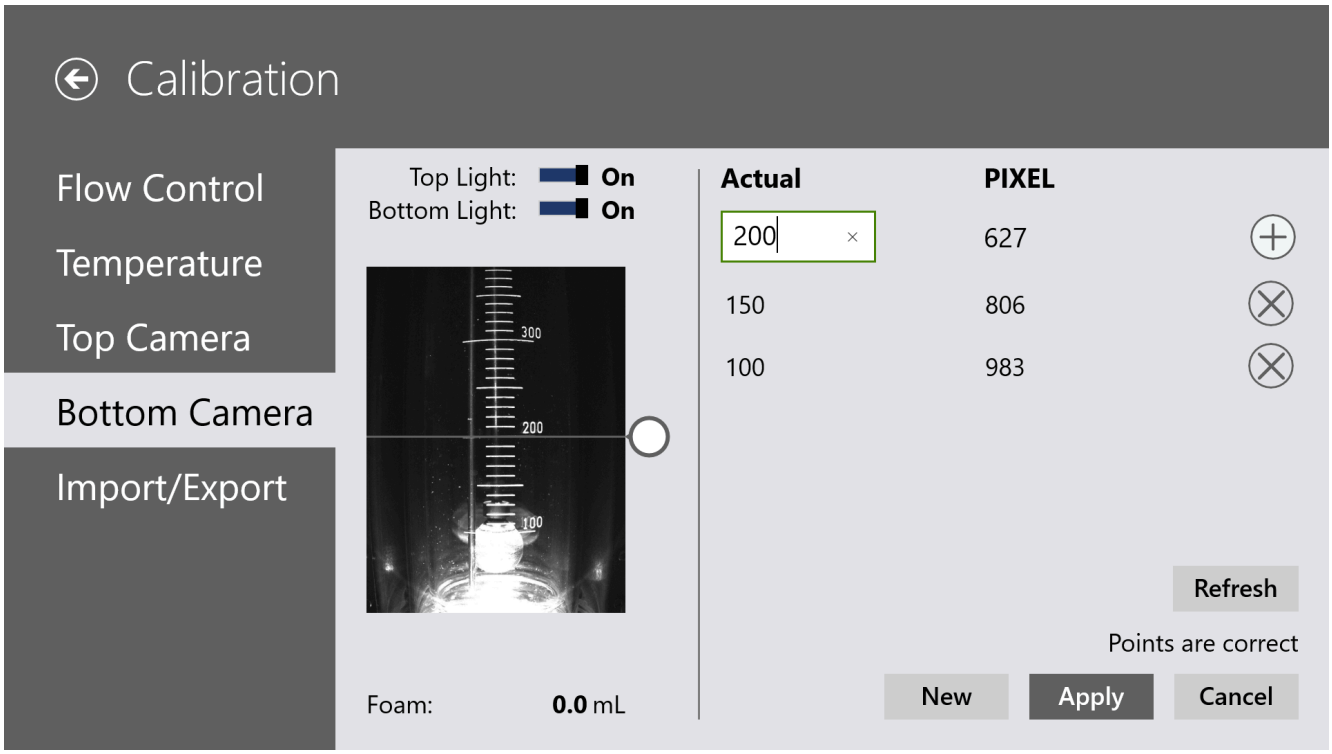
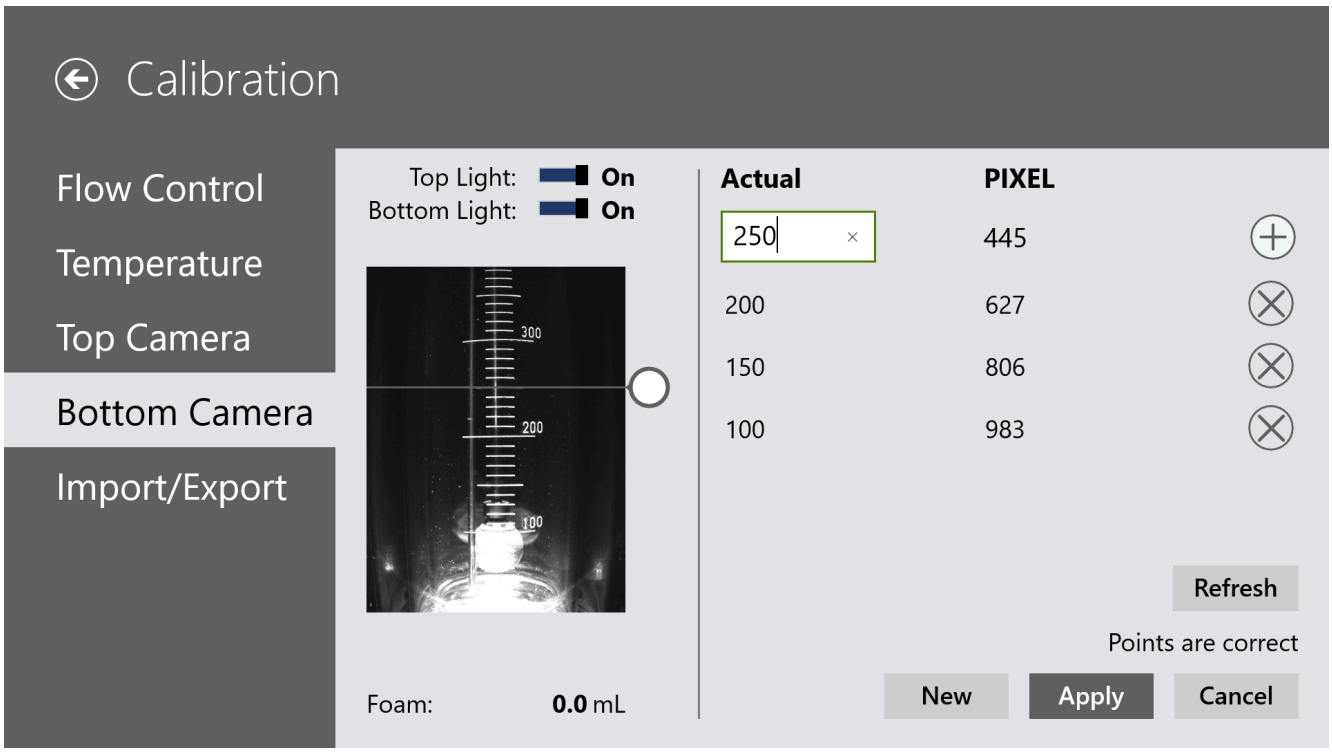


Continue until all desired set points have been added to the calibration on the right.



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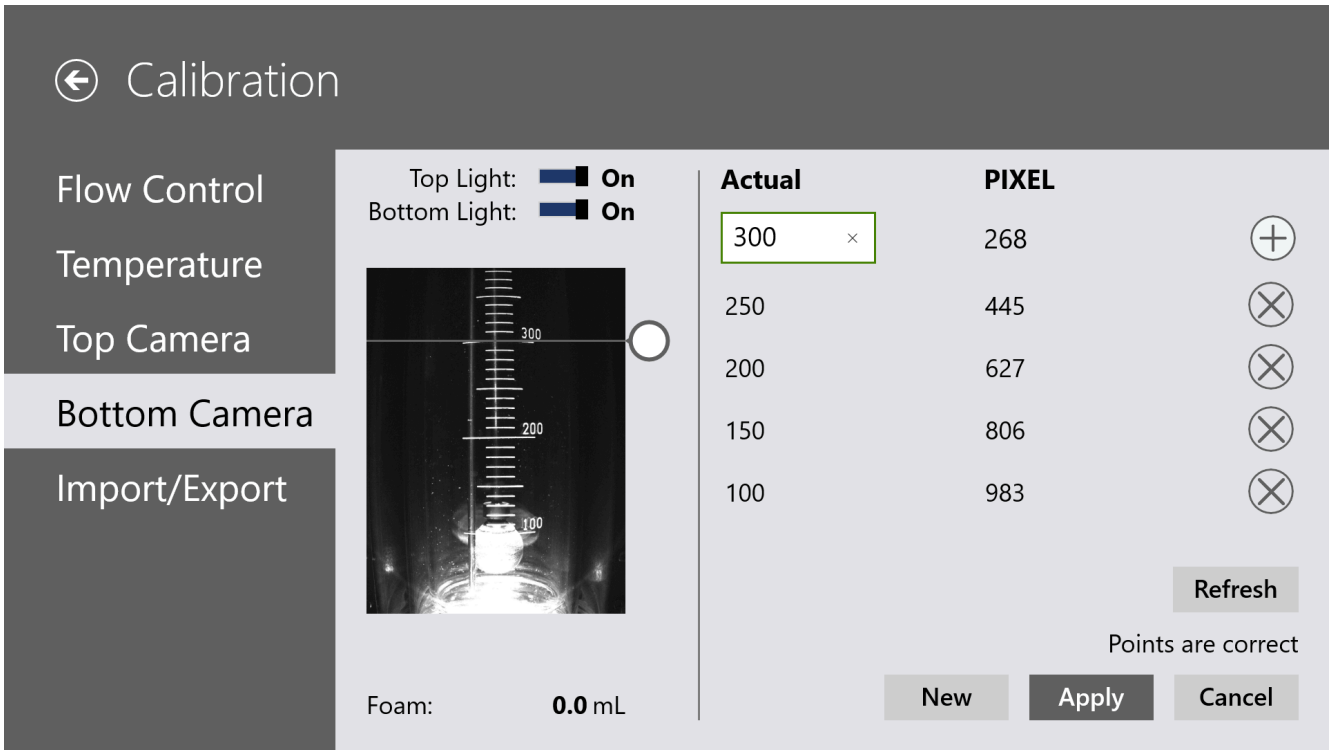

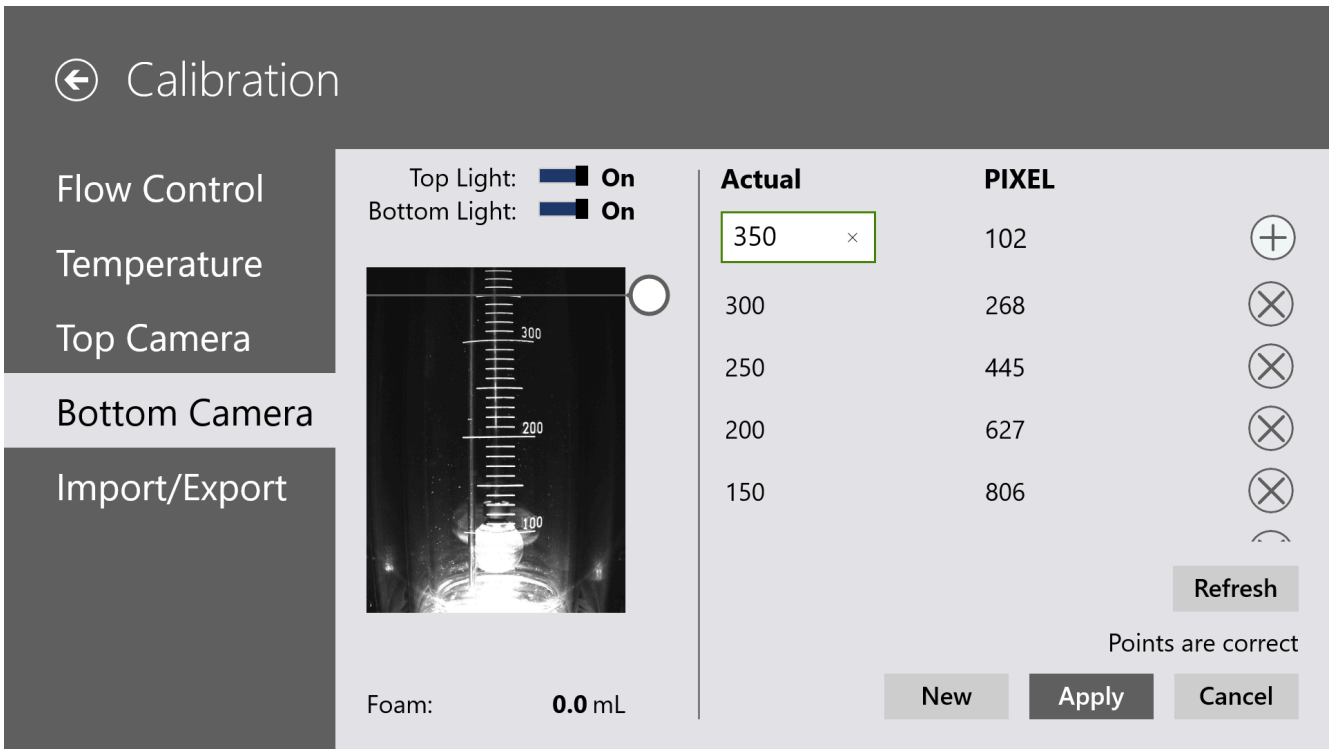

Once all graduations have been added, click Apply to confirm and save the calibration.

Note: Once applied and confirmed, the old calibration is disregarded.

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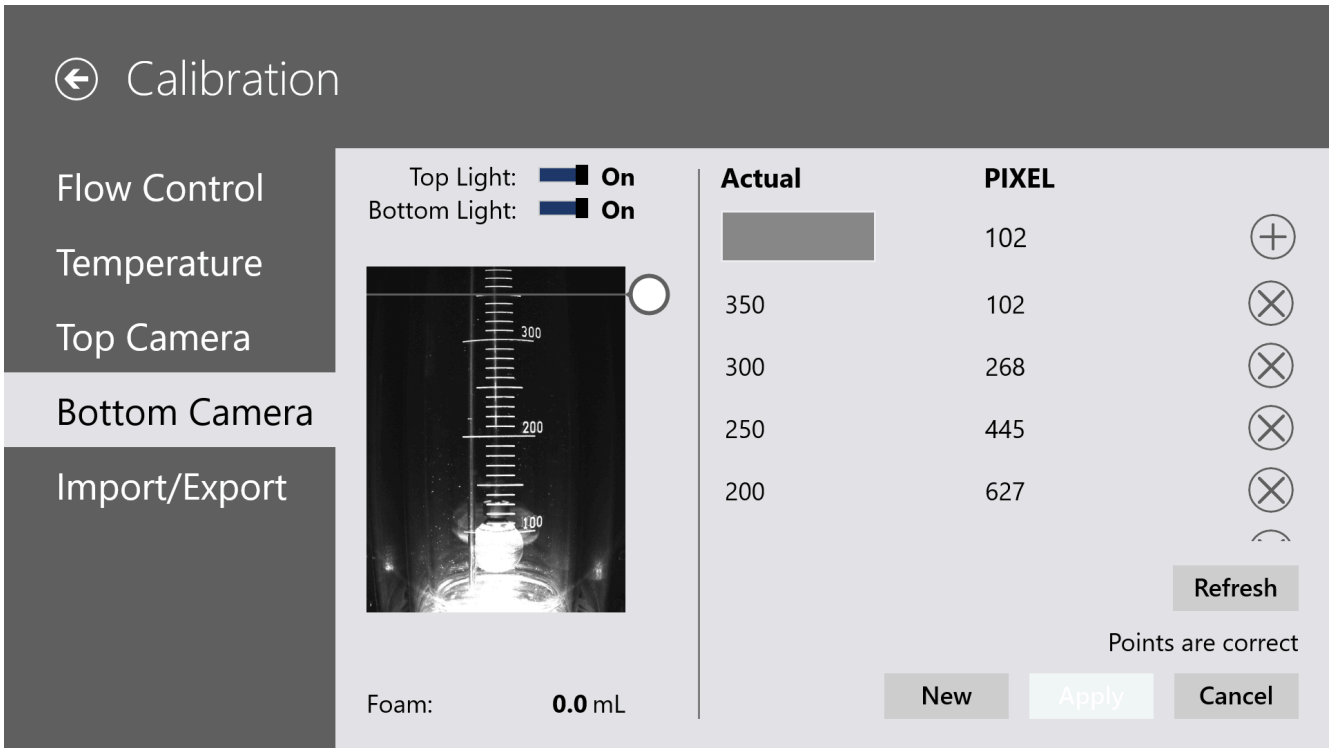


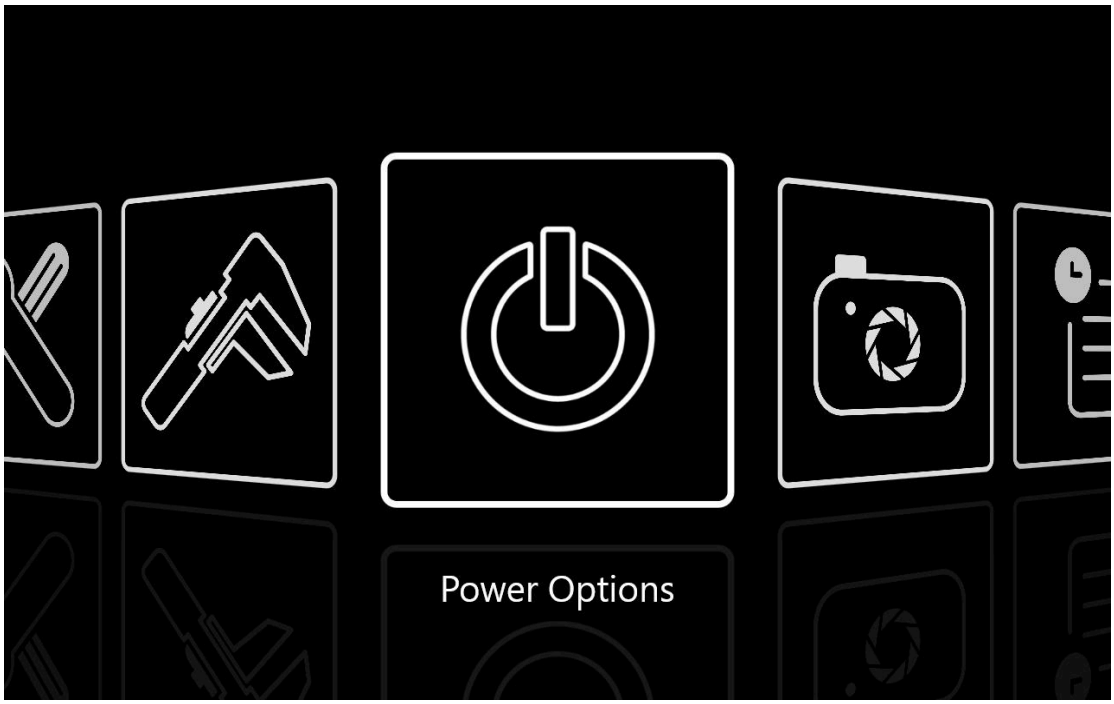
If the operator wants to return to the prior calibration, they can simply click cancel at any time *BEFORE* confirming the calibration with the Apply button.

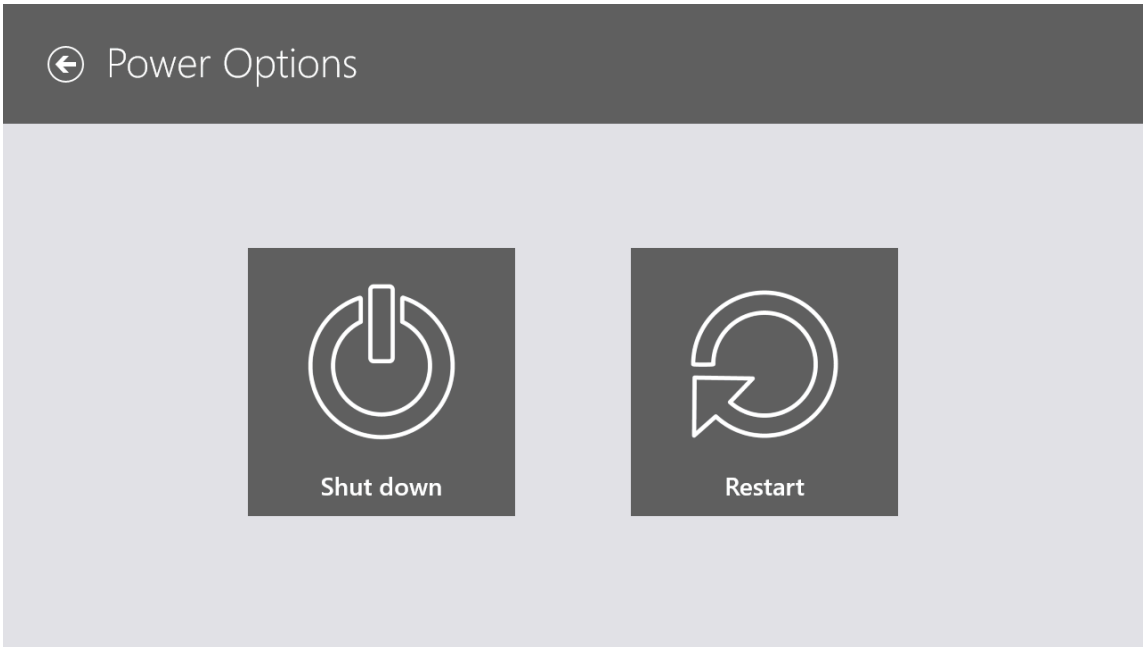
To delete only part of the calibration, select the X icon on the right side of the screen.

To undo a deletion, click Refresh.

5.7. Power Options

The Power Options menu allows users to restart or shut off the FoamDDI via the Main Screen. The FoamDDI can be easily powered down and restarted using the Power Options screen. Due to an embedded OS controlling the FoamDDI, it is important that the FoamDDI is properly shut down and restarted.

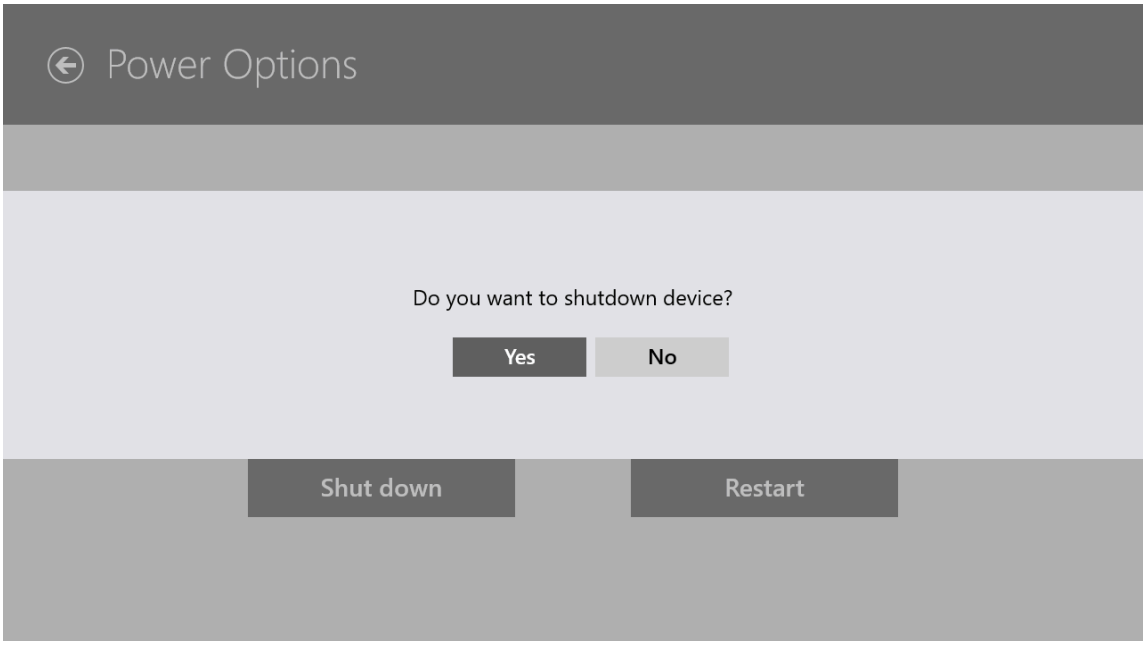




5.7.1. Shut Down

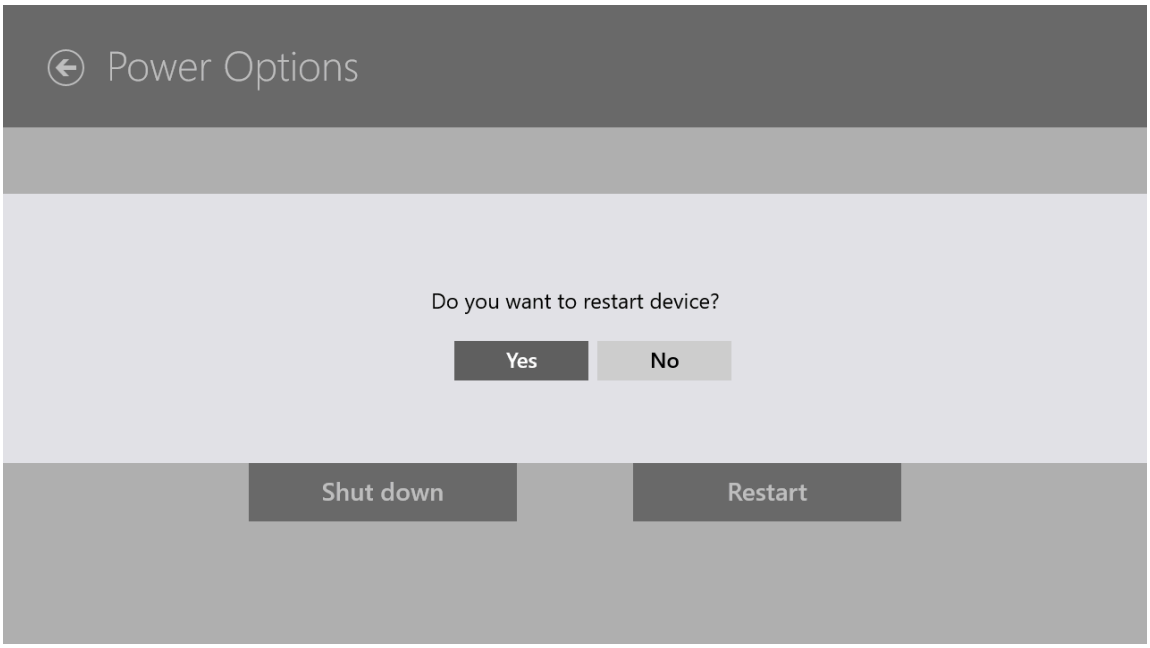
To shut down the FoamDDI, press the Shut Down button. You will be alerted with a screen to confirm you would like to shut down the unit. Press the OK button to proceed. This process takes approximately 10 seconds to complete. The rear Power button can now

safely be used to completely turn off the unit. To restart the FoamDDI, press the Power Reset button on the front panel



5.7.2. Restart

To restart the OS and software, simply press the Restart button. You will be prompted to confirm a restart. Press OK to proceed. This will restart the FoamDDI software.



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6. Troubleshooting

The FoamDDI continuously monitors hardware, software, and communication processes for any irregularities.

When a fault occurs, an error message will appear on-screen describing the issue.

The list below includes the most common messages, their meaning, and recommended actions.

Note: If an error persists after retrying or restarting the device, record the full message and timestamp from the Log screen and contact VISAYA Service or your authorized distributor for assistance.

6.1. Air Supply & Flow

Issue / Message	What It Means	What To Do
No Air Detected / Timeout	Air not enabled or below 20 PSI	Verify compressor on; check 20–40 PSI supply; restart sequence
PSI Too Low (< 2 PSI)	Insufficient pressure prevents air from exiting diffuser; no foam forms	Increase pressure to 20–40 PSI; inspect tubing for leaks or kinks
PSI Too High (> 60 PSI)	Pressure exceeds MFC range, causing timeout or hardware risk	Depressurize system; reset to 20–40 PSI; retry
Air Not Dry	Moisture affects foam characteristics	Replace desiccant in drying tower (pink = spent)
Air Not Oil-Free	Oil contamination restricts flow or damages MFC sensor	Replace inline air filter (red = spent)
Insufficient Flow	Flow < 2 SLPM prevents target rate	Check compressor capacity; verify fittings; restart
Vent Valve Not Operating	Back pressure not relieved; inaccurate foam	Toggle vent in Diagnostics; verify flow equalization
Air Jet Not Operating	Optional cool-down valve disabled or disconnected	Check valve connection to vBoard; enable in Settings

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6.2. Power System

Issue / Message	What It Means	What To Do
No Power During Startup	Module not powered; not detected by software	Check rear toggle switch, verify 24 V/12 V/5 V LEDs on vBoard
Power Loss While Idle / Testing	Module temporarily lost connection	Restore power; allow 30 s for LogicBox to reconnect
Voltage Too Low / High	Incorrect IR lamp or mains voltage	Confirm correct IR lamp type and regional power (110–125 VAC / 220–240 VAC)
No Ground Connection	Improper grounding affects module communication	Verify proper ground on all power cords
Module Not Found (USB)	No USB connection between vBoard ↔ LogicBox	Reseat VISAYA USB cable; use only factory cable
Incorrect USB Cable	Third-party cable may corrupt EEPROM	Replace with VISAYA cable; power down module before disconnecting
Incorrect Firmware	Board uncommissioned or outdated	Contact VISAYA Service for firmware update

6.3. Operating Environment

Issue / Message	What It Means	What To Do
Ambient Too Cold (< 15 °C)	Heating slow; may not reach target	Maintain 15–30 °C environment; avoid cold drafts
Ambient Too Warm (> 30 °C)	Cooling limited; may not reach setpoint	Move to cooler area; ensure ventilation
Unit Not Level	Vision misreads liquid curvature	Re-level using bubble or digital level; verify straight liquid line in live view

6.4. Sample Preparation and Setup

Issue / Message	What It Means	What To Do
Glassware Not Clean	Vision detects residue; image clarity low	Clean all glassware before use
Sample Level Too High / Low	Foam height cannot be measured accurately	Adjust sample to 180–190 mL

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Sample Too Dark	Camera contrast too low	Use Manual Override to rate foam; contact VISAYA for updates
Stopper Not Seated	Air leak detected (Flow Totalizer Test Failed)	Re-seat stopper; lightly lubricate O-ring if needed
Diffuser Not Clean / Certified	Residue alters foam rating	Clean and certify diffuser per ASTM D892
Inlet Tube O-Ring Missing	Leak between diffuser and inlet	Replace O-ring; verify seal
Graduations Visible to Camera	Vision misreads foam due to markings	Rotate graduations to side (left / right)
Air Not Connected / Not Sealed	Flow test fails or leaking O-ring	Verify quick-disconnects and replace damaged seals

7. Maintenance

Routine maintenance is essential to ensure accurate results and long service life of the FoamDDI. The following procedures should be performed regularly by trained operators.

Your FoamDDI is designed to operate worry-free for years. Use only VISAYA-certified parts and consumables. Never operate without the protective outer glass installed. In the event your FoamDDI needs to be replaced, a replacement can be done quickly in the field with a software-driven calibration procedure.

Ensure proper handling and refrain from dropping FoamDDI on the laboratory floor. In the event the machine is dropped, ensure there are no cracks or breaks, and that the machine is fully operational. One can also check the operation in the Diagnostic menu of the software to ensure mechanics and electronics are responding correctly.

7.1. Self Verification

Operators should verify basic functionality before daily operation:

- Confirm software boots normally.
- Run Diagnostics menu (Vision, Airflow, Temperature).
- Verify cylinder drains cleanly after rinsing (no drop formation).
- Ensure diffuser airflow is steady and free of blockages.

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7.2. Cleaning

Proper cleaning is crucial to prevent contamination and ensure reproducible FoamDDI results. Due to the requirement to use dangerous solvents, such as heptane and toluene (Warning: Flammable, vapor harmful), which often have low flash points, adequate protection (clothing and ventilation) must be provided, and operators must avoid the possibility of fire or explosion.

Furthermore, the instrument must not be sterilized using an autoclave or any other mechanical device.

Thorough cleansing of the test cylinder and gas diffuser, and air-inlet tube is essential after each use to remove any additives remaining from previous tests, which can seriously interfere with the results of subsequent tests.

7.2.1. Cylinder

Thorough cleaning of the graduated sample cylinder is essential after each use to prevent contamination and ensure reproducible results.

Criterion for Cleanliness: The interior walls of the cylinder must drain water cleanly, without drops forming.

Cleaning Steps:

- Rinse the cylinder with heptane.
- Wash the cylinder with a suitable detergent.
- Rinse sequentially with distilled water.
- Rinse with acetone (Warning: Extremely flammable).
- Dry with a current of compressed air or in a drying oven.

Note: If detergent residue adheres to the glass, several rinsings with water and acetone may be required.

7.2.2. Gas Diffuser and Air-Inlet Tube

The diffuser and air tube must have no visual evidence of residual material remaining from a previous analysis.

Note: DO NOT WIPE THE GAS DIFFUSER.

Air Tube Cleaning Steps:

- Disassemble the air tube from the diffuser.
- Immerse the connected diffuser in about 300 mL of toluene.
- Flush a portion of the toluene back and forth through the diffuser at least five times using vacuum and air pressure.
- Repeat the flushing process using heptane.
- Dry both components thoroughly by forcing clean air through them.
- Wipe the outside of the air inlet tube with a cloth moistened with heptane, then a dry cloth.

Note: Heptane and Toluene are dangerous chemicals and should be handled with care. Refer to the SDS sheets that accompanied the chemicals.

Diffuser Cleaning Steps:

- Attach the diffuser to a compressed air system.
- Submerge the diffuser in Heptane.
- Blow air through the diffuser for about five minute.
- Remove the diffuser above the surface of the Heptane.
- Blow air through the unsubmerged diffuser.
- Submerge the diffuser in IPA (Isoppropyl Alcohol).
- Blow clean, dry air through the diffuser once again.
- Remove the diffuser to just above the IPA surface.
- Blow air through the unsubmerged diffuser.
- Allow adequate and sufficient time for the diffuser to completely dry.
- To ensure dryness, wait 60 minutes before validation.

Note: There are other methods to clean both the tube and diffuser; these are the recommended procedures.

7.2.3. Exterior

General cleaning maintenance is recommended.

Instrument Exterior: The exterior enclosure can be cleaned using a mild detergent and a non-abrasive cloth. Acceptable methods include using a clean dry cloth, a cloth dampened with water, or a cloth dampened with soapy water (using domestic soap only; no other chemical detergent or chemical may be used).

Touchscreen: Use a soft, lint-free cloth that is dry or dampened with water, eyeglass cleaner, or a screen cleaning wipe. Do not use glass or other chemical cleaners on the touchscreen surface.

Camera Lens: If cleaning is necessary, first remove dust using a blower or a lens cleaning brush. Smudges can be cleaned using lens cleaning tissue, potentially dampened with reagent-grade isopropyl alcohol.

Lightbox Reflector Panels: Can be spot cleaned if necessary. The material is chemically resistant to solutions like ammonia, soap, and 409, but not compatible with bleach or caustic.

7.3. Maintenance Schedule

7.3.1. Daily

Before Each Test	Sample Prep	Verify sample cylinder and diffuser are clean	Clean glassware with lint-free cloth and approved solvent
Daily	Air Supply	Confirm 20–40 PSI, dry, oil-free air	Check compressor, desiccant (pink = replace), filter (red = replace)
Daily	Self Verification	Ensure air flow, temperature, and vision are operating properly	Test each spec in the Diagnostics panel
Daily	Air Connections	Ensure diffuser and return lines sealed	Inspect O-rings, tighten quick-connects

7.3.2. Weekly

Weekly	Vision	Check live camera image clarity/focus	If unfocused, clean lenses with lens tissue and isopropanol before readjusting focus
Weekly	Temperature	Verify heating/cooling via Diagnostics	Confirm IR lamp glows; cooling responds

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Weekly	Fans & Valves	Test Peltier, Chassis, Bath, and Vent valves	Toggle each in Diagnostics
Weekly	Air Supply	Inspect tubing and fittings	Look for kinks or leaks

7.3.3. Monthly

Monthly	Level & Environment	Ensure instrument is level and ventilated	Use bubble/digital level; keep vents clear
Monthly	Electrical	Inspect cables and connectors	Replace worn power or USB cables
Monthly	vBoard	Confirm 24 V / 12 V / 5 V LEDs lit	Verify proper grounding
Monthly	Sample Hardware	Replace worn O-rings and lubricate stopper	Use compatible oil for test sample

7.3.4. Annually

Quarterly	Air Supply	Replace desiccant and inline filter	Depressurize before service
Quarterly	Calibration	Verify flow and temperature calibration	Compare Diagnostics values to standards
Quarterly	Firmware	Check for updates	Contact VISAYA Service for assistance
Annually	System	Perform full calibration and diagnostics	Run system verification routine
Annually	Cameras	Recommission modules if replaced	Required if camera serials changed
Annually	Cooling System	Replace fans showing wear or noise	Check direction and RPM
Annually	Documentation	Record all maintenance	Log in LIMS or service binder

7.4. Alignment

Regularly inspect the sample cylinder alignment during installation.

Adjust stopper rings if the cylinder rotates under the weight of connections.

Check optical alignment through Diagnostics → Vision menu.

7.5. Lubrication

No lubrication is required for standard operation. If cylinders or stoppers become difficult to install or remove, contact VISAYA service before applying any lubrication.

However, the stopper can sometimes become stuck in the cylinder. To prevent this issue, you can lubricate the O-rings on the stopper with a light brush of the lubricant being tested. If you are conducting non-lubricant samples, do not lubricate the O-rings and rely only on twisting the stopper until it becomes free of the cylinder.

8. Service

Service on the analyzers should only be conducted by trained personnel. Reach out to the local distributor or Visaya personnel for any service.

8.1. Internal Components

8.1.1. Access

Voltages present inside the instrument are potentially dangerous.

If there is a problem with the instrument, the power cable should be removed until qualified service personnel have repaired it.

Cleaning, dismantling, maintenance, adjustment, and repair should generally only be performed by personnel trained in such work. It is recommended to contact VISAYA Technical Service for service procedures.

Note: Attempting to service the instrument without proper training can invalidate warranty claims.

Accessing the internal components of the FoamDDI module for service or maintenance is typically done by removing the side panels of the instrument.

The instrument (or similar VISAYA modules like the CuDDI and FeDDI, which share service access methods) is designed to function properly even without the side panels in place, allowing access to internal components.

Steps to Remove the Side Panels (Access):

The procedure for removing the side panels to access the internal components is as follows:

- Disconnect Power: Always disconnect the power cord of the equipment before handling internal components to ensure safe handling.
- Locate Screws: Locate and remove the four retaining screws holding both or one white, upper side panel in place.
- Use an M2.5 screwdriver to loosen the bottom two screws before removing the top two screws keeping ahold of the panel so it does not fall when the last screw is removed.
- If lower compartment access is required, continue to unscrew the black panel screws.
- Remove Panels: Gently pull the panels straight out. You can typically remove either side or both, as there are no internal linkages tying them together.

Once the side panels are removed, you gain access to internal components like the cameras for focus adjustment. For the FoamDDI module specifically, internal components include the Camera, IR lamp, fans, valves, and flow control systems.

Steps to Restore the Side Panels:

- Position the Unit: Turn the unit completely to its side so the mounting holes for the panel you are installing face towards you.
- Align Panels: Align the side panel with the chassis. You should feel a small lip on the edge of the panel notch into place when correctly positioned.
- Secure: Replace and tighten the retaining screws.

8.1.2. Identification

Internal assemblies include:

Power supply and fuse block

Embedded industrial PC (IPC)

Camera and vision system

Diffuser air system and pneumatic connections

Heating module and IR lamp

8.2. Electrical

The electrical configuration involves two main components: the FoamDDI module itself and the IPC Logic Box controller.

8.2.1. FoamDDI Module Power (Testing Unit)

- The FoamDDI Modules are powered using 110/120 VAC or 220/230 VAC at 50/60 Hz main power supply.
- The internal heating element, the Infrared (IR) Lamp, is specifically configured based on the input voltage, using either a 110VAC lamp or a 240VAC lamp. The IR Lamp is used to heat and maintain the sample temperature.

Note: Changing the voltage configuration must be performed by a trained VISAYA personnel, as connecting the wrong voltage may cause damage.

8.2.2. IPC Logic Box Power (Control Unit)

- The IPC Logic Box utilizes an external voltage converter that features a universal switching power supply.
- This converter is capable of accepting power across a wide range, from 90 ~ 264VAC and frequencies ranging from 47 ~ 63Hz.

8.2.3. Fuse Specification and Purpose

The fuse is a critical safety component used to protect the instrument from electrical damage.

- Required Fuse Rating: For proper and safe operation, you must only use the specific type of fuse described in the documentation: 4.0 Amp type "T" slow blow.
- Fuse Location and Function: A fast-acting fuse is generally placed near the power input on the rear of the module. This component is crucial because it protects the instrument from power surges and internal shorts.
- Safety Requirement: The fuse helps ensure the leakage current of the instrument remains within the limits allowed by international safety standards for laboratory equipment.

8.2.4. Safety and Operation Precautions

Hazardous Voltages: Voltages present inside the instrument are potentially dangerous.

Power Disconnection: If a problem occurs with the instrument, the power cable must be removed until qualified service personnel have repaired it. This prevents anyone from inadvertently using the instrument, which could cause harm or damage.

Ground Connection: An efficient ground connection is imperative for the physical protection of the user. Static buildup caused by a lack of proper ground connection can lead to false inputs on the touchscreen.

Service Recommendation: If the instrument malfunctions, it is recommended to contact VISAYA Technical Service. Service and repair should generally only be conducted by authorized personnel or the manufacturer.

The specialized 4.0 Amp type "T" slow blow fuse acts as a deliberate weak link in the electrical chain, much like a carefully calibrated dam that bursts exactly when the flood pressure (electrical surge) exceeds a safe level, ensuring that the critical components downstream (the entire machine) are protected from catastrophic damage.

8.3. Pneumatics

The FoamDDI modules require a controlled external air source:

Air Quality: The instrument requires dry, clean, instrument grade air. For materials specifications, the requirement is 40 psi Instrumental Grade air.

Pressure: The required regulated pressure range connected to the rear of the FoamDDI is typically 25-30 PSI or 25-35 PSI. The required air pressure range is also stated as 20-25 PSI Regulated Dry Air.

Dew Point: The air must be dry, ideally to a dew point of −60°C or lower.

Inlet Tubing: Air supply input utilizes 6mm OD rigid tubing connected to the rear of the FoamDDI module.

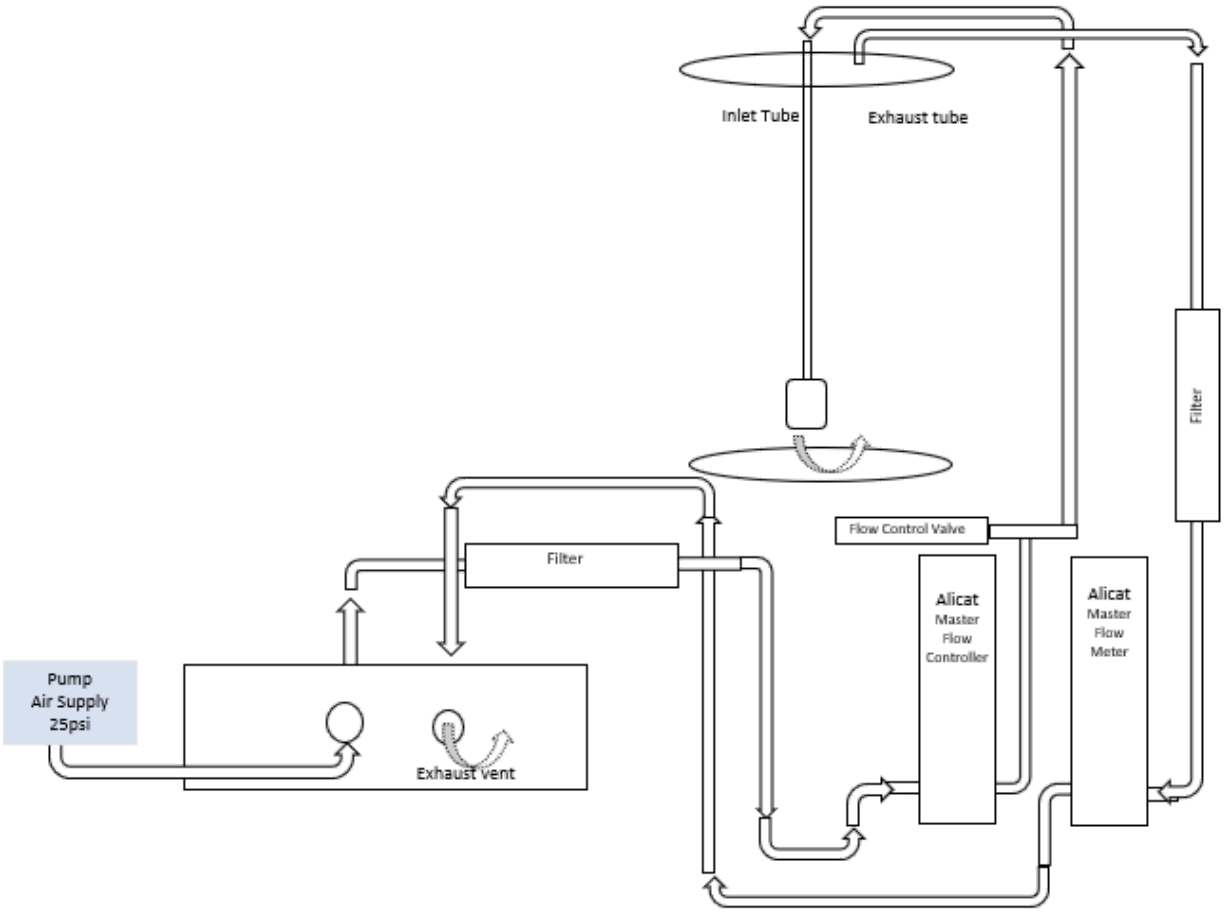
Air Supply Pump/Filter: Optionally, an Air Supply Pump and a Filter, Regulator and Dryer System (Part AYA-15-00055) can be purchased for the FoamDDI. The user supplied air supply may range from 50 to 150 PSI, which is then regulated.

8.3.1. Internal Pneumatic Components and Control

The FoamDDI module is a self-contained single position testing apparatus with internal automated air flow control.

The internal pneumatics system includes the following components:

- Air Flow Control: A Flow Control Valve and Alicat Master Flow Controller regulate the air supply.
- Air Measurement: An Alicat Master Flow Meter (used for mass flow monitoring) measures the return air (Totalizer).
- Piping: Air is supplied from the Pump (25 psi) and passes through an Inlet Tube and an Inline Air Filter.
- Valves: The system includes valves for venting and air jets (Valve(s) Vent and Air Jets). A valve is used to vent the air circuit (between the Mass Flow Controller and Diffuser) to the atmosphere.
- Quick Connectors: Air flow tubes connect to the RDX Stopper & Temperature Sensor Assembly via quick connection valves.



8.3.2. Diffusers and Foam Measurement

The pneumatic system works in conjunction with the selected diffuser and stopper assembly:

- Diffuser Types: The instrument can utilize either a Certified Cylindrical Gas Diffuser or a Certified Norton Spherical Gas Diffuser. The overall test method allows for both circular, Norton ceramic (stone) diffusers and Mott cylindrical metal diffusers.
- Specifications: Standard diffusers must meet specifications: a maximum pore diameter of not greater than 80 µm and a permeability between 3000 mL/min to 6000 mL/min.
- Stopper Assembly: The RDX Stopper Assembly is designed for leak-free operation and contains the quick-connect fittings for the air flow and return tubes, ensuring an airtight seal when inserted into the sample cylinder.

8.3.3. Exhaust and Totalizer

Air exhaust is monitored and managed by the system:

- Exhaust Tubing: Air exhaust uses 8mm OD tubing. Optionally, the exit air can be routed to a fume hood or extraction.
- Totalizer: The Totalizer function measures the accumulated air flow out and the air moving back in (Flow In) via the Mass Flow Monitor. The totalizer tracks both Flow Out and Flow In. Measuring the total volume of air passed through the system is critical to verify the system is airtight.

9. Version Updates

For both software updates and updates to documentation, please visit the VISAYA update center: https://visayaengineering.github.io/foamddi/

9.1. Software Updates

New software updates are supplied when reasonably necessary for same-generation FoamDDIs free of charge by VISAYA.

To acquire the new software update for your analyzer, perform the following steps:

Navigate to the VISAYA update center via the link above.

Go to the FoamDDI tab

Download the updated software in the Downloads section onto a USB drive

Update Only: FoamDDI software — the option customers will need 95% of the time to update their FoamDDI

Full Installer: Includes driver/camera/etc. software

To update your analyzer, perform the following steps:

Plug in the USB drive containing the updated software.

In the Visaya software, go to Settings → About, then click Exit Application.

Log in to VISAYA Admin using the password visaya.

Run the appropriate *.exe installer and follow the prompts (defaults are fine).

In Windows, go to Start → Power → Restart to reboot the LogicBox.

After reboot, verify the version in Settings → About.

To verify successful installation of the new software, navigate to the About screen (**Section 5.4.1**) and check the current version.

Note: It is highly recommended to perform a backup or download the result archive before the update is installed.

9.2. Updates to the Manual

New versions of the manual are updated when deemed necessary. VISAYA will not distribute new versions of the manual unless directly asked by the customer.

In order to see if there is a new version of the manual available, navigate to the VISAYA update center via the link above, and check the document version for the manual available for download under the Documents section.